GRAND CANYON NATIONAL PARK
GRAND CANYON, ARIZONA

Rehabilitate North Rim Infrastructure

GRCA 318719

SPECIFICATIONS

Volume B (Divisions 02-23)

CONSTRUCTION DOCUMENTS

May 2026

NATIONAL PARK SERVICE
DENVER SERVICE CENTER

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SEALS AND SIGNATURES - Volume B

It is a violation of applicable laws and regulations governing professional licensing and registration for any person, unless acting under the direction of the licensed and registered design professional(s) indicated above, to alter in any way the Specifications in this project manual.

	I hereby certify that the portion of the Specifications described below was prepared by me or under my direct supervision and that I am a duly registered Professional Engineer under the laws of the State of Arizona. Ryan Rhoades, PE 05/01/2026 Pages or Specifications divisions or sections covered by this seal: Sections 02 41 00
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DIVISION 02

EXISTING CONDITIONS

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SECTION 02 41 00
DEMOLITION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. General provisions applicable to all demolition and removals.
   2. Civil/site demolition and removals.
   3. Architectural and structural demolition and removals.
   4. Mechanical demolition and removals
   5. Electrical demolition and removals.
   6. Disposal of demolition debris, materials, and equipment.

B. Scope:
   1. Contractor shall provide all labor, materials, equipment, tools, and incidentals as shown, specified and required for demolition, removals, and disposal Work.
   2. The Work under this Specifications section includes, but is not necessarily limited to:
      a. Demolition and removal of existing materials and equipment as shown or indicated in the Contract Documents. The Work includes demolition of structural concrete, foundations, walls, doors, windows, structural steel, metals, roofs, masonry, attachments, appurtenances, piping, electrical and mechanical systems and equipment, pavement, sidewalks, gutters, fencing, and similar existing materials, equipment, and items.
   3. Demolitions and removals indicated in other Specifications sections shall comply with requirements of this Specifications section.
   4. Perform demolition Work within areas shown or indicated.
   5. Pay all costs associated with transporting and, as applicable, disposing of materials and equipment resulting from demolition and removals Work.

C. Related Requirements:
   1. Section 31 10 00 - Site Clearing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. National Fire Protection Association (NFPA):
      a. 241, Safeguarding Construction, Alteration, and Demolition Operations.

B. Regulatory Requirements:
   1. Demolition, removals, and disposal Work shall be in accordance with 29 CFR 1926.850 through 29 CFR 1926.860 (Subpart T – Demolition), and all other Laws and Regulations.
   2. Comply with requirements of authorities having jurisdiction.

C. Qualifications:
   1. Electrical Removals: Entity and personnel performing electrical removals shall be electrician(s) legally qualified to perform electrical construction and electrical work in the jurisdiction where the Site is located.
   2. Plumbing Removals: Entity and personnel performing plumbing removals shall be plumber(s) legally qualified to perform plumbing construction and plumbing work in the jurisdiction where the Site is located.

1.3 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Review procedures under this and other Specifications sections and coordinate the Work that will be performed with or before demolition and removals.

1.4 SUBMITTALS

A. Informational Submittals: Submit the following:
   1. Procedure Submittals:
      a. Demolition and Removal Plan: Not less than ten days prior to starting demolition Work, submit acceptable plan for demolition and removal Work, including:
         1) Plan for coordinating shut-offs, capping, temporary services, and continuing utility services.
         2) Other proposed procedures as applicable.
         3) Equipment proposed for use in demolition operations.
         4) Recycling/disposal facility(ies) proposed, including facility owner, facility name, location, and processes. Include copy of appropriate permits and licenses, and compliance status.
         5) Planned demolition operating sequences.
         6) Detailed schedule of demolition Work in accordance with the Schedule accepted by Contracting Officer.
   2. Notification of Intended Demolition Start: Submit in accordance with Paragraph 3.1.A of this Specifications Section.
   3. Field Quality Control Test Results:
      a. Results of megger-testing of existing motors to remain Governments's property.
   4. Qualifications Statements:
      a. Name and qualifications of entity performing electrical removals, including copy of licenses required by authorities having jurisdiction.
      b. Name and qualifications of entity performing plumbing removals,

1.5 SITE CONDITIONS

A. Government makes no representation of condition or structural integrity of area(s) to be demolished or where removals are required by the Contract Documents.

PART 2 - PRODUCTS - (NOT USED)

PART 3 - EXECUTION

3.1 PREPARATION

A. Notification:
   1. Not less than 48 hours prior to commencing demolition or removal, advise Contracting Officer in writing of planned start of demolition Work. Do not start removals without permission of Contracting Officer.
   2. Where demolition or removals has potential to affect adjacent properties, occupants, streets, or other public thoroughfare, transportation facilities, and utilities, furnish required notices to owners and occupants of properties, buildings, and structures that may be affected by the demolition of removal.
   3. In accordance with Laws and Regulations, furnish to authorities having jurisdiction, including emergency services as necessary, appropriate notices of planned demolition and removals.
   4. Submit to Contracting Officer copies of notices furnished to adjacent property owners, occupants, and authorities having jurisdiction.

B. Protection of Adjacent Areas and Facilities:
   1. Perform demolition and removal Work in manner that prevents damage and injury to property, structures, occupants, the public, and facilities. Do not interfere with use of, and free and safe access to and from, structures and properties unless allowed by the Contract Documents otherwise allowed in writing by Contracting Officer.

2. Closing or obstructing of roads, drives, sidewalks, and passageways adjacent to the Work is not allowed unless indicated otherwise in the Contract Documents. Conduct the Work with minimum interference to vehicular and pedestrian traffic.
3. Provide temporary partitions between demolition work areas and (a) areas that will be occupied during demolition and removals, and (b) areas accessible to the public or visitors. Temporary partitions shall be sturdy, braced plywood in good condition, of dimensions sufficient to adequately screen demolition work from view of occupants, public, and visitors. Maintain temporary partitions in place until demolition and removals work in the subject area is complete or until other Work requires removal of temporary partitions.
4. Provide appropriate temporary barriers, lighting, sidewalk sheds, and other necessary protection.
5. Repair damage to facilities that are to remain which such damages results from Contractor's operations.

C. Existing Utilities: In addition to requirements of the General Conditions, Supplementary Conditions, and Division 01 Specifications, perform the following:
1. Should unforeseen, unknown, or incorrectly shown or indicated Underground Facilities be encountered, Contractor responsibilities shall be in accordance with the General Conditions as may be modified by the Supplementary Conditions. Cooperate with utility owners in keeping adjacent services and facilities in operation.
2. Sanitary Sewerage: Before proceeding with demolition, locate and cap all sewer lines and service laterals discharging from the building or structure being demolished.
3. Storm Water Sewerage: Existing storm water system shall remain in place until demolition of existing building or structure is complete. Upon completing demolition, cut and cap storm sewerage at locations shown on the Drawings. Remove existing storm water piping and related structures between points of cutting, and backfill, restore to grade, and stabilize the area over the removed facilities in accordance with the Contract Documents.
4. Water Piping and Related Facilities: Before proceeding with demolition, locate and cap all potable and non-potable waterlines and service laterals serving the building or structure being demolished. Ensure compliance with Laws and Regulations regarding water quality.
5. Other Utilities: Before proceeding with demolition, locate and cap as required all other utilities, such as fuel and gas; compressed air; heating, ventilating, and air conditioning; electric; and communications; and service laterals serving the building or structure being demolished.
6. Shutdown of utility services shall be coordinated by Contractor, assisted by Contracting Officer as required relative to contacting utility owners.

D. Remediation:
1. If unanticipated Hazardous Environmental Condition is believed to be encountered during demolition and removals, comply with requirements of the General Conditions, as may be modified by the Supplementary Conditions.

3.2 DEMOLITION - GENERAL

A. Locate construction equipment used for demolition Work and remove demolished materials and equipment to avoid imposing excessive loading on supporting and adjacent walls, floors, framing, facilities, and Underground Facilities.

B. Pollution Controls:
1. Use water sprinkling, temporary enclosures, and other suitable methods to limit emissions of dust and dirt to lowest practical level. Comply with Section 01 50 00 - Temporary Facilities and Controls, and Laws and Regulations.
2. Do not use water when water may create hazardous or objectionable conditions such as icing, flooding, or pollution.
3. Clean adjacent structures, facilities, properties, and improvements of dust, dirt, and debris caused by demolition Work, in accordance with the General Conditions and Section 01 73 40 - Execution.

C. Explosives:
   1. Explosives are not allowed at the Site. Do not use explosives for demolition and removal Work.

D. Comply with Section 01 73 29 - Cutting and Patching and NFPA 241.

E. Building or Structure Demolition and Removals:
   1. Unless otherwise approved by Contracting Officer, proceed with demolition from top of building or structure to the ground. Complete demolition Work above each floor or tier before disturbing supporting members of lower levels.
   2. Demolish concrete and masonry in small sections.
   3. Remove structural framing members and lower to ground using hoists, cranes, or other suitable methods. Do not throw or drop to the ground.
   4. Break up and remove foundations, mats, and slabs-on-grade unless otherwise shown or indicated as remaining in place.
   5. Temporary Bracing and Supports:
      a. Provide temporary bracing and supports sufficient to maintain safety, stability, and resist all loads to which the structure may be subject during demolition and removals, until entirety is permanently removed or permanently stabilized.
      b. Temporary bracing and supports shall be sufficient for associated dead load, live load, transient loading, and dynamic loads such as wind, seismic, and other loads to which the temporary bracing or support may be subject.
      c. Where appropriate, retain a professional structural engineer, duly licensed and registered in the same jurisdiction as the Site, to design temporary bracing and supports.

F. Salvage and Ownership:
   1. Materials and equipment to remain Government's property shall be:
      a. Carefully removed and appropriately handled by Contractor to avoid damage and invalidation of warranties in effect. Brace motors attached to flexible mountings until reinstallation or delivery to Government's storage location. Fully remedy to pre-construction condition or replace items damaged during removal or handling by Contractor.
      b. Removed as functional units, together with all appurtenances required for operation.
      c. Cleaned, listed, and tagged for storage.
      d. Protected from damage.
      e. Delivered to designated storage location at the Site or other site indicated in the Contract Documents, at place designated by Contracting Officer or Government.

   2. Preparation of Government's existing equipment for storage:
      a. Where appropriate, identify each component with markings or tags to indicate its position in the assembly and the assembly of which it is part.
      b. Place small parts in appropriate, durable boxes and clearly mark contents on the outside of box or container.
      c. Remove oil from oil-lubricated bearings and gear boxes and replace with storage oil.
      d. Grease grease-lubricated bearings.
      e. Replace breather plugs with solid plugs.
      f. Megger-test motor windings: Attach report of the test results to the associated motor and submit copy to Contracting Officer's Representative.
      g. Attach unit to suitable crate bottom.
      h. Enclose unit in polyethylene film and seal all seams and the film to the base of the unit with tape.
      i. Construct crate of wood slats around top and sides of unit.
      j. Attach permanent instruction tag to outside of crate stating "This unit has been prepared for storage. Replace oil, vent plugs, and lubricant in accordance with manufacturer's instructions before start-up."

G. Finishing of Surfaces Exposed by Removals: Unless otherwise shown or indicated in the Contract Documents, surfaces of walls, floors, ceilings, and other areas exposed by removals, and that will remain as finished surfaces, shall be repaired and re-finished with materials that match existing adjacent surface, or as otherwise approved by Contracting Officer's Representative.

3.3 STRUCTURAL REMOVALS

A. Remove structures to lines and grades shown or indicated, unless otherwise directed by Contracting Officer. Where limits are not shown or indicated, limits shall be four inches outside item to be installed. Removals beyond limits shown or indicated shall be at Contractor's risk and expense and such excess removals shall be reconstructed to satisfaction of Contracting Officer without additional cost to Government.

B. Recycling and Reuse of Demolition Materials:
   1. All concrete, brick, tile, masonry, roofing materials, reinforcing steel, structural metals, miscellaneous metals, plaster, wire mesh, and other items contained in or upon building or structure to be demolished shall be removed, transported, and disposed of away from the Site, unless otherwise approved by Contracting Officer.
   2. Do not use demolished materials as fill or backfill adjacent to structures, in pipeline trenches, or as subbase under structures or pavement.

C. After removing concrete and masonry walls or portions thereof, mats, slabs, and similar construction that ties in to the Work or to existing construction, neatly repair the junction point to leave exposed only finished edges and finished surfaces.

D. Where parts of existing structures are to remain in service following demolition, remove the portions shown or indicated for removal, repair damage, and leave the building or structure in proper condition for the intended use.
   1. Remove concrete and masonry to the lines shown or indicated by sawing, drilling, chipping, and other suitable methods. Leave the resulting surfaces true and even, with sharp, straight corners that will result in neat joints with new construction and be satisfactory for the purpose intended.
   2. Do not damage reinforcing bars beyond the area of concrete and masonry removal. Do not saw-cut beyond the area to be removed.
   3. Reinforcing bars that are exposed at surfaces of removed concrete and masonry that will not be covered with new concrete or masonry shall be removed to 1.5 inches below the final surface. Repair the resulting hole, with repair mortar for concrete and grout for masonry, to be flush with the surface.
   4. Where existing reinforcing bars are shown or indicated to extend into new construction, remove existing concrete so that reinforcing bars are clean and undamaged.

E. Removal of Anchorages and Protruding Metals:
   1. Where equipment or material anchored to concrete or masonry are removed and anchors are not to be re-used, and where existing metals (and to be removed) protrude from concrete, remove the anchors and other metal to not less than 1.5 inches beneath surface of concrete or masonry member. Repair the resulting hole, using repair mortar for concrete and grout for masonry, to be flush with the surface.
   2. Alternately, when the anchor is stainless steel, the anchor may be cut flush with the surface of the concrete or masonry, when so approved by Contracting Officer's Representative.

F. Jambs, sills and heads of windows, passageways, doors, or other openings (as applicable) cut-in to the Work or to existing construction shall be dressed with masonry, concrete, or metal to provide smooth, finished appearance.

G. Where anchoring materials, including bolts, nuts, hangers, welds, and reinforcing steel, are required to attach the Work to existing construction, provide such materials under this Specifications section, unless specified elsewhere in the Contract Documents.

3.4 MECHANICAL REMOVALS

A. Mechanical demolition and removal Work includes dismantling and removing existing:
   1. Piping systems and ductwork systems.
   2. Mechanical equipment and appurtenances.
   3. Mechanical elements of instrumentation and control systems, such as sensors and transmitters and similar items.
   4. Mechanical removals include cutting and capping as required, except that cutting of existing piping and ductwork to make connections is included under Section 01 73 29 - Cutting and Patching; Specifications sections in which requirements for coordination with Government operations are indicated; and applicable Specifications of Division 21 - Fire Suppression, Division 22 - Plumbing, Division 23 - Heating, Ventilating, and Air Conditioning, Division 40 - Process Interconnections, and others as applicable.
   5. Mechanical removals as required herein apply to systems exposed to view, hidden from view, and Underground Facilities. Mechanical removals may require work in spaces that may be classified confined spaces.

B. Life-Safety Systems:
   1. Retain existing life-safety systems, including but not limited to fire suppression systems, in place for as long as possible prior to performing associated demolition and removals.
   2. Where demolishing buildings or structures equipped with life-safety systems, remove or deactivate life-safety systems only in the area where active demolition operations are in progress.

C. Demolition and Removals of Piping, Ductwork, and Similar Items:
   1. Scope:
      a. Safety purge piping and tanks (as applicable) of chemicals, fuel, solids, liquids, and gases (as applicable) and make safe for removal and capping. Discharge contents of existing piping appropriately while avoiding damaging property; restricting access to or use of property; and cresting unsafe, unsanitary, nuisances, and noisome conditions.
      b. To the extent shown or indicated, remove existing piping conveying water (potable and non-potable), waste and vent, fuel (liquids and gases), heating fluids (such as water-glycol solutions), chemicals, solids and slurries, sludge, wastewater, other fluids, and processes gases, and other piping.
      c. Remove piping to the nearest structurally sound (or “solid”) piping support, and provide caps on ends of remaining piping.
      d. Where piping to be demolished passes through existing walls to remain, cut off and cap pipe on each side of the wall.
   2. Caps, Closures, Blind Flanges, and Plugs – General (All Piping and Ducts):
      a. Provide closure pieces, such as blind flanges and caps, where shown or required to complete the Work.
      b. Where used in this Specifications section, the term “cap” means the appropriate type closure for the piping or ductwork being closed, including caps, blind flanges, and other closures.
      c. Caps shall be compatible with the piping or ductwork on which the cap is installed, fluid-tight and gastight, and appropriate for the fluid or gas conveyed in the pipe or duct.
      d. Unless otherwise shown or indicated, caps shall be mechanically fastened, fused, or welded to pipe or duct. Plug piping with means other than specified in this Specifications section only when expressly so shown or indicated in the Contractor Documents or when allowed by Contracting Officer’s Representative.
   3. Underground Facilities:
      a. When Underground Facilities are altered or removed, properly cut and cap piping left in place, unless otherwise shown or indicated.
   4. Waste and Vent Piping; Ductwork:
      a. Remove waste and vent piping, and ductwork to extent shown and cap as required.

b. Where demolished vent piping, stacks, and ductwork passes through existing roofing, patch the roof with the same or similar materials as existing, and fully compatible with ensign materials. Completed patch shall be watertight and comply with roofing manufacturer's recommendations.

5. Potable Water Piping; Plumbing; Fire Suppression Piping and Systems; Heating Piping:
   a. Modifications to potable water piping, fire suppression systems, other plumbing piping, and heating system piping shall comply with Laws and Regulations.
   b. All portions of potable water systems that have been modified or opened shall be hydrostatically tested and disinfected in accordance with the Contract Documents, and Laws and Regulations. Hydrostatically test other, normally-pressurized, plumbing and fire suppression piping and heating piping systems.

D. Equipment Demolition and Removals:
1. To the extent shown or indicated and as required for the Work, remove existing mechanical equipment, including (but not limited to):
   a. Facility equipment, such as food service equipment, laundry equipment, dumbwaiters, and similar facility items.
   b. Conveying equipment such as elevators, escalators, and similar general-use conveying systems.
   c. Fire suppression and plumbing equipment.
   d. Heating, ventilating, and air conditioning equipment.
   e. Standby power generators.
   f. Security systems equipment.
   g. Transportation-related equipment.
   h. Flow control gates and valves.
   i. Hoisting equipment.
   j. Bulk materials conveying equipment.
   k. Process heating and cooling equipment.
   l. Blowers, compressors, air filters, air dryers, and similar equipment.
   m. Pumps.
   n. Tanks.
   o. Process equipment, including purification equipment, pollution control and solid waste equipment, and treatment process equipment.
   p. Turbines.
   q. Appurtenances (including motors, drive systems, controls, cooling water and seal water systems) as shown, indicated, and required for completion of the Work.
2. Where required, disassemble equipment to avoid imposing excessive loading on supporting walls, floors, framing, facilities, and Underground Facilities. Disassemble equipment as required for access through and egress from building or structure. Disassembly and removal shall comply with Laws and Regulations. Provide required means to remove equipment from building or structure.
3. Remove control panels, operator stations, and instruments associated with equipment being removed, unless shown or indicated otherwise.
4. Tanks and Equipment Containing Process Material:
   a. Purge contents in accordance with Paragraph 3.5.A of this Specifications Section and other requirements of the Contract Documents, as applicable.
   b. When removing generators, remove associated fuel storage tanks unless otherwise indicated to remain.
   c. Where contents of tank or equipment item may pose a potential hazard, such as hydrocarbon fuels or chemicals, properly dispose of contents in accordance with Laws and Regulations and the Contract Documents.
   d. Where tank or equipment contains wastewater or liquid sludge, and the Site is a wastewater treatment facility, transport and dispose of stored contents onsite at location acceptable to Contracting Officerunless otherwise indicated in the Contract Documents. If Site is other than a wastewater treatment facility, dispose of contents appropriately in accordance with Laws and Regulations.

e. Where tank or equipment contains solid or slurry-type material, remove, handle, and transport the contents and appropriately dispose of the materials offsite in accordance with Laws and Regulations, unless otherwise indicated in the Contract Documents.

5. Remove equipment supports as applicable, anchorages, base, grout, and piping. Remove anchorage systems in accordance with the "Structural Removals" Article in this Specifications section.
6. Remove small-diameter piping back to header unless otherwise indicated.
7. Remove access platforms, ladders, and stairs related to equipment being removed, unless otherwise shown or indicated.
8. Instrumentation and Control Systems Removal:
   a. Remove instrumentation and controls equipment in accordance with this Specifications section's requirements for mechanical removals and electrical removals.
   b. Comply with this Specifications section's "Disposal of Demolition Debris" Article for restrictions on sales of removed items.

3.5 ELECTRICAL REMOVALS

A. Electrical demolition Work includes removing existing:
   1. Disconnecting cabling from motors, electrical sources, control panels, control stations, instrumentation and control items, and similar devices and equipment.
   2. Conduits, raceways, cable trays, hangers and supports, cabling, and related items.
   3. Switches, panelboards, control stations, and similar items.
   4. Transformers, distribution switchboards, control panels, motors, starters, variable speed controllers, and similar items.
   5. Lighting fixtures and related items.
   6. Utility poles, site lighting standards, and overhead cabling.
   7. Appurtenances and miscellaneous electrical equipment, as shown, specified, or required.

B. Electrical Removals – General:
   1. Comply with Laws and Regulations, including the National Electric Code.
   2. Lock Out and Tagging:
      a. Contractor shall lock out and tag circuit breakers and switches operated by Government and shall verify that affected cabling are de-energized to ground potential before commencing electrical removals Work.
      b. Upon completion of electrical removals Work, remove the locks and tags and promptly advise Contracting Officer that existing facilities are available for use.
   3. Remove existing electrical equipment, fixtures, and systems to avoid damaging systems to remain, to keep existing systems in operation, and to maintain integrity of grounding systems.
   4. Disconnect and remove motors, control panels, and other electrical gear where shown or indicated.
   5. Store removed motors, microprocessors and electronics, and other electrical gear to be reused in accordance with its manufacturer's recommendations and requirements of the Contract Documents.

C. Motor Control Centers and Switchgear:
   1. Remove or modify motor control centers and switchgear as shown or indicated.
   2. Modified openings shall be cut square and dressed smooth to dimensions required for installation of equipment.

D. Removal of Cabling, Conduits, Raceways and Similar Items:
   1. Verify the function of each cable before disconnecting and removing.
   2. Remove cabling, conduits, hangers and supports, and similar items back to the power source or control panel, unless otherwise shown or indicated.
   3. Remove cabling, conduits, and similar items where shown or indicated for removal. Abandoned conduits concealed in floor, ceiling slabs, or in walls shall be cut flush with the slab or wall (as applicable) at point of entrance, suitably capped, and the area repaired in a flush, smooth manner acceptable to Contracting Officer's Representative.

4. Disassemble and remove exposed conduits, junction boxes, other electrical appurtenances, and their supports.
5. Repair all areas of the Work to prevent rusting on exposed surfaces.
6. Underground Electric:
    a. Conduits in Underground Facilities not scheduled for reuse shall be suitably capped watertight where each enters building or structure to remain.
    b. Where shown or indicated, remove direct-burial cabling. Openings in buildings for entrance of direct-burial cabling shall be patched with repair mortar or other material approved by Contracting Officer's Representative for such purpose, and made watertight.

E. Electrical Service Entrances and Outdoor, Overhead Electrical Utilities:
1. Existing poles and overhead cabling shall be removed or abandoned as shown and specified.
2. Completely remove from the Site poles not owned by electric utility, including site lighting standards and appurtenances, shown or indicated for removal.
3. Existing substation(s) and poles owned by electric utility will be removed by the electric utility.
4. Make necessary arrangements with electric utility owner for removal of utility owner's transformers and metering equipment after new electrical system has been installed and energized.

F. Lighting fixtures, wall switches, receptacles, starters, and other miscellaneous electrical equipment, not designated as remaining as Government property, shall be removed and properly disposed off-Site as required in accordance with Laws and Regulations.

3.6 DEMOLITION OF SITE IMPROVEMENTS

A. Pavement, Sidewalks, Curbs, and Gutters:
1. Demolition of asphalt or concrete pavement, sidewalks, curbs, and gutters, as applicable, shall terminate at cut edges. Edges shall be linear and have a vertical cut face.
2. To cut pavement, sidewalks, curbs, and gutters, use machinery or tools that provides a smooth-cut edge, appropriate for the required. Where cut edges are not smooth, repair the cut edge to remain to provide a smooth, even appearance.

B. Fencing, Guardrails, and Bollards:
1. Remove to the limits shown or indicated on the Drawings.
2. Completely remove below-grade posts and concrete.

C. Manholes, Vaults, Chambers, and Handholes:
1. Remove to the limits shown or indicated on the Drawings.
2. If not shown or indicated on the Drawings, remove to not less than three feet below finished grade indicated on the Drawings.

D. Underground Facilities Other than Manholes, Vaults, Chambers, and Handholes:
1. Remove to the extent shown or indicated on the Drawings.
2. Unless otherwise shown or indicated, cap ends of piping to remain in place in accordance with the "Mechanical Removals" Article in this Specifications section.

E. Landscaping: Comply with Section 31 10 00 - Site Clearing.

F. Other Site Improvements: When the Contract Documents require removal of other site improvements not addressed above, copy with Contract requirements for removal of buildings or structures.

3.7 DISPOSAL OF DEMOLITION DEBRIS

A. Disposal – General:
1. Promptly remove from the Site all debris, waste, rubbish, material, and equipment resulting from demolition and removal operations. Promptly upon completion of demolition and removal operations, remove from the Site construction equipment used in demolition Work.

2. Do not sell at the Site demolition materials or removed equipment. If materials, equipment or debris will be sold by Contractor, remove the items from the Site and perform the sale or transaction elsewhere, in accordance with Laws and Regulations.
3. Cleaning and Removal of Debris: Comply with the General Conditions, Supplementary Conditions, and Section 01 74 19 - Construction Waste Management and Disposal.

B. Transportation and Disposal:
1. Non-Hazardous Materials, Equipment, and Debris: Properly transport and dispose of non-hazardous demolition materials, equipment, and debris at appropriate landfill or other suitable location, in accordance with Laws and Regulations. Non-hazardous material does not contain Constituents of Concern such as (but not limited to) asbestos, PCBs, petroleum, hazardous waste, radioactive material, or other material designated as hazardous in Laws or Regulations.
2. Hazardous Materials, Equipment, and Debris: When handling and disposal of items containing Constituents of Concern is included in the Work, properly transport and dispose of such items in accordance with the Contract Documents and Laws and Regulations.

C. Submit to Contracting Officer's Representative information required in this Specification Section on proposed facility(ies) where demolition materials, equipment, and debris will be recycled. Upon request, Contracting Officershall be allowed to visit recycling facility(ies) to verify adequacy and compliance status. During such visits, recycling facility operator shall cooperate and assist Contracting Officerand Government.

END OF SECTION

DIVISION 03

CONCRETE

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SECTION 03 01 30
REPAIR AND REHABILITATION OF EXISTING CONSTRUCTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Preparation and assessment of existing concrete for repair and rehabilitation:
      a. Concrete removal for repairs.
      b. Preparation of exposed reinforcing steel.
      c. Preparation for joint sealant installation.
   2. Repair of damaged (and deteriorated) concrete.
      a. Application of repair mortar.
      b. Repair of exposed items embedded in concrete.
   3. Expansion joint repair.

B. Scope:
   1. Contractor shall provide all materials, equipment, labor, tools, services, and incidentals necessary to repair and rehabilitate existing concrete, whether damaged or deteriorated, at locations shown on the Drawings or at locations indicated by Contracting Officer Representative, in accordance with the Contract Documents.
   2. Re-sealing concrete joints as shown on the Drawings, and at other locations directed by Contracting Officer Representative, and indicated in this Section.
      a. Included in the unit price(s) for joint sealant Work are repairs of existing concrete adjacent to the associated joint sealant Work, in accordance with this Section.
   3. Surface repairs of new concrete Work are addressed in Section 03 35 00 - Concrete Finishing and Repair of Surface Defects.

C. Related Requirements: Include but are not necessarily limited to:
   1. Section 03 15 19 - Anchorage to Concrete.
   2. Section 03 21 00 - Reinforcement.
   3. Section 03 35 00 - Concrete Finishing and Repair of Surface Defects.
   4. Section 07 92 00 - Joint Sealants.

1.2 PRICE AND PAYMENT PROCEDURES

A. Unit Prices:
   1. The Work of this Section is Unit Price Work unless otherwise shown or indicated. Work covered by this Section but not eligible for payment under Unit Price Work bid/pay items shall be performed at Contractor's expense.
   2. Bid/pay item classifications of Unit Price Work addressed in this Section are indicated in the Bid Form, the Agreement (or an exhibit thereto), and Section 01 27 00 – Definition of Contract Line Items.
   3. Unit Price Work of this Section is classified as follows:
      a. Concrete Surface Repair - Type I (Thin): Repair of spalled, delaminated, or deteriorated concrete up to 1.5 inches deep below concrete's original surface.
      b. Concrete Surface Repair - Type II (Moderate): Repair of spalled, delaminated, or deteriorated concrete greater than 1.5 inches deep to 3 inches below original concrete surface.
      c. Concrete Surface Repair - Type III (Severe): Repair of spalled, delaminated, or deteriorated concrete greater than 3 inches below original concrete surface.
      d. Expansion Joint Repair - Type A: Replacement of existing expansion joint material including removing existing backer rod, joint filler, and sealant and providing new materials and repair, in accordance with this Section. Expansion joint repair not

designated in the Contract Documents as other types is Type A unless directed otherwise by Contracting Officer Representative.

e. Expansion Joint Repair - Type B: Provide where shown or indicated on the Drawings or where directed by Contracting Officer Representative. Includes sealing existing expansion joints by providing new epoxy resin adhesive sealant system over existing expansion joints.

f. Expansion Joint Repair - Type C: Provide where shown or indicated on the Drawings or where directed by Contracting Officer Representative. Includes sealing existing expansion joints by providing new polyurethane joint sealing system over existing expansion joints.

B. Measurement:

1. Criteria for measurement for payment of this Section's Unit Price Work are in the General Conditions (as may be modified by the Supplementary Conditions), 01 27 00 – Definition of Contract Line Items, and this Section.
2. Quantities of this Section's Unit Price Work:
   a. Unit Price Work of this Section shall be measured for payment prior to commencement of the associated Work in each work area.
   b. Work not measured in advance for payment will not be eligible for payment by Contracting Officer.
   c. Contracting Officer Representative will observe the associated concrete repair and rehabilitation Work performed. Such Work shall be in accordance with the Contract Documents for to such Work to be eligible for payment by Contracting Officer, even when such Work was measured (for payment) in advance.
3. Repair of new concrete Work provided by Contractor is not eligible for payment under the Unit Price Work bid/pay items covered by this Section. Such repairs are included in the Work of the associated bid/pay item under which the subject new concrete Work was provided.

1.3 REFERENCES

A. Terminology:

1. This provision indicates terminology used in this Section and in other Contract Documents that coordinate with this Section. Such terminology may or may not be indicated using initial capital letters and, when used in relation to the Work of this Section, have the meanings indicated below.
2. "Existing concrete damage" means damage to existing concrete surfaces deeper than 1/8 inches, such as:
   a. Concrete corrosion.
   b. Corroded items embedded within concrete or through the concrete surface.
   c. Spalls.
   d. Cracking at depth and arrangement that, in Contracting Officer Representative's sole opinion, cannot be repaired in accordance with Section 03 35 00 – Concrete Finishing and Repair of Surface Defects.
3. "Installer" means the entity installing or applying repair materials at the Site. The terms "installer" and "applicator" have the same meaning. Installer or applicator may be Contractor or Subcontractor.
4. "MPII" means, "manufacturer's printed installation instructions".
5. "Rehabilitation" means repairing and restoring concrete to structurally-sound, durable condition suitable for the structure's intended purpose as determined by Contracting Officer Representative, including repair of existing concrete damage in accordance with this Section and other applicable provisions of the Contract Documents.
6. "Water-bearing structure" means concrete structure with a surface that is normally, or may be, in contact with water or process fluids or slurries during typical operation of the completed Project, including, but not limited to: tanks, channels, wet wells, distribution chambers, dams, and the like. Also, where specifically indicated on the Drawings, "water-

bearing structures" includes basements and structures extending below the ordinary, wet-season groundwater surface.

7. Other terminology used in this Section is consistent with terminology of ACI CT.

B. Reference Standards:

1. American Concrete Institute (ACI):
   a. CT, Concrete Terminology.
   b. 117, Specification for Tolerances for Concrete Construction and Materials.
   c. 308, Standard Practice for Curing Concrete.
2. ASTM International (ASTM):
   a. C150, Standard Specification for Portland Cement.
   b. C309, Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete.
   c. C881, Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete.
   d. C1315, Standard Specification for Liquid Membrane-Forming Compounds Having Special Properties for Curing and Sealing Concrete.
   e. D1682, Breaking Load and Elongation of Textile Fabric.
   f. D1876, Standard Test Method for Peel Resistance of Adhesives (T-Peel Test).
   g. D4060, Abrasion Resistance of Organic Coatings by the Taber Abraser.
   h. D4258, Standard Practice for Surface Cleaning Concrete for Coating.
   i. D4259, Standard Practice for Abrading Concrete.
   j. D4263, Indicating Moisture in Concrete by the Plastic Sheet Method.
   k. D7234, Standard Test Method for Pull-off Adhesion Strength of Coatings on Concrete Using Portable Pull-off Adhesion Tests.
3. International Concrete Repair Institute (ICRI).
   a. 310.1R, Guide for Surface Preparation.
   b. 310.1R, Guide for Surface Preparation for the Repair of Deteriorated Concrete Resulting from Steel Corrosion.
4. Society for Protective Coatings/NACE International (SSPC/NACE):
   a. SP 13/NACE No. 6, Surface Preparation of Concrete.

1.4 QUALITY ASSURANCE

A. Qualifications:

1. Installer:
   a. Installer of materials for rehabilitation of existing concrete shall possess not less than five years of relevant experience performing concrete rehabilitation of similar type, scope, and complexity to that required for this Project.
   b. Certification or Approval by Materials Manufacturer:
      1) Installer (as either business entity or individual) of materials associated with rehabilitation of existing concrete shall be certified or expressly approved in writing, by manufacturer of materials to be provided.
      2) As an option, installer may be certified or approved, in writing, at the Site during initial Work of this Section, in accordance with this Section's Paragraph 3.4.D.
   c. Submit documentation of qualifications and experience in sufficient detail to demonstrate to Contracting Officer Representative's satisfaction compliance with requirements of this Section's qualifications requirements.
2. Structural Concrete Repairer:
   a. This provision is in addition to qualifications requirements applicable to installers of the Work under this Section. This provision applies to business entities physically performing rehabilitation of structural concrete.
   b. Structural concrete repairer shall have not less than five years' current, relevant experience in repairing and rehabilitating concrete structures in facilities of generally similar environmental exposures as the Work of this Section for this Project
   c. Submit documentation of qualifications and experience in sufficient detail to demonstrate to Contracting Officer Representative's satisfaction compliance with requirements of this Section's qualifications requirements.

3. Joint Sealant System Installer:
    a. This provision is in addition to qualifications requirements applicable to all installers of the Work of this Section. This provision applies to entities performing concrete joint sealant system repair or rehabilitation Work.
    b. Entity performing joint sealant system Work shall possess not less than 10 years' relevant experience in waterproofing concrete joints, crack and leak repair, and concrete coating application on projects of similar size, complexity, and environment as the concrete joint sealing Work on this Project.
    c. Entity performing concrete joint sealant system Work shall have completed not less than five projects of size, type, and complexity similar to the Work under this Section within the most-recent period indicated in the paragraph immediately above, on structures that have been in service for not less than three years each. Projects shall have utilized the required type of joint sealant system materials indicated for the Work.
    d. Entity performing concrete joint sealant system Work shall be certified or approved, in writing, by joint sealant system manufacturer (whose product is used in the Work) to furnish and install the required concrete joint sealant system materials.
    e. Submit documentation of qualifications and experience in sufficient detail to demonstrate to Contracting Officer Representative's satisfaction compliance with requirements of this Section's qualifications requirements

B. Mock-Ups:
1. Mock-ups are Samples that, when approved by Contracting Officer Representative, indicate minimum standard of quality for the associated Work. Standard of quality of mock-ups shall be not less than that required by the Contract Documents.
2. If so approved by Contracting Officer Representative, mock-ups may become part of completed Work.
3. Maintain, segregate, identify, and protect mock-ups during performance of the Work to allow Contracting Officer Representative to readily compare the Work with approved mock-up.
4. When mock-up is not part of the completed Work, remove mock-ups when directed by Contracting Officer Representative. If mock-up is part of the completed Work, remove protection and indication of mock-up when so directed by Contracting Officer Representative.
5. Provide mock-up for each type of concrete rehabilitation required by this Section for which the Contract has an associated bid/pay item, including finish.
6. Size of each mock-up shall be acceptable to Contracting Officer. Concrete surface repair mock-ups shall be not less than 2 feet by 2 feet each. Expansion joint repair mock-ups shall be not less than 3 feet long, each.
7. Where mock-ups will be part of the completed Work, Contractor and Contracting Officer will jointly select the location of each mock-up.
8. Concrete surface repair mock-ups shall include:
    a. Sample of patched tie or bughole.
    b. Sample of all jointing specified.
    c. Sample of mortar repair.
9. Expansion joint repair mock-ups shall include:
    a. Repaired expansion joint.
    b. Expansion joint seal.
    c. Rehabilitation of adjacent concrete surfaces.
10. Also provide concrete surface repair mock-up of wall having polymer-modified cementitious coating.
    a. Mock-up shall be stepped to show surface preparation, repairs and coating in all stages of application.
11. Mock-up areas shall be readily identifiable during construction. Provide appropriate temporary signage indicating status as mock-up and protect the mock-up.

1.5 SUBMITTALS

A. Action Submittals: Submit the following:
   1. Shop Drawings:
      a. Schedule (table) indicating, for each type of concrete rehabilitation Work required by this Section, the material type and product manufacturer proposed for each application.
   2. Product Data:
      b. Manufacturer's published, technical data for each manufactured material proposed for use in the Work of this Section.
      c. Manufacturer's written certification that proposed materials comply with associated reference standards cited in this Section.
   3. Samples:
      a. Advise Contracting Officer in writing when required mock-ups will be provided and the location proposed for each.
   4. Test Procedures:
      e. Procedure for pre-repair condition survey required in this Section's Article 3.1, including method for recording results of survey.
      f. Written method of performing pull-off testing of repair material and testing equipment information and accessory materials proposed for use in testing.

B. Informational Submittals: Submit the following:
   1. Certifications:
      a. Laboratory test reports (for previously-tested materials identical to those to be furnished) and material manufacturer's certificates verifying that ingredients comply with the Contract Documents and have a minimum of six months' residual shelf life at the time of shipment to the Site.
      b. Certification from Supplier stating that material is suitable for the intended use on this Project.
      c. Certification that materials proposed for use are compatible with each other, when such materials will contact each other, and will not interfere with bonding of future floor or wall finishes.
   2. Test and Evaluation Reports:
      a. Results of pre-rehabilitation condition survey required in this Section's Article 3.1.
   3. Manufacturer's Instructions:
      a. Manufacturer's instructions for all concrete rehabilitation materials, for handling, storing, and installing materials.
   4. Field Quality Control Submittals:
      a. In-situ pull-off test results for joint sealant system.
   5. Qualifications Statements:
      a. Installer: Documentation of qualifications in accordance with this Section's "Quality Assurance" Article.
      b. Manufacturer's written approval of installer or certification of training performed at the Site in accordance with Paragraph 3.4.D of this Section.
         1) Affidavit, signed by either materials manufacturer or by installer's business entity, indicating that manufacturer of rehabilitation materials has instructed installer in proper handling and installation of each rehabilitation material to be used in the Work.
      c. Entity performing structural concrete repair Work.
      d. Entity performing joint sealant system Work.

C. Closeout Submittals: Submit the following:
   1. Affidavit of compliance from joint sealant system manufacturer, certifying that sealants were installed at the Site in accordance with manufacturer's written instructions and recommendations.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Comply with material manufacturer's written instructions and recommendations regarding delivery, handling, and storage or materials to be incorporated into the Work.

B. Storage:
   1. Store materials in tightly-sealed, original containers, off the ground and in dry location with humidity controls.
   2. Do not store in direct sunlight.
   3. Protect materials from temperature extremes and avoid freezing temperatures.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Anti-Corrosion Bonding Agent:
      a. Euclid Chemical Company.
      b. Or equal..
   2. Epoxy Bonding Adhesive:
      a. Euclid Chemical Company.
      b. Or equal.
   3. Repair Mortar:
      a. Euclid Chemical Company.
      b. Or equal.
   4. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Materials that will be in direct contact with potable water or water that will be treated to become potable shall be certified in accordance with ANSI/NSF 61 and suitable for prolonged immersive exposure to chlorinated water with a total residual of up to 5 mg/l.

B. Bonding Agents:
   1. Bonding agents and adhesives shall have pot life that allows proper placement of new material against existing material in accordance with manufacturer's written instructions.
   2. For repair of existing concrete damage, when no reinforcing steel is exposed, and where specifically shown or indicated on the Drawings, use epoxy bonding adhesive.
   3. For repair of existing concrete damage, when reinforcing steel is exposed, and where specifically shown or indicated on the Drawings, use anti-corrosion bonding agent.
   4. For repair of new concrete: Comply with Section 03 35 00 - Concrete Finishing and Repair of Surface Defects.

C. Water:
   1. Potable.
   2. Clean and free from deleterious substances.
   3. Free of oils, acids and organic matter.

D. Reinforcing Steel: In accordance with 03 21 00 - Reinforcement.

E. Epoxy Patch Seal:

1. For use as seal on patches at repair of existing concrete damage and as otherwise shown or specified in the Contract Documents.
    a. Sikadur 32 Hi-Mod LPL by Sika Corporation.
    b. Or equal.
2. Submit request for substitution in accordance with Specification Section 01 33 23.

F. Anti-Corrosion Bonding Agent:
1. Three-component, moisture tolerant, cementitious bonding agent manufactured for purpose of bonding fresh concrete to hardened concrete and providing anti-corrosion coating to embedded reinforcing materials.
    a. Duralprep A.C. by Euclid Chemical Company.
    b. Or equal.
2. Submit request for substitution in accordance with Specification Section 01 33 23.

G. Epoxy Bonding Adhesive:
1. For use where bonding new concrete or patch material to existing concrete.
2. Two-component, moisture insensitive adhesive manufactured for purpose of bonding fresh concrete to hardened concrete.
    a. Euco No. 452 MV by Euclid Chemical Company.
    b. Or equal.
3. Submit request for substitution in accordance with Specification Section 01 33 23.

H. Repair Mortar:
1. Pre-packaged cement-based, modified (polymer or latex) product specifically formulated for repair of concrete surface defects, with the following properties:

Physical Property	Value	ASTM Standard
Compressive strength (minimum)		C109
at one day	2000 psi	
at 28 days	6000 psi	
Bond strength (minimum)		C882 (*)
at 28 days	1800 psi	

(*) Modified for use with repair mortars.

2. Trowelable, selection based on horizontal, vertical, or overhead application.
3. Where the least dimension of the placement, in width or thickness, exceeds 1.5 inches, repair mortar shall be extended by addition of aggregate per MPII.
4. Acceptable Products:
    a. SikaRepair SHA, SikaTop 123 Plus, SikaTop 111 Plus by Sika Corporation.
    b. Or equal.
5. Submit request for substitution in accordance with Specification Section 01 33 23.

I. Epoxy Coating:
1. Use as a coating over concrete repairs and at repaired embedded items in concrete where shown or indicated on the Drawings.
2. Pigmented, two-component, 100% solids, moisture-tolerant epoxy resin specifically formulated to serve as a protective, corrosion-resistant coating to all common structural substrates.
3. Acceptable Products:
    a. Sikagard 62 by Sika Corporation.
    b. Or equal.

4. Submit request for substitution in accordance with Specification Section 01 33 23.

J. Epoxy Resin Adhesive Sealing System:
   1. Provide sealing system consisting of the following components:
      a. Epoxy Resin Adhesive:
         1) Two-component, 100% solids, high-strength, non-sag structural epoxy paste adhesive.
         2) Color: Gray.
         3) Minimum Pot Life: 30 minutes.
         4) Acceptable Products:
            a) Sikadur 31 Hi-Mod Gel .
            b) Or equal.
      b. Hypalon Sheeting:
         1) Composed of Hypalon rubber with perforations along bonding edges of sheeting to provide mechanical key.
            a) Shall have the ability to be vulcanized with aromatic hydrocarbon solvent to allow its adhesion to an epoxy resin adhesive.
            b) Sheet Dimensions: 8 inches wide and 40 mils thick.
            c) Provided with a removable center expansion strip.
      c. Activating Solvent:
         1) Aromatic hydrocarbon with a specific gravity of 0.86.
         2) All sealing system components shall be fully compatible with each other.
         3) Acceptable Products:
            a) Sikadur Combiflex System by Sika Corporation.
            b) Or equal.
   2. Submit request for substitution in accordance with Specification Section 01 33 23.
      a.

K. Polyurethane Joint Sealing System:
   1. Provide high-solids, two-component liquid, cold-applied, asphalt extended urethane elastomer that cures to durable abrasion-resistant film, forming flexible, water-impermeable barrier.
   2. Provide all components from a single manufacturer to ensure compatibility of components with each other, except for hydrophobic polyurethane grout material.
   3. Concrete Surface Primer: Acceptable products:
      a. CIM 61BG Epoxy Primer, by CIM Industries.
      b. Or equal.
   4. Tack Coat and Top Coat: Acceptable products:
      a. CIM 1061, by CIM Industries.
      b. Or equal.
   5. Joint/Crack Filler: Acceptable products:
      a. CIM 1000TG, by CIM Industries.
      b. Or equal.
   6. Cant Strip Material: Acceptable products:
      a. CIM 1000 TG Cartridges, by CIM Industries
      b. Or equal..
   7. Reinforcing Fabric (Scrim): Acceptable Products:
      a. CIM Scrim, by CIM Industries.
      b. Or equal.
   8. Bonding Agent (for re-application of CIM 1061 if recoat window exceeded):
      a. Acceptable products:
         1) CIM Bonding Agent, by CIM Industries.
         2) Or equal.
      b. Material:
         1) Organo-silane compound dispersed in isopropyl alcohol.
   9. Hydrophobic Polyurethane Grout: Acceptable products:

a. De Neef Hydro Active Flex LV, by GCP Applied Technologies.
b. Or equal.

10. Submit request for substitution in accordance with Specification Section 01 33 23.

L. Expansion Joint Sealant and Backer Rod:
1. Provide in accordance with Section 07 92 00 - Joint Sealants.

PART 3 - EXECUTION

3.1 PREPARATION AND ASSESSMENT

A. As indicated in Article 1.1 of this Section, repair of surface defects in new concrete provided under the Contract and repair of cracks in concrete are addressed in other Specification Sections.

B. Condition Survey:
1. Contractor and Contracting Officer Representative shall jointly perform condition survey of each existing concrete structure included in the scope of the concrete repair Work before scheduling and performing the associated repair Work.
2. Contracting and payment for specialty inspections and tests by third-parties will be by Contracting Officer.
3. Prior to Condition Survey:
a. Submit to Contracting Officer, and obtain Contracting Officer's acceptance, of procedure for performing condition survey. Indicate proposed date(s) of the condition survey and other procedures for inspecting and documenting extent of concrete repair Work to be performed.
b. Prior to the condition survey, power-wash all concrete surfaces within the scope of the concrete repair Work. Power-wash at not less than 4,000 psi using orbital nozzle.
4. Condition survey shall include, but is not necessarily limited to:
a. Visual inspection for:
1) Deficiencies in joints.
2) Cracks.
3) Leakage and efflorescence.
4) Scaling.
5) Spalling
6) Exposed reinforcing.
7) Previous repairs.
b. Extent of chemical attack using phenolphthalein.
c. Delamination survey.
d. Half-cell measurements for corrosion potential of reinforcing.
5. Results of Condition Survey:
a. Submit written results of condition survey to Contracting Officer Representative promptly following the condition survey at the Site.
b. Condition survey results shall clearly indicate the location, nature, size, length, width, and depth of all deficiencies in existing concrete in the area(s) surveyed.
c. Contracting Officer will use results of condition survey to determine extent of repair Work required.
6. Contracting Officer's Direction to Contractor:
a. After consulting with Contracting Officer Representative as necessary, Contracting Officer will transmit to Contractor written direction on extent of concrete repairs required.
b. Contracting Officer will issue such direction promptly after Contracting Officer Representative's receipt of acceptable results of condition survey. Allow in the Progress Schedule 14 days for issuance of Contracting Officer's written direction.
c. Where such Work would either exceed Contract quantities of associated Unit Price Work or requires Work not in the Contract's bid/pay items, an appropriate Contract modification will be issued.

C. Concrete Removal:
   1. Remove all loose and unsound concrete from areas to be repaired, in accordance with ICRI Guideline 310.1R, as modified by the Contract Documents.
   2. Removals:
      a. At areas of damage or deterioration of existing concrete, saw-cut the perimeter of unsound concrete surface areas, to depth of not less than 1/2 inches.
      b. Saw-cuts to be perpendicular to or slightly undercutting existing concrete surface. Concrete removal boundaries shall be straight and aligned parallel to opposite boundary edges resulting in repair areas that are approximately rectangular.
      c. Remove all existing concrete from within the saw-cut repair boundary to of not less than 1/2 inches.
      d. Featheredges are unacceptable.
   3. Clean surfaces of repair areas in accordance with ASTM D4258 to remove dust, dirt, grease, and other contaminants prior to abrasive blasting, chipping, grinding or wire brushing.
   4. Abrasive-blast surfaces in accordance with ASTM D4259 and SSPC SP 13/NACE No. 6 to completely open defects down to sound concrete and remove laitance.
   5. Chip concrete substrate to obtain a surface profile of 1/16 inches to 1/8 inches deep with new fractured aggregate surface. The area to be repaired shall not be less than 1/2 inches in depth.
   6. Concrete removal shall extend along any exposed existing reinforcing to locations along the bar that are free of bond inhibiting corrosion and where the bar is well-bonded to surrounding concrete.
   7. Rinse surface with clean water and allow surface water to evaporate prior to repairing the surface.

D. Preparing Exposed Steel Reinforcing:
   1. Clean and prepare exposed embedded steel reinforcing in accordance with ICRI Guideline 310.1R and the Contract Documents.
   2. Where one-half or more of the steel reinforcing diameter is exposed, either by existing conditions or concrete removal, bond between concrete and steel reinforcing is inhibited or lost completely, or corrosion is present, remove concrete to provide not less than 1 inch clearance around the entire perimeter and along the entire exposed length of the steel reinforcing.
   3. If existing, exposed steel reinforcing is cut through, cracked, or cross-sectional area is reduced by more than 20%, provide new steel reinforcing bar the same size as existing steel reinforcing. Lap the new bar with existing in accordance with ACI requirements. Coat all new and existing steel reinforcing with anti-corrosion bonding agent, as specified in this Section. Abrasive-blast exposed reinforcing to remove contaminants and corrosion to provide white-metal, bright steel finish.

E. Preparation for Joint Sealant System Installation:
   1. Provide adequate surface preparation in area of each joint sealant repair, to not less than 6 IN from face of joint, on each side of joint.
   2. Surface preparation shall remove existing concrete laitance, loose material, oil, grease, and shall expose tops of underlying aggregate to an ICRI concrete surface profile of 4-6. The following methods may be used:
      a. Abrasive-blasting in accordance with ASTM D4259.
      b. High-pressure water blasting at not less than 5,000 psi, in accordance with ASTM D4259.
      c. Shot-blasting in accordance with ASTM D4259, at horizontal surfaces only.
   3. Concrete joint areas to receive joint sealant system shall be fully dried prior to application of joint sealant system. Test substrate moisture at not less than three locations per structure to confirm moisture. The following methods may be used to determine moisture:
      a. Plastic Sheet Method (ASTM D4263): Pass/fail.
      b. Relative Humidity Test (ASTM F2170): Less than 85%.
      c. Calcium Chloride Test: Less than 5 pounds per 1000 square feet per 24 hours.
      d. Radio Frequency Test: Less than 5% moisture.

4. Perform adhesion tests at not less than three locations per structure in mock-up (non-joint) areas to verify adequacy of the joint surface preparation. Perform modified T-Peel (ASTM D1876) field test:
    a. Apply duct tape on substrate.
    b. Apply the coating top coat material to the substrate, fully lapping onto the duct tape.
    c. After coating material has fully cured, cut 1 inch wide strips through coating material, perpendicular to duct tape strip: a minimum of three pull tests per testing location.
    d. Peel up the outside edge of duct tape and provide tarp clip.
    e. Hook a fish weighing or similar scale through the tarp clip and pull perpendicular to the substrate to point of coating adhesion failure and record the load at failure. Scale used for this field measurement shall be calibrated immediately prior to testing.
    f. Minimum acceptable coating adhesion force at failure shall be 15 pounds; no test performed shall demonstrate adhesion below this threshold for surface preparation to be acceptable.
5. Inspect concrete surfaces in area of surface preparation and repair as follows:
    a. Blast/expose all bug-holes to eliminate blind side surfaces.
    b. Cracks less than 1/16 inches wide that extend through the joint coating areas do not require special treatment.
    c. Cracks 1/16 inches to 1/8 inches wide that extend through the joint coating areas shall be stripe-coated for not less than 2 inches on each side and filled with CIM (trowel grade) prior to application of the joint sealant system (after moisture content and surface preparation are acceptable in accordance with the Contract Documents)
    d. Cracks greater than 1/8 inches wide and cracks that experience movement (that extend through the joint coating areas) shall be reinforced with scrim material, similar to the joint coating, prior to sealing the joints.
    e. Cracks greater than 1/4 inches -wide: Provide backer rod, in addition to complying with the paragraph immediately above.
    f. In areas of joint coatings, patch surface spalls and other concrete defects in accordance with requirements for concrete repairs indicated in this Section.
6. Inspect existing joint filler material. If defects in existing joint filler material extend greater than 1/2 inches deep, or if determined by sealant installer to be unsatisfactory, remove not less than 1 inch of existing filler material and pack joint with appropriately sized backer rod soaked in polyurethane grout. Trim grout foam/backer rod flush with surface of concrete once cured. Joints to be coated shall be filled according to this provision and trimmed flush prior to application of coating system.
7. Provide and tool 1 inch minimum cant strips at wall-slab and slab-column joints to be coated. Allow cant strips to cure prior to applying coating system.
8. Perform surface preparation, substrate moisture field tests, adhesion test results, treatment of defects in joint coating areas, and filling (as required) of existing joints for approved by coating manufacturer's technical representative prior to installing joint coating system.

3.2 INSTALLATION AND APPLICATION

A. Environmental Conditions for Installation:
    1. Comply with material manufacturer's written instructions for substrate temperature and moisture content, ambient temperature, and ambient humidity, ventilation, and other conditions affecting performance of concrete repair materials.
    2. Do not repair existing concrete damage when ambient temperature is or is expected to be below 50 degrees F. If necessary to maintain the progress Schedule, enclose and heat area to between 50 and 70 degrees F during repair of surface defects and curing of patching material. Use only indirect fired heating using clean-burning fuel.
    3. If proper environmental conditions do not comply with the Contract Documents and manufacturer's instructions, do not perform the Work until such conditions are acceptable. Provide means to bring conditions into compliance by providing temporary environmental controls, enclosures, and other temporary construction and temporary facilities.
    4. Contractor is not eligible for changes in Contract Times or Contract Price for delays or costs incurred to bring environmental conditions for installation into compliance.

B. Existing Concrete Damage Repair:
   1. Type I Repair:
      a. Provide epoxy bonding adhesive and repair mortar.
   2. Type II Repair:
      a. Provide epoxy bonding adhesive, if no reinforcing steel is exposed. Use anti-corrosion bonding agent, if reinforcing steel is exposed.
      b. Prepare exposed reinforcing steel in accordance with Paragraph 3.1.D of this Section.
      c. Provide repair mortar:
         1) Provide 3/8 inches aggregate in accordance with MPII.
   3. Type III Repair:
      a. Provide anti-corrosion bonding agent.
      b. Prepare exposed steel reinforcement per the requirements of Paragraph 3.1.B.
      c. Provide new adhesive anchor dowels as shown on the Drawings and as indicated in Section 03 15 19 - Anchorage to Concrete.
      d. Provide repair mortar:
         1) Provide 3/8 inches aggregate in accordance with MPII.

C. Repair Mortar Application:
   1. Comply with MPII for mixing and placement of repair mortar.
   2. After initial mixing of repair mortar, do not introduce additional water to change consistency. Discard repair mortar if consistency becomes too stiff to place.
   3. Place repair mortar to not less than recommended minimum thickness indicated in the MPII and in no event less than 3/8 inches.
      a. Apply repair mortar in accordance with the following minimum requirements:
         1) Not less than 3/8 inches over existing sound, exposed coarse aggregate.
         2) Not less than 2 inches of cover (unless otherwise required) over exposed reinforcing steel.
   4. At horizontal applications, repair mortar shall be screeded and bullfloated to the proper elevation, to ensure all surface moisture will drain freely and properly without puddle areas.
   5. Provide repair mortar in even, uniform plane to restore the concrete member to its original surface finish and plane.
      a. Tolerance for being out-of-plane shall be such that gap between a 1 foot straight edge and repair mortar surface shall not exceed 1/4 inches, and gap between a 4 feet straight edge and repair mortar surface shall not exceed 3/8 inches. This shall apply to straight edges placed in any orientation at any and all location on the repair mortar surface.
   6. Prevention of Drying:
      a. Prevent exposed plastic mortar surfaces from drying. Provide windbreaks, foggers, and evaporation retarders, as necessary, during finishing.
      b. Foggers shall maintain humidity at height of 2 feet to 3 feet above surface of concrete.
      c. If necessary, apply evaporation retarder between finishing operations.
   7. Repair mortar shall receive smooth, steel troweled finish.

D. Repair of Exposed Embedded Items in Concrete:
   1. This provision addresses repair and rehabilitation of corroded metal items embedded in existing concrete and to other locations as expressly shown or indicated on the Drawings. Existing concrete damage by corrosion of embedded metal shall be repaired in accordance with this Section's Paragraph 3.2.B.
   2. Preparation:
      a. Fully expose extent of metal corrosion within each embedded item by chipping to sound material. Where specifically shown or indicated on the Drawings, completely remove exposed metal item to extent shown or indicated.
      b. Prepare exposed reinforcing steel attached to or adjacent to embedded, corroded metal items in accordance with this Section's Paragraph 3.1.D.
      c. If existing concrete has been removed during chipping and repair of metal item, prepare repair area in accordance with this Section's Paragraph 3.1.C.
      d. Remove corrosion on embedded metal item and corrosion on exposed reinforcing steel by abrasive blasting to a white-metal finish.

3. Repair:
    a. Where no existing concrete has been removed or damaged adjacent to embedded metal item:
        1) On surface of embedded metal item, provide two coats of epoxy coating in accordance with coating manufacturer's recommendations.
            a) Color of First Coating: Red.
            b) Color of Second Coating: Gray.
        2) Before applying second coat, allow first coat of epoxy coating to fully cure in accordance with coating manufacturer's recommendations.
    b. Where areas of existing concrete have been removed or damaged adjacent to embedded metal item:
        1) Patch area of removed or damaged concrete in accordance with this Section's Paragraph 3.2.B.
        2) Provide on surface of embedded metal item two coats of epoxy coating in accordance with coating manufacturer's recommendations.
            a) Color of First Coat: Red.
            b) Color of second Coat: Gray.
        3) Before applying second coating, allow first coat of epoxy coating to fully cure in accordance with coating manufacturer's recommendations.

E. Extend existing control, construction, and expansion joints through concrete repairs.

F. For repairs of existing concrete damage, finish of repaired areas shall match the finish of existing adjacent concrete surface.

G. Expansion Joint Repair – Type A:
1. Provide materials and installation methods in accordance with sealant manufacturer recommendations and the Contract Documents.
2. Provide sealant and backer rod at moderate, reasonably stable, concrete temperature to optimize repair performance.
3. Comply with Section 07 92 00 - Joint Sealants.

H. Expansion Joint Repair – Type B:
1. Provide epoxy resin adhesive sealing system, in accordance with joint sealant system manufacturer's instructions and the Contract Documents, at locations shown or indicated on the Drawings:
    a. Existing sealant, backer rod, and joint filler material in acceptable condition are to remain in-place at existing joint locations to receive sealing system.
    b. Concrete substrate on both sides of existing joint must be clean, dry, sound, and free of surface contaminants to not less than 10 inches wide strip centered on existing joint or greater when recommended by sealant manufacturer.
        1) Remove dust, laitance, grease, oils, curing compounds, form release agents, and foreign matter by sandblasting or other mechanical means acceptable to the Contracting Officer.
    c. Mix epoxy resin adhesive and install hypalon sheeting in accordance with adhesive manufacturer's printed installation instructions and the Contract Documents.
    d. Cleanup:
        1) Leave finished Work and work area in a neat, clean condition without evidence of spillovers on adjacent surfaces.
        2) Clean uncured epoxy resin adhesive with approved solvent appropriate for the application and area.
        3) Remove cured epoxy resin adhesive only by mechanical means.

I. Expansion Joint Repair – Type C (Joint Sealant System):
1. Installation of joint coating system shall comply with applicable material manufacturer's published recommendations and the Contract Documents.
2. Do not install concrete joint coating when substrate is in a rising-temperature mode.

3. Substrate, air, and coating material temperatures shall comply with applicable material manufacturer's recommendations and the Contract Documents.
4. Primer:
    a. Provide epoxy primer to concrete prior to installing final coating materials.
    b. Applied Thickness: 5 to 10 wet mils.
    c. Reapply coating as necessary to achieve pinhole/holiday-free surface
5. Comply with material manufacturer's recoat time. If recoat time(s) are exceeded, comply with material manufacturer's recommended procedures for surface abrasion, cleaning, and application of bonding agent.
6. Application of Coatings:
    a. Provide primer, base coat, scrim, and top coat in accordance with material manufacturer's requirements, as shown in the Drawings, and in accordance with other Contract Documents.
    b. Install primer (see item 4 above)
    c. Provide initial tack coat at thickens of 10 to 20 wet mils, to 6 inches on each side of joints to be coated.
    d. Push scrim material evenly into wet tack coat and allow to cure one to four hours, as recommended by coating manufacturer.
    e. Provide not less than thickness of 60 wet mils of top coat over scrim material.
    f. Protect and cure in accordance with material manufacturer's recommendations and the Contract Documents.
7. Should substrate temperatures be less than sealant system manufacturer's written recommended minimum temperature, comply with material manufacturer's modified procedures provide materials approved by the Contracting Officer and suitable for installation in cold conditions.

3.3 CURING

A. Curing of Repair Mortar:
    1. Perform curing of repair mortar immediately after final finishing.
    2. Perform curing by combination of covering repair Work with wet burlap and applying liquid membrane-forming curing compound.
    3. Employ methods and sequence to maintain moisture for not less than seven days.

3.4 FIELD QUALITY CONTROL

A. Field Tests:
    1. In-Situ Pull-Off Tests:
        a. Perform in-situ pull-off tests on repaired areas at locations indicated by Contracting Officer Representative.
            1) Perform not less than one pull-off test, in accordance with ASTM C1583, for each surface (wall, roof, floor, and other). In event of unacceptable pull-off test, repair the defective Work in accordance with the Contract Documents and perform additional test(s) at location(s) indicated by Contracting Officer Representative.
        b. Submit written results of testing, indicating: date of test, entity performing testing, personnel present during testing, time of test, location of test, pertinent results recorded, and other relevant observations.
        c. Criteria for Acceptance: Pull-off test will be deemed as passing (acceptable results) when failure occurs within the existing concrete substrate (cohesive concrete failure). If failure occurs at the joint surface with existing concrete (bond failure), within the repair material, or at connection to testing device, pull-out test will be deemed as passing (acceptable results) when the unit tensile stress acting on the core cross section at failure exceeds 300 psi. If failure occurs within the repair material, additional test samples of the repair material in the area of the test shall be taken for compressive testing to verify strength of repair material.
        d. Remedy all damage resulting from in-situ testing, in accordance with the Contract Documents.

B. Observations and Inspections:
   1. Contracting Officer Representative will witness surface preparation, substrate moisture conditions, and installation of materials indicated in this Section. Such observations do not relieve Contractor from obligation to comply with the Contract Documents.
   2. Contracting Officer-retained special inspector shall be present while material manufacturer's technical representatives are at the Site instructing Contractor's structural concrete repair personnel, Contractor's joint sealant system personnel, and installers in the use of the associated material(s).

C. Defective Work:
   1. Defective Repair:
      a. Any and all repairs are defective Work when one or more of the following occurs:
         1) Pull-off test fails.
         2) When grout cube tests yield results less than 3500 psi for repair mortars in seven days.
         3) Repair is not properly finished and in accordance with specified tolerances.
      b. Promptly remove and remedy defective concrete repair Work in accordance with the Contract Documents.
   2. Damaged Work:
      a. Before acceptance of the Work (following final inspection in accordance with the General Conditions and other Contract Documents), neatly repair damaged surfaces, corners of concrete, and finish.
      b. When performing surface remedial repairs, finish areas to smooth, dense watertight condition.
      c. Replace unsatisfactory concrete patching Work.
   3. Corrective Work:
      a. If correction of defective Work (under this Section) is necessary, remove defective Work. Key area to be remedied, clean, and soak surface with water and patch with approved materials. Patch concrete to match existing adjacent concrete surfaces.
      b. Clean surface cavities resulting from form ties, other holes, honeycomb spots, broken corners and edges, and other effects. Saturate with water and point with a mortar of patching material paste. Comply with patching material manufacturer's recommendations concerning placement, pot life, and curing.
      c. Prepare pointing material not more than 30 minutes prior to use. Cure mortar patches properly. Carefully tool contraction and articulated joints in completed Work and keep them free of concrete. Where necessary, leave joint filler exposed for its full length with clean and true edges.
      d. Tolerance deviations and other surface defects may also be corrected, when approved by Contracting Officer Representative, by grinding high areas of swales.
      e. Where remedial work is unsatisfactory, completely remove such Work and replace with new Work in accordance with the Contract Documents.
   4. Special inspection of remedial work is required. Special inspection will be performed after completion of surface preparation and during installation of remedial Work. Refer to Structural Statement of Special Inspections for additional requirements. .
   5. Defective Joint Sealant System Work:
      a. Any and all locations where joint sealant system has separated from concrete or exhibits cracking or tearing is defective Work.
      b. Remove defective joint sealant system Work, clean the surface, and re-provide new joint sealant system materials.
      c. Unless directed otherwise by Contracting Officer, when area of defective joint sealant Work exceeds 25% of total area of joint sealant system Work provided, replace all joint sealant system Work in accordance with the Contract Documents.

D. Suppliers' Services:
   1. Manufacturers' factory-trained technical representatives of concrete repair materials and joint sealant system shall be at the Site prior to and during first installation of the materials furnished under this Section to review surface preparation, surface moisture conditions and

adhesion testing, and proposed installation methods. Joint sealant system manufacturer's representative shall also inspect the entire completed installation and submit, through Contractor, an affidavit of compliance certifying that installed materials comply with manufacturer written instructions and recommendations. Compensation for which is part of the associated unit price for such Work.

3.5 POST-CONSTRUCTION OBLIGATIONS

A. Warranty Inspection:
   1. Perform warranty inspections during eleventh month after Substantial Completion of the Work of this Section.
   2. For the inspections, Contrtacting Officer will dewater and reasonably clean water-bearing structures.
   3. Operational constraints preclude removing all water-bearing structures from service simultaneously for warranty inspection. It is anticipated that ONE separate warranty inspections will be necessary due to operational constraints, each on different, non-consecutive days.
   4. One warranty inspection of each joint sealed is required, unless defective Work is evident. When defective joint sealant Work is apparent, an additional warranty inspection of the repaired joint is required in the eleventh month after the defective joint sealant is remedied.
   5. Contractor, joint sealant Subcontractor (if any), joint sealant system manufacturer's technical representative, Contracting Officer, shall be present at each warranty inspection.
   6. Contractor shall promptly remedy and replace defective Work, whether evident during the warranty inspection or otherwise, in accordance with the Contract Documents.
   7. If contractor is unable or unavailable to remedy defective Work promptly when water-bearing structures are dewatered and cleaned for the warranty inspection, Contracting Officer may place such structures back into service and Contractor will be responsible for appropriately dewatering and cleaning the structure to perform the necessary remedial Work.
   8. Regarding remedy of defective joint sealant system Work, refer to this Section's Paragraph 3.4.C.

B. Item 1 – Concrete Surface Repair - Type I (Thin):
   1. Measurement will be the area, in square feet, of existing, concrete surface area repaired, measured in one or more appropriate geometric shapes such as rectangles, squares, triangles, and trapezoids, at the concrete surface.
   2. Item Includes (all in accordance with the Contract Documents):
      a. As indicated in Section 03 01 30 - Repair and Rehabilitation of Existing Concrete.
   3. Not included in this bid/pay item:
      a. As indicated in Section 03 01 30 - Repair and Rehabilitation of Existing Concrete.
   4. Payment: Unit price per square foot for this item will be full compensation for Work for surface repair of the type indicated for existing concrete, and all related Work, performed under this item, complete in accordance with the Contract Documents, and not specifically included under other bid/pay items or contracts.

C. Item 2 – Expansion Joint Repair - Type A:
   1. Measurement will be the length, in linear feet, of existing, concrete expansion joint repaired, measured at the concrete surface.
   2. Item Includes (all in accordance with the Contract Documents):
      a. As indicated in Section 03 01 30 - Repair and Rehabilitation of Existing Concrete.
   3. Not included in this bid/pay item:
      a. As indicated in Section 03 01 30 - Repair and Rehabilitation of Existing Concrete.
   4. Payment: Unit price per linear foot for this item will be full compensation for all expansion joint repair Work of the type indicated, and all related Work, performed under this item, complete in accordance with the Contract Documents, and not specifically included under other bid/pay items or contracts.

END OF SECTION

SECTION 03 05 05
CONCRETE TESTING AND INSPECTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Contractor requirements for testing of concrete and grout.
   2. Definition of Government provided testing.
   3. Acceptance criteria for concrete.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 21 00 - Reinforcement.
   2. Section 03 31 30 - Concrete, Materials and Proportioning.
   3. Section 03 31 31 - Concrete Mixing, Placing, Jointing and Curing.
   4. Section 03 41 33 - Precast and Prestressed Concrete.

1.2 RESPONSIBILITY AND PAYMENT

A. Construction Management Representative will hire an independent Testing Agency/Service Provider to perform the following testing and inspection and provide test results to the Contracting Officer's Representative and Contractor.
   1. Testing and inspection of concrete and grout produced for incorporation into the work during the construction of the Project for compliance with the Contract Documents.
   2. Additional testing or retesting of materials occasioned by their failure, by test or inspection, to meet requirements of the Contract Documents.
   3. Strength testing on concrete required by the Contracting Officer's Representative or Special Inspector when the water-cement ratio exceeds the water-cement ratio of the typical test cylinders.
   4. In-place testing of concrete as may be required by Contracting Officer's Representative when strength of structure is considered potentially deficient.
   5. Other testing services needed or required by Contractor such as field curing of test specimens and testing of additional specimens for determining when forms, form shoring or reshoring may re-removed.
   6. Government will pay for services defined in Paragraph 1.2A.1.
   7. See Specification Section 01 42 00.

B. Hire a qualified testing agency to perform the following testing and provide test results to the Contracting Officer.
   1. Testing of materials and mixes proposed by the Contractor for compliance with the Contract Documents and retesting in the event of changes.
   2. Additional testing and inspection required because of changes in materials or proportions requested by Contractor.
   3. Pay for services defined in Paragraphs 1.2B.1. and 1.2B.2.
   4. Reimburse Government for testing services defined in Paragraphs 1.2A.2., 1.2A.3., 1.2A.4. and 1.2A.5.
   5. See Specification Section 01 42 00.

C. Duties and Authorities of Testing Agency/Service Provider:
   1. Any Testing Agency/Service Provider or agencies and their representatives retained by Contractor or Contracting Officer for any reason are not authorized to revoke, alter, relax, enlarge, or release any requirement of Contract Documents, nor to reject, approve or accept any portion of the Work.
   2. Testing Agency/Service Provider shall inform the Contractor and Contracting Officer's Representative regarding acceptability of or deficiencies in the work including materials

furnished and work performed by Contractor that fails to fulfill requirements of the Contract Documents.

3. Testing Agency to submit test reports and inspection reports to Contracting Officer's Representative and Contractor immediately after they are performed.
    a. All test reports to include exact location in the work at which batch represented by a test was deposited.
    b. Reports of strength tests to include detailed information on storage and curing of specimens prior to testing.
4. Contracting Officer retains the responsibility for ultimate rejection or approval of any portion of the Work.

1.3 QUALITY ASSURANCE

A. Referenced Standards:
    1. American Concrete Institute (ACI):
        a. 318, Building Code Requirements for Structural Concrete.
    2. ASTM International (ASTM):
        a. ASTM Cement and Concrete Reference Laboratory (CCRL).
        b. C31, Standard Practice for Making and Curing Concrete Test Specimens in the Field.
        c. C39, Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens.
        d. C42, Standard Test Method for Obtaining and Testing Drilled Cores and Sawed Beams of Concrete.
        e. C94, Standard Specification for Ready-Mixed Concrete.
        f. C143, Standard Test Method for Slump of Hydraulic-Cement Concrete.
        g. C172, Standard Practice for Sampling Freshly Mixed Concrete.
        h. C1019, Standard Test Method for Sampling and Testing Grout.
        i. C1218, Standard Test Method for Water-Soluble Chloride in Mortar and Concrete.
        j. E329, Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection.

B. Qualifications:
    1. Contractor's Testing Agency:
        a. Meeting requirements of ASTM E329 and ASTM C94.
        b. Provide evidence of recent inspection by CCRL of NBS, and correction of deficiencies noted.

C. Use of Testing Agency and approval by Contracting Officer's Representative of proposed concrete mix design shall in no way relieve Contractor of responsibility to furnish materials and construction in full compliance with Contract Documents.

1.4 DEFINITIONS

A. Testing Agency/Service Provider: An independent professional testing/inspection firm or service hired by Contractor or by Contracting Officer to perform testing, inspection or analysis services as directed, and as provided in the Contract Documents.

1.5 SUBMITTALS

A. Shop Drawings:
    1. Product technical data including:
        a. Concrete materials and concrete mix designs proposed for use.
            1) Include results of all testing performed to qualify materials and to establish mix designs.
            2) Place no concrete until approval of mix designs has been received in writing.
            3) Submittal for each concrete mix design to include:
                a) Sieve analysis and source of fine and coarse aggregates.
                b) Test for aggregate organic impurities.
                c) Proportioning of all materials.

d) Type of cement with mill certificate for the cement.
e) Brand, quantity and class of fly ash proposed for use along with other submittal data as required for fly ash by Specification Section 03 31 30.
f) Slump.
g) Brand, type and quantity of air entrainment and any other proposed admixtures.
h) Shrinkage test results.
i) Total water soluble chloride ion concentration in hardened concrete from all ingredients determined per ASTM C1218.
j) 28-day compression test results and any other data required by Specification Section 03 31 30 to establish concrete mix design.

2. Certifications:
   a. Testing Agency qualifications.

PART 2 - PRODUCTS - (NOT USED)

PART 3 - EXECUTION

3.1 TESTING SERVICES TO BE PERFORMED SERVICE PROVIDER/TESTING AGENCY

A. The following concrete testing will be performed by the Service Provider/Testing Agency:
   1. Concrete strength testing:
      a. Secure concrete samples in accordance with ASTM C172.
         1) Obtain each sample from a different batch of concrete on a random basis, avoiding selection of test batch other than by a number selected at random before commencement of concrete placement.
      b. For each strength test, mold and cure cylinders from each sample in accordance with ASTM C31.
         1) Record any deviations from requirements on test report.
         2) Cylinder size: Per ASTM C31.
            a) 4 inches cylinders shall not be used for concrete mixes with maximum aggregate size larger than 1 inch.
            b) Use the same size cylinder for all tests for each concrete mix.
         3) Quantity:
            a) 6 inches diameter by 12 inches high: Four cylinders.
            b) 4 inches diameter by 8 inches high: Six cylinders.
      c. Field cure one cylinder for the seven day test.
         1) Laboratory cure the remaining.
      d. Test cylinders in accordance with ASTM C39.
         1) 6 inches diameter cylinders:
            a) Test two cylinders at 28 days for strength test result and the one field cured sample at seven days for information.
            b) Hold remaining cylinder in reserve.
         2) 4 inches diameter cylinders:
            a) Test three cylinders at 28 days for strength test result and the one field cured cylinder at seven days for information.
            b) Hold remaining cylinders in reserve.
      e. Strength test result:
         1) Average of strengths of two, 6 inches diameter cylinders or three, 4 inches diameter cylinders from the same sample tested at 28 days.
         2) If one cylinder in a test manifests evidence of improper sampling, molding, handling, curing, or testing, discard and test reserve cylinder(s); average strength of remaining cylinders shall be considered strength test result.
         3) Should all cylinders in any test show any of above defects, discard entire test.
      f. Frequency of tests:

1) Concrete sand cement grout: One strength test for each 4 hour period of grout placement or fraction thereof.
   a) Test grout in accordance with ASTM C1019.
2) Concrete topping, concrete fill and lean concrete: One strength test for each 10 CUYD of each type of concrete or fraction thereof placed.
3) Precast concrete: Frequency per Specification Section 03 41 33.
4) All other concrete:
   a) One strength test to be taken not less than once a day, nor less than once for each 60 cubic yards or fraction thereof placed in any one day.
   b) Once for each 5000 square feet of slab or wall surface area placed each day
   c) If total volume of concrete on Project is such that frequency of testing required in above paragraph will provide less than five strength tests for each concrete mix, tests shall then be made from at least five randomly selected batches or from each batch if fewer than five batches are provided.

2. Slump testing:
   a. Determine slump of concrete sample for each strength test.
      1) Determine slump in accordance with ASTM C143.
   b. If consistency of concrete appears to vary, the Contracting Officer's Representative shall be authorized to require a slump test for each concrete truck.
      1) This practice shall continue until three consecutive batches are determined to be consistent and meet the slump requirements specified.
3. Air content testing: Determine air content of concrete sample for each strength test in accordance with ASTM C231 , ASTM C173 , or ASTM C138.
4. In-place concrete testing (if required).

3.2 SPECIAL INSPECTIONS

A. See Section 01 40 00.
   1. Special Inspections listed are for the Contractor reference only and is not part of the Contract Documents.
   2. It is included to assist the Contractor in understanding the Government-provided Services so that those services may be factored into the Contractor's pricing and schedule.

B. Formwork Special Inspections:
   1. Shape, location, and dimensions.
      a. Inspect in accordance with dimensions and details on Drawings.
      b. Frequency: Inspect prior to each concrete pour.

C. Reinforcing Special Inspections:
   1. Reinforcing size, spacing, lap length and concrete cover.
      a. Inspect in accordance with Drawings and Specification.
      b. Frequency: Inspect prior to each concrete pour.
   2. Reinforcing adhesive anchoring system:
      a. Inspect in accordance with ICC-ES report.
      b. Frequency:
         1) Inspect all adhesive anchors for the first 4 hours of installation.
         2) Inspect approximately 25% of adhesive anchors thereafter.
         3) Additional inspection will be required for different installer or if the quality of installation appears to vary.
   3. Mechanical splices:
      a. Inspect in accordance with ICC-ES report.
      b. Frequency:
         1) Inspect all mechanical splices prior to placing concrete .
         2) Inspect approximately 25% of mechanical splices thereafter.
         3) Additional inspection will be required for different installer or if the quality of installation appears to vary.

D. Mixing, Placing, Jointing, and Curing Special Inspections:

1. Perform concrete tests per the requirements of this Specification Section.
2. Verification of proper mix design.
    a. Frequency: Periodically, prior to each concrete pour.
3. Proper concrete placement techniques.
    a. Inspect per requirements of Section 03 31 31.
    b. Frequency: During each concrete pour.
4. Proper curing temperature and techniques.
    a. Inspect per requirements of Section 03 31 31.
    b. Frequency: Periodically, but not less than every third day.
5. Joints:
    a. Inspect joints for proper joint type, dimensions, reinforcing, dowel alignment, surface preparation and location.
    b. Frequency: Prior to each concrete pour.
6. Waterstops:
    a. Visually inspect waterstops for proper location, continuity, installation to prevent displacement, cleanliness and damage to waterstop.
    b. Frequency:
        1) Prior to each concrete pour.

E. Anchorage to Concrete Special Inspection:
1. Post installed anchors as required by the building code, ICC-ES Evaluation Reports, and as specified by the Contracting Officer's Representative.
    a. Frequency: Per ICC-ES Report.
2. Cast-in-place concrete anchors, including anchor size, embedment, material and location.
    a. Frequency: Prior to each concrete pour.

3.3 SAMPLING ASSISTANCE AND NOTIFICATION FOR OWNER

A. To facilitate testing and inspection, perform the following:
1. Furnish any necessary labor to assist Testing Agency in obtaining and handling samples at site.
2. Provide and maintain for sole use of Testing Agency adequate facilities for safe storage and proper curing of test specimens on site for first 24 hours as required by ASTM C31.
3. Take samples at point of placement into concrete member.

B. Notify Contractor's Testing Agency sufficiently in advance of operations (minimum of 24 hours) to allow for assignment of personnel and for scheduled completion of quality tests.

3.4 ACCEPTANCE

A. Completed concrete work which meets applicable requirements will be accepted without qualification.

B. Completed concrete work which fails to meet one or more requirements but which has been repaired to bring it into compliance will be accepted without qualification.

C. Completed concrete work which fails to meet one or more requirements and which cannot be brought into compliance may be accepted or rejected as provided in these Contract Documents.
1. In this event, modifications may be required to assure that concrete work complies with requirements.
2. Modifications, as directed by Contracting Officer's Representative, to be made at no additional cost to Government.

D. Dimensional Tolerances:
1. Formed surfaces resulting in concrete outlines smaller than permitted by tolerances shall be considered potentially deficient in strength and subject to modifications required by Contracting Officer's Representative.
2. Formed surfaces resulting in concrete outlines larger than permitted by tolerances may be rejected and excess material subject to removal.

a. If removal of excess material is permitted, accomplish in such a manner as to maintain strength of section and to meet all other applicable requirements of function and appearance.

3. Concrete members cast in wrong location may be rejected if strength, appearance or function of structure is adversely affected or misplaced items interfere with other construction.
4. Inaccurately formed concrete surfaces exceeding limits of tolerances and which are exposed to view, may be rejected.
   a. Repair or remove and replace if required.
5. Finished slabs exceeding tolerances may be required to be repaired provided that strength or appearance is not adversely affected.
   a. High spots may be removed with a grinder, low spots filled with a patching compound, or other remedial measures performed as permitted or required.

E. Appearance:
   1. Concrete surfaces exposed to view with defects which, in opinion of Contracting Officer's Representative, adversely affect appearance as required by specified finish shall be repaired by approved methods.
   2. Concrete not exposed to view is not subject to rejection for defective appearance unless, in the opinion of the Contracting Officer's Representative, the defects impair the long-term strength or function of the member.

F. High Water-Cement Ratio:
   1. Concrete with water in excess of the specified maximum water-cement ratio will be rejected.
   2. Remove and replace concrete with high water-cement ratio or make other corrections as directed by Contracting Officer's Representative.

G. Strength of Structure:
   1. Strength of structure in place will be considered potentially deficient if it fails to comply with any requirements which control strength of structure, including but not necessarily limited to following:
      a. Low concrete strength:
         1) Test results for standard molded and cured test cylinders to be evaluated separately for each mix design.
            a) Such evaluation shall be valid only if tests have been conducted in accordance with specified quality standards.
            b) For evaluation of potential strength and uniformity, each mix design shall be represented by at least three strength tests.
            c) A strength test shall be the average of two, 6 inches diameter cylinders or three, 4 inches diameter cylinders from the same sample tested at 28 days.
         2) Acceptance:
            a) Strength level of each specified compressive strength shall be considered satisfactory if both of the following requirements are met:
               (1) Average of all sets of three consecutive strength tests equal or exceed the required specified 28 day compressive strength.
               (2) No individual strength test falls below the required specified 28 day compressive strength by more than 500 psi.
      b. Reinforcing steel size, configuration, quantity, strength, position, or arrangement at variance with requirements in Specification Section 03 21 00 or requirements of the Contract Drawings or approved Shop Drawings.
      c. Concrete which differs from required dimensions or location in such a manner as to reduce strength.
      d. Curing time and procedure not meeting requirements of this Specification Section.
      e. Inadequate protection of concrete from extremes of temperature during early stages of hardening and strength development.

f. Mechanical injury, construction fires, accidents or premature removal of formwork likely to result in deficient strength.
g. Concrete defects such as voids, honeycomb, cold joints, spalling, cracking, etc., likely to result in deficient strength or durability.

2. Structural analysis and/or additional testing may be required when strength of structure is considered potentially deficient.
3. In-place testing of concrete may be required when strength of concrete in place is considered potentially deficient.
   a. Testing by impact hammer, sonoscope, or other nondestructive device may be permitted by Contracting Officer's Representative to determine relative strengths at various locations in the structure or for selecting areas to be cored.
      1) Such tests shall not be used as a basis for acceptance or rejection.
   b. Core tests:
      1) Where required, test cores will be obtained in accordance with ASTM C42.
         a) If concrete in structure will be dry under service conditions, air dry cores (temperature 60 to 80 degrees F, relative humidity less than 60%) for seven days before test then test dry.
         b) If concrete in structure will be wet or subjected to high moisture atmosphere under service conditions, test cores after immersion in water for at least 40 hours and test wet.
         c) Testing wet or dry to be determined by Contracting Officer's Representative.
      2) Three representative cores may be taken from each member or area of concrete in place that is considered potentially deficient.
         a) Location of cores shall be determined by Contracting Officer's Representative so as least to impair strength of structure.
         b) If, before testing, one or more of cores shows evidence of having been damaged subsequent to or during removal from structure, damaged core shall be replaced.
      3) Concrete in area represented by a core test will be considered adequate if average strength of three cores is equal to at least 85% of specified strength and no single core is less than 75% of specified strength.
      4) Fill core holes with non-shrink grout and finish to match surrounding surface when exposed in a finished area.
4. If core tests are inconclusive or impractical to obtain or if structural analysis does not confirm safety of structure, load tests may be required and their results evaluated in accordance with ACI 318, Chapter 27.
5. Correct or replace concrete work judged inadequate by structural analysis or by results of core tests or load tests with additional construction, as directed by Contracting Officer's Representative, at Contractor's expense.
6. Contractor to pay all costs incurred in providing additional testing and/or structural analysis required.

END OF SECTION

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SECTION 03 11 13
FORMWORK

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Formwork requirements for concrete construction.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 05 05 - Concrete Testing and Inspection.
   2. Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing.
   3. Section 03 35 00 - Concrete Finishing and Repair of Surface Defects.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Concrete Institute (ACI):
      a. CT-13, Concrete Terminology.
      b. 117, Specification for Tolerances for Concrete Construction and Materials.
      c. 347R, Guide to Formwork for Concrete.

B. Qualifications:
   1. Formwork, shoring and reshoring to be designed by a licensed professional engineer currently registered or having a minimum of three years of experience in this type of design work.
      a. Above qualifications apply to slabs and beams not cast on the ground.

C. Miscellaneous:
   1. Design and engineering of formwork, shoring and reshoring as well as its construction is the responsibility of the Contractor.
   2. Design requirements:
      a. Design formwork for loads, lateral pressures and allowable stresses outlined in ACI 347R and for design considerations, wind loads, allowable stresses and other applicable requirements of the controlling local building code.
         1) Where conflicts occur between the above two standards, the more stringent requirements shall govern.
      b. Design formwork to limit maximum deflection of form facing materials reflected in concrete surfaces exposed to view to 1/240 of span between structural members.
   3. For slabs and beams not cast on the ground, develop a procedure and schedule for removal of shores and installation of reshores and for calculating the loads transferred to the structure during this process in accordance with ACI 347R.
      a. Perform structural calculations as required to prove that all portions of the structure in combination with remaining forming and shoring system has sufficient strength to safely support its own weight plus the loads placed thereon. Calculations shall be performed by a licensed professional engineer.
      b. When developing procedure, schedule and structural calculations, consider the following at each stage of construction:
         1) The structural system that exists.
         2) Effects of all loads during construction.
         3) Strength of concrete.
         4) The influence of deformations of the structure and shoring system on the distribution of dead loads and construction loads.
         5) The strength and spacing of shores or shoring systems used, as well as the method of shoring, bracing, shore removal, and reshoring including the minimum time intervals between the various operations.

6) Any other loading or condition that affects the safety or serviceability of the structure during construction.

1.3 DEFINITIONS

A. Words and terms used in these Specifications are defined in ACI CT-13.

B. SCC: Self-Consolidating Concrete.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Manufacturer and type of proposed form ties.

B. Samples:
   1. A 12 inches SQ sample of each of the following form finishes.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 MATERIALS

A. Forms for Surfaces Exposed to View:
   1. Wood forms:
      a. 5/8 or 3/4 inches 5-ply structural plywood of concrete form grade.
      b. Built-in-place or prefabricated type panel.
   2. Metal forms:
      a. Metal forms may be used except for aluminum in contact with concrete.
      b. Forms to be tight to prevent leakage, free of rust and straight without dents to provide members of uniform thickness.

B. Forms for Surfaces Not Exposed to View:
   1. Wood or metal sufficiently tight to prevent leakage.
   2. Do not use aluminum forms.

2.2 ACCESSORIES

A. Form Ties:
   1. Commercially fabricated for use in form construction.
      a. Field fabricated ties are unacceptable.
   2. Constructed so that ends or end fasteners can be removed without causing spalling at surfaces of the concrete.
   3. Embedded portion of ties to be not less than 1-1/2 inches from face of concrete after ends have been removed.
   4. Cone size:
      a. 3/4 inches minimum diameter cones on both ends.
      b. Depth of cone not to exceed the concrete reinforcing cover.
   5. Provide ties with built-in waterstops in all walls that will be in contact with below grade soil.
   6. Through-wall ties that are designed to be entirely removed are not allowed in all walls that will be in contact with liquids during plant operation.

B. Form Release Material:
   1. If project contains self-consolidating concrete, provide reactive, vegetable-based product, not barrier type.

C. Void Forms:
   1. Constructed from double faced corrugated cardboard or fiberboard which is wax impregnated and laminated with moisture-resistant adhesive.

2. Capable of resisting moisture with no loss of load carrying strength or change in depth or configuration.

PART 3 - EXECUTION

3.1 PREPARATION

A. Form Surface Treatment:
   1. Before placing of reinforcing steel or concrete, cover surfaces of forms with an approved release material that will effectively prevent absorption of moisture and prevent bond with concrete, will not stain concrete or prevent bonding of future finishes.
      a. A field applied form release agent or sealer of approved type or a factory applied nonabsorptive liner may be used.
   2. Do not allow excess form release material to stand in puddles in forms nor in contact with hardened concrete against which fresh concrete is to be placed.

B. Apply form release material to minimize bugholes and pinholes. Follow manufacturer's printed installation instructions specific to the form facing material.

C. Provide temporary openings at base of column and wall forms and at other points where necessary to facilitate cleaning and observation immediately before concrete is placed, and to limit height of free fall of concrete to prevent aggregate segregation.
   1. Temporary openings to limit height of free fall of concrete shall be spaced no more than 8 feet apart.

D. Clean surfaces of forms, reinforcing steel and other embedded materials of any accumulated mortar or grout from previous concreting and of all other foreign material before concrete is placed.

3.2 ERECTION

A. Install products in accordance with manufacturer's instructions.

B. Tolerances:
   1. Conform to ACI 117.
   2. Variation from plumb:
      a. In lines and surfaces of columns, piers, walls, and in risers.
         1) Maximum in any 10 feet of height: 1/4 inches.
         2) Maximum for entire height: 1/2 inches.
      b. For exposed corner columns, control-joint grooves, and other exposed to view lines:
         1) Maximum in any 20 feet length: 1/4 inches.
         2) Maximum for entire length: 1/2 inches.
   3. Variation from level or from grades specified:
      a. In slab soffits, ceilings, beam soffits and in arises, measured before removal of supporting shores.
         1) Maximum in any 10 feet of length: 1/4 inches.
         2) Maximum in any bay or in any 20 feet length: 3/8 inches.
         3) Maximum for entire length: 3/4 inches.
      b. In exposed lintels, sills, parapets, horizontal grooves, and other exposed to view lines:
         1) Maximum in any bay or in 20 feet length: 1/4 inches.
         2) Maximum for entire length: 1/2 inches.
   4. Variation of linear structure lines from established position in plan and related position of columns, walls, and partitions:
      a. Maximum in any bay: 1/2 inches.
      b. Maximum in any 20 feet of length: 1/2 inches.
      c. Maximum for entire length: 1 inch.
   5. Variation in sizes and location of sleeves, floor openings, and wall openings: Maximum of +1/2 inches.

6. Variation in horizontal plan location of beam, column and wall centerlines from required location: Maximum of +1/2 inches.
7. Variation in cross sectional dimensions of columns and beams and in thickness of slabs and walls: Maximum of -1/4 inches, +1/2 inches.
8. Footings and foundations:
    a. Variations in concrete dimensions in plan: -1/2 inches, +2 inches.
    b. Misplacement or eccentricity:
        1) 2% of footing width in direction of misplacement but not more than 2 inches.
    c. Thickness:
        1) Decrease in specified thickness: 5%.
        2) Increase in specified thickness: No limit except that which may interfere with other construction.
9. Variation in steps:
    a. In a flight of stairs:
        1) Rise: +1/8 inches.
        2) Tread: +1/4 inches.
    b. In consecutive steps:
        1) Rise: +1/16 inches.
        2) Tread: +1/8 inches.
10. Establish and maintain in an undisturbed condition and until final completion and acceptance of Project, sufficient control points and benchmarks to be used for reference purposes to check tolerances.
11. Regardless of tolerances listed allow no portion of structure to extend beyond legal boundary of Project.
12. To maintain specified tolerances, camber formwork to compensate for anticipated deflections in formwork prior to hardening of concrete.

C. Make forms sufficiently tight to prevent loss of mortar from concrete.

D. Place 3/4 inches chamfer strips in exposed to view corners of forms to produce 3/4 inches wide beveled edges.

E. At construction joints, overlap contact surface of form sheathing for flush surfaces exposed to view over hardened concrete in previous placement by at least 1 inch.
    1. Hold forms against hardened concrete to prevent offsets or loss of mortar at construction joint and to maintain a true surface.
    2. Where possible, locate juncture of built-in-place wood or metal forms at architectural lines, control joints or at construction joints.

F. Where circular walls are to be formed and forms made up of straight sections are proposed for use, provide straight lengths not exceeding 2 feet wide.
    1. Brace and tie formwork to maintain correct position and shape of members.

G. Construct wood forms for wall openings to facilitate loosening, if necessary, to counteract swelling.

H. Anchor formwork to shores or other supporting surfaces or members so that movement of any part of formwork system is prevented during concrete placement.

I. Provide runways for moving equipment with struts or legs, supported directly on formwork or structural member without resting on reinforcing steel.

J. Provide positive means of adjustment (wedges or jacks) of shores and struts and take up all settlement during concrete placing operation.
    1. Securely brace forms against lateral deflection.
    2. Fasten wedges used for final adjustment of forms prior to concrete placement in position after final check.

K. Stay-In-Place Forms:
    1. Support stay-in-place forms as required to maintain the formwork in proper position.

2. Hold the edge of stay-in-place forms back a minimum of 2 inches from all smooth formed concrete surfaces.
3. Stay-in-place forms may be used at the Contractor's option at:
    a. Surfaces that will be backfilled with soil.
        1) Maintain a minimum of 3 inches of concrete cover over all reinforcing.
    b. Roughened construction joints.
    c. Other locations approved by Contracting Officer Representative.

3.3 REMOVAL OF FORMS

A. No construction loads shall be supported on, nor any shoring removed from, any part of the structure under construction except when that portion of the structure in combination with remaining forming and shoring system has sufficient strength to safely support its weight and loads places thereon.

B. When required for concrete curing in hot weather, required for repair of surface defects or when finishing is required at an early age, remove forms as soon as concrete has hardened sufficiently to resist damage from removal operations or lack of support.

C. Remove top forms on sloping surfaces of concrete as soon as concrete has attained sufficient stiffness to prevent sagging.
    1. Perform any needed repairs or treatment required on such sloping surfaces at once, followed by curing specified in Specification Section 03 31 31.

D. Loosen wood forms for wall openings as soon as this can be accomplished without damage to concrete.

E. Formwork for columns, walls, sides of beams, and other parts not supporting weight of concrete may be removed as soon as concrete has hardened sufficiently to resist damage from removal.

F. Where no reshoring is planned, leave forms and shoring used to support weight of concrete in place until concrete has attained its specified 28-day compressive strength.
    1. Where a reshoring procedure is planned, supporting formwork may be removed when concrete has reached the concrete strength required by the formwork designer's structural calculations.

G. When shores and other vertical supports are so arranged that non-load-carrying form facing material may be removed without loosening or disturbing shores and supports, facing material may be removed when concrete has sufficiently hardened to resist damage from removal.

3.4 RESHORING

A. No construction loads shall be supported on, nor any shoring removed from, any part of the structure under construction except when that portion of the structure in combination with remaining forming and shoring system has sufficient strength to safely support its weight and loads placed thereon.

B. While reshoring is underway, no superimposed dead or live loads shall be permitted on the new construction.

C. During reshoring do not subject concrete in structural members to combined dead and construction loads in excess of loads that structural members can adequately support.

D. Place reshores as soon as practicable after stripping operations are complete but in no case later than end of working day on which stripping occurs.

E. Tighten reshores to carry their required loads without overstressing.

F. Shoring, reshoring and supporting formwork may be removed when concrete has reached the concrete strength required by the formwork designer's structural calculations.

G. For floors supporting shores under newly placed concrete leave original supporting shores in place or reshore.

1. Reshoring system shall have a capacity sufficient to resist anticipated loads.
2. Locate reshores directly under a shore position above.

H. In multi-story buildings, extend reshoring over a sufficient number of stories to distribute weight of newly placed concrete, forms, and construction live loads in such a manner that design superimposed loads of floors supporting shores are not exceeded.

3.5 FIELD QUALITY CONTROL

A. Special Inspection:
1. See Section 01 40 00.
2. See Section 03 05 05.

END OF SECTION

SECTION 03 15 19
ANCHORAGE TO CONCRETE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Requirements for anchorages in concrete, including: cast-in-place anchor bolts, anchor rods, reinforcing anchorage adhesive, and post-installed concrete anchors required for the Project but not specified elsewhere in the Contract Documents.
   2. Delegated design requirements for concrete anchors not expressly shown or indicated on the Drawings or elsewhere in the Contract Documents, including, but not limited to, anchorages in concrete for the following structural and nonstructural components:
      a. Structural members and accessories.
      b. Metal, wood, and plastic fabrications.
      c. Architectural Work, including building components, finishes, specials, facility equipment and conveying equipment, furnishings, and special construction Work.
      d. Mechanical and electrical Work, including process-mechanical Work, site and infrastructure Work, electrical Work, communications Work, electronic safety and security systems Work, and others.
      e. Fire suppression, plumbing, and HVAC Work.
      f. Other components requiring anchorages to concrete.

B. Related Requirements: Include but are not necessarily limited to:
   1. Section 01 35 73 - Delegated Design Procedures.
   2. Section 01 81 10 - Wind and Seismic Criteria
   3. Section 03 05 05 - Concrete Testing and Inspection.
   4. Section 46 13 15 – Aluminum Covers – Formed Panel Type

1.2 REFERENCES

A. Definitions and Terminology:
   1. This provision presents definitions and terminology, which have the meanings indicated in this provision, applied to the singular or plural thereof, and without regard to use of initial capital letters.
      a. Adhesive Anchors:
         1) Post-installed anchors developing their strength primarily from chemical bond between the concrete and the anchor.
         2) Includes anchors using acrylics, epoxy and other similar adhesives.
      b. Anchor Bolt: Any cast-in-place anchorage that is made of a headed (i.e. bolt) material.
      c. Anchor Rod: Any cast-in-place or post-installed anchorage made from unheaded, threaded, rod or deformed bar material.
      d. Concrete Anchor: Generic term for either an anchor bolt or an anchor rod.
      e. Galvanizing: Hot-dip galvanizing in accordance with ASTM A123, ASTM A153 or ASTM F2329 with minimum coating of 2.0 ounces of zinc per square foot of metal (average of specimens) unless noted otherwise or dictated by reference standard.
      f. Hardware: As defined in ASTM A153.
      g. MPII: Manufacturer's printed installation instructions.
      h. Mechanical Anchors:
         1) Post-installed anchors developing their strength from attachment other than thru adhesives or chemical bond to concrete.
         2) Includes expansion anchors, expansion sleeve, screw anchors, undercut anchors, specialty inserts and other similar types of anchorages.
         3) Drop-in anchors and other similar non-ICC ES approved anchors are unacceptable.

i. Post-Installed Anchor: Adhesive or mechanical anchor installed into previously placed and adequately cured concrete.

B. Reference Standards:
1. American Concrete Institute (ACI):
a. 318, Building Code Requirements for Structural Concrete and Commentary.
b. 350, Code Requirements for Environmental Engineering Concrete Structures and Commentary.
2. American Concrete Institute/Concrete Reinforcing Steel Institute (ACI-CRSI):
a. Adhesive Anchor Installation Certification Program: Adhesive Anchor Installer.
3. American Institute of Steel Construction (AISC):
a. 303, Code of Standard Practice for Steel Buildings and Bridges.
b. 355.2, Seismic Testing of Post-Installed Concrete and Masonry Anchors in Cracked Concrete.
c. 355.4, Qualification of Post-Installed Adhesive Anchors in Concrete.
4. ASTM International (ASTM):
a. A36, Standard Specification for Carbon Structural Steel.
b. A108, Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished.
c. A123, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
d. A153, Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
e. A307, Standard Specification for Carbon Steel Bolts and Studs, 60,000 PSI Tensile Strength.
f. A496, Standard Specification for Steel Wire, Deformed, for Concrete Reinforcement.
g. A563, Standard Specification for Carbon and Alloy Steel Nuts.
h. A780, Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings.
i. F436, Standard Specification for Hardened Steel Washers.
j. F593, Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.
k. F594, Standard Specification for Stainless Steel Nuts.
l. F1554, Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength.
m. F2329, Standard Specification for Zinc Coating, Hot-Dip, Requirements for Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts, and Special Threaded Fasteners
5. International Code Council Evaluation Service (ICC-ES):
a. AC193, Acceptance Criteria for Mechanical Anchors in Concrete Elements.
b. AC308, Acceptance Criteria for Post-Installed Adhesive Anchors in Concrete Elements.

1.3 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
1. For post-installed anchors and other anchorages installed in existing concrete, comply with Section 01 73 29 - Cutting and Patching.
2. Comply with coordination requirements in accordance with Contract Documents.

B. Pre-Design Conference:
1. Pre-design conference for anchorages to concrete (where such anchorages are delegated design Work) is not required.

1.4 QUALITY ASSURANCE

A. Regulatory Requirements:
1. Code-required Special Instructions:
a. Special Inspection is required in accordance with the building code for all concrete anchorages.

b. Notify the Special Inspector that an inspection is required prior to concrete placement (or during post-installed anchorage installation).
c. See the "Field Quality Control" Article in "Part 3 – Execution" of this Section for additional requirements.

B. Qualifications:
1. Delegated Design Professional:
a. Delegated design professional's qualifications and responsibilities shall be in accordance with Contract Documents.
2. Installer:
a. Installer for post-installed anchors shall be trained by the manufacturer or certified by a training program approved by the Contracting Officer.
b. Installer for adhesive anchors installed in horizontal, upward incline, or overhead applications shall be certified by ACI-CRSI Adhesive Anchor Installation Certification Program.

1.5 SUBMITTALS

A. Action Submittals: Submit the following:
1. Shop Drawings:
a. Submit schedule (table or listing) of types, sizes (diameter, length, embedment length), material, finish, and proposed manufacturers of anchorages to concrete to be provided. Apportion by Project-specific application (for example, "Anchorages for cooling water pumps in basement") and indicate where anchorages are fully-designed by Contracting Officer's Representative and those for which final design was prepared by delegated design professional.
b. Contracting Officer's approval of such Shop Drawing will be only for anchorages fully designed by Engineer. For anchorages for which final design is by delegated design professional, include on such Shop Drawing delegated design professional's approval stamp.
2. Product Data: Submit as Action Submittals product data for anchorages to concrete fully designed by Contracting Officer's Representative. For anchorages to concrete for which final design is by delegated design professional, submit as Informational Submittals bearing approval stamp of delegated design professional.
a. Manufacturer's express, written acknowledgement that proposed items comply with referenced standards indicated in this Section and, as applicable, by delegated design professional.
b. Manufacturer published data and information for each anchor.
1) Clearly indicate items that are proposed for the Work. Neatly strike out or obscure materials and products not proposed.
c. Manufacturer's published installation instructions and instructions for code-required special inspections and tests.
d. Post-Installed Anchors: In addition, submit for each post-installed anchor system current ICC-ES report , indicating the following:
1) Manufacturer's certification that anchors comply with requirements indicated in the Contract Documents.
2) Performance data indicating that anchor is approved by its manufacturer for use in cracked concrete.
3) Seismic design categories for which anchor system is approved by ICC-ES report.
e. Anchorage layout drawings and details:
1) Drawings showing location, configuration, spacing and edge distance.
3. Delegated Design Professional's Instruments of Service Submittals: Submit, in accordance with this Section and Contract Documents, for each anchorage to concrete for which final design is by delegated design professional:
a. Certification of compliance, in accordance with Contract Documents. Indicate design tension and shear loads used for anchor design.
b. Delegated design professional's drawings.

c. Delegated design professional's specifications, if any, when prepared by delegated design professional.
d. Post-Installed Anchors (for which final design is by delegated design professional): Submittal shall also include the following for post-installed anchors for which final design was by delegated design professional:
1) Show diameter, embedment depth and edge distance of each anchor.
2) Indicate compliance with ACI 318 Chapter 17, or, ACI 350-20 Appendix E.
3) Type of post-installed anchor system used.
a) Provide manufacturer's ICC-ES report for the following:
(1) Mechanical anchorage per ICC-ES AC193.
(2) Adhesive anchorage per ICC-ES AC308.

4. Samples:
a. Submit representative Samples of anchorages to concrete, when requested by Contracting Officer. Contracting Officer's approval of Samples will be for type and finish only.

B. Informational Submittals: Submit the following:
1. Shop Drawings and Product Data Approved by Delegated Design Professional:
a. Submit with delegated design professional's approval stamp those Shop Drawings and product data Submittals indicated in this Article but for which final design was performed by delegated design professional.
2. Calculations by Delegated Design Professional:
a. Submit sealed and signed calculations for sizing and determining embedment length of anchorages to concrete not fully designed by Contracting Officer's Representative.
b. Post-Installed Anchors Designed by Delegated Design Professional: In addition, also submit design calculations:
1) Indicate design load to each anchor.
2) When design load is not indicated on the Drawings, include calculations to develop anchor forces based on performance and design criteria indicated in this Section.
3. Supplier's Instructions:
a. Submit manufacturer's published instructions for installation.
4. Field Quality Control Submittals:
a. Submit written results of required field quality control activities indicated in this Section.
5. Reports of Supplier's Visits to Site:
a. Submit each written report of visit to the Site by Supplier's factory trained representative and delegated design professional. For each, indicate date and time of visit, purpose of visit, observations made, decisions made, problems encountered, and other pertinent information.
6. Qualifications Statements:
a. Delegated design professional.
b. Each installer.

PART 2 - PRODUCTS

A. Product requirements are to conform with the following salient characteristics.

2.2 MATERIALS – ALL ANCHORAGES

A. Materials – General:
1. This Article applies to all anchorages to concrete, regardless of whether fully designed by Contracting Officer Representative or delegated design professional. Requirements for delegated designs are in the following Article.
2. Additional requirements for anchorages fully designed by Contracting Officer Representative are indicated in the Article following requirements for delegated design anchorages.

3. For structural applications, do not use powder actuated fasteners and other types of bolts or fasteners not specified in this Section unless approved by Contracting Officer Representative or otherwise required by the Contract Documents.

B. Description:
   1. Provide anchorages to concrete, of the types shown or indicated, to secure to concrete materials, equipment, and appurtenances installed as part of the Work.
   2. Locations where anchorages are required are generally shown or indicated on the Drawings. Where not shown or indicated on the Drawings provide anchorages or the types required for materials, equipment, and systems where such materials, equipment, and systems are shown on the Drawings.
   3. Anchorages required include those for materials, equipment, and systems shown on the structural Drawings and Drawing other than the structural Drawings.
   4. Design loads for concrete anchorages are shown or indicated on the Drawings for anchorages where design responsibility is delegated to Contractor's delegated design professional. For such anchorages, embedment depths are not shown or indicated.

C. Cast-in-place Concrete Anchors:
   1. Buildings, non-building structures, and equipment, unless otherwise specified:
      a. ASTM F1554, Grade 36 or Grade 55 with weldability supplement S1 for galvanized threaded rods.
      b. ASTM A307, Grade A for galvanized headed bolts.
   2. All other cast-in-place concrete anchors:
      a. Stainless steel with matching nut and washer.
      b. Submerged application: ASTM F593, Type 316, minimum yield strength of 45,000 psi.
      c. Non-submerged application: ASTM F593, Type 304 or Type 316, minimum yield strength of 45,000 psi.

D. Post-Installed Mechanical and Adhesive Concrete Anchors:
   1. Submerged application: ASTM F593, Type 316, minimum yield strength of 45,000 psi with matching nut and washer.
   2. Non-submerged application: ASTM F593, Type 304 or Type 316, minimum yield strength of 45,000 psi with matching nut and washer.
   3. Post-installed anchors and related materials shall be listed by ICC-ES or Contracting Officer Representative-approved equivalent.

E. Reinforcing: Comply with Section 03 21 00 - Reinforcement.

F. Headed Studs: ASTM A108 with a minimum yield strength of 50,000 psi and a minimum tensile strength of 60,000 psi.

G. Deformed Bar Anchors: ASTM A496 with minimum yield strength of 70,000 psi and a minimum tensile strength of 80,000 psi.

H. Washers:
   1. ASTM F436 unless indicated otherwise, finish to match bolt.
   2. When stainless steel anchorage is provided for cast-in-place anchorages, provide washers of the same material and alloy as in the associated anchorage.
   3. Plate washers: Minimum 1/2 inch thick fabricated ASTM A36 square plates as required.
   4. Comply with manufacturer's written instructions for all post-installed anchorages.

I. Nuts:
   1. ASTM A563 for cast-in-place anchorages.
   2. When stainless steel anchorages are provided for cast-in-place anchorages, nuts shall comply with ASTM F594 and shall match material and alloy of the associated anchorage.
   3. Follow manufacturer's requirements if using post-installed anchorage.

J. Galvanizing Repair Paint:
   1. High zinc dust content paint for regalvanizing welds and abrasions.
   2. ASTM A780.

3. Zinc content: Minimum 92 percent in dry film.
4. Products and Manufacturers: "ZRC Cold Galvanizing", by ZRC; or "High Performance Zinc Spray", by Clearco; or equal.

K. Dissimilar Materials Protection: Comply with Section 09 96 00 - High-Performance Industrial Coatings.

2.3 DELEGATED DESIGN ANCHORAGES TO CONCRETE

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Post-installed anchor systems for indicated manufacturers are acceptable only when a current ICC-ES evaluation report is furnished as a Submittal and the subject anchorage system is approved by a delegated design professional.
2. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.
3. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
a. Hilti.
b. Or equal.

B. Description: Perform delegated design for anchorages when one or more of the following applies:
1. Design load for concrete anchorage is shown or indicated on the Drawings and anchorage embedment depth is not shown or indicated.
2. When specifically required by the Contract Documents.
3. When an anchorage is necessary but is not shown or indicated on the Drawings.
4. Anchorages shown on the Drawings other than the structural Drawings.

C. Performance and Design Criteria for Delegated Design Anchorages:
1. Determine design loads, including wind and seismic loads, in accordance with applicable building code and other Laws and Regulations.
2. For anchorage of equipment and non-structural components, use actual dead load and operating loads obtained by Contractor or delegated design professional from manufacturer. Design loads shall include operating conditions when equipment or element of the Work is in operation, dynamic loads, and other loads as appropriate or required by the building code or other Laws or Regulations.
3. Design assuming cracked concrete.

2.4 ANCHORAGES FULLY DESIGNED BY ENGINEER

A. When size, length, and details of anchorage are shown or indicated on the structural Drawings, such anchorages are considered as fully designed by Engineer and delegated design of such anchorage is not required.

B. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:

:

1. For post-installed anchor systems regardless of whether proposed manufacturer is indicated below, furnish as Submittal current evaluation agency report and anchor system is certified by ICC-ES for cracked concrete conditions.
2. Mechanical Anchors:
   a. Hilti:
      1) Kwik Bolt TZ2 (ICC-ESR-4266).
   b. Or equal.
3. Adhesive Anchors for Concrete:
   a. Hilti:
      1) HIT RE 500 V3 (ICC ESR-3814).
   b. Or equal.
4. Screw Anchors for Concrete:
   a. Hilti:
      1) Kwik HUS-EZ Screw (ICC-ES ESR-3027).
   b. Or equal.
5. Requests, if any, for Contracting Officer's approval of "or-equals" or substitutes shall indicate proposed anchor has at least the same tension and shear strength as the associated anchorage products indicated by name in this Article.
6. Submit requests for substitution in accordance with Specification Section 01 33 23.

PART 3 - EXECUTION

3.1 PREPARATION

A. For cast-in-place concrete anchorages, allow adequate time for proper installation, inspection, and observation prior to placing concrete.

B. Prior to installation, inspect and verify areas and conditions under which concrete anchorages will be installed.
   1. Notify Contracting Officer Representative of conditions detrimental to proper and timely completion of the Work.
   2. Do not proceed with the Work until unsatisfactory conditions are properly remedied.

3.2 INSTALLATION

A. Installation Requirements - General:
   1. Install items in accordance with the Contract Documents, manufacturer's written instructions, and Laws and Regulations. Where such requirements conflict, obtain interpretation or clarification from Contracting Officer's Representative prior to commencing the associated Work.
   2. Perform the following unless shown or indicated otherwise:
      a. Provide stainless steel anchorages for connecting aluminum and steel members to concrete and masonry. Provide dissimilar materials protection in accordance with Section 09 96 00 – High-Performance Industrial Coatings.
      b. Provide washers for all anchorages.
      c. Where exposed, extend threaded anchorage a maximum of 0.75 inch and a minimum of 0.5 inch above top of fully-engaged nut. If anchorage is cut off to required maximum height, dress the threads to allow nuts to be removed without damage to nuts.
   3. Tightening of nuts: Do the following after nuts are snug-tightened down:
      a. Upset anchorage threads to prevent nuts from backing off. Provide double nut or lock nut in lieu of upset threads for items that may require future removal.
      b. For cast-in-place anchorages (excluding post-installed anchorages), tighten nuts an additional 1/8 turn beyond snug tight to prevent nuts from backing off.
      c. When two nuts are used per concrete anchor above the base plate, tighten top nut an additional 1/8 turn to "lock" the two nuts together.
      d. For post-installed anchorages, comply with MPII.

4. Secure architectural components to avoid aesthetic distortion and to avoid overstressing fasteners from expansion, contraction, or installation.

B. Cast-in-Place Anchorages:
   1. Provide where anchor rods or anchor bolts are indicated on the Drawings, unless another anchor type is approved by Contracting Officer's Representative.
   2. Provide concrete anchorages as shown on the Drawings or as required to secure the Work to concrete.
   3. Tie cast-in-place anchorage in position to embedded reinforcing steel using wire.
   4. Tack welding of anchorage is prohibited.
   5. Chase threads as required and coat projected portion of carbon steel anchors and nut threads with a heavy coat of clean grease after concrete has cured.
   6. Anchorage location Tolerance: in accordance with AISC 303.
   7. Provide steel or durable wood templates for all column and equipment anchorages. Place templates above top of concrete; do not impede proper concrete placement and consolidation.

C. Mechanical Anchorages:
   1. Use only where specifically indicated on the Drawings or when approved for use by the Contracting Officer's Representative.
   2. Do not use where subjected to vibration.
   3. May be used in overhead applications.
   4. Contact Contracting Officer's Representative for clarification when anchors will not be installed in compliance with the manufacturer's printed installation requirements.

D. Post-installed Anchorages:
   1. For post-installed anchors, comply with MPII regarding hole diameter and depth required to fully develop the tensile strength of anchor or reinforcing bar.
   2. Use hammer drills to create holes.
   3. Properly clean out holes in accordance with the associated ICC-ES report using non-metallic, fiber bristle brush and compressed air, or as otherwise necessary to remove all loose material from each hole prior to installing anchor in the presence of Special Inspector.
   4. Adhesive Anchorages:
      a. Provide only where specifically indicated on the Drawings or when approved for use by Contracting Officer's Representative.
      b. May be provided where subjected to vibration or at buried or submerged locations.
      c. Do not install for overhead applications or sustained tension loading conditions such as utility hangers.
      d. Install adhesive anchors in concrete aged not less than 21 days.

E. Finishes:
   1. Coat aluminum surfaces in contact with dissimilar materials in accordance with Section 09 96 00 – High-Performance Industrial Coatings.
   2. Repair of damaged galvanized surfaces:
      a. Prepare damaged surfaces by abrasive blasting or power sanding.
      b. Repair damaged galvanized surfaces in accordance with ASTM A780.
      c. Apply galvanizing repair paint to not less than 6 mils dry film thickness in accordance with galvanizing repair material manufacturer's instructions and ASTM A780.

F. Ensure that embedded items are protected from damage and are not filled in with concrete or related materials.

3.3 FIELD QUALITY CONTROL

A. Field Tests and Inspections:
   1. Special Inspections: Comply with Section 01 40 00 – Quality Requirements.
   2. Comply with Section 03 05 05 - Concrete Testing and Inspection.
   3. Tests and inspections of anchorages shall comply with ACI 355.2 and/or ACI 355.4 as applicable.

4. The Contracting Officer reserves the right to inspect and test completed anchorages at a minimum of 25 to 50 percent of provided anchorages.

B. Supplier's services:
   1. Post-installed anchor manufacturer's representative shall demonstrate and observe the proper installation procedures for the post-installed anchors.

C. Defective Work:
   1. Anchorages that do not successfully pass required field tests and inspections or that are otherwise deemed defective by the Contracting Officer's Representative shall be remedied, in accordance with the Contract Documents, at no cost to the Contracting Officer.

3.4 CLEANING

A. After concrete has been placed, remove protection and clean all anchorage of all concrete, dirt, and other foreign matter.

B. Provide surface acceptable to receive field applied paint coatings when specified in Specification Section 09 96 00.

END OF SECTION

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SECTION 03 21 00
REINFORCEMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Reinforcing bar requirements for concrete construction.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 05 05 - Concrete Testing and Inspection.
   2. Section 03 15 19 - Anchorage to Concrete.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Concrete Institute (ACI):
      a. SP 66, ACI Detailing Manual.
      b. 117, Specification for Tolerances for Concrete Construction and Materials.
      c. 315, Manual of Standard Practice for Detailing Reinforced Concrete Structures.
      d. 318, Building Code Requirements for Structural Concrete.
      e. 350, Code Requirements for Environmental Engineering Concrete Structures.
   2. ASTM International (ASTM):
      a. A36, Standard Specification for Carbon Structural Steel.
      b. A276, Standard Specification for Stainless Steel Bars and Shapes.
      c. A615, Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement.
      d. A706, Standard Specification for Low-Alloy Steel Deformed and Plain Bars for Concrete Reinforcement.
      e. A970, Standard Specification for Headed Steel Bars for Concrete Reinforcement.
      f. A1064, Standard Specification for Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete.
   3. Concrete Reinforcing Steel Institute (CRSI):
      a. Manual of Standard Practice.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Mill certificates for all reinforcing.
      d. Manufacturer and type of proprietary reinforcing mechanical splices.
   2. Qualifications of welding operators, welding processes and procedures.
   3. Reinforcing number, sizes, spacing, dimensions, configurations, locations, mark numbers, lap splice lengths and locations, concrete cover and reinforcing supports.
   4. Sufficient reinforcing details to permit installation of reinforcing.
   5. Reinforcing details in accordance with ACI SP 66 and ACI 315.
   6. Locations where proprietary reinforcing mechanical splices are required or proposed for use.
   7. Shop Drawings shall be in sufficient detail to permit installation of reinforcing without reference to Contract Drawings.
      a. Shop Drawings shall not be prepared by reproducing the plans and details indicated on the Contract Drawings but shall consist of completely redrawn plans and details as necessary to indicate complete fabrication and installation of all reinforcing steel.

b. Where multiple types of supports for reinforcing steel (such as chairs, runners, bolsters, and other types of supports) will be used in the Work, clearly indicate on the Shop Drawings the support types and materials of supports.

1.4 DELIVERY, STORAGE, AND HANDLING

A. Support and store all reinforcing above ground.

B. Ship to jobsite with attached plastic or metal tags with permanent mark numbers which match the Shop Drawing mark numbers.

PART 2 - PRODUCTS

2.1 MANUFACTURES

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Reinforcing adhesive anchors:
      a. See Specification Section 03 15 19.
   2. Reinforcing mechanical splices:
      a. Lenton Rebar Splicing by Erico, Inc.
      b. Or Equal.
   3. Submit requests for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Reinforcing Bars: ASTM A615, grade 60, deformed.

B. Reinforcing Bars to be Welded: ASTM A706, Grade 60, deformed.

C. Welded Wire Reinforcement: ASTM A1064.

D. Smooth Dowel Bars:
   1. Water containing structures: ASTM A276, Type 316.
   2. All other locations: ASTM A36, with metal end cap to allow longitudinal movement equal to joint width plus 1 inch.

E. Proprietary Reinforcing Mechanical Splices: To develop in tension and compression a minimum of 125% of the yield strength of the reinforcing bars being spliced.

F. Headed Deformed Bars:
   1. ASTM A970, Class A.

G. Reinforcing Adhesive Anchors:
   1. See Specification 03 15 19.

2.3 ACCESSORIES

A. Chairs, Runners, Bolsters, Spacers, Hangers, and Other Reinforcing Supports:
   1. Metal fabrications with plastic-coated tips in contact with forms.
      a. Plastic coating meeting requirements of CRSI Manual of Standard Practice.
   2. All plastic construction meeting the requirements of CRSI Manual of Standard Practice.
      a. 100% non-metallic, non-corrosive.

b. Required for all walls and elevated construction exposed to liquid containing structures.

B. Protective plastic caps at mechanical splices.

2.4 FABRICATION

A. Tolerances:
1. Conforms to ACI 117, expect as modified herein.
2. Sheared lengths: +1 inches.
3. Overall dimensions of stirrups, ties and spirals: +1/2 inches.
4. All other bends: +0 inches, -1/2 inches.

B. Minimum diameter of bends measured on the inside of the reinforcing bar to be as indicated in ACI 318 Paragraph 7.2.

C. Ship reinforcing to jobsite with attached plastic or metal tags.
1. Place on each tag the mark number of the reinforcing corresponding to the mark number indicated on the Shop Drawing.
2. Mark numbers on tags to be so placed that the numbers cannot be removed.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Tolerances:
1. Conform to ACI 117, except as modified herein.
2. Reinforcing placement:
a. Clear distance to formed surfaces: +1/4 inches.
b. Minimum spacing between bars: -1/4 inches.
c. Top bars in slabs and beams:
1) Members 8 inches deep or less: +1/4 inches.
2) Members between 8 inches and 2 feet deep: -1/4 inches, +1/2 inches.
3) Members more than 2 feet deep: -1/4 inches, +1 inches.
d. Crosswise of members: Spaced evenly within +1 inches.
e. Lengthwise of members: +2 inches.
3. Minimum clear distances between reinforcing bars:
a. Beams, walls and slabs: Distance equal to bar diameter or 1 inch, whichever is greater.
b. Columns: Distance equal to 1-1/2 times the bar diameter or 1-1/2 inches, whichever is greater.
c. Beam and slab reinforcing shall be threaded through the column vertical rebars without displacing the column vertical bars and still maintaining the clear distances required for the beam and slab reinforcing bars.

B. Minimum concrete protective covering for reinforcement: As shown on Drawings.

C. Unless indicated otherwise on Drawings, provide splice lengths for reinforcing as follows:
1. For reinforcing: As indicated on the Drawings.
2. For welded wire reinforcement:
a. Splice lap length measured between outermost cross wires of each fabric sheet shall not be less than one spacing of cross wires plus 2 inches, nor less than 1.5 x development length nor less than 6 inches.
b. Development length shall be as required for the yield strength of the welded wire reinforcement in accordance with ACI 318 and ACI 350.
3. Provide splices of reinforcing not specifically indicated or specified subject to approval of Contracting Officer's Represntative.
a. Mechanical proprietary splice connectors may only be used when approved or indicated on the Contract Drawings.

D. Welding:
1. Welding reinforcing is not permitted.

E. Placing Reinforcing:
1. Assure that reinforcement at time concrete is placed is free of mud, oil or other materials that may affect or reduce bond.
2. Reinforcement with rust, mill scale or a combination of both will be accepted as being satisfactory without cleaning or brushing provided dimensions and weights including heights of deformations on a cleaned sample is not less than required by applicable ASTM specification that governs for the reinforcing supplied.
3. Reinforcing support:
    a. Uncoated reinforcing:
        1) Support reinforcing and fasten together to prevent displacement by construction operations.
            a) Locate and support reinforcement with bar supports to maintain minimum concrete cover.
            b) Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.
            c) Reinforcement shown on the Contract Documents may not be repositioned for use a support for reinforcement. Additional drop bars may be provided for support of reinforcing,
        2) Reinforcing supported on ground:
            a) Slab on grade and other members with only one mat of reinforcing:
                (1) Provide metal bar supports with bottom plate.
                (2) Do not use concrete blocks to support slab-on-grade reinforcing.
            b) All other members: Provide supporting concrete blocks or metal bar supports with bottom plate.
        3) Reinforcing supported on formwork:
            a) Concrete surfaces in contact with or over process liquid: All-Plastic chairs, runners and bar supports.
            b) All other formed surfaces:
                (1) Provide plastic-coated metal chairs, runners, bolsters, spacers, hangers and other reinforcing support.
                (2) Only tips in contact with the forms need to be plastic coated.
4. Support reinforcing over cardboard void forms by means of concrete supports which will not puncture or damage the void forms during construction nor impair the strength of the concrete members in any way.
5. Where parallel horizontal reinforcement in beams is indicated to be placed in two or more layers, bars in the upper layers shall be placed directly above bars in the bottom layer with clear distance between layers to be 1 inch.
    a. Place spacer bars at 3 feet maximum centers to maintain the required 1 inch clear distance between layers.
6. Extend reinforcement to within 2 inches of concrete perimeter edges.
    a. If perimeter edge is formed by earth, extend reinforcement to within 3 inches of the edge.
7. To assure proper placement, furnish templates for all column vertical bars and dowels.
8. Do not bend reinforcement after embedding in hardened concrete unless approved by Contracting Officer's Representative.
    a. Do not bend reinforcing by means of heat.
9. Do not tack weld reinforcing.
10. Embed reinforcing into hardened concrete utilizing adhesive anchor system specifically manufactured for such installation:
    a. See Specification Section 03 15 19.

3.2 FIELD QUALITY CONTROL

A. Reinforcement Congestion and Interferences:
1. Notify Contracting Officer's Representative whenever the specified clearances between bars cannot be met.

2. Do not place any concrete until the Contracting Officer's Representative submits a solution to reinforcing congestion problem.
3. Reinforcing may be moved as necessary to avoid interference with other reinforcing steel, conduits, or embedded items.
4. If bars are moved more than one bar diameter, obtain Contracting Officer's Representative's approval of resulting arrangement of reinforcing.
5. No cutting of reinforcing shall be done without written approval of the Contracting Officer.

B. Special Inspection:
1. See Section 01 40 00.
2. See Section 03 05 05.

END OF SECTION

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SECTION 03 31 30
CONCRETE MATERIALS AND PROPORTIONING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Requirements for concrete materials, including:
      a. Materials, including various types of grout, admixtures, cement, sand, aggregate, and other materials.
      b. Concrete strength and proportioning, including design mixes for:
         1) Normal weight concrete.
         2) Flowable fill (Controlled low strength material).
         3) Grout.

B. Scope:
   1. Unless shown or indicated otherwise in the Contract Documents, concrete Work shall comply with:
      a. ACI 301.
      b. Laws and Regulations, including applicable building code.
   2. In this Section, material is apportioned into the following grouting types:
      a. Base plates for columns and equipment.
      b. As otherwise shown or indicated in the Contract Documents.
   3. This Section addresses materials for concrete. Other Specifications sections present other requirements for complete concrete Work, including, but not necessarily limited to:
      a. Section 03 05 05 - Concrete Testing and Inspection.
      b. Section 03 11 13 - Formwork.
      c. Section 03 21 00 - Reinforcement.
      d. Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing.
      e. Section 03 35 00 - Concrete Finishing and Repair of Surface Defects.
      f. Others as indicated in the Contract Documents.

C. Related Requirements: Include but are not necessarily limited to:
   1. Section 03 05 05 - Concrete Testing and Inspection.
   2. Section 03 15 19 - Anchorage to Concrete.
   3. Section 03 21 00 - Reinforcement.
   4. Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing.
   5. Section 03 41 33 - Precast and Prestressed Concrete.

1.2 REFERENCES

A. Abbreviations and Terminology:
   1. Abbreviations: The following abbreviations are used in this Section:
      a. “AAR” means deleterious “alkali-aggregate reaction”, resulting from either alkali-silica reactive (ASR) or alkali-carbonate reactive (ACR) aggregates.
      b. “SCM” means “supplementary cementitious materials”, with the meaning indicated below.
      c. “CLSM” means controlled low strength material in accordance with requirements of this Section.
   2. Terminology: Terminology indicated below are not defined terms and are not indicated with initial capital letters, but when used in this Section have the meanings indicated below:
      a. “Supplementary cementitious materials” (SCM) means fly ash, silica fume, and GGBFS.
      b. “Water-bearing concrete” is concrete surface to be in contact (whether continuously or intermittently) with water, process liquid, or slurries during intended operation of the

facility, including, but not limited to, concrete tanks, channels, wet wells, distribution chambers, and secondary containment structures.

c. Independent Laboratory:
   1) Testing shall be performed by an independent laboratory complying with requirements of the generally recognized accrediting entity for the jurisdiction where the Site is located.
   2) Testing laboratory shall obtain all concrete samples and waterproofing product samples from the manufacturer of the associated product or material.

B. Reference Standards:
1. American Concrete Institute (ACI):
   a. CT-13, Concrete Terminology.
   b. 117, Specification for Tolerances for Concrete Construction and Materials.
   c. 211.1, Standard Practice for Selecting Proportions for Normal, Heavyweight, and Mass Concrete.
   d. 212.3R, Chemical Admixtures for Concrete.
   e. 232.2R, Use of Fly Ash in Concrete.
   f. 301, Specifications for Structural Concrete for Buildings.
   g. 318, Building Code Requirements for Structural Concrete.
   h. 350, Code Requirements for Environmental Engineering Concrete Structures.
2. ASTM International (ASTM):
   a. C33, Standard Specification for Concrete Aggregates.
   b. C39, Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens.
   c. C94/C94M, Standard Specification for Ready-Mixed Concrete.
   d. C125, Terminology Relating to Concrete and Concrete Aggregates.
   e. C150, Standard Specification for Portland Cement.
   f. C157, Standard Test Method for Length Change of Hardened Hydraulic-Cement, Mortar, and Concrete.
   g. C192, Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory.
   h. C260, Standard Specification for Air-Entraining Admixtures for Concrete.
   i. C494, Standard Specification for Chemical Admixtures for Concrete.
   j. C595, Standard Specification for Blended Hydraulic Cements.
   k. C618, Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete.
   l. C1107, Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Non-shrink).
   m. C1116, Standard Specification for Fiber-Reinforced Concrete.
   n. C1399, Standard Test Method for Obtaining Average Residual-Strength of Fiber-Reinforced Concrete.
   o. C1609, Standard Test Method for Flexural Performance of Fiber-Reinforced Concrete (Using Beam With Third-Point Loading).
   p. C1602, Standard Specification for Mixing Water Used in the Production of Hydraulic Cement Concrete.
   q. C1778, Standard Guide for Reducing Risk of Deleterious Alkali-Aggregate Reaction in Concrete.
3. State of Arizona Department of Transportation (ADOT):
   a. Standard Specification for Road and Bridge Construction.

1.3 SUBMITTALS

A. Action Submittals: Submit the following:
1. Shop Drawings:
   a. Schedule (table) of concrete materials proposed, listed by each specified grouping of concrete Work, including, but not limited to, aggregates, sand, cement (by type), SCM,

admixtures, synthetic fibers, grouts, and other materials. For each separate material and product, indicate the manufacturer and type of material.

b. Mix Designs:
   1) Proposed mix design for each concrete grouping required. For each, indicate concrete designation (type) indicated in the Contract Documents, proposed materials and proportioning, and intended special uses, such as concrete intended for placement in cold weather or warm weather, concrete to be placed by pumping, concrete intended for specific locations in the Work, and others.
   2) Contracting Officer's approval of mix design Shop Drawing is only for limited purposes indicated in the Contract Documents, including the General Conditions, and in no way reduces or mitigates Contractor's responsibility for construction means, methods, techniques, procedures, and sequences.

2. Product Data:
   a. Written affidavit stating materials proposed comply with requirements of reference standards indicated in this Section and, where applicable, compliance with Arizona Department of Transportation standard specifications for highway and bridge construction in the jurisdiction of the Site. Clearly indicate specific reference standards applicable to each specific material.
   b. For aggregate and sand, indicate source (quarry) and gradation of materials proposed for use. Indicate the specific concrete mix design(s) proposed for each.
   c. For cement and SCM, indicate material source and submit manufacturer's technical data (except safety data sheets).
   d. For each proposed admixture and type of grout material (including non-shrink grouts, epoxy grout, and grout cure/seal compound), submit manufacturer's published technical data (except safety data sheets).

B. Informational Submittals: Submit the following:
1. Certifications: Submit concurrent with, but separate from, associated Shop Drawings and product data Submittals:
   a. Certification of standard deviation, in units of pounds per square inch, for ready mix plant furnishing concrete.
   b. SCM: Certification that SCM complies with quality requirements of this Section, and SCM Supplier's certified test reports of SCM delivered to concrete Supplier.
   c. ASTM C33: Certification that class of coarse aggregate complies with ASTM C33 for type and location of concrete Work.
   d. Aggregate:
      1) Certification of aggregate gradation.
      2) Certification of coarse aggregate impurities relative to alkali-aggregate reactivity in accordance with ASTM C1778.
   e. Certification of shrinkage test results.
2. Test Reports:
   a. Cement and SCM mill certificates for all materials to be supplied.
   b. Test results for AAR impurities of coarse aggregates within proposed mixes, in accordance with ASTM standards cited in this Section.
3. Supplier's Instructions: Submit concurrent with, but separate from, associated product data Submittals:
   a. Manufacturer's written instructions on proper storage, handling, mixing, and use of materials furnished.

1.4 DELIVERY, STORAGE AND HANDLING

A. Storage of Materials:
1. Admixtures:
   a. Store admixtures in manner that avoids contamination, evaporation, and damage.
   b. For admixtures used in form of suspensions or non-stable solutions, perform agitating as recommended by manufacturer to ensure uniform distribution of ingredients.

c. Protect liquid admixtures from freezing and temperature changes that adversely affect admixture characteristics and performance.

2. Cement and SCM:
   a. Store cement and SCM in containers in weathertight space that prevent contamination with moisture and other contaminants.
3. Aggregates:
   a. Store and access aggregates in manner avoiding excessive segregation and preventing contamination with other materials and other sizes of like aggregate.
   b. Do not use frozen or partially frozen aggregate.
4. Sand: Allow natural sand to drain until sand has relatively uniform moisture content, prior to use.
5. If stockpiled materials contact the ground, unless such materials are stored on a clean, firm, reasonably impervious surface such as concrete or asphalt paving, do not use in the concrete Work bottom six inches of stockpiled materials.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Cement:
   1. Provide ASTM C150 Type I/II cement or ASTM C595 Type IL, unless otherwise required by the Contract Documents.

B. SCM:
   1. Fly Ash:
      a. ASTM C618, Class F .
      b. Non-staining.
      c. Appropriate for providing hardened concrete of uniform, light-gray color.
      d. Compatible with all other concrete ingredients. Fly ash shall have no deleterious effect on hardened concrete Work.
      e. Produced by source approved, by the Arizona Department of Transporation for use in concrete for highway bridges.
      f. Evaluate and use in accordance with ACI 232.2R.
   2. SCM type used shall match SCM type used in associated approved mix design.

C. Admixtures:
   1. Admixtures – General:
      a. Provide admixtures of same type, manufacturer and quantity as used in establishing required concrete proportions in mix design approved by Contracting Officer.
      b. Provide admixtures certified by manufacturers as compatible with other admixtures proposed.
   2. Air Entraining Admixtures: Comply with ASTM C260.
   3. Water Reducing Admixtures:
      a. Provide water-reducing admixtures in all concrete mixes to provide and maintain required water-to-cement ratio without additional cement.
      b. Water Reducing, Retarding and Accelerating: Comply with ASTM C494 Types A through E, and ACI 212.3R.
      c. High Range Water Reducers (Superplasticizers):
         1) Required for pumped concrete.
         2) Comply with ASTM C494 Types F and G.
   4. Hydration Stabilizer:
      a. Comply with ASTM C494 Type D.
   5. Admixture Chlorides – General:
      a. Provide chloride-free admixtures.
      b. Do not use calcium chloride.

D. Crystalline Cementitious Waterproofing:

1. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
    a. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.
    b. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.
2. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
    a. Xypex Chemical Corporation, C-500.
    b. Or equal.
3. Submit requests for substitution in accordance with Specification Section 01 33 23.
4. Description: Factory blended dry powder compound consisting of Portland cement, treated aggregate, and active chemicals.
5. Dosage rate shall be not less than 2 percent by weight of cementitious materials or additional as necessary to comply with the performance requirements indicated below.
6. Required Locations: Provide crystalline cementitious waterproofing admixture in concrete as follows: all water-bearing concrete and at locations shown or indicated on the Drawings.
7. Performance Criteria:
    a. Mix waterproofing material in proportions recommended by manufacturer.
    b. Submit to Contracting Officer, together with product data Submittal, results of tests (previously performed for manufacturer) by independent third-party testing entity.
    c. Crystalline cementitious waterproofing material, identical to that proposed for use in the Work, shall have been tested in accordance with the following:
        1) DIN 1048/ EN 12390, Water Impermeability of Concrete:
            a) Prepare and test untreated samples ("control samples") and treated samples, all of the same concrete mix design. Test not less than three control samples and three treated samples.
            b) Each sample shall be not less than 120 mm thick.
            c) Subject samples to test pressure of water not less than design hydrostatic pressure to which concrete will be subject after Substantial Completion, for test duration of not less than 72 hours. However, test duration and pressure shall be sufficient for control samples to have not less than 100 mm of water penetration (average for all control samples tested).
            d) Treated samples shall demonstrate not less than 90 percent reduction in depth of water penetration (average of all treated samples tested compared with average of all untreated control samples tested).
        2) Permeability: Supplier shall retain independent testing entity to perform permeability testing in accordance with USCOE CRD-C48 - Mod Permeability of Concrete.
            a) Test samples of concrete, of identical mix design, using mix design similar to that used for the Work. Test untreated samples ("control samples") and samples treated with the admixture.
            b) Samples shall each be not greater than two inches thick.
            c) Subject samples to water at test pressure of 150 psi.
            d) When compared with control samples, the treated samples shall exhibit no measurable leakage. Control samples shall exhibit full saturation and measurable leakage.

E. Macro Synthetic Fibers:
1. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

a. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.
b. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

2. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   a. Master Builders Solutions, MasterFiber MAC Series.
   b. GCP Applied Technologies, Strux 90/40.
   c. Euclid Chemical Company, Tuf-Strand SF.
   d. Or equal.
3. Manufacturers and Products: Subject to compliance with the Contract documents, the following are acceptable:
   a. Master Builders Solutions, MasterFiber, MAC Series.
   b. Or equal
4. Comply with ASTM C1116.
5. Provide dosage to obtain average residual strength (at net deflection of L/150) of not less than 170 psi, in accordance with ASTM C1609 and ASTM C1399.
6. Submit requests for substitution in accordance with Specification Section 01 33 23.

F. Water:
1. Potable in accordance with Laws and Regulations.
2. Clean and free from deleterious substances.
3. Free of oils, acids, and organic matter. Comply with ASTM C1602.

G. Aggregates for Normal Weight Concrete:
1. Comply with ASTM C33.
2. Fine aggregates and coarse aggregates are separate ingredients.
3. Coarse Aggregate:
   a. Material shall be well-graded, washed aggregate, free of organic material, of the Arizona Department of Transportation.
   b. Gradation: In accordance with Table 03 31 30-A in this section's "Mixes" Article.
4. Alkali-Reactive Aggregates:
   a. Aggregates that may be deleteriously reactive, when combined with alkalis in cement, are unacceptable.
   b. Evaluate proposed aggregates for potential deleterious alkali-aggregate reaction in concrete in accordance with ASTM C1778.
      1) Submit to Contracting Officer results of source quality control testing for alkali-aggregate reactivity presenting the following:
         a) Analysis and classification of aggregates in accordance with ASTM C1778
         b) Results of source quality control analysis of aggregates.
         c) Include the flow from Figure 1 of ASTM C 1778 indicating test results sequence.
         d) Field performance history alone shall not be submitted to document acceptable aggregate performance.
         e) Size and exposure condition of the Work in Table 2 of ASTM C 1778: dry.
         f) Structure class for use in Table 3 of ASTM C1778: Class SC3.

H. Maximum total water-soluble chloride ion content for concrete mix including all ingredients measured as weight percent of cement in accordance with ASTM C1218:
1. Prestressed concrete: 0.06.
2. All other concrete: 0.10.

I. Sand Cement Grout:
1. Description: Typical use is for filling keyways in precast construction, and as otherwise required by the Contract Documents.
2. Performance Criteria:

a. Minimum 28 day compressive strength:
   1) 3,000 PSI.
b. Strength shall be not less than strength of adjacent concrete to which grout is applied.

3. Materials:
   a. Provide sand cement grout as approximately three parts sand to one part Portland cement, 6 +/-1% entrained air and water to provide slump allowing grout to completely fill required spaces and surround adjacent reinforcing.
   b. Provide sand in accordance with requirements for fine aggregate for concrete.

J. Non-shrink Grout:
1. Manufacturers and Products: Subject to compliance with the contract Documents, the following are acceptable:
   a. Master Builders Solutions, Masterflow, 713.
   b. Euclid Chemical Company, NS Grout.
   c. Sika, Sika Grout 212.
   d. Or equal.
2. Description:
   a. This provision requires non-shrink, non-metallic grout. Unless otherwise shown or indicated in the Contract Documents, references to "non-shrink grout" refer to non-shrink, non-metallic grout required by this provision.
3. Performance Criteria:
   a. Non-shrink grout shall produce a positive but controlled expansion.
   b. Mass expansion shall not be created by gas liberation or by other means.
   c. Minimum 28-day Compressive Strength: 7,000 PSI.
4. Material:
   a. Provide material that is non-shrink, non-metallic, non-corrosive, and non-staining.
   b. Comply with ASTM C1107, Grade B.
   c. Premixed with water only. Add water in accordance with manufacturer's written instructions.

K. Epoxy Grout:
1. Manufacturers and Products: Subject to compliance with the Contract Documents, the following are acceptable:
   a. Master Builders Solutions, Masterflow 648.
   b. Five Star Products, DP Five Start Epoxy Grout.
   c. Euclid Chemical Company, E3 Flowable.
   d. Sika, Sikadur 42, Grout Pak.
   e. Or equal.
   f. One manufacturer shall furnish both aggregate and adhesive.
2. Description:
   a. Three-component epoxy resin system, comprised of two liquid epoxy components and one inert aggregate filler component.
   b. Indication of locations where epoxy grout is required are indicated in the grout schedule in Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing.
   c. Furnish each component in separate package for mixing at the Site.
3. Performance Criteria:
   a. Minimum 28-day Compressive Strength: 13,000 psi.
4. Materials:
   a. Aggregate shall be compatible with adhesive.

2.2 MIXES

A. Mixes - General:
1. Provide concrete capable of: (a) being placed without segregation of aggregate from other components, and (b) developing all properties necessary and required.
2. Provide ready-mix concrete in accordance with ASTM C94/C94M.
3. Batching and other tolerances shall be in accordance with ACI 117.

4. All concrete shall be normal weight concrete, unless otherwise required by the Contract Documents. Normal weight concrete shall weigh approximately 145 to 150 pounds per cubic foot (without reinforcing steel), measured 28 days after placing.

B. Concrete Mixes:
   1. Mix design requirements are indicated in this Section's Table 03 31 30-A, below.

C. Air Entrainment:
   1. Provide air entrainment in concrete providing total air content, expressed as percent by volume, in accordance with this Section's Table 03 31 30-A, below.
   2. Adjust dosage rate as necessary to compensate for shrinkage reducing admixtures and concrete placing method.

D. Slump:
   1. Slump – General:
      a. Measure slump at point of discharge of wet concrete into final location.
      b. Compensate for slump loss due to placing method.
      c. Concrete with slump less than minimum required may be used provided such concrete can be properly placed and consolidated.
      d. Slump of Concrete to be Placed by Pumping:
         1) Provide water or water-reducing admixture at ready-mix plant for concrete to be placed by pumping, to allow for slump loss due to pumping.
         2) Provide additional water sufficient only so that slump of concrete at discharge end of pump hose does not exceed: (a) maximum allowable slump indicated, and (b) maximum specified water-to-cement ratio.
      e. Slump Adjustment at the Site:
         1) Slump may be adjusted at the Site by providing water reducers.
         2) Dosing shall be performed by experienced quality control technician employed by concrete Supplier. Concrete mixing thereafter shall be directed by the same technician.
      f. Slump tolerances shall comply with ACI 117.
   2. Concrete for Walls and Columns:
      a. 8 inches maximum; 4 inches minimum.
      b. Slump shall be obtained by use of mid-range or high-range water reducer complying with ASTM C494.
   3. All Other Members:
      a. Concrete using water reducer in accordance with ASTM C494: 8 inches maximum; 4 inches minimum.
      b. Concrete without water reducer in accordance with ASTM C494: 5 inches maximum, one inch minimum.

E. Proportioning:
   1. Proportioning - General:
      a. Proportion components of concrete to provide mixture that can be placed: (a) into corners and angles of forms, and around reinforcing, by placing and consolidation methods employed, (b) without component materials becoming segregated, and (c) without excessive, free water to collecting on concrete surface or other surfaces.
      b. Proportion component elements of concrete to provide proper concrete Work, including concrete durability, strength, and other necessary and required properties.
   2. Normal Weight Concrete:
      a. Normal weight concrete target cementitious materials contents and maximum water to-cementitious ratios shall be in accordance with this Section's Table 03 31 30-A, below.
      b. Target cementitious materials contents indicated in the Contract Documents are intended to provide crack-free, durable, finished concrete Work, rather than concrete Work of excessive strength.
   3. SCM:
      a. SCM – General:

1) Based on results of AAR testing by Supplier (performed in accordance with ASTM C1778) and alkali content of cement, SCM content (in accordance with this Section) may be adjusted in lieu of the indicated percentages to reduce risk associated with AAR.
2) Use only one type of SCM in a given mix unless expressly required or approved by Contracting Officer.
3) Water-to-cementitious (i.e, total of SCM plus cement) ratio shall not exceed required maximum water-to-cement ratio indicated in this Section.

b. Fly Ash:
1) For cast-in-place concrete only, maximum of 25 percent by weight of Portland cement content, per cubic yard, may be fly ash, at rate of one pound fly ash for one pound cement.
2) For the chlorine handling tank only, a minimum of 15 percent by weight of Portland cement content, per cubit yard, may be fly ash, at a rate of one pound fly ash for one pound cement.
3) Do not use fly ash in concrete required in Section 03 41 33 - Precast and Prestressed Concrete, and.

4. Water-Reducing, Water-Retarding, and Water-Accelerating Admixtures:
a. Provide in accordance with admixture manufacturer's written instructions.
b. Add to mix at ready-mix plant.
c. Use hydration stabilizer admixture, or AAR-inhibiting admixture, in concrete, as necessary and required, for placing and workability.
1) Water reducers are required to maintain required maximum water-to-cement ratios.

5. High Range Water Reducers (Superplasticizers):
a. Superplasticizers are required in:
1) Concrete to be pumped, except slabs-on-grade and mats.
2) Concrete for water-bearing structures.
3) Other concrete Work at Contractor's option.
b. Maximum concrete slump, before addition of admixture, shall be three inches. Maximum slump after addition of superplasticizer admixture shall be eight inches.
c. Comply with Section 03 31 31 - Concrete Mixing, Placing, Jointing, and Curing, relative to superplasticizers.

6. Macro Synthetic Fibers:
a. Uniformly disperse synthetic fibers into concrete mixtures. Provide synthetic fibers in mixer at ready-mix plant.
b. Required Locations: Provide macro synthetic fibers in the following concrete Work:
1) Water bearing concrete structures.
2) Concrete so shown or indicated on the Drawings.
c. Required Dosage:
1) Determined by Contractor and concrete Supplier as necessary to comply with concrete's required minimum average residual strength.
2) Provide in accordance with ASTM C1399 and ASTM C1609.
3) Under no circumstances shall dosage of macro synthetic fibers be less than:
a) Four pounds per cubic yard when used in concrete slabs on metal decking, in accordance with SDI 31.
b) Three pounds per cubic yard for all other locations where such fibers are required or used.

7. Normal Weight Concrete Mix Proportioning:
a. Method 1:
1) Use Method 1 when combination of concrete component materials and mixes will be evaluated and selected via trial-and-error.
2) Provide mixes with suitable proportions and properties in accordance with ACI 211.1, using not less than three different water-to-cementitious ratios providing a range of concrete compressive strengths, including required average compressive strength.

3) Trial mixes shall have slump within 0.75 inches of maximum allowed in the Contract Documents. For air-entrained concrete, air content of trial batches shall be within 0.5 percent of air entrainment required by the Contract Documents.
4) For each water-to-cementitious ratio:
a) Provide not less than three trial compressive strength tests for concrete test age required, and cure in accordance with ASTM C192.
b) Cylinder Size: In accordance with ASTM C31.
c) Test for compressive strength at 28 days, in accordance with ASTM C39.
(1) Quantity of cylinders shall comply with one of the following trial strength test:
(a) 6-inch diameter cylinders: Two.
(b) 4-inch diameter cylinders: Three.
5) From results of such required tests, plot curve showing relationship between water-to-cementitious ratio and compressive strength.
6) Based on required curve, select water-to-cementitious ratio for the Work, that will provide concrete of required average compressive strength.
7) Provide cementitious content and mixture proportions so maximum water cement ratio is not exceeded when slump is equal to maximum allowed in the Contract Documents.
8) Required average compressive strength is indicated below in this Section.

b. Method 2:
1) In lieu of trial mixes required by Method 1, field test results from prior projects, for concrete made using identical or substantively identical concrete component materials and proportioning, may be used by concrete supplier in determining proposed mix proportions, provided the test results are within a year of project start date.
2) Use of proposed concrete mix proportions based on field test results from prior projects are subject to approval by Contracting Officer. Contracting Officer's decision will be based on information in such Submittals and demonstrated ability, of such concrete successfully provided on such prior projects, to provide required average compressive strength.
3) Requirements for Submittals of Concrete Test Results from Prior Projects:
a) Submittals of field test results from prior projects shall clearly indicate all materials, proportions, and conditions, and clearly indicate where such matters are similar to those required for the concrete Work on the Project.
b) Changes in the materials, proportions, and conditions within submitted test results from prior projects shall have been not more restricted than those for the subject, proposed concrete mix.
c) Field test reports from prior projects shall be in accordance with ACI 350].
4) Concrete proportions for the concrete Work may be determined from test results of prior projects via interpolation (by Contractor and concrete Supplier) between compressive strengths and proportions of two or more test results from prior projects, each in accordance with requirements of the Contract Documents for this Project.

8. Required average compressive strength shall exceed required 28-day compressive strength by the extent determined in accordance with ACI 350, Chapter 5 using the standard deviation of concrete ready-mix plant proposed for the Work as described in ACI 350, Chapter 5.

F. Flowable Fill or Controlled Low Strength Material (CLSM):
1. CLSM and Flowable fill are interchangeable.
2. Provide mixture of cement, SCM, fine sand, water, and air, with consistency allowing flow under a very low pressure (low head).
3. Approximate quantities of each component per cubic yard of mixed material:
a. Cement (Type I or Type II): 50 pounds.
b. SCM: 200 pounds.

c. Fine Sand: 2,700 pounds.
d. Water (approximate): 420 pounds.
e. Air Content (approximate): 10 percent.

4. Adjust actual quantities to provide yield of one cubic yard with materials used.
5. Approximate compressive strength shall be 85 to 175 psi.
6. Sand Gradation: Fine sand shall be evenly graded material with not less than 95 percent passing No. 4 sieve and not more than five percent passing No. 200 sieve.

G. Allowable Shrinkage:
1. Provide in accordance with Table 03 31 30-A of this Section, tested in accordance with ASTM C157 at 28 days.
2. Continue testing for 64 weeks in accordance with ASTM C157 and submit results to Contracting Officer as Informational Submittals.
3. Perform for concrete Work for all water bearing or basement structures unless expressly indicated otherwise in the Contract Documents.

TABLE 03 31 30-A							
Type of Concrete	28-day Compressive Strength	Max. W/C Ratio	Target Total Cement (pounds)	SCM	ASTM C33 Size No.	Air Content (%)	Allowable Shrinkage Limit
Normal weight concrete fill utility encasement concrete	3000 PSI	0.50	517	Note 1	57	4.5 to 7.5	NA
Normal weight concrete topping	4000 PSI	0.45	564	Note 1	7	5.5 to 8	NA
Normal weight precast concrete	5000 PSI	0.42	611		57	4.5 to 7.5	NA
Normal weight water-bearing concrete	5000 PSI	0.40	564	Note 1	57 Note 2	4.5 to 7.5	0.042 percent
Normal weight all other concrete	4500 PSI	0.45	564	Note 1	57	4.5 to 7.5	0.048 percent

Table 03 31 30-A Notes:

1. If fly ash is proposed for use, the weight of fly ash plus weight of Portland cement shall be used to comply with total target cement content.
2. Unless otherwise indicated, larger aggregate (No. 467) is required for concrete slabs or foundations on grade; optional elsewhere
3. Red dye shall be used in buried utility encasements for duct banks.

PART 3 - EXECUTION

3.1 FIELD QUALITY CONTROL

A. Field Testing and Inspections:
   1. Code-Required Tests and Special Inspections:
      a. Section 01 40 00 - Quality Requirememts, indicates testing scope for code-required testing and special inspections by testing entity retained by Contracting Officer or entity for whom Government is responsible.
      b. Section 03 05 05 - Concrete Testing and Inspection, indicates required testing for concrete Work.
   2. Contractor-Performed Field Testing and Inspections:
      a. Where concrete testing and inspection is required by the Contract Documents and is not part of the code-required tests and special inspections by Contracting Officer or other entity for whom Government is responsible, such tests and inspections shall be by Contractor.
      b. Perform concrete testing and inspections in accordance with Section 03 05 05 - Concrete Testing and Inspection.
      c. Aggregates and Other Stockpiled Materials: To ensure stockpiles at the concrete mixing location are not contaminated and otherwise comply with Contract requirements, perform tests on such materials at the concrete ready-mix plant.
      d. Perform strength test on all concrete to which water or superplasticizer, above the amount stated in concrete mix design Submittal approved by Contracting Officer, has been added.
         1) Perform sampling after water or superplasticizer has been added and additional mixing has been performed.

END OF SECTION

SECTION 03 31 31
CONCRETE MIXING, PLACING, JOINTING, AND CURING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Mixing, placing, jointing, and curing of concrete construction.

B. Related Requirements: Include but are not necessarily limited to:
   1. Section 03 05 05 - Concrete Testing and Inspection.
   2. Section 03 11 13 - Formwork.
   3. Section 03 21 00 - Reinforcement.
   4. Section 03 31 30 - Concrete, Materials and Proportioning.
   5. Section 03 35 00 - Concrete Finishing and Repair of Surface Defects.
   6. Section 07 92 00 - Joint Sealants.

1.2 REFERENCES

A. Terminology:
   1. Terminology: Terminology indicated below are not defined terms and are not indicated with initial capital letters, but when used in this Section have the meanings indicated below:
      a. Unless obviously meant or intended otherwise, certain words and terms used in this Section have meanings indicated in ACI CT-13.
      b. "Water-bearing concrete" means concrete surface to be in contact (whether continuously or intermittently) with water, process liquid, or slurries during intended operation of the facility, including, but not limited to, concrete tanks, channels, wet wells, distribution chambers, and secondary containment structures.

B. Reference Standards: Standards referenced in this Section include, but are not necessarily limited to, the following:
   1. American Concrete Institute (ACI):
      a. CT-13, Concrete Terminology.
      b. 117, Specification for Tolerances for Concrete Construction and Materials.
      c. 304R, Guide for Measuring, Mixing, Transporting and Placing Concrete.
      d. 304.2R, Placing Concrete by Pumping Methods.
      e. 305R, Guide to Hot Weather Concreting.
      f. 305.1, Specification for Hot Weather Concreting.
      g. 306R, Guide to Cold Weather Concreting.
      h. 306.1, Standard Specification for Cold Weather Concreting.
      i. 308.1, Specification for Curing Concrete.
      j. 309R, Guide for Consolidation of Concrete.
      k. 318, Building Code Requirements for Structural Concrete and Commentary.
      l. 350, Code Requirements for Environmental Engineering Concrete Structures.
      m. 360R, Guide to Design of Slabs-on-Ground.
   2. ASTM International (ASTM):
      a. C94/C94M, Standard Specification for Ready-Mixed Concrete.
      b. C309, Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete.
      c. C1315, Standard Specification for Liquid Membrane-Forming Compounds Having Special Properties for Curing and Sealing Concrete.
      d. D994, Standard Specification for Preformed Expansion Joint Filler for Concrete (Bituminous Type).
      e. D1056, Standard Specification for Flexible Cellular Materials-Sponge or Expanded Rubber.

f. D1751, Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types).

3. National Ready Mixed Concrete Association (NRMCA):
   a. Checklist for Certification of Ready Mixed Concrete Production Facilities.
4. United States Army Corps of Engineers (USACE):
   a. CRD-C572, Specifications for Polyvinylchloride Waterstop.

1.3 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Coordinate concrete placement Work with other Work, and work of other contractors (if any), that must be installed prior to placement of concrete.
   2. Inform other contractors, if any, affected by the concrete Work of intended date(s) of placement of concrete. Provide sufficient advance notice to allow other contractors to perform their work that is to be completed prior to placing of concrete.

B. Pre-Installation Conference:
   1. A meeting to review the detailed requirements of the Contractor's proposed concrete design mixes, to determine the procedures for producing proper concrete construction, and to clarify the roles of the parties involved shall be held no later than 30 days after Notice to Proceed.
      a. Schedule the meeting to occur no later than five days in advance of the first scheduled date of concrete placement.
   2. All parties involved in the concrete work shall attend the conference, including: Required Participants:
      a. Contractor's Representative
      b. Testing laboratory representative/inspectors.
      c. Concrete subcontractor.
      d. Reinforcing steel installer.
      e. Reinforcing steel detailer.
      f. Concrete supplier.
      g. Admixture manufacturer's representative.
      h. Contracting Officer.
      i. Contracting Officer's representative.
      j. Building Code Official.
   3. The conference shall be held at a mutually agreed upon time and location.
   4. Participants shall be prepared to discuss the topics indicated below, and shall be empowered by their employer to make binding decisions concerning the conference topics:
      a. Designation of responsible personnel.
      b. Schedule for concrete Work and subsequent Work.
      c. Status of other construction requiring coordination with subject concrete Work.
      d. Status of required Submittals for concrete Work.
      e. Contractor's concrete pre-placement plan checklist.
      f. Mock-ups as a quality standard.
      g. Subbase preparation and status (as applicable).
      h. Formwork.
      i. Reinforcing.
      j. Jointing, waterstops (as applicable), and related Work.
      k. Sampling and testing for quality control and compliance with building codes and other Laws and Regulations.
         1) Contractor responsibility for coordinating with testing entity and special inspectors.
         2) Storage and protection of Samples.
         3) Additional test cylinders to be made for concrete deliveries in which water exceeding the water/cement ratio of approved mix design was added at the Site.
         4) Additional test cylinders to be made for concrete structural elements Contractor intends to subject to live loading prior to 28 days after placement of concrete.
         5) Duties and authority of testing and inspection agency.

l. Logistics for concrete delivery (Contractor's responsibility) and placing, including:
    1) Travel time between ready-mix plant and Site, maximum time concrete to be in delivery vehicles, and exceptions to limits.
    2) Review of approved concrete mix(es), including limits of water addition and person authorized to add water, if water has been withheld at the plant.
    3) Traffic flow and delivery vehicle cleanout (Contractor's responsibility).
m. Concrete placing and consolidation means, methods, procedures, techniques, and sequences, and related safety and protection measures (Contractor's responsibility).
n. Concrete finishing.
o. Temperature, weather, and atmosphere during placing and curing (Contractor's responsibility).
p. Curing procedures (Contractor's responsibility).
q. Removal of formwork and patching (Contractor's responsibility).
r. Criteria for approval and rejection of work.
s. Other, related, subsequent Work.
t. Other necessary topics.

5. Discussion, whether at the pre-installation conference or otherwise, of Contractor's means, methods, procedures, techniques, and sequences of construction, and associated safety and protection measures, in no way changes or mitigates Contractor's sole responsibility for such matters.
6. Record of Conference: Within two days following the pre-installation conference, Contractor shall prepare and distribute to all participants of pre-installation conference, and others as appropriate, a reasonably complete record of topics discussed, identified problems, decisions made, and other matters pertinent to the subject concrete Work.
7. Should conference reveal matters that will not be suitably addressed by the scheduled date for the subject concrete placement, after taking steps to address the matter(s) in question, reconvene the conference. If necessary, reschedule concrete placement without change in the Contract Times and Contract Price.

1.4 DEFINITIONS

A. Words and terms used in this Specification Section are defined in the ACI CT-13.

1.5 SUBMITTALS

A. Action Submittals: Submit the following:
    1. Shop Drawings:
        a. Scaled (minimum 1/8 inch per foot) drawings showing proposed locations of construction joints, control joints, expansion joints (as applicable) and joint profile dimensions for each joint type.
        b. Drawing indicating location, size and type of prefabricated waterstop joints.
    2. Product Data including:
        a. Manufacturers and types:
            1) Joint fillers.
            2) Curing agents.
            3) Construction joint bonding adhesive.
            4) Prefabricated waterstops.
        b. Acknowledgement that products submitted meet requirements of standards referenced.
    3. Monitoring Procedures:

B. Informational Submittals: Submit the following:
    1. Delivery Tickets:
        a. Copies of concrete delivery tickets.
    2. Supplier Instructions:
        a. Procedure for adding high-range water reducer at the Site.
    3. Instructions for handling and installing products approved by Contracting Officer's Representative. Supplier Site Visit Reports:

a. Report of each visit to the Site by Supplier, summarizing purpose of visit, activities while onsite, problems encountered, advice given to Contractor or Subcontractor, and actions taken.

4. Special Procedure Submittals:
   a. Description of proposed curing methods.
5. Qualifications:
   a. Ready-mix plant certification, in accordance with this Section's "Quality Assurance" Article.
   b. Waterstops: Certificate of training from waterstop manufacturer for individuals performing installation of waterstops and field welds and splices of waterstops.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Concrete Delivery:
   1. Prepare a delivery ticket for each load of ready mixed concrete.
   2. Truck operator shall hand ticket to Contractor at the time of delivery.
   3. Ticket to show:
      a. Mix identification.
      b. Quantity delivered.
      c. Quantity of material in each batch.
      d. Outdoor temperature in the shade at location of concrete placement.
      e. Time at which cement was added.
      f. Time of delivery.
      g. Time of discharge.
      h. Quantity of water that may be added at the site without exceeding the specified water-cement ratio.
      i. Quantity of any water added at the Site.

1.7 PROJECT CONDITIONS

A. Adjust concrete mix design when material characteristics, job conditions, weather, strength test results or other circumstances warrant.
   1. Do not use revised concrete mixes until submitted to and approved by Contracting Officer.

1.8 SEQUENCING AND SCHEDULING

A. Do not begin concrete production until proposed concrete mix design has been approved by the Contracting Officer.
   1. Approval of concrete mix design does not relieve Contractor of responsibility to provide concrete that meets the requirements of the Contract Documents.

PART 2 - PRODUCTS

2.1 PRODUCTS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements.
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the manufacturers listed in this article are acceptable.

C. Neoprene Expansion Joint Fillers:
   1. Acceptable manufacturers:
      a. Permaglaze.
      b. Or equal.

2. Materials:
    a. Closed cell neoprene.
    b. ASTM D1056, Type 2, Class A or C.
    c. Grade: Compression deflection as required to limit deflection to 25 percent of joint thickness under pressure from concrete pour height.

D. Asphalt Expansion Joint Fillers:
1. Acceptable manufacturers:
    a. W.R Meadows.
    b. Or equal.
2. Materials: ASTM D994.

E. Fiber Expansion Joint Fillers:
1. Materials: ASTM D1751.

F. Waterstops, PVC Type:
1. Acceptable manufacturers:
    a. Sika Greenstreak Plastic Products.
    b. Or equal.
2. Materials:
    a. Virgin polyvinyl chloride compound not containing scrap or reclaimed materials or pigment.
    b. Cast-in-place type: USACE CRD-572.
3. Approved profiles as listed.
    a. Construction joints:
        1) Ribbed: 6 inches wide by 3/8 inch.
        2) Sika Greenstreak Plastic Products Style #679, or equal.
    b. Control joints:
        1) 6-inch wide by 3/8 inch thick with ribs and center bulb.
        2) Sika Greenstreak Plastic Products Style #705, or equal.
    c. Expansion joint:
        1) 9 inches wide by 3/8 inch thick center bulb two inch outside diameter.
        2) Sika Greenstreak Plastic Products Style #739, or equal.
    d. Retro-fit Waterstop:
        1) Use at interface between existing construction and new concrete.
        2) Sika Greenstreak style 609, or equal.
4. Provide factory-made waterstop fabrications at all changes in direction, intersections and transitions, leaving only straight butt splices for the field. Butt welds to be a minimum six inches clear of the intersection.
5. Factory prepunched (less than 1.5 feet on centers, each edge, staggered) for wire supports.
    a. Provide hog rings or grommets at all punched holes along the length of the waterstop.
6. See Drawings for application and other requirements.

G. Waterstops, Preformed Strip Type:
1. Acceptable manufacturers:
    a. Adeka Ultra Seal USA (MC-2010MN).
    b. Or equal.
    c.
2. Hydrophilic, non-bentonite composition.
3. Manufactured solely for the purpose of preventing water from traveling through construction joints.
4. Volumetric expansion limited to three times maximum.
5. See the Drawings for locations, applications, and other requirements.

H. Water Swelling Sealant:
1. Required adhesive for use with strip-type waterstop.
2. Compatible with strip-type waterstop.
3. Single component, gun applied.

4. Moisture cured.
5. Minimum 70 percent volumetric expansion swelling capability.

I. Evaporation Retarder:
1. Waterborne, monomolecular film forming, manufactured for application to fresh concrete.
2. Manufacturer and Products:
   a. Euclid Chemical Company, Eucobar
   b. Or equal.

J. Curing Materials shall comply with one or more of the following:
1. Absorbent Covers.
2. Moisture Retaining Covers.
3. Dissipating curing compound:
   a. Fugitive dye, waterborne, membrane-forming.
   b. ASTM C309, Type 1D, Class A or B, shall be composed of hydrocarbon resins, and dissipating agents that begin to break down upon exposure to UV light, and traffic, approximately four to six weeks after applications, providing a film that is removable with standard degreasing agents, and mechanized scrubbing actions to not impair the later addition and performance of applied finishes.
   c. Products and Manufacturers:
      1)
      2) Euclid Chemical Company; Kurez DR VOX.
      3) Or equal.
4. Clear, water-borne, membrane-forming curing and sealing compound:
   a. ASTM C1315, Type 1, Class A.
   b. Moisture loss shall be not more than 0.40 kilograms per square meter when applied at 300 square feet per gallon.
   c. Manufacturer's certification is required.
   d. Subject to Project requirements, provide one of the following products:
      1) Solvent-based:
         a) Euclid Chemical Company; Super Diamond Clear, Luster Seal 300 (exterior), Super Rez-Seal (interior).
         b) Or equal.
      2) Water-based:
         a) Euclid Chemical Company; Super Diamond Clear VOX.
         b) Or equal.

K. Vapor Retarder: Comply with Section 07 26 00 - Under Slab Vapor Retarder.

L. Sand cement grout, non-shrink grout and epoxy grout: Comply with Section 03 31 30 - Concrete Materials and Proportioning.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Installing the Work constitutes Contractor’s approval of underlying work, substrates, and field conditions prevailing at the time of the Work.

3.2 PREPARATION

A. Preparation – General:
1. Materials and construction shall comply with the tolerances as specified in ACI 117.
2. Complete formwork.
   a. Perform formwork in accordance with Section 03 11 13 - Formwork.
3. Remove earth, snow, ice, water, foreign matter, and other extraneous materials from areas that will receive concrete.
4. Secure reinforcement in place.
   a. Provide reinforcing in accordance with Section 03 21 00 - Reinforcement.

5. Provide expansion joint material, anchors and other embedded items.
6. Prior to placing concrete, obtain Contracting Officer's concurrence that associated formwork and reinforcing appear to be in accordance with the Contract Documents. However, in no event will Contracting Officer's observations, comments, or concurrence that any part of the Work appears to be in accordance with the Contract Documents modify or reduce Contractor's sole responsibility for providing Work in accordance with the Contract Documents.
7. Do not place concrete during rain, sleet, snow, or other adverse ambient or environmental conditions unless adequate protection is provided by Contractor and such placement is acceptable to Contracting Officer's.
    a. Provide sufficient quantity of competent, experienced workers with due regard for effects of concrete temperature and atmospheric conditions on rate of hardening of concrete as required to obtain good surfaces and avoid unplanned cold joints.
    b. Do not allow precipitation or storm water runoff to increase mixing water nor to damage surface finish.
8. Remove hardened concrete and foreign materials from inner surfaces of conveying equipment and formwork.
9. Provide slabs and beams of minimum indicated required depth when sloping structural foundation base slabs and elevated slabs to drains.
    a. For floor slabs on grade, slope top of subgrade to provide slab of required uniform thickness.

B. Preparation of Subgrade for Slabs On Ground:
1. Subgrade shall be wetted without standing water immediately prior to placing concrete.
2. Obtain approval of subgrade compaction density prior to placing slabs on ground.

C. Edge Forms and Screeds:
1. Set accurately to produce designated elevations and contours of finished surface.
2. Sufficiently strong to support vibrating screeds or roller pipe screeds, if required.
3. Use strike off templates, or acceptable vibrating type screeds, to align concrete surfaces to contours of screed strips.

3.3 CONCRETE MIXING

A. Concrete Mixing - General:
1. Provide concrete materials from one ready-mix batch plant qualified in accordance with this Section's "Quality Assurance" Article.
2. Batch, mix, and transport in accordance with ASTM C94/C94M.

B. Control of Admixtures:
1. Control at ready-mix plant:
    a. Admixtures shall be introduced at the ready-mix plant in accordance with manufacturer's recommendations.
    b. Charge admixtures into mixer as solutions.
        1) Measure by means of an appropriate mechanical dispensing device.
        2) Liquid considered a part of mixing water.
        3) Admixtures that cannot be added in solution may be weighed or measured by volume if so recommended by manufacturer.
    c. Add separately, when two or more admixtures are used in concrete, to avoid possible interaction that might interfere with efficiency of either admixture, or adversely affect concrete.
    d. Complete performing addition of retarding admixtures within one minute after addition of water to cement has been completed, or prior to beginning of last three quarters of required mixing, whichever occurs first.
2. Control of Admixtures at the Site:
    a. Additional quantities of admixtures (with the exception of retarders) may be added at the Site provided:

1) Addition of admixtures shall be under the supervision of the ready-mix plant's quality control representative.
2) Addition of each admixture shall be documented on the delivery ticket.
3) Provide additional mixing in accordance with ASTM C94.

C. Tempering and Control of Mixing Water:
1. Mix concrete only in quantities for immediate use.
2. Discard concrete which has set.
3. Discharge concrete from ready-mix delivery vehicles within time limit stated in ASTM C94.
4. Addition of water at the Site:
a. Comply with Section 03 31 30 - Concrete Materials and Proportioning, for specified water cement ratio and slump.
b. Do not exceed maximum specified water cement ratio or slump.
c. Incorporate water by additional mixing equal to at least half of total mixing required.

3.4 PLACING OF CONCRETE

A. Placing of Concrete - General:
1. Place concrete as a rate such that concrete, which is being integrated with fresh concrete, is "workable".
a. Provide sufficient, competent, skilled workers for timely delivery of concrete into forms to avoid unintended cold joints and placement consolidation issues.
2. Contractor may choose to submit an equivalent self-consolidating concrete mix to reduce consolidation and placement issues.
3. Tined rakes are prohibited as a means of conveying fiber reinforced concrete.
4. Provide additional equipment to place mass concrete if deemed required to avoid placement issues.
5. Comply with ACI 304R,ACI 304.2R, and ACI 301.
6. Do not begin placing concrete during rain, sleet, snow, or other adverse weather or environmental conditions that may be detrimental to concrete placement.
a. Protect fresh concrete from ensuing inclement weather and other environmental conditions that may be detrimental to concrete.
7. Time between concrete batching and final placement shall not exceed 90 minutes.
a. Selection of time limit to end of discharge should consider ambient conditions, types of cementitious materials and admixtures used, placement procedures, and projected transportation time between ready-mix plant and the point of delivery.
b. If discharge is acceptable after more than 90 minutes have elapsed since batching, verify that air content of air-entrained concrete, slump, and temperature of concrete are as specified.
8. Do not deposit concrete which has partially hardened or has been contaminated by foreign materials.
9. Begin work only when work of other trades affecting concrete is complete.
10. When pumping concrete, do not use excess grout or mortar to lubricate concrete conveyance piping.
11. Do not use excess water for workability or any reason when placing concrete by freefall.
12. Deposit concrete continuously to avoid cold joints.
13. Provide construction joints at locations shown or indicated in the Contract Documents, unless otherwise approved by Contracting Officer via Shop Drawing or Contract modification.
a. Plan size of crews with due regard for effects of concrete temperature and ambient conditions to avoid unplanned cold joints.
14. Spreaders:
a. Temporary: Remove as soon as concrete placing renders their function unnecessary.
b. Embedded:
1) Obtain approval of Contracting Officer for their use.
2) Materials: Concrete or metal.
3) Ends of metal spreaders coated with plastic coating two inches from each end.

15. Do not begin placing of concrete in supported elements until concrete previously placed in supporting members is no longer plastic and has been in place at least a minimum of 2 hours.
16. Deposit concrete as nearly as practicable in its final position to avoid segregation.
    a. Maximum free fall: 4 feet.
    b. Place concrete by means of hopper, "elephant trunk" or tremie pipe extending to within four feet of surface.
17. Perform the following operations before bleeding water has an opportunity to collect on surface:
    a. Spread.
    b. Consolidate.
    c. Straightedge.
    d. Darby or bull float.
    e. Evaporation Retarder.
        1) See manufactures printed instruction for required waiting period prior to finishing or troweling.
18. No water shall be added to the concrete surface to ease finishing operation.
19. Do not discharge water into forms.
20. Consider use of form vibrators for certain placement situations.
21. Deposit concrete on void forms by methods which will not crush or damage void forms in any way.

B. Cold Weather Concrete Placement:
    1. Comply with ACI 306.1.
    2. Do not place concrete on forms or subgrades that are below 32 degrees F or contain frozen material.
    3. Maintain all materials, forms, reinforcement, subgrade and any other items which concrete will come in contact with free of frost, ice or snow at time of concrete placement.
    4. Temperature of concrete when discharged at site: in accordance with ACI 306.1.
    5. Heat subgrade forms, embedments and reinforcement to between 45 and 70 degrees F, when temperature of surrounding air is 40 degrees F or below at time concrete is placed.
        a. Remove frost from subgrade, forms and reinforcement before concrete is placed.
    6. Combine water with aggregate in mixer before cement is added, if water or aggregate is heated above 90 degrees F.
    7. Do not mix cement with water or with mixtures of water and aggregate having a temperature greater than 90 degrees F.
    8. Comply with ACI 306R for specific requirements dealing with elevated steel troweled slabs that will be exposed to freeze-thaw cycles.

C. Hot Weather Concrete Placement:
    1. Comply with ACI 305.1.
    2. Cool ingredients before mixing, or add flake ice or well crushed ice of a size that will melt completely during mixing for all or part of mixing water if high temperature, low slump, flash set, cold joints, or shrinkage cracks are encountered.
    3. Temperature of concrete at point of delivery (i.e. delivery vehicle discharge) when placed:
        a. Not to exceed 90 degrees F.
        b. Not to exceed:

Section Thickness (Least Dimension, feet)	Maximum Temperature (degrees F)	Concrete Placement Size (Volume, CU YD limit)
Up to 2.0	90	Unlimited
2.0 to 3.0	76	Unlimited
3.0 to 6.0	72	250

Greater than 6.0	72	Review by Contracting Officer's Representative

c. Not so high as to cause:
   1) Shrinkage cracks.
   2) Difficulty in placement due to loss of slump.
   3) Flash set.

4. Temperature of forms and reinforcing when placing concrete:
   a. Not to exceed 90 degrees F.
   b. May be reduced by spraying with water to cool below 90 degrees F.
      1) Leave no standing water to contact concrete being placed.
5. Prevent plastic shrinkage cracking, slab curling, or both due to evaporation.

D. Consolidating:
1. Consolidate in accordance with ACI 309R except as modified herein.
2. Consolidate by vibration so that concrete is thoroughly worked around reinforcement, embedded items and into corners of forms.
   a. Ensure no displacement of reinforcing or other embeds from final position.
   b. Eliminate:
      1) Air or stone pockets.
      2) Honeycombing, pitting, or both.
      3) Planes of weakness.
3. Use suitable form vibrators located just below top surface of concrete, where internal vibrators cannot be used in areas of congested reinforcing.
   a. Size and coordinate external vibrators to specifically match forming system used.
4. Internal vibrators:
   a. Minimum frequency of 8000 vibrations per minute.
   b. Insert and withdraw at points approximately 1.5 feet apart.
      1) Allow sufficient duration at each insertion to consolidate concrete but not sufficient to cause segregation.
   c. Use in:
      1) Beams and girders of framed slabs.
      2) Columns and walls.
      3) Vibrating concrete around all waterstops.
   d. Size of vibrators shall be in accordance with ACI 309R, Table 5.1.5.
5. Obtain consolidation of slabs with internal vibrators, vibrating screeds, roller pipe screeds, or other appropriate means.
6. Do not use vibrators to transport concrete within forms.
7. When placing self-consolidating concrete, the use of form or pencil vibrators is acceptable, provided such methods do not cause aggregate segregation, or otherwise adversely affect the quality of the Work.
8. Do not excessively vibrate or power screed concrete that contains "traprock" aggregate.
9. Provide sufficient spare vibrators at the Site during all concrete placing operations to ensure continuous vibration.
10. Bring a full surface of mortar against form by vibration supplemented if necessary by spading to work coarse aggregate back from formed surface, where concrete is to have an as-cast finish.
11. Prevent construction equipment and machinery, construction operations, and personnel from introducing vibrations into freshly placed concrete after the concrete has been placed and consolidated.

E. Handle concrete from mixer to place of final deposit by methods which will prevent segregation or loss of ingredients and in a manner which will assure that required quality of concrete is maintained.
1. Use truck mixers, agitators, and non-agitating units in accordance with ASTM C94.
2. Temporary horizontal belt conveyors:

a. Mount at a slope which will not cause segregation or loss of ingredients.
b. Protect concrete against undue drying or rise in temperature.
c. Use an arrangement at discharge end to prevent segregation.
d. Do not allow mortar to adhere to return length of belt.
e. Discharge conveyor runs into equipment specially designed for spreading concrete.

3. Temporary metal or metal lined chutes:
   a. Slope not exceeding 1 vertical to 2 horizontal and not less than 1 vertical to 3 horizontal.
   b. Chutes more than 20 feet long and chutes not meeting slope requirements may be used provided they discharge into a hopper before distribution.
   c. Provide end of each chute with a device to prevent segregation.
4. Temporary pumping or pneumatic conveying equipment:
   a. Designed for concrete application and having adequate pumping capacity.
   b. Control pneumatic placement so segregation is avoided in discharged concrete.
   c. Loss of slump in pumping or pneumatic conveying equipment shall not exceed 1.5 inches.
   d. Do not convey concrete through pipe of aluminum or aluminum alloy.
   e. Provide pumping equipment without Y sections.

3.5 JOINTS AND EMBEDDED ITEMS

A. Construction Joints - General:
1. Provide joints at locations as shown or indicated on the Drawings or as shown on approved Shop Drawings approved by Contracting Officer or in an appropriate Contract modification.
   a. Where construction joint spacing shown on the Drawings exceeds the joint spacing indicated in Paragraph B. below, submit proposed construction joint location in conformance with this Section.
2. Unplanned construction joints are unacceptable.
   a. If concrete cannot be completely placed between planned construction joints, then it must be removed.
3. In general, locate joints near middle of spans of slabs, beams and girders unless a beam intersects a girder at this point, in which case, offset joint in girder a distance equal to twice the width of the beam.
4. Locate joints in walls and columns at underside of floors, slabs, beams, or girders, and at tops of foundations or floor slabs, unless shown otherwise.
   a. At Contractor's option, beam pockets may be formed into concrete walls.
   b. Size pockets to allow beam reinforcing to be placed as detailed on the Drawings.
5. Place beams, girders, column capitals and drop panels at same time as slabs.
6. Place corbels monolithically with their supporting members.
   a. Locate wall vertical construction joints midway between corbels.
   b. Where only a single corbel is located, place it also monolithically with wall and locate wall vertical construction joint a minimum of three feet from face of corbel.
7. Make joints perpendicular to main reinforcement with all reinforcement continuous across joints.
8. Provide the following joints unless shown or indicated otherwise on the Drawings:
   a. Roughen joints: horizontal construction joints.
   b. Keyed joints: vertical construction joints.
9. Roughen construction joints:
   a. Clean the previously hardened concrete interface and remove all laitance.
   b. Intentionally roughen the interface to a full amplitude of 0.25 inch.
10. Keyways:
    a. Construction joint keyways shall have the following dimensions, unless shown or indicated otherwise on the Drawings or as directed by Contracing Officer. .
    b. Wall keys:
       1) Keyway width, not less than one-third and not more than one-half the wall thickness measured perpendicular to wall faces.

2) Keyway depth shall be not less than 1.5 inches.
3) Continuous along length of wall.
4) Place keyway in wall center unless otherwise shown or indicated on the Drawings.

c. Keyways in footings, foundations, base slabs, and structural or elevated slabs:
1) Keyway height not less than one-third and not more than one-half the footing or slab thickness.
2) Keyway depth not less than 1.5 inches.
3) Continuous along footing or slab.
4) Keyway in footing or slab center unless shown otherwise on the Drawings.

d. Beam keyways:
1) Full width of beam.
2) Keyway height not less than 5.5 inches.
3) Keyway depth not less than 1.5 inches.
4) Keyway located in initial beam pour, directly above the bottom reinforcing, unless shown otherwise on the Drawings.

11. Minimum time before placement of adjoining concrete construction:
a. All concrete: 60 hours, unless otherwise noted.

B. Construction Joints - Spacing Unless Otherwise Specified:
1. Structures not intended to contain liquid:
a. Wall vertical construction joints:
1) 50 feet maximum on centers.
2) At wall intersections, four feet minimum from corner.

b. Base slab, floor, and roof slab construction joints:
1) Placements to be approximately square and not to exceed 4,000 square feet.
2) Maximum side dimension of a slab pour shall be 90 feet.

2. Water bearing structures:
a. Wall vertical construction joints:
1) 30 feet maximum on centers.
2) At wall intersections, six feet minimum from corner.

b. Wall horizontal construction joints: 18 feet on centers.

c. Floor slab, construction joints:
1) Placements to be approximately square and not to exceed 2,000 square feet.
2) Maximum side dimension of a slab pour to be less than:
a) Twice the length of the short side.
b) 60 feet.

d. Elevated slab construction joints:
1) Placements to be approximately square and not to exceed 4,000 square feet.
2) Maximum side dimension of a slab pour to be less than:
a) Twice the length of the short side.
b) 70 feet.

C. Construction Joints - Bonding:
1. Obtain bond between concrete pours at construction joints by thoroughly cleaning and removing all laitance from construction joints.
2. Before new concrete is placed, all construction joints shall be coated with cement grout, or dampened, as indicated below:
3. Roughen construction joints:
a. Roughen the surface of the concrete to expose the coarse aggregate uniformly with 0.25 inch minimum amplitude.
1) Remove laitance, loosened particles of aggregate or damaged concrete at the surface.

4. Keyed construction joints:
a. Thoroughly clean construction joints and remove all laitance.
b. Dampen the hardened concrete immediately prior to placing of fresh concrete.

D. Slab On Grade Joints:

1. Locate construction and control joints in slabs on grade as shown or indicated on the Drawings.
2. Time cutting properly with set of concrete, if saw cut joints are required or allowed.
    a. Start cutting as soon as concrete has hardened sufficiently to prevent aggregates being dislodged by saw.
    b. Complete before shrinkage stresses become sufficient to produce cracking.

E. Expansion Joints:
1. Do not allow reinforcement or other embedded metal items bonded to concrete (except smooth dowels bonded on only one side of joint) to extend continuously through an expansion joint.
2. Use neoprene expansion joint fillers, unless shown or indicated otherwise on the Drawings.
3. Seal expansion joints as shown or indicated on the Drawings.
    a. Comply with Section 07 92 00 - Joint Sealants.

F. Waterstops - General:
1. Waterstop shall be continuous with splices in accordance with manufacturer's instructions and create water tight joints.
2. Do not mix different types of waterstop materials in the same structure without approval from the Contracting Officer unless shown or indicated on the Drawings.
3. Preformed strip type:
    a. Locate waterstop at center of wall, unless shown or indicated otherwise on the Drawings.
        1) Maintain at least three inches from edge of concrete or as recommended by manufacturer.
    b. Install in a bed of swelling sealant on smooth surface of hardened concrete by use of nails, adhesive or other means as recommended by manufacturer to prevent movement of waterstop during placement of concrete.
    c. Roughened joints shall be specially prepared during concrete placement to provide smooth surface for proper water stop installation.
    d. Use in joints against existing concrete where shown or indicated on the Drawings.
4. PVC waterstops:
    a. Pre-position waterstop accurately in joints, with adequate clearance from all reinforcing. Do not push waterstop into wet concrete.
    b. Secure waterstops in correct position using hog rings or grommets spaced no more than 1.5 feet maximum staggered along each edge full length and passed through the edge of the waterstop.
        1) Tie wire to adjacent reinforcing.
    c. Hold horizontal waterstops in place with continuous supports.
    d. Install according to manufacturer's instructions.
        1) Do not displace reinforcement from required location.
    e. Splice ends and intersections with perpendicular butt splice using electrical splicing iron in accordance with manufacturer's instructions.
        1) Use factory fabricated "T" and corner intersection fittings.
        2) Field splice straight runs of material.
    f. Unless otherwise noted, provide for construction joints in new construction for all structures indicated on the Drawings.

G. Other Embedded Items:
1. Place sleeves, inserts, anchors, and embedded items required for adjoining work or for its support, prior to initiating concreting.
2. Do not route electrical conduit, drains, or pipes in concrete slabs, walls, columns, foundations, beams or other structural members unless approved by Contracting Officer’s Representative.

H. Placing Embedded Items:
1. Support against displacement.

2. Fill voids in sleeves, inserts and anchor slots temporarily with readily removable material to prevent entry of concrete into voids.
3. Provide adequate means for anchoring waterstop in concrete.
    a. Provide means to prevent waterstops in the forms from being folded over by the concrete as it is placed.

3.6 FINISHING

A. Comply with Section 03 35 00 - Concrete Finishing and Repair of Surface Defects.

B. Coordinate mixing and placing with finishing.

3.7 INSTALLATION OF GROUT

A. Grout Schedule:
1. Sand cement grout:
    a. Fill keyways in precast HCU.
    b. Construction joint bedding (base of wall pours with comparable compressive strength to wall).
    c. General use.
    d. As shown or indicated on the Drawings.
2. Non-shrinking non-metallic grout:
    a. Filling form tie holes.
    b. Under column and beam base plates.
    c. Other uses shown or indicated on the Drawings.
3. Epoxy grout:
    a. Patching cavities in concrete.
    b. Grouting of dowels and anchor bolts into existing concrete.
    c. Grouting of rotating or oscillating equipment base plates.
    d. As shown or indicated on the Drawings.

B. Grout Installation:
1. Sand cement grout:
    a. Fill wetted keyways between precast concrete hollow core slabs with sand cement grout.
    b. Consolidate grout by rodding or by other means to assure complete filling of keyways.
    c. Cure grout by one of methods specified.
2. Non-shrink non-metallic grout:
    a. Clean concrete surface to receive grout.
    b. Saturate concrete with water for 24 hours prior to grouting.
    c. Mix in a mechanical mixer.
    d. Use no more water than necessary to produce flowable grout.
    e. Place in accordance with manufacturer's instructions.
    f. Provide under beam, column, and equipment base plates, in joints between precast concrete and cast-in-place slabs, and in other locations shown or indicated on the Drawings.
    g. Completely fill spaces and cavities below the top of base plates.
    h. Provide forms where base plates and bed plates do not confine grout.
    i. Where exposed to view, finish grout edges smooth.
    j. Except where a slope is shown or indicated on the Drawings, finish edges flush at the base plate, bed plate, member or piece of equipment.
    k. Coat exposed edges of grout with cure or seal compound recommended by the grout manufacturer.
3. Epoxy grout:
    a. Mix and place in accordance with manufacturer's written instructions.
    b. Apply only to clean, dry, sound surface.
    c. Completely fill cavities and spaces around dowels and anchors without voids.
    d. Grout base and bed plates as specified for non-shrinking, non-metallic grout.

e. Obtain manufacturer's field technical assistance as necessary to ensure proper placement.

3.8 CURING AND PROTECTION

A. Protect concrete from premature drying, excessively hot or cold temperatures, and mechanical damage immediately after placement, and maintain with minimal moisture loss at relatively constant temperature for period necessary for hydration of cement, hardening, and compressive strength gain.
   1. Comply with ACI 308.1 except as modified herein.
   2. Do not impose loads by foot traffic, wheeled traffic, and other loads until concrete has sufficiently cured to carry imposed loads without adversely affecting the concrete. In no event shall concrete be subject to loading or traffic during initial 48 hours of curing, unless otherwise approved by Contracting Officer.
   3. Evaporation Retarder:
      a. Apply evaporation retarder to uniformed concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 0.2 pounds per square foot per hour (one kilogram per square meter per hour) before and during finishing operations.
         1) Apply according to manufacturer's written instructions one or more times after placing, screeding, and bull floating or darbying concrete, but before float finishing.

B. For surfaces of non-water bearing structures, apply one of the following curing procedures immediately after completion of placement and finishing (surfaces not in contact with forms).
   1. Ponding or continuous sprinkling. Avoid eroding the surface of freshly placed concrete.
   2. Application of wet Absorbent Covers:
      a. Minimum lap: 12 inches.
      b. Provide continuous uniform supply of moisture, such as sprinklers or soaker hoses as necessary to keep concrete surface continuously wet.
      c. Monitor Absorbent Covers as required to prevent cover materials or concrete surface from drying out.
   3. Continuous application of steam (not exceeding 150 degrees F) or mist spray.
   4. Application of Moisture Retaining Cover sheet materials.
      a. Place as soon as possible after final finishing and without marring the surface.
      b. Minimum lap: 12 inches.
      c. Seal edges to make water-tight.
      d. Place Moisture Retaining Cover in intimate contact with the concrete surface, without wrinkles and weighted to hold in place.
      e. Hold cover and edges in place as necessary to prevent wind from displacing the cover.
      f. Moisture Retaining Fabric:
         1) Install in accordance with manufacturer’s written recommendations.
         2) Saturate concrete surface and fabric side of cover immediately prior to placing.
      g. Monitor continuously during the curing period:
         1) Repair holes, tears or displaced cover.
         2) Rewet as necessary to keep concrete moist under cover.
   5. Application of other moisture retaining covering acceptable to Contracting Officer..
   6. Water used for curing shall be within 20 degrees F of the concrete temperature.
   7. Application of a curing compound.
      a. Apply curing compound in accordance with manufacturer's written recommendations immediately after any water sheen, which may develop after finishing, has disappeared from concrete surface.
      b. Do not use on surface against which additional concrete or other material is to be bonded unless it is proven that curing compound will not prevent bond.
      c. Where a vertical surface is cured with a curing compound, the vertical surface shall be covered with a minimum of two coats of the curing compound.
         1) Apply the first coat of curing compound to a vertical surface immediately after form removal.

2) The vertical concrete surface at the time of receiving the first coat shall be damp with no free water on the surface.
3) Allow the preceding coat to completely dry prior to applying the next coat.
4) A vertical surface: Any surface steeper than 1 vertical to 4 horizontal.

8. Surfaces In Contact with Forms:
   a. Formed surfaces: Cure formed concrete surfaces utilizing final curing methods per ACI 308.1, including underside of beams, supported slabs, and other similar surfaces,
      1) See Section 03 11 13 - Formwork.
   b. Minimize moisture loss from and temperature gain of concrete placed in forms exposed to heating by sun by keeping forms wet and cool until they can be safely removed.
   c. Make provisions to keep concrete wall moist while stripping forms and until curing measures are in place.
   d. After form removal, cure concrete until end of time prescribed.
   e. Use one of the methods listed above.
   f. Forms left in place shall not be used as a method of curing in hot weather.
   g. The term "hot weather", where used in these Specifications, is defined in ACI 305.1.
   h. In hot weather, remove forms from vertical surfaces as soon as concrete has gained sufficient strength so that the formwork is no longer required to support the concrete.

C. For Surfaces of Water Bearing Structures:
   1. Protect concrete from premature drying, excessively hot or cold temperatures, and mechanical damage immediately after placement, and maintain with minimal moisture loss at relatively constant temperature for period necessary for hydration of cement, hardening, and compressive strength gain.
      a. Follow recommendations of ACI 308.1 except as modified herein.
      b. Use Evaporation Retardant to reduce surface moisture evaporation of slabs during concrete placement. Comply with the manufacturer’s written instructions of use as required to obtain the intended results.
   2. Apply one of the following moist curing procedures immediately after completion of placement and finishing, for concrete surfaces not in contact with forms.
      a. Ponding or continuous sprinkling.
      b. Application of absorptive mats or fabric kept continuously wet and in intimate contact with concrete.
      c. Surfaces shall be covered with a double layer of absorptive mats or fabric, wetted before placing, and overlapped at least 6 inches.
      d. Application of sand kept continuously wet.
      e. Continuous application of steam (not exceeding 150 degrees F) or mist spray.
      f. Ponding and sprinkling in conjunction with application of waterproof sheet materials, conforming to ASTM C171 and only with a program as approved by the Engineer that will keep the surface continuously wet.
      g. Ponding and sprinkling in conjunction with application of other moisture retaining covering as approved by the Engineer that will keep the surface continuously wet.
   3. After seven days of moist curing, application of a curing compound conforming to ASTM C309 may be substituted for moist curing.
      a. Apply curing compound in accordance with manufacturer's written recommendations immediately after any water sheen which may develop during moist curing has disappeared from concrete surface.
      b. Do not use on surface against which additional concrete or other material is to be bonded unless it is proven that curing compound will not prevent bond.
      c. Where a surface is cured with a curing compound, the surface shall be covered with a minimum of two coats of the curing compound, 30 mils thick each coat.
      d. Apply the first coat of curing compound immediately after form removal or discontinued moist curing and before the surface displays water loss. Apply in one direction only, covering uniformly to a minimum thickness of 30 mils.
      e. The concrete surface at the time of receiving the first coat shall be damp with no free water on the surface.

f. Allow the preceding coat to completely dry prior to applying the next coat.
g. Apply second coat in direction perpendicular to the first coat application direction, covering uniformly to a minimum thickness of 30 mils.
h. Curing compounds used for concrete that will be in contact with potable water or water that will be treated to become potable shall be non-toxic, taste and odor-free, and be compliant with NSF/ANSI 61.
i. Alternately, remove curing compound, non-compliant with NSF/ANSI 61, from surfaces of concrete that will be in contact with potable water or water that will be treated to become potable, without damaging the concrete finish.

4. Curing Concrete In Contact with Forms:
   a. Minimize moisture loss of concrete placed in forms by keeping forms wet and cool until they can be safely removed.
   b. Moist cure the top surface of concrete placed in forms as specified.
   c. After form removal, cure concrete until end of time prescribed.
      1) Use one of methods indicated above.
      2) When acceptable to the Engineer, placement of the second pour at joints may occur prior to the end of the curing period.
   d. Forms left in place shall not be used as a method of curing in hot weather.
   e. The term "hot weather", where used in these Specifications, is defined in ACI 305R.
   f. In hot weather, remove forms from vertical surfaces as soon as concrete has gained sufficient strength so that the formwork is no longer required to support the concrete and commence moist curing as specified.

D. Curing Period:
   1. Continue curing for at least seven days for all concrete except Type III, high early strength concrete for which period shall be at least three days.
      a. If one of curing procedures indicated above is used initially, it may be replaced by one of other procedures indicated any time after concrete is two days old, provided concrete is not permitted to become surface dry during transition.

E. Cold Weather:
   1. Comply with of ACI 306.1.
   2. Maintain temperature of concrete in accordance with ACI 306.1 for a minimum of 72 hours after concrete is placed, when outdoor temperature is 40 degrees F, or less.
      a. Maximum temperature rate of decrease: In accordance with ACI 306.1.
   3. Provide temporary heating, covering, insulating, or housing of the concrete Work to maintain required temperature without damage due to concentration of heat.
   4. Do not use combustion heaters unless precautions are taken to prevent exposure of concrete to exhaust gases which contain carbon dioxide.
   5. Interior slabs in areas intended to be heated shall be adequately protected so that frost does not develop in the supporting subgrade.

F. Hot Weather:
   1. Comply with ACI 305.1 and ACI 308.1.
   2. Provide as necessary cooling forms, reinforcement and concrete, windbreaks, shading, fog spraying, sprinkling, ponding, or wet covering with a light colored material.
   3. Provide protective measures as quickly as concrete hardening and finishing operations will allow.
   4. Maximum temperature rate of decrease: In accordance with ACI 305.1.

G. Rate of Temperature Change:
   1. Maintain temperature of air immediately adjacent to concrete as uniform as possible, during and immediately following curing.

H. Protection from Mechanical Damage:
   1. Protect concrete from damage of all types, whether or not mechanically imparted including load stresses, heavy shock, and excessive vibration.

2. Protect finished concrete surfaces from all types of damage, including damage by construction equipment and machinery, construction means and methods, precipitation, and running water.
3. Do not load self-supporting structures in such a way as to overstress concrete.

3.9 FIELD QUALITY CONTROL

A. Tests and Inspections:
   1. Special Inspections in accordance with building code:
      a. Comply with Section 01 40 00 and Statement of Structural Tests and Special Inspections from for special inspection requirements.
      b. Concrete tests required per Specification Section 03 05 05.

END OF SECTION

SECTION 03 35 00
CONCRETE FINISHING AND REPAIR OF SURFACE DEFECTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Concrete finishing and repair of surface defects.
   2. Chemical Sealers.
   3. Polymer Modified Cementitious Coating.
   4. Resurfacing Mortar.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 11 13 - Formwork.
   2. Section 03 31 30 - Concrete, Materials and Proportioning.
   3. Section 03 31 31 - Concrete Mixing, Placing, Jointing and Curing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Concrete Institute (ACI):
      a. CT-13, Concrete Terminology.
      b. 117, Specification for Tolerances for Concrete Construction and Materials.
      c. 303R, Guide to Cast-in-Place Architectural Concrete Practice.
      d. 308, Standard Practice for Curing Concrete.
   2. ASTM International (ASTM):
      a. C109, Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or 50-mm Cube Specimens).
      b. C150, Standard Specification for Portland Cement.
      c. C157, Standard Test Method for Length Change of Hardened Hydraulic-Cement Mortar and Concrete.
      d. C309, Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete.
      e. C666, Standard Test Method for Resistance of Concrete to Rapid Freezing and Thawing.
      f. C779, Standard Test Method for Abrasion Resistance of Horizontal Concrete Surfaces.
      g. C1315, Standard Specification for Liquid Membrane-Forming Compounds Having Special Properties for Curing and Sealing Concrete.
      h. D4258, Standard Practice for Surface Cleaning Concrete for Coating.
      i. D4259, Standard Practice for Abrading Concrete.
      j. E1155, Standard Test Method for Determining F(F) Floor Flatness and F(L) Floor Levelness Numbers.
      k. E1486, Standard Test Method for Determining Floor Tolerances Using Waviness, Wheel Path and Levelness Criteria.
   3. International Concrete Repair Institute (ICRI):
      a. 310.2R, Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, and Polymer Overlays.
   4. National Council Highway Research Program (NCHRP):
      a. 244, Concrete Sealers for the Protection of Bridge Structures.
   5. The Society for Protective Coatings/NACE International (SSPC/NACE):
      a. SP 13/NACE No. 6, Surface Preparation of Concrete.

B. Qualifications:
   1. Chemical Sealer CS-2:

a. Applicator shall be factory trained and approved, in writing, by the manufacturer to apply the product.
b. Applicator shall have a minimum of five years of experience successfully applying materials specified.

C. Mock-Ups.
1. General:
a. Construct additional mock-ups as required until accepted.
b. Mock-ups constitute minimum standard of quality for actual construction.
c. Maintain mock-up during construction.
d. Remove when directed by Contracting Officer Representative.
2. Construct mock-up for each type of wall finish specified for review and acceptance by Contracting Officer Representative.
a. Minimum 4 x 4 feet area for each different wall finish specified.
b. Mock-ups shall include:
1) Sample of patched tie hole.
2) Sample of all jointery being used in the walls.
c. Include mock-up of wall having polymer modified cementitious coating.
1) Mock-up shall be stepped to show surface preparation, repairs and coating in all stages of application.
3. Construct mock-up floor slab for review and acceptance by Contracting Officer Representative.
a. Minimum 10 x 10 feet.

1.3 DEFINITIONS

A. Vertical Surface Defects:
1. Any void in the face of the concrete deeper than 1/8 inches, such as:
a. Tie holes.
b. Air pockets (bug holes).
c. Honeycombs.
d. Rock holes.
2. Scabbing:
a. Scabbing is defect in which parts of the form face, including release agent, adhere to concrete.
3. Foreign material embedded in face of concrete.
4. Fins 1/16 inches or more in height.

B. Installer or Applicator:
1. Installer or applicator is the person actually installing or applying the product in the field at the Project site.
2. Installer and applicator are synonymous.

C. Other words and terms used in this Specification Section are defined in ACI CT-13.

1.4 SUBMITTALS

A. Shop Drawings:
1. Product technical data including:
a. Acknowledgement that products submitted meet requirements of standards referenced.
b. Manufacturer's installation instructions.
2. Certifications:
a. Certification of aggregate gradation.
b. Certification of manufacturer experience qualifications and performance history.
c. Certification of applicator's qualifications.
1) Refer to Qualifications paragraph.
2) Provide manufacturer's written approval of applicators.
3) Provide references substantiating specialty experience.

B. Informational Submittals:

1.5 DELIVERY, STORAGE, AND HANDLING

A. Comply with manufacturer's recommendations and requirements for materials used.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Bonding Agents:
      a. Euclid Chemical Co.
      b. Or equal
   2. Chemical Sealers:
      a. Euclid Chemical Co.
      b. Or equal.
   3. Polymer Modified Cementitious Coating:
      a. Euclid Chemical Co.
      b. Or equal.
   4. Patching Mortar:
      a. Euclid Chemical Co.
      b. Or equal.

C. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Chemical Sealer CS-1:
   1. High solids, water-based solution containing acrylic copolymers.
      a. ASTM C1315, Type I, Class A.
      b. Non-yellowing UV resistant.
      c. VOC Content: <200 G/L.
   2. USDA approved as a concrete floor sealer.
   3. Euclid Chemical Super Diamond Clear VOX.

B. Chemical Sealer CS-2:
   1. Water based chemical solution containing a blend of silicate and siliconate polymers designed to seal, harden and dustproof concrete floors.
   2. VOC Content: 0 G/L.
   3. Performance of treated concrete floor:
      a. Coefficient of Friction:
         1) Dry: 0.81.
         2) Wet: 0.72.
      b. Liquid repellency, RILEM Method 11.4:
         1) ≥ 1 mL.
   4. Euclid Chemical Euco Diamond Hard.

C. Chemical Sealer CS-3:
   1. Clear, penetrating, breathable, waterborne silane-siloxane solution.
   2. VOC content: ≤50 G/L.
   3. Odorless.

4. Flash point: >200 degrees F.
5. Water absorption: 85% reduction per NCHRP 244.
6. Chloride penetration: 82% reduction per NCHRP 244.
7. Euclid Chemical Baracade WB 244.

D. Patching Mortar: Trowelable cementitious repair mortar for vertical, overhead, and horizontal repairs.
   1. Portland cement-based, rapid set repair mortar for interior or exterior use.
   2. Compressive Strength, ASTM C109:
      a. Minimum 3000 psi at 7 days.
      b. Minimum 5000 psi at 28 days.
   3. Freeze Thaw Durability, ASTM C666: 96.75% at 300 Cycles.
   4. Shrinkage, ASTM C157: 0.069%.
   5. Euclid Chemical Speed Crete Red Line.

E. Bonding Agents:
   1. For use only on concrete surfaces not receiving liquid water repellent coating:
      a. High solids acrylic latex base liquid for interior or exterior application as a bonding agent to improve adhesion and mechanical properties of concrete patching mortars.
         1) Master Builders MasterEmaco A 660.
         2) Euclid Chemical Co. Flex-Con.
         3) Laticrete L&M Everbond.
   2. For use only on concrete surface receiving liquid water repellent:
      a. Non-acrylic base liquid for interior or exterior application as a bonding agent to improve adhesion and mechanical properties of concrete patching mortars.

F. Cement:
   1. ASTM C150, Type II, Portland, or ASTM C595, Type IL-HS Blendfor areas exposed to sewage.
   2. ASTM C150, Type I Portland or ASTM C595, Type IL elsewhere.

G. Aggregate:
   1. Sand: Maximum size #30 mesh sieve.
   2. For exposed aggregate finish surfaces: Same as surrounding wall.

H. Water: Potable.

I. Polymer modified cementitious coating:
   1. Polymer modified Portland cement based coating for concrete and masonry.
      a. Waterproof.
      b. Resistant to both positive and negative hydrostatic pressure.
      c. Breathable.
   2. Master Builders Solutions MasterSeal 581 or Euclid Chemical Tamoseal.
      a. Color:
         1) Interior surfaces: Standard gray.
         2) Exterior surfaces: Standard gray.
      b. Texture: Fine.

J. Nonshrink Grout: See Specification Section 03 31 30 and Specification Section 03 31 31.

2.3 MIXES

A. Bonding Grout: One part cement to one part aggregate.

B. Patching Mortar:
   1. One part cement to 2-1/2 parts aggregate by damp loose volume.
      a. Substitute white Portland cement for a part of gray Portland cement to produce color matching surrounding concrete.

PART 3 - EXECUTION

3.1 PREPARATION

A. For methods of curing, see Specification Section 03 31 31.

B. Surface Preparation:
   1. Clean surfaces in accordance with ASTM D4258 to remove dust, dirt, form oil, grease, or other contaminants prior to abrasive blasting, chipping, grinding or wire brushing.
   2. Prepare surfaces in accordance with ASTM D4259 and SSPC SP 13/NACE No. 6 to completely open defects down to sound concrete and remove laitance.
      a. Provide concrete surface profile (CSP) in accordance with ICRI 310.2:
         1) Areas to receive Repair Mortar:
            a) Areas larger than 1 SF or deeper than 1/4 inches Abrasive blast, scarify or needle scale to CSP No. 6-8.
      b. If additional chipping or wire brushing is necessary, make edges perpendicular to surface or slightly undercut.
      c. No featheredges will be permitted.
      d. Rinse surface with clean water to remove all dust, dirt, debris, loosened concrete, laitance, and other contaminants.

C. Preparation of Bonding Grout Mixture:
   1. Mix cement and aggregate.
   2. Mix bonding agent and water together in separate container in accordance with manufacturer's instructions.
   3. Add bonding agent/water mixture to cement/aggregate mixture.
   4. Mix to consistency of thick cream.
   5. Bonding agent itself may be used as bonding grout if approved by manufacturer and Contracting Officer Representative.

D. Preparation of Patching Mortar Mixture:
   1. Mix specified patching mortar per manufacturer’s published recommendations.
   2. For repairs exceeding 2 inches in depth, mix with clean, pre-dampened 3/8 inches pea gravel in accordance with the manufacturer’s recommendations.

E. Polymer modified cementitious coating:
   1. Mix in accordance with manufacturer's recommendations using bonding agent acceptable to coating manufacturer.

3.2 INSTALLATION AND APPLICATION

A. Do not repair surface defects or apply wall or floor finishes when temperature is or is expected to be below 50 degrees F.
   1. If necessary, enclose and heat area to between 50 and 70 degrees F during repair of surface defects and curing of patching material.
      a. Use only clean fuel, indirect fired heating apparatus.
      b. Exhaust combustion byproducts outside of work area.

B. Chemical Sealer Application:
   1. General:
      a. Immediately prior to Substantial Completion, thoroughly clean floor in accordance with ASTM D4258 and prepare to receive chemical sealer.
         1) Remove previously applied membrane curing compounds.
         2) Remove soil, oils, stains, discoloration, or any other imperfection having a negative impact on the appearance of the finished floor.
      b. Apply product to floor areas indicated on the Drawings.
      c. Apply in accordance with manufacturer's published installation instructions.
   2. Chemical Sealer (CS-1):
      a. Apply two uniform coats at rate recommended by manufacturer.
         1) Apply using manufacturer’s recommended equipment with a fan-tip nozzle.

2) Do not allow material to puddle.
b. Allow first coat to completely dry before applying second coat.
c. Spotted or mottled appearances will not be accepted.

3. Chemical Sealer (CS-2):
   a. Apply two uniform coats at rate recommended by manufacturer.
      1) Scrub the material into the floor using a mechanical scrubber.
         a) Keep the surface wet for not less than 30 minutes.
         b) Continue scrubbing in accordance with manufacturer's application instructions.
         c) After material has thickened, but not more than 60 minutes after application, remove all excess liquid.
      2) Thoroughly rinse with clean water to remove all residue.
         a) Damp mop with clean water to remove any streaks.
         b) Do not allow residue to dry on floor surface.
      3) Do not track material onto untreated surfaces.
   b. After rinsing, allow floor to dry completely and apply second coat following the same procedures.
   c. Final floor finish shall have uniform sheen without streaking, stains or white residue.
4. Chemical Sealer (CS-3):
   a. Apply uniform coats at rate recommended by manufacturer.
      1) Apply with fine, uniform spray or microfiber pad.
   b. Allow floor to dry completely and remove any dried residue using hot water and mild citric acid.
   c. Final floor finish shall be uniform, free of residue, and shall repel water.
   d. Apply additional coat(s) as necessary to achieve water repellent finish.

C. Repairing Surface Defects:
1. This method is to be used on vertical concrete surfaces as indicated in the Concrete Finishes for Vertical Wall Surfaces paragraph of this Specification Section and similar concrete surfaces not otherwise specified to receive another finish or coating.
   a. For surfaces indicated to receive finish or coating other than those specified herein; refer to the applicable Specification Section for surface preparation requirements:
      1) Fluid Applied Waterproofing: See Specification Section 07 14 00.
      2) High Performance Industrial Coatings: See Specification Section 09 96 00.
2. Fill and repair surface defects and tie-holes using patching mortar mix specified in the MATERIALS Article in PART 2.
   a. Prime exposed reinforcing steel, embeds or other steel surfaces with primer as recommended by patching mortar manufacturer.
   b. Scrub bond coat:
      1) Wet substrate to a saturated surface dry (SSD) condition.
      2) Mix patching mortar to a scrub coat or slurry consistency per manufacturer's published recommendations and apply to entire area.
   c. As an alternate to the scrub bond coat, concrete may be primed with manufacturer's recommended epoxy primer.
   d. Patching Mortar Application:
      1) Mix and apply Patching Mortar per manufacturer's recommendations within the open time of the product scrub coat or any bonding agents.
      2) Finish to level of surrounding concrete surface utilizing techniques recommended by manufacturer.
3. Consolidate patching mortar into place and strike off so as to leave patch slightly higher than surrounding surface.
4. Leave undisturbed until mortar has stiffened before finishing level with surrounding surface.
   a. Do not use steel tools in finishing a patch in a formed wall which will be exposed to view.
5. Cure patching mortar in accordance with ACI 308.

D. Concrete Finishes for Vertical Wall Surfaces:

1. General:
    a. Give concrete surfaces finish as specified below after removal of formwork and repair of surface defects.
    b. Finish numbers not listed are "Not Used".
2. Finish #1 - As cast rough form finish:
    a. Selected forming materials are not required.
    b. Prepare surface in accordance with the PREPARATION Article in PART 3 of this Specification Section.
    c. Repair the following surface defects using patching mortar specified in PART 2:
        1) Tie holes.
        2) Honeycombs deeper than 1/4 inches.
        3) Air pockets deeper than 1/4 inches.
        4) Rock holes deeper than 1/4 inches.
    d. Chip or rub off fins exceeding 1/4 inches in height.
    e. Provide at unexposed surfaces such as:
        1) Foundations.
        2) Below-grade walls not to be waterproofed.
        3) Concealed surface of concrete back-up wythe in cavity wall construction.
3. Finish #2 - As cast form finish:
    a. Form facing material shall produce a smooth, hard, uniform texture.
        1) Use forms specified for surfaces exposed to view in accordance with Specification Section 03 11 13.
    b. Prepare surface in accordance with the PREPARATION Article in PART 3 of this Specification Section.
        1) Chip or rub off fins exceeding 1/8 inches in height.
        2) Abrasive blast surfaces in accordance with ASTM D4259 and SSPC SP 13/NACE No. 6 to completely open defects down to sound concrete and remove laitance.
            a) Provide ICRI 310.2 Concrete Surface Profile (CSP) No. 3, minimum across the entire surface.
                (1) For contiguous repair areas larger than 1 SF or deeper than 1/4 inches Abrasive blast, scarify or needle scale to CSP No. 6-8.
            b) If additional chipping or wire brushing is necessary, make edges perpendicular to surface or slightly undercut.
            c) No feather edges will be permitted.
        3) Rinse surface with clean water and allow surface water to evaporate prior to repairing surface defects.
        4) Repair the following surface defects using patching mortar specified in PART 2:
            a) Tie holes.
            b) Honeycombs deeper than 1/4 inches or larger than 1/4 inches diameter.
            c) Air pockets deeper than 1/4 inches or larger than 1/4 inches diameter.
            d) Rock holes deeper than 1/4 inches or larger than 1/4 inches diameter.
            e) Scabbing.
        5) Brush blast repaired areas to match adjacent surface texture.
    c. Provide this finish for:
        1) Interior walls of pipe galleries , pump rooms , or tipping floors.
        2) Underside of horizontal elements adjacent to the finished surface.
        3) Exposed surfaces not specified to receive another finish.
4. Finish #4 - Polymer modified cementitious coating:
    a. Form facing material shall produce a smooth, hard, uniform texture.
        1) Use forms specified for surfaces exposed to view in accordance with Specification Section 03 11 13.
        2) Comply with ACI 303R for formwork accuracy and form joint handling to prevent grout leakage.
    b. Prepare surface in accordance with the PREPARATION Article in PART 3 of this Specification Section.
        1) Chip or rub off fins exceeding 1/8 inches in height.

2) Abrasive blast and repair surface defects in accordance with Concrete Finish #2.

c. Apply decorative coating to entire surface.
   1) As a mixing liquid for the coating, use bonding agent and water mixture as recommended by coating manufacturer.
   2) Apply two (2) coats at 2 pounds per square yard per coat.
      a) During application of first coat, complete fill all voids, depressions or other surface imperfections.

d. When second coat is set, float to a uniform texture with a sponge float.

e. Provide this finish on all exposed to view:
   1) Exterior building surfaces not otherwise indicated to receive an Architectural Abrasive Blast Finish.
   2) Interior walls, columns and similar vertical surfaces where indicated on Room Finish Schedule on the Drawings.
   3) Underside of horizontal elements adjacent to the finished surface.

f. Construct mock-up per the Mock-Ups paragraph in the QUALITY ASSURANCE Article in PART 1 of this Specification Section.

E. Related Unformed Surfaces (Except Slabs):
   1. Strike smooth and level tops of walls or buttresses, horizontal offsets, and similar unformed surfaces occurring adjacent to formed surfaces after concrete is placed.
   2. Float surface to a texture consistent with that of formed surfaces.
      a. If more than one finish occurs immediately adjacent to unformed surface, provide surface with most stringent formed surface requirement.
   3. Continue treatment uniformly across unformed surfaces.

F. Concrete Finishes for Horizontal Slab Surfaces:
   1. General:
      a. Tamp concrete to force coarse aggregate down from surface.
      b. Screed with straightedge, eliminate high and low places, bring surface to required finish elevations; slope uniformly to drains.
      c. Dusting of surface with dry cement or sand during finishing processes not permitted.
   2. Unspecified slab finish:
      a. When type of finish is not indicated, use following finishes as applicable:
         1) Surfaces intended to receive bonded applied cementitious applications: Scratched finish.
         2) Surfaces intended to receive roofingor waterproofing membranes: Floated finish.
         3) Floors : Troweled finish.
         4) Garage floors and ramps: Broom or belt finish.
         5) Exterior slabs, sidewalks, platforms, steps and landings, and ramps, not covered by other finish materials: Broom or belt finish.
         6) All slabs to receive a floated finish before final finishing.
   3. Scratched slab finish: After concrete has been placed, consolidated, struck off, and leveled to a Class B tolerance, roughen surface with stiff brushes or rakes before final set.
   4. Floated finish:
      a. After concrete has been placed, consolidated, struck off, and leveled to a Class B tolerance, do no further work until ready for floating.
      b. Begin floating when water sheen has disappeared and surface has stiffened sufficiently to permit operations.
         1) Use wood or cork float.
      c. During or after first floating, check planeness of entire surface with a 10 feet straightedge applied at not less than two different angles.
   5. Cut down all high spots and fill all low spots to produce a surface with Class B tolerance throughout.
      a. Refloat slab immediately to a uniform texture.
   6. Troweled finish:
      a. Float finish surface to true, even plane.
      b. Power trowel, and finally hand trowel.

c. First troweling after power troweling shall produce a smooth surface which is relatively free of defects, but which may still show some trowel marks.
d. Perform additional trowelings by hand after surface has hardened sufficiently.
e. Final trowel when a ringing sound is produced as trowel is moved over surface.
f. Thoroughly consolidate surface by hand troweling.
g. Finish in accordance with the FIELD QUALITY CONTROL Article in PART 3 of this Specification Section.
1) Leave finished surface essentially free of trowel marks, uniform in texture and appearance.
h. On surfaces intended to support floor coverings, remove any defects that would show through floor covering.

7. Broom or belt finish: Immediately after concrete has received a float finish as specified, give it a transverse scored texture by drawing a broom or burlap belt across surface.
8. Underside of concrete slab finish:
a. Match finish as specified for adjacent vertical surfaces.
b. If more than one finish occurs immediately adjacent to underside of slab surface, provide surface with most stringent formed surface requirement.

3.3 FIELD QUALITY CONTROL

A. Tolerances:
1. Finished floor slabs:
a. Provide Floor Flatness (F_F) and Floor Levelness (F_L) in accordance with ACI 117.
1) Measure in accordance with ASTM E1155.
b. Slabs not indicated to be sloped:
1) F_F: Equal or greater than 35.
2) F_L: Equal or greater than 25.
c. Slabs indicated to be sloped or curved:
1) Measure in accordance with ASTM E1486.
2) Provide slopes or curves as indicated on the Drawings.
d. Slabs indicated to receive polished concrete floor:
1) F_F: Equal or greater than 45.
2) F_L: Equal or greater than 35.
3) Refer to Room Finish Schedule on Drawings.
2. Horizontal surfaces other than finished floor slabs, including but not limited to, top of footings, top of walls, concrete fill in tankage, channels and similar applications:
a. Gap between a 10 feet straightedge placed anywhere and the finished surface shall not exceed:
1) Class A tolerance: 1/4 inches.
2) Class B tolerance: 3/8 inches.
3) Class C tolerance: 1/2 inches.
b. Accumulated deviation from intended true plane of finished surface shall not exceed 1/2 inches.

B. Unacceptable finishes shall be replaced or, if approved in writing by Contracting Officer Representative, may be corrected provided strength and appearance are not adversely affected.
1. High spots to be removed by grinding and/or low spots filled with a patching compound or other remedial measures to match adjacent surfaces.

C. Provide services of manufacturer's technical representative:
1. A certified manufacturer's representative experienced in the use of the products used shall be present on a full-time basis to observe and oversee all operations associated with the installation.
2. Contractor, along with manufacturer, shall be fully responsible for the proper application, including all means and methods incidental thereto necessary for a sound, secure and complete installation.
3. Manufacturer's representative shall be present for installation of:
a. Dry-shake Hardener.

b. Heavy-duty Metallic Aggregate Topping.

3.4 PROTECTION

A. All horizontal slab surfaces receiving chemical sealer shall be kept free of traffic and loads for minimum of 72 hours following installation of sealer.

END OF SECTION

SECTION 03 41 33
PRECAST AND PRESTRESSED CONCRETE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Precast and prestressed concrete.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 05 05 - Testing.
   2. Section 01 35 73 – Delegated Design Procedures
   3. Section 09 96 00 - High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Association of State Highway and Transportation Officials (AASHTO):
      a. HB, Standard Specifications for Highway Bridges.
   2. American Concrete Institute (ACI):
      a. 211.2, Standard Practice for Selecting Proportions for Structural Lightweight Concrete.
      b. 318, Building Code Requirements for Structural Concrete.
   3. ASTM International (ASTM):
      a. A36, Standard Specification for Carbon Structural Steel.
      b. A108, Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished.
      c. A416, Standard Specification for Steel Strand, Uncoated Seven-Wire for Prestressed Concrete.
      d. A496, Standard Specification for Steel Wire, Deformed, for Concrete Reinforcement.
      e. A1064, Standard Specification for Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete.
      f. C33, Standard Specification for Concrete Aggregates.
      g. C150, Standard Specification for Portland Cement.
      h. C330, Standard Specification for Lightweight Aggregates for Structural Concrete.
      i. D2240, Standard Test Method for Rubber Property-Durometer Hardness.
      j. E329, Standard Specification for Agencies Engaged in Construction Inspection and/or Testing.
   4. American Welding Society (AWS):
      a. A5.1/A5.1M, Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding.
      b. A5.5/A5.5M, Specification for Low-Alloy Steel Electrodes for Shielded Metal Arc Welding.
      c. D1.1, Structural Welding Code - Steel.
      d. D1.4, Structural Welding Code - Reinforcing Steel.
      e. D1.6, Structural Welding Code – Stainless Steel.
   5. Occupational Safety and Health Administration (OSHA).
   6. Precast/Prestressed Concrete Institute (PCI):
      a. MNL 116, Manual for Quality Control for Plants and Production of Precast and Prestressed Concrete Products.
      b. MNL 120, Design Handbook - Precast and Prestressed Concrete.

B. Qualifications:
   1. Provide precast and prestressed concrete units produced by an active member of PCI.
   2. Plant to be certified by the Precast/Prestressed Concrete Institute, Plant Certification Program, as applicable:
      a. Certification Code C1: Precast Concrete Products.

b. Certification Code C2: Precast Hollow Core and Repetitive Products.
c. Certification Code C3: Prestressed Straight Strand Structural Members.
d. Certification Code C4: Prestressed Deflected Strand Structural Members.
e. Plant shall have been certified within past year from bid date.

3. Plant shall be certified by IAS and shall be acceptable to the Building Code Official to assure compliance with approved fabricator Special Inspection requirements in accordance with the building code.
   a. Plants that are not certified by IAS or not acceptable to the Building Code Official may be acceptable to work on the Project, provided:
      1) Plant meets all remaining qualifications.
      2) Contractor reimburses the Government the cost of Special Inspection services.
4. Provide units manufactured by plant which has regularly and continuously engaged in manufacture of units of same type as those required for a minimum of three years.
5. Assure manufacturer's testing facilities meet requirements of ASTM E329.
6. Welding operators and processes to be qualified in accordance with:
   a. AWS D1.1 for welding steel shapes and plates.
   b. AWS D1.4 for welding reinforcing bars.
7. Welding operators to have passed qualification tests for type of welding required during the previous 12 months prior to commencement of welding.
8. Engineer for all precast or prestressed members: Professional Engineer licensed in the State of Arizona.
   a. Engineer to have minimum five years of experience in design of precast and prestressed members with scope similar to this Project.
9. Precast erector:
   a. Minimum five years of experience with projects of similar size and complexity.

1.3 DEFINITIONS

A. Slabs: May refer to hollow core slabs or solid flat slabs, prestressed or non-prestressed.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Sizes, types and manufacturer of bearing pads.
      d. Hardware to be utilized to support suspended appurtenances.
   2. Shop Drawings and erection plans for precast units, their connections and supports showing:
      a. Member size and location.
      b. Size, configuration, location and quantity of reinforcing bars and prestressing strands.
      c. Initial prestress forces.
      d. Size and location of openings verified by Contractor.
      e. Size, number, and locations of embedded metal items and connections.
      f. Required concrete strengths.
      g. Identification of each unit using same standard marking numbers as used to mark actual units.
   3. Calculations for members and connections designed by fabricator.
      a. Calculations to be sealed by a professional Structural Engineer registered in the State of Arizona.
      b. Perform calculations using the dead load of the members plus the superimposed uniform and concentrated loads shown on the Drawings and indicated in this Specification Section.
      c. Design wet well structure for the surcharge loading from the retaining wall footings as applicable.
      d. Indicate the following:
         1) Design for maximum moment, maximum shear and maximum torsion.

2) Final top and bottom flexural stresses resulting from the stresses due to maximum moment and prestress force.
3) Ultimate moment capacity.
4) Final top and bottom flexural stresses, ultimate moment capacity, and ultimate shear capacity, if affected, for members with reduced cross sections due to openings or penetrations.
5) When required on Drawings, a check for no tension in top and bottom of members due to prestress force and member dead load plus superimposed loads indicated on Drawings and in this Specification Section.
6) Column design for maximum axial load and maximum moment.

4. Submit test results, when so required on Drawings, showing that embedded connection items will adequately support the indicated loads.
   a. Connection items to have an ultimate load capacity of at least two times the required indicated load.
5. Concrete mix design(s) including submittal information defined in Specification Section 03 31 30.
6. Fabricator's quality control documentation for special inspections as required by the building code.
7. Copies of source quality control tests.
8. Certification of manufacturer's testing facility qualifications.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Headed studs and deformed bar anchors:
      a. Nelson Stud Welding Div., TRW, Inc.
      b. Or Equal
   2. Bearing pads:
      a. Korolath of New England.
      b. Or Equal.

C. Submit requests for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Embedded Steel Plates and Shapes:
   1. ASTM A36.

B. Bearing Pads:
   1. Under slabs:
      a. Plastic bearing strips.
      b. Minimum compressive strength: 8,000 psi with no fracture at 26,000 psi.
      c. Korolath of New England, Inc., or equal.
   2. For all other locations:
      a. Random, fiber-reinforced elastomeric pads.
      b. Preformed, randomly oriented synthetic fibers set in elastomer.

c. Capable of supporting a compressive stress of 3000 psi with no cracking, splitting, excessive bulging or delaminating in the internal portions of the pad.
d. Size pad to keep a minimum stress 200 psi under minimum dead load.
e. Masticord as manufactured by JVI, Inc., or equal.

C. Cement:
1. Comply with ASTM C150, Type I or III; or ASTM C595, Type IL
2. Type II cement to be used in the following precast units:
a. Precast weir troughs.

D. Aggregates for Normal Weight Concrete:
1. ASTM C33 with coarse aggregate meeting the gradation for Size 67 as stated in ASTM C33.
2. Provide aggregates approved for bridge construction by the State Highway Department in the state where the precast units are fabricated or in the state where the Project is located.
3. All fine aggregate to be natural not manufactured.

E. Aggregates for Lightweight Concrete:
1. ASTM C330 with 3/4 inches maximum size coarse aggregate.
2. All fine aggregate to be natural not manufactured.

F. Water:
1. Potable, clean.
2. Free of oils, acids, and organic matter.

G. Maximum total chloride ion content contributed from all ingredients of concrete including water, aggregates, cement and admixtures measured as a weight percent of cement to not exceed 0.06 for prestressed concrete and 0.10 for all other precast concrete.

H. Prestressing Strands:
1. Either 250K or 270K high tensile strength uncoated seven wire strand.
2. Manufacture and test strands in accordance with ASTM A416.

I. Reinforcing Steel and Welded Wire Reinforcement: See Specification Section 03 21 00.

J. Headed Studs:
1. ASTM A108.
2. Minimum yield strength: 50,000 psi.
3. Minimum tensile strength: 60,000 psi.

K. Deformed Bar Anchors:
1. ASTM A496 or ASTM A1064.
2. Minimum tensile strength: 80,000 psi.
3. Minimum yield strength: 70,000 psi.

L. Electrodes:
1. E70 series conforming to AWS A5.1/A5.1M or AWS A5.5/A5.5M for welding steel shapes and plates.
2. E90 series conforming to AWS A5.5/A5.5M for welding rebar.

M. Concrete sand cement grout in keyways between slabs.
1. See Specification Section 03 31 30.

2.3 DESIGN

A. General Design Requirements:
1. Design units and connections in strict accordance with ACI 318 and the PCI MNL 120.
2. Design units for spans, dead load of members, dead and live loads indicated on the Drawings with concentrated loads placed in their actual locations.
a. Verify weights and locations of concentrated loads.
3. Design units taking into account reduced cross section at openings and penetrations.
4. Provide all reinforcing in units as indicated.

a. Where not indicated, design and provide all reinforcing and prestressing strands subject to approval of the Contracting Officer's Representative.

5. Due to presence of corrosive atmosphere, design prestressed members where indicated on Drawings for no tension in top and bottom of members resulting from loads indicated on Drawings and in this Specification Section.
6. Design connections to allow rotation and/or movement as appropriate to avoid damage to connections, supporting members, joint sealants and other building components.

B. Specific Design Requirements:
1. Refer to Drawings for loadings and hoist supports.

2.4 MIXES

A. See Specification Section 03 31 30.

B. Do not begin fabrication of units until concrete mix design(s) have been approved by the Contracting Officer.

2.5 FABRICATION

A. Do not fabricate units until Shop Drawings have been approved by the Contracting Officer and returned to Contractor and support locations have been field verified by Contractor.

B. Manufacture, quality, dimensional and erection tolerances of all units to be in accordance with both PCI MNL 116 and PCI MNL 120.

C. Cast all members in smooth rigid forms which will provide straight, true members of uniform thickness and uniform color and finish.

D. Use sand cement grout mixture to fill all air pockets and voids, and to repair chipped edges.

E. Finish all repairs smooth and to match adjacent surface texture and color.

F. Where units are to receive concrete topping, provide units having heavy broom finish on top surface for bond.
1. Provide roughness of top surface to provide bond with topping and design for horizontal shear at topping and unit interface in accordance with requirements of ACI 318, Horizontal Shear Strength paragraph.
2. Make all other surfaces smooth.

G. Incorporate embedded plates, angles, and flange welding strips into members at time of manufacture.
1. Provide embedded items as shown on the Drawings unless prior approval is received from Contracting Officer to do otherwise.
2. Provide flange welding strips on all flanged edges of all double tee units as indicated on Drawings.
3. Space strips as shown on Drawings.
4. Cast lifting handles into units at or near support points.
a. Remove lifting handles after units are erected.

H. Cast openings larger than 6 inches SQ or 6 inches diameter in units at time of manufacture.
1. Make smaller openings by neat cutting or neat drilling by trades requiring them.
2. Coordinate sizes and locations of all openings before fabrication of units.

I. Make provisions for support of suspended ceilings, lighting fixtures, ducts, piping, conduits and other suspended work.
1. When drilled expansion bolts or powder-driven fasteners are approved for use, coordinate prestress strand location with prestress concrete member supplier so that drilled expansion bolts or powder-driven fasteners do not hit or are drilled or driven into prestress strands.
2. Install powder-driven fasteners by means of a low velocity powder-actuated tool complying with requirements of OSHA.

a. Assure that the load to be supported by each in place drilled expansion bolt or powder-driven fastener does not exceed the maximum allowable load recommended by the bolt or fastener manufacturer for the concrete strength encountered and for the type, size and embedment length of expansion bolt or driven fastener installed.

J. Automatically weld headed studs and deformed bar anchors to members to provide full penetration weld between studs, bar anchors and members they are attached to.

K. Weld steel shapes and plates per AWS D1.1 and reinforcing steel per AWS D1.4.

L. Minimum concrete compressive strength at time of strand release: 3500 psi.

M. Mark each unit as indicated on the erection plans.
1. Place mark on non-exposed-to-view surface.

N. Coat or finish ends of exposed prestressing strands to prevent rusting.

O. Fabricate the following types of precast and prestressed units (all units to be made with normal weight concrete unless noted otherwise on Drawings):
1. Prestressed hollow core slabs of sizes indicated.
a. Weight of hollow core slabs not to exceed the following:

DEPTH	NORMAL WEIGHT CONCRETE
8 inches	70 psf
12 inches	105 psf

2. Precast items shown on Drawings including but not limited to:
a. Lintels.
b. Sills.
c. Splash blocks.
3. Precast concrete beams and columns as shown on Drawings:
a. Reinforce as indicated.
1) If reinforcement is not indicated, design and provide reinforcement as required to support maximum torsion, shear, moment and axial loads.
b. See the Specific Design Requirements paragraph in the DESIGN Article in PART 2 of this Specification Section
c. Provide beam and column connections as indicated on Drawings and as required to support all loads subject to Contracting Officer's approval.

2.6 SOURCE QUALITY CONTROL

A. During production of precast concrete units, conduct strength tests of concrete placed in units as required in Specification Section 03 05 05 for concrete placed during fabrication.
1. Results of strength tests to be sent immediately to the Contractor and Contracting Officer.
2. Test reports to indicate units they represent.

B. When approved by Contracting Officer, strength tests may be made by precast manufacturer after he has submitted certification that his testing facilities meet the requirements of ASTM E329.

C. Conduct tests on precast concrete using the following procedures:
1. If the precast manufacturer's quality control program requires more frequent or more stringent testing requirements, the manufacturer's quality control program will take precedence over the specific type of test.
a. Precast manufacturer to employ services of an independent testing laboratory to perform concrete testing for manufacturer's production procedures (not listed below) and quality control program.
2. If the precast fabrication plant is not certified by IAS and acceptable to the Building Code Official, Government will employ and pay for precast concrete production special inspection.

a. Coordinate with Owner's special inspector.
    1) Provide minimum 7 calendar days notice prior to the start of fabrication.
    2) Provide minimum 24 hours notice prior to fabrication of any precast members.
3. If precast fabrication plant is certified by IAS and acceptable to the Building Code Official, perform concrete tests as specified in Section 03 05 05. Frequency of tests: Per PCI MNL-116 or PCI MNL-117 as applicable.

PART 3 - EXECUTION

3.1 PREPARATION

A. Verify acceptability and location of supports to receive units.
    1. Check bearing surfaces to determine that they are level and uniform.

B. Verify compressive strengths of concrete and masonry supports.
    1. Do not start erection of units until supports have reached their 28 day required compressive strengths.

3.2 ERECTION

A. Sequence erection to provide a balance of loads across beams and columns.

B. Give consideration to possible lack of stability or capacity of partially completed frame or structure.

C. Contractor to be responsible for guying, shoring, and bracing of frame, walls and individual members as necessary to resist forces due to wind, erection, or any other source that may occur before structure is completed.

D. Use only erection equipment adequate for placing units at lines and elevations indicated on Drawings.
    1. Do not damage units or existing construction during erection.
    2. Erect units using lifting handles cast into the units.

E. Place each leg of all double tees on a 3/8 inches thick bearing pad held 1 inch back from edge of support.
    1. Pad dimensions equal to length of bearing -1 inches x bearing width +2 inches.

F. Place slabs on continuous 1/4 inches thick bearing pad so that width equals bearing length -1 inches.

G. Provide a 1/2 inches thick bearing pad on the top of all precast concrete columns.

H. Pad to cover entire top surface of column except hold pad back 1 inch from face of column all around.

I. Weir Trough:
    1. Anchor weir troughs to supports as indicated on Drawings.
    2. Provide continuous 3/8 inches thick bearing pad under troughs at support.
    3. Hold back pads 1 inch from edge of support.

J. After erection, verify that there is no direct contact between bottom of units and supporting members.
    1. Where direct contact occurs, install additional layers of bearing material to raise units off supports.

K. Lintels:
    1. Length of lintel bearing on supports to be as indicated on Drawings.
        a. If not indicated, minimum length of lintel bearing to be 8 inches.
    2. Fill masonry cells under lintel bearing with masonry grout and reinforce cells as indicated.
    3. Provide minimum 3/8 inches thick full bed joint of masonry mortar between underside of lintel and top surface of grouted masonry for complete lintel bearing length.

L. Weld steel shapes and plates per AWS D1.1 and reinforcing steel per AWS D1.4.

M. Fill all keyways between slabs with concrete sand cement grout.
   1. See Specification Section 03 31 30.

N. After all precast units are erected and all precast unit connections have been made, coat all exposed surfaces of the connections.
   1. See Specification Section 09 96 00.

3.3 FIELD QUALITY CONTROL

A. Testing and Special Inspections: See Section 01 40 00 and Statement of Special Inspections.

B. Causes for rejection of units include, but are not necessarily limited to the following:
   1. Cracked units.
   2. Chipped, broken, or spalled edges.
   3. Units not within allowable casting tolerances.
   4. Voids or air pockets which, in opinion of Contracting Officer's Representative, are too numerous or too large.
   5. Non-uniform finish or appearance.
   6. Low concrete strength.
   7. Improperly placed embedded items and/or openings.
   8. Exposed wire mesh, reinforcing or prestressing strands.

END OF SECTION

DIVISION 04

MASONRY

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SECTION 04 01 20
MASONRY CLEANING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Masonry cleaning.

B. Related Sections include but are not necessarily limited to:
   1. Section 04 22 00 - Concrete Masonry.

1.2 QUALITY ASSURANCE

A. Qualifications:
   1. Use experienced workmen familiar with product and its application.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Manufacturer's application instructions.
      b. Manufacturer's dilution recommendations.
      c. Manufacturer's recommendations on neutralizing rinse.

B. Certifications:
   1. Certification that Contractor is experienced in this type of masonry cleaning.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Cleaning solution, detergent type:
      a. PROSOCO, Inc. – Sure Klean #600 detergent masonry cleaner.
      b. Or Equal.
   2. Cleaning solution for manganese or vanadium-stained masonry:
      a. PROSOCO, Inc. – Vanatrol.
      b. Or Equal.

2.2 MATERIALS

A. Water: Potable.

B. Neutralizing rinse as required by manufacturer.

2.3 MIXES

A. Dilute cleaning solution with potable water at rate which will provide for the weakest solution allowable for cleaning wall.

B. If project conditions require solution of greater than 5 percent acid, obtain permission from Contracting Officer Representative in writing prior to applying solution to wall surface.

PART 3 - EXECUTION

3.1 PREPARATION

A. Allow 28 days after completion of masonry work before start of cleaning.

B. Remove excess mortar using wooden paddles and scrapers.

C. Protect adjacent surfaces not to be cleaned.

3.2 APPLICATION

A. Protect adjacent surfaces subject to potential damage by cleaning solution.

B. Apply masonry cleaner to exposed-to-view masonry surfaces.
   1. Do not use wire brushes.
   2. Use only tools free of rust.
   3. Apply solution using fibered wall-washing brush.

C. Thoroughly rinse and pre-soak walls.

D. Flush all loose mortar and dirt from surface.

E. Wet to prevent "run-off" streaking.

F. Scrape off mortar and reapply cleaning solution.

G. After scrubbing, clean thoroughly with pressurized water.

H. Apply neutralizing rinse as recommended by manufacturer.

END OF SECTION

SECTION 04 05 13
MASONRY MORTAR AND GROUT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Masonry mortar.
   2. Masonry grout.
   3. Integral water repellent admixture.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 04 22 00 - Concrete Masonry.
   3. Section 09 06 00 – Color Schedule.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. C143/C143M, Standard Test Method for Slump of Hydraulic-Cement Concrete.
      b. C144, Standard Specification for Aggregate for Masonry Mortar.
      c. C150/C150M, Standard Specification for Portland Cement.
      d. C207, Standard Specification for Hydrated Lime for Masonry Purposes.
      e. C270, Standard Specification for Mortar for Unit Masonry.
      f. C404, Standard Specification for Aggregates for Masonry Grout.
      g. C476, Standard Specification for Grout for Masonry.
      h. C1019, Standard Test Method for Sampling and Testing Grout.
      i. C1093, Standard Practice for Accreditation of Testing Agencies for Masonry.
      j. C1384, Standard Specification for Admixtures for Masonry Mortars.
   2. The Masonry Society (TMS):
      a. 602, Specification for Masonry Structures.

B. Qualifications:
   1. Preconstruction Testing Laboratory shall be an independent agency qualified in accordance with ASTM C1093 for performing the testing indicated.
      a. Testing Laboratory shall have a minimum of 10 years of experience in the testing of mortar and grout.
      b. Technician conducting tests shall have minimum of five years of experience in the testing of mortar and grout.

C. Mock-Ups:
   1. Provide mortar and grout for mock-up specified in Specification Section 04 22 00.

1.3 DEFINITIONS

A. Coarse grout and fine grout are defined by the aggregate size used in accordance with ASTM C476.

B. Coarse aggregate and fine aggregate are defined in ASTM C404, Table 1.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. General:
         1) Product data for cementitious materials.

2) Source or producer of aggregates and gradation.
3) Integral water repellent manufacturer's dosage rate.

c. Proposed mortar mix design:
d. Proposed masonry grout mix design.

2. Test results:
   a. Preconstruction mortar test results.
   b. Preconstruction masonry grout test results.
3. Colored mortar samples for color approval.
   a. Refer to "Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals" for submission procedures and requirements of Color Samples.
   b. Color cards and plastic simulations are not acceptable.

B. Informational Submittals:
   1. Qualifications of testing lab and technician.
   2. Test results and inspection reports per Specification Section 01 40 00.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Store cementitious materials on elevated platforms, under cover, and in a dry location.
   1. Do not use cementitious materials that have become damp.

B. Store aggregates where grading and other required characteristics can be maintained and contamination avoided.

C. Deliver preblended, dry mixes in moisture-resistant containers.
   1. Store preblended, dry mixes in delivery containers on elevated platforms, under cover, and in a dry location or in a metal dispensing silo with weatherproof cover.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:

C. Masonry Mortar for Factory Shot Blast-Face Masonry:
   a. Ash Grove – Pro Mortar.
   b. Or Equal.

2.2 MATERIALS

A. Cement:
   1. Portland ASTM C150/C150M, Type I or II; or Blended Hydraulic ASTM C595, Type IL.
   2. No air entrainment.
   3. Natural color.
   4. Maximum percent of alkalis: 0.60 in accordance with ASTM C150/C150M, Table 2.

B. Hydrated Lime:
   1. ASTM C207, Type S.
   2. Type SA not acceptable.
   3. Lime substitutes are not acceptable.

C. Mortar Aggregate: ASTM C144, free of gypsum.

D. Grout Aggregate: ASTM C404.

E. Water: Potable.

F. Mortar Pigments:
   1. Commercial colorants suitably compounded for use in mortar mixes.
   2. Do not exceed manufacturer's recommended pigment-to-cement ratios.

G. Integral Water Repellent Admixture:
   1. Liquid polymeric admixture: ASTM C1384.
   2. Verify compatibility with liquid water repellent admixture being used in the fabrication of concrete masonry units.

2.3 MIXES

A. Mortar and grout shall comply with TMS 602 and building code.

B. Type "S" mortar shall be used:
   1. Comply with ASTM C270, Table No. 1, Cement-Lime Mortar.
      a. Do not use masonry cement or mortar cement.
      b. Fly ash shall be limited to 20 percent replacement of the cement content.
   2. Mix materials minimum of three minutes and maximum of five minutes.
   3. Adjust consistency to satisfaction of mason.
   4. Do not use admixtures unless otherwise indicated.
   5. Provide colored mortar for the exterior “Factory Shot Blast-Face Masonry”.
      a. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
   6. Provide integral water repellent admixture in mortar used for:
      a. Exterior concrete masonry work.
      b. Interior concrete masonry work in wet areas.
   7. Do not use integral water repellent admixture in mortar for brick.

C. Masonry Grout:
   1. ASTM C476.
      a. Minimum 28-day compressive strength: 2,000 psi.
      b. Slump: 8 to 11 inches.
   2. Mix 5 minutes minimum.
   3. No admixtures allowed.
   4. At Contractor's option, premixed or preblended grout meeting the above minimum requirements may be used.

2.4 SOURCE QUALITY CONTROL

A. Perform preconstruction laboratory tests on proposed masonry mortar and grout prior to start of masonry work.
   1. Perform tests far enough in advance so that any necessary retesting can be accomplished before masonry construction begins.
      a. Test mortar per ASTM C270.
      b. Test grout per ASTM C1019.

B. Source Limitations for Mortar Materials:
   1. Obtain mortar ingredients of a uniform quality, including color for exposed masonry, from a single manufacturer for each cementitious component and from one source or producer for each aggregate.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions and TMS 602.

B. Mortar:

1. If standard gray mortar begins to stiffen, it may be retempered by adding water and remixing unless prohibited by water repellent admixture manufacturer.
    a. Standard gray mortar shall not be retempered more than one time.
    b. Colored mortar shall not be retempered.
2. All mortar must be used within 2-1/2 hours maximum after initial mixing per TMS 602.
3. Contracting Officer reserves right to alter mix design based on initial rate of absorption of masonry units.
4. Set Prefaced masonry using type S mortar.
    a. Rake mortar from joint as recommended by the unit manufacturer.
    b. Tuckpoint raked joints using pointing grout.
        1) Install pointing grout in accordance with ANSI A108.10 and masonry unit manufacturer's published instructions.
        2) Use polymer modified sanded pointing grout for joints in:
            a) Exterior masonry.
            b) Interior dry areas.
        3) Use epoxy pointing grout for joints in interior areas subject to exposure to corrosive or caustic chemicals.

C. Masonry Grout:
1. Use grout within 1-1/2 hours maximum after initial mixing.
2. Use no grout after it has begun to set.
3. Do not retemper grout after initial mixing.
4. Place grout in lifts not exceeding 4 feet.
5. Use coarse grout in spaces with least dimension over 2 inches.
6. Consolidate all grout while installing.
    a. Consolidate grout pours 12 inches or less in height by mechanical vibration or by puddling.
    b. Consolidate grout pours exceeding 12 inches in height by mechanical vibration and reconsolidate by mechanical vibration after initial water loss and settlement has occurred.

3.2 FIELD QUALITY CONTROL

A. Masonry Mortar and Grout Testing and Inspection:
1. Testing and inspection services will be provided by the Governments special masonry inspector.
    a. Do not include in the bid price the cost of these services.
2. Testing and inspection shall include, but is not limited to:
    a. Observe proportions of site-prepared mortar and grout.
    b. Observe grout space prior to grouting.
    c. Grout compressive strength sampling, testing and reporting per ASTM C1019.
        1) One strength test shall be the average of three specimens from the same sample, tested at 28 days.
    d. Grout slump test sampling, testing, and reporting per ASTM C143/C143M.
    e. Frequency of sampling: One sample (three specimens) collected each grouting operation during masonry construction.
3. Reporting: Special inspector to submit test results and inspection reports per Specification Section 01 40 00.

END OF SECTION

SECTION 04 05 23
MASONRY ACCESSORIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Masonry accessories.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 21 00 – Reinforcement.
   2. Section 04 22 00 – Concrete Masonry.
   3. Section 05 50 00 – Metal Fabrications.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. A480, Standard Specification for General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip,
      b. A580, Standard Specification for Stainless Steel Wire.
      c. A666, Standard Specification for Annealed or Cold-Worked Austenitic Stainless-Steel Sheet, Strip, Plate, and Flat Bar.
      d. A951, Standard Specification for Steel Wire for Masonry Joint Reinforcement.
      e. C920, Standard Specification for Elastomeric Joint Sealants.
      f. D412, Standard Test Method for Vulcanized Rubber and Thermoplastic Elastomers - Tension.
      g. D624, Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers.
      h. D638, Standard Test Method for Tensile Properties of Plastics.
      i. D790, Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials.
      j. D1781, Standard Test Method for Climbing Drum Peel for Adhesives.
      k. D2000, Standard Classification System for Rubber Products in Automotive Applications.
      l. D2240, Standard Test Method for Rubber Property—Durometer Hardness.
      m. E2016, Standard Specification for Industrial Woven Wire Cloth.

B. Mock-Ups:
   1. Provide specified products for inclusion into mock-up panels required by Specification Section 04 22 00.
   2. Coordinate with built-in items and veneer coursing.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Detailed drawings of all factory or field formed stainless steel thru wall flashing.
      d. Tear resistance of flashing material.
      e. Manufacturer's recommendations for flashing adhesive.
      f. Manufacturer's data sheet on each product.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Brick veneer anchoring system:
      a. Heckmann Building Products.
         1) #75 POS-I-TIE.
      b. Or Equal.
   2. Thru wall flashing and stainless-steel drip:
      a. Heckmann Building Products.
      b. Or Equal.
   3. Mortar net protection system:
      a. Mortar Net USA, Ltd.
         1) MortarNet with Insect Barrier.
      b. Or Equal.
   4. Weep vents for cavity wall construction:
      a. Mortar Net USA, Ltd.
         1) Mortar Net Weep Vents.
      b. Or Equal.
   5. Rebar Positioner Masonry Anchors:
      a. Hohmann & Barnard, Inc.
         1) RB Rebar Positioner
      b. Or Equal.
   6. Masonry horizontal joint reinforcing and miscellaneous anchors:
      a. Heckmann Building Products
         1) #1100 Ladder Type Wire Reinforcement.
      b. Or Equal.
   7. Mesh Wall Ties:
      a. Hohmann & Barnard, Inc.
         1) MWT – Mesh Wall Tie
      b. Or Equal.
   8. Preformed rubber control joint inserts:
      a. Hohmann & Barnard, Inc.
         1) RS – Series – Rubber Control Joint
      b. Or Equal.
   9. Reglets:
      a. Hohmann & Barnard, Inc.
         1) For masonry construction: Type #MR - Masonry Reglet.
         2) For concrete construction: Type #CR - Concrete Reglet.
      b. Or Equal.
   10. Grout screen:
      a. Hohmann & Barnard, Inc.
         1) MGS – Mortar/Grout Screen.
      b. Or Equal.

2.2 MANUFACTURED UNITS

A. Veneer Anchorage System for Concrete and Metal Stud Back-up:

1. Barrel Anchors:
    a. One piece barrel screw, 92 percent Zamac 2 Zinc Barrel, steel/EPDM washer, flanged head and eye to receive thermal clip designed to seat barrel directly on structural portion of backup with flanged head and washer sealing the fastener hole.
        1) Screws shall be self-drilling or screwed thru per-drilled holes depending on substrate and per the manufacturer's recommendations.
        2) Lengths shall be determined by thickness of the substrates between concrete masonry unit backup or metal stud backup and the air space.
2. Pintle Wire Ties:
    a. 3/16-inch diameter, type 304 stainless steel.
3. Thermal Clip:
    a. Composite material to provide a thermal break between anchors and pintle
4. Continuous Insulation Washers:
    a. 1-1/2-inch diameter, 18-gauge steel washer with EPDM backing.

B. Thru Wall Flashing and Stainless-Steel Drip:
1. 40 mil EPDM manufactured specifically for thru wall flashing.
    a. Tear resistance: ASTM D624, 150 pound/IN minimum.
    b. Width as necessary.
        1) Provide single piece full width, no horizontal joints will be allowed unless approved in writing by Contracting Officer Representative.
    c. Factory precut wherever possible.
    d. Factory fabricated inside corners, outside corners, and end dams.
2. Stainless steel drip:
    a. ASTM A666, Type 316.
    b. Finish: ASTM A480, 2D.
    c. Minimum 26 GA.
    d. Maximum lengths of 10 feet.
        1) Extend horizontally the full depth of veneer.
    e. Factory fabricated.
    f. Factory fabricated inside and outside corners with a minimum return of 16 inches on each leg.
        1) Weld all joints and grind smooth.
    g. Provide 1/2-inches drip leg on exterior side of wall.
    h. Refer to the Drawings for profile.
3. Lap sealant: Two component polyurethane sealant, complying with ASTM C920, Type M, Grade NS, Class 25 (non-sag).
4. Flashing Adhesive: As recommended by flashing manufacturer for sealing laps, sealing to vertical masonry and concrete surfaces and sealing to stainless steel surfaces.

C. Mortar Net Protection System:
1. 40 percent post – consumer recycled content.
2. Structural strand polymer mesh.
3. Provided with an insect barrier.
4. Minimum 10 inches high by full width of air cavity.
5. Trapezoidal shape for dual levels.

D. Weep Vents:
1. 90 percent open mesh vent designed to be placed in vertical mortar joint.
2. Color: Gray.

E. Rebar Positioner Masonry Anchors:
1. "Z" shaped - 9 gauge wire diameter with ends turned 1-inch.
2. Length is 2-inches less than the concrete block width.
3. 1-1/8-inch inside diameter loops in center of rebar positioner.
4. Stainless Steel: Complying with ASTM A580/A580M - AISI Type 304.

F. Horizontal Joint Reinforcing:

1. General - single wythe wall joint reinforcing: Truss Type.
    a. Conform to ASTM A951.
    b. Stainless steel, ASTM A580, Type 304.
    c. 9 GA side rods.
    d. 9 GA cross rods.
    e. Prefabricated corner and tee sections with minimum length of 30 inches from point of intersection.

G. Mesh Wall Ties:
1. Stainless steel, ASTM E2016, Type 304.
2. 16 gauge, 1/2-inches square mesh.
3. Width: 2-inches less than nominal wall thickness.
4. Length: As necessary to embed minimum 6 inches into each wall.

H. Preformed Rubber Control Joint Inserts:
1. ASTM D2000, M2AA-805.
2. Hardness: ASTM D2240, Shore A Durometer, 80 +/-5.
3. Ultimate elongation: 575 percent, ASTM D412.
4. Tensile strength: 1000 psi, ASTM D412.

I. Reglets:
1. Impact-resistant, rigid polyvinyl chloride tested in conformance with:
    a. ASTM D1781 (Cell Classification)
    b. ASTM D2240 (Hardness Shore D)
    c. ASTM D638 (Tensile Yield & Modulus)
    d. ASTM D790 (Flexural Strength & Modulus)
2. Masonry Reglets: Embedded into mortar joint backup to accept flashing.
    a. Built-in face piece keeps mortar out as construction progresses yet easily tears off prior to installation of flashing.
    b. Reglets shall be made of rigid PVC.
        1) Fits into 3/8-inch mortar joint.
        2) 8-feet long lengths.
        3) 5/8-inch internal depth, 1-1/4-inch depth.
3. Concrete Reglets: Attaches easily to form for poured concrete conditions.
    a. Built-in face piece keeps mortar out as construction progresses yet easily tears off prior to installation of flashing.
4. Reglets shall be made of PVC.
    a. 3/4-inch-deep by 8 feet long.

J. Grout Screen:
1. Polypropylene monofilament tested in conformance with ASTM D5034.
2. 1/4-inch by 1/4-inch mesh.
3. Width of grout screen to be 2-inches less than nominal width of CMU.
4. Micro-thin material with high tensile and burst strength yet will not interfere with mortar bond.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Veneer Anchoring System:
1. Anchor veneer to new construction using barrel type anchoring system.
    a. Self-Drilling Screw: Use a standard drill with a variable clutch adjustment and a chuck adapter designed for installing the barrel type system. Place the barrel end of the screw in the chuck adapter, drill through the substrates into the metal stud. No pre-drilling is required.

b. Concrete Masonry Unit Screw: Use a standard hammer drill with a 3/16-inch masonry drill bit. Set the drill to Hammer mode and drill a 2-inch deep hole into the backup. Using a drill with the chuck adapter designed for installing the barrel type anchors in non-hammer mode, place the barrel end of the screw in the chuck adapter. Drill the screw into the hole.

2. Insert the composite thermal clip tab into the anchor barrel eye from the bottom. Fold over the top tab of the composite thermal clip and squeeze the clip together until a distinct click of the two tabs connecting is heard.
3. Insert the pintle wire ties into the composite thermal clip and adjust to fit into the mortar joint.
   a. The length of the pintle shall be sized so that the front of the tie is held back a minimum of 5/8-inches from the front face of the concrete masonry veneer and extends at least 1-1/2-inches beyond the back face of the concrete masonry veneer.
4. Steel with plastic washers to be slid onto the barrel screw and prior to drilling into the substrates and backup.
5. Provide veneer anchorage at not more than 16 inches on-center vertically and 16 inches on-center horizontally.

C. Thru Wall Flashing and Stainless-Steel Drip:
1. Install to provide positive drainage of cavity moisture.
2. Extend stainless steel drip beyond the exterior face of the wall to minimum distance possible while still allowing drip to perform intended purpose.
3. Extend flashing horizontally beyond each edge of lintel or sills to next vertical mortar joint but not less than 4 inches and turn up edge one full veneer course.
   a. Seal all joints.
4. Where thru wall flashing and stainless steel stainless-steel drip steps up or down in the wall, provide end dam at step.
   a. End dam shall extend up or down to tie into thru wall flashing step.
   b. Seal all joints for continuous watertight barrier.
5. Lap stainless steel drip minimum of 2-inches and bond two pieces together using stainless steel pop rivets and two beads of lap sealant.
6. At concrete masonry unit back-up, install upper edge of flashing a minimum of 6 inches above top of the mortar net protection system and into a block joint 2 inches deep.
7. At metal stud back-up, secure upper edge of flashing with metal termination bar and bead of sealant a minimum of 6-inches above the top of the mortar net protection system.
8. Adhere vertical surface of flashing to back-up wall with adhesive recommended by flashing manufacturer.
9. Extend flashing minimum of 6-inches above top of weep joint mortar protection system.
10. Lap and seal flashing at all inside and outside corners to provide continuous uninterrupted barrier.
11. Use the following if specifying stainless steel thru wall flashing.

D. Mortar Net Protection System:
1. Install directly on thru wall flashing at base of wall, over all wall openings, shelf angles, sills and lintels.
2. Install continuous row(s) of material after the first or second courses of concrete masonry units.
3. Provide multiple thicknesses of material compressed as necessary to completely fill the entire air cavity.
4. Thickness to be at least 10 percent wider than air cavity being filled.
5. Flashing to extend at least 6-inches above the top of mortar net protection system as should any other materials used to fill space between mortar net protection system and inside cavity surface.

E. Weep Vents:

1. Base of wall, over all wall openings, shelf angles, sills and lintels: Provide open weep vents in joints at maximum 16-inches on-center in head joint of first course of veneer immediately above thru wall flashing.
    a. Omit mortar bed on top of thru wall flashing at each open weep joint location to allow moisture an unobstructed path to the exterior.
    b. Set weep vents back away from face of veneer slightly so the front edge of the vent is contained within the mortar joint.
    c. Weep joints shall be not more than 4-inches high.
2. Top of wall: Provide weep vents maximum 16-inches on-center in top of head joint of top course of veneer or as indicated on Drawings.
    a. Set weep vents back away from face of veneer slightly so the front edge of the vent is contained within the mortar joint.
    b. Weep joints shall be not more than 4-inches high.

F. Rebar Positioner Masonry Anchors:
1. Rebar positioners are to assist in positioning rebar in center of block that are to be reinforced with rebar as shown on drawings or as specified in Section 03 21 00. The Z-shaped wire bridges cell of block while the bends rest on the shell of the concrete masonry unit.
2. Structural reinforcing in concrete masonry:
    a. Install reinforcing bars where indicated on Drawings.
    b. Provide means necessary to ensure position of vertical steel reinforcing meets requirements of the building code.
    c. At intersecting load-bearing walls, provide rigid steel anchors 16 inches on-center vertically, embed ends in grout filled cores.
        1) Alternate rigid steel anchors with horizontal joint reinforcing.
    d. At intersecting non-load bearing walls or at intersecting load bearing/non-load bearing walls provide mesh wall ties in mortar joint at 16 inches on-center vertically.
        1) Extend minimum 6-inches into each wall.
        2) Alternate mesh wall ties with horizontal joint reinforcing.
    e. Anchor intersecting concrete masonry to intersecting cast-in-place or precast concrete using dovetail slots and anchors.
        1) Provide dovetail anchors at 16-inches on center or as noted on Drawings.
    f. Repair all galvanized coatings damaged as a result of welding.
3. See Specification Section 05 50 00 for galvanizing repair system.
4. Reinforcing concrete veneer:
    a. Reinforce veneer with joint reinforcement placed in veneer mortar joints:
    b. In concrete masonry back-up construction alternate veneer horizontal joint reinforcing with the concrete back-up horizontal joint anchoring system.

G. Horizontal Reinforcing Masonry:
1. General:
    a. Provide continuous horizontal joint reinforcing in all concrete masonry wall construction.
        1) Embed longitudinal side rods in mortar for entire length with minimum cover of 5/8-inches on exterior side of walls and 1/2-inches at other locations.
            a) For interior partitions, the "exterior" side of the wall is considered the side having the most corrosive atmosphere or the corridor side of the wall.
        2) Lap reinforcement minimum of 12-inches at ends.
        3) Remove cross wires on one side of the lap splice and bend the side rods slightly so the lap is provided with 12-inches of uninterrupted wire lap occurring in the same plane.
        4) Do not bridge control joints with horizontal joint reinforcing.
        5) Do not bridge expansion joints with horizontal joint reinforcing.
        6) At corners and wall intersections use prefabricated "L" and "T" horizontal joint reinforcing sections.

7) Cut and bend as necessary.

b. Install reinforcing at 16-inches on-center vertically unless noted otherwise on Drawings.

c. Install reinforcing 8-inches on-center vertically for a minimum of 24-inches at starter courses.
   1) Do not install horizontal joint reinforcing in veneer mortar joint having through-wall flashing.

d. In concrete masonry back-up construction, install additional horizontal joint reinforcing and adjustable pintle veneer anchors 16 inches on-center in courses on each side of vertical control joints and on each jamb of openings for full height of joint or opening.
   1) Alternate with normal wall horizontal joint reinforcing.
   2) Extend reinforcing minimum 32-inches beyond joint or jambs of opening.

e. In concrete masonry back-up construction, reinforce masonry openings over 12-inches wide with horizontal joint reinforcing and adjustable pintle veneer anchors placed in three horizontal joints above lintel and two horizontal joints below sill.
   1) Extend minimum of 32-inches beyond jambs of opening.

H. Rubber control joints:
   1. Insert rubber control joints vertically in concrete masonry control joints and fill cell solid between masonry units with grout.
   2. Seal interior and exterior joint with backer rod and sealant.

I. Reglets:
   1. Install reglets as walls are being constructed.
   2. Set reglets true with wall, plumb and at consistent depth.
   3. Remove all excess mortar and grout from reglets as walls are being constructed and protect reglet openings from filling with mortar, grout and other construction debris.

J. Grout Screen:
   1. Install in designated areas requiring reinforcing of concrete block that will isolate flow of grout.

END OF SECTION

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SECTION 04 05 50
COLD AND HOT WEATHER MASONRY CONSTRUCTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Cold weather protection.
   2. Hot weather protection.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 04 22 00 – Concrete Masonry.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Brick Industry Association (BIA):
      a. Technical Note 1, Cold and Hot Weather Construction.
   2. Concrete Masonry and Hardscapes Association (CMHA) .
      a. TEK 03-01C, All Weather Concrete Masonry Construction.
   3. The Masonry Society (TMS):
      a. 602, Specification for Masonry Structures.

1.3 DEFINITIONS

A. As defined in TMS 602.

PART 2 - PRODUCTS - (NOT USED)

PART 3 - EXECUTION

3.1 ERECTION AND APPLICATION

A. General:
   1. Comply with CMHA TEK 03-01C and BIA Tech Note 1 recommendations and practices.
   2. Do not use frozen or ice coated materials.
   3. At end of each day or at shutdown, cover tops of all walls not enclosed or sheltered with clear polyethylene minimum 6 mil thick.
      a. Extend down each side of wall minimum of 16 inches and secure.

B. Temporary Facilities:
   1. Construct and maintain temporary protection required to permit continuous and orderly progress of work.
   2. Provide and maintain heat sufficient to assure temperature above 32 degrees F within protected areas.
   3. Remove all temporary facilities after completion of work.

C. Cold Weather Construction and Protection Requirements:
   1. Prior to and during installation:
      a. Air temperature 32 to 40 degrees F: Heat mixing water or aggregate to produce mortar temperatures between 40 and 120 degrees F.
      b. Air temperature 25 to 32 degrees F:
         1) Heat mixing water or aggregate to produce mortar temperatures between 40 and 120 degrees F.
         2) Maintain mortar temperatures above freezing until used.
      c. Air temperature below 25 degrees F:

1) Heat mixing water and aggregate to produce mortar temperatures between 40 and 120 degrees F.
2) Maintain mortar temperatures above freezing until used.
3) Maintain temperature of units until laid at not less than 40 degrees F.
4) Provide heat on both sides of walls under construction to maintain air temperature above freezing.
5) Provide windbreaks or shelters when wind is in excess of 15 mph.
a) Wind breaks or shelters shall be translucent.

2. After installation:
a. Air temperature 32 to 40 degrees F: Protect from rain or snow for not less than 24 hours by covering with weather-resistive translucent membrane.
b. Air temperature 25 to 32 degrees F: Completely cover with translucent weather-resistive membrane for not less than 24 hours.
c. Air temperature 20 to 25 degrees F: Completely protect with insulating blankets for not less than 24 hours or provide other protection approved by Contract Officer Representative.
d. Air temperature below 20 degrees F:
1) Provide enclosed translucent shelters and heating to maintain air temperature on each side of wall above 32 degrees F for 24 hours.
2) Do not allow rapid drop in temperature after removal of heat.
e. Promptly repair all tears, holes, etc., to translucent membrane and shelter using compatible patching material and tape as recommended by membrane manufacturer.

D. Hot Weather Construction and Protection Requirements:
1. Comply with requirements of CMHA, BIA and TMS 602.
2. Storage and preparation of materials.
a. Cover or shade masonry units and mortar materials from direct sun.
b. Maintain sand in a damp loose condition.
1) Sand moisture shall be maintained at minimum 8%.
2) Sprinkle with cool water as required to maintain moisture content.
c. Use cool water for mixing mortars.
d. Avoid using tools and equipment that have been sitting in the sun.
1) Sprinkle mortar boards, mortar pans, wheel barrows, mixers, etc., with cool water.
e. Do not wet concrete masonry units prior to use.
3. Installation:
a. Place masonry units within one minute of the spreading of the mortar.
1) Mortar beds shall not be spread more than 4 feet ahead of the masonry unit being placed.
b. Provide wind screens and shading partitions as required to eliminate direct sunlight exposure.
c. Wet installed units using fog spray of clean water.
d. Cover installed work immediately after installation to slow rate of loss of moisture from units.
e. Fog-spray new masonry work until damp.
1) Repeat fog spraying minimum of three times per day until masonry work has cured for 72 hours.
2) In high humidity conditions, Contract Officer Representative reserves the right to discontinue fog spraying if operation is found to be introducing excessive amounts of moisture into the Work.

END OF SECTION

SECTION 04 22 00
CONCRETE MASONRY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Concrete masonry construction (CMU), including:
      a. Standard concrete masonry.
      b. Pre-colored masonry.
      c. Factory Shot Blast-face masonry.
      d. Cast stone caps.
   2. Integral water repellent admixture.
   3. Penetrating water repellent sealer system. (Referenced as CMU – Concrete Masonry Units Sealed, on Room Finish Schedules, found on Architectural Drawings).
   4. Masonry special inspection.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 21 00 - Reinforcement.
   2. Section 01 33 23 – Submittal Procedures.
   3. Section 03 31 30 - Concrete, Materials and Proportioning.
   4. Section 03 35 00 - Concrete Finishing and Repair of Surface Defects.
   5. Section 04 01 20 - Masonry Cleaning.
   6. Section 04 05 13 - Masonry Mortar and Grout.
   7. Section 04 05 23 - Masonry Accessories.
   8. Section 04 05 50 - Cold and Hot Weather Masonry Construction.
   9. Section 07 14 00 – Fluid Applied Waterproofing.
   10. Section 07 21 00 – Building Insulation.
   11. Section 07 27 46 – Vapor Permeable Air Barrier.
   12. Section 07 46 10 – Cement Board Siding.
   13. Section 07 92 00 - Joint Sealants.
   14. Section 09 06 00 – Color Schedule.

1.2 REFERENCES

A. Terminology:
   1. Terminology indicated below are not defined terms and are not indicated with initial capital letters, but when used in this section have the meaning indicated below:
      a. Terminology used in this Section are in accordance with “Concrete Masonry Unit Shapes, Sizes, Properties, and Specifications” Table 2, CMU-TEC-001-23.

B. Reference Standards:
   1. ASTM International (ASTM):
      a. C33, Standard Specification for Concrete Aggregates.
      b. C55, Standard Specification for Concrete Building Brick.
      c. C90, Standard Specification for Loadbearing Concrete Masonry Units.
      d. C140, Standard Test Methods for Sampling and Testing Concrete Masonry Units and Related Units.
      e. C150, Standard Specification for Portland Cement.
      f. C173, Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method.
      g. C231, Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method.
      h. C260, Standard Specification for Air-Entraining Admixtures for Concrete.

i. C423, Standard Test Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method
j. C426, Standard Test Method for Linear Drying Shrinkage of Concrete Masonry Units.
k. C494, Standard Specification for Chemical Admixtures for Concrete.
l. C595, Standard Specification for Blended Hydraulic Cements.
m. C618, Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete.
n. C979, Standard Specification for Pigments for Integrally Colored Concrete.
o. C989, Standard Specification for Slag Cement for Use in Concrete and Mortars.
p. C1194, Standard Test Method for Compressive Strength of Architectural Cast Stone.
q. C1195, Standard Test Method for Absorption of Architectural Cast Stone.
r. C1357, Standard Test Methods for Evaluating Masonry Bond Strength.
s. C1364, Standard Specification for Architectural Cast Stone.
t. E514, Standard Test Method for Water Penetration and Leakage Through Masonry.

2. Concrete Masonry and Hardscapes Association:
   a. CMU-TEC-001-23, Concrete Masonry Unit Shapes, Sizes, Properties, and Specifications.
   b. TEK 03-4C, Bracing Concrete Masonry Walls During Construction.
   c. TEK 08-02A, Removal of Stains from Concrete Masonry.
   d. TEK 08-03A, Control and Removal of Efflorescence.
3. The Masonry Society (TMS):
   a. 602, Specification for Masonry Structures.

1.3 QUALITY ASSURANCE

A. Regulatory Requirements:
   1. Applicable construction codes including building code are indicated in Section 01 42 00 - References.

B. Qualifications:
   1. Concrete masonry unit manufacturer shall be licensed or qualified, in writing, by manufacturer of integral water repellent admixture to produce masonry units containing manufacturer's admixture.
      a. Concrete masonry unit manufacturer shall have not less than five years' experience manufacturing masonry units containing admixture manufacturer's products.

1.4 MOCK-UPS

A. Mock-ups Requirements:
   1. Prior to performing concrete masonry Work, provide mockup at the Site of concrete masonry in conjunction with the following sections:
      a. Section 04 22 00 - Concrete Masonry.
      b. Section 07 14 00 – Fluid Applied Waterproofing.
      c. Section 07 21 00 – Building Insulation.
      d. Section 07 27 46 – Vapor Permeable Air Barrier.
      e. Section 07 46 10 – Cement Board Siding.
      f. Section 07 92 00 – Joint Sealants.
   2. Mockups – General: Prior to permanent wall construction, construct mockup.
      a. Construct mockup on a concrete slab as necessary to demonstrate construction details.
         1) Minimum slab thickness: four inches.
      b. Mockup shall show full color range, texture and bond pattern(s) of each type of masonry required.
      c. Size: As large as necessary to properly display all conditions required by masonry construction.
         1) Not less than six feet high by eight feet long.
            a) Return corners and intersections not less than four feet.
         2) Mockup shall demonstrate:
            a) Outside corner condition.

b) Inside corner condition.
c) Jamb condition demonstrating lintel bearing and flashing.
d) Masonry control joint with sealant.

d. Include all special corners and other special CMU detailing shown on the Drawings.
e. Include all types of masonry shown on the Drawings, including:
1) Pre-colored masonry.
2) Factory Shot Blast-face masonry.
3) Cast stone caps.
f. Mockup shall include:
1) Each type of masonry required for the Work.
a) Each type of special shape.
b) Each type of back-up wall system(s).
2) Vertical wall reinforcing with grouted cell.
3) Typical bond beam construction.
4) Typical lintel construction.
5) Positioning, securing and lapping of reinforcing steel.
6) Masonry accessories:
a) Horizontal joint reinforcing.
(1) Positioning and lapping of joint reinforcing.
b) Veneer anchorage system(s).
c) Thru wall flashing and drip edge.
(1) Demonstrate inside and outside corner conditions showing thru wall flashing lapping, jointing and sealing.
d) Weep joint mortar protection system.
e) Weep joints and weep vents.
f) Typical control joint construction.
g) Mesh wall ties.
h) Rigid steel masonry anchors.
7) Pre-Finished Fiber-Cement Board Lap Siding.
8) Insulation.
9) Cleaning of masonry work.

3. Step construction of mockup to allow observation of components.
4. Following acceptance of mockup by Contracting Officer Representative, shall constitute minimum standard of quality for the Work.
a. Maintain and safeguard mockup until Substantial Completion.
5. If not acceptable as determined by Contracting Officer Representative, provide additional mockups as necessary.
6. Remove mockups when directed by Contracting Officer Representative.

1.5 SUBMITTALS

A. Action Submittals: Submit the following:
1. Shop Drawings:
a. Scaled (minimum 1/8 inches per foot) plans showing proposed locations of masonry control joints.
b. Wall elevations and sections, indicating special shapes, shape part numbers, applicable dimensions.
c. Reinforcing number, sizes, spacing, dimensions, configurations, locations, mark numbers, lap splice lengths and locations, masonry cover, and reinforcing supports.
1) Reference Section 03 21 00 for additional requirements.
d. Detail drawings for:
1) Precast concrete lintels.
a) Show profiles, cross-sections, reinforcement and steel components.
2. Product Data:
a. Manufacturer's information on aggregate and cement type used in manufacture.
b. Data sheet on each type of masonry unit required, including:

1) Pre-colored masonry.
2) Factory Shot Blast-face masonry.

c. Data sheets on integral water repellent admixture being used in masonry unit manufacturing.

d. Technical bulletins on cleaning masonry containing integral water repellent.

3. Samples:

a. Concrete Masonry Finish Samples:

1) Refer to "Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals" for submission procedures and requirements of Color Samples.

2) Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.

B. Informational Submittals: Submit the following:

1. Certifications:

a. Certification that concrete masonry units meet or exceed requirements of standards referenced.

b. Certification that fire-resistive rated units meet the requirements of applicable building code.

c. Certification that integral water repellent admixture will not affect the use of coloring processes or alter actual colors of factory colored masonry units.

d. Certification of integral water repellent admixture dosage rates from concrete masonry unit Supplier.

e. Concrete masonry Supplier shall certify that integral liquid water repellent admixture was furnished at dosage rate recommended by admixture manufacturer for use in exterior (outdoor) wall construction.

2. Supplier Instructions:

a. Instructions for handling, storing, and installation.

3. Source Quality Control Submittals:

a. Results of tests, inspections, and other quality control activities required by the Contract Documents and performed at the place of production or fabrication.

4. Field Quality Control Submittals:

a. Results of tests, inspections, and other quality control activities required by the Contract Documents and performed at the Site.

5. Qualifications:

a. Supplier of masonry units when requested by Contracting Officer Representative.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Delivery and Acceptance Requirements:

1. Comply with Section 01 65 50 - Material Transport Helicopters.
2. Inspect units upon delivery, to verify color match with mockup and approved samples, dimensional quality, and trueness of unit.

B. Storage and Handling Requirements:

1. Comply with Section 01 67 00 - Product Requirements.
2. Covering material shall be weather-proof but vapor permeable to prevent accumulation of moisture under cover.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Standard masonry units:
      a. Any manufacturer capable of meeting the requirements of this Specification Section.
   2. Factory Shot Blast-face masonry units:
      a. Block-Lite Co. Inc.
      b. Or Equal.
   3. Cast Stone Caps:
      a. Artisan Stone.
      b. Or Equal.
   4. Integral water repellent admixture:
      a. GCP Applied Technologies, Inc.
      b. Or Equal.
   5. Penetrating Water Repellent Sealer System:
      a. Tenemec – "Prime-A-Pell Plus – Series 662.

2.2 MATERIALS

A. Cement: Type I or II Portland, ASTM C150.

B. Aggregate: ASTM C33.

C. Reinforcing Bars: Refer to Section 03 21 00.

D. Mortar: Refer to Section 04 05 13.

E. Masonry Grout: Refer to Section 04 05 13.

F. Masonry Accessories: Refer to Section 04 05 23.

G. Insulation: Refer to Section 07 21 00.

H. Sealants: Refer to Section 07 92 00.

I. Integral Concrete Masonry Water Repellent:
   1. Liquid polymeric admixture.

J. Penetrating Water Repellent Sealer System: (Referenced as CMU – Concrete Masonry Units Sealed, on Room Finish Schedules, found on Architectural Drawings).
   1. Modified Siloxane/Silane with Diffused Quartz Carbide.
      a. Clear, filmless, penetrating water repellent for block masonry.
      b. Sealer system shall be chloride ion intrusion, stain damage, freeze/thaw spalling, efflorescence and rust damage.
      c. The solution penetrates the masonry block substrate and chemically reacts to create a powerful barrier against water penetration. The barrier is resistant to ultraviolet and weather deterioration.

2.3 MANUFACTURED UNITS

A. General:
   1. Masonry units of each type, color, or face style shall be from a single production run by Supplier.
   2. Factory fabricate special shapes unless otherwise required.
   3. For the purposes of manufacturing concrete masonry units using integral water repellent admixture, all concrete masonry units are considered to be installed in an exterior (outdoor) environment.
   4. Fire resistive units: Fabricate to comply with applicable building code.
   5. Fabricated in manufacturing facility.
   6. Provide square corners unless required otherwise.

B. Concrete Masonry Units:
   1. Modular units: ASTM C90.

a. Normal weight units: Minimum of 125 pound/cubic foot.
b. Interior units: Standard gray.
c. Light weight or medium weight units are not acceptable.

2. Concrete bricks:
   a. Structural units: ASTM C55.
      1) Same material, texture, color and density as modular units.
3. Exterior Architectural Veneer Units: ASTM C1634, Normal weight.
   a. Texture:
      1) Factory Shot Blast-Face.
   b. Color:
      1) Exposed exterior units: Pre-colored. Refer to Specification Section 09 06 00 for color selection.
   c. Design compressive strength: f' m=2,000PSI minimum.
      1) Determine in accordance with TMS 602.
      2) Unit strength method, sampled and tested in accordance with ASTM C140.
4. Provide masonry units manufactured with integral water repellent admixture for the following exposures:
   a. Exterior “Factory Shot Blast-Face” veneer.
   b. Interior areas defined as wet and/or corrosive.
      1) See Specification Section 07 92 00 for definition of wet and/or corrosive areas.
5. Special shapes and faces:
   a. Corner units.
      1) Corner units used in veneer wythe shall have a finished return leg one-half the length of a standard modular stretcher unit.
      2) Corner units shall maintain regular modular masonry coursing.
   b. Finished end units.
   c. Other special shapes as indicated on Drawings or necessary to maintain coursing.
6. Comply with current version ASTM C1364.
7. Physical properties: Provide the following:
   a. Compressive Strength - ASTM C1194: 6,500 pounds per square inch minimum at 28 days.
   b. Absorption – ASTM C1195: 6.0 pecent maximum at 28 days.
   c. Air Content – Provide sufficient air content to meet the freeze-thaw requirements for wet cast products, when the air content is tested in accordance with Test Method C173/C173M or Test Method C231/C231M. Air entrainment is not required for Vibrant Dry Tamp (VDT) products.
   d. Freeze-thaw – ASTM C666/C666M in accordance with ASTM C1364. The CPWL shall be less than 5.0 percent after 300 cycles of freezing and thawing.
   e. Linear Drying Shrinkage – ASTM C426: Test and report in accordance with ASTM C1364.
8. Raw Materials:
   a. Portland cement – Type I or Type III, white and/or grey, ASTM C150 or ASTM C595 Blended Hydraulic Cement (Type 1L).
   b. Coarse aggregates - Granite, quartz or limestone, ASTM C33, except for gradation, and are optional for the Vibrant Dry Tamp (VDT) casting method.
   c. Fine aggregates - Manufactured or natural sands, ASTM C33, except for gradation.
   d. Colors - Inorganic iron oxide pigments, ASTM C979 except that carbon black pigments shall not be used.
   e. Admixtures- Comply with the following:
      1) ASTM C260 for air-entraining admixtures.
      2) ASTM C494/C495M Types A - G for water reducing, retarding, accelerating and high range admixtures.
      3) Other admixtures: Integral water repellents and other chemicals, for which no ASTM Standard exists, shall be previously established as suitable for use in concrete by proven field performance or through laboratory testing.

4) ASTM C618 mineral admixtures of dark and variable colors shall not be used in surfaces intended to be exposed to view.
5) ASTM C989 granulated blast furnace slag may be used to improve physical properties. Tests are required to verify these feature.

9. Water – Potable
10. Curing:
    a. Cure units in a warm curing chamber approximately 100-degree F at 95 percent relative humidity for approximately 12 hours or cure in a 95 percent moist environment at a minimum 70-degree F for 16 hours after casting. Additional yard curing at 95 percent relative humidity shall be 350 degree-days (i.e. 7 days 50-degree F or 5 days 70-degree F prior to shipping. Form cured units shall be protected from moisture evaporation with curing blankets or curing compounds after casting.
11. Manufacturing Tolerances:
    a. Minimum Thickness shall be 2.5-inches to facilitate testing for compressive strength and absorption as specified in ASTM C-1364 Standard Specification for Architectural Cast Stone.
    b. Cross section dimensions shall not deviate by more than ±1/8-inch from approved dimensions.
    c. Length of units shall not deviate by more than length/ 360 or ±1/8-inch., whichever is greater, not to exceed ±1/4-inches.
    d. Maximum length of any unit shall not exceed 15 times the average thickness of such unit unless otherwise agreed by the manufacturer.
    e. Warp, bow or twist of units shall not exceed length/ 360 or ±1/8-inch, whichever is greater.
    f. Location of dowel holes, anchor slots, flashing grooves, false joints and similar features – On formed sides of unit, 1/8-inch., on unformed sides of unit, 3/8-inch. maximum deviation.
12. Delivery, Storage And Handling:
    a. Mark production units with the identification marks as shown on the shop drawings.
    b. Package units and protect them from staining or damage during shipping and storage.
13. Color and Finish:
    a. Exposed Front and Side Edges:
        1) Chiseled Faced.
    b. Color:
        1) See Section 09 06 00 – Color Schedule.

2.4 PERFORMANCE AND DESIGN REQUIREMENTS:

A. Integral Concrete Masonry Water Repellent:
1. Water permeance of masonry: Capable of achieving a Class E Rating when evaluated using ASTM E514 with the test extended to 72 hours, using the rating criteria specified in ASTM E514.
2. Flexural bond strength of masonry: An increase of 10%, minimum, in masonry flexural bond strength shall occur as a result of adding integral water-repellent concrete masonry and mortar admixtures when compared to a control (containing no admixtures) concrete masonry and mortar tested in accordance with ASTM C1357.
3. Compressive strength validation shall be per unit strength method.
4. Drying shrinkage of masonry: Maximum 5% increase in drying shrinkage of the concrete masonry units shall occur as a result of adding integral water repellent concrete masonry admixture when compared to a control (containing no admixtures) concrete masonry when tested in accordance with ASTM C426.
5. Grout shear bond strength: Maximum 5% decrease in grout shear bond strength shall occur as a result of adding integral water repellent admixture to the concrete masonry units when compared to a control (containing no admixtures).

PART 3 - EXECUTION

3.1 PREPARATION

A. Verify that anchors and flashings are correct.

B. Lay out walls in advance for uniform and accurate spacing of bond patterns and joints.
   1. Properly locate openings, movement type joints, returns, and offsets weep joints and weep vents.

3.2 MASONRY INSTALLATION

A. General:
   1. Build in flashing, reinforcing, and related accessory items.
      a. See Specification Section 04 05 23 for installation of accessory items.
   2. Perform all cutting using masonry saw blades with power tools that produce straight and true edges.
   3. Drill holes using masonry drill bits or core drill.
      a. Holes made by chipping unit will not be accepted.
   4. Install field units in running bond, unless noted otherwise.
      a. Provide special coursing where indicated on the Drawings.
   5. Cut as required to maintain bond pattern.
   6. Use solid units where cutting or laying would expose holes and as noted on Drawings.
   7. Avoid use of less than half size units, whenever possible.
   8. Do not use chipped, cracked, spalled, stained or imperfect units exposed in finish work.
   9. Provide units of uniform color, within the range demonstrated on the approved mock-up.
   10. Do not wet concrete masonry units.
   11. Build chases and recesses as indicated and required for work of other trades.
      a. Provide not less than 8 inches of masonry between chase or recess and jamb of openings, and between adjacent chases and recesses unless detailed otherwise on the Drawings.
   12. In fire-resistive rated wall construction, install fire resistive units in accordance with the building code.

B. Concrete Masonry Units:
   1. Grout solid all cells containing steel reinforcing and as indicated on Drawings.
      a. Refer to Specification Section 04 05 13 for grouting.

C. Laying and Tooling:
   1. Lay masonry units with completely filled bed and head joints.
      a. Provide full mortar bed on all block cross webs and completely fill head joints.
         1) Do not slush head joints.
         2) Protect cells requiring grout fill from mortar droppings.
         3) Omit mortar from head joint at weep joint opening.
   2. Maintain nominal 3/8-inches joint widths.
      a. Cut joints flush where concealed.
      b. Tool exposed joints concave.
      c. Compress mortar in below ground joints and in joints concealed by insulation in cavity wall construction.
      d. Provide wider joints where noted on Drawings.
         1) In no case shall any mortar joint be more than 3/4 inches wide.
      e. Where masonry sits on top of steel support omit the mortar joint on top of the support and sit masonry directly on top of the thru wall flashing or the steel support member unless a mortar joint is required to maintain coursing.
   3. During tooling of joints, enlarge any voids or holes except weeps, and completely fill with mortar.
   4. Point-up all joints at corners, openings, and adjacent work to provide neat, uniform appearance.
   5. Remove masonry disturbed after laying.

a. Clean and relay in fresh mortar.
b. Do not pound units to fit.
c. If adjustments are required, remove units, clean, and reset in fresh mortar.

6. Where work is stopped and later resumed, rack back 1/2 masonry unit length in each course.
   a. Remove loose units and mortar prior to laying fresh masonry.
7. As work progresses, build in items indicated on Drawings and specified.
   a. Fill in solidly with mortar around built-in items.
   b. Where built-in items are to be embedded in cores of hollow masonry units, place grout screen in joint below and fill core solid with mortar.

D. Control Joints and Sealants:
1. Provide vertical expansion, control and isolation joints where indicated on Drawings.
2. Where not indicated on Drawings, submit proposed control joint locations in accordance with the following requirements:
   a. Provide control joints at maximum 24 feet OC.
   b. Provide at all T intersections.
   c. Locate joints so as to allow lintels and bond beams above and below openings to extend beyond the opening as indicated on the Drawings without control joints thru the lintel or bond beam.
   d. Provide preformed rubber control joint inserts in accordance with Specification Section 04 05 23.
3. Rake out mortar in joint.
4. Refer to Specification Section 07 92 00 for sealant installation requirements.
   a. Seal control and expansion joints.

E. Tolerances:
1. Maximum variation from plumb in vertical lines and surfaces of columns, walls, and arises:
   a. 1/4 inches in 10 feet.
   b. 3/8 inches in a story height not to exceed 20 feet.
   c. 1/2 inches in 40 feet or more.
2. Maximum variation from plumb for external corners, expansion joints, and other conspicuous lines:
   a. 1/4 inches in any story or 20 feet maximum.
   b. 1/2 inches in 40 feet or more.
3. Maximum variation from level of grades for exposed lintels, sills, parapets, horizontal grooves, and other conspicuous lines:
   a. 1/4 inches in any bay or 20 feet.
   b. 1/2 inches in 40 feet or more.
4. Maximum variation from plan location of related portions of columns, walls, and partitions:
   a. 1/2 inches in any bay or 20 feet.
   b. 3/4 inches in 40 feet or more.
5. Maximum variation in cross-sectional dimensions of columns and thicknesses of walls from dimensions shown on Drawings:
   a. Minus 1/4 inches.
   b. Plus 1/2 inches.
6. Maximum variation in mortar joint width:
   a. Bed joints: 3/32 inches in 10 feet.
   b. Head joints:
      1) Minus 1/8 inches.
      2) Plus 1/8 inches.

F. Protect against weather when work is not in progress.
1. During inclement weather conditions, cover top of walls with translucent waterproof membrane.
2. See Specification Section 04 05 50.

G. Protect against cold/hot weather as specified in Specification Section 04 05 50.

3.3 CAST STONE CAP INSTALLATION

A. Examination:
   1. Installing contractor shall check Cast Stone materials for fit and finish prior to installation.
   2. Unacceptable units shall not be set.

B. Setting Tolerances:
   1. Comply with Cast Stone Institute Technical Manual.
   2. Set stones 1/8-inch. or less, within the plane of adjacent units.
   3. Joints, plus - 1/16- inch., minus - 1/8-inch.

C. Jointing:
   1. Joint size: 3/8-inch.

D. Setting:
   1. Wet units with clean water prior to setting.
   2. Fill dowel holes and anchor slots completely with mortar or non-shrink grout.
   3. Set units in full bed of mortar, unless otherwise detailed.
   4. Remove excess mortar from unit faces immediately after setting.
   5. Tuck point unit joints to a slight concave profile.

E. Repair and Cleaning:
   1. Repair chips with touchup materials furnished by manufacturer.
   2. Saturate units to be cleaned prior to applying an approved masonry cleaner.
   3. Consult with manufacturer for appropriate cleaners.

3.4 PENETRATING WATER REPELLENT SEALER SYSTEM INSTALLATION

A. Penetrating Water Repellent Sealer System Installation: (Referenced as CMU – Concrete Masonry Units Sealed, on Room Finish Schedules, found on Architectural Drawings).
   1. Install and provide coverage in accordance with manufacturer's written instructions and coverage rates.

3.5 FIELD QUALITY CONTROL

A. Bracing Concrete Masonry Walls During Construction:
   1. At a minimum, provide bracing in accordance with CMHA TEK 03-04C.
   2. Contractor is responsible for adequately bracing all masonry during construction.

B. Remove and replace loose, stained, damaged and other unacceptable units as directed by Contracting Officer Representative.
   1. Provide new units to match.
   2. Install in fresh mortar.
   3. Point to eliminate evidence of replacement.

C. Special Masonry Inspection:
   1. Masonry inspection services will be provided during the following construction activities:
      a. Cost of masonry inspection services will be paid by Government.
      b. Comply with Statement of Structural Special Inspections 01 40 00.A
      c. During laying of units:
         1) During the first day of the masonry construction, inspect proportions of site prepared mortar, construction of mortar joints, location of all reinforcing and connectors, size and location of structural elements, type, size and location of anchors, protection of masonry during cold weather.
         2) Inspection to be continuous the first full day of masonry construction which requires special inspection.
            a) Thereafter, a minimum of 3 hours every third day of construction until the concrete masonry work is complete.
         3) Inspection while laying masonry units may be made concurrently with other inspection duties provided all inspection duties are adequately performed.

4) When deficiencies are found, additional inspection shall be provided as required until deficiencies have been corrected.
5) If masonry crews change, an additional full day of inspection is required during the first day the new crew is on-site.

d. Placement of reinforcing steel:
1) Verification of all reinforcing including size, grade, lap lengths, and type.
2) Inspection may be periodic as required to verify all reinforcing.
3) Inspector to be present during the concrete pour in which any dowels connecting concrete to masonry are cast.
a) Inspector to verify proper location of dowels.

e. Prior to each grouting operation, verify that grout space is clean, reinforcing is clean, and connectors are properly placed, proportions of site-prepared grout are correct and mortar joints have been properly constructed.
1) Inspection may be periodic as required to verify proper grout space.

f. Verify compliance with building code and Specifications continuously during all grouting operations.

g. Provide special inspection in accordance with and including observation of masonry work for conformance to the Contract Documents:
1) Provide inspection reports to the Contracting Officer Representative, Building Official and Government.
a) Notify Contracting Officer Representative, Building Official and Government, in writing, when discrepancies have been satisfactorily corrected.
2) Submit final signed report stating that work requiring special inspection was, to the best of the inspector's knowledge, in conformance to the Contract Documents and the applicable workmanship previsions of the building code.

3.6 CLEANING

A. Clean concrete masonry as the wall is being constructed using fiber brushes, wooden paddles and scrapers.
1. Do not use metal tools or wire brushes.
2. No acid-based cleaning solutions shall be used unless approved in writing by Contracting Officer Representative.

B. Remove dirt and stains in accordance CMHA TEK 08-02A.

C. Remove primary efflorescence in accordance with CMHA TEK 08-03A.

D. After wall construction has been completed, completely clean wall using detergent recommended by factory shot blast-face unit manufacturer.
1. See Specification Section 04 01 20.

END OF SECTION

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DIVISION 05

METALS

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SECTION 05 12 00
STRUCTURAL STEEL

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Structural steel, including the fabrication and erection of support and bracing members, including connections.
   2. Connection detail design as required and noted on the drawings.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 15 19 - Anchorage to Concrete.
   2. Section 09 96 00 - High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Institute of Steel Construction (AISC):
      a. 303, Code of Standard Practice for Steel Buildings and Bridges.
      b. 360, Specifications for Structural Steel Buildings.
      c. Quality Certification Program for Fabricators.
   2. American Society of Mechanical Engineers (ASME):
      a. B18.21.1, Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series).
   3. ASTM International (ASTM):
      a. A2, Standard Specification for Carbon Steel Girder Rails of Plain, Grooved, and Guard Types.
      b. A6/A6M, Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling.
      c. A36/A36M, Standard Specification for Carbon Structural Steel.
      d. A53/A53M, Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
      e. A108, Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished.
      f. A123/A123M, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
      g. A153/A153M, Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
      h. A307, Standard Specification for Carbon Steel Bolts and Studs, 60,000 psi Tensile Strength.
      i. A500/A500M, Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes.
      j. A563, Standard Specification for Carbon and Alloy Steel Nuts.
      k. A992/A992M, Standard Specification for Structural Steel Shapes.
      l. A1064/A1064M, Standard Specification for Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete.
      m. F436, Standard Specification for Hardened Steel Washers.
      n. F959, Standard Specification for Compressible-Washer-Type Direct Tension Indicators for Use with Structural Fasteners.
      o. F1554, Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength.
      p. F3125, Standard Specification for High Strength Structural Bolts, Steel and Alloy Steel, Heat Treated, 120 ksi (830 MPa) and 150 ksi (1040 MPa) Minimum Tensile Strength, Inch and Metric Dimensions.
   4. American Welding Society (AWS):

a. A5.1/A5.1M, Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding.
b. A5.5/A5.5M, Specification for Low-Alloy Steel Electrodes for Shielded Metal Arc Welding.
c. A5.17/A5.17M, Specification for Carbon Steel Electrodes and Fluxes for Submerged Arc Welding.
d. A5.18/A5.18M, Specification for Carbon Steel Electrodes and Rods for Gas Shielded Arc Welding.
e. A5.20/A5.20M, Specification for Carbon Steel Electrodes for Flux Cored Arc Welding.
f. A5.23/A5.23M, Specification for Low-Alloy Steel Electrodes and Fluxes for Submerged Arc Welding.
g. A5.28/A5.28M, Specification for Low-Alloy Steel Electrodes and Rods for Gas Shielded Arc Welding.
h. A5.29/A5.29M, Specification for Low-Alloy Steel Electrodes for Flux Cored Arc Welding.
i. D1.1/D1.1M, Structural Welding Code - Steel.
1) Steel stud connectors and their installation to comply with requirements of AWS D1.1/D1.1M.

5. National Institute of Steel Detailing (NISD).
6. Research Council on Structural Connections (RCSC):
a. Specification for Structural Joints Using High-Strength Bolts.

B. Qualifications:
1. Steel fabricator:
a. Minimum of 10 years of experience in fabrication of structural steel or participate in the AISC Certification program and is designated an AISC Certified Plant, Category at time of bid.
b. Fabricator plant quality control and inspection program: Meet requirements of the building code and/or be an Approved Fabricator.
c. Plants that are not an Approved Fabricator may be acceptable, provided:
1) Plant meets all remaining qualifications.
2) Contractor reimburses the Government the cost of required Special Inspection services.

2. Steel erector:
a. Minimum of 10 years of experience in erection of structural steel similar in the scope of this project or certified as CSE under the AISC Quality Certification Program.
b. With an active and enforced quality assurance program in place, as described in the applicable Codes.
3. Qualify welding procedures and welding operators in accordance with AWS.

1.3 DEFINITIONS

A. Government: May mean the Owner's Designated Representative for Construction as defined by the AISC 303 or the Contracting Officer.

B. Galvanizing: Hot-dipped galvanizing per ASTM A153/A153M and/or ASTM A123/A123M with minimum coating of 2.0 oz of zinc per square foot of metal (average of specimens) unless noted otherwise or dictated by aforementioned standards.

C. Approved Fabricator: Approved by the Building Official to perform the building code required Special Inspections.

1.4 SUBMITTALS

A. Shop Drawings:
1. Product technical data including:
a. Acknowledgement that products submitted meet requirements of standards referenced.
b. Manufacturer's installation instructions.

c. Detailed supplemental specification relating to load indicator washers or high-strength bolts.
    1) Alternate design for Contracting Offcer approval (submitted at Contractor's option if desired by Contractor for use).
d. Source and certification of quality for high-strength bolts, nuts and washers.

2. Fabrication and/or layout drawings:
    a. Prepare Shop Drawings under NISD Quality Procedures Program certification.
    b. Complete Shop Drawings for all of the work showing clearly all pieces, sizes, dimensions, details, connections materials and shop coatings.
        1) All Shop Drawings must be checked and signed "approved" before submittal.
        2) Show all cuts, copes, and holes.
        3) Indicate all shop and field bolts.
        4) Indicate all shop and field welds using AWS symbols.
    c. Prepare complete erection drawings showing the location and marks of all pieces.
        1) Copies of up-to-date erection drawings shall accompany the Shop Drawings.
        2) Use match marks on the erection drawings to indicate the sheet number on which each particular member is detailed.
    d. Correct any incorrect or unacceptable material or fabrication due to incorrect detailing, shop work, or erection, without additional charge.

3. Certifications:
    a. Certificates of compliance with standards specified for all major components and fasteners incorporated into work.
    b. Copies of current welding certificates for each welder assigned to perform welding indicating compliance with testing specified by AWS.
    c. Welder qualification data and prequalified procedures.
    d. Special Inspections reports.
    e. Source Quality Control Documentation, including certificate of compliance stating that the work performed in the fabrication shop was done in accordance with the approved construction documents.
        1) Certification is required only if the fabricator is fabricator approved by the Building Official.

4. Test reports:
    a. Certified copies of mill tests.
    b. Manufacturer's load test and temperature sensitivity data for post-installed anchor bolts.
    c. Test reports for all structural steel work.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Handle and store steel members above ground on skids or other supports.
    1. Keep free of dirt and other foreign material and protect against corrosion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
    1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
    1. High-strength bolts:
        a. Portland Bolt and Manufacturing Company.

b. Or Equal.
2. Load indicator washers for high-strength bolts:
   a. Portland Bolt and Manufacturing Company.
   b. Or Equal.
3. Alternate design high-strength bolts:
   a. LeJuene Bolt Co.
   b. Or Equal.
4. Headed studs and deformed bar anchors:
   a. Nelson Stud Welding Division, TRW, Inc.
   b. Or Equal.
5. Mechanical anchor bolts:
   a. See Section 03 15 19.
6. Adhesive anchors bolts:
   a. See Section 03 15 19.
7. Anchor bolt sleeves:
   a. Sinco/Wilson.
   b. Or Equal.

C. Submit requests for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Steel, Structural Shapes and Plate (unless noted otherwise on Drawings):
1. All W-shapes and WT-shapes: ASTM A992/A992M.
2. All other plates, bars and rolled shapes: ASTM A36/A36M.

B. Pipe: ASTM A53/A53M, Grade B (Type E or S) (Fy=35).

C. Hollow Structural Sections (HSS):
1. Round: ASTM A500/A500M, Grade B (Fy=42).
2. Square or rectangular: ASTM A500/A500M, Grade C (Fy=50).

D. High-Strength Bolts, Nuts and Washers:
1. ASTM F3125, Grade A325, Grade A325TC, Grade F1852 (tension controlled bolts where noted on the Contract Documents) with ASTM A563 nuts:
2. High-strength bolts:
   a. Provide two ASTM F436 washers for all bolts.
   b. Provide beveled washers at connections of sloped/tapered sections.
3. High-strength bolts with compressible washer type direct tension indicators (DTI), ASTM F959.
   a. Provide at Contractor's option and subject to approval of Contracting Officer.

E. Alternate high-strength design: Provide at Contractor's option and subject to approval of Contracting Officer. Bolts, Non-high Strength: ASTM A307, Grade A.

F. Threaded Rod: ASTM F1554, Grade 36.

G. Washers, Plain (for Non-high Strength Bolts): ASME B18.22.1, Type B.

H. Welding Electrodes:
1. Shielded metal arc: AWS A5.1/A5.1M or AWS A5.5/A5.5M, E70XX or E801X-X.
2. Submerged arc: AWS A5.17/A5.17M or AWS A5.23/A5.23M, F7XX-EXXX or F8XX-EXXX-XX.
3. Gas metal arc: AWS A5.18/A5.18M, E70S-X or E70U-1 or AWS A5.28/A5.28M, ER80S-XX, E80C-XXX.
4. Flux cored arc: AWS A5.20/A5.20M, E7XT-X (except 2, 3, 10, GS), AWS A5.29/A5.29M, E7XT-X or E8XTX-X, E8XTX-XM.

I. Anchor Rods and Bolts:
1. See Section 03 15 19.

J. Headed Studs and Deformed Bar Anchors:
   1. Headed studs:
      a. ASTM A108, complying with AWS D1.1/D1.1M, Section 7, Type B; minimum yield strength 50,000 psi, minimum tensile strength 60,000 psi.
      b. Uniform diameter.
      c. Heads: Concentric and normal to shaft.
      d. Weld end: Chamfered and solid flux.
   2. Deformed bar anchor:
      a. ASTM A1064/A1064M, complying with AWS D1.1/D1.1M, Section 7, Type C.
      b. Minimum yield strength: 70,000 psi.
      c. Minimum tensile strength: 80,000 psi.
      d. Straight, unless indicated otherwise.
      e. Solid flux.
   3. After welding, remove ceramic ferrules and maintain free from any substance which would interfere with function, or prevent bonding to concrete.

K. Nonshrink Grout: See Specification Section 03 31 30.

L. Crane Rails:
   1. Controlled-cooled, open-hearth carbon steel ASCE (American Society of Civil Engineers) rails per ASTM A2, Class A, #1 rails, unless noted otherwise, of size and weight indicated.
      a. Furnish rails with milled tight end joints suitable for crane service, with standard drilling, removable end stops and all related accessories required, including:
         1) Joint bars: Match rail section and properties, drilled to match rail drilling.
         2) Joint bar bolts and nuts: High strength.
         3) Hardened washer: ASTM F436 for bar bolts.
         4) Except as indicated otherwise, two-bolt type fixed or floating rail clamps to suit the conditions, of forged or pressed steel, complete with Grade A325 bolts, reversible fillers, and self-locking nut or nut and lock washer.

M. Mechanical and Adhesive Anchor Bolts for Fastening to Concrete:
   1. See Specification Section 03 15 19.

2.3 FABRICATION

A. Comply with requirements of applicable building code and AISC 360 with modifications and additional requirements specified herein.
   1. Identify high-strength steel material in fabricated members in accordance with ASTM A6/A6M.

B. Minimize the amount of field welding.
   1. Shop assemble components into largest size possible commensurate with transportation and handling limitations.
   2. Shop connections: Bolted with high-strength bolts or welded.

C. Connection Details:
   1. Provide as a minimum, two, 3/4 inches diameter, high-strength bolts for all bolted connections unless otherwise specified.

D. Provide bearing type connections for all bolted connections, unless otherwise noted.

E. Field Connections:
   1. Provide bolts for all field connections except where shown otherwise on the Drawings.
   2. Use high-strength bolts unless shown or specified otherwise.
   3. Use of high-strength bolts: Conform to RCSC Specification for Structural Joints Using High-Strength Bolts.
   4. Unfinished bolts may be used for attaching stair treads to stringers.
   5. If structural steel details (field welds versus shop welds, etc.) shown on design Drawings are not compatible with selected erection procedures, submit proposed modifications for review.

6. Connections to structural steel provided by others: Provide all connectors and coordinate location of bolt holes to match connection holes in steel provided by others.

F. Accurately mill column end bearing surfaces to true plane.

G. Fabricate and erect beams with non-specified camber in accordance with AISC 360, Chapter L1.

H. Cut, drill, or punch holes at right angles to surface of metal.
   1. Do not make or enlarge holes by burning.
   2. Make holes clean cut, without torn or ragged edges.
   3. Remove outside burrs resulting from drilling or reaming operations with tool making 1/16 IN bevel.
   4. Provide holes in members to permit connection of work of other trades or contractors.

I. Make allowance for draw in all cross bracing to provide small amount of initial tension in members.

J. Make splices only where indicated or where approved.

K. Wall Girts:
   1. Extend past columns and miter ends unless noted otherwise.
   2. Connect girts to each other at corners unless noted otherwise.

L. Cope at 45 degrees, corners of stiffener plates at junction of member flanges with webs.

M. Flame cut bevels for welds, provided such cutting is done automatically.
   1. Leave free of burrs and slag by grinding or planning the cut edges.

N. Grind smooth all rough welds and sharp steel edges shall be ground to approximately 1/8 inches radius.

O. Tolerances (unless noted otherwise on Drawings):
   1. When material received from the mill does not satisfy ASTM A6/A6M tolerances for camber, profile, flatness or sweep, Contractor is permitted to perform corrective work by the use of controlled heating, and mechanical straightening, subject to the limitations of the AISC 360.
   2. Fabrication tolerance:
      a. Member length:
         1) Both ends finished for contact bearing: 1/32 inches.
         2) Framing members: 30 feet or less: 1/16 inches.
      b. Member straightness:
         1) Compression members: 1/1000 of axial length between points laterally supported.
         2) Non-compression members: ASTM A6/A6M tolerance for wide flange shapes.
      c. Specified member camber (except compression members):
         1) 50 feet or less: -0/+1/2 inches.
         2) Over 50 feet: -0/+1/2 inches (+1/8 inches per 10 feet over 50 feet).
         3) Members received from mill with 75% of specified camber require no further cambering.
         4) Fabricate beams/trusses without specified camber so after erection, camber is upward.
         5) Measure camber in fabrication shop in unstressed condition.
      d. Use filler plates at bolted splices to take up depth deviation.
         1) At welded joints, adjust weld profile to conform to variation in depth.
         2) Slope weld surface per AWS requirements.
      e. Free finished members from twists, bends and open joints.
         1) Sharp kinks, bends and deviation from the above tolerances are cause for rejection of material.

2.4 WELDING

A. Comply with AWS D1.1/D1.1M, and other requirements indicated herein, for all welding, techniques of welding employed, appearance and quality of welds, and methods used to correct defective work.
   1. Qualify joint welding procedures or test in accordance with AWS qualification procedures.

B. Test and qualify welders, welding operators and tackers in compliance with AWS D1.1/D1.1M for position and type of welding to which they will be assigned.
   1. Conduct tests in presence of approved testing agency.
   2. Certification within previous 12 months will be acceptable, provided samples of the welder's work are satisfactory.

C. Before Starting Welding:
   1. Carefully plumb and align members in compliance with specified requirements.
   2. Fully tighten all bolts.
   3. Comply with AWS D1.1/D1.1M, Section 5 for assembly and surface preparation.
   4. Preheat base metal to temperature stated in AWS D1.1/D1.1M.
      a. When no preheat temperature is given in AWS D1.1/D1.1M and base metal is below 50 degrees F, preheat base metal to at least 70 degrees F.
      b. Maintain temperature during welding.
      c. Preheat surface of all base metal within distance from point of welding equal to thickness of thicker part being welded or 3 inches, whichever is greater, to specified preheat temperature.
      d. Maintain this temperature during welding.
   5. Mark welds with an identifying mark unique to each welder.

D. Make flange welds before making web welds.

E. Where groove welds have back-up plates, make first three passes with 1/8 inches round electrodes.
   1. Use backup plates in accordance with AWS D1.1/D1.1M, extending minimum of 1 inch either side of joint.

F. Flame cut edges of stiffener plates at shop or field butt weld.
   1. Do not shear.

G. Grind flush web fillets at webs notched to receive backup plates for flange groove welds.

H. Low Hydrogen Electrodes: Dry and store electrodes in compliance with AWS D1.1/D1.1M.

I. Do not perform welding when ambient temperature is lower than 0 degrees F or where surfaces are wet or exposed to rain, snow, or high wind, or when welders are exposed to inclement conditions.

J. Headed Studs and Deformed Bar Anchors:
   1. Automatically end welded in accordance with the AWS D1.1/D1.1M and manufacturer's recommendations.
   2. Fillet welding of headed studs and deformed bar anchors is not allowed unless approved by Contracting Officer’s Representative.

K. Test in-place studs in accordance with requirements of AWS D1.1/D1.1M to ensure satisfactory welding of studs to members.
   1. Replace studs failing this test.

L. When headed stud-type shear connectors are to be applied, clean top surface of members to receive studs in shop to remove oil, scale, rust, dirt, and other materials injurious to satisfactory welding.
   1. Do not shop paint or galvanize metal surfaces to receive field applied studs.

2.5 SHOP COATING

A. Refer to Specification Section 09 96 00 and coordinate shop primer, surface preparation and coating with field applied primers and coatings where specified.

B. Provide suitable methods of handling and transporting painted steel to avoid damage to coating.

C. Do not coat following surfaces:
   1. Machined surfaces, surfaces adjacent to field welds, and surfaces fully embedded in concrete.
   2. All other members for which no coating is specified.
   3. Contact surfaces at bolted slip-critical connections, unless surface condition conforms to the RCSC Specification for Structural Joints Using High-Strength Bolts, Part 3.2.2.

D. Clean thoroughly all surfaces not coated before shipping.
   1. Remove loose mill scale, rust, dirt, oil and grease.
   2. Protect machined surfaces.

2.6 SOURCE QUALITY CONTROL

A. Special Inspection and Testing:
   1. See Specification Section 01 40 00 and the Spc.
   2. If the fabricator is not an Approved Fabricator, the Contractor will employ the services of an independent testing agency to inspect and test structural steel shop work for compliance with Specifications.
      a. Contractor provides sufficient notification and access so inspection and testing can be accomplished.
   3. Contractor responsible for testing to qualify shop welders and as needed for Contractor's own quality control to ensure compliance with Contract Documents.

B. Approved Fabricator or Testing Agency Responsibilities:
   1. Inspect shop and field welding in accordance with AWS D1.1/D1.1M, Section 6 including the following non-destructive testing:
      a. Visually inspect all welds.
      b. In addition to visual inspection, test 50% of full penetration welds and 20% of fillet welds with liquid dye penetrant.
      c. Test 20% of liquid dye penetrant tested full penetration welds with ultrasonic or radiographic testing.
   2. Inspect high-strength bolting in accordance with the RCSC Specification for Structural Joints Using High-Strength Bolts, Section 9.
      a. Verify proper pretension for slip-critical bolted connection only.
      b. Verify direct tension indicator gaps for slip-critical bolted connection only.
   3. Inspect stud welding in accordance with AWS D1.1/D1.1M, Section 7.8.
   4. Prepare and submit inspection and test reports to Contracting Officer.

2.7 GENERAL

A. Contractor is solely responsible for safety.
   1. Construction means and methods and sequencing of work is the prerogative of the Contractor.
   2. Take into consideration that full structural capacity of many structural members is not realized until structural assembly is complete; e.g., until slabs, decks, bracing or rigid connections are installed.
   3. Partially complete structural members shall not be loaded without an investigation by the Contractor.
   4. Until all elements of the permanent structure and lateral bracing system are complete, provide temporary bracing designed, furnished, and installed by the Contractor for the partially complete structure.

B. Adequate temporary bracing to provide safety, stability and to resist all loads to which the partially complete structure may be subjected, including wind, construction activities, and operation of equipment, is the responsibility of the Contractor.
   1. Use temporary guys, braces, shoring, connections, etc., necessary to maintain the structural framing plumb and in proper alignment until permanent connections are made, the succeeding work is in place, and temporary work is no longer necessary.
   2. Use temporary guys, bracing, shoring, and other work to prevent injury or damage to adjacent work or construction from stresses due to erection procedures and operation of erection equipment, construction loads, and wind.
   3. Design of the temporary bracing system and consideration of the sequence and schedule of placement of such elements and effects of loads imposed on the structural steel members by partially or completely installed work, including work of all other trades, is the Contractor's responsibility.
      a. If not obvious from experience or from the Drawings, confer with the Contracting Officer’s Representative to identify those structural steel elements that must be complete before the temporary bracing system is removed.
   4. Remove and dispose of all temporary work and facilities off-site.

C. Examine work-in-place on which specified work is in any way dependent to ensure that conditions are satisfactory for the installation of the work.
   1. Report defects in work-in-place which may influence satisfactory completion of the work.
   2. Absence of such notification will be construed as acceptance of work-in-place.

D. Field Measurement:
   1. Take field measurements as necessary to verify or supplement dimensions indicated on the Drawings.
   2. Contractor is responsible for the accurate fit of the work.

E. Check the elevations of all finished footings or foundations and the location and alignment of all anchor bolts before starting erection.
   1. Notify Contracting Officer of any errors or deviations found by such checking.

PART 3 - EXECUTION

3.1 ERECTION

A. Framing member location tolerances after erection shall not exceed the framing tolerances listed in the FIELD QUALITY CONTROL Article in PART 3 of this Specification Section.

B. Erect plumb and level; introduce temporary bracing required to support erection loads.

C. Use light drifting necessary to draw holes together.
   1. Drifting to match unfair holes is not allowed.

D. Welding:
   1. Conform to AWS D1.1/D1/1M and requirements of this Specification Section.
   2. Join two (2) sections of steel of different ASTM designations using welding techniques in accordance with a qualified AWS D1.1/D1.1M procedure.

E. Shore existing members when unbolting of common connections is required.
   1. Use new bolts for rebolting connections.

F. Clean stored material of all foreign matter accumulated during erection period.

G. Clean bearing and contact surfaces before assembly.

H. Set beam and column base and bearing plates accurately, as indicated, on nonshrink grout.
   1. Set and anchor each base plate to proper line and elevation.
   2. Use metal wedges, shims or setting nuts as required and tighten anchor bolts.
      a. Use same metal as base plate.

b. Cut off protrusions of wedges and shims flush with edge of base plate.
3. Fill sleeves around anchor bolts with nonshrink grout.
4. Pack grout solidly between bottom of plate and bearing surface.
5. Refer to Specification Section 03 31 30 for nonshrink grout requirements.

I. Cast-in-place Anchor Bolts:
1. See Specification Section 03 15 19.

J. Install high strength bolts with hardened washers.
1. Install and tighten in accordance with the RCSC Specification for Structural Joints Using High-Strength Bolts, Section 8.
2. Coordinate installation with inspection.
a. Do not start installation until coordination with Testing Agency is complete.
3. Bearing-type connections: High-strength bolts shall be tightened to snug-tight condition.
4. Slip-critical connections:
a. Perform calibration testing for all methods of installation of high-strength bolts in accordance with RCSC Specification for Structural Joints Using High-Strength Bolts, Section 8.2.
b. Turn-of-nut tightening:
1) Inspector shall observe the pre-installation verification testing.
2) Subsequently, ensure by routine observation that the bolting crew properly rotates the turned element relative to the unturned element by the amount specified.
3) Alternatively, when fastener assemblies are match-marked after the initial fitup of the joint but prior to pretensioning, visual inspection after pretensioning is permitted in lieu of routine observation.
c. Calibrated wrench tightening: Calibrate on a daily basis.
d. Direct tension indicator tightening: If previously approved by Contracting Officer.
e. Installation of alternate design bolts: If previously approved by Contracting Officer.
5. In the event any bolt in a connection is found to be defective, check and retighten all bolts in the connection.

K. Do not use gas cutting to correct fabrication errors.
1. In case members do not fit or holes do not match, ream out the holes and insert the next larger size bolt.
a. Drill new holes if the connections require new holes.
b. Make no such corrections without prior approval of the Engineer.
2. Burning of holes is not permitted.

L. Prior to making field connections to existing structural steel, remove completely all paint from existing steel which will be in contact with new steel and new welds.

M. Tighten and leave in place erection bolts used in welded construction.

N. Provide beveled washers to give full bearing to bolt head or nut where bolts are to be used on surfaces having slopes greater than 1 in 20 with a plane normal to bolt axis.

O. After bolts are tightened, upset threads of non-high strength bolts and anchor bolts to prevent nuts from backing off.

P. After Erection:
1. Grind smooth all sharp surface irregularities resulting from field cutting or welding.
2. Power tool clean welds, bolts, washers and abrasions to shop coat removing all rust and foreign matter.

Q. Mechanical Anchor Bolts and Adhesive Anchor Bolts:
1. See Specification Section 03 15 19.

3.2 FIELD QUALITY CONTROL

A. Special Inspection and Testing:
1. See Specification Section 01 40 00.

2. Special Inspection to be in accordance with the building code.
3. Special Inspection is required for:
    a. Material verification of high-strength bolts, nuts, and washers.
        1) Frequency: All high-strength bolts, prior to being covered up or substantial completion.
    b. Inspection of high-strength boltings:
        1) Frequency:
            a) All high-strength bolts, prior to being covered up or substantial completion.
            b) Pretensioned and slip-critical joints using turn-of-nut without match marking or calibrated wrench methods of installation require continuous inspection.
    c. Material verification of structural steel.
        1) Frequency: Prior to being covered up or substantial completion,
    d. Material verification of weld filler materials.
        1) Frequency:
            a) Prior to welding on site.
            b) Randomly thereafter.
    e. Inspection of welding.
        1) Frequency:
            a) Visually inspect all welds.
            b) In addition to visual inspection, test 50% of full penetration welds and 20% of fillet welds with liquid dye penetrant or magnetic particle.
            c) Test 20% of liquid dye penetrant tested full penetration welds with ultrasonic or radiographic testing.
    f. Inspect structural steel which has been erected.
        1) Frequency: Prior to members being covered up or substantial completion.
    g. Inspect stud welding in accordance with AWS D1.1/D1.1M, Section 7.8.

B. Erected Framing Tolerance, unless noted otherwise on the Drawings:
    1. Do not exceed cumulative effect of rolling, fabrication and erection tolerance for overall finished dimensions.
    2. Erection tolerances are defined relative to member working points and working lines as follows:
        a. Actual centerline of top flange or surface at each end for horizontal members.
        b. Actual center of member at each end for all other members.
        c. Other points may be used, providing they are based on these definitions.
        d. Working line is straight line connecting member working points.
    3. Tolerances on position and alignment are as specified in the Code, unless otherwise modified.
        a. Provide "adjustable items" such as lintels, wall supports, curb angles, window mullions and similar members with adjustable connections to supporting structural framing.
    4. Certification by steel erector:
        a. Certify the location of erected structural steel is acceptable for plumbness, level and aligned within tolerances specified.
        b. Provide certification upon completion of any part of work.
        c. Provide certification prior to start of work by other trades that may be supported; attach to structural steel work.

3.3 CLEANING AND REPAIR OF SHOP PRIMER PAINT

A. After erection, clean all steel of mud or other foreign materials, and repair any damage.
    1. Touchup coatings to comply with Specification Section 09 96 00.

END OF SECTION

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SECTION 05 14 00
STRUCTURAL ALUMINUM

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Structural aluminum including the fabrication and erection of framing and bracing members, including connection design as required.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 15 19 - Anchorage to Concrete.
   2. Section 05 52 46 – Mechanically Fastened Aluminum Railings.
   3. Section 09 96 00 - High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Aluminum Association (AA):
      a. ADM 1, The Aluminum Design Manual.
      b. DAF 45, Designation System for Aluminum Finishes.
   2. American Institute of Steel Construction (AISC):
      a. 325, Manual of Steel Construction.
   3. ASTM International (ASTM):
      a. B209, Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate.
      b. B221, Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes.
      c. B247, Standard Specification for Aluminum and Aluminum-Alloy Die Forgings, Hand Forgings, and Rolled Ring Forgings.
      d. B308, Standard Specification for Aluminum-Alloy 6061-T6 Standard Structural Profiles.
      e. B429, Standard Specification for Aluminum-Alloy Extruded Structural Pipe and Tube.
      f. F467, Standard Specification for Nonferrous Nuts for General Use.
      g. F468, Standard Specification for Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use.
      h. F593, Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.
      i. F594, Standard Specification for Stainless Steel Nuts
   4. American Welding Society (AWS):
      a. D1.2, Structural Welding Code - Aluminum.

B. Qualifications:
   1. Minimum of 10 years of experience in fabrication of structural aluminum.
   2. For welding aluminum:
      a. Qualify welding procedures and welding operators in accordance with AWS D1.2.
      b. Welding operators to have been qualified during the 12-month period prior to commencement of welding.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Fabrication and/or layout drawings:
      a. Erection plans and details of each piece including connection details:
         1) Show all cuts, copes and holes.
         2) Indicate all shop welds using AWS symbols.
         3) Indicate all shop and field bolts.

b. Complete shop drawings for all of the work showing clearly all pieces, details, connections, materials and shop-applied coatings.
c. Prepare complete erection drawings showing the location and marks of all pieces.

2. Product technical data including:
   a. Acknowledgement that products submitted meet requirements of standards referenced.
   b. Manufacturer's installation instructions.
3. Certifications.
   a. Certificates of compliance with standards specified for all major components and fasteners incorporated into work.
   b. Copies of current welding certificates for each welder assigned to perform welding indicating compliance with testing specified by AWS.
4. Test reports.
   a. Certified copies of material mill tests.
   b. Manufacturer's load test and temperature sensitivity data for expansion anchor bolt and adhesive anchor bolts.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Mechanical anchor bolts: See Specification Section 03 15 19.
   2. Adhesive anchor bolts: See Specification Section 03 15 19.

2.2 MATERIALS

A. Alloy 6061-T6, 32,000 psi tensile yield strength minimum.
   1. ASTM B209 for sheets and plates.
   2. ASTM B221 and ASTM B308 for shapes: Beams , channels , and angles.
   3. ASTM B247 for forgings.

B. Alloy 6063-T5 or T615,000 psi tensile yield strength minimum. Nonferrous Bolts and Nuts: ASTM F467 and ASTM F468 of alloy 2024-T4 (60,000 psi tensile strength minimum).

C. Stainless Steel Bolts and Nuts : ASTM F593 and ASTM F594, Type 304 or 316 with a minimum yield strength of 30,000 psi and a minimum tensile strength of 75,000 psi.

D. Washers: Same material and alloy as found in bolts and nuts with which the washers are to be used.

E. Electrodes for Welding Aluminum: AWS D1.2 filler alloy 5356.

F. Mechanical and Adhesive Anchor Bolts for Fastening to Concrete:
   1. Where approved by Contracting Officer’s Representative.
   2. Stainless steel, Type 304 or 316.
   3. See Specification Section 03 15 19.

2.3 DESIGN

A. All design shall consider effect of welds on material properties.
   1. Eliminate the use of field welding.
   2. Observe locations of any specified shop splice welds.

B. Where final design of connections for any portion of structure is not indicated, perform final design of such connections in accordance with the building code.

C. Final designs of connections shall conform to AA ADM 1 and to details and requirements stated in the Contract Documents.
   1. Design loads for connections to be designed by the Contractor shall be as shown on the Drawings.

2. Where design loads are not specified, connections shall be detailed to develop the full capacity of the member.

2.4 FABRICATION

A. Fabrication of bolted and welded connections of aluminum work shall be in accordance with AA ADM 1.

B. Contractor to be solely responsible for correctness of all shop and field fabrication and fit. Verify field conditions and dimensions prior to fabrication.

C. Fabricate aluminum work and assemble in shop to greatest extent possible.
   1. Make splices only where indicated or approved by the Contracting Officer's representative.

D. Provide connections as indicated.
   1. Where not indicated, design and provide connections in accordance with requirements of this Specification Section.
   2. One-sided or other types of eccentric connections are not acceptable unless indicated on Contract Drawings or approved on Shop Drawings.

E. Drill or punch holes at right angles to surface of metal.
   1. Do not make or enlarge holes by burning.
   2. Provide holes clean and free of torn or ragged edges.
   3. Use tools which will make a 1/16 inches bevel to remove outside burrs resulting from drilling or punching operations.
   4. Punch or drill for field connections and for attachment of work by other trades.

F. Cope at 45 degrees corners of stiffener plates at junction of member flanges with webs.

G. Welding:
   1. Weld connections to members in shop and bolt connections in field.
   2. Perform welding using electrodes of filler alloy 5356.
   3. Perform welding in accordance with AWS D1.2.
   4. Use only welding procedures and welding operators qualified in accordance with requirements of Paragraph 1.2B. Qualifications.

H. All full penetration welds shall be nondestructively tested by liquid penetrant or ultrasonic methods per AWS standards.

I. Form to shapes indicated with straight lines, true angles and smooth curves.
   1. Grind smooth all rough welds and sharp edges.
   2. Round all corners to approximately 1/8 inches radius.

J. Finish: Mill finish as fabricated.

2.5 SOURCE QUALITY CONTROL

A. Special Inspection and Testing:
   1. Contractor responsible for testing to qualify shop welders and as needed for Contractor's own quality control to ensure compliance with Contract Documents.
   2. Contractor to inspect and test fabrication in accordance with the fabricators quality control procedures and in accordance with ADM-1 2015 or later.
   3. Contractor inspect shop welding in accordance with AWS D1.2, Section 5 including the following non-destructive testing:
      a. Visually inspect all welds.
      b. In addition to visual inspection, test 50% of full penetration welds and 20% of fillet welds with liquid dye penetrant.
      c. Test 20% of liquid dye penetrant tested full penetration welds with ultrasonic or radiographic testing.

PART 3 - EXECUTION

3.1 ERECTION

A. Contractor is solely responsible for safety.
   1. Construction means and methods and sequencing of work is prerogative of the Contractor.
   2. Take into consideration that full structural capacity of many structural members is not realized until structural assembly is complete; e.g., until decks and diagonal bracing or rigid moment connections are installed.
   3. Partially complete structural members shall not be loaded without an investigation by the Contractor.
   4. Until all elements of the permanent structure and lateral bracing system are complete, provide temporary bracing designed, furnished and installed by the Contractor.

B. Bolting:
   1. Protect bolt threads from damage.
   2. Rest bolt heads and nuts squarely against surfaces.
   3. Where bolt heads or nuts rest on beveled surfaces having slope greater that 1 in 20 with plane normal to bolt axis, use beveled washers to provide full bearing to head and nut.
   4. Correct poor matching of holes by drilling to next larger size and use larger diameter bolt.
   5. Unless otherwise specified, connect aluminum members to:
      a. Aluminum members using 3/4 inches diameter nonferrous bolts of alloy 2024-T4 or stainless steel bolts (ASTM F593 and ASTM F594).
      b. Carbon and low alloy steel using 3/4 inches diameter stainless steel bolts (ASTM F593 and ASTM F594).
         1) Provide dissimilar materials protection.
      c. Concrete or masonry using stainless steel expansion anchor bolts or adhesive anchor bolts unless shown otherwise.
         1) Provide dissimilar materials protection.

C. Welding: Field welding of aluminum is not allowed unless indicated on Drawings.

D. Correct fabrication errors and damaged members in shop.
   1. Do not use cutting torch in shop or in field to cut any members, to correct fabrication errors, or to cut openings.

E. Provide templates for anchors, bolts, and other items to be installed in other work.

F. Field Assembly:
   1. Tolerances shall comply with AISC 325 and the ADM-1.
      a. Before members are assembled, thoroughly clean all bearing surfaces and surfaces that will be in permanent contact.
      b. After assembly, carefully align all members of each frame or assembly and accurately adjust until final, correct and true location is achieved.
         1) As work progresses, securely fasten in place.
   2. Provide full length members without splices.
   3. Securely tighten and leave in place all erection bolts used in welded construction, unless removal is required.

G. Set beam and column baseplates accurately, as indicated on nonshrink grout, in accordance with Division 03.
   1. If not indicated, provide minimum of 1 inch grout thickness under base plates.
   2. Set and anchor each base plate to proper line and elevation.
      a. Use aluminum wedges, shims, or setting nuts for leveling and plumbing columns and beams.
         1) Tighten anchor bolts.
      b. Fill space between bearing surface and bottom of base plate with nonshrink grout.
         1) Fill space until voids are completely filled and base plates are fully bedded on wedges, shims, and grout.

c. Do not remove wedges or shims and where they protrude, cut off flush with edge of base plate.

H. Temporary Protection:
1. Suitably protect aluminum surfaces against lime mortar stains, discoloration, surface abrasion and other construction abuses.
2. Remove protection closeout Period.

I. Contact with Dissimilar Materials: Where aluminum surfaces will be embedded in concrete, built into masonry, or in contact with steel, concrete, grout, masonry, or other dissimilar materials, coat the aluminum surfaces as described in Specification Section 09 96 00.

3.2 FIELD QUALITY CONTROL

A. Special Inspection and Testing:
1. See Section 01 40 00.
2. Special Inspection is required to:
a. Inspect structural aluminum delivered on site, prior to erection.
1) Visually inspect all welds, bolts and general fabrication
b. Verify proper identification markings for bolts, nuts and washers per applicable standards. Verify proper hole and bolt size.
1) Frequency: All bolts, prior to being covered up or substantial completion.
c. Inspect structural aluminum which has been erected.
1) Inspect the following:
a) Details such as bracing and stiffening.
b) Member locations.
c) Erection tolerances.
d) Application of joint details at each connection.
2) Frequency: Prior to members being covered up or substantially complete.

END OF SECTION

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SECTION 05 21 00
STEEL JOISTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Manufactured open-web steel joists and joist accessories.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 40 00 - Quality Requirements.
   2. Section 05 12 00 - Structural Steel.
   3. Section 05 30 00 - Metal Decking.
   4. Section 09 96 00 - High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Institute of Steel Construction (AISC):
      a. 341, Seismic Provisions for Structural Steel Buildings.
      b. 360, Specifications for Structural Steel Buildings (referred to herein as AISC Specification).
   2. ASTM International (ASTM):
      a. A36, Standard Specification for Carbon Structural Steel.
      b. A307, Standard Specification for Carbon Steel Bolts and Studs, 60,000 psi Tensile Strength.
      c. A563, Standard Specification for Carbon and Alloy Steel Nuts.
      d. F436, Standard Specification for Hardened Steel Washers Inch and Metric Dimensions.
      e. F3125, Standard Specification for High Strength Structural Bolts, Steel and Alloy Steel, Heat Treated, 120 ksi (830 MPa) and 150 ksi (1040 MPa) Minimum Tensile Strength, Inch and Metric Dimensions.
   3. American Welding Society (AWS):
      a. D1.1, Structural Welding Code - Steel.
   4. Corps of Engineers (COE):
      a. CRD-C621, Standard Specification for Packaged, Dry, Hydraulic-Cement Grout (Nonshrink).
   5. Steel Joist Institute (SJI):
      a. Recommended Code of Standard Practice for Steel Joists and Joist Girders.
      b. Standard Specification for Open Web Steel Joists, K-Series (ANSI SJI-K-1.1).
      c. Standard Specification for Longspan Steel Joists LH-Series and Deep Longspan Steel Joists DLH-Series (ANSI SJI-LH-DLH-1.1).
      d. Standard Specification for Joist Girders (ANSI SJI-JG-1.1).
      e. Standard Specifications Load Tables and Weight Tables for Steel Joists and Joist Girders:
         1) Standard Load Table Open Web Steel Joists, K-Series.
         2) Standard Load Table Longspan Steel Joists, LH-Series.
         3) Standard Load Table Deep Longspan Steel Joists, DLH-Series.

B. Qualifications:
   1. Manufacturer: Member of SJI.
      a. Structural design calculations and details of manufactured joists shall be prepared by a qualified professional engineer retained by the manufacturer and registered in the state where the project is located.
   2. Qualification of welding work:

a. Qualify welding processes, operations, and operators in accordance with requirements of AWS D1.1.
b. Welding operators to have been qualified during the 12-month period prior to commencement of welding.

1.3 SUBMITTALS

A. Shop Drawings:
1. Fabrication and/or layout drawings, signed by a professional engineer registered in the State of Arizona:
a. Detailed Shop Drawings showing size and layout of each joist unit, bridging, connections, and accessories. Include mark, number, type, location, and spacing of joists and bridging.
b. Show joining splice and connection to other work details.
c. Provide templates or location drawings for installation of anchor bolts.
d. Provide details of bridging, method of attachment to joists, and joist end anchorage and other details required for joist installation. Indicate beveled end plates for joist roof pitch where required.
e. Show shop-applied coatings.
f. Shop Drawings shall not be reproductions of the Contract Drawings.
2. Product technical data including:
a. Joist manufacturer's load tables, Standard Specifications and installation instructions for each type of joist and its accessories. Include product data describing materials, shop coating, bridging, and accessories.
3. Certifications:
a. Manufacturer's certification that steel joists and accessories comply with specified requirements.
b. Manufacturer member of SJI.
c. Joist material, shop welding and testing, manufacturing and shop inspection and testing are in accordance with SJI requirements.

1.4 DELIVERY, STORAGE, AND HANDLING

A. Deliver, store and handle steel joists as recommended by SJI.
1. Exercise care to avoid damage to joists.

B. Store joists clear of soil on platforms, skids or other durable supports.
1. Protect joists after delivery to prevent rust and deterioration.

C. Provide anchor bolts and other items to be embedded in concrete or masonry, with templates as required, in time for incorporation into the work.

PART 2 - PRODUCTS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Nucor Corporation Vulcraft Divisions.
2. Or Equal.

C. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Steel: Comply with SJI and AISC Specifications for joist series indicated.

B. High-Strength Bolts, Nuts and Washers:
   1. Bolts: ASTM F3125, Grade A325 or Grade A490 as required, heavy hexagon structural bolts.
   2. Nuts: ASTM A563.
   3. Washers (hardened): ASTM F436.

C. Bolts and Nuts, Unfinished: ASTM A307, Grade A, regular hexagon type, low carbon steel, with carbon steel washers.

2.3 STEEL JOISTS

A. Design of steel joists proposed shall have been checked by the SJI and found to conform to the standard specifications and load tables.

B. Design and fabricate joists and accessories in accordance with SJI Specifications and as follows:
   1. Make shop connections and splices using either arc or resistance welding.
      a. Do not shop bolt connections.
   2. Design and fabricate for maximum deflection of 1/360 of clear span under design live load.
   3. Shop holes, field holes, and enlargement of holes will not be permitted unless approved by Contracting Officer.
   4. Fabricate bearing ends to provide following minimum bearing unless a longer bearing length is indicated on the Contract Documents.

	K SERIES	LH AND DLH SERIES	SPECIAL CUSTOM JOISTS
On masonry or concrete:	4 inches min	6 inches min	6 inches min
On steel:	2-1/2 inches min	4 inches min	4 inches min

   5. With steel angle tops and bottom chord members.

C. Loading Criteria:
   1. Refer to Drawings for joist schedule and loading.

D. Special or Concentrated Loads:
   1. Where special or concentrated loads are indicated or required by other trades, reinforce joist and develop details as necessary for support of such loads.
   2. Design connections to resist uplift loads per building code, unless otherwise specified.
   3. Submit written certification that the joists requiring special loads have been designed, fabricated, and are capable of supporting the required design loadings.
      a. Written certification to be signed and sealed by a registered professional Structural Engineer.
      b. Submit certification along with the Shop Drawings for the joist(s).

E. Sloped joists shall have bearing ends that are flush to supporting element.

F. Provide extended bottom chords where indicated.
   1. Comply with SJI and AISC requirements and load tables.

G. Provide extended top chords where indicated.
   1. Comply with SJI and AISC requirements and load tables.

H. Provide ceiling extensions in areas having ceilings attached directly to joist bottom chord.

1. Provide either an extended bottom chord or a separate unit of sufficient strength to support ceiling construction.
2. Extend ends to within 1/2 inches of wall surface.

I. Provide nailers bolted to top chord where indicated.

J. Prepare and paint steel joists and accessories in compliance with Section 09 96 00.

K. Comply with SJI Specifications:
1. Joist designations indicated on the Drawing are minimum requirements; increase as required to comply with design requirements specified.
2. Wherever possible, increased joists shall have the same depth as joist indicated on Drawings.
3. Where necessary to increase joist depths to meet design requirements, coordinate all project changes required due to the increased depth.
4. Make all required joist revisions at no additional cost to Government.

2.4 SOURCE QUALITY CONTROL

A. The Contracting Officer reserves right to inspect joists or manufacturer's shop during joist fabrication.

B. Identify each joist type, size and manufacturer.
1. Provide tagging or other suitable (permanent) means.
2. Maintain identification continuously.

PART 3 - EXECUTION

3.1 PREPARATION

A. Examine areas and conditions under which steel joists are to be installed for conditions detrimental to proper and timely completion of work.

B. Do not proceed with work until unsatisfactory conditions have been corrected.

C. Do not start placement of steel joists until supporting work is in place and secured.

D. Joists will be subject to rejection if:
1. Joists do not comply with requirements of SJI and AISC Specifications and requirements herein.
2. Joists are improperly manufactured, welded, painted or installed.
3. Joists are damaged so that strength is impaired.
4. Joists are not installed as indicated on Drawings.
5. Chords are not installed straight within a tolerance of plus or minus 0.0028 times the length of the joist or the distance between points of lateral support.

3.2 INSTALLATION

A. Install products in accordance with manufacturer's instructions.
1. Where not specifically indicated otherwise, place and secure steel joists in accordance with SJI and AISC Specifications and as herein specified.

B. Splice any joist delivered to the site in more than one piece.

C. Do not overload joists.
1. Note: Joists may not be stable or able to carry design loads until bridging and deck is fully installed.
2. Contractor is solely responsible for safety, construction methods and sequencing of the Work.
3. Do not install joists damaged so that strength is impaired.

D. Place joists on supporting work, adjust and align in accurate location and spacing before permanently fastening.
   1. Provide end bearing and anchorages to secure all joists to supporting members or walls in accordance with SJI Specifications, unless otherwise indicated.
   2. When joists are misfabricated and do not bear flush on supporting member or wall, take corrective measures to ensure full bearing.
      a. Provide steel shims as required for uniform flush bearing.

E. Field weld joists to supporting steel framework in accordance with SJI, AISC and AWS Specifications for type of joists used.
   1. Coordinate welding sequence and procedure with placing of joists.
   2. Observe any special welding requirements on erection drawings.

F. Bearing on Masonry or Concrete Bearing Surfaces:
   1. Secure joists resting on masonry or concrete bearing surfaces by bedding in mortar and anchoring to masonry or concrete construction as specified in SJI Specifications for type of steel joist used.
      a. Masonry or concrete required to support joists to have reach required 28-day compressive strength prior to placing joists thereon.
      b. Area under joist bearing shall be solidly filled with grout.
   2. Furnish anchor bolts or steel bearing plates to be built into concrete and masonry construction.
      a. Furnish templates as may be necessary for accurate location of anchors. Steel bearing plates to conform to ASTM A36.
   3. Bedding mortar:
      a. Sand cement grout:
         1) Approximately 3 parts sand, 1 part Portland cement, 6 plus/minus 1% entrained air and water to produce a slump which allows grout to completely fill required areas and surround adjacent reinforcing.
         2) Minimum 28-day compressive strength: 3000 psi.
      b. Non-shrink grout complying with COE CRD-C621.

G. Provide type, size, spacing, and attachment of bridging in accord with SJI and AISC Specifications, where not specifically indicated otherwise, except as modified herein.
   1. Clearly specify bridging on erection drawings.
   2. Provide diagonal type bridging unless otherwise noted.
   3. Do not use sag rods as substitute for bridging.

H. Install bridging completely, immediately after erection, and before any loads are applied.
   1. Anchor ends of bridging lines at top and bottom chords of each joist and where terminating at walls or beams.
   2. Provide bridging connections at top and bottom chords capable of safely resisting a force specified by SJI Specifications for open web, long span, deep long span joists, and joist girders respectively.
   3. Where five rows of bridging are required in spans over 40 feet, laterally brace each joist before erecting next joist or applying loads.
   4. Do not release hoisting cables before installing center row of diagonal bridging and anchoring bridging line to prevent lateral movement.
   5. During construction period, Contractor is responsible for any loads placed on joists.
      a. Contractor's attention is directed the fact that joists may be unstable and cannot carry their design load until steel deck and bridging are completely installed.

I. Remove or repair damaged joists or other work, to satisfaction of Contracting Officer at no additional expense to the Government.

J. After installation, touch up paint or field paint as specified in Section 09 96 00.

3.3 FIELD QUALITY CONTROL

A. Special Inspection and Testing:
   1. See Section 01 40 00.
   2. See Statement of Structural Tests and Special Inspections.

END OF SECTION

SECTION 05 30 00
METAL DECKING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Manufactured acoustical and standard roof decking.
   2. Manufactured metal form decking.

B. Related Requirements: Include but are not necessarily limited to:
   1. Section 01 40 00 - Quality Requirememts.
   2. Section 05 21 00 - Steel Joists.
   3. Section 07 21 00 - Building Insulation.
   4. Section 09 96 00 - High Performance Industrial Coatings.

1.2 REFERENCES

A. Reference Standards:
   1. American Iron and Steel Institute (AISI):
      a. S100, Specification for the Design of Cold-Formed Steel Structural Members.
   2. ASTM International (ASTM):
      a. A36/A36M, Standard Specification for Carbon Structural Steel.
      b. A653/A653M, Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
      c. A780/A780M, Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings.
      d. A1008/A1008M, Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable.
      e. C1513, Standard Specification for Steel Tapping Screws for Cold Formed Steel Framing Connections.
      f. D746, Standard Test Method for Brittleness Temperature of Plastics and Elastomers by Impact.
      g. D1056, Standard Specification for Flexible Cellular Materials-Sponge or Expanded Rubber.
   3. American Welding Society (AWS):
      a. D1.1/D1.1M, Structural Welding Code - Steel.
      b. D1.3/D1.3M, Structural Welding Code - Sheet Steel.
   4. Steel Deck Institute (SDI):
      a. 31, Design Manual for Composite Decks, Form Decks and Roof Decks.
   5. Underwriters Laboratories, Inc. (UL):
      a. UL Fire Resistance Directory.

1.3 QUALITY ASSURANCE

A. Qualifications:
   1. Decking Manufacturer:
      a. Manufacturer shall possess current, valid, full membership in SDI; associate membership is unacceptable. Submit to Contracting Officer documentation of such membership.
      b. When decking is not specified on the Drawings, structural system design of manufactured decking shall be prepared by qualified professional engineer, employed by or retained by decking manufacturer. When such services are required, submit to Contracting Officer evidence of current, valid license and registration in the same jurisdiction as the Site.

2. Welders:
    a. Qualify welding processes, operations, and personnel in accordance with AWS D1.1 and AWS D1.3, for decking manufacturing and welding at the Site.
    b. Welders shall be qualified, as indicated in the paragraph immediately above, during the 12-month period preceding commencement of welding for the Work and be experienced in welding light-gage metal.
    c. Promptly submit documentation of qualifications when requested by the Contracting Officer.

1.4 SUBMITTALS

A. Action Submittals: Submit the following:
    1. Shop Drawings:
        a. Complete framing and erection plans and details.
        b. Location, length, type, cross section, thickness, and markings of metal decking units.
        c. Size and location of each opening in decking and conditions requiring closure panels and supplementary framing. Show and indicate necessary accessories and reinforcing.
        d. Sequence and procedure to be followed for erecting, fastening, and securing decking. Contracting Officer's approval or commenting on Shop Drawings does not in any way reduce or mitigate Contractor's responsibility for means, methods, procedures, techniques, and sequences of construction or for safety and protection measures incident thereto.
        e. Show and indicate units and surfaces to receive shop-applied finishing, as applicable.
        f. Location of required shoring for form decking. Contracting Officer's approval or comments on Shop Drawings showing or indicating such information in no way reduces or modifies Contractor's obligations for construction means, methods, sequences, procedures, and techniques and safety and protection measures incident thereto.
        g. Details and gages of accessories and miscellaneous items, showing sump pans, cant strips, ridge and valley plates, closure and filler strips and insulation supports.
        h. Fasteners and welding details for installation, including size, spacing, type, and location of fasteners necessary and required to secure decking.
        i. Recommended welding rod size, type, burn-off rate, and welder setting for deck thickness to be joined. Contracting Officer's approval or comments on Shop Drawings showing or indicating such information in no way reduces or modifies Contractor's obligations for construction means, methods, sequences, procedures, and techniques and safety and protection measures incident thereto.
        j. Define welds by indicating standard AWS welding symbols.
        k. Shop Drawings shall not be developed from or reproductions of the Drawings prepared by Contracting Officer's Representative.
    2. Product Data:
        a. Manufacturer's specifications and published catalog data for:
            1) Welds and welding procedure.
            2) Galvanizing repair paint.
            3) Screws.
            4) Powder actuated fasteners.
            5) Joint sealing compound.
        b. Manufacturer's load tables for decking to be provided, including:
            1) Allowable gravity load for metal roof and floor decking.
            2) Allowable diaphragm shear values for metal roof and floor deck.
            3) Allowable metal decking superimposed load.
            4) Allowable unshored span lengths for form decking.

B. Informational Submittals: Submit the following:
    1. Certifications:
        a. Manufacturer's certification that metal decking provided complies with specified requirements, including:

1) Deck material, manufacturing, and shop testing and inspection are in accordance with SDI requirements.

2. Supplier's Instructions:
   a. Manufacturer's written instructions for storing, handling, and installing materials furnished.
3. Test and Evaluation Reports:
   a. Powder driven fastener ICC-ES Evaluation Report(s).
4. Source Quality Control Submittals:
   a. Manufacturer's certified test and mill reports for actual materials furnished.
5. Qualifications Statements:
   a. Evidence of manufacturer's current, valid SDI membership.
   b. Welder qualifications and certifications, when requested by the Contracting Officer.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Delivery and Handling:
   1. Comply with the Contract Documents, SDI requirements and recommendations, and decking manufacturer's written instructions.

B. Storage:
   1. Comply with the Contract Documents, SDI requirements and recommendations, and decking manufacturer's written instructions.
   2. Store metal decking off the ground, entirely covered by waterproof covering, with one end elevated for drainage. Properly vent waterproof covering to avoid condensation.
   3. Prevent corrosion, deterioration, and accumulation of foreign matter.
   4. Prevent formation of "white rust" on surfaces with galvanized finish.
   5. Do not overload supporting members.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. 1.5-inch deep metal roof decking:
      a. Vulcraft.
      b. Or Equal.
   2. 2-inch deep dovetail acoustical roof decking:
      a. Vulcraft, Type 2.0DA
      b. Or Equal.

C. Submit requests for substitutions in accordance with Specification Section 01 33 23.

2.2 METAL ROOF DECKING

A. Design of the metal decking shall be checked by decking manufacturer's professional engineer and found by such professional engineer to comply with SDI's standard specifications and load tables.
   1. Allowable superimposed live uniform loading per square foot for metal roof decking provided for spans shown or indicated shall equal or exceed allowable superimposed live uniform load per square foot for same spans as indicated in SDI's tables.

2. Maximum Deflection: Less than 1/240 of span under live load.

B. Provide decking configurations complying with SDI 31 and as shown and indicated.
   1. Painted Decking: ASTM A1008, shop-applied.
   2. Galvanized Decking: ASTM A653 G90 zinc coating.

C. Metal Roof Decking, 1-1/2 Inch Deep:
   1. Rib type B, sheet steel, 20-gage, minimum, unless noted otherwise on Contract Documents, with minimum uncoated thickness of 0.0358-inch, galvanized.
   2. Wide Rib Deck: Ribs spaced approximately 6 inches on centers; width of rib opening at top surface maximum 2.5 inches; width of bottom rib surface minimum 1.75 inches.

D. Acoustic Decking:
   1. 2-inch deep, 24-1/2" wide, 20-gage, minimum, acoustical base deck, dovetail pattern, and shall be perforated with 1/8-inch diameter holes at 3/8-inch on centers.
   2. .

2.3 FABRICATION

A. Standard Deck Profiles:

DEPTH	TYPE (Vulcraft designations, UNO)	RIB SPACING	TOP SURFACE MAXIMUM RIB OPENING	MINIMUM BOTTOM OF RIB WIDTH
1-1/2 inches	Roof Deck - Wide Rib ('1.5BI')	6 inches	2-5/8IN	1-5/8 inches
1-1/2 inches	Form Deck ('1.5C')	6 inches	4-3/8 inches	2-5/8 inches (max)
2 inches	Dovetail Roof Deck ('2.0DA')	6 1/8 inches	1 1/2 inches	5 3/8 inches

* Indicates produced by DACS

B. Minimum Deck Thickness:
   1. Where gage of metal is indicated, provide the not less than uncoated thickness required by SDI.
      a. Delivered thickness of uncoated steel: Not less than 95 percent of the stated thickness.
   2. Use steel with minimum yield stress of 33 ksi.
   3. Comply with deflection limits, especially when proposing steel with greater strength than required in this provision.

C. Fabrication:
   1. Fabricate decking units in lengths to span three or more support spacings with flush, telescoped or nested end laps not less than 2 inches.
      a. End laps shall occur on supporting members.
      b. Provide decking units with overlapping male and female type side laps or joints to provide positive vertical and lateral alignment of adjacent decking units.

2.4 ACCESSORIES

A. Metal Closures:
   1. Form to configuration required for tight-fitting closures at open ends and sides of decking.
   2. Minimum thickness before galvanizing: 0.0358-inch (20-gage).

B. Ridge and Valley Plates:
   1. Minimum Width: 4.5 inches.
   2. Bend to provide tight-fitting closure with deck units.
   3. Provide plates in 10-foot length where possible.
   4. Minimum thickness before galvanizing: 0.0747-inch (12-gage).

C. Welding Washers:
   1. 16-gage bent steel plate with 3/8-inch or as required by weld size center hole.
   2. Provide at all deck units thinner than 22-gage.

D. Filler Sheet:
   1. Flat or formed 20-gage galvanized steel.

E. Roof Sump Pans:
   1. Provide roof sump pans at locations shown.
   2. Fabricate each from single piece of galvanized sheet steel with level bottom and sloping sides to direct water flow to drain.
   3. Provide sump pans of adequate size to receive roof drains with bearing flanges not less than three inches wide.
   4. Recess pans not less than 1.5 inches below roof deck surface, unless otherwise shown or indicated or required by deck configuration.
   5. Minimum thickness before galvanizing: 0.0747-inch (14-gage).

F. Cant Strips:
   1. Provide bend cant strips to form 45-degree slope not less than five inches wide, with top and bottom flanges not less than three inches wide.
   2. Minimum thickness before galvanizing: 0.0358-inch (20-gage).

G. Insulation Supports:
   1. Provide insulation supports that do not impair quality or finish of decking.

H. Venting:
   1. Provide slotted openings in bottom flutes in accordance with manufacturer's standards.

I. Metal Pour Stops:
   1. Provide and form to configuration required for mortar-tight closures at open sides and ends of decking.

J. Primer Paint:
   1. Unless otherwise required in the Contract Documents, provide decking manufacturer’s baked-on, rust-inhibitive paint, factory-applied to chemically cleaned and phosphate treated metal surfaces.
   2. Paint is not required for galvanized surfaces, unless otherwise indicated.

K. Galvanized coating for metal decking accessories: Comply with ASTM A653 (or equivalent) G90 zinc coating.

L. Galvanizing Repair Materials:
   1. Galvanized repair paint shall be in accordance with Section 09 96 00 - High Performance Industrial Coatings.

M. Screws:
   1. Self-drilling, self-tapping, no. 12 size minimum, hex washer head metal screws.
   2. Protected carbon steel, Hilti S-MD KF, or equal.
      a. Kwik-Cote finish.
      b. ASTM C1513 or equal.

N. Powder Actuated Mechanical Fasteners:
   1. Material: AISI 1070 modified.
   2. Hardness: Minimum Rockwell Hardness C 54.5.
   3. Strength: Minimum tensile strength 285 ksi; minimum shear strength 175 ksi.
   4. Design and Manufacture: Knurled shank with forged ballistic point. Manufacturing process shall ensure steel ductility and prevent development of hydrogen embrittlement.
   5. Washers:
      a. For Steel bar Joist Framing: Minimum 12 mm (0.472-inch) steel washers.
      b. For Structural Steel Framing: Minimum 15 mm (0.591inch) steel washers.
   6. Corrosion Resistance:
      a. For steel roof decks with waterproofing membrane: Five-micron zinc electroplated in accordance with ASTM B633 SC1 Type III.

b. For exposed steel roof decks: Minimum AISI 304 stainless steel sealing caps with bonded neoprene washer shall be provided over each fastener.

7. Design Requirements:
   a. ICC-ES AC43 or SDI method for diaphragm shear strength and stiffness.
   b. FM wind uplift resistance.
   c. UL fire classification.
8. Products and Manufacturers:
   a. For use with steel bar joist and light structural steel framing supports with top chord or flange thickness 1/8-inch to 3/8-inch:
      1) Hilti X-HSN24 (1/8-inch up to and including 3/8-inch).
      2) Or equal.
   b. For use with structural steel framing supports with top flange thickness 1/4-inch or thicker:
      1) Hilti X-ENP-19 L15 (1/4-inch or thicker).
      2) Or equal.

O. Miscellaneous Steel Shapes:
   1. Comply with ASTM A36.

P. Sheet Metal Accessories:
   1. Provide of same material and finish as decking panels.

Q. Flexible Closure Strips for Decking:
   1. Vulcanized, closed cell expanded chloroprene elastomer, complying with ASTM D1056, Grade SCE 41.
   2. Brittleness Temperature: -40 degrees F, ASTM D746.
   3. Flammability Resistance: Self-extinguishing.
   4. Provide with adhesive in accordance with manufacturer/s written instructions.
   5. Ensure complete closure in the Work.

2.5 SOURCE QUALITY CONTROL

A. Tests and Inspections:
   1. Before shipping materials from the shop or fabrication facility to the Site or offsite storage, perform manufacturer's or fabricator's ordinary inspections, testing, and other source quality control activities.
   2. Do not ship materials that are damaged or otherwise defective.

PART 3 - EXECUTION

3.1 PREPARATION

A. Examination:
   1. Examine areas and conditions under which metal decking will be installed for conditions detrimental to proper and timely completion of the Work.
   2. Do not proceed with the Work until unsatisfactory conditions are remedied.

B. Do not start installing metal decking until supporting Work is in place and properly secured.

3.2 INSTALLATION

A. Installation – General:
   1. Install decking in accordance with the Contract Documents, SDI requirements and recommendations, decking manufacturer's written instructions and approved Submittals. In event of conflict among such requirements, obtain Contracting Officer Representative's written interpretation or clarification before proceeding with the associated Work.
   2. Loading During Handling and Installation:
      a. Do not overload supporting structural members or architectural elements.

b. Until entire assembly is substantially complete, structural elements may be unstable or incapable of supporting loads required by code or design loads indicated in the Contract Documents.
c. Locate deck bundles to prevent overloading of structure.
d. Ensure decking construction loads are less than recommendation of SDI 31, except where adequate temporary supports are provided.

3. Provide decking manufacturer's standard accessories as needed to complete decking Work installation.
4. Install each decking unit on supporting structural frame, adjust to final position and accurately align with ends properly bearing on supporting members.
5. Interlock decking units at sides without stretching, contracting, or deforming.
6. Install deck units flat and square and secure to framing without warp or excessive deflection.
7. Install decking units in accurate and close alignment for entire length of run and with close alignment of flutes of one unit with those of abutting unit.
8. Refer to requirements, below in this Article, applicable to all decking, including:
   a. Cutting, fitting, and openings.
   b. Temporary supports.
   c. Welding – general.
   d. Fastening (mechanical and welding).
   e. Pour stops for concrete.
   f. Accessories.

B. Roof Decking:
1. Install roof decking units and accessories as required, in accordance with SDI 31.
2. Lap roof deck units at ends not less than 2 inches.
3. Refer to requirements, below in this Article, for roof decking provisions on:
   a. Insulation supports.
   b. End closures.
   c. Ridge and valley plates.
   d. Roof sump pans.
   e. Cant strips.
   f. Others.

C. Cutting, Fitting, and Openings:
1. Cut and fit deck units and accessories around other construction projecting through or adjacent to decking.
2. Comply with Section 01 73 29 - Cutting and Patching.
3. Make cutting and fitting neat, square, and trim.
4. Cut decking by mechanical means, not by burning.
5. Neatly and accurately install reinforcing at openings, except:
   a. Circular openings less than six-inch diameter.
   b. Rectangular openings having no side dimension greater than 6 inches.
6. Reinforcing at Openings:
   a. Reinforce openings that are not framed between 6 inches and 1 foot with 20-gage flat steel sheet, 1-foot greater in each dimension than opening.
   b. Provide sheeting around opening. Fusion-weld to top surface of deck at each corner and midway along each side.

D. Fastening (Mechanical and Welded):
1. Prevent over-torqueing of screw fasteners by using a tool with depth-limiting nosepiece and clutch.
2. Fastening of 1.5-inch Deep Metal Roof Deck:
   a. Unless otherwise shown or indicated, secure decking units to supporting frame and side laps as follows:
      1) Fasten edge ribs of panels at each support.
      2) At interior supports and at ends of deck, use:

a) For 2-foot wide deck: Three, 5/8-inch round plug welds per deck unit.
b) For 2.5-foot and 3-foot wide deck: Four, 5/8-inch round plug welds per deck unit.
c) Install 20-gage and thinner deck with welding washers at all specified weld locations.

3) At perimeter supports, provide 5/8-inch round plug welds at 1-foot on centers.
a) At side laps, provide no. 12 hex-head screws at 1.5 feet on centers.

3. Fastening of Two-inch Deep Metal Acoustical Roof Deck:
a. Unless otherwise shown or indicated, secure decking units to supporting frame and side laps as follows:
1) Fasten edge ribs of panels at each support.
2) At interior supports and at ends of deck provide:
a) Four, 7/8-inch round arc spot welds per decking unit (2-foot unit).
b) Install 20-gage and thinner deck with welding washers at specified weld locations.
3) At perimeter supports 7/8-inch round plug welds at 1-foot on centers.
4) At side laps, provide no. 12 hex-head self-drilling, self-tapping screws at 6 inches on centers.

E. Insulation Supports:
1. Provide insulation supports as necessary and required.
2. Provide insulation supports where top surface of roof decking does not extend as necessary and required to completely support roof insulation.
3. Weld insulation supports into position.

F. Closure Strips:
1. Provide metal closure Strips at open, uncovered ends and edges of roof decking, and in voids between deck and other construction.
2. Weld closure strips into position to provide complete decking.
3. Flexible Closure Strips:
a. Provide flexible closure strips lieu of metal closures, at Contractor's option and when approved by Contracting Officer, wherever use will ensure complete closure.
b. Provide with elastomeric type adhesive in accordance with written directions and recommendations of manufacturers of closure strips and adhesives.

G. Ridge and Valley Plates:
1. Provide weld ridge and valley plates at top surface of roof decking.
2. Lap end joints not less than three inches with laps in direction of water flow.

H. Roof Sump Pans:
1. Provide over openings in roof decking.
2. Weld to top deck surface. Space welds at not more than 1-foot on center with not less than one weld at each corner and each side midway between each corner.
3. Cut opening in bottom of roof sump to accommodate drain size shown or indicated.

I. Cant Strips:
1. Provide cant strips welded to top surface of roof decking at 1-foot on centers.
2. Lap end joints not less than 3 inches.

J. Accessors:
1. Provide metal accessories to close all openings and gaps between decking and other construction, at objects projecting through decking, at locations where decking changes direction, and at open ends of deck units where deck units terminate.
2. Weld into position to provide a complete installation.

K. Pour Stops:
1. Provide pour stops continuous around decking perimeter.
2. Locate so that slab terminates beyond perimeter support centerline distance indicated on the Drawings.

3. Weld into position adequately to resist forces due to placement and finishing of concrete and in accordance with manufacturer's written recommendations.

L. Cleaning and Touch-Up:
1. After installation, remove surplus materials and debris from decking surfaces. Comply with Section –01 73 40 - Execution.
2. Field-repair shop-painted surface in accordance with paint or coating manufacturer's written instructions and applicable provisions of the Contract Documents.
3. Repair damaged galvanized surfaces in accordance with Section 09 96 00 - High Performance Industrial Coatings.
4. Repair damaged galvanized surfaces in accordance with ASTM A780.
   a. Prepare damaged surface by abrasive blasting or power standing.
   b. Apply galvanized repair paint to minimum 6 miles DFT in accordance with manufacturer's instructions.

3.3 FIELD QUALITY CONTROL

A. Code-Required Special Inspection:
1. Comply with Section 01 40 00 - Quality Requirememts.
2. Special inspection is required for:
   a. Verifying proper decking materials and layout.
      1) Frequency: Prior to attaching decking to structure.
   b. Verifying proper weld filler materials, screws, powder actuated mechanical fasteners, weld testing and fastener spacing.
      1) Frequency:
         a) First four hours during the first day of decking attachment; and
         b) Inspection prior to covering decking Work with concrete, insulation, or other materials.
   c. Weld Testing:
      1) Perform the following test in special inspector's presence on first deck panel installed:
         a) Place one end of decking panel over perimeter support and attach only to that support with two welds as specified six inches apart.
         b) Move opposite end of decking panel in plane parallel to span of decking panel until shear distress is observed or apparent in weld.
         c) Provide welds of sufficient quality to cause local distortions in decking panel around welds and show good perimeter contact between welds and decking panel.
         d) When results of test are satisfactory and approved by special inspector, install remaining decking using same weld rod size and type, welding amperage setting, and procedures used in the tested deck.
         e) Weld test shall be performed for each metal deck welder.
      2) Visually inspect remainder of welds.
         a) When, in special inspector's opinion, one or more welds are defective or of insufficient quality, provide additional, acceptable weld adjacent to rejected weld.
         b) Provide new weld on sound, unburned decking sufficient distance from rejected weld.

B. Defective Work:
1. Metal decking Work will be deemed defective Work and will be rejected by Contracting Officer when it:
   a. Do not comply with SDI requirements and recommendations.
   b. Does not comply with the Contract Documents.
   c. Is improperly: manufactured, finished, and installed.
   d. Is damaged resulting in impaired strength, defacing, or inappropriate appearance.

2. Remedy defective Work in accordance with the Contract Documents. Replace damaged decking with new decking in accordance with the Contract Documents
3. Remove and replace decking that:
   a. Is structurally weak or unsound.
   b. Has burn holes due to improper welding; or
   c. Is deemed defective Work by Contracting Officer (unless Contracting Officer expressly allows repair or other remedy).

END OF SECTION

SECTION 05 50 00
METAL FABRICATIONS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Custom fabricated metal items and certain manufactured units not otherwise indicated to be supplied under work of other Specification Sections.
      a. Bollards.
      b. Loose Lintels.
      c. Exterior Equipment Screens.
   2. Delegated design systems and components, including but not limited to:
      a. Aluminum Grating.
      b. Aluminum Non-Punched Planks
      c. Ladders.
      d. Ladder gates.
      e. Modular framing system.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 01 35 73 – Delegated Design Procedures.
   3. Section 03 15 19 - Anchorage to Concrete.
   4. Section 03 31 30 - Concrete, Materials and Proportioning.
   5. Section 03 31 31 – Concrete Mixing, Placing, Jointing, and Curing.
   6. Section 05 12 00 - Structural Steel.
   7. Section 05 14 00 - Structural Aluminum.
   8. Section 05 52 46 – Mechanically Fastened Aluminum Railings
   9. Section 06 82 00 - Fiberglass Reinforced Plastic Fabrication.
   10. Section 09 06 00 – Color Schedule.
   11. Section 09 96 00 - High Performance Industrial Coatings.

1.2 REFERENCES

A. Definitions
   1. Fasteners: As defined in ASTM F1789.
   2. Galvanizing: Hot-dip galvanizing per ASTM A123/A123M or ASTM A153/A153M with minimum coating of 2.0 oz of zinc per square foot of metal (average of specimens) unless noted otherwise or dictated by standard.
   3. Hardware: As defined in ASTM A153/A153M.
   4. Installer or Applicator:
      a. Installer or applicator is the person actually installing or applying the product in the field at the Project site.
      b. Installer and applicator are synonymous.
   5. PQR: Procedure Qualification Record
   6. MIC: Microbiologically induced corrosion.

B. Referenced Standards: Standards referenced in this section include, but are not necessarily limited to, the following:
   1. Aluminum Association (AA):
      a. ADM 1, Aluminum Design Manual.
   2. American Association of State Highway and Transportation Officials (AASHTO):
      a. HB, Standard Specifications for Highway Bridges.
   3. American Institute of Steel Construction (AISC):
      a. 325, Manual of Steel Construction.

b. 360, Specifications for Structural Steel Buildings (referred to herein as AISC Specification).

4. The American Ladder Institute (ALI):
   a. A14.3, Ladders - Fixed - Safety Requirements.
5. American Society of Civil Engineers (ASCE):
   a. 7, Minimum Design Loads for Buildings and Other Structures.
6. ASTM International (ASTM):
   a. A6, Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling.
   b. A36, Standard Specification for Carbon Structural Steel.
   c. A47, Standard Specification for Ferritic Malleable Iron Castings.
   d. A48, Standard Specification for Gray Iron Castings.
   e. A53, Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
   f. A108, Standard Specification for Steel Bar, Carbon and Alloy, Cold Finished.
   g. A123/A123M, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
   h. A153/A153M, Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
   i. A197, Standard Specification for Cupola Malleable Iron.
   j. A269, Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service.
   k. A276, Standard Specification for Stainless Steel Bars and Shapes.
   l. A307, Standard Specification for Carbon Steel Bolts and Studs, 60,000 psi Tensile Strength.
   m. A312, Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes.
   n. A380/A380M, Standard Practice for Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment, and Systems.
   o. A500, Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes.
   p. A501, Standard Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing.
   q. A536, Standard Specification for Ductile Iron Castings.
   r. A554, Standard Specification for Welded Stainless Steel Mechanical Tubing.
   s. A572, Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel.
   t. A563, Standard Specification for Carbon and Alloy Steel Nuts.
   u. A666, Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar.
   v. A668, Standard Specification for Steel Forgings, Carbon and Alloy, for General Industrial Use.
   w. A780, Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings.
   x. A786, Standard Specification for Hot-Rolled Carbon, Low-Alloy, High-Strength Low-Alloy, and Alloy Steel Floor Plates.
   y. A992, Standard Specification for Steel for Structural Shapes.
   z. A1064, Standard Specification for Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete.
   aa. A1011, Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength.
   bb. B26, Standard Specification for Aluminum-Alloy Sand Castings.
   cc. B209, Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate.

dd. B221, Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes.
ee. B308, Standard Specification for Aluminum-Alloy 6061-T6 Standard Structural Profiles.
ff. B429, Standard Specification for Aluminum-Alloy Extruded Structural Pipe and Tube.
gg. B632, Standard Specification for Aluminum-Alloy Rolled Tread Plate.

hh. F436, Standard Specification for Hardened Steel Washers Inch and Metric Dimensions.
ii. F467, Standard Specification for Nonferrous Nuts for General Use.
jj. F468, Standard Specification for Nonferrous Bolts, Hex Cap Screws, and Studs for General Use.
kk. F593, Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.
ll. F835, Standard Specification for Alloy Steel Socket Button and Flat Countersunk Head Cap Screws.
mm. F879, Standard Specification for Stainless Steel Socket Button and Flat Countersunk Head Cap Screws.
nn. F1789, Standard Terminology for F16 Mechanical Fasteners.
oo. F3125, Standard Specification for High Strength Structural Bolts, Steel and Alloy Steel, Heat Treated, 120 ksi (830 MPa) and 150 ksi (1040 MPa) Minimum Tensile Strength, Inch and Metric Dimensions.

7. American Welding Society (AWS):
   a. A5.1/A5.1M, Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding.
   b. D1.1, Structural Welding Code - Steel.
   c. D1.2, Structural Welding Code - Aluminum.
   d. D1.6/D1.6M, Structural Welding Code - Stainless Steel.
8. National Association of Architectural Metal Manufacturers (NAAMM):
   a. AMP 555, Code of Standard Practice for the Architectural Metal Industry (Including Miscellaneous Iron).
   b. MBG 531, Metal Bar Grating Manual.
9. NACE International (NACE).
10. Nickel Development Institute (NiDI):
    a. Publication 11 007, Guidelines for the Welded Fabrication of Nickel-Containing Stainless Steels for Corrosion Resistant Services.
11. Occupational Safety and Health Administration (OSHA):
    a. 29 CFR 1910, Occupational Safety and Health Standards, referred to herein as OSHA Standards.

1.3 SUBMITTALS

A. Action Submittals: Submit the following:
   1. Shop Drawings:
      a. Fabrication and/or layout drawings and details:
         1) Submit drawings for all fabrications and assemblies.
            a) Include erection drawings, plans, sections, details and connection details.
         2) Identify materials of construction, shop coatings and third-party accessories.
   2. Product technical data (for fabrications not delegated design) including:
      a. Acknowledgement that products submitted meet requirements of the referenced standards.
      b. Provide Manufacturer’s data and information on all products specified.
      c. Manufacturer's installation instructions.
      d. Provide manufacturer's standard allowable load tables for the following:
         1) Grating and checkered plate.
         2) Castings, trench covers and accessories.
         3) Modular framing systems.
   3. Delegated Design Professional’s “Instruments of Service” Submittals:

a. Design drawings sealed and signed by delegated design professional, including:
    1) Design drawings deemed appropriate and necessary by delegated design professional.
    2) Show or indicate necessary details establishing the method of fastening throughout system and attachments to supporting primary structure including anchorage to concrete as required.
    3) Coordinate drawings and details with delegated design professional's calculations required in this Section.
    4) At option of Contractor or fabricator, design drawings may be combined with the Shop Drawings required in this Section, although delegated design professional's seal and signature is required on the Shop Drawings in the event of a combined Submittal.
b. Design drawings and specifications shall sealed and signed by delegated design professional licensed in the State of Arizona.
c. Certification of compliance in accordance with Contract Documents, including:
    1) List of design loads and load combinations used to design structural systems and components.
    2) List of loads transmitted to the supporting structural elements.
    3) Compliance with project performance and design criteria specified in the Contract Documents.
    4) Expressly indicate where structural element reaction forces to designed supporting elements, verify reaction forces indicated on Contract Documents, or existing construction differ from reaction forces indicated in the Contract Documents.

B. Color Samples:
1. Refer to "Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals" for submission procedures and requirements of Color Samples.
2. Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.

C. Informational Submittals:
1. Certifications
    a. NACE certification of surface preparation.
    b. NACE certification of paint application.
2. Supplier's Instructions:
    a. Fabricator's instructions for handling, storing, and installing structural elements furnished.
    b. Galvanizing repair paint manufacturer's installation instructions.
3. Qualifications Statements:
    a. NACE inspector qualifications.
    b. Delegated design professional.
    c. Fabricator, when requested by Contracting Officer Representative, including:
        1) Qualifications and experience.
        2) Qualifications of welders to be employed in fabricating the structural elements for the Project.
        3) Qualifications of welders to be employed during erection of the structural elements for the Project.
    d. Installer, when requested by Contracting Officer Representative, including fabricator's approval of installer.
4. Delegated design Shop Drawings approved by delegated design professional:
    a. Detailed structural elements erection drawings, shop details, erection detail.
    b. Indicate member type, location, spacing, size, and gauge of members; method of attachment to supporting members; and necessary erection details.
    c. Show and indicate supplemental bracing, strapping, splices, bridging, accessories, and details necessary for proper installation.

d. Structural element Placement Diagram: Diagram showing and indicating assumed location of each structural element and references the corresponding structural element design drawing.

5. Delegated design product data approved by delegated design professional:
   a. Fabricator's published product data for each type of structural element and accessory required.
   b. Expressly indicate compliance with applicable reference standards. If not addressed in fabricator's published information, submit as special certification addressing compliance with reference standards indicated in this Section.
   c. Published product data for galvanizing repair paint.
6. Delegated design calculations in accordance with Contract Documents:
   a. Complete set of design calculations, indicating:
      1) Basis of design, including a list of codes and standards used for design.
      2) List of design loads and load combinations.
      3) Design moments, shear forces, and axial compression/tension forces.
      4) Design of connections, including splices, if required.
      5) Bridging and bracing design, including attachments to trusses and anchorage to supporting structural elements, if required.
      6) Special requirements, if any.

1.4 QUALITY ASSURANCE

A. Regulatory Requirements:
   1. Applicable construction codes including building code, are indicated in Section 01 42 00 - References.

B. Qualifications:
   1. Qualify welding procedures and welding operators in accordance with AWS.
      a. All welders shall be qualified in all positions that will be utilized during welding. Welders shall utilize qualified PQR for testing.
   2. Fabricator shall have minimum of 10 years of experience in fabrication of metal items specified.
   3. Engineer for contractor-designed systems and components: Professional structural engineer licensed in the State of Arizona
   4. NACE certified inspector shall have minimum of two years of experience performing inspections as indicated.
      a. Have a current Level III coating inspector certification.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver and handle fabrications to avoid damage.

B. Store above ground on skids or other supports to keep items free of dirt and other foreign debris and to protect against corrosion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.
   2. Headed studs and deformed bar anchors:
      a. Nelson Stud Welding Div., TRW Inc.
      b. Stud Welding Products, Inc.

3. Mechanical anchor bolts:
    a. See Section 03 15 19.
4. Epoxy adhesive anchor bolts:
    a. See Section 03 15 19.
5. Concrete screw anchors:
    a. See Section 03 15 19.
6. Aluminum Grating:
    a. McNichols
    b. Or Equal.
7. Aluminum Non-Punched Planks:
    a. IKG
    b. Or Equal.
8. Aluminum ladders:
    a. O'Keeffe's Inc. – Model 501.
    b. Or Equal.
9. Ladder safety extension posts:
    a. Bilco - "LadderUp" safety post.
    b. Or Equal.
10. Ladder safety system:
    a. Miller Equipment - "Saf-T-Climb" system.
    b. Or Equal.
11. Exterior Equipment Screens:
    a. PalmSHIELD
    b. Or Equal.
12. Galvanizing repair paint:
    a. Aervoe Industries, Inc.
    b. ZRC Products.
13. Modular framing system:
    a. Unistrut Building Systems.
    b. B-Line Systems.
    c. Kindorf.
    d. Superstrut.

2.2 MATERIALS

A. Steel:
1. Structural:
    a. W-shapes and WT-shapes: ASTM A992, Grade 50.
    b. All other plates and rolled sections: ASTM A36.
2. Pipe: ASTM A53, Types E or S, Grade B or ASTM A501.
3. Structural tubing:
    a. ASTM A500, Grade B (46 ksi minimum yield).
4. Bolts, high strength:
    a. ASTM F3125, Grade A325.
5. Nuts, high strength:
    a. ASTM A563.
6. Washers (hardened):
    a. ASTM F436.
    b. Provide two (2) washers with all bolts.
7. Bolts and nuts (unfinished):
    a. ASTM A307, Grade A.
8. Welding electrodes: AWS D1.1, E70 Series.
9. Steel forgings: ASTM A668.

B. Iron:
1. Ductile iron: ASTM A536.
2. Gray cast iron: ASTM A48 (minimum 30,000 psi tensile strength).

3. Malleable iron: ASTM A47, ASTM A197.

C. Stainless Steel:
   1. Stainless steel in welded applications: Low carbon 'L' type.
   2. Minimum yield strength of 30,000 psi and minimum tensile strength of 75,000 psi.
      a. Bars, shapes: ASTM A276, Type 304.
      b. Tubing and pipe: ASTM A269, ASTM A312 or ASTM A554, Type 304 or 316.
      c. Strip, plate and flat bars: ASTM A666, Type 304 or 316.
      d. Bolts and nuts: ASTM F593, Type 304 or 316.
   3. Minimum yield strength of 25,000 psi and minimum tensile strength of 70,000 psi.
      a. Strip, plate and flat bar for welded connections, ASTM A666, Type 304L or 316L.
   4. Welding electrodes: In accordance with AWS for metal alloy being welded.

D. Aluminum:
   1. Alloy 6061-T6 or Alloy 6061-T6, 32,000 psi tensile yield strength minimum.
      a. ASTM B221 and ASTM B308 for shapes including beams, channels, angles, tees and zees.
      b. Weir plates, baffles and deflector plates, ASTM B209.
   2. Alloy 6063-T5 or T6, 15,000 psi tensile yield strength minimum.
      a. ASTM B221 and ASTM B429 for bars, rods, wires, pipes and tubes.
   3. ASTM B26 for castings.
   4. ASTM F468, alloy 2024 T4 for bolts.
   5. ASTM F467, alloy 2024 T4 for nuts.
   6. Electrodes for welding aluminum: AWS D1.2, filler alloy 4043 or 5356.
   7. Sheet: Alloy 5005-H3 to comply with ASTM B209.

E. Washers: Same material and alloy as found in accompanying bolts and nuts.

F. Embedded Anchor Bolts:
   1. See Specification Section 03 15 19.

G. Mechanical Anchor Bolts and Adhesive Anchor Bolts:
   1. See Specification Section 03 15 19.

H. Headed Studs: ASTM A108 with a minimum yield strength of 50,000 psi and a minimum tensile strength of 60,000 psi.

I. Deformed Bar Anchors: ASTM A1064 with a minimum yield strength of 70,000 psi and a minimum tensile strength of 80,000 psi.

J. Iron and Steel Hardware: Galvanized in accordance with ASTM A153/A153M when required to be galvanized.

K. Galvanizing Repair Paint:
   1. High zinc dust content paint for regalvanizing welds and abrasions.
   2. ASTM A780.
   3. Zinc content: Minimum 92% in dry film.
   4. ZRC "ZRC Cold Galvanizing" or Aervoe " Crown Cold Galvanize Coating 93% Zine Rich."

L. Dissimilar Materials Protection: See Specification Section 09 96 00.

2.3 MANUFACTURED UNITS

A. Ladders:
   1. General:
      a. All ladders of a particular material shall have consistent construction and material shapes and sizes unless noted otherwise on the Drawings.
      b. Design ladder in accordance with OSHA Standards, ANSI A14.3, ASCE 7 and the building code.
      c. Ladders shall be designed to support a minimum concentrated live load of 300 pounds at any point to produce the maximum stress in the member being designed.

1) Apply additional 300 pound loads for each section of ladder exceeding 10 feet.
d. Maximum allowable stresses per AA ADM 1.
e. Maximum lateral deflection: Side rail span/240 when lateral load of 100 pounds is applied at any location.

2. Material:
a. Aluminum.
1) Finish: Mill.

3. Brackets:
a. Angle or bent plate brackets welded to side rails:
1) 3/16-inch thick extruded aluminum bracket on each side of ladder.
a) "Z" shaped brackets shall be minimum 9-inches high to fasten to wall and back side of rails with fasteners/bolts per manufacturer.
2) Provide punched holes per manufacturers standards for bolts or anchors.
3) Minimum distance from centerline of rung to wall or any obstruction: 7-inches.
4) Maximum spacing: 4 feet on center.
b. Floor supported ladders shall have 3/16-inch thick aluminum angles 4-inch by 1-3/8-inch legs by 2-inch wide fastened to inside face of rails and to floors.
1) Provide one on each side of ladder rails.
2) Provide wall brackets on floor supported units if vertical run is over 4 feet.

4. Ladder safety extension post:
a. Telescoping tubular aluminum section that automatically locks into place when fully extended.
b. Non-ferrous corrosion-resistant spring and hardware.
c. Factory assembled with all hardware necessary for mounting to ladder.

5. Ladder safety system:
a. Extruded aluminum safety rail.
1) Safety rail shall extend from within 3-feet of base of ladder to top of ladder side rails.
2) Provide rung clamps at not more than 4-feet on center.
3) Provide dismount extensions as appropriate for application.
b. Aluminum shuttle.
1) Provide (2) shuttles per ladder.
c. Full body harness with adjustable leg straps, backpad, and front dee ring.
1) Provide (2) harnesses per ladder.
d. Provide all components required for a complete OSHA compliant ladder safety system.

B. Bollards:
1. 8 inches diameter extra strength steel pipe, ASTM A53.
a. Galvanized.
b. See Specification Section 09 96 00 for painting requirements.
c. Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.

C. Aluminum Grating:
1. NAAMM MBG 531.
2. Minimum depth: 1-1/2 inches.
3. Minimum rectangular bearing bar size:
a. 3/16 inches thick.
b. Maximum 1-3/16 inches on-center spacing.
4. Minimum I-bar flange width: 1/4 inches.
5. For removeable panels:
a. Weight of panel not to exceed 50 lbs.
6. Design live load, unless noted otherwise on Drawings:
a. 100 psf, uniform load.
b. 300 pounds concentrated load on 4 inches square area.
c. All components to be adequate for the uniform load or the concentrated load, whichever requires the stronger component.

d. Maximum deflection: 1/300 of span under a superimposed live load of 50 psf.
7. Cross bars:
   a. Welded, swaged or pressure locked to bearing bars.
   b. Maximum 4 inches on-center spacing.
8. Top edges of bars: Grooved or serrated.
9. Removable grating sections: Not wider than 3 feet and not more than 100 pounds.
   a. Provide (2) flush lifting handles for each section designated as removable.
10. Standard mill finish.
11. Ends and perimeter edges: Banded.
12. Openings through grating: Reinforced to provide required load carrying capacity and banded with 4 inches high toe plate.
13. Provide openings at joints between individual grating sections.
14. Clips and bolts: Stainless steel.
15. Seat angles: Aluminum.
16. At Contractor's option, aluminum plank type grating with skid-resistant surface may be used.
    a. Grating to have a minimum 35 percent up to a maximum of 45 percent open area and meet the design loads and deflection specified in this Specification Section.

D. Aluminum Non – Punched Plank:
1. Minimum depth: 1-1/2 inches.
2. Standard length: 26 feet.
3. Standard width: 6 inches.
4. Material: Extruded Aluminum 6063 per ASTM B221
   a. Fs = 12,000 psi
   b. E = 10,000,000 psi
5. Bearing bars spaced 1.2 inches on center.
6. Ten (10) bearing bars per foot of grating.
7. Seat angles: Aluminum.

E. Loose Lintels:
1. Steel, ASTM A36 or ASTM A572 Grade 50, sizes as indicated on Drawings.
2. Hot-dip galvanized per ASTM A123/A123M.

F. Modular Framing System:
1. Materials:
   a. Steel: ASTM A1011, carbon steel, Grade 33.
      1) Hot-dipped galvanized, ASTM A123 or ASTM A153.
   b. Aluminum: ASTM B221 or ASTM B209.
   c. Stainless steel: ASTM A666.
   d. Fiberglass: See Specification Section 06 82 00.
2. Channels and inserts:
   a. Steel or stainless steel: Minimum 12 GA.
   b. Aluminum: Minimum 0.080 inches.
   c. Channels to have one side with a continuous slot with in-turned lips.
      1) Width: 1-5/8 inches.
      2) Depth and configuration as necessary for loading conditions.
3. Fittings: Same material as system major components.
4. Fasteners:
   a. Nuts: Toothed groves in top of nuts to engage the in-turned lips of channel.
   b. Bolts: Hex-head cap screws.
   c. Same material as system major components.
5. End caps:
   a. At each exposed end of each piece mounted on walls, or guardrails, or suspended from framing 7 feet or less above the floor or platform.
      a) Plastic for all exposed ends 7 feet or more above floor or platform.
      b) Plastic or metallic for all other exposed ends.

6. Schedule:
    a. Interior wet areas: Stainless steel.
        1) Including the following rooms or areas:
            a) Water Treatment Room
    b. Interior corrosive areas: Stainless steel.
        1) Including the following rooms or area:
            a) Water Treatment Room
    c. Exterior areas: Stainless steel.
    d. All other areas not listed above: Hot-dipped galvanized steel.
7. Provide dissimilar materials protection in accordance with Specification Section 09 96 00.
8. Repair all cut ends or otherwise damaged areas of galvanized steel in accordance with ASTM A780.

G. Exterior Equipment Screen:
1. Louvered exterior equipment screening metal louvers for providing ventilation while keeping items out of sight.
2. Exterior equipment screens shall be configured as a series of screens, and swing gates.
3. Exterior equipment screens are custom fabricated with 0.125-inch-thick extruded aluminum sloped louvers placed horizontally with a profile of 2-inches x 2.873-inches tall.
    a. Each louver is welded to 3-inch x 3-inch x 1/4-inch-thick aluminum angle vertical supports.
    b. Each louvered panel shall have a 3-inch x 3-inch x 1/4-inch-thick aluminum angle horizontal top and bottom cap. The horizontal top and bottom cap are welded to the vertical supports for a picture frame design.
    c. Each louver panel shall have a 1-inch x 1/4-inch-thick aluminum flat bar not to exceed 24-inches to support the louvers between the frameworks. The support is to prevent intruders from spreading the louvers apart.
    d. Posts shall be 4-inch x 4-inch square x 3/8-inch-thick extruded tubular aluminum section wall posts with solid aluminum caps
    e. Hinges shall be the manufacturer’s extra heavy duty barrel hinges. Each hinge shall be 3/4-inch rod, ball bearing, and grease zert. Hinge plates to be 1/2-inch thick plates offset to create a 5/8-inch gap. Hinges shall be bolted to gate frame and field welded to steel gate posts.
    f. Gate Latch:
        1) Provide with an internal lock with exterior grab handles. Lock to be keyed to existing Grand Canyon National Park keying system. Existing System is a Figure 8, small format, WH Keyway.
    g. Gate Drop Rods:
        1) Provide each gate with a 1-inch diameter center cane bolt assembly and strike.
4. Fasteners:
    a. All nuts, bolts and fasteners are stainless steel.
5. Finish:
    a. Manufacturer shall sweep blast all materials and provide a polyester powder coating system. electrostatically applied.
        1) Minimum Hardness – ASTM D3363 (2H).
        2) Direct Impact Resistance – ASTM D2794 (Withstand 160-inch-pounds).
        3) Weatherability – ASTM D822 (No film failure and 88 percent gloss retention after 1 year exposure in South Florida with test panels tilted 45 degrees).
    b. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

2.4 FABRICATION

A. Verify field conditions and dimensions prior to fabrication.

B. Form materials to shapes indicated with straight lines, true angles, and smooth curves.
1. Grind smooth all rough welds and sharp edges.

a. Round all corners to approximately 1/32 - 1/16 inches nominal radius.

C. Provide drilled or punched holes with smooth edges.
   1. Punch or drill for field connections and for attachment of work by other trades.

D. Weld Shop Connections:
   1. Stainless steel connections subject to corrosive environments or immersion service will be subject to the following enhanced visual and welding requirements:
      a. Fabricator
         1) Fabricator shall have QC program and procedures in place to avoid cross contamination with carbon steel products and material.
      b. Weld Procedures
         1) Weld procedures shall be qualified for all positions that will be welded, qualified by testing in accordance with applicable welding code AWS D1.6 requirements.
         2) Procedure Qualification Record (PQR) shall document all essential variables to perform consistent quality welds. PQR shall include maximum heat input per pass and shall be tested for CVN properties, macro hardness, and macro/micro hardness testing.
   2. Welds shall be continuous fillet type unless indicated otherwise.
   3. Full penetration butt weld at bends in ladder side rails.
   4. Weld structural steel in accordance with AWS D1.1 using Series E70 electrodes conforming to AWS A5.1/A5.1M.
   5. Weld aluminum in accordance with AWS D1.2.
   6. Weld stainless steel in accordance with AWS D1.6.
   7. All headed studs to be welded using automatically timed stud welding equipment.
   8. Grind smooth welds that will be exposed.

E. Make provisions to prevent carbon steel/free iron contamination of stainless steel surfaces (i.e., contact between carbon steel or iron and stainless steel component). Do not use cutting/grinding/drilling tools, wire brushes, or wire wheels on stainless steel that have been previously used on carbon steel or iron. Do not support or hang stainless steel weldments or fixtures with carbon steel cables or dunnage.

F. Passivate stainless steel items and stainless steel welds after they have been ground smooth, in accordance with ASTM A380/A380M.

G. Passivation Requirements:
   1. For components/assemblies to be used only in dry air service environments:
      a. No post-fabrication passivation treatments are required for external corrosion control.

H. Conceal fastenings where practicable.

I. Fabricate work in shop in as large assemblies as is practicable.

J. Tolerances:
   1. Rolling:
      a. ASTM A6.
      b. When material received from the mill does not satisfy ASTM A6 tolerances for camber, profile, flatness, or sweep, the Contractor is permitted to perform corrective work by the use of controlled heating and mechanical straightening, subject to the limitations of the AISC Specification.
   2. Fabrication tolerance:
      a. Member length:
         1) Both ends finished for contact bearing: 1/32 inches.
         2) Framed members:
            a) 30 feet or less: 1/16 inches.
            b) Over 30 feet: 1/8 inches.
      b. Member straightness:
         1) Compression members: 1/1000 of axial length between points laterally supported.

2) Non-compression members: ASTM A6 tolerance for wide flange shapes.

c. Specified member camber (except compression members):
   1) 50 feet or less: -0/+1/2 inches.
   2) Over 50 feet: -0/+1/2 inches (+1/8 inches per 10 feet over 50 feet).
   3) Members received from mill with 75% of specified camber require no further cambering.
   4) Beams/trusses without specified camber shall be fabricated so after erection, camber is upward.
   5) Camber shall be measured in fabrication shop in unstressed condition.

d. At bolted splices, depth deviation shall be taken up by filler plates.
   1) At welded joints, adjust weld profile to conform to variation in depth.
   2) Slope weld surface per AWS requirements.

e. Finished members shall be free from twists, bends and open joints.
   1) Sharp kinks, bends and deviation from above tolerances are cause for rejection of material.

K. Fabricate grating, checkered plate, ladders and accessories using aluminum unless shown otherwise on Drawings.
   1. Finish:
      a. Mill, unless noted otherwise.
      b. Coat surfaces in contact with dissimilar materials.
         1) See Specification Section 09 96 00.

L. Fabricate grating in accordance with NAAMM MBG 531.
   1. Maximum tolerance for difference in depth between grating depth and seat or support angle depth: 1/8 inches.
   2. Distance between edge of grating and face of embedded seat angle or face of wall or other structural member: 1/4 inches.
      a. Tolerance: NAAMM MBG 531.
   3. Removable sections: Not wider than 3 feet and not heavier than 100 pounds.
   4. Ends and perimeter edges: Banded, with alternate bearing bars welded to band.
      a. Provide full depth banding unless noted otherwise.
      b. Banding at trenches and sumps to be 1/4 inches less than grating depth to allow for drainage.
   5. Openings through grating: Reinforced to provide required load carrying capacity and banded with 4 inches high toe plate.
   6. Provide joints at openings between individual grating sections.
   7. Fabricate grating so that bearing bars and cross bars in adjacent sections are aligned.

M. Fabricate checkered plate and miscellaneous metals in accordance with NAAMM AMP 555.
   1. Workmanship: Class 2 unless noted otherwise.

N. See Specification Section 09 96 00 for preparation and painting of ferrous metals and other surfaces.

2.5 SOURCE QUALITY CONTROL

A. Surface Preparation:
   1. Refer to Specification Section 09 96 00 for surface preparation requirements.
   1. All miscellaneous metal fabrication item surfaces shall be observed and approved, prior to application of shop-applied coatings, by a NACE Certified Coatings Specialist (CIP-3) with at least 4 years of experience in similar inspections.
      a. Inspection shall be performed to determine depth of blast profile and cleanliness of surface.
      b. Fabricator shall reblast and or re-clean surfaces as required until acceptable.

B. Shop Applied Coating Application:
   1. Refer to Specification Section 09 96 00 for coating requirements.

2. After surface has been accepted in writing by NACE certified coatings inspector, fabricator may proceed with application of coatings.
3. Application of coatings shall be observed and certified by NACE certified coatings inspector.

C. Shop Inspection and Testing:
1. Contractor will employ and pay for the services of a qualified independent testing agency to inspect and test all structural steel work for compliance with Contract Documents.
2. Contractor responsible for testing to qualify shop and field welders and as needed for Contractor's own quality control to ensure compliance with Contract Documents.
3. Independent testing agency shall have a minimum of five years performing similar work and shall be subject to Government's approval.

D. Responsibilities of Testing Agency:
1. Inspect shop and field welding in accordance with AWS Code including the following non-destructive testing:
a. Visually inspect all welds.
b. In addition to visual inspection, test 50% of full penetration welds and 20% of fillet welds with liquid dye penetrant or mag particle.
c. Test 20% of liquid dye penetrant tested full penetration welds with ultrasonic or radiographic testing.
d. For all surfaces intended to be immersed or intended to be water tight:
1) 100% of all fillet welds shall be PT tested to ensure water-tight surfaces are free of surface defects that would create areas for MIC.
2) 100% of all completed SS welds and material shall be passivated in accordance with ASTM A380/A380M.
2. Inspect high-strength bolting in accordance with the RCSC Specification for Structural Joints Using High-Strength Bolts, Section 9.
a. Verify direct tension indicator gaps, if applicable.
3. Inspect structural steel which has been erected.
4. Inspect stud welding in accordance with AWS Code.
5. Prepare and submit inspection and test reports to Contracting Officer Representative.
a. Assist Contracting Officer Representative to determine corrective measures necessary for defective work.

PART 3 - EXECUTION

3.1 PREPARATION

A. Provide items to be built into other construction in time to allow their installation.
1. If such items are not provided in time for installation, cut in and install.

B. Prior to installation, inspect and verify condition of substrate.

C. Correct surface defects or conditions which may interfere with or prevent a satisfactory installation.
1. Field welding aluminum is not permitted unless approved in writing by Contracting Officer Representative.

3.2 INSTALLATION

A. Set metal work level, true to line, plumb.
1. Shim and grout as necessary.

B. Contractor is solely responsible for safety.
1. Construction means and methods and sequencing of work is the prerogative of the Contractor.

2. Take into consideration that full structural capacity of many structural members is not realized until structural assembly is complete; e.g., until slabs, decks, and diagonal bracing or rigid connections are installed.
3. Partially complete structural members shall not be loaded without an investigation by the Contractor.
4. Until all elements of the permanent structure and lateral bracing system are complete, temporary bracing for the partially complete structure will be required.

C. Adequate temporary bracing to provide safety, stability and to resist all loads to which the partially complete structure may be subjected, including construction activities and operation of equipment is the responsibility of the Contractor.
1. Plumb, align, and set structural steel members to specified tolerances.
2. Use temporary guys, braces, shoring, connections, etc., necessary to maintain the structural framing plumb and in proper alignment until permanent connections are made, the succeeding work is in place, and temporary work is no longer necessary.
3. Use temporary guys, bracing, shoring, and other work to prevent injury or damage to adjacent work or construction from stresses due to erection procedures and operation of erection equipment, construction loads, and wind.
4. Contractor shall be responsible for the design of the temporary bracing system and must consider the sequence and schedule of placement of such elements and effects of loads imposed on the structural steel members by partially or completely installed work, including work of all other trades.
    a. If not obvious from experience or from the Drawings, confer with the Contracting Officer Representative to identify those structural steel elements that must be complete before the temporary bracing system is removed.
5. Remove and dispose of all temporary work and facilities off-site.

D. Examine work-in-place on which specified work is in any way dependent to ensure that conditions are satisfactory for the installation of the work.
1. Report defects in work-in-place which may influence satisfactory completion of the work.
2. Absence of such notification will be construed as acceptance of work-in-place.

E. Field Measurement:
1. Take field measurements as necessary to verify or supplement dimensions indicated on the Drawings.
2. Contractor responsible for the accurate fit of the work.

F. Check the elevations of all finished footings or foundations and the location and alignment of all anchor bolts before starting erection.
1. Use surveyor's level.
2. Notify Contracting Officer Representative of any errors or deviations found by such checking.

G. Framing member location tolerances after erection shall not exceed the frame tolerances listed in the FIELD QUALITY CONTROL Article in PART 3 of this Specification Section.

H. Erect plumb and level; introduce temporary bracing required to support erection loads.

I. Use light drifting necessary to draw holes together.
1. Drifting to match unfair holes is not allowed.

J. Welding:
1. Comply with AWS D1.1, AWS D1.2, and AWS D1.6 (as applicable for the material welded) and requirements of this Section’s “Fabrications” Article in “Part 2 - Products”.
2. When joining two sections of steel of different ASTM designations, welding techniques shall be in accordance with a qualified AWS D1.1 procedure.
3. 100% of all completed stainless steel welds and material shall be passivated in accordance with ASTM A380/A380M.

K. Shore existing members when unbolting of common connections is required.

1. Use new bolts for rebolting connections.

L. Clean stored material of all foreign matter accumulated prior to the completion of erection.

M. Bolt Field Connections: Where practicable, conceal fastenings.

N. Field Welding:
   1. Follow AWS procedures.
   2. Grind welds smooth where field welding is required.
   3. 100% of all completed stainless steel welds and material shall be passivated in accordance with ASTM A380/A380M.

O. Field cutting grating to correct fabrication errors is not acceptable.
   1. Replace entire section.

P. Remove all burrs and radius all sharp edges and corners of miscellaneous plates, angles, framing system elements, etc.

Q. Unless noted or specified otherwise:
   1. Connect steel members to steel members with 3/4 inches diameter ASTM F3125, Grade A325 high strength bolts.
   2. Connect aluminum to aluminum with 3/4 inches diameter stainless bolts.
   3. Connect aluminum to structural steel using 3/4 inches diameter stainless steel bolts.
      a. Provide dissimilar metals protection.
   4. Connect aluminum and steel members to concrete and masonry using stainless steel mechanical anchor bolts or adhesive anchor bolts unless shown otherwise.
      a. Provide dissimilar materials protection.
   5. Provide washers for all bolted connections.
   6. Where exposed, bolts shall extend a maximum of 3/4 inches and a minimum of 1/2 inches above the top of installed nut.
      a. If bolts are cut off to required maximum height, threads must be dressed to allow nuts to be removed without damage to the bolt or the nuts.

R. Install and tighten ASTM F3125, Grade A325 high-strength bolts in accordance with the AISC 325, Allowable Stress Design (ASD).
   1. Provide hardened washers for all Grade A325 bolts.
      a. Provide the hardened washer under the element (nut or bolt head) turned in tightening.

S. After bolts are tightened, upset threads of ASTM A307 bolts or anchor bolts to prevent nuts from backing off.

T. Secure metal to wood with lag screws of adequate size with appropriate washers.

U. Do not field splice fabricated items unless said items exceed standard shipping length or change of direction requires splicing.
   1. Provide full penetration welded splices where continuity is required.

V. Provide each fabricated item complete with attachment devices as indicated or required to install.

W. Anchor such that work will not be distorted nor fasteners overstressed from expansion and contraction.

X. Set beam and column base plates accurately on nonshrink grout as indicated on Drawings.
   1. See Division 03 Specification Sections for non-shrink grout and anchorage.
   2. Set and anchor each base plate to proper line and elevation.
      a. Use metal wedges, shims, or setting nuts for leveling and plumbing columns and beams.
         1) Wedges, shims and setting nuts to be of same metal as base plate they support.
         2) Tighten nuts on anchor bolts.
      b. Fill space between bearing surface and bottom of base plate with nonshrink grout.

1) Fill space until voids are completely filled and base plates are fully bedded on wedges, shims, and grout.

c. Do not remove wedges or shims.
   1) Where they protrude, cut off flush with edge of base plate.

d. Fill sleeves around anchor bolts solid with non-shrink grout.

Y. Tie anchor bolts in position to embedded reinforcing steel using wire.
   1. Tack welding prohibited.
      a. Coat projecting bolt threads and nuts with heavy coat of clean grease.
   2. Anchor bolt location tolerance:
      a. Per Section 03 15 19.

Z. Install bollards as detailed on Drawings.
   1. Fill pipe with concrete and round off at top.

AA. Accurately locate and place frames for openings or for floor grating angle supports before casting into floor slab so top of plate is flush with surface of finished floor.
   1. Keep screw holes clean and ready to receive screws.
   2. Set framing and supports level to prevent rocking of covers, grating and floor planks. Shims are to installed only where necessary.

BB. Attach grating to end and intermediate supports with grating saddle clips and bolts.
   1. Maximum spacing: 2 feet on-center with minimum of two per side.
   2. Attach individual units of aluminum grating together with clips at 2 feet on-center maximum with a minimum of two clips per side.

CC. Coat aluminum surfaces in contact with dissimilar materials in accordance with Specification Section 09 96 00.

DD. Repair damaged galvanized surfaces in accordance with ASTM A780.
   1. Prepare damaged surfaces by abrasive blasting or power sanding.
   2. Apply galvanizing repair paint to minimum 6 mils DFT in accordance with manufacturer's instructions.

EE. Anchor ladder to concrete structure with minimum 3/4-inch stainless steel anchor bolts with minimum 6-inches embedment.

FF. Install ladder safety extension post in accordance with manufacturer's instructions.
   1. Mount device opposite the climbing side.
   2. Provide ladder safety extension device for all ladders unless noted otherwise.

GG. Mount ladder fall protection system with rail offset from ladder side rail approximately 3-inches.

HH. Install loose lintels as shown on drawings.

II. Install Exterior Equipment Screen in accordance with the manufacturer’s written instructions:
   1. Minimum requirements - posts to be anchored to concrete slab with 10-inch x 10-inch x 3/4-inch aluminum base plates welded to bottom of posts and anchored with four (4) 3/4-inch epoxy anchors into concrete slab.

3.3 FIELD QUALITY CONTROL

A. Tolerances shall meet structural requirements of Specification Section 05 12 00 for erecting items of structural nature.

B. Tolerances (unless otherwise noted on the Drawings):
   1. Frame placement, after assembly and before welding or tightening.
      a. Deviation from plumb, level and alignment: 1 in 500, maximum.
      b. Displacement of centerlines of columns: 1/2-inches maximum, each side of centerline location shown on Drawings.

3.4 CLEANING

A. After fabrication, erection, installation, or application, clean all miscellaneous metal fabrication surfaces of all dirt, weld slag and other foreign matter.

B. Provide surface acceptable to receive field applied paint coatings specified in Specification Section 09 96 00.

END OF SECTION

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SECTION 05 52 46
MECHANICALLY FASTENED ALUMINUM RAILINGS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Aluminum guardrail.
   2. Aluminum guardrail gates.

B. Delegated design systems and components, including but not limited to:
   1. Mechanically fastened aluminum railings and accessories.

C. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 35 73 – Delegated Design Procedures.
   2. Section 03 15 19 – Anchorage to Concrete.
   3. Section 05 50 00 - Metal Fabrications.
   4. Section 09 96 00 - High Performance Industrial Coatings.

1.2 REFERENCES

A. Referenced Standards:
   1. Aluminum Association (AA):
      a. ADM 1, Aluminum Design Manual.
   2. ASTM International (ASTM):
      a. B108, Standard Specification for Aluminum-Alloy Permanent Mold Castings.
      b. B209, Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate.
      c. B221, Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes.
      d. B247, Standard Specification for Aluminum and Aluminum-Alloy Die Forgings, Hand Forgings, and Rolled Ring Forgings.
      e. B308, Standard Specification for Aluminum-Alloy 6061-T6 Standard Structural Profiles.
      f. B429, Standard Specification for Aluminum-Alloy Extruded Structural Pipe and Tube.
   3. Fenestration and Glazing Industry Alliance (FGIA):
      a. AAMA 609.1, Cleaning and Maintenance Guide for Architecturally Finished Aluminum.
   4. National Association of Architectural Metal Manufacturers (NAAMM):
      a. AMP 521, Pipe Railing Systems Manual.
   5. U.S. Department of Justice, Architectural and Transportation Barriers Compliance Board (Access Board):
      a. Americans with Disabilities Act (ADA):
         1) Accessibility Guidelines for Buildings and Facilities (ADAAG).
   6. Occupational Safety and Health Administration (OSHA):
      a. 29 CFR 1910, Occupational Safety and Health Standards, referred to herein as OSHA Standards.

1.3 DEFINITIONS

A. Guardrail: A system of building components located near the open sides of elevated walking surfaces for the purpose of minimizing the possibility of an accidental fall from the walking surface to the lower level.

B. Railing: Generic term referring to guardrail.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Fabrication and/or layout drawings:
      a. Drawings showing profile, location, and fabrication details.
      b. Type and details of anchorage.
      c. Location and type of expansion joints.
      d. Materials of construction, shop coatings and all third-party accessories.

B. Product technical data including:
   1. Acknowledgement that products submitted meet requirements of standards referenced.
   2. Manufacturer's installation details.

C. Certifications:
   1. Provide certifications that railings have been designed and fabricated to meet the loading requirements specified.

D. Delegated Design Professional's "Instruments of Service" Submittals:.
   1. Design drawings sealed and signed by delegated design professional, including:
      a. Design drawings deemed appropriate and necessary by delegated design professional.
   2. Show or indicate necessary details establishing the method of fastening throughout system and attachments to supporting primary structure including anchorage to concrete as required.
      a. Coordinate drawings and details with delegated design professional's calculations required in this Section.
      b. At option of Contractor or fabricator, design drawings may be combined with the Shop Drawings required in this Section, although delegated design professional's seal and signature is required on the Shop Drawings in the event of a combined Submittal.
      c. Design drawings and specifications shall be sealed and signed by delegated design professional licensed in the State of Arizona.
   3. Calculations:
      a. Calculations shall be performed, sealed, signed and dated by a registered professional structural engineer licensed in the State of Arizona
      b. Calculations shall be specific to this Project and shall include all assumptions, references and design interpretations used to achieve the results obtained by the Manufacturer's Professional Structural Engineer.
      c. Reduction in load criteria is not acceptable as reason for deviation from sizes indicated in the Specification.
   4. Refer to Section 01 35 73 – Delegated Design Procedures for additional requirements.

E. Cleaning and Maintenance Data:
   1. Provide cleaning and maintenance data per Voluntary Guide Specification for Cleaning and Maintenance of Architectural
      Anodized Aluminum (Publication No. AAMA 609.1-85)

1.5 QUALITY ASSURANCE

A. Manufacturer Qualifications: Minimum 5 years' documented experience producing systems specified in this section.

B. Field Measurements & Coordination:
   1. Take field measurements prior to preparation of shop drawings and fabrication.

C. Shop Assembly: Preassemble items in shop to the greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation.

1.6 DELIVERY, STORAGE AND HANDLING

A. Deliver and handle railings to preclude damage.

B. Store railings on skids, keep free of dirt and other foreign matter which will damage railings or finish and protect against corrosion.

1.7 WARRANTY:

A. Provide Government with manufacturer's standard limited warranty for materials and installation - (1-year standard).

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:

1. Mechanically fastened component railing systems.
   a. Moultrie Manufacturing Company – Wesrail II.
   b. Or Equal.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:

1. Self – Closing Swing Gates.
   a. Benko Products.
   b. Or Equal.

2.2 MATERIALS

A. Alloy 6061-T6 or Alloy 6061-T6.

1. ASTM B209 for sheets and plates.
2. ASTM B221 and ASTM B308 for shapes - beams, channels, angles, tees, and zees.
3. ASTM B247 for forgings.

B. Aluminum Alloy 6063-T5 or T6.

1. ASTM B221 and ASTM B429 for bars, rods, wires, pipes and tubes.

C. Fittings:

1. Cast aluminum: ASTM B108.
2. Machined aluminum: 6063-T5 or T6 alloy.

D. Shims: Aluminum of same alloy as component being shimmed.

E. Fasteners: 304 Stainless steel.

F. Expansion and Adhesive Anchors: See Specification Section 03 15 19.

2.3 FABRICATION

A. General:

1. Verify field conditions and dimensions prior to fabrication.
2. For fabrication of items which will be exposed to view, use only materials which are smooth and free of surface blemishes including pitting, seam marks, roller marks, rolled trade names and roughness.
3. Form exposed work with smooth, short radius bends, accurate angles and straight edges.
   a. Ease exposed edges to a radius of approximately 1/32-inches.

b. Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing work.
c. Drill or punch holes with smooth edges.

4. Form exposed connections with flush, smooth, hairline joints, using stainless steel or aluminum splice locks to splice sections together.
   a. Ease the edges of splices and expansion joints and remove all burrs left from cutting.
5. Provide for anchorage of type indicated on Drawings or as required by field conditions.
   a. Drill or punch holes with smooth edges.
6. Design railings and anchorage system in accordance with NAAMM AMP 521 to resist loading as required by the building code.
   a. Maximum allowable stresses per AA ADM 1.

B. Custom fabricate railings to dimensions and profiles indicated.
   1. Guardrails:
      a. Posts: Minimum 1-1/2-inches nominal diameter Schedule 80, 6105-T5 and 6005-T5 aluminum pipe.
         1) Space vertical posts as required by loading requirements but not more than 4-feet-0-inches on center.
         2) Post spacing may exceed 4-feet dimension if supported with an engineered analysis for justification and remain OSHA compliant.
      b. Rails: Minimum 1-1/2-inches nominal diameter Schedule 40, 6105-T5 and T1, 6005-T5 and T1 aluminum pipe.
         1) Where details are not indicated, space intermediate rails to requirements of the building code or OSHA Standards, whichever requires the more restrictive design.
   2. Mounting:
      a. Provide manufacturer's standard cast or machined flanges or brackets as necessary for conditions shown on Drawings.
         1) Railing posts shall be secured using socket-head stainless steel set screws.
            a) Bolts, TEK screws, rivets, or adhesive are not acceptable.
      b. Flanges and brackets shall allow for removal of railing sections where removable railings are indicated on the Drawing.
         1) Completed assembly shall be designed to withstand the loading capacity specified.
   3. Toeboards:
      a. 1/4-inches thick by 4-inches high, above finished floor, extruded toeboard with stiffener ribs.
   4. Guardrail gates:
      a. Self - Closing Safety Guardrail Gates: Fully assembled gate shall be capable of swinging in either direction by inverting installation position. Gate size shall be laterally adjusted from minus 3-inches to plus 3-inches. System shall have top, mid and bottom rail in accordance with OSHA Standards - 29 CFR 1910.23 (a)(2). Gates shall swing in direction shown on drawings.
      b. Constructed to fit the guardrail system.
      c. Width of gate is 24-inches or as shown on Drawings.
      d. Hinges:
         1) Stainless Steel.
         2) Self-closing.
            a) Stainless steel spring actuated self-closing.
      e. Gate latch and stop:
         1) Single Leaf Opening:
            a) Stainless Steel stop plate fastened to leaf.
         2) Double Leaf Openings:
            a) Stainless Steel stop plate fastened to one of the double leaves.
            b) Provide a 1/2-inch x 12-inch stainless steel Cane Bolt on each leaf. (National #348-516 or Equal).
            c) Provide a 5-inch stainless steel barrel bolt latch. (National #N342-477 or Equal).

f. Height:
    1) Top Rail: 42-inches, minimum.
    2) Intermediate Rail: 21 inches.
    3) Bottom Rail with Toe Board: Each safety gate shall be provided with a 4-inch-high toe board on bottom rail.

C. Railing Fabrication:
1. All railings are to be mechanically fastened component system.
2. Railing system shall be an engineered system designed specifically for use as guardrail system.
    a. Fittings shall be internally connected, flush-fitting aluminum or stainless steel.
    b. Fasteners shall be 302 series stainless steel Allen head set screws.
        1) Rivets, adhesive or headed screws are not acceptable.
3. Fit exposed ends of guardrails with solid terminations.
    a. Where guardrail terminates at a wall, provide a vertical post or end-loop 4-inches off the wall to center of vertical member.
    b. Provide and install end caps on all railings where ends of pipe are exposed.
4. Preassemble items in shop to the greatest extent possible to minimize field splicing and assembly of units at project site.
5. Provide weeps to drain water from hollow sections of railing at exterior and high humidity conditions.
    a. Drill 1/4-inches weep hole in railings closed at bottom:
        1) 1-inches above walkway surface at bottom of posts set in concrete.
        2) 1-inches above base flange or reinforcing spud where applicable.
        3) At low point of intermediate rails.
    b. Do not drill weep holes:
        1) In bottom of base flange.
6. Expansion joints:
    a. Joints to be designed to allow expansion and contraction of railing and still meet design loads required.
        1) Top rail splices and expansion joints shall be located within 8-inches of post or other support.
        2) Where railings span building expansion joints; provide a railing expansion joint in the span crossing the building expansion joint.
    b. Provide expansion joints in any continuous run exceeding 20-feet in length.
        1) Space expansion joints at not more than 40-feet on center.
    c. Provide minimum 0.10-inches of expansion joint for each 20-feet length of top rail for each 25 Degree F differential between installation temperature and maximum design temperature.
        1) Maximum expansion joint width at time of installation shall not exceed 3/8 inches.
            a) Provide additional expansion joints as required to limit expansion joint width.
    d. Provide slip-joint with internal sleeve.
        1) Extend slip joint min 2-inches beyond joint at maximum design width.
        2) Fasten internal sleeve securely to one side.
            a) Provide Allen-head set screw located in bottom of rail.
            b) Rivets or exposed screw heads are not acceptable.

D. Guardrail Finish:
    a. 215-R1, Architectural Class 1 coating, AA-M10C22A41, clear anodized.

PART 3 - EXECUTION

3.1 PREPARATION

A. Prior to installation, inspect and verify condition of substrate.

B. Correct surface defects or conditions which may interfere with or prevent a satisfactory installation.

3.2 INSTALLATION

A. Install guardrails to meet loading requirements of the building code and OSHA.

B. Install products in accordance with manufacturer's instructions.

C. Set work accurately in location, alignment and elevation; plumb, level and true.
   1. Measure from established lines and items which are to be built into concrete, masonry or similar construction.

D. Align railings prior to securing in place to assure proper matching at butting and expansion joints and correct alignment throughout their length.
   1. Provide shims as required.

E. Install proper sized expansion joints based on temperature at time of installation and differential coefficient of expansion of materials in all railings as recommended by manufacturer.
   1. Lubricate expansion joint splice bar for smooth movement of railing sections.

F. Provide removable railing sections where indicated on Drawings.

G. Anchor railings to concrete with a minimum of (4) bolts on each base plate, and with stainless steel adhesive anchors with stainless steel bolts, nuts and washers unless noted otherwise in the Contract Documents.
   1. Where exposed, bolts shall extend minimum 1/2-inches and maximum 3/4-inches above the top nut.
      a. If bolts are cut off to required height, threads must be dressed to allow nuts to be removed without damage to the bolt or the nut.
      b. Bevel the top of the bolt after cutting to provide a smooth surface.

H. Anchor railings to metal structure with stainless steel bolts, nuts and washers.

I. Provide toeboards on the walkway side of all elevated walkways, platforms and stair landings, and where indicated on the Drawings or required by OSHA Standards.
   1. Install toeboards to fit tight to the walking surface.
      a. Bottom of toeboard shall not exceed 1/4-inches above walking surface.
   2. Attach to railing vertical post with manufacturer's standard mounting clamp:
      a. Adjustable.
      b. Designed to engage in extruded slot on back of toeboard.
   3. Provide splice bars, corner splices and brackets:
      a. Manufacturer's standard items as required for a complete installation.
   4. Provide spacer bar and U-bracket where necessary for toeboard to clear mounting flange.

J. Coat aluminum in contact with dissimilar metal or concrete in accordance with Specification Section 09 96 00.

K. Provide railings as required for stair construction identified in Specification Section 05 50 00.

L. Install guardrail gate plumb and level in location shown on Drawings.
   1. Center gate/s in openings.
   2. Top of gate to match top of guardrail.
   3. Fasten hinges to gate and jamb post:
      a. Minimum three (3) 1/4-inches stainless steel countersunk machine screws per leaf.
      b. Drill and tap into railing and gate vertical posts.
   4. Provide not less than two hinges per gate.
   5. Install gate latch and stop on strike side of opening.
      a. Fasten to gate with 1/4-inches stainless steel countersunk machine screws.
      b. Drill and tap into gate vertical post.
      c. Drill hole in railing vertical post to receive latch pin.
   6. Adjust to provide smooth operation:

a. Self-closing and self-latching.

END OF SECTION

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DIVISION 06

WOOD, PLASTICS, AND COMPOSITES

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SECTION 06 10 00
ROUGH CARPENTRY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Rough Carpentry.
   2. Composite Wood Decking.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 15 19 – Anchorage to Concrete
   2. Section 07 61 13 - Metal Roofing.
   3. Section 07 62 00 - Flashing and Sheet Metal.
   4. Section 07 92 00 - Joint Sealants.
   5. Section 09 22 16 – Non-Structural Metal Framing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. The Engineered Wood Association (APA):
      a. PRP-108, Performance Standards and Qualification Policy for Structural Use Panels.
      b. U450, Storage and Handling of APA Trademarked Panels.
      c. Y510, Plywood Design Specification.
   2. ASTM International (ASTM):
      a. A153/A153M, Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
      b. D1037, Standard Test Methods for Evaluating Properties of Wood-Base Fiber and Particle Panel Materials.
      c. D1413, Standard Test Method for Wood Preservatives by Laboratory Soil-Block Cultures.
      d. D1761, Standard Test Methods for Mechanical Fasteners in Wood and Wood-Based Materials.
      e. D2898, Standard Practice for Accelerated Weathering of Fire-Retardant-Treated Wood for Fire Testing.
      f. D4442, Standard Test Methods for Direct Moisture Content Measurement of Wood and Wood-Base Materials.
      g. D4444, Standard Test Method for Laboratory Standardization and Calibration of Hand-Held Moisture Meters.
      h. E84, Standard Test Method for Surface Burning Characteristics of Building Materials.
   3. American Wood Protection Association (AWPA):
      a. E1-72, Laboratory Methods for Evaluating the Termite Resistance of Wood-based Materials: Choice and No-choice Tests.
      b. M2, Standard for Inspection of Preservative Treated for Industrial Use.
      c. M3, Standard for the Quality Control of Preservative Treated Products for Industrial Use.
      d. M4, Standard for the Care of Preservative-Treated Wood Products.
      e. T1 Processing and Treatment Standard.
      f. U1, Use Category System: User Specification for Treated Wood.
   4. American National Standards Institute/Single Ply Roofing Industry (ANSI/SPRI):
      a. ES-1, Wind Design Standard for Edge Systems Used with Low Slope Roof Systems.
   5. Environmental Protection Agency (EPA).
   6. FM Global (FM):
      a. 1-49, Property Loss Prevention Data Sheets - Perimeter Flashing.
   7. National Institute of Standards and Technology (NIST):

a. PS 1, Quantitative NMR (Benzoic Acid).
b. PS 20, American Softwood Lumber Standard.

8. Underwriters Laboratories, Inc. (UL):
   a. 723, Standard for Test for Surface Burning Characteristics of Building Materials.

B. Qualifications:
1. Wood Treatment Plant: AWPA M3.
2. Treated Wood Inspection: AWPA M2.

C. Miscellaneous:
1. Factory marking:
   a. Lumber:
      1) Identify type, grade, moisture content, inspection service, producing mill, and other qualities specified.
      2) Marking may be omitted, as allowed by the building code, if certificate of inspection is provided for each shipment.

1.3 SUBMITTALS

A. Shop Drawings:
1. Fabrication drawings of all fabricated items.
2. Product technical data including:
   a. Acknowledgement that products submitted meet requirements of standards referenced.
   b. Manufacturer's installation instructions for all products specified.
3. Certifications:
   a. Chemicals used in treatment process are registered with and approved by EPA.
   b. Moisture content of material prior to treatment: 25% maximum.
   c. Material has been kiln-dried after treatment (KDAT) to the moisture content specified.
4. Documentation of treatment of treated material in accordance with standards referenced.
5. Warranties:
   a. Provide signed manufacturers warranties.

1.4 DELIVERY AND STORAGE

A. Delivery, storage and handling of untreated wood products:
1. Lumber: As recommended by the grading agency indicated on the grade stamp.
2. Plywood: APA U450.

B. Delivery, storage, handling and disposal of treated wood products: AWPA M4.

1.5 WARRANTY

A. Interior fire-retardant treated wood: Manufacturer’s standard 20-year warranty.

B. Composite Wood Decking: Manufacturer’s standard commercial 10-year warranty for Fade Resistance, Physical Integrity and Stain Resistance.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:

1. Fire-Retardant Treated Material (FRTM):
    a. Hoover Treated Wood Products, Inc.:
    b. Or Equal.
2. Plywood sheathing:
    a. Hoover Treated Wood Products "Exterior Fire-X."
    b. Or Equal.
3. Composite Wood Decking:
    a. Trex “Transcend Topicals”
    b. Or Equal.

2.2 MATERIALS

A. General:
1. Lumber (for framing, blocking, nailers, furring, grounds and similar members):
    a. NIST PS 20.
    b. Species:
        1) Treated material: As indicated in the appropriate AWPA standard.
            a) Provide species of FRTM as necessary to achieve UL rating listed.
    c. Grade:
        1) For nominal sizes up to and including 2 x 4: Standard and better.
        2) For nominal sizes up to 2-inches thick and wider than 4-inches: #2 and better.
2. Structural plywood:
    a. NIST PS 1, NIST PS-2.
    b. APA PRP-108, APA Y510.
    c. Exterior Sheathing: 5/8-inch thick unless shown otherwise on the Drawings.
3. Non-structural plywood:
    a. NIST PS 1.
    b. C-D plugged:
        1) Exposure INT.
        2) Thickness: 3/4-inch thick unless shown otherwise on the Drawings.
        3) Touch sanded.
4. Moisture content:
    a. Kiln-dry, ASTM D4442 and ASTM D4444.
    b. Lumber: 19 percent maximum.
    c. Plywood: 15 percent maximum.

B. Preservative Treated Material:
1. Moisture content:
    a. Prior to treatment: 25 percent.
    b. Kiln-dry after treatment (KDAT), ASTM D4442 and ASTM D4444:
        1) Lumber: 19 percent maximum.
        2) Plywood: 18 pecent maximum.
2. Preservative:
    a. Waterborne: AWPA T1.
    b. As indicated in the appropriate AWPA standard.
3. Pressure-treat material in accordance with AWPA U1.
4. Wherever practicable, material to be treated shall be manufactured in its final form prior to treatment.

C. Fire-Retardant Treated Material (FRTM):
1. Maximum moisture content:
    a. Prior to treatment: 25 percent.
    b. Kiln-dry after treatment (KDAT), ASTM D4442 and ASTM D4444:
        1) Lumber: 19 percent (KDAT).
        2) Plywood: 15 percent (KD-15).
2. Fire-retardant preservative:
    a. Provide protection against decay:

1) EPA registered for use as a wood preservative.
b. Shall not bleed-through or adversely affect bond of any finish.
3. Pressure-treat material in accordance with AWPA U1.
4. UL Classified:
a. FR-S, UL 723.
b. Exterior: No increase in classification when subjected to the Standard Rain Test, ASTM D2898.
c. Provide UL mark on each piece of FRTM.
5. Maximum flame spread rating: 25, ASTM E84.
6. Wherever practicable, material to be treated shall be manufactured in its final form prior to treatment.

D. Fasteners and Anchors:
1. Nails and screws:
a. Dry, non-corrosive exposure: Hot dipped galvanized meeting ASTM A153 or Type 304 stainless steel.
b. Wet, corrosive, marine, and/or below grade: Type 316 stainless steel.
2. Adhesive anchors, expansion anchors, self-tapping concrete anchors, bolts, nuts, and washers: See Specification Section 03 15 19.

E. Exterior Wall Sheathing:
1. Plywood sheathing:
a. Fire retardant treated plywood, UL classified FR-S, UL 723.
b. 5/8-inch thick unless shown otherwise on the Drawings.
c. Provide minimum B/C face with "B" face installed on the exterior side.

F. Composite Wood Decking:
1. Material consisting of recycled Linear Low-Density Polyethylene (LLDPE) and recycled wood.
2. Decking is extruded into shapes and sizes as follows:
a. Decking Boards (nominal dimensions); 2-inch x 3-1/2-inch.
b. Spacer Decking Boards (nominal dimensions); 1-inch x 5-1/2-inches. Spacer boards are to be ripped to 2-3/4-inches wide.
c. Lengths – 16-feet.
d. Color – Spiced Rum.
3. Physical and Mechanical Properties as follows:

	Test	Test Method	Value
a.	Flame Spread Index	ASTM E84	Class B
b.	Thermal Expansion	ASTM D1037	1.9 x 10^5 inch/inch/degree F
c.	Moisture Absorption	ASTM D1037	less than 1 percent
d.	Screw Head Pull-Through	ASTM D1761	161 foot pound per screw Fastener used in testing: #8 x 2-1/2-inches approved SS Screw.
e.	Fungus Resistance	ASTM D1413	Rating - no decay
f.	Termite Resistance	AWPA E1-72	Rating = 9.6

4. Fasteners:
a. Carriage bolts shall be 1/4-inch diameter stainless steel.
b. Screws shall be as recommended by composite wood decking manufacturer and shall be color matched to decking color.

PART 3 - EXECUTION

3.1 PREPARATION

A. Verify measurements, dimensions, and shop drawing details before proceeding.

B. Coordinate location of studs, nailers, blocking, grounds and similar supports for attached work.

C. Eliminate sharp projections which would puncture roofing, flashing or underlayment material.

3.2 ERECTION AND INSTALLATION

A. General:
   1. Provide preservative treated material for all wood used:
      a. Outside building.
      b. Below grade.
      c. Roofing nailers.
   2. Provide fire-retardant treated material for all wood used:
      a. Inside building for blocking in walls and exposed backing panels for electrical panels, devices, etc.
      b. Exterior building wall sheathing.

B. Attach work securely by anchoring and fastening as indicated or required to support applied loading.
   1. Anchor wood to concrete using adhesive or expansion anchors as specified in Specification Section 03 15 19.
      a. Separate wood from direct contact to concrete with polyethylene foam gasket strip.
         1) Size: 1/4-inches by width of wood member.
         2) Owens Corning "SillSealR".
   2. Anchor wood to metal using bolts and nuts as specified in Specification Section 03 15 19.
   3. Provide flat washers under all bolt heads and nuts.
   4. Fasten plywood in accordance with APA recommendations.
   5. Use fasteners of size that will not penetrate members where opposite side will be exposed to view or receive finish materials.
   6. Install fasteners without splitting of wood; predrill as required.
   7. Do not drive threaded friction type fasteners.
   8. Tighten bolts and lag screws at installation and retighten as required.

C. Set work to required levels and lines, plumb, true.
   1. Shim as required.
   2. Cut and fit accurately.

D. Provide wood nailers or blocking where required for attachment of other work and surface applied items.
   1. Form to shapes indicated or required.
      a. FRTM lumber:
         1) Do not rip or mill.
         2) Cross-cutting and drilling are allowable in accordance with manufacturer's recommendations and UL requirements.
         3) Resurfacing, planing or fabrication of special shapes or profiles shall be done prior to treatment.
      b. FRTM plywood:
         1) Cross-cutting, ripping and drilling are allowable in accordance with manufacturer's recommendations and UL requirements.
      c. Light sanding of FRTM as permitted by UL to remove raised grain or prepare for finishing is allowable.
      d. Field treat cuts and holes in preservative treated material in accordance with AWPA M4 and manufacturer's published recommendations.
   2. Install roofing nailers as necessary for attachment of flashing, curbs, fascia, coping, and related accessories:
      a. Match height of nailers to insulation.
      b. Anchor nailers to resist force of 300 PLF unless required otherwise by FM Global or roofing manufacturer.
         1) Metal decking attachment:

a) Attach base nailer to metal roof deck using self-tapping stainless steel sheet metal screws (STSMS) with plate washers or with minimum 3/8 inches Type 304 stainless steel hex head bolts with nuts and washers.
b) Countersink heads of bolts flush with top of nailer.

2) Concrete decking attachment:
a) Attach base nailer to concrete roof deck using minimum 3/8 inches stainless steel adhesive anchors with minimum 3 inches embedment.
b) Countersink heads of bolts flush with top of nailer.

3) Provide size and spacing of anchorage as required to meet loading criteria specified.
a) Fasten blocking for perimeter flashing in accordance with ANSI/SPRI ES-1 and FM Global 1-49.

c. Provide 1/2-inches vent spaces between lengths of nailers.
d. Install nailers over vapor retarder.

E. When wood has been exposed to moisture allow to completely dry out prior to covering with additional wood or another material.

F. Correct or replace wood which shows bowing, warping or twisting to provide a straight, plumb and level substrate for applications of other materials.

G. Exterior Wall Sheathing Installation:
1. Install sheathing in accordance with manufacturer's installation guidelines and fastening requirements for loading requirements noted in the Contract Documents.
2. Use maximum lengths possible.
3. If sheathing surface varies more than 1/8-inches from any one panel to an adjoining panel, remove the panels and reset.
a. If the condition persists, remove the panels and correct sub framing as required so panels align properly.
4. Drive fasteners to bear tight against and flush with surface of sheathing.
a. Do not countersink, fracture core or puncture facers with head of fastener.

H. Composite Wood Decking Installation:
1. Install according to drawings and manufacturer’s instructions.

END OF SECTION

SECTION 06 41 00
ARCHITECTURAL WOOD CASEWORK

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Architectural cabinetwork.
   2. Solid Surface Material window stools and countertops.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 07 92 00 - Joint Sealants.
   3. Section 09 06 00 – Color Schedule.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Americans with Disability Act (ADA):
      a. Accessibility Guidelines for Buildings and Facilities (ADAAG).
   2. American National Standards Institute (ANSI):
      a. A208.1, Particleboard.
      b. A208.2, Medium Density Fiberboard (MDF) for Interior Applications.
   3. American National Standards Institute/American Hardboard Association (ANSI/AHA).
      a. A135.4, Basic Hardboard.
   4. American National Standards Institute/Builders Hardware Manufacturers Association (ANSI/BHMA):
      a. A156.11, American National Standard for Cabinet Locks.
   5. ASTM International (ASTM):
      a. F547, Standard Terminology of Nails for Use with Wood and Wood-Base Materials.
   6. Architectural Woodwork Institute (AWI):
      a. Architectural Woodwork Quality Standards.
   7. Hardwood Plywood and Veneer Association (HPVA):
      a. HP-1, Standard for Hardwood and Decorative Plywood.
   8. National Electrical Manufacturers Association (NEMA):
      a. LD 3, High-Pressure Decorative Laminates (HPDL).

B. Qualifications:
   1. Fabricators shall have a minimum of 10 years' experience in design and fabrication of architectural cabinetwork with minimum of three successfully completed projects with similar scope in the last 2 years.

C. Miscellaneous:
   1. Construction details, fastening, tolerances and workmanship: AWI – "Custom" grade standards with exceptions indicated.

1.3 DEFINITIONS

A. Architectural Cabinetwork: Millwork.

B. Exposed Surfaces:
   1. All surfaces visible when doors and drawers are closed, inside of doors, and:
      a. Door and drawer fronts and their edges.
      b. Exposed end.
      c. Countertop and backsplash and their exposed edges.
      d. Face frame (if used).
      e. Interior of open cabinets.

f. Toe kick not to be covered by resilient base or coved flooring.
g. Wall mounted adjustable shelves.
h. Bottom of wall case over 4-feet above floor.
i. Top of wall and tall cases below 6-feet above floor.

C. Concealed Surfaces:
1. Surfaces not visible after installation, and:
a. Web frames.
b. Dust panels.

D. Semi-Exposed Surfaces:
1. All other surfaces not exposed or concealed.

1.4 SYSTEM DESCRIPTION

A. Fabricated cabinets including all hardware, countertops, and finishing thereof.

1.5 SUBMITTALS

A. Product Data:
1. Product technical data including:
a. Acknowledgement that products submitted meet the requirements of standards referenced.
b. Manufacturer's installation instructions.
c. Schedule of window sill and countertop locations indicating sizes, shape etc.
2. Certifications: Solid Surface Material fabricator's letter of authorization from material manufacturer.

B. Shop Drawings:
1. Provide shop drawings for all items of casework / millwork to be constructed. Shop drawings will include plans, elevations and sections of all casework / millwork; however, typical sections may be provided for repeated or similar items. Plan and elevation views will be drawn at minimum scale 3-inches=1-foot-0-inches. Sections will be drawn at minimum scale of-3 inches=1-foot-0-inches. Shop drawings shall indicate thickness of all materials, method of assembly and jointing, limits of plastic laminate surfacing, limits and location of backing sheets, base details, limits of self - edged surfaces, location and method of attachment for hardware and accessories, equipment, fixtures and tops to be supplied by others, scribing methods, and any other details required to clearly show the quality, construction and installation of work included under this section.

C. Samples:
1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections unless noted in this section of the specifications.
3. Plastic Laminate, Edge Banding and Solid Surface Material:
a. Refer to “Room Finish Schedule and Interior Finish Schedules” found on Architectural Finish Schedules drawing.
4. Samples of hardware:
a. Drawer Pulls.
b. Hinges.
c. Shelf Supports.

D. Informational Submittals:
1. Millwork fabricator experience qualifications.
2. Listing of millwork fabricators projects within the last two years with similar scope.

1.6 SINGLE SOURCE

1. Insofar as possible, all plastic laminate casework shall be constructed by one firm. If part of the work is subcontracted to another firm or firms, one and only one set of shop drawings

shall be made to show all of the pieces and their intersections. Identification of who makes what part(s) shall be included in the shop drawings.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Deliver all millwork items windowsills, and countertops to the Project Site and store in the area in which items will be installed.
   1. Building areas to receive millwork items shall be enclosed, weather tight and conditioned to a relative humidity between 25 percent and 55 percent before, during and after installation.
   2. Remove any plastic packaging or wrapping from millwork upon delivery to Project Site.
   3. Protect stored items from damage with vapor-permeable covering during storage.
   4. Allow material to acclimate to the surrounding environment a minimum of 96 hours prior to installation.
   5. Install temporary coverings necessary to prevent damage during construction.

1.8 JOB CONDITIONS

1. If it is necessary to vary from arrangements or profiles indicated because of structural or other considerations, request approval from Contracting Officer Representative. Make approved variations at no expense to the Government.
2. Data indicated on drawings is reasonably accurate, but absolute accuracy is not guaranteed. Exact locations, distances, and levels will be governed by conditions at the building. The Contractor will be responsible for obtaining exact measurements of openings as they exist in the field prior to completion of plastic laminate clad cabinetry construction. The Contractor retains sole responsibility for ensuring cabinetry items are constructed to fit field conditions. Poorly fitted cabinetry will constitute cause for rejection.
3. Contractor is responsible for fitting all recesses. Include all trim pieces, fillers and closures in sizes that are required by job conditions.

1.9 COORDINATION OF WORK

1. Field Dimensions: Where plastic laminate clad casework / millwork is indicated to be fitted to other construction, check actual dimensions of other construction by accurate field measurements before manufacturing woodwork; show recorded measurements on final shop drawings. Coordinate manufacturing schedule with construction progress to avoid delay of work.
2. The Contractor shall be responsible for the coordination of the installation of plastic laminate clad casework/millwork units and shall provide all labor, material, equipment, transportation, unloading of trucks, delivery into position, etc., required to furnish and install the work shown on the drawings, and as specified herein, except those items specifically marked as not being in contract.
3. Plastic laminate clad casework and millwork shall be installed only under supervision of the Contractor's superintendent.
4. Examine all surfaces to which units are to be applied, and indicate, in writing, any conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions have been corrected.
5. Coordinate temporary openings in walls or floors which may be required for passing large sections of equipment into the building or individual spaces.
6. Plumbing: Provide all openings required through tops and other exposed plastic laminate clad casework/millwork components after thorough coordination with the mechanical contractor.
7. Electrical: Coordinate all holes necessary to mount recessed electrical devices in or on the plastic laminate clad casework/millwork with the electrical contractor. Provide the various cuts and holes in concealed spaces, needed for the passage of conduit.
8. Government Furnished Equipment: Openings required to accommodate equipment which is recessed into plastic laminate clad cabinetry shall be provided by the Contractor and coordinated with the Contracting Office Representative or the equipment representative.
9. Hardware Coordination: Provide cabinet hardware and coordinate cabinet shop drawings and fabrication with hardware requirements.

1.10 SEQUENCING AND SCHEDULING

1. Work shall be coordinated with other trades. Units shall not be installed in any room or space until painting, and ceiling installations are complete within the room where the units are located. Floor cabinets shall be installed before finished flooring materials are installed.

1.11 WARRANTY

1. Furnish a written two-year manufacturer's warranty covering all items furnished under this section against defective materials and workmanship. Warranty period shall begin at the date of substantial completion as defined by the Contract Documents.
2. Upon receipt of notice from the Government of failure of any part of the guaranteed equipment during the warranty period, the affected part or parts shall be replaced promptly with new parts by and at the expense of the Contractor.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Plastic laminate:
      a. Wilsonart Engineered Surfaces.
      b. Or Equal.
   2. Plastic overlay panel products:
      a. Panolam by Domtar Decorative – Panel Plastic.
      b. Or Equal.
   3. Cabinet hardware:
      a. Hinges:
         1) Hafele – Advantage 5 K.
         2) Or Equal.
      b. Catches:
         1) Ives 338.
         2) Or Equal.
      c. Slides:
         1) KV 8400 series.
         2) Or Equal.
      d. Pulls:
         1) Amerock.
         2) Or Equal.
      e. Hanging File Rails:
         1) Kine Flex - KI34 AL 24, Wood Workers Hardware.
         2) Or Equal.
      f. Hanging File Brackets:
         1) Kine Flex - KIKHFBFM ZC, Wood Workers Hardware.
         2) Or Equal.
      g. Drawer and Door Locks:
         1) CompX, National, Wood Workers Hardware.
         2) Or Equal.
      h. Adjustable Shelf Supports:

1) Hafele shelf supports #282.11.707.
2) Or Equal.

i. Countertop Support Brackets:
1) Federal Brace – FB-07442.
2) Or Equal.

j. Backsplash and Side Splash Two Piece Panel Clips:
1) Panelclip SE - Brooklyn Hardware LLC.
2) Or Equal.

4. Solid Surface Material:
a. Wilsonart Engineered Surfaces.
b. Or Equal.

5. Edge Banding:
a. Doellken - Woodtape.
b. Or Equal.

2.2 MATERIALS

A. General: Provide materials that comply with requirements of AWI woodworking standard for each type of plastic laminate clad woodwork and quality grade indicated and, where the following products are part of woodwork, with requirements of the referenced product standards, that apply to product characteristics indicated.

B. Plastic Laminate:
1. NEMA LD 3 high-pressure laminate, matte finish.
2. Countertops and backsplashes and their edges: Grade GP50, 0.050-inches thick.
3. All other exposed surfaces: Grade GP28, 0.028-inches thick.
4. Semi-exposed backer sheet: Grade CL20, 0.020-inches thick; color to match plastic overlay.
5. Concealed backer sheet: Grade BK20, 0.020-inches thick.
6. Colors: See Interior Finish Legend shown on Drawings.

C. Plastic Overlay:
1. Resin impregnated paper overlay hot press cured onto substrates with backer/balance sheet.
2. Conform to NEMA requirements for "general purpose" decorative board (not "light duty" liner type).
3. Satin finish: Opaque color.
4. Resin: Polyester; phenolic resin may be used on concealed surfaces.
5. Color: Standard White.

D. Particleboard:
1. Three-ply, medium density industrial particleboard.
2. ANSI A208.1: Grade M-2 or M-3.
3. Density: 45 pounds per cubic feet minimum.
4. Particleboard used for countertops:
a. ANSI A161.2.
b. Moisture resistant.

E. Fiberboard:
1. Medium density wood fiberboard.
2. ANSI A208.2: Interior Grade M.

F. Hardboard:
1. Tempered, smooth on both sides.
2. ANSI/AHA A135.4: Class 1, S2S.

G. Hardwood Plywood:
1. Species: White birch.
2. Plain sliced.
3. Veneer or particle core.
4. HPVA HP-1, Grade AA.

H. Plywood:
   1. Softwood Plywood, A grade and PS1.

I. Hardwood:
   1. Solid, S4S.
   2. AWI Section 300, premium grade for exposed surfaces.
   3. Exposed: Species to match veneer plywood unless noted otherwise on Drawings.
   4. Other locations: Any hardwood.

J. Edge Banding:
   1. Edge banding for casework doors and drawer fronts shall be PVC vinyl and shall be 0.020-inch thick. Material width shall be 1-5/16-inches or as required by material being banded.
      a. Color and pattern shall match the exposed door and drawer front laminate pattern. Refer to “Room Finish Schedule found on Architectural Finish Schedules” drawing and “Interior Finish Legend found on Architectural Finish Legend and Notes” drawing.
   2. T-edging will not be allowed.

K. Plastic Laminate Adhesive:
   1. High pressure plastic laminate shall be bonded to core with a liquid polyvinyl acetate or a urea resin adhesive at temperatures above 65 degrees F at a pressure of no less than 15-pounds per square inch.
   2. Laminate and core are not to exceed 10 percent moisture content and is to be laminated and cured in a controlled environment between 40 percent and 60 percent relative humidity.
   3. Plastic laminate adhesives shall meet V.O.C. compliant regulations.
   4. PVC edge banding shall be adhered using a polymer - based hot melt glue.

L. Adhesives:
   1. Adhesives shall be of a formula and type recommended by AWI. Adhesives shall be selected for their ability to provide a durable, permanent bond and shall take into consideration such factors as materials to be bonded, expansion and contraction, bond strength, fire rating, and moisture resistance.
   2. Adhesives shall meet local regulations regarding VOC emissions and off - gassing. Adhesives used to bond wood members shall be a Type II for interior use urea - formaldehyde resin formula or polyvinyl acetate resin emulsion.
   3. Adhesives shall withstand a bond test as described in WDMA I.S. 1 - A.

M. Fasteners:
   1. Nails, screws, and other suitable fasteners shall be the size and type best suited for the purpose and shall conform to ASTM F547 where applicable.

N. Solid Surface Material:
   1. Homogeneous non-porous, polymeric material.
      a. ICPA/ANSI SS-1.
      b. NSF 51 approved.
   2. Flammability: Class 1.
   3. Joint adhesive: Manufacturer's recommended adhesive.
   4. Colors: Refer to “Specification Section 09 06 00 – Color Schedule” for color selections and see Interior Finish Legend shown on Drawings.

O. Sealant:
   1. Silicone.
   2. See Specification Section 07 92 00.

2.3 CASEWORK HARDWARE

A. General:
   1. Provide handles, pulls, latches, locks, and other operating devices in accordance with the ADAAG.

B. Hardware for Hinged Doors and Drawers:

1. Hinges:
    a. Five-knuckle, wraparound type with hospital type tips.
    b. Not less than 2-1/2-inches long.
    c. Minimum two screws each leaf.
    d. For doors up to 48-inches high: Two hinges.
    e. For doors over 48-inches high: Three hinges.
    f. Finish:
        1) Standard 304 Stainless steel.
2. Catch:
    a. Heavy duty, roller catch.
    b. Case and strike: Wrought steel.
    c. Roller: Rubber.
    d. Provide one per door.
    e. Finish: Standard Bright nickel plated, clear coated.
3. Slides:
    a. 100-pound capacity, precision steel ball bearings, positive closing and pull-out stops, removable drawer without use of tools; telescoping full extension slides, epoxy-coated with standard white finish.
4. Pulls:
    a. Steel
    b. 4-inches centers.
    c. 5/16-inches diameter.
    d. Provide one per door and drawer.
        1) Provide two pulls on drawers over 36-inches wide.
    e. Finish:
        1) Standard Brushed Chrome.
5. Hanging File Rails:
    a. 3/4-inch x 24-inch, aluminum file rails.
    b. Provide 2 per Drawer.
6. Hanging File Brackets:
    a. Flush mount, Zinc hanging file brackets.
    b. Provide 4 per Drawer.
7. Drawer and Door Locks:
    a. Locks must be BHMA Certified Grade 1 to ANSI/BHMA 156.11 for cabinet door and drawer.
        1) BHMA designation E07121 for cabinet doors.
        2) BHMA designation E07041for cabinet drawers.
    b. Locks must be manufactured in the United States
    c. Lock plug must be five-disc tumbler
    d. Cylinders shall be threaded type with cam.
        1) Cam to have ability, if necessary, for spacer installation providing multiple dimensional variations.
        2) Insert spacer to cam in hole provided. Spacer insures better installation, removes “play” from door or drawer.
    e. Cylinder mounting hole measurement must be 3/4-inch for wood applications.
    f. Mounting hole measurement for metal material must be 0.635 vertical flats by 0.760 diameter.
    g. Maximum wood material thickness of 1-13/16-inches.
    h. The combined thickness of the front and frame cannot exceed the cylinder length.
        1) Provide 90 degree clockwise key rotation.
    i. Master keyed; keyed alike. Each lock to be supplied with (2) keys
    j. Finish:
        1) Standard BHMA Certified Grade 1 US26D – Dull Chrome
8. Adjustable Shelve Supports:
    a. Holes bored at 1-inch on center.

b. Provide 4 supports per shelf.
c. Finish:
1) Standard Nickel plated.

C. Hardware for Countertops:
1. Countertop support brackets:
a. 7 Gauge Steel.
b. Size:
1) Nominal 18-inches long by 18-inches high.
2) 2-inches wide.
3) Load Rating: 500 pounds.
c. Finish:
1) Corrosion resistant powder coating.
2) Color: Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
2. Backsplash and Side Splash Two Piece Panel Clips:
a. 2-5/8-inch-high x 1/4-inch wide two piece panel clips.
b. 6063 high strength aluminum, heat treated to a hardness of T6.

2.4 FABRICATION

A. General:
1. Perform all work to AWI “Custom” grade standards unless more stringent requirements are shown on drawings or specified herein. Factory assemble all parts and pre - finish. Items too large for entrance into area may be made in attachable sections. Fabricate each section complete in itself, with provisions for job connection.
2. Plastic laminated casework / millwork details shall conform to "Flush Overlay" as defined by AWI except that exposed ends of cases shall have a finish end piece extending from back of case to a line flush with the front of the door(s) / drawer(s), and except as otherwise shown on the drawings.
3. Finish panel edges to match exterior surfaces. Assemble body panel joints without use of exposed fasteners. Perform assembly with nickel plated hardware so a section can be replaced by disengaging hardware and removing screws from back of cabinet when required on the drawings.
4. Provide cutouts and access panels when installed work covers mechanical or electrical items requiring access. Cabinetry shall be fabricated to allow accessibility to electrical wiring.
5. Provide materials with ample allowance for cutting and fitting on site. Provide trim for scribing and site cutting.
6. Self-edged doors and drawer fronts shall be fabricated from a separate piece of plastic laminate or edge band tape.
7. Fabricate base cabinet ends and dividers with corner joints between frame members fully lock jointed, glued and screwed. Dado and glue cabinet backs into sides and bottom. Screw countertops to cabinets from underside.
8. Backs on wall hung cabinets shall be screwed to top and bottom members so units can be installed without angles or rails.
9. Exposed - to - view corners on all plastic laminate cabinetry shall be mitered to form a single hairline joint at corner. Corners formed by overlapping panels will not be acceptable.
10. Tolerances: Flushness, flatness, and joint tolerances of laminated surfaces shall meet the AWI “Custom” grade requirements.
11. Color: See Interior Finish Legend shown on Drawings.

B. Casework Fabrication:
1. Style: Reveal overlay doors and drawer fronts overlapping case body with uniform reveal at all edges.
2. Case body:
a. All joints glued.
b. Top and bottom (and fixed horizontals):

1) Lock jointed and screwed.
2) Dado'd or rabbeted into ends/dividers.
3) Doweled at approximately 2-1/2-inches on-center.

c. Back: Dado'd into top, sides, and bottom.
d. Fixed small compartment dividers: Dado'd.

3. Drawers (with subfront):
   a. All joints glued.
   b. All corners:
      1) Dovetailed or doweled.
      2) Front corners dovetailed and back corners lock jointed.
      3) Sides dadoed for front and back and all joints nailed, stapled or screwed.
   c. Bottom: Dado'd into all four sides.
   d. Front: Screwed onto subfront.
   e. Top edges of drawer box rounded.
   f. Install hanging file rails and brackets in all file base cabinet drawers.
4. Allow for two different cabinet face colors
   a. Base cabinet face colors – Color 1.
   b. Upper cabinet face colors – Color 2.
      1) Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.
   c. Thickness:
      1) Case body: 0.020-inch
      2) Door and drawer edges: 0.020-inch.
   d. Colors:
      1) Match adjacent plastic laminate.
      2) Allow for two different edging colors.
         a) Refer to "Specification Section 09 06 00 – Color Schedule" for plastic laminate color selections.
5. Use no blocking or fasteners in exposed or semi-exposed locations.

C. Plastic Laminate Casework Fabrication:
   1. Finishes for cabinets:
      a. All exposed surfaces.
         1) Doors - Plastic laminate.
         2) Casework fronts - Plastic laminate.
         3) Exposed casework partitions, filler panels, sides and bottoms – Plastic laminate.
         4) Vallances – Plastic laminates.
      b. All semi-exposed surfaces not covered with plastic laminate (except hardwood).
         1) Casework interior partitions, bottoms, and backs - Plastic overlay.
         2) Drawer sides, back, bottoms and subfronts - Plastic overlay.
   2. Edges:
      a. Self - Edges of case body members, doors, drawer fronts, drawers and shelving: 0.020-inch thick PVC.
   3. Case body members (except backs not exposed):
      a. Minimum 3/4-inches thick particleboard.
      b. Base unit top:
         1) Use either full sub top or web frames.
         2) Web frames: Hardwood.
      c. Provide drawer lock rails at all drawers.
      d. Provide backs on all cabinets.
   4. Unexposed case back:
      a. Minimum 1/4-inch thick particleboard.
      b. Full bound: Capture and glue back panel in dado at top, sides and bottom of case.
   5. Shelves:
      a. Minimum 3/4-inches thick particleboard self – edged.
      b. 1-inch thick if over 36-inches between supports self – edged.
   6. Doors:

a. Particleboard:
    1) Up to 26-inches wide or 48-inches high: 3/4-inches thick.
    2) Up to 36-inches wide or 66-inches high: 1-1/4-inches thick.
7. Drawers:
    a. Fronts: 3/4-inches thick particleboard.
    b. Subfront, sides and back: 1/2-inches thick particleboard.
    c. Bottom:
        1) Minimum 1/4-inches thick particleboard, captured four sides with 3/8 inches standing shoulder.
        2) Over 18-inches wide provide intermediate reinforcing rails.
8. Case base:
    a. Separate ladder style base.
    b. Preservative treated lumber.
    c. Provide concealed fastening of cabinet body to base.
        1) Fasteners through the bottom of casework will not be accepted.
9. Filler panels and scribe pieces: Particleboard – Plastic Laminate; provide as required to fit standard size units to space.
10. Colors: See Interior Finish Legend shown on Drawings.

D. Solid Surface material windowsills and countertops:
1. Countertops:
    a. Use maximum size sheet to eliminate joints.
    b. Joints shall be not closer than 24-inches to sinks or other cut-outs.
    c. Refer to Drawings for special edge treatment. Ease all exposed edges.
    d. 1/2-inches thick built up to 1-1/4-inches minimum at exposed edges unless shown otherwise on drawings.
2. Back splashes and side splashes:
    a. Material shall be the same as the countertop.
    b. 3/4-inches thick x 4-inches high unless shown otherwise on drawings.
    c. Provide wherever top abuts wall.
3. Colors: Refer to “Specification Section 09 06 00 – Color Schedule” for color selections. See Interior Finish Legend shown on Drawings.

E. Case Configuration:
1. Similar reveal, approximately 1/4-inches at all sides, top and bottom of doors and drawer fronts, and between doors and drawer fronts in same unit.
2. Double door units: No vertical rail or divider between doors unless called for or unless locks are called for.
3. Toe space:
    a. 4-inches high by approximately 3-inches deep.
    b. Provide on front of each base unit unless otherwise noted.
4. Countertop: Overhang front and exposed ends 1-inch unless shown otherwise on drawings.
5. Hardware mounting:
    a. Drawers: Center pull in front, horizontally.
    b. Drawers with two pulls: Set pulls at 1/4 points.
    c. Swinging doors:
        1) Set pull in swing side corner:
            a) Base units: Vertically at top of door.
            b) Wall units: Vertically at bottom of door.
            c) Tall units: Vertically, centered 40-inches above finished floor.
6. Adjustable shelves:
    a. Use drilled hole support.
    b. Depth: 1/2-inches less than inside cabinet depth.

PART 3 - EXECUTION

3.1 PREPARATION

A. Verify dimensions at site.

B. Verify locations of items specified in other Specification Sections.

3.2 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Install products in accordance with the ADAAG.
   1. Provide sizes, heights and clearances required for accessibility compliance.

C. The Contractor shall provide an experienced and competent supervisor who will be present at the job site during installation of all plastic laminated casework/millwork items. The supervisor shall examine all items before installation to ensure quality workmanship and shall be responsible for ensuring that all items are installed in-line, true and even, with no damage to the adjacent work of other trades. Upon inspection by the Contracting Officer Representative, items found to exhibit faulty workmanship, or which were improperly installed, shall be removed and replaced at no additional cost to the Project.

D. Provide balanced construction on each plastic laminated item.

E. Use manufacturer's printed instructions or drawings in all cases where items or details are not indicated.

F. Provide all trim, fillers, closures, stands, supports, sleeves, collars, escutcheons, brackets, braces or other miscellaneous items required for complete installation.

G. Test and adjust for satisfactory operation.

H. Seal components with silicone sealants in accordance with AWI Standards.
   1. Seal joints in plastic laminate countertops before assembly.
   2. Seal joints between back splashes and side splashes and countertops.
   3. Seal joints where back splashes and side splashes meet adjoining surfaces.

I. Adjust hinges so doors hang straight, and door fronts made level.

J. All plastic laminated casework/millwork items shall be delivered to the building, placed in proper location, assembled, neatly fit and secured in place.

K. Anchor plastic laminated casework/millwork to anchors or blocking built in or directly attached to substrates. Secure to grounds, stripping and blocking with a countersunk, concealed fasteners and blind nailing as required for a complete installation. Except where prefinished matching fastener heads are required, use fine finishing nails for exposed nailing, countersunk and filled flush with woodwork and matching final finish where transparent finish is indicated.

L. Anchor backsplashes to walls with aluminum two-piece panel clips. Panel clips shall be continuous the entire length of the backsplash and shall stop 1-1/2 inches from all finished exposed ends. Fasten lower piece to metal studs or blocking and upper piece to back of backsplash. Use fasteners that do not extend through face of backsplash.

M. CLEAN AND ADJUST
   1. All debris, rubbish and dirt created as a result of this work shall be removed by the manufacturer / supplier.
   2. After installation, casework shall be thoroughly cleaned, and all operating parts shall be adjusted and left in perfect operating condition.
   3. Provide final protection and maintain conditions, in a manner acceptable to manufacturer and installer, and that ensures the plastic laminated casework / millwork is without damage or deterioration at time of Substantial Completion.

N. WARRANTY

1. Provide written warranty at Substantial Completion of project.

END OF SECTION

SECTION 06 82 00
FIBERGLASS REINFORCED PLASTIC FABRICATIONS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Fiberglass reinforced plastic (FRP) fabrications including but not limited to:
      a. Grating.
      b. Grating support framing.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Standards Institute (ANSI):
      a. ANSI/NSF Standard 61.
   2. ASTM International (ASTM):
      a. E84, Standard Test Method for Surface Burning Characteristics of Building Materials.

1.3 DEFINITIONS

A. Skid-resistant:
   1. Manufacturer's standard applied abrasive grit coating.
   2. Abrasive coated tape is not acceptable.

B. FRP: Fiberglass Reinforced Plastic.

1.4 SYSTEM DESCRIPTION

A. All fiberglass reinforced plastic and support systems shall be designed by a registered professional structural engineer licensed in the State of Arizona.

1.5 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Manufacturer's recommendations on reinforcing field cut openings.
   2. Fabrication and/or layout drawings.
      a. Plan showing profile, location, section and details of each item including anchorage or support system(s).
      b. Locations and type of expansion joints.
      c. Materials of construction including shop applied coatings.
      d. Listing of all accessory items being provided indicating material, finish, etc.
   3. Certifications:
      a. Certification of Delegated Design Structural Engineer's qualifications.
      b. Certification that all components and systems have been designed and fabricated to meet the loading requirements specified.
   4. Manufacturer's full line of colors available for each component.

B. Informational Submittals:
   1. Complete design calculations of all supporting structure, fastening conditions, and anchorage to support structures or concrete.
      a. Design calculations to be for information only.
      b. Contracting Officer will not review or take any action on submittal.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Deliver and handle each item to preclude damage.

B. Store all items on skids above ground.
   1. Keep free of dirt and other foreign matter which will damage items or finish and protect from corrosion and UV exposure.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Compressible sealant:
      a. Grating and solid plates;
      b. Strongwell Corporation – Duradek.
      c. Or Equal.
   2. Structural Shapes:
      a. Strongwell Corporation
      b. Or Equal.

C. Submit requests for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Fiberglass Reinforced Plastic (FRP):
   1. Vinyl ester with fiberglass reinforcing.
      a. Pultruded.
      b. Type V.
   2. Fire retardant.
      a. Flame spread: ASTM E84, 25 or less.
   3. Color: Yellow.

B. Fasteners, Clips, Saddles, and Miscellaneous Components:
   1. Stainless steel 316L. Provide 2 at each support with a minimum of 4 per panel.
   2. Adhesive: Recommended by manufacturer.

C. Skid-resistant Surfacing: Manufacturer-applied abrasive grit coating.

D. FRP Framing, Columns, and Angles :
   1. Polyester resin with fiberglass reinforcing.
   2. U.L. Certified to ANSI/NSF Standard 61.
   3. Color: Grey.

2.3 FABRICATION

A. General:
   1. Verify field conditions and dimensions prior to fabrication.
   2. Preassemble items in shop to greatest extent possible.
   3. All components shall be treated with UV inhibitor.
   4. Drill or punch holes with smooth edges.

B. Grating and Solid Plate Material:
   1. Design live load:
      a. 100-pounds per square foot uniform live load.
      b. 300-pounds concentrated load.
      c. Maximum deflection of l/300 of span under a superimposed live load.
      d. Design for the most severe loading condition noted above.
      e. Equipment loads as indicated on the Drawings.
   2. Grating shape: “I”.
   3. Minimum grating depth: 1-1/2-inches.
   4. Bar span: Maximum of 1-3/4-inches center to center.
   5. The bearing bars shall be joined into panels by passing continuous length fiberglass pultruded cross-rods through the web of each bearing bar. A continuous fiberglass pultruded bar shaped section shall be wedged between the two cross rod spacers mechanically locking the notches in the cross-rod spacers to the web of the bearing bars. Continuous adhesive bonding shall be achieved between the cross-rod spacers and the bearing web and between the bar shaped wedge and the two cross rod spacers locking the entire panel together to give a panel that resists twist and prevents internal movement of the bearing bars.
   6. All bearing bars that are to be exposed to UV shall be coated with polyurethane coating to provide additional UV protection.
   7. Walking surface: Manufacturer's standard applied abrasive grit coating.
   8. After fabrication, all cut ends, holes and abrasions of FRP shapes shall be sealed with a compatible resin coating.

C. Embedded Grating Supports:
   1. Fiberglass.
   2. Size to suit depth of grating.
   3. Provide leg or strap for embedding and anchoring into concrete.

D. Structural Members:
   1. Provide structural members having the same resin composition as the item being supported.
   2. Factory fabricate with all required connection holed and holes for work of other trades.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Set work accurately in location, alignment and elevation, plumb, level, and true.
   1. Measure from established lines and levels.
   2. Provide temporary bracing or anchors in formwork for items which are to be built into concrete, masonry or similar construction.
   3. Tolerances:
      a. Maximum variation from plumb in vertical line: 1/8-inches in 3-feet.
      b. Maximum variation from level of horizontal line: 1/4-inches in 20-feet.
      c. Maximum variation from plan location: 1/4-inches in 20-feet.

C. Attach grating to each end and intermediate support clip or saddle with bolts, nuts and washers.
   1. Maximum spacing: 2-feet on-center with minimum of two per side.
   2. Attach clips or saddles to bearing bars only.
   3. Reinforce all field cut openings in accordance with manufacturer's recommendations.

D. All field cut or drilled edges shall be coated per the manufacturer’s recommendations to prevent fiber blooming or fraying.

E. All of the clips required for installation shall be supplied by the FRP manufacturer.

F. Field modifications (cuts, copes, holes, etc.) unless shown on the shop drawings are not allowed without the manufacturer’s written approval.

END OF SECTION

DIVISION 07

THERMAL AND MOISTURE PROTECTION

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SECTION 07 14 00
FLUID APPLIED WATERPROOFING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Fluid applied waterproofing, below grade applications.
   2. Protection course.
   3. Specific concrete finishing requirements.

B. Related Specification Sections include but are not necessarily limited to:
   1. Division 03 35 00 - Concrete Finishing and Repair of Surface Defects.
   2. Section 07 21 00 - Building Insulation.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. C836/C836M, Standard Specification for High Solids Content, Cold Liquid-Applied Elastomeric Waterproofing Membrane for Use with Separate Wearing Course.
      b. D4258, Standard Practice for Surface Cleaning Concrete for Coating.
   2. International Concrete Repair Institute (ICRI):
      a. 310.2R, Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, Polymer Overlays, and Concrete Repair.
   3. The Society for Protective Coatings/NACE International (SSPC/NACE):
      a. SP 6/NACE No. 3, Commercial Blast Cleaning.
      b. SP 13/NACE No. 6, Surface Preparation of Concrete.

B. Qualifications:
   1. Applicator(s) licensed or approved in writing by manufacturer.
   2. Applicator(s) shall have minimum of seven years' experience in application of cold liquid-applied elastomeric waterproofing membranes with minimum of two years installation of products specified or accepted for use on this Project.
      a. Provide list of projects completed in last two years using products proposed for use.
         1) Include name of structure, area waterproofed (SQFT) and name of contact with phone number.

C. Miscellaneous:
   1. Manufacturer's authorized representative shall review substrate preparation and provide written approval of substrate prior to installation of product.

1.3 DEFINITIONS

A. Installer or Applicator:
   1. Installer or applicator is the person installing or applying the product in the field at the Project site.
   2. Installer and applicator are synonymous.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Details showing flashing of penetrations, terminations, expansion joints, protection course attachment and other special conditions.
   2. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.

3. Certification of Applicator qualifications.
4. Applicator's experience record.
5. Listing of projects completed in last two years.

B. Informational Submittals:
1. Manufacturer's written approval of substrate.
2. Warranty – See Specification Section 01 67 00 – Product Requirements.
a. The warranty shall warrant that the products will be free from manufacturing defect at the time of purchase, will remain in a watertight condition and will perform as warranted in the manner specified for the stated term of (2) two years measured from the Date of Project Substantial Completion, The total liability of the Company over the life of the warranty shall not exceed the aggregate dollar value of the original cost of the products specified.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Fluid Applied Waterproofing system:
a. Carlisle Coatings & Waterproofing – "MiraSEAL".
b. Or Equal.

2.2 MATERIALS

A. Primer: Manufacturer's recommended primer appropriate to substrate.

B. Waterproofing System:
1. One or two components, moisture curing polyurethane elastomer meeting requirements of ASTM C836/C836M.
2. Flowing type for surfaces up to 5 percent slope.
3. Non-flow type for surfaces exceeding 5 percent.

C. Adhesive: Manufacturer's standard.

D. Flashing Reinforcement: Woven uncoated fiberglass mesh.

E. Sealant: Manufacturer's recommended sealant.

F. Protection Course:
1. Material capable of protecting cured membrane from damage caused by rocks and other debris in the backfill material.
2. Acceptable to waterproofing manufacturer.

G. Backer Rod: Closed cell polyurethane foam rod.

PART 3 - EXECUTION

3.1 PREPARATION

A. Cure concrete and masonry in accordance with manufacturer's recommendations.
1. Verify moisture content does not exceed manufacturer's maximum allowable.

2. Ensure that curing agents used are compatible with coating system.

B. Remove surface contamination by high pressure water cleaning per ASTM D4258.

C. Verify that concrete has been troweled and broomed, free of fins, ridges or voids.
   1. Verify that all tie holes and honeycomb areas, holes and voids have been patched in accordance with Specification Section 03 35 00 and coating manufacturer's recommendations.

D. Prepare substrate per manufacturer's published instructions and this Specification Section.
   1. Concrete surfaces:
      a. Abrasive blast in accordance with SSPC SP 13/NACE No. 6 to provide a profiled surface.
         1) Profile: ICRI 310.2R, CSP 3 minimum.
   2. Metal surfaces:
      a. Abrasive blast in accordance with SSPC SP 6/NACE No. 3.
         1) Minimum 1-mil surface profile.
      b. Prime coat all metal surfaces.
   3. Flash all penetrations and other areas in accordance with manufacturer's instructions.
   4. Clean and seal cracks and joints in accordance with manufacturer's instructions.

E. Protect adjacent surfaces.

3.2 APPLICATION AND INSTALLATION

A. Apply fluid applied waterproofing system in accordance with manufacturer's printed instructions and this Specification Section.
   1. Provide minimum 60 mil dry film thickness.
   2. Apply waterproofing to the exterior side of vertical concrete wall surfaces (surfaces against earth) where finished interior building space occurs on the opposite side of the wall and where indicated on the Drawings.
      a. Terminate top of waterproofing in accordance with manufacturer's instructions approximately 4-inches below finished grade unless shown otherwise on Drawings.

B. Extend coating over all previously flashed areas.

C. Allow vertical applications to cure minimum of 12 hours at 75 degrees F or as recommended by manufacturer, prior to backfilling.

D. Protection Course:
   1. Protection course is to be installed prior to any perimeter insulation specified in Specification Section 07 21 00.
      a. Secure protection course to prevent displacement during backfilling.
         1) Adhere to cured waterproofing membrane.
         2) Mechanical fasteners are not acceptable.

END OF SECTION

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SECTION 07 21 00
BUILDING INSULATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Building insulation.
      a. Does not include roof insulation or roof vapor retarder; see Specification Section 07 61 13 – METAL ROOFING.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 31 31 - Concrete Mixing, Placing, Jointing and Curing
   2. Section 04 22 00 - Concrete Masonry.
   3. Section 07 46 10 - Cement Board Siding
   4. Section 07 92 00 - Joint Sealants.
   5. Section 09 29 00 - Gypsum Board.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. C272/C272M, Standard Test Method for Water Absorption of Core Materials for Structural Sandwich Constructions
      b. C423, Standard Test Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method.
      c. C518, Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus.
      d. C578, Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation.
      e. C665, Standard Specification for Mineral-Fiber Blanket Thermal Insulation for Light Frame Construction and Manufactured Housing.
      f. D1621, Standard Test Method for Compressive Properties of Rigid Cellular Plastics.
      g. E84, Standard Test Method for Surface Burning Characteristics of Building Materials.
      h. E96/E96M, Standard Test Methods for Water Vapor Transmission of Materials.
   2. Underwriters Laboratories, Inc. (UL):
      a. Building Materials Directory.

B. Mock-Ups:
   1. Provide insulation for mock-ups required in Specification Section 04 22 00 – Concrete Masonry.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Manufacturer's recommendations on sealants, tapes and mastics.

B. Informational Submittals:
   1. Certification from insulation manufacturer stating that insulation proposed is acceptable for intended use per the Drawings.

1.4 SITE CONDITIONS

A. For purposes of this Specification Section, design frost line for this Project is 36 inches below grade.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:

1. Rigid extruded polystyrene board insulation:
   a. Dupont – Styrofoam Brand Square Edge XPS.
   b. Or Equal.
2. Rigid polyisocyanurate foam board insulation:
   a. Hunter Panels – Xci CG panels.
   b. Or Equal.
3. Blanket or batt thermal insulation:
   a. CertainTeed – Sustainable Insulation.
   b. Or Equal.
4. Sound control and Fire Safing insulation:
   a. Thermafiber by Owens Corning - SAFB.
   b. Or Equal.
5. Rigid Insulation with Integral Nail Board:
   a. Hunter Panels – Xci Ply, Nailable Polyiso Insulation (Nail Base).
   b. Or Equal.
6. Vapor retarder:
   a. Reef Industries, Inc – Griffolyn TX-1200FR
   b. Or Equal.

2.2 MATERIALS

A. General:

1. Foam plastic insulation used in buildings and structures shall comply with the requirements of the building code.
   a. Surface burning characteristics: ASTM E84.
   b. Flame spread index: Maximum 75.
   c. Smoke developed: Maximum 450.

B. Rigid Polystyrene Board Insulation:

1. Extruded: ASTM C578, Type IV.
   a. Water vapor transmission: ASTM E96/E96M, 1.5 perm-IN maximum.
   b. Water absorption: ASTM C272/C272M, 0.3% maximum.
   c. Thermal resistance: ASTM C518 at 75 degrees F mean temperature, 5.0 per inch.
2. Provide insulation designed for intended use.
   a. Perimeter insulation and protection board.
      1) Compressive strength: ASTM D1621, 25 psi.
      2) Thickness:
         a) Perimeter insulation: 2 inches (R-10).
         b) Protection board: 3/4 inch (R-4).
      3) Edges:
         a) Long edge: Square.
         b) Short edge: Square.

C. Rigid Polyisocyanurate Foam Board Insulation with coated glass facers on both sides:

1. ASTM C1289, Type 2, Class 2, Grade 3.

2. Compressive Strength: ASTM D1621, Grade 3, Minimum 25 psi.
3. CFC and HCFC free.
4. Ozone depletion potential: 0.
5. Thickness shall be 2.5 inches (R-15.3) unless noted otherwise on Drawings.
6. Fire resistance: ASTM E84.
    a. Flame spread: Less than 75.
    b. Smoke developed: Less than 450.
7. Provide insulation designed for intended use.

D. Sealant and Mastic (for setting polystyrene and/or polyisocyanurate insulation board): Manufacturer's recommended standard.

E. Blanket or Batt Thermal Insulation:
1. Glass or other inorganic fibers and resinous binders formed into flexible blankets or semi-rigid sheets.
2. Unfaced:
    a. ASTM C665, Type 1.
    b. Thermal Resistance - ASTM C518 - 6 inches (R-21).
    c. ASTM E136 – Non-combustible.
3. Minimum thickness as noted on Drawings.

F. Rigid Insulation with Integral Nail Board:
1. Rigid Insulation with Integral Nail Board shall comply with the following requirements:
    a. One layer of polyisocyanurate (polyiso) insulation bonded to 5/8-inch fire treated plywood nail board. (Overall thickness of insulation and nail base as shown on drawings).
    b. Rigid polyisocyanurate foam board properties:
    c. ASTM C1289, Type V (nail base).
        1) Compressive strength – ASTM D1621: Grade 3 (25 PSI minimum).
        2) Flame Spread – ASTM E84/UL 723: less than or equal to 75.
        3) Smoke Development – ASTM E84/UL 723: less than or equal to 450
        4) Water vapor transmission: ASTM E96, less than 1 perm at 1inch.
        5) Water absorption: ASTM C209, less than 1 percent by volume.
2. Fasteners for Rigid Insulation with Integral Nail board - Polyiso Rigid Sheathing: Several factors are involved with the proper fastening of the rigid insulation with integral nail board polyiso rigid sheathing. These include overall thickness of the panel, the weight of the specified cladding, substrate panels being fastened to and the type of support provided at the base of the wall assembly. Contact the manufacturer for assistance with fastening type and rate.

G. Vapor Retarder:
1. Fire rated, reinforced, 3 ply, Class 1 material.
2. Perm rating: ASTM E96, Not exceeding 0.036 grains/HR-FT^2-IN-Hg.
3. Puncture Strength: ASTM D-4833, 48 foot pounds.
4. Flame Spread: ASTM E-84, not less than 5.
5. Smoke Developed: ASTM E-84, not less than 45.

H. Vapor Retarder Tape: As recommended by vapor retarder manufacturer.

I. Sound Control and Fire Safing Insulation:
1. Mineral wool batts.
    a. ASTM C665, Type I.
    b. UL listed when used in fire rated construction.
2. Density: Minimum 2.4 pounds per cubic foot.
3. Sound Reduction, ASTM C423.
    a. Minimum NRC for 3 inches thick material: 1.05.
4. Thickness: As noted on Drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. General:
   1. Insulate full thickness over surfaces to be insulated.
   2. Fit tightly around obstructions, fill voids.
   3. Cover all penetrations (electrical junction boxes, switch boxes, piping, conduits, etc.) with insulation, taking care not to compromise the workings of the device.
   4. Fit butted joints of batt or blanket insulations tightly together.
   5. Apply single or double layer to achieve total thickness.
      a. If double layer is provided, stagger all joints minimum 12 inches.
   6. Do not use broken or torn pieces of insulation.
   7. Install so that completed installation is vapor tight.
      a. Seal all joints.
      b. Seal to abutting materials to maintain vapor retarder integrity.
      c. Provide manufacturer’s recommended vapor retarder tape for use with faced batt insulation or separate vapor retarder.
         1) If vapor retarder tape fails to adhere to any surface, apply sprayed-on adhesive as recommended by tape manufacturer to promote adhesion.
      d. Provide manufacturer’s recommended solvent-free sealant compatible with insulation board for rigid board insulation.
         1) Tape is not acceptable for use with rigid board insulation.

C. Blanket or Batt Insulation using Separate Vapor Retarder Sheet in Exterior Stud Wall Systems:
   1. Verify that all piping, conduit, electrical box and other in-wall work is complete prior to installing insulation and vapor retarder.
   2. Install insulation friction fit between studs.
   3. Tightly butt ends.
   4. Install vapor retarder to warm side of building exterior wall.
      a. Completely seal each wall area to surrounding construction.
   5. Install vapor retarder vertically.
      a. Use widest practical sheet.
      b. Install in continuous sheets, floor to structure above, without horizontal joints.
      c. Fold flaps of vapor retarder over studs.
      d. Tape flaps together continuously.
      e. Tape bottom and top edges to structure continuously.
      f. After installation of any additional conduit, boxes, piping or other items within wall system, repair all tears or penetrations of vapor retarder with vapor retarder tape prior to installation of gypsum board.

D. Rigid Board Polystyrene Insulation on Interior side of Exterior Walls behind Acoustical Cementitious Wood Fiber Panel:
   1. Install insulation on interior surfaces of exterior walls behind acoustical cementitious wood fiber panels.
   2. Set solidly in mastic.
   3. Make all joints tight.
   4. Cut and fit insulation at corners, and at penetrations, connections, etc.
   5. Seal all joints with sealant applied continuously to edges of pieces before installation or apply sealant bead continuously to joint after installation.

E. Rigid Board Insulation in Cavity Walls:
   1. Do not proceed with installation until subsequent work which conceals insulation is ready to be performed.
   2. Set each piece of insulation flush with the abutting piece to eliminate ledges in the face of the insulation.

3. Install mastic on face of concrete or masonry back-up in accordance with mastic and insulation manufacturer's recommendation.
4. Press courses of insulation between wall ties (horizontal reinforcing) with edges butted tightly both ways.
5. Set units firmly into mastic.
6. Seal all horizontal and vertical joints with sealant recommended by insulation manufacturer.
7. Do not use damaged insulation.

F. Rigid Insulation at Perimeter Below Grade and Under Slab:
1. Install insulation below grade on outside face of foundation walls.
   a. Install in mastic with tight joints.
2. Where footings are located below the design frost line, extend insulation down to the design frost line.
   a. Where indicated on the Drawings, extend beyond the design frost line.
3. Where footings are located at the design frost line, extend insulation down to top of footing or as indicated on Drawings.
4. Protect insulation from damage and/or displacement during backfilling and/or pouring of floor slab.

G. Rigid Insulation with Integral Nail Base:
1. Install on exterior side of exterior walls in accordance with the manufacturer’s recommendations.
2. Installer shall take precautions for fire safety when roof insulation is installed. Protect foam from flame cutting and welding operations, etc.

H. Sound Control Insulation:
1. Install friction fit between studs.
2. Do not obstruct ventilation spaces.
3. Fill all miscellaneous voids unless noted otherwise on Drawings.
4. After installation of conduit, boxes, piping or other items within wall system, reposition displaced insulation and fill all voids.

3.2 FIELD QUALITY CONTROL

A. Repair or replace damaged insulation and/or vapor retarder as directed by Contracting Officer Representative.

END OF SECTION

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SECTION 07 26 00
UNDER SLAB VAPOR RETARDER

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Under slab vapor retarder.

B. Related Specification Sections include but are not necessarily limited to:

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Concrete Institute (ACI):
      a. 302.2R, Guide for Concrete Slabs that Receive Moisture-Sensitive Flooring Materials.
   2. ASTM International (ASTM):
      a. E1643, Standard Practice for Selection, Design, Installation, and Inspection of Water Vapor Retarders Used in Contact with Earth or Granular Fill Under Concrete Slabs.
      b. E1745, Standard Specification for Plastic Water Vapor Retarders Used in Contact with Soil or Granular Fill Under Concrete Slabs.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Product data sheet on vapor retarder sheet and vapor retarder tape.
      c. All accessories proposed for use.
      d. Manufacturer's installation instructions.

B. Samples:
   1. Provide two, 6 inches x 6 inches samples of vapor retarder material taped together using the vapor retarder tape proposed.
   2. Provide two samples of all accessories proposed for use.

C. Informational Submittals: Manufacturer's recommendation on vapor retarder tape.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Vapor retarder:
      a. Stego Industries, LLC.
      b. Or Equal.

2.2 PERFORMANCE REQUIREMENTS

A. Vapor Retarder:
   1. ASTM E1745, Class A.
   2. Thickness: Minimum 15 mil.
   3. Water vapor permeance: 0.02 maximum.

2.3 ACCESSORIES

A. Pipe Boots: Manufacturer's standard boot fabricated to maintain the integrity of the vapor retarder system.

B. Vapor Retarder Tape: As recommended by vapor retarder manufacturers.

C. Mastic: Manufacturers liquid vapor retardant water-based, trowel-grade asphaltic emulsion.

D. Flexible, Double-Sided Tape: Manufacturers flexible and moldable double sided adhesive tape used for perimeter sealing.

E. Termination Bar: Manufacturers semi-flexible plastic or metallic termination bar for securing vapor retarder to concrete, masonry or wood. Termination bar shall have predrilled holes every 6 inches on center.

PART 3 - EXECUTION

3.1 PREPARATION

A. Base material over which vapor retarder is to be installed shall be level, compacted and free of debris, foreign objects or other deleterious materials.

B. Surfaces at perimeter and penetrations of vapor barrier shall be clean, smooth and free of sharp objects, fins or projections.

3.2 INSTALLATION

A. Install products in accordance with manufacturer's instructions, ASTM E1643 and ACI 302.2R.

B. Provide vapor retarder where indicated on the Drawings.
   1. Place continuous vapor retarder above granular fill subgrade material, unless noted otherwise.
   2. Unroll vapor barrier with the longest dimension parallel with the direction of the concrete placement and face laps away from the expected direction of the placement whenever possible.
   3. Extend vapor barrier to the perimeter of the slab and extend up wall. Terminate at the top of the slab or where obstruct impediments, such as dowels, water stops, or any other site condition requiring early termination of the vapor barrier. Terminate and seal the vapor barrier to the entire perimeter foundation wall or footing/grade beam with double sided tack tape or a continuous bead of mastic between the face of the substrate and the back side of the vapor retarder. Place the vapor retarder over the tack tape or mastic bead and roll to flatten and provide a seal. Install the termination bar on the exposed face of the vapor retarder and fastened per manufacturer’s instructions. Ensure the concrete is clean and dry prior to adhering tape.

C. Lap minimum 6 inches at seams and seal in accordance with ASTM E1643 and manufacturer's recommendations.

D. Extend to extremities of area and seal to adjacent elements.

E. Seal all penetrations per manufacturers written instructions: Provide pipe boot for all pipes or conduit penetrating the floor slab.

F. Use reinforcing bar supports with base sections that eliminate or minimize the potential for puncture of the vapor barrier.

G. Repair damaged areas with vapor barrier material of similar (or better) permeance, puncture and tensile.

3.3 FIELD QUALITY CONTROL

A. Ensure proper precautions are implemented to prevent damage to installed vapor retarder membrane prior to and during pouring of concrete floor slab.

B. Inspect vapor retarder immediately prior to placement of concrete.
   1. Patch all punctures, tears, holes, etc.
      a. Patch small punctures with vapor retarder tape as allowed by ASTM E1643 and manufacturer's recommendations.
      b. Repair larger damage with additional layer of vapor retarder.
         1) Lap repairs minimum 6 inches beyond extent of damage in all directions.
         2) Seal perimeter of patch with vapor retarder tape or as recommended by manufacturer.

END OF SECTION

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SECTION 07 27 46
VAPOR PERMEABLE AIR BARRIER

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Vapor Permeable Air Barrier where identified on drawings as Fluid Applied Air and Moisture Barrier, in accordance with provisions of Contract Documents.

B. Completely coordinate with work of other trades.

1.2 REFERENCES

A. ASTM International (ASTM):
   1. ASTM C920, Standard Specification for Elastomeric Joint Sealants that are cured single or multi-component, cold applied elastomeric sealants.
   2. ASTM E96, Test Methods for Water Vapor Transmission of Materials.
   3. ASTM E331, Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference.
   4. ASTM E2178, Standard Test Method for Air Permeance of Building Materials.

B. National Fire Protection Association (NFPA):
   1. NFPA 285, Standard Fire Test Method for Evaluation of Fire Propagation Characteristics of Exterior Non-Load-Bearing Wall Assemblies Containing Combustible Components.

1.3 SUBMITTALS

A. Product Data:
   1. Manufacturer's literature indicating specified material and required components.

B. Shop Drawings:
   1. Show locations and extent of fluid applied air and moisture barrier.
   2. Include details for substrate joints and cracks, counterflashing strip, penetrations, inside and outside corners, terminations, and tie-ins with adjoining construction.
   3. Details and isometrics at openings showing layouts of air barrier flashings and sequence of installation.

C. Project Information:
   1. Written documentation demonstrating applicator's qualifications.
   2. Minutes of Preinstallation Conference.
   3. Mockup and Field Test:
      a. Install fluid applied air and moisture barrier on exterior mockup wall panel showing application and detailing of joints, corners and seams.

D. Contract Closeout Information:
   1. Installer Warranty.

1.4 QUALITY ASSURANCE

A. Manufacturer Qualifications:
   1. Fluid Applied Air and Moisture Barrier systems shall be manufactured by firm with minimum of 20 years of experience in production of waterproofing.
   2. Obtain primary air barrier materials and air barrier accessories from single source from single manufacturer.

B. Installer Qualifications:
   1. Minimum five years of continued experience in successful installation of vapor and water resistive air barrier products on similar project applications.

1.5 MOCK-UP

A. Provide Fluid Applied Air and Moisture Barrier for mock-ups required in Specification Section 04 22 00 – Concrete Masonry.

1.6 WARRANTY

A. Installer five-year warranty for Fluid Applied Air and Moisture Barrier and accessories have been installed in accordance with manufacturer's recommendations, and that components used in this section have been sourced from one manufacturer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Fluid Applied Air and Moisture Barrier (Vapor Permeable):
      a. Henry a Carlisle Company - Fire Resist Barritech VP.
      b. Or Equal.

2.2 DESIGN CRITERIA

A. Fluid Applied Air and Moisture Barrier System:
   1. Fluid applied air and moisture barrier system shall be capable of performing as a continuous, vapor permeable air barrier.
   2. Design system to accommodate substrate movement and to seal to expansion and control joints.
   3. Select components to accommodate construction material changes, penetrations, and transitions at perimeter conditions without deterioration and air leakage exceeding specified limits.
   4. Made from inherently fire-resistant materials.

B. Air Permeance:
   1. M 0.004 cubic feet per minute per square foot 0.3 water pressure differential, as tested in accordance with ASTM E2178.

C. Vapor Permeance:
   1. Minimum 10 Perms when tested in accordance with ASTM E96.

D. Water Intrusion Resistance on Sheathing Wall Assembly:
   1. No leaking observed after 2 hours at 6.24 pounds per square foot and after 15 minutes at 15 pounds per square foot per ASTM E331.

E. Wall Assembly Burn Test:
   1. Pass up to 3 inches of polyiso or XPS insulation per NFPA 285.

F. Exposure Rating:
   1. Provide material with appropriate exposure rating to suit construction schedule.

G. Application Window:
   1. Provide standard or low temperature air and moisture barrier system based upon ambient weather conditions at time of installation.

a. Low temperature products shall meet or exceed design criteria of standard product.

2.3 MATERIALS

A. Fluid applied air and moisture barrier membrane:
   1. 35 dry MIL elastomeric membrane.

B. Transition Membrane:
   1. Self-adhesive.
   2. Compatible with air and moisture barrier material and adjacent substrate.
   3. As recommended by air and moisture barrier manufacturer.

C. Primer: Spray applied adhesive as directed by manufacturer for substrate.

D. Termination Sealant:
   1. Silicone sealant – One Component complying with ASTM C920, Type S, Grade NS, Class 25, Use NT, G, A, O.

PART 3 - EXECUTION

3.1 PREPARATION

A. Examine conditions for compliance with requirements for installation, tolerances and other specific conditions affecting performance of fluid applied air and moisture barrier.

B. Substrate to be smooth and free of voids, spalled areas, and sharp protrusions.

C. Masonry Substrates:
   1. Fill voids, holes, and mortar joints, with lean mortar mix, non-shrinking grout or parge coat.
   2. Verify that concrete has cured and aged for minimum period recommended by air barrier manufacturer.
   3. Verify that concrete is visibly dry and free of moisture. Test for capillary moisture by plastic sheet method according to ASTM D4263.
   4. Strike masonry joints smooth and completely fill with mortar.

D. Exterior Plywood Sheathing Panels:
   1. Pre-treat joints with reinforced self-adhesive tape or mesh style wallboard tape.
   2. Caulk gaps greater than 1/4-inches.

E. Remove deleterious materials from surfaces to be covered.

F. Do not proceed with installation until unsatisfactory conditions have been corrected.

G. Installation indicates acceptance of substrates and responsibility for performance.

3.2 INSTALLATION

A. Install according to manufacturer’s directions.

B. Perform work only when existing and forecast weather conditions are within limits established by manufacturer.

C. Join fluid applied air and moisture barrier material in an airtight and flexible manner to air and moisture barrier materials of adjacent systems, allowing for relative movement of systems due to thermal and moisture variations and creep, between:
   1. Foundation and walls.
   2. Walls and windows or doors.
   3. Different wall systems.
   4. Wall and roof.
   5. Wall and roof over unconditioned space.
   6. Walls, floors and roof across construction, control, and expansion joints.
   7. Walls, floors and roof to utility, pipe, and duct penetrations.

D. Installation of the Fluid Applied Air and Vapor Barrier Membrane System:
   1. Primer: Apply at the rate recommended by manufacturer prior to membrane installation. Allow primer to dry completely before membrane application.
   2. Fluid Applied Air and Vapor Barrier Membranes:
      a. Spray or trowel apply continuous uniform film using multiple, overlapping passes to ensure even thickness and coverage.
      b. Carry membrane into openings at a minimum of 3 inches.
      c. Seal brick ties and penetrations as work progresses.
      d. Overlap the below grade fluid applied waterproofing, where the first course of concrete masonry is set on concrete foundation wall, a minimum of 6 inches, with the fluid applied air and moisture barrier membrane.

E. Installation of Transition Membrane:
   1. Install mastic at terminations, substrate transitions, penetrations and overlaps according to manufacturer's standard details.
   2. Overlap fluid applied membrane onto each surface at beams, columns, and joints.
   3. Tie into window and door frames, spandrel panels, roof and floor intersections, and changes in substrate.
   4. Seal top edge of flashing with termination mastic.

3.3 FIELD QUALITY CONTROL

A. Mockup:
   1. Construct typical exterior wall section, incorporating back-up wall, cladding, window, door frame and sill, insulation, flashing and other critical junctions.
   2. Allow adequate time to inspect and test mock-up before proceeding with work.

3.4 PROTECTION

A. Schedule work to ensure that the system is covered as soon as practicable.
   1. Protect the system from damage during subsequent operations.

B. Protect fluid applied air and moisture barrier from exposure to UV light and harmful weather exposure as required by manufacturer.
   1. Apply temporary UV protection if the system cannot be covered within the prescribed period.
   2. Replace air barrier if determined be damaged by UV exposure by Manufacturer.
   3. Replace air barrier exposed to UV for longer than recommended period.

C. Clean spills, stains, and soiling from construction that would be exposed in completed work as recommended by manufacturer of affected construction.

D. Remove masking materials after installation.

END OF SECTION

SECTION 07 46 10
CEMENT BOARD SIDING

PART 1 - GENERAL

1.1 SUMMARY

A. Pre-Finished Fiber-Cement Board Lap Siding, trim, fascia, and accessories.

1.2 RELATED SECTIONS

A. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 - Submittal Procedures.
   2. Section 04 22 00 - Concrete Masonry.
   3. Section 06 10 00 - Rough Carpentry.
   4. Section 07 21 00 - Building Insulation.
   5. Section 07 27 46 - Vapor Permeable Air Barrier.
   6. Section 09 06 00 - Color Schedule.

1.3 REFERENCES

A. References:
   1. ASTM International (ASTM):
      a. C209, Standard Test Methods for Cellulosic Fiber Insulating Board
      b. C1289, Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board
      c. D882, Standard Test Method for Tensile Properties of Thin Plastic Sheeting
      d. D903, Standard Test Method for Peel or Stripping Strength of Adhesive Bonds
      e. D1177, Standard Test Method for Freezing Point of Aqueous Engine Coolants
      f. D1621, Standard Test Method for Compressive Properties of Rigid Cellular Plastics
      g. D4073, Standard Test Method for Tensile-Tear Strength of Bituminous Roofing Membranes
      h. E84, Standard Test Method for Surface Burning Characteristics of Building Materials
      i. E96, Standard Test Methods for Water Vapor Transmission of Materials
      j. E331, Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference
      k. E2178, Standard Test Method for Determining Air Leakage Rate and Calculation of Air Permeance of Building Materials
   2. International Building Code
      a. IBC – 2018 Edition

1.4 PERFORMANCE REQUIREMENTS

A. Wind Load Design: Basic Wind Speed (Ultimate Wind Speed) 111 Miles per Hour with an Exposure “C” Category, Risk Factor “IV”.

1.5 SUBMITTALS

A. Product Data: Manufacturer's data sheets on each product to be used, including:
   1. Preparation instructions and recommendations.
   2. Storage and handling requirements and recommendations.
   3. Installation methods, including nailing patterns.

B. Maintenance and periodic inspection recommendations.

C. Shop Drawings: Provide detailed Drawings of atypical non-standard applications of cementitious siding materials which are outside the scope of the standard details and specifications provided by the manufacturer.

D. Selection Samples:
   1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
   2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

1.6 QUALITY ASSURANCE

A. Installer Qualifications: Minimum of 3 years’ experience with installation of similar products.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Store products in manufacturer's unopened packaging until ready for installation.

B. Store siding on edge or lay flat on a smooth level surface. Protect edges and corners from chipping. Store sheets under cover and keep dry prior to installing.

C. Store and dispose of solvent-based materials, and materials used with solvent-based materials, in accordance with requirements of local Authorities Having Jurisdiction.

1.8 PROJECT CONDITIONS

A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.

1.9 MOCK-UP

A. Provide pre-finished fiber-cement board lap siding for mock-ups required in Specification Section 04 22 00 – Concrete Masonry.

1.10 SEQUENCING

A. Ensure that products of this section are supplied to affected trades in time to prevent interruption of construction progress.

B. Install specified sheathing substrates, fluid applied weather barrier and continuous insulation with integral nail base in accordance with applicable specifications prior to installing hardboard exterior system. Sequence of the hardboard exterior system is as follows:
   1. Install Exterior Plywood Sheathing or Concrete Masonry Substrate.
   2. Install Water - Resistant Fluid Applied Air and Moister Barrier.
   3. Install Continuous Rigid Insulation with Nail Base.
   4. Install Fluid Applied Air and Moisture Barrier.
   5. Install Hardboard Trim Boards.
   6. Install Hardboard Starter Strips.
   7. Install Pre-Finished Fiber-Cement Board Lap Siding.

1.11 WARRANTY

A. Product Warranty: Limited, non-pro-rated product warranty.
   1. Pre-Finished Fiber-Cement Board Lap Siding for 30 years.

B. Workmanship Warranty: Application limited warranty for 2 years.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.

B. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.
   1. Pre-finished Fiber - Cement Board Siding and Trim Boards
      a. Basis of Design: James Hardie Building Products, Inc., which is located at: 26300 La Alameda Suite 400; Mission Viejo, CA 92691; Toll Free Tel: 866-274-3464; Tel: 949-367-4980; Fax: 949-367-4981.
      b. Or Equal.
   2. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 LAP SIDING & TRIM BOARDS

A. Pre-finished Fiber Plank lap siding:
   1. Style: Plank Lap Siding.
   2. Width: 7.25-inches with 6-inch exposed surface.
   3. Color: Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
   4. Finish: Smooth.

B. Pre-finished Fiber Trim Boards:
   1. Trim Boards Style/Finish: 4/4 Smooth.
   2. Width: 5.5-inches.
   3. Thickness: 0.75-inches.
   4. Color: Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
   5. Hardboard trim boards shall come from the manufacturer with a protective laminate slip sheet that shall remain on the trim boards during cutting and fastening to walls to reduce damage to the boards. Remove the protective sheet only after installing the boards and filling the fastener holes in accordance with the manufacturer’s recommendations.

2.3 FASTENERS

A. Fasteners for Lap Siding shall be corrosion resistant, hot - dipped zinc - coated galvanized steel complying with the IBC 2018 edition code.
   1. Consult product evaluation ICC - ES, ESR - 2290 for correct fastener type and placement to achieve specified design wind load.
   2. Exposed fasteners shall be prefinished color coordinated with the color of the lap siding.

B. Fasteners for Trim Boards shall be corrosion resistant, hot - dipped zinc - coated galvanized steel complying with the IBC 2018 edition code.
   1. Consult product evaluation ICC - ES, ESR - 2290 for correct fastener type and placement to achieve specified design wind load.
   2. Exposed fasteners shall be prefinished color coordinated with the color of the trim boards.

2.4 FINISHES

A. Hardboard Lap Siding and Trim boards shall be factory - primed and factory - pre - finished with a factory - applied oven - baked finish by the manufacturer.

B. Color:
   1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
   2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Do not begin installation until substrates have been properly prepared.

B. Notify Contractor and Contracting Officer Representative of unsatisfactory preparation before proceeding.

3.2 PREPARATION

A. Clean surfaces thoroughly prior to installation.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

C. Install a water-resistive barrier is required in accordance with local building code requirements.

3.3 MOCKUP

A. Construct typical exterior wall section, incorporating back-up wall, cladding, window, door frame and sill, insulation, air and moisture barrier, flashing, pre-finished fiber-cement board lap siding and other critical junctions.

B. Mockup shall be constructed in a stepped method allowing each material to be seen in the full wall assembly.

C. Allow adequate time to inspect and test mock-up before proceeding with work.

3.4 INSTALLATION TRIM BOARDS

A. Hardboard Trim Boards shall be used for Outside and Inside Corners, Band Boards, Windows, and Door Trim Applications.
   1. All Hardboard Trim boards shall be installed with flat tabs and corner tabs which provide a concealed fastening system. Only flat tabs and corner tabs provided by the hardboard trim manufacturer shall be used with the hardboard trim boards to create the concealed fastening system. Provide additional framing as required to ensure the flat tabs and corner tabs are fastened properly to the structure.
   2. Only install siding to side of trim by butting to it to the trim. Do not install any trim product over siding.
   3. All weather cut joints shall be touched up prior to installation per the hardboard manufacturers written instructions.

B. Outside and Inside Corner Trim Installation: Corners made from hardboard trim boards shall be pre - assembled before they’re installed. Pre - assembled corners look better and are easy to install. Inside corners shall be made with one board on each wall. All corners shall be attached with a concealed fastening system unless noted otherwise on drawings.
   1. To join two pieces of hardboard trim boards for a corner, drive 2 inches long by 14-gauge corrosion - resistant finish nail 1/2 inches from the edge and spaced at 16 inches apart along the edge. Pre - assemble corners of hardboard trim boards by positioning the finish nailer parallel with the trim.
   2. When walls are more than 10 feet high, splice corner boards together using weather cuts of at least a 22.5-degree angle. The angle of the weather cut must slope downward and away from the building. Nail both boards to the building with the same attachment specified for pre - assembled corners. Keep weather cuts at least 12-inches apart on adjacent corner boards.
   3. Corner Trim Board Installation: Hardboard trim boards shall be installed around corners by pre - building the corner off the wall with corner tabs providing a concealed fastening system.
   4. Attach concealed corner tabs to the back side of the trim with (8) 18-gauge 1/2-inch-long x 1/4-inch-wide narrow crown corrosion resistant staples using a pneumatic staple gun. Ensure the corner tabs are fastened tight and straight to the trim boards.
   5. Attach the trim to the building with galvanized siding fasteners, fastened through the corner tabs.
   6. Attach a corner tab 1 inch from each edge and every 20 inches on center.
   7. Creating a jig for the workstation is recommended to ensure the corners are fastened securely and straight.

8. When concealed fasteners are not specified to be installed fasten 4-inch-wide trim corners to the wall by driving a pair of fasteners (one fastened into each face of the corner) spaced 16 inches apart except that 4-inch-wide hardboard trim boards shall have two fasteners per side at splices. For 6-inch-wide trim corners, drive a pair of fasteners (two into each face of the corner) spaced 16-inches apart and 6-inch-wide hardboard trim boards shall have two nails per side at splices. Fasteners used to fasten trim to walls shall be kept ¾-inch from the edges of the board and 1 inch from the ends.

C. Band Board Installation: When installing a band board, pay special attention to flashing details and allow for potential shrinkage of solid rim joists in the walls that the band board may be attached to. Terminate ends of the Band Board into trim or siding or miter cut the edges of the trim at the corners of the building. Hardboard trim boards shall be installed with corner tabs providing a concealed fastening system.
   1. Caulk between the underside of the band board and the siding below. Do not caulk between the flashing and siding above the band board and maintain a 1/4-inch gap between the two. Also make sure the flashing covers over the concealed flat tab fasteners and that the water - resistive weather barrier laps over the flashing for a continuous drainage plane. When running lap siding above the band board, a 1-1/4 inch wide by 5/16-inch-thick starter strip shall be installed first to maintain the correct siding angle. Small periodic gaps shall be left in the starter strip to provide an escape route for excess moisture that may drain down behind the siding. Use bevel - cut splice joints of at least 22.5 degrees to join long lengths of hardboard trim boards.
   2. Provide concealed fastening by installing a flat tab at the end of each trim board and one tab at every stud with a maximum of 16 inches on center. The concealed fastening flat tabs shall be attached to the trim in an alternating pattern to the top and bottom of the band board. Attach Flat Tabs to the back side of the trim with (4) 18-gauge 1/2-inch-long x 1/4-inch-wide narrow crown corrosion resistant staples, equally spaced in one row, positioned no closer than 1/2-inch from trim edges, using a pneumatic staple gun. Attach the trim to the building using siding fasteners, fastened through the flat tabs into metal stud framing.
   3. When concealed fasteners are not specified to be installed fasten band boards to the building by driving two fasteners every 16-inches for 4-inch and 6-inch-wide boards.

D. Windows and doors shall be installed per the manufacturer's instructions. Window flanges or flashings must be properly installed and lapped correctly under the water - resistive weather barrier prior to the installation of hardboard trim boards. Once the hardboard trim boards are installed, proper flashing shall be installed above the trim and lapped under the water - resistive weather barrier correctly.

E. Install hardboard trim boards around all doors and windows using the "Cap Over" method, which means that the header or horizontal top piece of the trim extends over and caps the vertical jamb pieces on both sides of the opening. For windows, the bottom trim piece or sill trim fits in between the jambs. For cap - over trim installation:
   1. Start by measuring the length of the bottom edge of the window, not including the flange.
   2. Cut a piece of trim to that length and install it.
   3. Next measure from the bottom of the installed trim to the top of the window.
   4. Cut two pieces of trim to that length and install them on either side of the window.
   5. For the cap, measure the distance between the outside edges of the side trim pieces. Cut a piece of trim to length and install it.

F. For doors the process is the same except that it starts with the side pieces, step three.
   1. Measure from the bottom of the door frame to the top of the window frame.
   2. Cut two pieces of trim to that length and install them on either side of the door frame.
   3. For the cap, measure the distance between the outside edges of the side trim pieces. Cut a piece of trim to length and install it at the head of the door frame and tops of the jamb trims.

G. For trimming around windows and doors with attachment flanges, install a shim strip to build out the wall even with the flange. This strip lets the trim sit flat and parallel with the wall.

3.5 INSTALLATION –PRE-FINISHED FIBER-CEMENT BOARD LAP SIDING

A. General Installation:
   1. Install in accordance with manufacturer's written instructions and drawing details.
      a. Read warranty and comply with terms necessary to maintain warranty coverage.
      b. Install in accordance with conditions stated in model code evaluation service report.
   2. Allow products to adjust to ambient conditions before starting installation.
   3. Do not install to green wood or crooked structural framing. Do not install over rain soaked or buckled materials. Do not install if excessive moisture is present in the interior, including that from wet gypsum board materials.
   4. Rigid Insulation: Fasten siding through insulation and sheathing into studs.
   5. Install sheet metal flashing above door and window casings and horizontal trim in field of siding.
   6. Do not install siding less than 6-inches from ground nor closer than 1 inch to roofs, patios, porches, and other surfaces where water may collect.
   7. Do not cut siding to fabricate trim; use trim components manufactured for their intended purpose.
   8. After installation, seal joints except lap joints of lap siding. Seal continuous around penetrations.

B. Specific Installation Requirements:
   1. Layout: For accurate fastening, snap vertical chalk lines on the water - resistive barrier at the center of every stud location.
   2. Installation of Starter Strip: Hardboard lap siding requires a starter strip beneath the first course to set it on the proper angle and to create a proper drip edge at the bottom of the siding. Starter strips are easily made by ripping 1-1/4-inch pieces of the 5/16-inch hardboard siding either from full or partial planks.
      a. The bottom of the starter strip shall be installed even with the bottom of the bottom edge of the base substrate. The strip shall be installed on the exterior side of the water - resistive weather barrier, with occasional gaps left in the starter strip to allow any accumulated moisture behind the siding to drain out.
   3. Installing First Course of Lap Siding Plank: The first course of hardboard siding is critical to the proper installation of the plank on the rest of the building. The first course shall start at the lowest point of the wall and within required ground clearances set by the hardboard siding manufacturer. Special attention shall be made to ensure that all siding planks are straight and level. Snap a chalk line guide for the first course to follow. Attention shall also be paid to staggering butt joints in the siding planks so that the installation is attractive while making efficient use of material. Install factory ends of planks at butt joints with ends landing over a stud. Stagger butt joints a minimum of 2 stud bays for 16-inches on center framing.
      a. Use a 4 foot or longer level or chalked level line to assure that the first course is level. As installation proceeds up the wall, periodically check the level and straightness of the courses. When correcting for flatness over products such as exterior insulation, use drywall shims. Snap a chalk line every 3 to 5 courses to keep the planks straight and level.
      b. Position the bottom edge of the first course of siding a minimum 1/4-inch below the edge of the starter strip (maintain required clearances) and secure. The bottom of the siding shall be kept even with the bottom of the trim, but the siding shall never hang below the trim.
      c. Run the siding to a trim board leaving a 1/8-inch gap between the siding and trim and fill with sealant.
   4. Plank Alignment at Corners: Provide for the heights of the plank courses to match on both sides of a corner. Use a framing square, speed square and / or a level to match up and transfer the plank elevation heights. Check every few courses to make sure proper elevation heights are being maintained.
      a. End plank courses 1/8 inch from vertical trim and fill gap with sealant.

5. Blind Fastening (Fastening Through Top of Lap Siding Plank): Blind fastening shall be used in the installation of the hardboard lap siding plank which provides each course to cover the fasteners on the course below.
    a. For blind fastening hardboard lap siding, it is recommended that fasteners are driven 1-inch from the top edge of the plank. Additionally, fasteners shall be placed no closer than 3/8-inch from the ends of the plank. All fasteners shall be driven through the continuous insulation, sheathing and into the metal stud. The hardboard lap siding shall be manufactured with a nail line that shall be used as a guide for proper fastening placement for blind fastening. This nail line shall not be used as a lap line. Avoid placing fasteners near the top edge of the plank. This practice, called “high nailing”, may lead to loosen planks, unwanted gaps or rattling. Pinning of butt joints with a finish nail may be done for aesthetic purposes only. The finish nail shall be nailed flush to the surface (not countersunk) and shall be fully corrosion resistant “hot - dipped galvanized” and does not provide any structural support.
6. Staggering Butt Joints: For walls longer than 12-feet it is necessary to butt joint additional lengths of hardboard lap siding. These butt joints shall be staggered to avoid noticeable patterns, which is determined by the placement of the first course. Butt joints between consecutive courses shall be spaced apart by at least two stud bays. Provide random placement of the plank joints to provide an aesthetically pleasing effect.
7. Joint Flashing: The following joint treatment shall be required where butt joints are made in the lap siding to comply with the building code. Provide a 6-inch-wide piece of flashing behind butt joints that overlaps the lap siding course below by 1inch for an extra level of protection against the entry of water at the joint. The joint - flashing material shall be cut from a piece of the water - resistant weather barrier material used over the insulation substrate. Use light - colored joint flashing when using light - colored pre - finished lap siding and dark - colored joint flashing when using dark colored pre - finished lap siding.
8. Joint Placement and Treatment: Butt joints in Hardie Plank lap siding shall always land on a stud. Butt joints between studs are not allowed. Factory finished ends shall be used at butt joints. Place cut ends where the siding meets a corner, door, window trim, or other break in the wall where the joint is to be sealed with sealant. All cut ends shall be sealed with manufacturer’s colored edge seal finish.
    a. Once the initial course of hardboard siding is fastened to the wall, continue installing successive courses with full 12-foot pieces until a window, door or other opening interrupts the course. Notch planks as needed to fit around windows and doors and seal all cut edges. Stagger butt joints to produce an attractive (non - linear) pattern and avoid placing butt joints directly above or below windows or above doors. Separate all joints from an opening by at least one course of siding above or below the opening. Pre - finished hardboard lap siding is shipped with a protective laminate slip sheet that shall be left in place during cutting and fastening to reduce marring and scratching. The sheet shall be removed immediately after each plank is installed. When installing hardboard lap siding that is pre - finished, position the plank in the immediate area where the plank is to be fastened. Do not place the plank on the course below and slide into position. Doing so may scuff or scratch the pre - finished surface of the lower installed piece.
    b. Butt joints shall land on a stud, make sure there is enough stud space for plank on both sides of the joint to land properly. Both sides of a butt joint need to land in the middle of a stud with 3/4-inch landing space for each side. The minimum stud space for a plank to land on the face of a stud is 3/8-inch. Butt lap siding planks with moderate edge contact.
    c. Window, doors, and corners shall have trim installed before the hardboard lap siding is installed. Vertical trim boards that cover the king studs beside windows or doors or cover up corner studs leaving no room for nailing the siding shall have an extra stud added as needed. Where corners are trimmed with hardboard trim boards, it may be necessary to measure and cut the first pieces of siding to make sure that the butt joints land on studs.

9. Fastening: Hardboard lap siding products shall be hand or fastened with a tool when approved by manufacturer. Fastener shall be driven snug with the surface of the siding.
    a. Drive fasteners perpendicular to siding and framing.
    b. Fastener heads shall fit snug against siding (no air space).
    c. Do not over - drive fastener heads or drive fasteners at an angle.
10. Do not use aluminum fasteners, staples, or clipped head nails.

3.6 FINISHING

A. Touching Up Hardboard Lap Siding Planks and Trim Boards: Care shall be taken when handling and cutting pre - finished hardboard lap siding and trim board products. During installation use a wet soft cloth or soft brush to gently wipe off any residue or construction dust left on the product, then rinse with a garden hose.
    1. Touch up nicks, scrapes and nail heads using the manufacturers touch - up applicator. Touch - up should be used sparingly. If large areas require touch - up, replace the damaged area with new hardboard lap siding and trim boards.
    2. Laminate sheet must be removed immediately after installation of each course.
    3. Treat all non - factory cut edges using the manufacturers edge coaters.

3.7 PROTECTION

A. Protect installed products until completion of project.

B. Touch-up, repair or replace damaged products before Substantial Completion.

C. Clean dirt from surface of installed products, using mild soap and water.

D. After completing installation of hardboard products, remove from project site excess materials and debris resulting from installation of the hardboard products and accessories.

END OF SECTION

SECTION 07 61 13
METAL ROOFING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
1. Vapor retarder.
2. Thermal barrier sheathing.
3. Roof insulation.
4. Roof Sheet Weather Barrier membrane.
5. Standing seam metal roofing.
6. Standing seam fascia system.
7. Soffit panels.
8. Snow retention system (Only on Roaring Springs Pumphouse)
9. Lightning protection system clamps.
10. Sheet metal work required for roofing.
11. Roofing accessories.

B. Related Specification Sections include but are not necessarily limited to:
1. Section 01 33 23 - Submittal Procedures.
2. Section 01 78 23 - Operation and Maintenance Data.
3. Section 04 05 23 - Masonry Accessories.
4. Section 05 30 00 Metal Decking.
5. Section 06 10 00 - Rough Carpentry.
6. Section 09 06 00 - Color Schedule.

1.2 SCOPE

A. Work under this Section consists of providing all labor, materials, tools and equipment necessary for and incidental to the complete and proper providing and installation of all preformed sheet metal roofing as shown on drawings or specified herein in accordance with all applicable requirements of the Contract Documents and completely coordinated with work of all other trades.

B. Sheet metal roofing manufacturer shall provide all components required for a complete metal roofing system to include panels, trims / flashings, fascias, ridges, closures, sealants, fillers and any other required items.

1.3 QUALITY ASSURANCE

A. Referenced Standards:
1. American Architectural Manufacturers Association (AAMA):
a. 621, Voluntary Specifications for High Performance Organic Coatings on Coil Coated Architectural Hot Dipped Galvanized (HDG) and Zinc-Aluminum Coated Steel Substrates.
2. ASTM International (ASTM):
a. A653/A653M, Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
b. A792/A792M, Standard Specification for Steel Sheet, 55% Aluminum-Zinc Alloy-Coated by the Hot-Dip Process.
c. C209, Standard Test Methods for Cellulosic Fiber Insulating Board.
d. C1289, Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board.
e. D882, Standard Test Method for Tensile Properties of Thin Plastic Sheeting.

f. D1970/D1970M, Standard Specification for Self-Adhering Polymer Modified Bituminous Sheet Materials Used as Steep Roofing Underlayment for Ice Dam Protection.
g. D4833/D4833M, Standard Test Method for Index Puncture Resistance of Geomembranes and Related Products.
h. E96/E96M, Standard Test Methods for Water Vapor Transmission of Materials.
i. E1592, Standard Test Method for Structural Performance of Sheet Metal Roof and Siding Systems by Uniform Static Air Pressure Difference.
j. E1646, Standard Test Method for Water Penetration of Exterior Metal Roof Panel Systems by Uniform Static Air Pressure Difference.
k. E1680, Standard Test Method for Rate of Air Leakage Through Exterior Metal Roof Panel Systems.
l. E1745, Standard Specification for Water Vapor Retarders Used in Contact with Soil or Granular Fill under Concrete Slabs.
m. F593, Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.

3. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA):
   a. Architectural Sheet Metal Manual.
4. Underwriters Laboratories, Inc. (UL):
   a. Building Materials Directory.
   b. Fire Resistance Directory.
   c. 580, Standard for Tests for Uplift Resistance of Roof Assemblies.
   d. 790, Standard Test Methods for Fire Tests of Roof Coverings.
   e. 1256, Standard for Fire Test of Roof Deck Constructions.

B. Qualifications:
1. Manufacturer shall have minimum of 10 years of experience in the production of structural standing seam metal roofing.
   a. All structural components of the roof system shall be designed and sealed by registered professional structural engineer licensed in the State of Arizona.
2. Installing contractor shall be licensed or approved in writing by manufacturer.
3. Contractor and installer shall have minimum of seven years of experience in the installation of structural standing seam metal roof systems similar to system specified.
4. Contractor and installer shall have successfully completed two projects of similar size, scope and complexity within past two years.

C. Mock-Ups:
1. Prior to start of permanent roof construction construct mock-ups of roofing system.
   a. Mock-ups shall be of sufficient size to properly display all components required by the roofing system.
   b. Mock-ups shall be a minimum 5-feet x 5-feet in size.
   c. Provide multiple mock-ups as required.
2. Mock-ups shall incorporate all components, specified and/or required but not specified, needed for a complete water and airtight roofing fascia, and soffit system.
   a. Components include, but are not limited to:
      1) Vapor retarder, thermal barrier sheathing, roof insulation, roof sheet weather barrier membrane.
      2) Roofing panels, including mounting system and seaming.
      3) All fascia and soffit conditions.
      4) All flashing and counterflashing conditions, including:
         a) Eave, rake, hip and ridge conditions.
         b) Roof/vertical wall intersections.
         c) Roof penetrations.
         d) All reglet conditions.
      5) All miscellaneous clips, angles, plates, brackets, closures and sealants.
      6) Snow retention system. (Only on Roaring Springs Pumphouse).
3. Panels shall be same panels as specified or approved for Project.

a. Exact color is not necessary; however, Contractor is to label each exposed component to identify final installed color of component.

4. Step construction to allow observation of all components.
5. Construct additional mock-ups or rework existing mock-ups until acceptable to Contracting Officer Representative.
6. Maintain mock-ups at project site until Contracting Officer Representative approves removal of mock-ups.
7. Approved mock-ups to constitute minimum acceptable standard of quality for actual construction.

D. Completed roof system to be inspected by roof manufacturer's authorized factory trained representative prior to issuance of roof warranty.

1.4 DEFINITIONS

A. Installer or Applicator:
1. Installer or applicator is the person installing or applying the product in the field at the Project site.
2. Installer and applicator are synonymous.

B. Steep Slope: Having a pitch of 3:12 or greater.

C. Low Slope: Having a pitch less than 3:12 but greater than 1/4:12.

D. PVDF: Polyvinylidene fluoride.

1.5 SYSTEM DESCRIPTION

A. Prefinished, field-insulated, structural standing seam metal roof system, including but not limited to:
1. Vapor retarder.
2. Thermal barrier sheathing.
3. Roof insulation.
4. Roof sheet weather barrier membrane.
5. Standing seam metal roof panels.
a. Roof panel support and attachment system to be determined by standing seam roof manufacturer.

B. All flashing and miscellaneous trim required for a complete water and airtight system, including but not limited to:
1. Reglets.
2. Flashing.
3. Counterflashing.
4. Sealants.

C. Standing seam fascia system.

D. Snow retention system.

1.6 SUBMITTALS

A. Shop Drawings:
1. Fabrication and/or layout drawings:
a. Manufacturer prepared computer-generated Drawings showing anchorage, flashing, jointing and all other accessories required, and all special detailing required by the system.
1) Minimum plan scale: 1-inch = 8-feet.
2) Minimum detail scale: 1-1/2-inches = 1-foot.
b. Provide complete erection plan for each building structure with all details and sections referenced, all penetrations shown, expansion joints shown, detailed and referenced, and all special conditions identified, referenced and detailed.
c. Erection plan to identify limits of each different substrate material (decking).

d. Provide distinction between factory and field assembled work.

2. Product technical data including:
   a. Manufacturer data sheets on each component, including masonry reglets used in the roof system.
   b. Acknowledgement that products submitted meet requirements of standards referenced.
      1) Certification by manufacturer that roofing assembly being supplied has been successfully tested under UL 580 procedures and has achieved a Class 90 rating.
3. Test results:
   a. UL 580, Class 90 test data.
   b. UL 1897, Wind uplift
   c. UL 790, Class A fire rated
   d. UL 263, Fire resistance rated
   e. UL 2218, Impact resistance rated
   f. ASTM E1592 test results.
      1) Provide results of tests conducted in accordance with ASTM E1592 for panel size and gage and clip type and spacing similar to panels and clips being used.
   g. ASTM E1646 and ASTM E1680 test results.
   h. Concentrated load test data.
      1) Load test to be conducted on panel size, gage and with clip spacing as required.
4. Qualifications:
   a. Manufacturer: Provide structural engineer qualifications.
   b. Contractor:
      1) Certification of approval or license to install product from manufacturer.
      2) Certification of experience.
      3) Listing of projects completed in the past two years with similar scope.
      4) Completed projects information to include, square footage of roofing installed, dollar value of roofing installed, manufacturer and type of roofing installed and contact name and telephone number of building.
   c. Installer: Provide qualifications of all personnel expected to be working on the Project.
5. Roofing manufacturer's letter of approval for insulation proposed for use.
6. Warranty: Sample language of manufacturer's warranty to be provided on this Project.

B. Samples:
1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
3. General: Tag, identify and provide statement regarding use for all fasteners, anchor clips, closures and sealants.
4. Roof panel:
   a. Two samples, full width, 24-inches long.
5. Fasteners.
6. Anchor clips.
7. Closures, (both metal and non-metallic).
8. Masonry reglet.
9. Factory and field applied sealants.

C. Contract Closeout Information:
1. Operation and Maintenance Data:
   a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

D. Informational Submittals:
1. Sample of Warranty as a part of the original product data submittal.
2. Final Warranty as a part of O and M submittals.

1.7 WARRANTY

A. Provide 20 year minimum complete system warranty, including material for air and weather tightness of entire roof assembly signed by manufacturer.
   1. Warranty limits shall meet the minimum load capacity requirements of ASTM E1592.

B. Material substrate warranty is available only for galvalume substrates. Default substrate is galvalume; if galvanized substrate is specified, delete substrate from warrently requirements.

C. Provide manufacturer's 20-year - non-prorated - warranty on panel finish against fading, chipping, cracking and peeling of the panel exterior finish and erosion of substrate metal.

PART 2 - PRODUCTS

2.1 CONDITIONS

A. Contracting Officer will consider Contractor’s request for comparable product when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this specification Section, are expected to provide a system (roofing panels, fascia panel, soffit panels) with similar profile, standing seam height, snapping construction, spacing, construction and factory applied finish.

2.2 MATERIALS

A. Roof Panels, Fascia Panels and Soffit Panels:
   1. General:
      a. Galvalume steel, ASTM A792/A792M, Class SS, Grade 50B.
         1) Painted surfaces: AZ50.
         2) Unpainted surfaces: AZ55.

B. Insert the following hidden text if using a straight fascia instead of a standing seam fascia.

C. Insert the following hidden text if using aluminum soffit panels.

D. Perimeter Trim, Panel Closures, Flashing and Counterflashing: Material and factory applied finish to match roof panels.

E. Fasteners:
   1. 300 series stainless steel, ASTM F593.
   2. Hexhead
   3. Self-drilling or self-tapping
   4. EPDM or PVC washer

F. Sealant: Manufacturer's standard non-curing butyl.

G. Masonry Reglets: See Specification Section 04 05 23.

2.3 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Metal Roofing Panels:
      a. Basis of Design: Petersen, Pac-Clad - Snap Clad.
      b. Or Equal.
   2. Metal Fascia:
      a. Basis of Design: Petersen, Pac-Clad –Metal Soffit Panels.
      b. Or Equal.
   3. Metal Soffit Panels:
      a. Basis of Design: Petersen, Pac-Clad – Flush Soffit Panels.
      b. Or Equal.
   4. Thermal Barrier Sheathing:
      a. Basis of Design: Georgia-Pacific – DensDeck Roof Board.
      b. Or Equal.
   5. Roof Vapor Retarder.
      a. Basis of Design: Carlisle – VapAir Seal 725TR.
      b. Or Equal.
   6. Roof Insulation (2 Layer System):
      a. Basis of Design: Hunter Panels:
         1) H-Shield NB (NailBase Top Layer).
         2) H-Shield CG (Flat Base Layer).
      b. Or Equal.
   7. Roof Sheet Weather Barrier:
      a. Basis of Design: CMI – Secure Grip PS Max 100 HT.
      b. Or Equal.
   8. Snow Retention System (Only on Roaring Springs Pumphouse):
      a. Basis of Design: Metal Roof Innovations, Ltd. S-5! "ColorGard" system.
      b. Or Equal.
   9. Roof Penetration Flashing:
      a. Basis of Design: DEKS North America, Inc. "DEK-TITE."
      b. Or Equal.
   10. Lightning Protection System:
      a. Basis of Design: East Coast Lightning Equipment Inc.
      b. Or Equal.
   11. Flashing Curb:
      a. Basis of Design: Provided by metal roofing manufacturer.
      b. Or Equal.

C. Submit request for substitution in accordance with Specification Section 01 33 23.

2.4 FABRICATION

A. General:
   1. Fabricate with square, true corners, mitered and welded.
   2. Fabricate trim, flashings and closure pieces to match panel profile and finish.
   3. Hem all edges.
   4. Fabricate panels in full length with no end laps.

B. Standing Seam Metal Roof Panels:
   1. Panels:
      a. 1-3/4-inch-high seam x 16-inch-wide panels, 22-gauge pre-finished steel with longitudinal stiffening elements “striations”.
   2. Seam configuration shall consist of a female cap fitting over male leg which provides an integral continuous interlocking standing seam.
   3. Panel seams shall interlock entire length of seam without use of field seaming machines.
   4. Panel Manufacturer shall engineer standing seam to lock up and resist joint disengagement during design wind uplift conditions as calculated according to local building codes.

5. Provide factory sealant inside female cap seam to aid in resistance of leaks and to provide panel - to - panel seal while allowing expansion and contraction movement.
6. Exposed fasteners in roofing panels will not be permitted. Intermediate “Swag” joints will not be allowed. Panels shall run full length of roof slope with no seams and shall be true to shape, accurate in size, square, and free from distribution or manufacturing defects.
7. The system shall be designed as a true standing seam shape.
8. Provide vertical striations (shadow lines) on sloped roof panels.
9. Concealed fasteners:
    a. Provide concealed fasteners in all locations.
        1) Fasteners: Manufacturers recommended self - drilling self - tapping pancake head Phillips drive screws for plywood; non - corrosive base material to meet the required snow and wind loads.
    b. If exposed fasteners are required by the roof panel manufacturer, because of location, constructability issues or other critical design requirement, finish of fastener shall match roof panel finish.
        1) Exposed fasteners are to be approved by Contracting Officer Representative.
    c. The use of deflection limiter devices is not allowed.

C. Metal Fascia:
1. Metal fascia - 22 gage steel, width - as shown on drawings, flush appearance and shall match metal roof panels finish and color. Provide with trim and closer accessories as required for a complete finished metal fascia system.

D. Metal Soffit Panels:
1. Metal soffit panels - 22 gage steel, 1-inch deep, 12-inches wide.
2. Panels shall be a factory-formed, interlocking, architectural metal panel designed for soffit applications. The panel shall be a solid flush or flat appearance by utilizing an interlocking leg and concealed fastening system.
3. Provide with trim and closer accessories as required for a complete finished soffit system.
4. Metal soffit panels and trims shall match metal roof panels finish and color.

2.5 ACCESSORIES

A. Thermal Barrier Sheathing:
1. Commercial roofing board used as a cover board, thermal barrier, or substrate board.
2. Provides enhanced Wind uplift resistance.
3. Resists punctures that occur from things like maintenance foot traffic or dropped tools.
4. Thermal Barrier Sheathing shall comply with the following requirements:
5. 5/8-inch-thick exterior gypsum sheathing panels
6. R Value – 0.67
7. ASTM E84 – 0 Flame Spread and 0 Smoke Development
8. Fire Rating – Type X
9. Flexural Strength, Parallel - less than or equal to 100-foot pounds.
10. Flute Spanability – 8-inches.
11. Permeance – 32 perms.
12. Water Absorption – 10 percent maximum.

B. Roof Vapor Retarder:
1. Composite vapor retarder consisting of self-adhering rubberized asphalt laminated to woven polypropylene film with a one-piece silicone poly release liner complying with the following requirements:
    a. Thickness: ASTM D1970 – 40 mils.
    b. Air permeance: ASTM E2178 - 0.000Liters meters squared @ 75Pa.
    c. Tensile strength: ASTM D412 - 250 pounds per square inch.
    d. Puncture strength: ASTM E154 - 60 pounds.
    e. Elongation: ASTM D412 – 250 percent.
    f. Permeability: ASTM D1970 – 0.015 perms.
    g. Vapor retarder tape: As recommended by vapor retarder manufacturer.

C. Roof Insulation (2 Layer System).
   1. Roof Top Layer Insulation: Rigid polyisocyanurate foam board (Nailbase Panels).
      a. Nailbase to be installed on top layer of multi-layer (2-layer) system.
      b. Comply with ASTM C1289, Type V.
      c. Compressive strength: 20 psi minimum, Grade 2.
      d. Water vapor transmission: ASTM E96/E96M, less than 1.0 perms.
      e. Water absorption: ASTM C209, less than 1.0 percent.
      f. 5/8-inch, 5-ply CDX plywood adhered to foam with an overall 3.2-inch thickness, R-value of 15.0, and flute spanability of 4-3/8-inch.
   2. Roof Base Rigid polyisocyanurate foam board layer (Flat Panels).
      a. Flat board to be installed on bottom layer of multi-layer (2-layer) system. Insulation shall be closed-cell flat foam roof insulation with coated high performance glass fiber facer bonded on both sides of panel.
      b. Rigid polyisocyanurate insulation board to comply with the following requirements:
      c. Conforming to ASTM C 1289, Type II, Class 2 with square edges.
      d. Manufactured using 3rd Generation Zero ODP, EPA Compliant Blowing Agent; Contains zero CFCs, HCFCs, or HFCs; Virtually no Global Warming Potential (GWP)
      e. Fire Ratings:
         1) UL 1256, No. 120 and 123.
         2) UL 790 (ASTM E 108), Class A.
         3) UL 263 (ASTM E 119).
      f. Compressive Strength: 20 pounds per square inch, Grade 2.
      g. R Value: Provide Insulation with LTTR (Long Term Thermal Resistance) in accordance with ASTM C 1289. Minimum thickness of panels shall be as follows:
      h. 3-inches thick, R-17.4 and flute spanability of 4-3/8-inches.
   3. Fasteners shall be FM Approved by roof insulation manufacturer for SIP/SD Panel Fasteners and steel deck application. Fasteners have a 3/16-inch shank and are corrosion resistant with oversized heads. Length of fasteners shall be as recommended by roof insulation manufacturer. Use of 2-inch round plates is not required. See the roof insulation manufacturer's written application guide for instructions.
      a. See roof insulation manufacturers written information for penetration and fastening schedule requirements.
      b. Penetration of fastener into bottom flute is not acceptable.

D. Roof Sheet Weather Barrier:
   1. Self-Adhered Roof Sheet Weather Barrier - polypropylene surface, underlayment with UV and weather resistive coatings manufactured to be a secondary water barrier in sloped roof system construction that is an air, water and vapor retarder complying with the following requirements:
      a. Thickness: ASTM D5147 – 40 mils.
      b. Nail Seal ability: ASTM E1970 - Pass.
      c. Tensile strength: ASTM D1970 - 95 pounds per inch.
      d. Liquid Water Transmission: ASTM 4869 - Pass.
      e. Thermal Stability: ASTM D1970 – 240 degrees F.
      f. Tear Resistance: ASTM 1970 – 25-foot pounds.
      g. Permeability: ASTM E96 – 0.03 perms.
      h. Fire Resistance: ASTM E108 – Class A.

E. Snow Retention System (Only on Roaring Springs Pumphouse):
   1. Snow retention system shall be a mechanical non-penetrating system for sloped metal roof systems to prevent ice and snow from sliding off roof.
   2. Provide splice fittings for a continuous installation.
   3. Snow retention system clamps:
      a. Manufactured from 6061 - T6 aluminum extrusions conforming to ASTM B 221 or aluminum castings conforming to ASTM B 85 and to AA Aluminum Standards and Data.

b. Shall be secured to the standing seam with non-penetrating stainless steel set screws “300 Series stainless steel, 18 - 8 alloy, 3/8-inch diameter, with round nose point”.
c. Attachment bolts shall be “300 Series stainless steel, 18 - 8 alloy, 10 mm diameter, with flat washers”.
d. Finish: Manufacturer's standard finish.

4. Snow retention brackets:
   a. Manufactured from 6061 - T6 alloy and temper aluminum extrusions conforming to ASTM B 221 and AA Aluminum Standards and Data or cast aluminum.
   b. Screws for attachment of brackets to roof: Type best suited and recommended by snow retention manufacturer for application.
5. Cross Members:
   a. Manufactured from 6061 - T6 alloy and temper aluminum extrusions conforming to ASTM B221 and AA Aluminum Standards and Data.
   b. Receptacle in face to receive prefinished sheet metal strip to match finish and color of metal roofing panels.
6. Snow and Ice Clips: Provide snow retention system manufacturer’s standard aluminum snow and ice clips that are a minimum 3-inches wide. Provide 1 per panel and install in center between seams.
   a. Provide all necessary components: Clamps, Brackets, Cross Members, Color Strips, Set Screws, Tek Screws, and End Caps required for a complete system.

F. Roof Penetration Flashing:
1. Pre - manufactured roof jack shall be a one-piece EPDM pleated rubber cone designed for 1/4-inch to 24-inch diameter pipes. The base of the jack shall be made of flexible, corrosion resistant aluminum that conforms to any metal roof panel configuration with a non-absorbent and non-porous sponge gasket that seals by compression. The base conforms to the panels profile to provide a watertight seal.

G. Lightning Protection System:
1. Rods and cables shall be provided with non - destructive and non -penetrating clamps that comply with the following requirements:
2. Clamps shall match the roofing profile and shall be supplied as a part of the Lightning Protection System.

H. Flashing Curb:
1. One-piece completely seal welded prefabricated roof curb, including vertical flashing, and counter flashing, cricket on high side of penetration and flat pan fabricated to replace standing seam metal roof panel.
   a. Size as required for penetration.
   b. Bottom sloped to match roof.
   c. Level on top.
   d. Minimum 16 GA galvannealed steel.
   e. Finish to match roof panel.

I. Foam and metal closures, sealant, gaskets, fasteners, washers, clips, angles, and all miscellaneous trims shall be provided by roofing manufacturer, fabricated for the specific condition as required.

2.6 SOURCE QUALITY CONTROL

A. Roof assembly to be Class A roof covering assembly per UL 1256.

B. Structural Testing:
1. The system shall be designed to safely resist the positive and negative loads per the building code and as shown on Drawings.
2. Structural-uniform uplift load capacity of the panel system shall be determined in accordance with ASTM E1592.
   a. The factor of safety on the test results shall be 1.65 for the panel, batten or clip ultimate loads with no increase for wind.

b. The factor of safety for fasteners shall be 3.0 for one single fastener per clip, 2.25 for two fasteners per clip and 4.0 inches masonry.
c. Design uplift capacity for conditions of gage, span or loading other than those tested may be determined by interpolation of test results.
1) Extrapolation of conditions outside the range of the tests is not acceptable.
d. Deflection shall be L/180 for positive loading.

C. Water Penetration: No uncontrollable leakage at minimum 6.24 pounds per square foot when tested in accordance with ASTM E1646.

D. Air Infiltration: Maximum 0.037 cubic feet per minute per square feet when tested at 1.57 pounds per square foot differential pressure when tested in accordance with ASTM E1680.

E. Fire Resistance/Wind Uplift Rating:
1. UL 790, Class 1.
2. UL 580, Class 90.

F. The panels shall withstand a 250-pound concentrated load applied to a 4-square inches area at the center of the panel at mid span between supports with no panel deformation, rib buckling, or panel sidelap separation which will adversely affect the weather tightness of the system.

G. Support roofing panels on top of roof insulation using bearing plates attached to the structural frame or connect to manufacturer-provided intermediate support system.
1. Bearing plate and standing seam roof panel anchor clip attachment is to be determined by the roofing manufacturer and shall take precedent over this Specification.
a. Provide attachment to roof structural frame or deck as required for loading criteria specified.
2. Roof panel anchor clips shall be designed to allow thermal movement of the panels except where specific fixed points are indicated.
a. Roof panel manufacturer shall be responsible for determining fixed point locations unless otherwise indicated.
3. Wood blocking indicated on drawings is strictly for attachment of miscellaneous flashings and shall not be used for any structural value.
4. Maximum spacing of roof clips shall be determined by manufacturer.

H. Roof panel manufacturer shall be responsible for designing and installing all necessary expansion joints in the roof system.
1. Where roof expansion joints occur, provide corresponding expansion joints in fascia and soffit.

2.7 FINISH AND COLORS

A. Pre-finished metal standing seam roofing panels, break metal flashing, metal fascia's, metal soffit panels, and metal accessories, finish and color selections shall comply with the following:
1. Paint shall be PVDF based with minimum 70 percent resin.
2. Color:
a. Refer to "Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals" for submission procedures and requirements of Color Samples.
b. Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.

2.8 MAINTENANCE MATERIALS

A. Provide Government with 12 ounces of touch-up paint to match each different color used in the system.

PART 3 - EXECUTION

3.1 PRE - ROOFING CONFERENCE

A. Prior to installation of roofing and associated work, General Contractor shall schedule a meeting at project site with Metal Standing Seam Roofing Installer, Metal Standing Seam Roofing Manufacturer, Installers of related work such as metal coping, lightning protection system, roof top mechanical equipment suppliers, and other entities concerned with roofing performance including the Contracting Officer Representative. General Contractor shall record discussions and agreements and provide a copy to each participant including the Contracting Officer Representative.

3.2 INSPECTION

A. Inspect substrate where roofing is to be installed and verify that the substrate tolerances, as required, have been met. Examine substrate to insure it is properly secured and prepared to receive metal roofing. Do not start work until all unsatisfactory conditions have been corrected.

B. Snow Retention System: Inspect roof system for properly being attached and installed to withstand additional loadings incurred. Notify General Contractor of any deficiencies before installing snow retention system.

3.3 PREPARATION

A. Verify that substrate layout complies with shop drawing layout.

B. Report any variations and potential problems to the Contractor and Contracting Officer Representative.

C. Do not start work until all unsatisfactory conditions have been corrected.

D. Field Measurements: Take measurements prior to fabrication of prefabricated panels, take field measurements of structure or substrates to receive panel system. Allow for trimming panel units where final dimensions cannot be established prior to fabrication.

3.4 INSTALLATION

A. General:
   1. Install products in accordance with manufacturer's instructions, SMACNA (where referenced) and details shown on Drawings for a complete weathertight installation without waves, warps buckles or distortions.
      a. Provide all closures, trim, angles, plates, sealant, gaskets, fasteners, washers, etc., as necessary.
   2. Attachments shall allow for thermal expansion and contraction.
   3. Seal all joints as required for watertight installation.
   4. Touch-up paint all damaged surfaces.

B. Thermal Barrier Sheathing:
   1. Butt all board edges and ends tightly together.
   2. Install to a structural metal deck in a roof board application, support the edges by installing the edge joints on and parallel to the top flutes.
   3. Refer to system manufacturer and FM Global requirements for fastener type and spacing,

C. Vapor Retarder:
   1. Install on top of thermal barrier sheathing which shall be completely dry, smooth, and free of voids, spalled areas, sharp protrusions and form release agents.
   2. Artificial drying methods such as torches are not acceptable.
   3. Install adhesives approved by the vapor retarder manufacturer per written instructions by the vapor retarder manufacturer.
   4. Apply vapor retarder from low to high points in a shingle overlap fashion so the laps will shed water. Overlap all edges by at least 2-inches. End laps shall be staggered. Position the membrane carefully to avoid fish – mouths and wrinkles.

5. Roll the vapor retarder membrane immediately after placement with a 100 – 150-pound roller wrapped in a resilient material. Vertical surfaces shall be prepared in the same method but shall be hand rolled with a 2-inch or 4-inch hand roller.
6. Apply a 2-inch-long bead of lap sealant internally along any T-Joint or Step Offs. Use a hand roller or stand-up seam roller to mate the entire seam together.
7. If seam surface is contaminated clean and prime with vapor retarder manufacturer's approved primer.

D. Roof Insulation:
   1. Install in accordance with manufacturer's recommendations.
      a. See the roof insulation manufacturer's written application guide for instructions.
         1) See roof insulation manufacturers written information for penetration and fastening schedule requirements.
         2) Penetration of fastener into bottom flute is not acceptable.
      b. Do not begin installation until substrates have been properly prepared.
      c. Verify deck, adjacent materials, and structural backing is dry and ready to receive insulation.
      d. Verify deck surface is flat, free of fins, protrusions, and irregularities.
      e. If deck preparation is the responsibility of another installer, notify Contractor of unsatisfactory preparation before proceeding.
      f. Clean surfaces thoroughly prior to installation.
      g. Apply vapor retarder, as specified or required by the local building code, to decking prior to the installation.
      h. For multiple layered installations, install the base layer of panels loose laid, and stagger the joints of subsequent layers a minimum of 12-inches in each direction and in accordance with good roofing practice. Fasten panels through the top nailable surface and through the wood block panel spacers using roof insulation approved threaded fasteners.
      i. Only install enough roof insulation per day that can be covered the same day by a completed roof covering material. Cover the top and edges of unfinished roof panel work to protect it from the weather and to prevent accumulation of water in the cores of the panels.

E. Roof Sheet Weather Barrier Membrane:
   1. Install per manufacturer's recommendations in areas indicated on Drawings.
   2. Provide roof sheet weather barrier membrane over the entire roof deck.
   3. Install on a dry roof deck when air temperatures, membrane and substrate is over 40 degrees F. For "cold weather" applications below 40 degrees F the use of a primer is required and the laps blind nailed. Blind nail with 1-inch or longer roofing nails with 3/8-inch head in the marked lap area.
   4. For steep slope installation (6:12 or greater) or when installing at temperatures greater than 100 degrees F it is required to blind nail the overlap areas as noted above in the cold weather applications.
   5. Roof sheet weather barrier membrane is to be laid out horizontally, parallel to the roof eave with printed side up, using 3-inch horizontal laps and 6-inch vertical laps. Roof sheet weather barrier membrane may also be installed vertically. For vertical installations, anchor with 3/8-inch head roofing nails along the upper most edges 3- inches down with 6 equally spaced nails across the 36-inch sheet width. Ensure that the roofing nails are installed perpendicular to the roof deck so the head lays at and seals well into the top of the surface of the membrane.
   6. Vertical overlaps are to be 6-inch and 3-inches on horizontal laps if required.
   7. Always work from the low point up to the high point of the roof surface. Apply the roof sheet weather barrier in valley areas before the membrane is applied to the eaves.
   8. It is recommended to cut the membrane into manageable lengths. Peel back 1 to 2- feet of the release liner, align the membrane and then continue to peel the release liner from the membrane. Hand press or walk the membrane onto the surface, then follow with a 40 pound

or heavier pressure roller to smooth and secure the membrane, especially at overlap areas. If a roller is not available or considered safe in the circumstance, walk-in all laps and as much of the field area as possible to secure the adhesive into the pores of the roof deck and overlap areas.

9. For eave installation roof sheet weather barrier membrane extend the membrane over the perimeter edges of the roof a minimum of 6-inches behind the metal fascia.
10. Roof sheet weather barrier membrane shall never be suspended or “bridge” a valley.
11. Repair holes, “fish mouths”, tears, and any other penetration damage to the membrane with a rounded patch of membrane extending past the damaged area by 6 inches in all directions. If fasteners are removed, leaving holes in the membrane or other penetrations are accidentally created, they must also be patched.
12. Do not install fasteners through the membrane over any unsupported areas of the roof deck structure such as over joints between adjacent structural roof panels. •

F. Standing Seam Metal Roofing Panels:
1. Install metal roof panels over solid substrate and a layer of moisture waterproofing underlayment.
2. General: Comply with panel fabricator's and material manufacturers' instructions and recommendations for installation, as applicable to project conditions and supporting substrates. Anchor panels and other components of the work securely in place with provisions for thermal / structural movement.
3. Field cutting of exterior panels by torch is not permitted.
4. Install panels with concealed fastening system.
5. The roofing shall be installed plumb, straight and true to adjacent work.
6. Secure neoprene, imbedded in sealant with metal closures where required.
7. Standing Seam Roof Panel System: Fasten roof panels to support with panel anchors in accordance with the manufacturer's instructions.
8. Panel anchors shall be installed to meet the required snow, and wind loads per the referenced International Building Code requirements and manufacturer's recommendations.
9. No perforations, cuts or penetrations shall be made in the panels, except as noted on the drawings. The use of a hot saw shall not be allowed at any time.
10. Exposed fasteners in panels shall not be allowed.
11. Installation tolerances: Shim and align panel units within installed tolerance of 1/4-inch in 20-feet-0-inches on level / plumb / slope and location / line as indicated, and within 1/8-inch offset of adjoining faces and of alignment of matching profiles or as required by roofing manufacturer.

G. Snow Retention System (Only on Roaring Springs Pumphouse):
1. Install starting 2-feet from the eave edge of the roof or as shown on drawings.
2. Install system in continuous lengths using manufacturer provided splice fittings.
3. Mount to metal roofing utilizing clamps fastened to standing seam with non-penetrating bullet-nosed set-screws.
    a. Fasteners shall be compatible with roof panel system and shall not void any roof warranties.
    b. Fasteners shall not damage panel finish.
4. Install metal color strip in face of extrusion.
    a. Color strip to match color of standing seam roofing.
5. Provide snow/ice clips in each standing seam panel laterally across the roof or as recommended by manufacturer.
    a. Provide single clip for seam spacing of less than-16-inches and two clips for seam spacing 16-inches and greater.

H. Metal Roof Panel Accessories and Sealants:
1. Install components required for a complete roof panel system, including trim, sills, cap, closures, clips, seam covers, flashings, sealants, gaskets, fillers, closure strips, and similar items.

2. Joint Sealers: Install gaskets, joint fillers and sealants where indicated and where required for weatherproof performance of panel systems. Provide types of gaskets, sealants and fillers indicated or, if not otherwise indicated, types recommended by panel manufacturer.
    a. Provide weatherseal under all cap flashing, and rake flashing closures. Flash and seal roof panels at cap termination with rubber, neoprene, or other closures to exclude weather.
    b. Apply a continuous ribbon of sealant tape to clean, dry surface of the weather side of fastenings on end laps, and on side laps of nesting type panels and elsewhere as needed to make roof sheets weatherproof to driving rains.

3.5 LIGHTNING PROTECTION SYSTEM INSTALLATION

A. Lightning Protection System:
    1. Rods and cables shall be fastened to the metal roof panel seams with non - destructive and non - penetrating clamps that are attached per the manufactures written instructions. Penetration through the metal pans is not acceptable and will be grounds for rejection and total replacement of the roof panels. Roofing Contractor shall coordinate with Lightning Protection System Installer on actual locations and quantities of clamps to be installed.

3.6 ROOFING MANUFACTURER'S FINAL INSPECTION

A. Field Tests
    1. Roofing Manufacturer's Representative shall make a final inspection at end of project to verify roofing installation complies with shop drawings.

3.7 DISSIMILAR METALS

A. Where sheet metal is in contact with dissimilar metals, execute juncture to facilitate drainage and minimize possibility of galvanic actions.

B. At point of contact with dissimilar metal, coat metal with protective paint or tape which can be placed between metals.

3.8 CLEANING AND PROTECTION

A. Damaged Units: Replace panels or other components of the work which have been damaged or have deteriorated beyond successful repair by means of finish touch - up or similar minor repair procedures.

B. Cleaning: Remove protective coverings and strippable films (if any) at time in project construction sequence which will afford greatest protection of work. Clean finished surfaces as recommended by panel manufacturer and maintain in a clean condition during construction.

END OF SECTION

SECTION 07 62 00
SHEET METAL FLASHING AND TRIM

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
1. Architectural flashing and sheet metal work not identified as a part of metal roofing.
2. Miscellaneous Sheet Metal Accessories.
3. Stainless Steel Clamping Rings.
4. Metal Flashings and Counter flashings.
5. Termination Bars.

B. Related Specification Sections include but are not necessarily limited to:
1. Section 01 33 23 - Submittal Procedures.
2. Section 04 05 23 - Masonry Accessories.
3. Section 06 10 00 – Rough Carpentry
4. Section 07 61 13 - Metal Roofing.
5. Section 07 92 00 - Joint Sealants.
6. Section 09 06 00 - Color Schedule.
7. Section 09 96 00 - High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
1. American Architectural Manufacturers Association (AAMA):
a. 2605, Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels.
2. American National Standards Institute/Single Ply Roofing Industry (ANSI/SPRI):
a. ES-1, Wind Design Standard for Edge Systems Used with Low Slope Roof Systems.
3. ASTM International (ASTM):
a. A653/A653M, Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy Coated (Galvannealed) by the Hot-Dip Process.
b. A666, Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar.
c. B32, Standard Specification for Solder Metal.
d. B209, Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate.
4. FM Global (FM).
5. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA):
a. Architectural Sheet Metal Manual.

B. Qualifications:
1. Sheet metal fabricator shall have minimum 10 year’s experience in fabrication of sheet metal items similar to items specified.
2. Sheet metal installer shall have minimum five year’s experience installing sheet metal items specified.

1.3 DEFINITIONS

A. Installer or Applicator:
1. Installer or applicator who are the persons actually installing or applying the product in the field at the Project site.
2. Installer and applicator are synonymous.

B. PVDF: Polyvinylidene fluoride.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Manufacturer's product data for all items.
      b. Acknowledgement that products submitted meet requirements of standards referenced.
      c. Manufacturer's installation instructions.
   2. Fabrication and/or layout drawings.
      a. Scaled drawing showing expansion joint locations, special conditions, profile, fastening and jointing details.
         1) Minimum plan scale: 1/8-inches = 1-foot.
         2) Minimum detail scale: 1-1/2-inches = 1-foot.
   3. Fabricator qualifications.
   4. Installer qualifications.
   5. Samples - color for pre-finished flashings, counter flashings and sheet metal:
      a. Refer to "Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals" for submission procedures and requirements of Color Samples.
      b. Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.

B. Informational Submittals:
   1. Warranty: Manufacturer's sample warranty language.
   2. Final Warranty as a part of the O and M submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Pre-finished flashings, counter flashings and sheet metal:
      a. To be provided by the Sheet Metal Roofing Manufacturer to be able to match materials, finishes and colors as required of Metal Roofing System.
   2. Reglets:
      a. Hickman - "DLR - 625 In - Wall Reglet".
      b. Or Equal.
   3. Termination Bar:
      a. Metal Era - "Termination Bar".
      b. Or Equal.
   4. Butyl sealant:
      a. Pecora Corporation.
      b. Or Equal.

2.2 MATERIALS

A. Zinc - Coated Steel: Commercial quality with 0.20 percent copper, ASTM A 526 except ASTM A 527 for lock-forming, G90 hot - dip galvanized, mill phosphatized where indicated for painting. Metal not otherwise identified on drawings shall be zinc - coated steel.

B. Stainless Steel: AISI Type 302 / 304, ASTM A 167, 2D annealed finish, soft except where harder temper required for forming or performance.

C. Sheet Aluminum: ASTM B 209, alloy 3003, temper H14.

D. Extruded Aluminum: Manufacturer's standard extrusions of sizes and profiles indicated, 6063 - T52; 0.080-inch minimum thickness for primary legs of extrusions.

E. Fasteners: Non-ferrous compatible with sheet metal.

F. Solder: For use with steel or copper, provide Alloy 50 B solder (ASTM B 32), with rosin flux.

G. Solder: For use with stainless steel, provide Alloy 60 B solder (ASTM B 32), with acid - chloride type flux, except use rosin flux over tinned surfaces.

H. Fasteners: Same metal as flashing/sheet metal or, other non - corrosive metal as recommended by sheet manufacturer. Match finish of exposed heads with material being fastened. (See Table 1 at the end of this section for additional requirements).

I. Bituminous Coating: FS TT-C-494 or SSPC - Paint 12, solvent type bituminous mastic, nominally free of sulfur, compounded for 15 mil dry film thickness per coat.

J. Felt: ASTM D 226, Type 1, No.15, unperforated organic felt, asbestos free.

K. Mastic Sealant: Polyisobutylene; non - hardening, non - skinning, nondrying, non - migrating sealant.

L. Elastomeric Sealant: Generic type recommended by manufacturer of metal and fabricator of components being sealed and complying with requirements for joint sealants as specified elsewhere in Division 07.

M. Epoxy Seam Sealer: Two part non - corrosive metal seam cementing compound, recommended by manufacturer for exterior/interior nonmoving joints including riveted joints.

N. Adhesives: Type recommended by flashing sheet manufacturer for waterproof/weather-resistant seaming and adhesive application of flashing sheet.

O. Metal Accessories: Provide sheet metal clips, straps, anchoring devices and similar accessory units as required for installation of work, matching or compatible with material being installed, non - corrosive, size and gauge required for performance.

P. Elastic Flashing Filler: Closed - cell polyethylene or other soft closed - cell material recommended by elastic flashing manufacturer as filler under flashing loops to ensure movement with minimum stress on flashing sheet.

Q. Sealants:
   1. Non-curing Butyl Sealant:
      a. See Specification Section 07 92 00.

R. Dissimilar Metal Protection: Comply with Specification Section 09 96 00.

S. Reglets: See Specification Section 04 05 23.

2.3 FABRICATED ITEMS

A. FABRICATED UNITS, GENERAL
   1. Shop - fabricate work to greatest extent possible. Comply with details shown, and with applicable requirements of SMACNA "Architectural Sheet Metal Manual" and other recognized industry practices. Fabricate for waterproof and weather - resistant performance; with expansion provisions for running work, sufficient to permanently prevent leakage, damage or deterioration of the work. Form work to fit substrates. Comply with material manufacturer instructions and recommendations. Form exposed sheet metal work without excessive oil - canning, buckling and tool marks, true to line and levels as indicated, with exposed edges folded back to form hems.
   2. Seams: Fabricate nonmoving seams in sheet metal with flat - lock seams. For metal other than aluminum, tin edges to be seamed, form seams, and solder. Form aluminum seams with epoxy seam sealer; rivet joints for additional strength where required.

3. Expansion Provisions: Where lapped or bayonet - type expansion provisions in work cannot be used, or would not be sufficiently water / weatherproof, form expansion joints of intermeshing hooked flanges, not less than 1 inch deep, filled with mastic sealant (concealed within joints).
4. Sealant Joints: Where movable, non - expansion type joints are indicated or required for proper performance on work, form metal to provide for proper installation of elastomeric sealant, in compliance with industry standards.
5. Separations: Provide for separation of metal from non - compatible metal or corrosive substrates by coating concealed surfaces at locations of contact, with bituminous coating or other permanent separation as recommended by manufacturer / fabricator.
6. Aluminum Extrusion Units: Fabricate extruded aluminum running units with formed or extruded aluminum joint covers for installation behind main members where possible. Fabricate mitered and welded corner units.

B. STAINLESS STEEL CLAMPING RINGS
1. At roof and wall penetrations for vent pipes provide 0.0312 inch – 22 gage stainless steel clamping rings to secure tops of pre - molded or prefabricated (non - metallic) pipe seals.

C. METAL FLASHING AND COUNTERFLASHING
1. Fabricate 24 gage zinc coated pre-finished steel flashing and counterflashing. Colors to be selected as outlined in color submittal process.

D. REGLETS
1. Reglet bars shall have no rough edges which might damage base flashing. Flashings shall be shop fabricated where shown indicated on drawings. Contractor shall have the option to use one of the following pre - manufactured products:

E. TERMINATION BARS

F. Termination bars shall be 6063 - T-5 or T6, 0.040-inch-thick minimum extruded aluminum 1-inch-high minimum and 10-feet lengths with a flat face and inverted drip along top edge. Termination bars shall have slotted mounting holes at 6-inches on center maximum.

PART 3 - EXECUTION

3.1 GENERAL

A. Except as otherwise indicated, comply with manufacturer's installation instructions and recommendations, and with SMACNA "Architectural Sheet Metal Manual". Anchor units of work securely in place by methods indicated, providing for thermal expansion of metal units; conceal fasteners where possible, and set units true to line and level as indicated. Install work with laps, joints and seams which will be permanently watertight and weatherproof.

B. Sheet metalwork shall be accomplished to form weather tight construction. Work shall be installed without waves, warps, buckles, fastening stresses or distortion and shall allow for expansion and contraction. Cutting, fitting, drilling, and other operations in connection with sheet metal required to accommodate the work of other trades shall be performed by sheet metal mechanics. Exposed edges shall be hemmed. Bottom edges of exposed vertical surfaces shall be angled to form drips. Flashing at the end of a run shall be formed into a three - dimensional configuration to direct water to the outside of the system. Accessories and other items essential to complete the sheet metal instillation, though not specifically indicated or specified, shall be provided.

C. Items not covered in this specification shall be installed as required to provide a complete installation.

D. Do not proceed with the work until curb and substrate construction, cant strips, blocking, reglets and other construction to receive the work is completed.

E. Examine the substrate and the conditions under which work is to be performed, and do not proceed until unsatisfactory conditions have been corrected. Surfaces to receive sheet metal are to be clean, even, smooth, dry and free from defects and projections which might adversely affect the application. Verify slope prior to instillation.

F. Cleating: For size and spacing, see SMACNA "Architectural Sheet Metal Manual." Unless otherwise specified, use 2-inches wide by 3-inches long cleats of the same material and thickness of metal being installed.

G. A continuous cleat shall be provided where indicated or specified to secure loose edges of the sheet metalwork. Butt joints shall be spaced approximately 1/8-inch apart. The cleat shall be fastened to the supporting construction with nails evenly spaced not over 12-inches on center or with a minimum of 2 fasteners. Where the fastening is to be made to concrete or masonry, screws shall be used and shall be driven in expansion shields set in concrete or masonry. The cleat for fascia anchorage shall be installed to extend below the supporting construction to form a drip and to allow the flashing to be hooked over the lower edge at least 3/4-inch. The cleat shall be of sufficient width to provide adequate bearing area to insure a rigid installation. Where horizontal nailer is vented for insulation and the cleat is placed over the masonry or concrete, the cleat shall be installed over 1/16-inch-thick metal washers placed at screws. Washers shall be of metal that is electrolytically compatible with the continuous cleat.

3.2 SOLDERING

A. Soldering shall apply to copper, copper clad stainless steel, and stainless steel items. Edges of sheet metals, except lead coated material shall be pretinned before soldering is begun. Soldering shall be done slowly with well heated soldering irons so as to thoroughly heat the seams and completely sweat the solder through the full width of the seam. Edges of lead coated material to be soldered shall be scraped or wire - brushed to produce a bright surface, and seams shall have a liberal amount of flux brushed in before soldering is begun. Edges of stainless steel to be pretinned shall be treated with soldering acid flux. Soldering shall follow immediately after the application of the flux. Upon completion of soldering, the acid flux residue shall be thoroughly cleaned from the sheet metal with a solution of washing soda in water and rinsed with clean water.

B. Flux: Use type of flux compatible with the sheet metal. Remove acid flux residue by neutralizing with ammonia or baking soda and rinsing with water.

3.3 RIVETING AND SEALING

A. Joints in aluminum sheets 0.040 inch or less in thickness shall be made mechanically and sealed.

3.4 SEAMS

A. Seams: Comply with SMACNA "Architectural Sheet Metal Manual", Plate 99 and 100 and other plates applicable to specific installations. Orient seams properly for direction of water flow.
   1. Standing Seams: Finish not less than 1-inch high unless otherwise specified.
   2. Flat - Lock Seams: Finish not less than 1-inch wide.
   3. Lap Seams Where Soldered: Finish not less than 1-inch wide.
   4. Lap Seams, No Soldered: Overlap 4-inches unless otherwise noted.

3.5 UNDERLAYMENT

A. Where stainless steel or aluminum is to be installed directly on cementitious or wood substrates, install a course of paper slip sheet and a course of polyethylene underlayment.

B. Bed flanges of work in a thick coat of bituminous roofing cement where required for waterproof performance.

3.6 FLASHING AND COUNTERFLASHING

A. Flashings shall be installed at at shelf angles, lintels, ledges, window sills, window and door openings, spandrels, parapet walls, and similar obstructions to the downward flow of water so as to divert such water to the exterior and drip away from side of building.

B. Cap flashing shall be installed in a sawn reglet joint where shown on drawings. Cap flashing shall be factory formed to provide spring action in the sawn reglet joint. Back horizontal edge of flashing shall have a longitudinal continuous V - type crimp. Cap flashing terminations shall be sealed along the reglet joint in wall after flashing has been installed.

3.7 TERMINATION BARS

A. Termination bars shall be set level and shall be anchored securely to vertical surface or wall. Install anchors at 6 inches on center.

B. Install elastomeric sealant over heads of anchors and continuously along top edge of termination bar.

3.8 DISSIMILAR MATERIALS

A. Caution must be exercised when aluminum is placed in contact with metals or materials not compatible with aluminum. Aluminum mails or fasteners must be used when installing soffit and accessories.

B. Aluminum shall not be used where it will be in contact with copper or where it will contact water which flows over copper surfaces. Aluminum that will be in contact with wet or pressure-treated wood, mortar, concrete, masonry, or ferrous metals shall be protected against galvanic or corrosive action by one of the following methods:
   1. Paint the aluminum with a non - lead pigmented paint.
   2. Paint the aluminum with an alkali - resistant coating such as a heavy - bodied bituminous paint.

C. Aluminum surfaces to be protected shall be solvent cleaned and given a coat of zinc - molydbate primer and one coat of aluminum paint. Aluminum paint shall conform with painting materials found elsewhere in Division 09.

D. Nonabrasive tape or gasket shall be placed between the adjoining surfaces and shall be cemented to the aluminum surface using a cement compatible with aluminum.

E. To prevent corrosion, see Table 1 which indicates each fastener materials which shall be used with each sheet metal.

F. Dissimilar materials shall be painted or otherwise protected when in contact with aluminum or when drainage from them passes over aluminum.

3.9 CLEANING AND PROTECTION

A. Clean exposed metal surfaces, removing substances which might cause corrosion of metal or deterioration of finishes.

B. Protection: Installer shall advise Contractor of required procedures for surveillance and protection of flashings and sheet metal work during construction, to ensure that work will be without damage or deterioration, other than natural weathering, at time of substantial completion.

TABLE 1 – FASTENER MATERIALS

SHEET METAL	NAILS	SCREWS	RIVETS	BOLTS
Aluminum	Aluminum	Aluminum	Aluminum	Aluminum
Copper	Copper	Bronze	Copper	Bronze
Copper Clad Stainless Steel	Copper or Stainless Steel	Bronze or Stainless Steel	Bronze or Stainless Steel	Bronze or Stainless Steel
Lead - Coated Copper	Copper	Bronze	Bronze	Bronze
Stainless Steel	Stainless Steel	Stainless Steel	Stainless Steel	Stainless Steel
Zinc - Coated Steel	Zinc - Coated Steel	Zinc - Coated Steel	Zinc - Coated Steel	Zinc - Coated Steel

END OF SECTION

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SECTION 07 84 00
FIRESTOPPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
1. Firestopping of joints in fire resistance-rated construction, including through-wall penetrations, and membrane penetrations.
2. Smoke stopping of joints, through-wall penetrations, and membrane penetrations of smoke-rated construction.
3. Firestopping assemblies.
4. Engineering Judgments furnished by manufacturers of firestopping materials.
5. Special Inspections.
6. Labeling of Firestopping Assemblies.

B. Related Requirements: Include but are not necessarily limited to the following:
1. Section 01 32 33 - Photo Documentation.
2. Section 01 35 73 – Delegated Design Procedures.
3. Section 01 40 00 - Quality Requirements.
4. Section 01 67 00 - Product Requirements.
5. Section 01 73 20 - Openings and Penetrations in Construction.
6. Section 01 77 00 - Closeout Procedures.
7. Section 09 96 00 - High-Performance Industrial Coatings.
8. Section 10 14 00 - Identification Devices.

1.2 REFERENCES

A. Terminology:
1. Terminology indicated below are not defined terms and are not indicated with initial capital letters, but when used in this section have the meaning indicated below:
   a. “Engineering judgement” means an evaluation of the anticipated performance of a firestopping materials or assembly that has not been fire tested (including but not limited to firestopping materials or systems that do not have a UL listing). Engineering judgements shall be governed by the International Firestop Council (IFC) Guidelines referenced in this Section and are based on the performance of listed firestopping materials and assemblies and generally accepted engineering principles. As indicated in the IFC guidelines, engineering judgements are issued by a firestopping product manufacturer’s technical engineering employee. Relative to this term, engineering judgements are not issued by and are not the responsibility of Contracting Officer Representative.
   b. “Firestopping” means material or combination of materials used to retain integrity of fire-rated construction by maintaining an effective barrier against the spread of flame, smoke, and hot gases through penetrations in fire rated wall and floor assemblies. The word “firestop” has the same meaning as “firestopping”.
   c. “Smoke stopping” means material or combination of materials used to retain integrity of smoke-rated construction by maintaining an effective barrier against the spread of smoke, and hot gases through penetrations in smoke-rated partitions and barrier assemblies. The word “smoke stop” has the same meaning as “smoke stopping”.

B. Reference Standards: Standards referenced in this section include, but are not necessarily limited to, the following:
1. ASTM International (ASTM):
   a. E814, Standard Method of Fire Tests of Through Penetration Fire Stops.

b. E1399, Standard Test Method for Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems.
c. E1966, Standard Test Method for Fire Resistive Joint Systems.
d. E2307,

2. Firestop Contractors International Association's (FCIA):
   a. Firestop Manual of Practice (MOP).
3. FM Global (FM):
   a. MF 4991, Approval Standard for Firestop Contractors.
   b. FM Approvals Guide (online product directory).
4. International Building Code (IBC):
   a. 2018 Edition.
5. International Firestop Council (IFC):
   a. Recommended IFC Guidelines for Evaluating Firestop Systems in Engineering Judgments (EJs), referred to herein as IFC Recommended Guidelines.
   b. Firestop Inspection Pocket Guide.
6. Underwriters Laboratories (UL):
   a. 263, Fire Tests of Building Construction and Materials.
   b. 1479, Fire Tests of Through Penetration Fire Stops.
   c. 2079, Standard for Tests for Fire Resistance of Building Joint Systems.
7. UL Solutions Product iQ (Online Product Directory):
   a. Through Penetration Firestop Systems (XHEZ).
   b. Joint Systems (XHBN).
   c. Fill, Void, or Cavity Materials (XHHW).
   d. Firestop Devices (XHJI).
   e. Forming Materials (XHKU).
   f. Wall Opening Protective Materials (CLIV).
   g. Fire Resistance Ratings (BXRH).

1.3 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Coordinate the Work required by this Section with other construction that must be performed prior to or concurrent with the Work of this Section.
   2. Materials and assemblies furnished as part of the Work of this Section shall be furnished by a single manufacturer. Supplier shall ensure compatibility of all materials and assemblies furnished under this Section and compatibility with other materials with which the Work of this Section will come into contact.

B. Sequencing and Scheduling:
   1. Perform installation of firestopping and smoke stopping after completing installation of penetrating items and prior to covering or concealing of openings.

1.4 QUALITY ASSURANCE

A. Regulatory Requirements:
   1. For required firestopping applications for which no qualified previously tested materials or system is commercially available through Supplier, submit to authority having jurisdiction an engineering judgment derived from similar qualified tested system designs or other tests for approval by the authority having jurisdiction prior to installation. Upon Contracting Officer Representative's request, submit to Contracting Officer Representative copy of such engineering judgements and approval by the authority having jurisdiction.
   2. Engineering judgment documents shall comply with requirements of International Firestop Council.
   3. Identification Labels for Firestop Assemblies and Penetrations:
      a. Follow guidelines set in Chapter 7 of the International Building Code (IBC 2018 Edition).

B. Qualifications:

1. Delegated Design Professional's "Instruments of Service" Submittals:
    a. Design drawings sealed and signed by delegated design professional, including:
    b. Design drawings deemed appropriate and necessary by delegated design professional.
    c. Show or indicate necessary details establishing the method of fastening throughout system and attachments to supporting primary structure.
    d. At option of Contractor or fabricator, design drawings may be combined with the Shop Drawings required in this Section, although delegated design professional's seal and signature is required on the Shop Drawings in the event of a combined Submittal.
    e. Design drawings and specifications shall be sealed and signed by delegated design professional licensed in the State of Arizona.
2. Design Professional:
    a. Where Contractor proposes providing firestopping materials or assemblies that have not been fire tested (including but not limited to, firestopping materials or systems that do not have a UL listing), Supplier shall prepare engineering judgement, submit to, and obtain approval or acceptance by authority having jurisdiction. Such engineering judgement shall be prepared by, or under the direct, personal, supervision and control of design professional registered and licensed in the same jurisdiction as the Site, appropriate for the type of firestopping required.
    b. Design professional shall be qualified and experienced in accordance with Section 01 73 40 - Execution, and shall possess or be covered by current, valid professional liability insurance in accordance with the Contract Documents. Required limits of professional liability insurance coverage shall not have been previously eroded by prior claims under the policy.
    c. Submit evidence of design professional's compliance with Contract requirements regarding licensure, registration, and insurance.
    d. Engineering judgements shall bear design professional's seal, signature, and date.
3. Installer:
    a. Entity that selects and installs firestopping materials and assemblies shall be an organization regularly engaged in and experienced in firestopping construction of the types and complexities similar to that required for the Work. Such entity shall be appropriately experienced in the type of Work required and shall possess current, valid, licensure and qualifications, in accordance with Laws and Regulations (where applicable) for the type of firestopping required.
    b. Entities that select and install firestopping materials and assemblies shall possess one or more of the following as current, valid accreditations:
        1) Firestop Contractors International Association (FCIA) Levels 1 – 4.
    c. In addition, entity selecting and installing firestopping materials and systems shall possess one or more of the following as current, valid certifications:
        1) Approved Firestop Contractor in accordance with FM4991.
        2) Qualified Firestop Contractor by UL Solutions
    d. Should Contractor propose as entity selecting and installing firestopping an organization with alternative qualifications, accreditation, or certification submit for Contracting Officer Representative's approval appropriate substitution request in accordance with the Contract Documents.
    e. Submit to Contracting Officer Representative documentation demonstrating installer's compliance with qualifications requirements of this provision.

1.5 SUBMITTALS

A. Action Submittals: Submit the following:
1. Shop Drawings:
    a. Firestopping schedule (table) indicating the following:
        1) Numerical designation for each penetration.
        2) Type of construction penetrated, such as wall, floor, shaft, head of wall, or other appropriate designation.
        3) F and T ratings for each penetration.

4) For each penetration, indicate appropriate designation by UL, FM, Intertek, or other applicable rating organization.
5) Product name and manufacturer for each product used at each penetration.

b. Plan-view drawings indicating the location of fire-rated and smoke-rated construction and the location of each penetration with serialized numerical designator for each penetration aligning with the Firestopping Schedule.
c. Firestopping detail drawings for each type of firestopping assembly proposed.
d. Installation detailed drawings for each type of Work element.

2. Product Data:
a. Manufacturer's listed firestopping product "design number".
b. For each item required under this Section, submit manufacturer's published product information, including data sheet (as applicable), clearly indicating materials, limitations, optional features that will be furnished and those options or features that will not be furnished.
c. Manufacturer's specifications:

B. Informational Submittals: Submit the following:
1. Engineering Judgements:
a. When UL or third-party listing is not available, provide a written engineering judgment in accordance with IFC Recommended Guidelines and requirements of this Section. Submit to Contracting Officer Representative copy of engineering judgement, sealed, signed, and dated by Supplier's design professional, together with documentation of approval or acceptance by authority having jurisdiction.
2. Supplier Instructions:
a. Serial numbers of items furnished.
b. Instructions for handling, installing, and startup.
3. Field Quality Control Submittals:
a. Results of tests, inspections, and other quality control activities required by the Contract Documents and performed at the Site.
4. Suppliers Site Visit Reports:
a. Report of each visit to the Site by Supplier, summarizing purpose of visit, activities while onsite, problems encountered, advice given to Contractor or Subcontractor, and actions taken.
5. Qualifications Statements:
a. Entity selecting and installing firestopping, in accordance with this Section's "Quality Assurance" Article.
b. Design professional responsible charge of preparing engineering judgements, if any.

C. Closeout Submittals: Submit the following:
1. Record Documentation:
a. Written statement that all fire-rated penetrations were sealed using products in accordance with UL or FM requirements for required rating.
b. Documentation of listed materials and systems installed and engineering judgments.
c. Indicate appropriate revisions for firestopping on record Drawings required in accordance with Section 01 77 00 – Closeout Procedures.
d. Firestopping Schedule (table), submitted as part of the Shop Drawings, revised to show and indicate as-constructed conditions.
e. Photographs:
1) Photograph of each penetration through fire resistance-rated construction showing pre-firestopping and Substantial Completion conditions.
2) Photographs shall be in accordance with Section 01 32 33 - Photo Documentation. Such photographs are not part of quantities of still photographs required in Section 01 32 33 – Photo Documentation.
3) Pre-firestopping and Substantial Completion condition photographs shall be taken in a manner to ensure similar photograph orientation to allow for photographic comparison.

f. Building code sections relevant to firestop systems.

1.6 DELIVERY, STORAGE AND HANDLING

A. Delivery and Acceptance Requirements:
1. In addition to requirements of Section 01 67 00 - Product Requirements, properly label packaging of firestopping and smoke stopping materials and assemblies with associated UL label where applicable.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Through Penetration Firestop Systems:
   a. Hilti.
   b. Or Equal.
2. Fire Resistive Joint Systems:
   a. Hilti.
   b. Or Equal.
3. Firestop Track Seal or Gasket:
   a. Hilti.
   b. Or Equal.
4. Firestop Devices:
   a. Hilti.
   b. Or Equal.
5. Intumescent Pads, Wall Opening Protective Materials:
   a. Hilti.
   b. Or Equal.
6. Fire Rated Cable Pathways:
   a. Hilti.
   b. Or Equal.
7. Firestop Putty:
   a. Hilti.
   b. Or Equal.
8. Firestop Blocks and Plugs:
   a. Hilti.
   b. Or Equal.
9. Firestop Pillows:
   a. 3M.
   b. Or Equal.
10. Forming Materials:
   a. Rockwool - Roxul.
   b. Or Equal.
11. Composite Sheet:
   a. Hilti.
   b. Or Equal.
12. Duct Wrap.

a. UniFrax – FyreWrap.
b. Or Equal.

2.2 MATERIALS

A. Materials - General:
1. Installer of the Work under this Section shall select specific materials and systems appropriate for each specific firestopping and smoke stopping application required, in accordance with the Contract Documents and Laws and Regulations.
2. Provide firestopping materials and assemblies in accordance with ASTM E814, ASTM E1399, ASTM E2307, UL 1479 and UL 2079, as applicable.
3. Provide tested firestopping materials and assemblies that provide a fire rating equal to that of the construction being penetrated.
4. Firestopping materials and assemblies shall comply with Laws and Regulations, including applicable building code.
5. Previous Testing Required: Furnish only materials and assemblies where identical items were previously tested, under controlled conditions for specific fire-rated construction conditions conforming to construction assembly type, penetrating item type, annular space requirements, and fire resistance rating involved for each separate item. Such testing shall be in accordance with the following as applicable:
a. Openings and Penetrations: UL 1479 or ASTM E814.
b. Building Joint Systems: ASTM E1966 and ASTM E1399, or UL 2079.
c. Perimeter Fire Barrier Systems: ASTM E2307.
6. Where firestopping is not exposed to view, provide materials and assemblies of manufacturer's standard color.

B. Forming Materials:
1. Comply with ANSI/UL 263.
2. UL listed.
3. Product category BZJZ, Batts and Blankets.

C. Firestop Silicone Elastomer:
1. UL-listed.
2. Fill, Void or Cavity Materials: Product category XHHW.
3. Through Penetration Firestop Systems: Product category XHEZ.

D. Firestop Sealant:
1. UL-listed.
2. Provide as required by the item’s associated UL-listed design.

E. Firestop Putty:
1. UL-listed.
2. Product category QCSN or CLIV, Wall Opening Protective Materials.

F. Firestop collars:
1. UL-Listed.
2. Provide as required by the item’s associated UL-listed design.

G. Fire Rated Cable Pathways:
1. UL-Listed.
2. Steel raceway and intumescent pads with adjustable smoke seal sleeve.
3. Fire resistance rating equal to rating of barrier penetrated.
4. Pathway devices:
a. Allow 0 to 100 percent fill of cables.
b. Firestopping shall adjust automatically to cabling installation and removal.
5. Size of firestopping shall accommodate quantity and size of cabling required by the Contract Documents plus and equal quantity of cabling installed in the future.
6. Provide cable management devices with gang plates for single or multiple devices.

H. Fire Barrier Pillows:

1. UL listed.
2. Provide pillow that are removable and reusable.
3. Provide pillows have self-locking feature.

I. Backer rod and compressible filler: UL listed, product category XHHW, Fill, Void or Cavity Materials.

J. Identification Labels:
1. Comply with Section 10 14 00 - Identification Devices.
2. Comply with IBC Chapter 7 (2018 Edition).
3. Identify through-penetration firestop systems by providing pressure-sensitive, self-adhesive, preprinted labels complying with the following.
   a. Tamper proof.
   b. Permanent and weatherproof.
   c. UV stabilized labelling, text, and colors.
   d. Provide labels with preprinted text. Handwritten text on labels is unacceptable.
4. Each firestopping label shall clearly indicate the following:
   a. Name and telephone number of Contractor and installer.
   b. Through-Penetration firestop listing designation or EJ number.
   c. Date of Installation, in the form of month, date, year.
   d. Through-Penetration firestopping materials and system manufacturer.
   e. Installing Technician's Name.
   f. Firestop Joint Systems:
      1) The words "Warning-Firestop System-Do Not Disturb. Notify Building Management of Any Damage."
   g. Through-Penetration Firestop Systems:
      1) The words "Warning-Through Penetration Firestop System-Do Not Disturb. Notify Building Management of Any Damage."
   h. Fire Walls, Barriers and Partitions:
      1) 3-inch-high stenciled fire rating based on "HOUR RATING" of walls/partitions. Provide on both sides of walls/partitions unless in an unoccupied space, IE: A chase wall. Locate near top of the wall/partition.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verification of Conditions:
1. Verify existing conditions and substrates before starting the Work of this Section. Remedy unsatisfactory conditions before proceeding.

B. Installing the Work constitutes Contractor's approval of underlying work, substrates, and field conditions prevailing at the time of the Work.

3.2 INSTALLATION

A. Installation - General:
1. Provide the Work of this Section in accordance with the Contract Documents, Laws and Regulations (including applicable building code), and material, system manufacturer's written installation instructions, and UL listing requirements to provide required fire-resistance rating. When UL listing is not available, install in accordance with approved written engineering judgment. When any such requirements conflict, obtain written interpretation or clarification from Contracting Officer Representative before proceeding further.
2. Provide firestopping materials and systems in accordance with ASTM E814, ASTM E1399, ASTM E2307, UL 1479 and UL 2079, as applicable.
3. Where firestopping will be exposed to view, provide materials and systems of colors matching adjacent construction. Alternatively, if approved by manufacturer of materials and

systems and following inspection and approval by authority having jurisdiction, field-paint materials and systems to match color of adjacent construction. Such field-painting shall be in accordance with Section 09 91 00 - Architectural Painting and Section 09 96 00 - High-Performance Industrial Coatings.

4. Do not install firestopping or smoke stopping materials or assemblies when the ambient temperature of the installation location does not comply with manufacturer's written installation instructions for duration of installation and curing.
5. During installation, provide masking and drop cloths to prevent firestopping materials from contaminating or marring adjacent surfaces.

B. Openings and Penetrations:
   1. Provide firestopping materials and assemblies for each opening, through-penetration, and membrane penetration in fire-resistant rated construction.
   2. Provide Flame (F), Thermal (T) rated assemblies as necessary to comply with Laws and Regulations (including applicable building code).
   3. Comply with Section 01 73 20 - Openings and Penetrations in Construction.

C. Building Joint Systems:
   1. Provide firestopping materials and assemblies for each joint assembly in or between fire-resistance-rated walls, floors or floor/ceiling assemblies and roofs or roof/ceiling assemblies.

D. Identification
   1. Install firestopping labelling permanently to surfaces of penetrated construction on both sides of each firestop system installation where labels will be readily visible.
   2. Provide identification in accordance with FCIA MOP.

3.3 FIELD QUALITY CONTROL

A. Field Tests and Inspections:
   1. Special Inspections of firestopping and smoke stopping Work will be in accordance with Chapter 17 of the applicable building code and Section 01 40 00 – Quality Requirements.

END OF SECTION

SECTION 07 92 00
JOINT SEALANTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Sealing all joints which will permit penetration of dust, air or moisture.

1.2 SCOPE

A. Work required under this Section includes all labor, materials, tools and equipment necessary to complete all work relative to the provision and installation of all exterior and interior sealants where the words caulk, caulking, or sealant are shown on the drawings and at additional locations as required by the specifications. Included are all joint fillers, backer rods and gaskets. Work within this section shall be completely coordinated with work of other trades.

B. Although such work may not be specifically shown or specified, provide and install all supplementary or miscellaneous items, appurtenances and devices incidental to or necessary for a sound, secure and complete installation.

C. The material and installation shall conform to the applicable local building code requirements, and all authorities having jurisdiction.

D. Notwithstanding reference to any manufactured or produced item or material in this specification that contains asbestos, materials or items that contain asbestos are not acceptable. In the event that any of the materials or items specified herein contain asbestos, the Contracting Officer Representative shall be promptly notified in writing and the Contracting Officer Representative will select a suitable substitute material or item which shall be furnished at no additional cost to the Government.

1.3 RELATED WORK

A. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 04 22 00 - Concrete Masonry.
   3. Section 06 41 00 - Architectural Wood Casework.
   4. Section 07 84 00 - Firestopping.
   5. Section 08 81 00 - Glass and Glazing.
   6. Section 09 06 00 – Color Schedule.

1.4 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Concrete Institute (ACI):
      a. 302.1R, Guide for Concrete Floor and Slab Construction.
   2. ASTM International (ASTM):
      a. C834, Standard Specification for Latex Sealants.
      b. C920, Standard Specification for Elastomeric Joint Sealants.
      c. C1311, Standard Specification for Solvent Release Sealants.
      d. C1521, Standard Practice for Evaluating Adhesion of Installed Weatherproofing Sealant Joints.
   3. National Fire Protection Association (NFPA).
      a. 820, Fire Protection in Wastewater Treatment and Collection Facilities.

B. Qualifications: Sealant applicator shall have minimum five year's experience using products specified on projects with similar scope.

C. Mock-Ups:
   1. Before sealant work is started, a mock-up of each type of joint shall be sealed where directed by the Contracting Officer Representative .
      a. The approved mock-ups shall show the workmanship, bond, and color of sealant materials as specified or selected for the work and shall be the minimum standard of quality on the entire project.
      b. Each sample shall cure for a minimum of seven days at which time the sealant manufacturer's authorized factory representative shall perform adhesion tests on each sample joint.
         1) Perform adhesion tests per ASTM C1521.
         2) If mock-up is not acceptable or if adhesion test fails, provide additional mock-up and adhesion testing as required until acceptable to Contracting Officer Representative .
   2. See Specification Section 04 22 00 – Concrete Masonry for additional Mock-Up requirements.

1.5 DEFINITIONS

A. Corrosive Areas Include:

B. Defective: Failure of watertightness or airtightness.

C. Finish sealant: Sealant material per this specification applied over face of compressible sealant or expanding foam sealant specified, to provide a finished, colored sealant joint.

D. Installer or Applicator:
   1. Installer or applicator is the person installing or applying the product in the field at the Project site.
   2. Installer and applicator are synonymous.

E. "Interior wet areas": Toilets, showers, laboratories and similar areas.

F. "Seal," "sealing" and "sealant": Joint sealant work.

1.6 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Manufacturer's recommendations for joint cleaner, primer, backer rod, tooling and bond breaker.
   2. Certification from sealant manufacturer stating product being used is recommended for and is best suited for joint in which it is being applied.
   3. Certification of applicator qualification.

B. Test Results:
   1. Provide adhesion test results for each sealant sample including adhesion results compared to adhesion requirements.
   2. Manufacturer's authorized factory representative recommended remedial measures for all failing tests.

C. Samples:
   1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
   2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
   3. Actual bead samples of sealants are required. Printed color charts are not acceptable.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Deliver material in manufacturer's original unopened containers with labels intact: Labels shall indicate contents and expiration date on material.

1.8 PROJECT CONDITIONS

A. Schedule installation of sealant work after completion of penetrating item installation but prior to covering or concealing of openings.

B. Verify existing conditions and substrates before starting work. Correct unsatisfactory conditions before proceeding.

C. Do not proceed with the installation of sealant materials when the ambient temperature is outside the manufacturer's recommended limitations for installation and curing times as printed on the product label and product data sheet.

D. During installation provide masking and drop cloths to prevent sealant materials from contaminating any adjacent surfaces.

E. Perimeter of, and all penetrations through, walls separating unclassified rooms and rooms classified per NFPA 820, shall be gas-tight to meet the definition of "Physically Separated".

F. For purposes of this Specification Section, the entire building/spaces of the following buildings will be considered as corrosive environments:
   1. Building 052 – Roaring Springs Pumphouse.
   2. Building 062 – Water Treatment.

G. Notwithstanding reference to any manufactured or produced item or material in this specification that contains asbestos, materials or items that contain asbestos are not acceptable.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Compressible sealant:
      a. Schul "HydroStop".
      b. Or Equal.
   2. Expanding foam sealant:
      a. Schul "Sealtite B".
      b. Or Equal.
   3. Polyether sealants:
      a. Master Builders Solutions "MasterSeal 150".
      b. Or Equal.
   4. Polysulfide rubber sealant:
      a. Pecora "Synthacalk GC2+" .
      b. Or Equal.
   5. Polyurea joint filler:
      a. Master Builders Solutions "MasterSeal CR100 control joint filler".
      b. Or Equal.
   6. Polyurethane sealants:
      a. Master Builders Solutions "MasterSeal NP-1, NP-II, SL-1 SL-2".
      b. Or Equal.
   7. Silicone sealants:
      a. Dow "7565, 790, 791, 795".

b. Or Equal.

8. Mildew resistant silicone sealant:
   a. Dow “786”.
   b. Or Equal.
9. Siliconized acrylic latex sealant:
   a. Pecora “AC-20+”.
10. Butyl sealant:
    a. Pecora “BA-98”.
    b. Or Equal.
11. Backer rod, compressible filler, primer, joint cleaners, bond breaker:
    a. As recommended by sealant manufacturer.
12. Fire-Resistant Sealant:
    a. See Specification Section 07 84 00 – Firestopping.

2.2 MATERIALS

A. Sealants - General:
   1. Provide colors matching materials being sealed.
   2. Where compound is not exposed to view in finished work, provide manufacturer's color which has best performance.
   3. Non-sagging sealant for vertical and overhead horizontal joints.
   4. Sealants for horizontal joints: Self-leveling pedestrian/traffic grade.
   5. Joint cleaner, primer, bond breaker: As recommended by sealant manufacturer.
   6. Bond Breaker Tape: Polyethylene tape or other plastic tape as recommended by sealant manufacturer, to be applied to sealant - contact surfaces where bond to substrate or joint filler must be avoided for proper performance of sealant. Provide self - adhesive tape where applicable.
   7. Sealant backer rod and/or compressible filler:
      a. Closed cell polyethylene, polyethylene jacketed polyurethane foam, or other flexible, nonabsorbent, non-bituminous material recommended by sealant manufacturer to:
         1) Control joint depth.
         2) Break bond of sealant at bottom of joint.
         3) Provide proper shape of sealant bead.
         4) Serve as expansion joint filler.
   8. Masking Tape: Non - staining, nonabsorbent material compatible with joint sealants and surfaces adjacent to joints.

B. Compressible Sealant:
   1. For joints exposed to wastewater fumes or vapor:
      a. Closed cell ethylene vinyl acetate (EVA) foam with epoxy adhesive.
   2. For all other applications: Foamed polyurethane strip saturated with polymerized polybutylene waterproofing coated on front face with nonreactive release agent that will act as bond breaker for applied sealant.
   3. Adhesive: As recommended by sealant manufacturer.

C. Expanding Foam Sealant:
   1. One or two component moisture cured expanding urethane.
   2. Shall not contain formaldehyde.
   3. Density: Minimum 1.5 pounds per cubic foot.
   4. Closed cell content: Minimum 70 percent.
   5. R-value: Minimum 5.0 per inch.
   6. Flame spread: Less than 25.
   7. Smoke developed: Less than 25.

D. Polyether Sealant:
   1. Silyl-terminated polyether polymer.
   2. ASTM C920, Type S, Grade NS, Class 50, Use NT, M, A, and O.

E. Polysulfide Rubber Sealant:
   1. One or two component.
   2. Meet ASTM C920.

F. Polyurea Joint Filler:
   1. Two component, semi-rigid material for filling formed or saw-cut control joints in interior concrete slabs.
   2. Comply with ACI 302.1R performance recommendations regarding control and construction joints.
   3. Color: Gray.

G. Polyurethane Sealant:
   1. One or two components.
   2. Paintable.
   3. Meet ASTM C920 Type S or Type M, Grade NS or P, Class 25, Use NT, T, M, A and O.

H. Silicone Sealant:
   1. One component.
   2. Meet ASTM C920, Type S, Grade NS, Class 25, Use NT, G, A, O.

I. Mildew Resistant Silicone Sealant (for sealing around plumbing fixtures):
   1. One component ready to use silicone rubber sealant.
   2. Meet ASTM C920, Type S, Grade NS, Class 25, Use NT.

J. Siliconized Acrylic Latex Sealant:
   1. One component.
   2. Accommodates joint movement of not more than 12.5 percent of the minimum joint width.
   3. Meet ASTM C834.

K. Butyl Rubber and Polyisobutylene Sealant:
   1. One component.
   2. Meet ASTM C1311.

L. Additional materials for stock.
   1. Provide stock/ surplus materials for each product used at each individual location. Minimum (2) large tubes or 1 gallon for pourable material.

PART 3 - EXECUTION

3.1 PREPARATION

A. Before use of any sealant, investigate its compatibility with joint surfaces, fillers and other materials in joint system.

B. Use only compatible materials.

C. Where required by manufacturer, prime joint surfaces.
   1. Limit application to surfaces to receive sealant.
   2. Mask off adjacent surfaces.

D. Provide joint depth for joints receiving polyurea joint filler in accordance with manufacturer's recommendations.

3.2 INSTALLATION

A. Employ only proven installation techniques, which will ensure that sealants will be deposited in uniform, continuous ribbons without gaps or air pockets, with complete "wetting" of joint bond surfaces equally on opposite sides. Except as otherwise indicated, fill sealant rabbet to a slightly concave surface, slightly below adjoining surfaces. Where horizontal joints are between a horizontal surface and vertical surface, fill joint to form a slight cove, so that joint will not trap moisture and dirt.

B. Install products in accordance with manufacturer's instructions and UL requirements.

C. Clean all joints.

D. Make all joints water and airtight.

E. At changes in direction of joints, joint intersections and where sealant joints interface with other construction, install continuous sealant as necessary to ensure a weather-tight seal.

F. Make depth of sealing compounds, except expanding foam and polyurea sealant, not more than one-half width of joint, but in no case less than 1/4 inches nor more than 1/2 inches unless recommended otherwise by the manufacturer.

G. Provide correctly sized backer rod, compressible filler or compressible sealant in all joints to depth recommended by manufacturer:
   1. Take care to not puncture backer rod and compressible filler.
   2. Provide joint backer rod as recommended by the manufacturer for polyurea joint filler.

H. Apply bond breaker where required.

I. Tool sealants using sufficient pressure to fill all voids.

J. Upon completion, leave sealant with smooth, even, neat finish.

K. Where piping, conduit, ductwork, etc., penetrate wall, seal each side of wall opening.

L. Install compressible sealant to position at indicated depth.
   1. Size so that width of material is twice joint width.
   2. Take care to avoid contamination of sides of joint.
   3. Protect side walls of joint (to depth of finish sealant).
   4. Install with adhesive faces in contact with joint sides.
   5. Install finish sealant where indicated.

M. Install expanding foam sealant to minimum 4-inches depth or thickness of wall being penetrated if less than 4-inches or as indicated on Drawings.
   1. Provide adequate backing material as necessary.
   2. Hold material back from exposed face of wall as necessary to allow for installation of backer rod and finish sealant.
      a. Allow expanding foam sealant to completely cure prior to installing backer rod and finish sealant.
   3. Trim off excess material flush with surface of the wall if not providing finished sealant.

N. Spillage: Do not allow sealants or compounds to overflow or spill onto adjoining surfaces, or to migrate into voids of adjoining surfaces. Clean adjoining surfaces by whatever means may be necessary to eliminate evidence of spillage.

O. Do not overheat hot - applied sealants.

P. Recess exposed edges of gaskets and joint fillers slightly behind adjoining surfaces, unless otherwise shown, so that compressed units will not protrude from joints.

Q. Bond ends of gaskets together with adhesive or "weld" by other means as recommended by manufacturer to ensure continuous watertight and airtight performance. Miter - cut and bond ends at corners unless molded corner units are provided.

R. For sealants applied to joints which occur in conjunction with waterproofing, install sealant prior to application of waterproofing.

S. Masking Tape:
   1. Use masking tape where required to prevent contact of sealant with adjoining surfaces that otherwise would be permanently stained or damaged by such contact or by cleaning methods required to remove sealant smears. Remove tape immediately after tooling without disturbing joint seal.

3.3 SEALANT WORK

A. General:
   1. Work includes but is not limited to: Sealing all joints which will permit penetration of dust, air, or moisture.
   2. Refer to SCHEDULE for materials to be used.
   3. See Specification Section 07 84 00 for firestopping.

B. Concrete joints:
   1. Flooring joints.
   2. Isolation joints.
   3. Joints between paving or sidewalks and building.
   4. Construction, control and expansion joints.

C. Masonry:
   1. Masonry control joints.
   2. Brick expansion joints.
   3. Cast stone / precast coping and sill head joints.
   4. Between masonry and other materials.

D. Flashing, reglets and retainers.

E. Pre-Finished Fiber-Cement Board Lap Siding and trim.

F. Openings:
   1. Perimeters of door and window frames, louvers, grilles, etc.
   2. Door thresholds shall be set in a full bed of sealant.
   3. Glass and glazing: See specification Section 08 81 00.

G. Interior finishes:
   1. Perimeter and penetrations of sound insulated walls.
   2. Casework and millwork: See Specification Section 06 41 00.
   3. Expansion and control joints in tile work.

H. Plumbing fixtures.

I. Penetrations of walls, floors and decks.

J. Other joints where sealant, expanding foam sealant or compressible sealant is indicated.

3.4 FIELD QUALITY CONTROL

A. Adhesion Testing:
   1. Perform adhesion tests in accordance with ASTM C1521 per the following criteria:
      a. Water bearing structures: One test per every 1000 linear foot of joint sealed.
      b. Exterior precast concrete wall panels: One test per every 2000 linear foot of joint sealed.
      c. Chemical containment areas: One test per every 1000 linear foot of joint sealed.
      d. Building expansion joints: One test per every 500 linear feet of joint sealed.
      e. All other type of joints except butt glazing joints: One test per every 3000 linear feet of joint sealed.
      f. Manufacturer's authorized factory representative shall recommend, in writing, remedial measures for all failing tests.

3.5 CURE AND PROTECTION

A. Cure sealants and caulking compounds in compliance with manufacturer's instructions and recommendations, to obtain high early bond strength, internal cohesive strength and surface durability. Advise Contractor of procedures required for cure and protection of joint sealers during construction period, so that they will be without deterioration or damage (other than normal wear and weathering) at time of acceptance.

3.6 SCHEDULE

A. Furnish sealant as indicated for the following areas:
   1. Exterior areas unless scheduled otherwise:
      a. Above grade: Polyether.
      b. Below grade: Polyurethane.
   2. Exterior precast concrete panels:
      a. Two Component Polyurethane.
   3. Exterior joints at door and window frames:
      a. Two Component Polyurethane.
   4. Exterior horizontal joints at perimeter of building where abutting pavements, walks, etc:
      a. One Component Moisture Curing Modified Polyurethane Sealant.
   5. Interior areas:
      a. Noncorrosive areas:
         1) Wet exposure: Polyether.
            a) Toilet rooms, locker rooms, janitor closets or similar areas: Mildew resistant silicone.
         2) Dry exposure: Polyether, unless noted otherwise.
            a) Sound insulated construction: Acoustical sealant.
      b. Corrosive areas:
         1) Wet exposure: Polysulfide.
         2) Dry exposure: Polyurethane.
      c. Sealant exposed to or having the potential of being exposed to concentrated chlorine gas or chlorine liquid: Polysulfide.
      d. Casework, countertops and solid surface materials: Silicone.
      e. Sinks, fixtures or other areas subject to potential splash, spillage or condensation: Mildew Resistant Silicone.
   6. Immersion:
      a. Prolonged contact with or immersion in:
         1) Potable water:
            a) Polysulfide.
            b) NSF 61 approved.
         2) Non-potable water, wastewater or sewage: Polysulfide.
   7. Compressible sealant:
      a. Where indicated.
   8. Exterior wall penetrations: Expanding urethane foam, with finish sealant.
      a. Finish sealant:
         1) Exterior side:
            a) Above grade: Polyether.
            b) Below grade: Polyurethane.
         2) Interior side:
            a) Noncorrosive area:
               (1) Wet exposure: Silicone.
               (2) Dry exposure: Polyether, unless noted otherwise.
            b) Corrosive area:
               (1) Wet exposure: Polysulfide.
               (2) Dry exposure: Polyurethane.
   9. Interior concrete slab formed or saw-cut control joints:
      a. Polyurea joint filler.
   10. Interior miscellaneous interior joints:
      a. Siliconized Acrylic Latex Sealant.
   11. Under door thresholds:
      a. Butyl Rubber and Polyisobutylene Sealant.
   12. Exterior joints between dissimilar metal materials:
      a. Butyl Rubber and Polyisobutylene Sealant.
   13. Use sealant types as specified in other specifications sections.

END OF SECTION

DIVISION 08

OPENINGS

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SECTION 08 11 00
HOLLOW METAL DOORS AND FRAMES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Metal doors and frames.
   2. Metal borrowed lite window frames.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 - Submittal Procedures.
   2. Section 01 78 23 - Operation and Maintenance Data.
   3. Section 08 70 00 - Finish Hardware.
   4. Section 08 81 00 - Glass and Glazing.
   5. Section 09 06 00 - Color Schedule.
   6. Section 09 91 00 - Architectural Painting.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. A653, Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
      b. E283, Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Skylights, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen.
   2. National Association of Architectural Metal Manufacturers (NAAMM):
      a. Hollow Metal Manufacturers Association (HMMA).
   3. National Fire Protection Association (NFPA):
      a. 80, Standard for Fire Doors and Fire Windows.
      b. 105, Standard for Smoke Door Assemblies and Other Opening Protectives.
   4. National Fire Protection Association (NFPA):
      a. 80, Standard for Fire Doors and Other Opening Protectives.
   5. Steel Door Institute (SDI):
      a. 100, Specifications for Standard Steel Doors and Frames
      b. 111, Recommended Details for Standard Steel Doors, Frames, Accessories and Related Components.
      c. 117, Manufacturing Tolerances for Standard Steel Doors and Frames.
      d. All SDI publications.
   6. Steel Door Institute/American National Standards Institute (SDI/ANSI):
      a. A250.4, Test Procedure and Acceptance Criteria for Physical Endurance for Steel Doors, Frames and Frame Anchors.
      b. A250.6, Recommended Practice for Hardware Reinforcing on Standard Steel Doors and Frames.
      c. A250.7, Nomenclature for Standard Steel Doors and Steel Frames.
      d. A250.8, Specifications for Standard Steel Doors and Frames.
      e. A250.10, Test Procedure and Acceptance Criteria for Prime Painted Steel Surfaces for Steel Doors and Frames.
      f. A250.11, Recommended Erection Instructions for Steel Frames.
   7. Underwriters Laboratories, Inc. (UL):
      a. Building Materials Directory.
      b. 1784, Standard for Air Leakage Tests of Door Assemblies and Other Opening Protectives.

B. Qualifications: Manufacturer must be current member of SDI, and NAAMM (HMMA).

C. Wipe coat galvanized steel is not acceptable as substitute for galvanizing finish specified.

1.3 DEFINITIONS

A. As identified in SDI/ANSI A250.7.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
   2. Schedule of doors and frames using same reference numbers as used on Drawings.
   3. SDI certification.
   4. FM approval.
   5. UL certification.

B. Color Samples:
   1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
   2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Store doors and frames in accordance with SDI/ANSI A250.11 and as follows:
   1. Doors shall be shipped individual packed. Ship frames with angle spreaders at door opening bottoms. Store doors and frames on the building site, in an upright position, under cover, on wood sills or floors, in a manner that prevents rust or damage. Ventilate canvas or plastic covers to prevent moisture traps. After installation, protect doors and frames from damage during subsequent construction activities. Damaged work will be rejected and shall be replaced with new work.
   2. Provide 1/4-inch space between stacked doors to allow air circulation.

1.6 JOB CONDITIONS

A. Examine structure, substrates and conditions under which work is to be installed for conditions detrimental to correct and timely completion. Do not proceed until unsatisfactory conditions have been corrected. Installation assumes responsibility for performance.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Metal Doors and Frames
      a. Basis of Design: Curries by ASSA ABLOY.
      b. Or Equal.
   2. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Steel Sheet: Hot-dipped galvannealed steel, ASTM A653, A60 coating.

B. Frames: Hot-dipped galvannealed steel, ASTM A653, A60 coating.

C. Supports and Reinforcing: Hot-dipped galvannealed steel, ASTM A653, A60 coating.

D. Inserts, Bolts and Fasteners: Manufacturer's standard.

E. Primer: Manufacturer's standard coating meeting SDI/ANSI A250.10.

F. Galvannealed Coating Repair: See Specification Section 09 96 00.

G. Thermal Insulation: Polyurethane, CFC free.

H. Sound Insulation: Fiberglass batt insulation or impregnated Kraft honeycomb.

2.3 ACCESSORIES

A. Glass Panels in Doors:
   1. Fixed, integral stops on exterior face and removable stops on interior face.
      a. Reinforce cut-out in door panel and frames with minimum 16 Gauge galvannealed or galvanized steel channel.
   2. Stops and Moldings:
      a. Provide stops and moldings around glazed lites where indicated. Install trim into the door as a four-sided welded assembly with mitered, reinforced and welded corners. Trim to be identical on both sides of the door.
      b. Form corners of stops and moldings with butted or mitered hairline joints at fabricator’s shop.
      c. Fixed and removable stops to allow multiple glazed lites each to be removed independently.
      d. Stops secured with countersunk Phillips head machine screws on interior face.
      e. Labeled and non - labeled doors shall use the same trim.
   3. Reinforce cut-out in door panel with minimum 20 GA galvannealed or galvanized steel channel.
   4. Glass: See Specification Section 08 81 00 Glass and Glazing.
   5. Acceptable mounting methods:
      a. Fit into a formed area of the door face, not extending beyond the door face, and interlocking into the recessed area
      b. Cap the cutout does not extend more than 1/16 inch from the door face.
      c. Other: Per Steel Door Institute Standards unless noted otherwise in this Section of the specification.

B. Frame Anchors:
   1. Jamb anchors:
      a. Masonry corrugated strap anchors: Minimum 16 Gauge galvanized.
      b. Stud partition, existing opening and base anchors: Minimum 18 Gauge, galvanized.

2.4 FABRICATION

A. General:
   1. SDI/ANSI A250.8.
   2. Fabricate rigid, neat in appearance and free from defects.
   3. Form to sizes and profiles indicated on Drawings.

a. Beveled edge.
4. Fit and assemble in shop wherever practical.
5. Mark work that cannot be fully assembled in shop to assure proper assembly at site.
6. Continuously wire weld all joints, dress exposed joints smooth and flush.
7. Fabricate doors and frames to tolerance requirements of SD I117.
8. Fit doors to SDI clearances.
9. Fit doors to SDI and NFPA 80 clearances if fire doors are used.
10. Provide fire rated doors which are approved and labeled by UL.
11. All doors shall be handed.
12. Hinge cut-out depth and size on doors and frames shall match hinge specified in Specification Section 08 70 00.
13. Design and fabricate doors to requirements of the building code.
14. Doors, windows and frames shall be finished field painted.
    a. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
    b. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
    c. Refer to “Room Finish Schedule Interior Finish Legend found on Architectural Drawings”.
    d. Refer to “Specification Section 09 91 00 – Architectural Painting” for paint material requirements.

B. Hollow Metal Doors:
1. General:
    a. 1-3/4 inches thick.
    b. Provide doors with corrosion resistant factory prime paint.
    c. Door Performance: Standard 3 feet by 7 feet door shall meet requirements of ANSI A250.4 procedure for level C doors for 250,000 cycles.
    d. Fabricate with flush top caps.
        1) Thickness and material to match door face.
        2) Interior doors:
            a) Attach top cap to door with concealed fasteners or by welding.
            b) Factory seal if attached with fasteners.
            c) No exposed fasteners will be accepted.
    e. Continuously wire weld all joints and dress, smooth and flush.
    f. Fire - rated Interior Doors: Provide SDI Physical Performance Level A, Extra Heavy - Duty 1 3/4-inch, Model 2 Seamless, hot dipped A60 galvanized, constructed and installed to comply with NFPA No. 80, bearing UL Label indicating rating, the letter “S”, manufacturer’s name and third-party inspection service.
2. Fire rated:
    a. SDI/ANSI A250.8, Level 3, and physical performance level A, Model 2.
    b. Face sheet minimum thickness: 16 Gauge.
    c. UL fire labeled.
    d. Comply with UL 10B and UL 10C.
    e. Maximum transmitted temperature:
        1) Fire doors shall have a maximum transmitted temperature end point of not more than 250 Degree F above ambient at the end of 30 minutes of standard fire test exposure.
    f. Comply with NFPA 80.
3. Interior:
    a. Doors 48 inches wide, or less: SDI/ANSI A250.8, Level 2, and physical performance level "B", Model 2.
        1) Face sheet minimum thickness: 18 Gauge.
    b. Sound insulated, minimum STC-35.

C. Hollow Metal Frames:
1. Door frames:

a. Provide 2-inches face at all heads, jambs and mullions for frames in stud walls.
b. Provide 4-inches face at head where noted on Drawings or required by wall construction.
c. 26 GA galvannealed steel boxes welded to frame at back of all hardware cutouts.
d. Steel plate reinforcement welded to frame for hinge, strikes, closers and surface-mounted hardware reinforcing.
    1) All plate reinforcement shall meet size and thickness requirements of SDI/ANSI A250.8.
e. Split type frames not acceptable.
    1) All horizontal and vertical mullions and transom bars shall be welded to adjacent members.
f. Conceal all fasteners.
g. Frames shall be set up with all face joints continuously wire welded and dressed smooth.
h. Fire rated:
    1) 16 Gauge.
    2) Comply with UL 10B and UL 10C.
    3) Comply with NFPA 80.
i. Interior: 16 Gauge.
j. Provide removable spreaders at bottom of frame.
k. All “Interior - Fire Rated, hollow metal frames” shall have an 1/8-inch integral kerf formed into the frame soffit to receive a gasket composed of a cellular modified foam core and is clad in an embossed, non-vinyl, paint - resistant liner which is UV stabilized. Frames shall comply with NFPA 105 smoke and draft control door assemblies, UL 1784 air leakage test of door assemblies, ASTM E283 air infiltration, and can be UL listed fire door frames up to and including 3-hour ratings. Frames to be mortised, reinforced, and drilled and tapped for mortise hardware compatible with kerf frame profile.Provide all interior fire rated frames with continuous smoke seal gasket on both sides and head of frames.
l. In Interior - non - fire rated frames drill stops to receive 3 silencers on strike jambs of single - swing frames and 2 silencers on heads of double - swing frames. Install plastic plugs to keep holes clear during construction. Provide and install silencers.

D. Prepare for finish hardware in accordance with hardware schedule, templates provided by hardware supplier, and SDI/ANSI A250.6.
    1. Locate finish hardware in accordance with SDI/ANSI A250.8.
    2. See Specification Section 08 70 00 for hardware.
    3. Prepare doors for swing direction indicated.
    4. Preparing doors for non-handed hinges is not acceptable.
    5. Frame Reinforcing for Interior Openings:
        a. Hinge: Minimum 3/16-inch-thick steel bar reinforcement extending continuous one piece from top to bottom of door frame. High frequency hinge reinforcement at all hinges will also be acceptable.
        b. Closer / Holder: Minimum 12-gage steel plate.
        c. Strike: 14-gauge steel.

E. Coordinate installation of electrical connections to electrical hardware items in both frames and doors.

F. After fabrication, clean off mill scale and foreign materials and prime with rust inhibiting primer.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install doors and frames in accordance with SDI/ANSI A250.11, the building code and manufacturer's instructions.

B. Plumb, align, and brace frames securely until permanently anchored.
   1. After completion of walls, remove temporary braces and spreaders.
   2. Anchor frames with minimum of three anchors per jamb.
      a. Number and location of anchors shall be in accordance with SDI and frame manufacturer's recommendations.

C. Place frames prior to construction of enclosing walls and ceilings. Plumb, align and brace securely until permanent anchors are set. After wall construction is completed, remove temporary braces and spreaders leaving surfaces smooth and undamaged. Comply with SDI - 105 and SDI - 110 for proper installation procedures.

D. Locate and install minimum of 3 wall anchors per jamb at hinge and strike levels. Anchors shall be of type appropriate to wall construction. Minimum acceptable anchors 16 gauge, 1-inch-wide corrugated steel. Provide (2) 12 gauge, adjustable 3-inch wide, flanged floor clips per frame jamb, punched for anchoring. Provide removable spreaders at bottom and mid - height of frame. Coordinate building - in of anchors and grouting of frames with other trades.

E. Install and fit labeled (fire - rated) frames in accordance with NFPA No. 80.

F. Fit doors to ANSI / SDI 100 clearances except for undercuts. Undercuts shall be coordinated with floor finishes and thresholds. Undercuts shall not be greater than 1/4 inch above floor finishes. Undercuts at thresholds shall provide for a complete tight seal with bottom of door.

G. Immediately after erection, sand smooth areas of prime coat which are rusted or damaged and apply touch - up of compatible primer. Leave smooth for finish painting.

H. Verify that door frames are rigidly anchored and are not more than 1/16 inch out of level, plumb, planar alignment ore member twist.

3.2 TESTING FOR ALL DOORS

A. When building environmental systems are operating as specified, conduct simple tests to ensure each completed door system will close and latch or otherwise secure properly, without assistance. Closer tests (for doors so equipped):
   1. Full open test: Open door to full width. Release door. Door to completely close and latch fully. If door is equipped with magnetic locks, magnetic lock armature to completely bond to lock with bonding sensor indicating secure.
   2. Mid - point close to latch test: Open door halfway. Release door. Door to completely close and latch fully. If door is equipped with magnetic locks, magnetic lock armature to completely bond to lock with bonding sensor indicating secure.
   3. Close to latch test: Open door to full width, then permit door to re - close until lock’s latch bolt touches strike lip and hold. When magnetic locks are used, permit door to return so that strike armature is within one inch of magnetic lock. Release door. Door to completely close and latch fully. If door is equipped with magnetic locks, magnetic lock armature to completely bond to lock with bonding sensor indicating secure.

B. Latch Bolt Release Test: Measure force to release latch bolt using a door force gage located 1 inch from end of lever handle or located at center of exit device push pad.
   1. For interior fire doors, latch shall release when lever is subjected to a force not more than 15 pounds.

C. Latch Bolt Security Test: With door closed and properly latched, pull sharply on door, in direction of swing. Door to remain secure.
   1. Door pairs only: With both doors closed and properly latched, pull sharply on active leaf in direction of swing while simultaneously pushing on inactive door. Door to remain secure.

D. Latch Bolt Dead Latch Test:
   1. With door closed and latched, insert dead latch test tool and engage latch bolt. Pull sharply. Latch bolt should not retract.
   2. With door open, and latch bolt and dead latch in view, hold dead latch in with your finger and rapidly and intermittently push (rattle) latch bolt. Latch bolt should not retract.

E. Lock Function Test: With proper key to operate lock cylinder, perform all functions lockset is capable of, and insure they operate properly.

F. Closer Tests: Preparation 1 - Adjust spring speed from 1.5 to 2 seconds when closing from 70 degrees to closed position. Preparation 2 - Adjust closer speed from 5 to 6 seconds when closing from 90 degrees to 12 degrees. Preparation 3 - Adjust closer power to the lowest value within the following ranges that consistently passes the following tests, measuring with a door force gage located 1 inch from latch edge of door.
   1. Interior non - fire doors - Force required to push door to open position: not more than 5 pounds
   2. Interior fire doors: Force required to unlatch door: not more than 15 pounds; Force required to set door in motion: not more than 30 pounds; Force required to push door to open position: not more than 15 pounds.

3.3 ADJUSTMENT, DEFECTIVE WORK, CLEANING AND PROTECTION

A. Verify that hardware supplier checks and readjusts operating finish hardware items just prior to final inspection. Leave work in complete and proper operating condition. Remove and replace defective work.

B. Doors or frames which are not manufactured to specified size and planar tolerances to be removed from site and replaced without charge to Government.

C. Frames that are not installed within specified plumb, level and planar tolerances shall be removed from site and replaced without charge to Government.

D. After installation protect doors and frames from damage during subsequent construction activities. Damaged work will be rejected and shall be replaced with new work. All doors and frames shall be properly cleaned, etc., and left in a suitable and acceptable condition for the finished painting work by others. Where doors where galvanized finish has been scratched shall have a zinc - rich primer applied to the galvanized coating area that has been damaged prior to the finish field paint coating being applied.

E. At completion of project, verify frames, doors and hardware are installed as required by Contract Documents and by approved changes during construction. Correct if noncompliant.

F. Inspect new door frames as soon after installation as possible, for not more than 1/16 inches out of square, plumb, planar alignment or member twist. Inspect anchoring for rigid attachment. If any parameter does not meet this specification, contact installer to rectify and re-inspect, after corrections are made.

G. Inspect all doors and frames upon which work is performed for loose or poor - fitting hardware or components. Tighten or otherwise adjust or replace as required for proper operation. Inspect meeting stiles, head jambs and thresholds for evidence of binding or poor fitting doors and frames.

END OF SECTION

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SECTION 08 11 16
ALUMINUM DOORS AND FRAMES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
1. Aluminum Storefront Exterior Doors and Aluminum Frames. (Doors SF, FAP – Frames AL, FAP)
2. Aluminum and Aluminum/Fiber Reinforced Plastic Hybrid Exterior Doors and Aluminum Frames. (Doors AL, FAP/FRP – Frames AL, FAP)
3. Aluminum and Aluminum/Fiber Reinforced Plastic Hybrid Exterior Transom Panel.
4. Aluminum/Fiber Reinforced Plastic Interior Doors and Aluminum Frames. (Doors AL, FRP Frames AL, FAP)
5. Aluminum Exterior and Interior Window Framing.

B. Related Specification Sections include but are not necessarily limited to:
1. Section 01 33 23 – Submittal Procedures.
2. Section 01 78 23 – Operation and Maintenance Data.
3. Section 07 92 00 - Joint Sealants.
4. Section 08 70 00 - Finish Hardware.
5. Section 08 81 00 - Glass and Glazing.
6. Section 09 96 00 - High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
1. American Architectural Manufacturers Association (AAMA):
   a. AAMA - 2605, Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels.
2. American National Standards Institute/National Fenestration Rating Council (ANSI/NFRC):
   a. ANSI/NFRC - 100, Procedure for Determining Fenestration Product U-Factors.
   b. ANSI/NFRC - 400, Procedure for Determining Fenestration Product Air Leakage.
3. American National Standards Institute/Steel Door Institute (ANSI/SDI):
   a. ANSI/SDI - A250.8, Specifications for Standard Steel Doors and Frames.
4. ASTM International (ASTM):
   a. ASTM - B209, Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate.
   b. ASTM - B221, Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes.
   c. ASTM - D2794, Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
   d. ASTM - E84, Standard Test Method for Surface Burning Characteristics of Building Materials.
   e. ASTM - E90, Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements.
   f. ASTM - E283, Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Skylights, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen.
5. International Building Code (IBC):
   a. 2018 Edition
6. National Fire Protection Association (NFPA):
   a. NFPA - 80, Standard for Fire Doors and Other Opening Protectives.
   b. NFPA - 252, Fire Tests of Door Assemblies.

c. NFPA - 275, Standard Method of Fire Tests for the Evaluation of Thermal Barriers.

7. Underwriters Laboratory (UL):
   a. UL - 10B, Standard for Fire Tests of Door Assemblies.
   b. UL - 10C, Standard for Positive Pressure Fire Tests of Door Assemblies

B. Qualifications:
   1. Door and frame fabricator must have a minimum of 10 years of experience in fabrication of flush aluminum doors, flush aluminum/fiber reinforced plastic hybrid doors and associated aluminum frames.
   2. Installation: Door and frame installation shall be performed by experienced workmen having a minimum five years of experience installing products similar to those specified.
   3. Door and frame components must be fabricated by the same manufacturer.

1.3 DEFINITIONS

A. Installer or Applicator:
   1. Installer or applicator is the person actually installing or applying the product in the field at the Project site.
   2. Installer and applicator are synonymous.

B. PVDF: Polyvinylidene fluoride.

1.4 SUBMITTALS

A. Shop Drawings:
   a. Product technical data including fabrication methods.
   b. Finishing.
   c. Hardware.
   d. Accessories.
   e. Applicable test reports.
   f. Acknowledgement that products submitted meet the requirements of standards referenced.
   g. Manufacturer's installation instructions.
   2. Schedule of doors and frames and window frames using same reference numbers as indicated on Drawings. Include the following:
      a. Elevations.
      b. Detail sections of typical composite members indicating thickness, sizes, and thermal breaks.
      c. Hardware mounting heights.
      d. Hardware schedule.
      e. Anchorages and reinforcements.
      f. Expansion provisions.
      g. Glazing details.
   3. Testing and Evaluation Reports:
      a. Submit testing reports and evaluations provided by manufacturers conducted by and accredited independent testing agency certifying doors and frames comply with specified performance requirements.
   4. Provide Manufacturer’s Certificates stating that aluminum has been given specified thickness of finish listed below:
      a. Painting
   5. Certifications:
      a. Certification of manufacturer qualifications.
      b. Certification of installer qualifications.
   6. Warranty.

B. Samples:
   1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.

a. 6 x 6 inches sample of door section showing edge construction, core, impact reinforcement and face sheet with finish as specified.

2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

C. Contract Closeout Information:

1. Operation and Maintenance Data:

a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.5 PROJECT CONDITION

A. Field Measurements: Check openings by field measurement before fabrication to ensure proper fitting of work; show measurements on final shop drawings. Coordinate fabrication schedule with construction progress to avoid delays in the work. Where necessary, proceed with fabrication without field measurements, and coordinate fabrication tolerances to ensure proper fit.

1. When construction schedule permits, verify field measurements with drawing dimensions prior to fabrication of glass products.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Store doors and frames in a dry, weather protected area.

1. Place units on wood skids providing a minimum 6 inches air space above the ground.
2. Do not store units flat, set frames and doors on edge providing minimum 1/2 inches air circulation space between each unit.
3. Provide covering which will ensure air flow around each unit to prevent trapping of moisture.
4. If door wrapper becomes wet immediately remove and provide dry protection equivalent to wrapper removed.

B. Storage recommendations by manufacturer shall take precedent over the above requirements.

1.7 WARRANTY

A. Total Door System

1. The contractor shall assume full responsibility and warrant for ten years the satisfactory performance of the total door and frame installation which includes that of the door, frame, hardware, anchorage and setting system, sealing, flashing, etc, as it relates to air, water and structural adequacy as called for in the specifications and approved shop drawings.
2. Any deficiencies due to such elements not meeting the specifications shall be corrected by the contractor at his expense during the warranty period.

B. Factory applied high performance organic coatings utilizing PVDF resins shall be provided with manufacturer's standard 10-year warranty against color fade, chalk and film integrity.

1. Thresholds do not have a finish warranty.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Aluminum exterior/interior storefront doors. (Doors SF, FAP – Frames AL, FAP).
      a. Basis of Design: Special-Lite; SL-15 Wide Stile Monumental Door and SL450TB Thermal Broke Aluminum Framing.
      b. Or Equal.
   2. Aluminum\fiber reinforced plastic exterior hybrid doors. (Doors AL, FAP/FRP – Frames AL, FAP).
      a. Basis of Design: Special-Lite; SL-16/SL-20 Aluminum/Fiber Reinforced Plastic Hybrid and Special-Lite and SL-450TB Thermal Broke Aluminum Framing.
      b. Or Equal.
   3. Aluminum\fiber reinforced plastic exterior hybrid transom panels.
      a. Basis of Design: Special-Lite; SL-16/SL-20 Aluminum/Fiber Reinforced Plastic Hybrid.
      b. Or Equal.
   4. Fiber reinforced plastic interior doors and frames. (Doors AL, FRP Frames AL, FAP).
      a. Basis of Design: Special-Lite: SL-20 Fiber Reinforced Plastic and Special-Lite: SL-245 FG Aluminum Framing.
      b. Or Equal.
   5. Aluminum exterior/interior window framing.
      a. Basis of Design: SL-450TB Thermal Broke Aluminum Framing.
      b. Or Equal.
   6. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 PERFORMANCE AND DESIGN REQUIREMENTS

A. Thermal Performance:
   1. Storefront and Hybrid Exterior Doors and Thermally Broke Frame Assembly shall meet the requirements of the Energy Code.
   2. Air Infiltration: NFRC 400, ASTM E283.
      a. Hybrid Door and Thermally Broke Frame Assembly (Less than 50 percent glass)
         1) 0.01 cfm/sq ft @ 1.57psf and 0.02 cfm/sq ft @ 6.24 psf.
      b. Storefront Entrance Door and Thermally Broke Frame Assembly (Greater than 50 percent glass)
         1) 0.39cfm/sq ft @1.57psf and 0.78cfm/sq ft @ 6.24psf.
   3. Thermal Transmittance: NFRC 100.
      a. Hybrid Door and Thermally Broke Frame Assembly (Less than 50 percent glass): U = 0.33 maximum.
      b. Storefront Entrance Doors and Thermally Broke Frame Assembly (Greater than 50 percent glass): U = 0.62 maximum.

B. Acoustic Performance, Door and Thermally Broke Aluminum Frame:
   1. Sound Transmission: ASTM E90.
      a. Opaque Swinging Hybrid Door and Thermally Broke Frame Assembly (Less than 50 percent glass):
         1) STC: 26 minimum.
         2) OITC: 26 minimum.
      b. Storefront Entrance Doors and Thermally Broke Frame Assembly (Greater than 50 percent glass):
         1) STC: 30 minimum.
         2) OITC: 28 minimum.

2.3 MATERIALS

A. Aluminum Members:
   1. Aluminum extrusions made 6061 or 6063 aluminum alloys.
   2. Sheet and plate to conform to ASTM-B209.

3. Alloy and temper to be selected by manufacturers for strength, corrosion resistance, and application of required finish, and control of color.

B. Fasteners.
   1. All exposed fasteners will have a finish to match material being fastened.
   2. 410 stainless steel or other non-corrosive metal.
   3. Must be compatible with items being fastened.

C. FRP Sheet:
   1. Extruded, Sandstone texture, through color FRP sheet.
   2. Fire Classification - Class A.

2.4 MANUFACTURED UNITS

A. General:
   1. Fabricate rigid, neat in appearance and free from defects.
   2. Form to indicated sizes and profiles.
   3. Fit and assemble in shop wherever practical.
   4. Mark work that cannot be fully assembled in shop to assure proper assembly at site.
   5. Factory prepare finish hardware, in accordance with the hardware schedule and templates provided by hardware supplier.
      a. Locate hardware per ANSI/SDI A250.8.
         1) See Specification Section 08 70 00 for hardware.
   6. Conceal fastenings wherever practical.
      a. Exposed fasteners to be countersunk Phillips or Jackson flat head screws and bolts.

B. Doors:
   1. Nominal 1-3/4-inches thick.
   2. Storefront Exterior Doors Stiles and Rails: (Doors SF, FAP).
      a. Stiles:
         1) 4-3/4-inch wide with integral glass stop on exterior side, no snap or applied stops allowed.
         2) Aluminum extrusions made from 6063 aluminum alloys with a minimum temper of T5.
            a) Screw or snap in place applied caps are not acceptable.
         3) Meeting stiles to include integral pocket to accept pile brush weather seal.
      b. Rails:
         1) Top Rail Height – 6-1/2-inches.
         2) Bottom Rail Height – 10-inches.
         3) Integral glass stops on exterior side, no snap or applied stops allowed.
         4) Aluminum extrusions made from 6063 aluminum alloys with a minimum temper of T5.
         5) Screw or snap in place applied caps are not acceptable.
   3. Hybrid Aluminum/Fiber Reinforced Plastic Exterior Doors and Fiber Reinforced Plastic Interior Doors Stiles and Rails: (Exterior Doors AL, FAP/FRP & Interior Doors AL, FRP).
      a. Stiles:
         1) Aluminum extrusions made from 6063 aluminum alloys with a minimum temper of T5.
         2) Minimum 2-5/16-inch-deep one-piece extrusion with have integral reglets to accept face sheet on both interior and exterior side of door which secure face sheet into place and permit flush appearance.
         3) Screw or snap in place applied caps are not acceptable.
      b. Rails:
         1) Top rails must have integral legs for interlocking continuous extruded aluminum flush cap.
         2) Bottom rails must have integral legs for interlocking continuous weather bar with single nylon brush weather stripping door bottom with two nylon brush weather stripping.

3) Meeting stiles to include integral pocket to accept pile brush weather seal.

4. Corners:
   a. Hybrid Aluminum/Fiber Reinforced Plastic Exterior Doors and Fiber Reinforced Plastic Interior Doors (Exterior Doors AL, FAP/FRP & Interior Doors AL, FRP) - Mitered.
      1) Secured with 3/8-inch diameter full-width steel tie rod through extruded splines top and bottom which are integral to standard tubular shaped rails.
      2) 1-1/4-inch x 1-1/4-inch x 3/16-inch 6061 aluminum angle reinforcement at corner to give strong, flat surface for locking hex nut to bear on.
      3) Weld, glue, or other methods of corner joinery are not acceptable.
   b. Storefront Exterior Doors (Doors SF, FAP) – True mortise and tenon joints.
      1) Secured with 3/8-inch diameter full-width steel tie rod through extruded splines top and bottom which are integral to standard tubular shaped rails.
      2) 1-1/4-inch x 1-1/4-inch x 3/16-inch 6061 aluminum angle reinforcement at corner to give strong, flat surface for locking hex nut to bear on.
      3) Weld, glue, or other methods of corner joinery are not acceptable.
5. Face Sheets:
   a. Hybrid Aluminum/Fiber Reinforced Plastic Exterior Doors Exterior Doors AL, FAP/FRP):
      1) Exterior face:
         a) 0.125-inches aluminum sheet.
            (1) Rabbet edges of sheet to be fully captured by perimeter extrusions.
         b) Texture: Smooth.
      2) Interior face:
         a) 0.120-inches FRP sheet.
            (1) Rabbet edges of sheet to be fully captured by perimeter extrusions.
         b) Texture: Sandstone.
   b. Storefront Exterior Doors (Doors SF, FAP):
      1) Exterior and Interior faces
         a) 0.125-inches aluminum sheet.
            (1) Rabbet edges of sheet to be fully captured by perimeter extrusions.
         b) Texture: Smooth.
   c. Interior Fiber Reinforced Plastic Doors Interior Doors (AL, FRP):
      1) Both faces:
         a) 0.120-inches FRP sheet.
            (1) Rabbet edges of sheet to be fully captured by perimeter extrusions.
         b) Bonded to core with adhesive according to manufactures listing.
         c) Texture: Sandstone.
6. Core:
   a. All exterior and interior doors:
      1) Poured-in-place polyurethane foam.
      2) Laid in foam cores are not acceptable.
      3) Foam Plastic Insulated Doors: IBC 2603.4.
         a) Foam plastic shall be separated from the interior of a building by an approved thermal barrier.
         b) Approved thermal barrier must meet the acceptance criteria of the Temperature Transmission Fire Test and Integrity Fire Test as stated in NFPA 275.
         c) IBC 2603.4.1.7 foam plastic insulation, having a flame spread index less than 75 and a smoke developed index of not more than 450 shall be permitted as a door core when the face is metal minimum 0.032-inch aluminum.
            (1) Standard door assembly can be tested to show it meets these requirements without the use of thermal barrier. If no independent testing conducted all doors with foam plastic core must have a thermal barrier.
7. Exterior Transom Panels:

a. Panels shall be made to be removable.
b. Panels shall match the construction of Hybrid Aluminum/Fiber Reinforced Plastic Exterior Doors.

8. Cutouts.
   a. Manufacture doors with cutouts for required vision lites and panels.
9. Hardware:
   a. Factory install door hardware.
   b. Pre-machine doors in accordance with templates from specified hardware manufacturers.
   c. Surface mounted closures will be reinforced for but not prepped or installed at factory
10. Hardware reinforcement:
    a. Aluminum extrusions made from 6061 or 6063 aluminum alloys.
    b. Sheet and plate to conform to ASTM-B209.
    c. Alloy and temper to be selected by manufacturer for strength, corrosion resistance, and application of required finish, and control of color.
    d. Bars and tubes to meet ASTM-B221.

C. Frames:
1. General:
   a. Minimum thickness: 0.120-inches.
   b. Site-line: Maximum: 2-1/4-inches.
2. Exterior aluminum door and window frames and interior aluminum window frames:
   a. Thermally broken extruded aluminum with applied stops.
      1) Weatherstripping: Resilient bulb.
   b. Depth: 4-1/2-inches.
   c. Perimeter Frame Members.
      1) Storefront frame with thermally broken pocket filler.
      2) Factory fabricated.
      3) Open-back framing is not acceptable.
   d. Thermal Strut.
      1) Fiber reinforced plastic, no other materials will be accepted.
   e. Applied Doorstops.
      1) 5/8-inch x 1-3/4-inch, 0.125-inch wall thickness, with screws and weather-stripping.
      2) Provide a solid 1/2-inch aluminum bar behind door stop for closer shoe attachment.
      3) Pressure gasketing for weathering seal.
      4) Counterpunch fastener holes in door stop to preserve full-metal thickness under fastener head.
      5) Minimum 1/2-inch aluminum bar reinforcement under doorstop for required hardware attachments, aluminum to meet ASTM-B221.
   f. Caulking.
      1) Caulk joints before assembling frame members.
   g. Frame Member to Member Connections.
      1) Secure joints with fasteners.
      2) Provide hairline butt joint appearance.
      3) Shear block construction only, no screw spline allowed.
   h. Hardware
      1) Pre-machine and reinforce frame members for hardware in accordance with manufacturer's standards and door hardware schedule.
      2) Surface mounted closures will be reinforced for but not prepped or installed at factory.
      3) Factory install door hardware.
   i. Anchors:
      1) Anchors appropriate for wall conditions to anchor framing to wall materials.
      2) Door Jamb and Header Mounting Holes: Maximum of 24-inch centers.
      3) Secure head and sill members of transom, side lites, and similar conditions.

3. Interior aluminum door frames:
    a. Tubular and Flush-glazed extruded aluminum with applied stops.
        1) Weatherstripping: Manufacturer's standard pile.
    b. Depth: 4-1/2-inches.
    c. Perimeter Frame Members.
        1) Heavy wall aluminum storefront system.
        2) Box type with 4 enclosed sides.
        3) Factory fabricated.
        4) Open-back framing is not acceptable.
    d. Applied Door Stops.
        1) 0.625-inch high, with screws and weatherstripping.
        2) Pressure gasketing for weathering seal.
        3) Counterpunch fastener holes in door stop to preserve full-metal thickness under fastener head.
    e. Caulking. Caulk joints before assembling frame members.
    f. Frame Member to Member Connections.
        1) Secure joints with fasteners.
        2) Provide hairline butt joint appearance.
    g. Hardware:
        1) Pre-machine and reinforce frame members for hardware in accordance with manufacturer's standards and door hardware schedule.
        2) Factory install door hardware.
    h. Anchors:
        1) Anchors appropriate for wall conditions to anchor framing to wall materials.
        2) Door Jamb and Header Mounting Holes: Maximum of 24-inch centers.
        3) Secure head and sill members of transom, sidelites, and similar conditions.

2.5 FABRICATION

A. Factory Assembly.
    1. Door and frame components from the same manufacturer.
    2. The required size for door and frame units shall be as indicated on the drawings.
    3. Complete cutting, fitting, forming, drilling, and grinding of metal before assembly.
    4. All cut edges to be free of burrs.
    5. Welding of doors or frames is not acceptable.
    6. Maintain continuity of line and accurate relation of planes and angles.
    7. Secure attachments and support at mechanical joints with hairline fit at contact surfaces.

B. Shop Fabrication
    1. All shop fabrication to be completed in accordance with manufactures process work instructions.
    2. Quality control to be performed before leaving each department.

2.6 FINISHES

A. Doors.
    1. Paint.
        a. Aluminum perimeter and aluminum face sheet.
            1) Fluropon®.
                a) Topcoat – 70 percent polyvinylidene difluoride (PVDF) resin, meets or exceeds all AAMA 2605 specifications.
                b) Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
                c) Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
    2. Fiberglass.
        a. Two-component acrylic urethane enamel Gloss topcoat.

1) Unique, high-solids, high-build, multifunctional coating.
2) Low VOC, Gloss coating.
3) Impact Resistance, ASTM-D2794: 80-102 in/lb (direct), 40-80 in/lb (reverse) @ 1.5 mils thickness.
4) Color retention: Δ E < 5 (CIE Lab.), Florida Exposure: 18 months
5) Excellent chemical resistance.
6) Refer to "Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals" for submission procedures and requirements of Color Samples.
7) Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.

B. Frames.
1. Coatings:
a. Aluminum.
1) Fluropon®.
a) Topcoat. 70 percent polyvinylidene difluoride (PVDF) resin, meets or exceeds all AAMA 2605 specifications
b) Refer to "Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals" for submission procedures and requirements of Color Samples.
c) Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.

2.7 ACCESSORIES

A. Weatherstripping:
1. Provide complete set of Weathering System including top, sides, astral, and bottom weatherstrip that comply with the air infiltration requirements of this Specification Section. Weatherstripping shall be made to be removable for replacement.

B. Sill Sweeps:
1. Rubber blade gasket sweep strip in an aluminum extrusion applied to the interior exposed surface of the bottom rail with concealed fasteners. Sill sweeps shall be provided that comply with the air infiltration requirements of this Specification Section.

C. Thresholds:
1. Extruded aluminum one piece per door opening with ribbed surface, complete with anchors and clips, coordinated with bottom weatherstrip. Threshold shall have extruded raised lip to provide a continuous contact surface for the bottom weatherstrip. Overall height of threshold shall meet Americans with Disabilities Act Accessibility Guidelines. Provide grout under thresholds.

D. Hardware Schedule:
1. Refer to Finish Hardware in section 08 70 00 for requirements for hardware items other than those indicated herein.

E. Lights (Glass Panels):
1. Fixed, integral stops on exterior face.
2. Screw-less snap-in stops or stops secured with countersunk Phillips head machine screws on interior face.
3. Glass: See Specification Section 08 81 00 Glass and Glazing.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install doors and frames in accordance with manufacturer's instructions.

B. Set units plumb, level, and true to line, without warp or rack of framing members, doors, or panels. Provide proper support and anchor securely in place.

1. Separate aluminum and other corrodible metal surfaces from sources of corrosion of electrolytic action at point of contact with other materials. Comply with requirements specified under paragraph "Dissimilar Materials" in the Appendix to AAMA 101.
    a. Coat the dissimilar metal with two coats of heavy - bodied alkali resistant bituminous paint.
    b. Place caulking compound, or non - absorptive tape, or gasket between the aluminum and the dissimilar metal.
    c. Paint aluminum in contact with mortar, concrete and plaster, with a coat of aluminum paint primer.

C. Allowable Installation Tolerances: Install work plumb and true, in alignment and in relation to lines and grades shown. Variation of 1/8-inch in eight feet, non - accumulative, is maximum permissible for plumb, level, warp, bow and alignment.

D. Anchor aluminum frames to adjoining construction at heads, jambs and bottom and to steel supports, and bracing. Anchor frames with stainless steel or aluminum countersunk flathead screws, expansion bolts or machine screws as applicable. Use aluminum clips for internal connections of adjoining frame sections.
    1. Number and location of anchors shall be in accordance with frame manufacturer's recommendation based on the frame size, wind, seismic and thermal loads - with a minimum of three anchors per jamb.
    2. In steel stud partitions, attach anchors with self-tapping screws.

E. Drill and tap frames and doors and apply surface - mounted hardware items, complying with hardware manufacturer's instructions and template requirements. Use concealed fasteners wherever possible.

F. Special care shall be taken by the installer to ensure that the impact side of the door faces the threat axis.

G. Set sill members and other members in bed of sealant to provide weathertight construction. Comply with sealant requirements of Section 07 92 00 Joint Sealants, for compounds, fillers, and gaskets. All joints between framing and the building structure shall be sealed in order to secure a watertight installation. Where work is installed in concrete openings, place no parts other than built - in anchors until after the concrete work is completed.

H. Attach entrance system to structure to permit sufficient adjustment to accommodate construction tolerances and other irregularities.

I. Hardware: See Specification Section 08 70 00 Finish Hardware.

3.2 FIELD QUALITY CONTROL

A. Repair all damaged finishes or replace framing member or door as directed by Contracting Officer Representative.
    1. Use only materials and finishes as recommended or furnished by door and frame manufacturer.
    2. Final repaired finish shall match surrounding original finish or item being repaired shall be replaced with new item.

3.3 ADJUST AND CLEANING

A. Adjust operating hardware to function properly, without binding, and to provide tight fit at contact points and weatherstripping.

B. Clean doors and frames as recommended by manufacturer prior to acceptance by Contracting Officer Representative.

C. Clean completed system, inside and out, promptly after installation. Remove excess glazing and sealant compounds, dirt, and other substances from aluminum surfaces.

3.4 PROTECTION

A. Protect doors and frames during construction.

B. Institute protective measures and other precautions required to assure that entrances, windows, and framing will be without damage or deterioration, other than normal weathering, until time of completion of project.

END OF SECTION

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SECTION 08 14 16
FLUSH WOOD DOORS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Field finished flush wood doors.
   2. Metal louvers in wood doors, including frames, fasteners, and accessories necessary for a complete assembly.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 - Submittal Procedures.
   2. Section 08 11 00 - Hollow Metal Doors and Frames
   3. Section 08 11 16 - Aluminum Doors and Frames.
   4. Section 08 70 00 - Finish Hardware.
   5. Section 09 06 00 - Color Schedule.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Standards Institute (ANSI):
      a. A208.1, Particleboard.
   2. ASTM International (ASTM):
   3. Architectural Woodwork Institute (AWI):
      a. Quality Standards:
         1) Section 1700, Installation of Woodwork.
   4. Window and Door Manufacturers Association (WDMA):
      a. I.S. 1A, Interior Architectural Wood Flush Doors.

B. Qualifications:
   1. Door manufacturers must have a minimum of 10 years of experience in manufacturing of wood veneer doors.
   2. Manufacturers shall be current member of AWI.
   3. Installer shall be experienced personnel regularly engaged in door and hardware installation.
   4. All doors to be provided by the same manufacturer.

C. All doors to be fabricated using Hot Press 5 Ply construction.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet the requirements of standards referenced.
      b. Manufacturer's door installation instructions.
      c. Schedule of doors shall use the same reference numbers as indicated on Drawings.
         1) Schedule shall include size, type, swing, rating, frame type and size, and hardware set required.
      d. Specified wood species and veneer cut of doors.
      e. Manufacturer’s data for metal louvers, including materials, dimensions, free area percentage, and installation instructions.
      f. Louver Shop Drawings: Indicate louver size, location in door, fastener type, and compatibility with door core.
      g. Louver Warranty: Manufacturer’s standard (5) five-year warranty as specified.

B. Samples:

1. Provide two, 12-inches x 12-inches samples showing door construction in veneer specified.
    a. Samples are to show interior door construction including core, blocking and edge treatment.
2. Stain Color:
    a. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
    b. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

C. Informational Submittals:
1. Certification of AWI membership.
2. Door Warranty.
3. Louver Warranty: Manufacturer’s standard (5) five-year warranty.

1.4 COORDINATION

A. Coordinate louver size and placement with door hardware, vision panels, and structural requirements.

1.5 DELIVERY STORAGE AND HANDLING

A. Store and protect doors in accordance with manufacturer's recommendations and WDMA.

B. Protect units from moisture damage according to AWI Quality Standards, Section 1700, Installation.

C. Protect doors with resilient packaging (sealed with heat shrunk plastic). Do not store in damp or wet areas; or in areas where sunlight might bleach veneer. Seal top and bottom edges upon receipt. Break seals on - site to permit ventilation.

D. Louvers: Deliver louvers in manufacturer’s original packaging.
1. Store materials in a dry, protected area.
2. Protect louvers and doors from damage during handling and installation.

1.6 WARRANTY

A. Warrant doors in writing for life of installation against defects including:
1. Veneer delamination.
2. Bow or twist of 1/4-inches or more.
3. Telegraphing of any part of core through face veneer.
4. Surface variation exceeding 1/100 inches in 3-inches span.
5. Any other defect which may impair or affect the performance of door for purpose for which it is intended.

B. Warranty to include:
1. Removal and replacement of defective door(s).
2. Removal of existing hardware and refitting to new door.

C. Metal Louvers shall be warranted to be free of defects in material and workmanship for a period of (5) five-years from manufacturer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.

2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Flush Wood Doors:
      a. Basis of Design: VT Industries, Inc.
      b. Or Equal.
   2. Commercial Metal Louvers:
      a. Air Louvers a brand of Activar Construction Products Group, Inc.
         1) Model: 900A1.
      b. Or Equal
   3. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Face Veneer:
   1. All interior wood doors are to receive a transparent finish. Provide manufacturer's standard thickness face veneers of AWI premium grade with “AA” grade face veneers of Plain Sawn, Select White Maple on both sides of doors. Book - match (match for color and grain) at veneer joints, unless otherwise indicated. Provide exposed edges and other exposed solid wood components of the same species as or closely matching face veneers.

B. Thickness:
   1. 1-3/4-inch.

C. Core:
   1. Non-fire-rated doors:
      a. Solid particle core (PC-5), ANSI A208.1.
      b. Grade: LD-2.
      c. Density: 30-35 pounds/CUFT.
      d. Provide solid wood reinforcing at all hardware locations and around all cut-outs.

D. Hardwood Edges:
   1. Match face veneer species.

E. Adhesives:
   1. Type 1 water resistant glue.

F. Glass Vision Panels:
   1. Factory cut.
   2. Size as indicated on Drawings.
   3. Glass Vision Frames shall be factory cut and shipped with door.
   4. Fixed integral wood stops on exterior face of door.
   5. Stops secured with countersunk Phillips head screws on interior face.
   6. Miter all corners of stops and metal moldings.
   7. Wood stops to match door wood species.
   8. See Section 08 81 00 for glass.

G. Metal Louvers:
   1. Material: Cold rolled steel (sizes of louvers as scheduled on drawings).
   2. Gauge: 18 gauge steel.
   3. Blade Type: Fixed - Standard Z-Blade.
   4. Blade Spacing: 1 inch
   5. Frame: Self-attaching, vandal-proof design with flanged frame welded to corridor side and separate removable flanged frame on non-corridor side. Frame shall be mitered and welded corners.
   6. Free Area: Minimum 43 percent minimum.
   7. Fasteners:
      a. Manufacturer’s standard, tamper resistant.

b. Compatible with door core and louver material.

2.3 FABRICATION

A. General:
1. All doors shall be fabricated in accordance with and shall meet the requirements of WDMA I.S. 1A Premium Grade standards.
2. Hardware preparation:
a. Factory machine doors for application of hardware specified.
b. Provide lock blocks at lock edge and top of door for closer for hardware reinforcement.
c. Top and Bottom Rails: Factory sealed.
d. Bevel vertical edges 1/8-inches in 2-inches.
e. Clearance at bottom of door: 1/2-inches.
f. Clearance at top of door: 1/8-inches.
g. Tolerances:
1) Width: +1/32-inches.
2) Height: +1/16-inches.
3) Thickness: +1/16-inches.
4) Hardware location: +1/32-inches.
5) Locks and hinges: +1/32-inches.
h. Refer to Specification Section 08 70 00 for hardware requirements and template provided by hardware manufacturer.
3. Cut-outs.
a. Make all cut-outs in the factory.
b. Seal edges of all openings.

B. "Hot Press" bond hardwood veneer cross-banding to core using Type II glue.

C. Identify doors for proper location.

D. Metal Louvers:
1. Louvers shall be factory fabricated with welded corners.
2. Provide integral flanges sized for the door thickness.
3. Provide sleeves or channels to protect door core during installation.

2.4 FINISH

A. Doors shall be Field Stained.
1. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

B. Metal Louvers: Factory applied powder coat.
1. Color: Flat Black.

2.5 SOURCE QUALITY CONTROL

A. Inspect finished door units at factory and repair damage in accordance with AWI.

PART 3 - EXECUTION

3.1 PREPARATION

A. Installer must examine door frames and verify that frames are the correct type and have been installed as required for proper hanging of corresponding doors and notify the Contractor in writing of conditions detrimental to the proper and timely installation of wood doors. Do not proceed with installation until unsatisfactory conditions have been corrected in a manner acceptable to the installer. Installation assumes responsibility for performance.

B. Verify door type, core, and thickness are compatible with louver installation.

3.2 INSTALLATION

A. Condition doors to average prevailing humidity in installation area prior to hanging.

B. Hardware: For installation found in Section 08 70 00 Finish Hardware.

C. Manufacturer's Instructions: Install wood doors in accordance with manufacturer's instructions, and per referenced Architectural Woodworking Institute - AWI Premium Quality Standard and as indicated.

D. Set and secure materials and components in place, plumb and level.

E. Trim non - rated door width by cutting equally on both edges.

F. Pilot drill screw and bolt holes.

G. Machine cut for hardware, Core for handsets and cylinders.

H. Coordinate installation of doors, frames and accessories.

I. Pre-fit Doors: Fit to frames and machine for hardware to whatever extent not previously worked at factory as required for proper fit and uniform clearance at each edge.

J. Clearance: For non - fire doors provide clearances of 1/8-inch at jambs and heads; and 1/2-inch from bottom of door to top of decorative floor finish or covering.

K. Restore finish on edges of shop - finished doors before installation, if fitting or machining is required at the job site.

L. Install louvers per manufacturer’s instructions and approved shop drawings.
   1. Cut door openings cleanly and accurately; seal exposed core material.
   2. Install louvers to be level, straight and plumb with door panel.
   3. Secure louver frames tightly on both sides of the door.
   4. Ensure blades are oriented correctly (downward toward secure side of door).
   5. Maintain clearances and tolerances required for door operation, hardware and door kickplates.

3.3 ADJUST AND CLEAN

A. Conform to Architectural Woodworking Institute - AWI Standards for fit and clearance tolerance.

B. Conform to Architectural Woodworking Institute - AWI Standards and Tests for warp and telegraphing.

C. Finished Doors: Refinish or replace doors damaged during installation.

D. Protection and Completed Work: Advise Contractor of proper procedures required for protection of installed wood doors from damage or deterioration until acceptance of the work.

E. Inspect installation for alignment, secure fastening, and proper operation of metal louvers.

F. Clean exposed surfaces after installation of metal louvers.

G. Protect louvers from damage until project completion.

3.4 TESTING FOR ALL DOORS

A. When building environmental systems are operating as specified, conduct simple tests to ensure each completed door system will close and latch or otherwise secure properly, without assistance. Closer tests (for doors so equipped):
   1. Full open test: Open door to full width. Release door. Door to completely close and latch fully. If door is equipped with magnetic locks, magnetic lock armature to completely bond to lock with bonding sensor indicating secure.
   2. Mid - point close to latch test: Open door halfway. Release door. Door to completely close and latch fully. If door is equipped with magnetic locks, magnetic lock armature to completely bond to lock with bonding sensor indicating secure.
   3. Close to latch test: Open door to full width, then permit door to re - close until lock’s latch bolt touches strike lip and hold. When magnetic locks are used, permit door to return so that

strike armature is within one (1) inch of the magnetic lock. Release door. Door to completely close and latch fully. If door is equipped with magnetic locks, magnetic lock armature to completely bond to lock with bonding sensor indicating secure.

B. Latch Bolt Release Test: Measure force to release latch bolt using a door force gage located 1 inch from end of lever handle or located at center of exit device push pad.
   1. For interior fire doors and exterior doors, latch shall release when the lever is subjected to a force not more than 15 pounds.

C. Latch Bolt Security Test: With door closed and properly latched, pull sharply on door, in direction of swing. Door to remain secure.

D. Latch Bolt Dead Latch Test:
   1. With door closed and latched, insert dead latch test tool and engage latch bolt. Pull sharply. Latch bolt should not retract.
   2. With door open, and latch bolt and dead latch in view, hold dead latch in with your finger and rapidly and intermittently push (rattle) latch bolt. Latch bolt should not retract.

E. Lock Function Test: With proper key to operate lock cylinder, perform all functions lockset is capable of, and insure they operate properly.

F. Closer Tests: Preparation 1 - Adjust spring speed from 1.5 to 2 seconds when closing from 70 degrees to closed position. Preparation 2 - Adjust closer speed from 5 to 6 seconds when closing from 90 degrees to 12 degrees. Preparation 3 - Adjust closer power to the lowest value within the following ranges that consistently passes the following tests, measuring with a door force gage located 1 inch from latch edge of door.
   1. Interior non-fire doors- Force required to push door to open position: not more than 5 pounds

G. Operation: Rehang or replace doors which do not swing or operate freely.

END OF SECTION

SECTION 08 31 00
ACCESS DOORS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Aluminum - Heavy Duty Floor Access Doors.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Specification Section 01 78 23 – Operation and Maintenance Data.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Association of State Highway and Transportation Officials (AASHTO).
   2. ASTM International (ASTM):
      a. A123/A123M, Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.
      b. A153/A153M, Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
      c. A240/A240M, Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications.
      d. A653/A653M, Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
      e. A666, Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar.
   3. Occupational Safety and Health Administration (OSHA):
      a. OSHA 29 CFR 1910, Laws and Regulations.

1.3 DEFINITIONS

A. Clear Opening Size: Space within the opening having no obstructions. Furnish model that will provide the minimum clear opening indicated.

B. Heavy Duty: Will support live load of 300 pounds per square foot.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.5 WARRANTY

A. Warranty:
   1. 25-year manufacturer’s warranty.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Aluminum – Heavy Duty Vault Doors:
      a. Basis of Design: The BILCO Company - J-AL Series
      b. Or Equal.
   2. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Aluminum:
   1. Sheet and Plate: ASTM B209.
   2. Extruded shapes: ASTM B221.

B. Steel:
   1. Sheet Metal: G90 Galvanized, ASTM A653/A653M.
   2. Fabrications: Hot-Dip Galvanized, ASTM A123/A123M.
   3. Hardware: Hot-Dip Galvanized, ASTM A153/A153M.

C. Stainless Steel: ASTM A240/A240M or A666.

2.3 MANUFACTURED UNITS

A. General:
   1. All access vault doors shall be provided by the same manufacturer when possible.
   2. Coat all aluminum components in contact with concrete or masonry with manufacturer's standard bituminous coating.

B. Aluminum - Heavy Duty Floor Access Doors:
   1. Frame:
      a. 1/4-inches mill finish aluminum channel with anchor tabs.
      b. 1-1/2-inches diameter drain coupling. Drains are not required to be connected.
   2. Cover:
      a. 1/4-inches mill finished diamond plate aluminum.
      b. Reinforce cover with aluminum stiffeners.
         1) Live load: 300 pounds per square foot.
         2) Deflection: Maximum 1/150 of span.
   3. Hardware:
      a. All hardware to be stainless steel.
      b. Positive hold open arm that engages automatically when door reaches full 90 degrees open position.
      c. Slam lock and removable key handle.
   4. Size(s):
      a. Refer to the "Vault Door Schedule" on "Architectural Door Schedules and Details" Drawing.
   5. Hardware:

a. Hinges: All hardware to be Type 316 stainless steel with 1/4-inch type 316 stainless steel pins.
b. Positive hold open arm that engages automatically when door reaches full 90 degree open position.
c. Latch: Type 316 stainless steel slam lock with fixed interior handle and removable exterior turn/lift handle. Latch release is protected by a flush, gasketed, removable screw plug.
d. Lift Assistance: Compression spring operators enclosed in telescopic tubes. Automatic hold-open arm with release handle automatically locks cover(s) in the open position.

6. Size(s):
   a. Refer to the "Vault Door Schedule" on "Architectural Door Schedules and Details" Drawing.

2.4 ACCESSORIES

A. Secondary Fall Protection System:
   1. Design and install system such that when in the open position, no part of the system obstructs the clear opening size listed in the "Vault Door Schedule" on "Architectural Door Schedules and Details" Drawing.
   2. Platform: Design grating to meet OSHA 29 CFR 1910 requirements for protection for floor openings.
   3. Finish:
      a. Powder coated.
      b. Color: Standard Safety Orange or Safety Yellow.
   4. Hardware:
      a. Stainless steel Type 316.
      b. Tamper proof Type 316 stainless steel bolts.
   5. Provide positive latch to hold grating in upright position.
   6. Size: Allow 6-inches clear space on each unhinged side for visual observation.
   7. Provide padlock hasp for Government provided padlock.
   8. Install secondary fall protection system at the factory.

B. Load Rating Plates:
   1. Minimum 18 gauge, Type 316 stainless steel, ASTM A666.
   2. Engraved with maximum design live load allowed for unit on which it will be mounted.
   3. Display load in English units.
   4. Size as required for text as needed.
   5. Text:
      a. Font: Helvetica Narrow, all caps.
      b. Size: 1/4-inches height.
      c. Depth of engraving: 3 mils.
   6. Finish:
      a. Text:
         1) Black epoxy baked on paint.
         2) Plate to have finish conductive to paint application.
      b. Coat entire plate with baked on clear coat on front and back side.
   7. Attach to top of all vault access doors using stainless steel screws in location determined by manufacturer.
      a. Provide a neoprene gasket under the plate to separate the stainless steel from the aluminum cover or frame.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Vault Doors do not need to be drained or extended to a drain.

3.2 SCHEDULES

A. Vault Access Door Schedule:
   1. Refer to the “Vault Door Schedule” on “Architectural Door Schedules and Details” Drawing.

END OF SECTION

SECTION 08 33 22
ALUMINUM ROLLING OVERHEAD DOORS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Aluminum rolling overhead doors.
   2. Motor operators.
   3. Control Systems.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 01 78 23 – Operation and Maintenance Data.
   3. Section 08 70 00 - Finish Hardware.
   4. Section 09 06 00 – Color Schedule.
   5. Section 09 96 00 – High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Aluminum Association (AA):
      a. DAF 45, Designation System for Aluminum Finishes.
   2. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE):
      a. 90.1, Energy Standard for Buildings Except Low-Rise Residential Buildings.
   3. ASTM International (ASTM):
      a. B209, Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate.
      b. B221, Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes.
      c. E90, Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements.
      d. E330, Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference.
      e. F467, Standard Specification for Nonferrous Nuts for General Use.
      f. F468, Standard Specification for Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use.
   4. Door and Access Systems Manufacturers Association (DASMA):
      a. 105, Test Method for Thermal Transmittance and Air Infiltration of Garage Doors.
   5. International Code Council (ICC):
      a. International Energy Conservation Code (IECC).
   6. Underwriter's Laboratories (UL):
      a. 325, Standard for Door, Drapery, Gate, Louver, and Window Operators and Systems.

B. Qualifications:
   1. Installer to be licensed or approved in writing by door manufacturer.

1.3 DEFINITIONS

A. Installer or Applicator:
   1. Installer or applicator is the person actually installing or applying the product in the field at the Project site.
   2. Installer and applicator are synonymous.

1.4 SUBMITTALS

A. Product Data:
   1. Product technical data including:

a. Acknowledgement that products submitted meet requirements of standards referenced.
b. Manufacturer's installation instructions.

2. Motor operator and accessories technical data including complete wiring and control diagram for motor operators and control stations.

B. Shop Drawings:
1. Provide drawings indicating guide details, head and jamb conditions, anchorage, accessories, finish colors, patterns and textures, operator mounts and other related information.
2. Schedule of doors using the same reference number for openings as indicated on Drawings.

C. Certifications:
1. Certification of Installer's qualifications.

D. Samples:
1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
3. Refer to “Specification Section 09 96 00 – High Performance Industrial Coatings” for specified material.

E. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Specification Section 01 78 23 for requirements for the content of Operation and Maintenance Manual submittals.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Utilize an authorized installer of door manufacturer who has demonstrated experience on projects of similar size and complexity.

B. Manufacturer Qualifications: Company with a minimum of five-years’ experience in producing the specified type of doors.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Store products in manufacturer's unopened packaging until ready for installation.

B. Store and dispose of solvent-based materials, and materials used with solvent-based materials, in accordance with requirements of local authorities having jurisdiction.

1.7 PROJECT CONDITIONS

A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.

1.8 WARRANTY

A. Manufacturer shall warrant the doors and its component parts for one (1) year against defects in material and workmanship.

B. Design doors of construction for high cycle use of up to 300,000 cycles for the full life of the product.

C. Manufacturer shall warrant the electrical operator and its component parts for two (2) years against defects in material and workmanship.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Aluminum Rolling Overhead Doors:
      a. Basis of Design: Raynor Garage Doors.
      b. Or Equal.
   2. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 PERFORMANCE AND DESIGN REQUIREMENTS

A. Design Wind Load: ASTM E330/E330M.
   1. Exterior doors: See Structural Drawings.
   2. Interior doors: 20 pounds per square feet minimum.

B. Air Infiltration:
   1. Meet ASHRAE 90.1 and IECC C402.4.3.
   2. Air leakage: <1.00 cubic feet per minute/square feet.

C. Thermal performance:
   1. DASMA 105: R Factor: 6.41 Minimum.

D. Sound Rating:
   1. ASTM E90 Minimum assembly rating: STC-21 minimum.

2.3 MATERIALS

A. Aluminum:
   1. Shapes, plate: ASTM B221.
   2. Sheet: ASTM B209, alloy 5005, temper H34.
   3. Bolts and nuts: ASTM F467 and F468.

B. Insulation:
   1. Polyisocyanurate.

C. Weatherproofing:
   1. Resilient: Neoprene or vinyl.
   2. Brush type: Nylon or polypropylene.

D. Miscellaneous Fasteners: Aluminum or stainless steel.

2.4 MANUFACTURED UNITS

A. Door Curtain Slats:
   1. Curtain: Interlocking roll-formed slats as specified below. Endlocks shall be attached to each alternate slat to prevent lateral movement.
   2. Insulated flat profile:
      a. 2-9/16-inches high by 13/16-inch thick.
      b. Interlocking face sheets:
         1) Aluminum 16 gauge (0.050-inch minimum thickness on both faces).

c. Galvanized endlocks.
d. Core: Insulated with polyisocyanurate insulation.

3. Bottom bar:
   a. Two structural angles bolted back-to-back.
      1) Minimum 1-1/2-inches by 1-1/2-inches by 1/8-inches thick with single-contact type bottom astragal.
   b. Aluminum (clear anodize finish) or stainless-steel (mill finish) angles.

B. Operation:
1. As scheduled on Door Schedule.

C. Guides:
1. Manufacturer's standard structural angle guide system for size of door specified.
   a. Guides shall be Structural Stainless-Steel with mill finish.
   b. Provide curtain wind locks as necessary for design wind load.
      1) Material:
         a) Exterior doors:
            (1) Doors requiring wind locks: Stainless steel.
   c. Guide Weatherseal: Seals to inhibit air infiltration between the guide and the curtain.
      1) Brush seal with an aluminum retainer.

D. Brackets:
1. 3/16-inch, minimum, steel plate, attached to wall angle of guide assembly with 1/2-inch diameter Grade 5 bolts. Inside of drive bracket fitted with sealed ball bearing.
2. Finish:
   a. Finish and Color to match curtain door slats.

E. Counterbalance Assembly:
1. Barrel: Minimum 4-1/2-inches O.D. and 0.120-inch wall thickness structural steel pipe. Deflection of pipe under full load shall not exceed 0.03-inch per foot of span.
2. Counterbalance: Provide counterbalance mechanism with helical torsion springs, grease packed and mounted on a continuous steel torsion shaft.
   a. Standard 15,000 cycles.

F. Hood:
1. Type: Square Hood.
2. Bracket Covers: Covers to enclose door mechanisms.
3. Material: 18-gauge aluminum (0.040-inch minimum thickness) in accordance with ASTM B209, alloy 5005, temper H34.
4. Finish and Color to match curtain door slats.

G. Hood Baffle:
1. Provide hood baffle with a rubber seal to inhibit air infiltration through hood cavity.

H. Header Seal:
1. Provide a “Z” shape aluminum retainer with EDPM rubber to inhibit air infiltration between the header and the curtain.

I. Finish:
1. Curtain Door Slats:
   a. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
   b. Refer to “Specification Section 09 96 00 – High Performance Industrial Coatings” for specified material.
2. Trim Pieces:
   a. Material and finish to match door curtain slats.

J. Locking:
1. Motor operated doors: Integral motor brake.
2. Manually Hand Chain Lock operated doors: To receive padlock provided by Government; for doors operated with hand chain..

2.5 ACCESSORIES

A. Motor / Hand Chain Hoist Operator: (Provide at Doors 62-109B and 62-112B).
   1. Control Hoist Standard:
      a. Type: Jackshaft with manual chain hoist.
      b. Motor Horsepower Rating: Per the manufacturer's recommended size for door.
      c. Electrical Requirements: 208-230 volt three phase.
      d. Duty Cycle: 30 cycles/hour or 300 cycles/day.
      e. Control Wiring: Solid state circuitry with provisions for connection of a monitored reversing device, external radio control hook-up and maximum run timer. Provisions for timers to close, monitored reversing devices, mid stop and lock bar sensor capability.
      f. Provide three button momentary contact "open-close-stop" at each motor operated door.
   2. Entrapment protection:
      a. Wired Monitored electric reversing edge extending full width of door.
         1) Comply with UL 325.
            a) Instant reversing.
            b) When activated it shall prevent the operator from closing an open door.

2.6 MAINTENANCE MATERIALS

A. Provide 2 ounces of touch-up paint properly labeled for each different color of door.
   1. Touch-up paint shall be formulated to be compatible with the finished specified and shall be capable of being applied without special equipment or tools.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Installation shall be done by the manufacturer's authorized representative.

C. Provide all required trim, weatherstripping, closures etc., for complete weather tight installation.

D. Adjust for proper counterbalance.

E. Seal along bottom of vertical track (guides), seal the vertical joint between the two separate track angles (if not filled by welding) and seal all holes in vertical track (not being used for fasteners) to provide a completely weather tight track and door system.
   1. At fastener locations provide aluminum washers under bolt head to completely cover the slotted holes in the vertical guide.

F. Electrical disconnect and conduit and wiring from standard three pushbutton control to motor operator is provided as indicated on the Electrical Drawings.

G. Provide aluminum or stainless-steel bracing for motor operator to eliminate vibration.

H. Provide dissimilar materials protection on all surfaces coming in contact with dissimilar materials.

3.2 ADJUSTING

A. Following completion of installation, including related work by others, lubricate, test, and adjust doors for ease of operation, free from warp, twist, or distortion.

3.3 FIELD QUALITY CONTROL

A. Site Test: Test doors and perforated security insect screens for normal operation and automatic closing. Coordinate with authorities having jurisdiction to witness test and sign Drop Test Form.

3.4 CLEANING

A. Clean surfaces soiled by work as recommended by manufacturer.

B. Remove surplus materials and debris from the site.

3.5 DEMONSTRATION

A. Demonstrate proper operation, testing and reset procedures to Contracting Officer Representative. Instruct Contracting Officer Representative in maintenance procedures.

END OF SECTION

SECTION 08 70 00
FINISH HARDWARE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Finish hardware.
      a. Mechanical door hardware.
      b. Electromechanical door hardware.
      c. Cylinders specified for doors in other sections.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 - Submittal Procedures.
   2. Section 01 77 00 - Closeout Procedures.
   3. Section 01 78 23 - Operation and Maintenance Data.
   4. Section 05 50 00 - Metal Fabrications.
   5. Section 07 92 00 - Joint Sealants.
   6. Section 08 11 00 - Hollow Metal Doors and Frames.
   7. Section 08 11 16 - Aluminum Doors and Frames.
   8. Section 08 14 16 - Flush Wood Doors.
   9. Section 08 33 22 - Aluminum Rolling Doors Overhead Doors.

1.2 QUALITY ASSURANCE

A. All door hardware shall be provided by a single hardware supplier.

B. Manufacturers Qualifications:
   1. Engage qualified manufacturers with a minimum of 5-years of documented experience in producing hardware and equipment similar to that indicated for this Project and that have a proven record of successful in-service performance.
   2. Certified Products: Where specified, products must maintain a current listing in the Builders Hardware Manufacturers Association (BHMA) Certified Products Directory (CPD).

C. Installer Qualifications:
   1. A minimum of 3-years' documented experience installing both standard and electrified door hardware similar in material, design, and extent to that indicated for this Project and whose work has resulted in construction with a record of successful in-service performance.

D. Door Hardware Supplier Qualifications:
   1. Experienced commercial door hardware distributors with a minimum of 5-years' documented experience supplying both mechanical and electromechanical hardware installations comparable in material, design, and extent to that indicated for this Project.
   2. Supplier recognized as a factory direct distributor by the manufacturers of the primary materials with a warehousing facility in Project's vicinity. Supplier to have on staff a certified Architectural Hardware Consultant (AHC) available during the course of the Work to consult with Contractor and Contracting Officer Representative concerning both standard and electromechanical door hardware and keying.

E. Source Limitations: Obtain each type and variety of door hardware specified in this section from a single source unless otherwise indicated.
   1. Electrified modifications or enhancements made to a source manufacturer's product line by a secondary or third-party source will not be accepted.
   2. Provide electromechanical door hardware from the same manufacturer as mechanical door hardware, unless otherwise indicated.
   3. Each unit to bear third party permanent label indicating compliance with the referenced testing standards.

F. Keying Conference: Contractor shall conduct a keying conference to incorporate the following criteria into the final keying schedule document:
   1. Function of building, purpose of each area and degree of security required.
   2. Plans for existing and future key system expansion.
   3. Requirements for key control storage and software.
   4. Installation of permanent keys, cylinder cores and software.
   5. Address and requirements for delivery of keys.

1.3 REFERENCED STANDARDS:

A. Americans with Disabilities Act (ADA):
   1. Accessibility Guidelines for Buildings and Facilities (ADAAG).

B. American National Standards Institute/Builders Hardware Manufacturers Association (ANSI/BHMA):
   1. A156.1, Butts and Hinges.
   2. A156.2, Bored and Preassembled Locks and Latches.
   3. A156.3, Exit Devices.
   4. A156.4, Door Controls -Closers.
   5. A156.5, Cylinders and input devices for locks.
   6. A156.6, Architectural Door Trim.
   7. A156.13, Mortise Locks.
   8. A156.16, Auxiliary Hardware.
   9. A156.18, Materials and Finishes.
   10. A156.26, Continuous Hinges.
   11. A156.36, Auxiliary Hardware.

C. American National Standards Institute/Steel Door Institute (ANSI).
   1. ANSI A117.1, Accessible and Usable Buildings and Facilities.
   2. ANSI/SDI A250.8, Specifications for Standard Steel Doors and Frames (SDI-100).
   3. ANSI UL10C, Positive Pressure Fire Tests of Door Assemblies.
   4. ANSI/UL 294, Access Control System Units.
   5. ANSI/UL 305, Panic Hardware.
   6. ANSI/UL 437, Key Locks.
   7. ANSI/UL 1784, Standard for Air Leakage Tests of Door Assemblies and Other Opening Protectives.

D. Door and Hardware Institute (DHI):
   1. TDH-007-20, Installation Guide for Doors and Hardware.

E. International Building Code (ICC/IBC):

F. National Fire Protection Association (NFPA):
   1. NFPA 80, Fire Doors and Windows.
   2. NFPA 101, Life Safety Code.
   3. NFPA 105, Installation of Smoke Door Assemblies.
   4. NFPA 252, Standard Methods of Fire Tests of Door Assemblies.

G. Qualifications:
   1. Installation shall be inspected by a certified Architectural Hardware Consultant (AHC).

1.4 DEFINITIONS

A. AHC: Architectural Hardware Consultant, certified by DHI.

B. Installer or Applicator:
   1. Installer or applicator is the person actually installing or applying the product in the field at the Project site.
   2. Installer and applicator are synonymous.

C. All weather: Capable of operation from -50 to +120 degrees F.

D. Active Leaf: Right-hand leaf when facing door from keyed side unless noted otherwise on Drawings.

1.5 SUBMITTALS

A. Product Data:
   1. Manufacturer's product data sheets including installation details, material descriptions, dimensions of individual components and profiles, operational descriptions and finishes.

B. Door Hardware Schedule:
   1. Prepared by or under the supervision of supplier, detailing, fabrication and assembly of door hardware, as well as procedures and diagrams.
   2. Coordinate the final Door Hardware Schedule with doors, frames, and related work to ensure proper size, thickness, hand, function, and finish of door hardware.
   3. Format:
      a. Comply with scheduling sequence and vertical format in DHI's "Sequence and Format for the Hardware Schedule."
   4. Organization:
      a. Organize the Door Hardware Schedule into door hardware sets indicating complete designations of every item required for each door or opening.
      b. Organize door hardware sets in same order as in the Door Hardware Sets at the end of this specification section.
   5. Submittals that do not follow the same format and order as the Door Hardware Sets will be rejected and subject to resubmission.
   6. Content: Include the following information:
      a. Type, style, function, size, label, hand, and finish of each door hardware item.
      b. Manufacturer of each item.
      c. Fastenings and other pertinent information.
      d. Location of door hardware set, cross-referenced to Drawings, both on floor plans and in door and frame schedule.
      e. Explanation of abbreviations, symbols, and codes contained in schedule.
      f. Mounting locations for door hardware.
      g. Door and frame sizes and materials.
      h. Warranty information for each product.
   7. Submittal Sequence:
      a. Submit the final Door Hardware Schedule at earliest possible date, particularly where approval of the Door Hardware Schedule must precede fabrication of other work that is critical in the Project construction schedule.
      b. Include Product Data, Samples, Shop Drawings of other work affected by door hardware, and other information essential to the coordinated review of the Door Hardware Schedule.

C. Shop Drawings:
   1. Details of electrified access control hardware indicating the following:
      a. Wiring Diagrams:
         1) Upon receipt of approved schedules, submit detailed system wiring diagrams for power, signaling, monitoring, communication, and control of the access control system electrified hardware.
         2) Differentiate between manufacturer-installed and field-installed wiring. Include the following:
         3) Elevation diagram of each unique access-controlled opening showing location and interconnection of major system components with respect to their placement in the respective door openings.
         4) Complete (risers, point-to-point) access control system block wiring diagrams.
         5) Wiring instructions for each electronic component scheduled herein.
      b. Electrical Coordination:

1) Coordinate with related sections the voltages and wiring details required at electrically controlled and operated hardware openings.

D. Keying Schedule:
   1. After a keying meeting with the Contracting Officer Representative has taken place prepare a separate keying schedule detailing final instructions.
   2. Submit the keying schedule in electronic format. Include keying system explanation, door numbers, key set symbols, hardware set numbers and special instructions. Contracting Officer Representative must approve submitted keying schedule prior to ordering permanent cylinders/cores.

E. Informational Submittals:
   1. Product Test Reports: Indicating compliance with cycle testing requirements, based on evaluation of comprehensive tests performed by manufacturer and witnessed by a qualified independent testing agency.
   2. Warranty information for all hardware devices having extended warranties.

F. Certifications:
   1. Certification from AHC stating all door hardware has been provided per approved Shop Drawings, has been installed in accordance with manufacturer's recommended installation instructions and all doors have been inspected and tested and found to be in proper working order.
   2. Door assemblies required to swing in the direction of egress have been inspected and tested in accordance with NFPA 101.
   3. All door hardware has been reviewed by AHC and verified to be compatible with doors and frames.
   4. All electrified door hardware has been reviewed by AHC and has been coordinated with power supply and access control system.

G. No submittals will be reviewed until Contracting Officer Representative has received AHC certification

H. Closeout Submittals:
   1. Operating and Maintenance Manuals:
      a. Provide manufacturers operating and maintenance manuals for each item comprising the complete door hardware installation in quantity as required in 01 78 23 Operation and Maintenance Data.
   2. Project Record Documents:
      a. Provide record documentation of record documents for door hardware sets in digital format (.pdf, .docx, .xlsx, .csv) and as required in Section 01 77 00 Closeout Procedures.

1.6 DELIVERY, STORAGE AND HANDLING

A. Inventory door hardware on receipt and provide secure lock-up and shelving for door hardware delivered to Project site. Do not store electronic access control hardware, software or accessories at Project site without prior authorization.

B. Tag each item or package separately with identification related to the final Door Hardware Schedule and include basic installation instructions with each item or package.

C. Deliver, as applicable, permanent keys, cylinders, cores, access control credentials, software and related accessories directly to Government via registered mail or overnight package service. Instructions for delivery to the Government shall be established at the "Keying Conference".

1.7 COORDINATION

A. Templates: Obtain and distribute to the party’s involved templates for doors, frames, and other work specified to be factory prepared for installing standard and electrified hardware. Check Shop Drawings of other work to confirm that adequate provisions are made for locating and installing hardware to comply with indicated requirements.

B. Door and Frame Preparation: Doors and corresponding frames are to be prepared, reinforced and pre-wired (if applicable) to receive the installation of the specified electrified, monitoring, signaling and access control system hardware without additional in-field modifications.

1.8 WARRANTY

A. Warranty Period:
   1. Written warranty, executed by manufacturer(s), agreeing to repair or replace components of standard and electrified door hardware that fails in materials or workmanship within specified warranty period after final acceptance by the Government. Failures include, but are not limited to, the following:
      a. Structural failures including excessive deflection, cracking, or breakage.
      b. Faulty operation of the hardware.
      c. Deterioration of metals, metal finishes, and other materials beyond normal weathering.
      d. Electrical component defects and failures within the systems operation.

B. Warranty Period: Unless otherwise indicated, warranty shall be one year from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Butt Hinges:
      a. Basis-of-Design: McKinney Manufacturing Co. - TA/T4A Series, 5-knuckle.
      b. Or Equal.
   2. Continuous Hinges:
      a. Basis-of-Design: Pemko – DFM_HD1 PT.
      b. Or Equal.
   3. Power Transfer:
      a. Basis-of-Design: Securitron – EL-CEPT Series.
      b. Or Equal.
   4. Electric Door Wire Harnesses:
      a. Basis-of-Design: McKinney – QC-C Series
         1) McKinney – Electrical Connecting Kit – QC-R001.
         2) McKinney – Connector Hand Tool – QC-R003.
      b. Or Equal.
   5. Flush Bolts:
      a. Basis-of-Design: Rockwood - 2845
      b. Or Equal.
   6. Coordinators:
      a. Basis-of-Design: Rockwood - 2600
      b. Or Equal.
   7. Cylinders and Keying:
      a. Basis-of-Design: dormakaba BEST – to Match Existing Key System.
   8. Mortise Locksets and Latchsets:

a. Basis-of-Design: Sargent Manufacturing - 8200 Series.
b. Or Equal.

9. Exit Devices:
   a. Basis-of-Design: Sargent Manufacturing - 80 Series.
   b. Or Equal.
10. Surface Door Closers:
   a. Basis-of-Design: Norton Rixson – 2800ST Series.
   b. Or Equal.
11. Door Protective Trim, Kick Plate:
   a. Basis-of-Design: Rockwood – K1050.
   b. Or Equal.
12. Door stops and holders:
   a. Basis-of-Design: Rockwood – 466-RKW.
   b. Or Equal.
13. Weatherstripping, Rain Guards, Sweeps and Thresholds:
   a. Basis-of-Design: Pemko.
   b. Or Equal.
14. Silencers:
   a. Basis-of-Design: Rockwood – 608.
   b. Or Equal.
15. Key Control:
   a. Basis-of-Design: Lund Equipment.
   b. Or Equal.
16. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Hardware shall not have any visible manufacturer names on exposed materials, except cylinders, when the door is in a closed position.

B. BUTT HINGES
   1. Hinges: ANSI/BHMA A156.1 butt hinges with number of hinge knuckles and other options as specified in the Door Hardware Sets.
   2. Quantity: Provide the following hinge quantity:
      a. Two Hinges: For doors with heights up to 60-inches.
      b. Three Hinges: For doors with heights 61 to 90-inches.
      c. Four Hinges: For doors with heights 91 to 120-inches.
      d. For doors with heights more than 120-inches, provide 4 hinges, plus 1 hinge for every 30-inches of door height greater than 120-inches.
   3. Hinge Size: Provide the following, unless otherwise indicated, with hinge widths sized for door thickness and clearances required:
      a. Widths up to 3-feet-0-inches: 4-1/2-inches standard or heavy weight as specified.
      b. Sizes from 3-feet-1-inch to 4-feet-0-inches: 5-inches standard or heavy weight as specified.
   4. Hinge Weight and Base Material: Unless otherwise indicated, provide the following:
      a. Exterior Doors: Heavy weight, non-ferrous, ball bearing or oil impregnated bearing hinges unless Hardware Sets indicate standard weight.
      b. Interior Doors: Standard weight, steel, ball bearing or oil impregnated bearing hinges unless Hardware Sets indicate heavy weight.
   5. Hinge Options: Comply with the following:
      a. Non-removable Pins: With the exception of electric through wire hinges, provide set screw in hinge barrel that, when tightened into a groove in hinge pin, prevents removal of pin while door is closed; for all out-swinging lockable doors.

C. CONTINUOUS HINGES
   1. Continuous Geared Hinges: ANSI/BHMA A156.26 Grade 1-600 continuous geared hinge. with minimum 0.120-inch thick extruded 6063-T6 aluminum alloy hinge leaves and a

minimum overall width of 4-inches. Hinges are non-handed, reversible and fabricated to template screw locations. Factory trim hinges to suit door height and prepare for electrical cut-outs.

D. POWER TRANSFER DEVICES
   1. Concealed Quick Connect Electric Power Transfers: Provide concealed wiring pathway housing mortised into the door and frame for low voltage electrified door hardware. Furnish with standardized plug connectors and sufficient number of concealed wires (up to 12) to accommodate the electrified functions specified in the Door Hardware Sets. Connectors plug directly to through-door wiring harnesses for connection to electric locking devices and power supplies. Wire nut connections are not acceptable.
   2. Electric Door Wire Harnesses: Provide electric/data transfer wiring harnesses with standardized plug connectors to accommodate up to twelve (12) wires. Connectors plug directly to through-door wiring harnesses for connection to electric locking devices and power supplies. Provide sufficient number and type of concealed wires to accommodate electric function of specified hardware. Provide a connector for through-door electronic locking devices and from hinge to junction box above the opening. Wire nut connections are not acceptable. Determine the length required for each electrified hardware component for the door type, size and construction, minimum of two per electrified opening.

E. DOOR OPERATING TRIM
   1. Flush Bolts and Surface Bolts: Provide products conforming to ANSI/BHMA A156.3 and A156.16, Grade 1.
      a. Flush bolts to be furnished with top rod of sufficient length to allow bolt retraction device location approximately six feet from the floor.
      b. Furnish dust proof strikes for bottom bolts.
      c. Surface bolts to be minimum 8-inches in length and U.L. listed for labeled fire doors and U.L. listed for windstorm components where applicable.
      d. Provide related accessories (mounting brackets, strikes, coordinators, etc.) as required for appropriate installation and operation.
   2. Coordinators: ANSI/BHMA A156.3 door coordinators consisting of active-leaf, hold-open lever and inactive-leaf release trigger. Model as indicated in hardware sets.
   3. General: Cylinder manufacturer to have minimum (10) years experience designing secured master key systems and have on record a published security keying system policy.
   4. Cylinder Types: Original manufacturer cylinders able to supply the following cylinder formats and types:
      a. Threaded mortise cylinders with rings and cams to suit hardware application.
      b. Rim cylinders with back plate, flat-type vertical or horizontal tailpiece, and raised trim ring.
      c. Bored or cylindrical lock cylinders with tailpieces as required to suit locks.
      d. Tubular deadlocks and other auxiliary locks.
      e. Mortise and rim cylinder collars to be solid and recessed to allow the cylinder face to be flush and be free spinning with matching finishes.
   5. Keyway: Match the Facility Standard which is a BEST, 7-Pin, “WH” Keyway.
   6. Small Format Interchangeable Cores: Provide small format interchangeable cores (SFIC) as specified, core insert, removable by use of a special key; usable with other manufacturers' cylinders. Cores shall match the Facility Standard which is a BEST, 7-Pin, Figure 8 (SFIC).
   7. Keying System: Each type of lock and cylinders to be factory keyed.
      a. Supplier shall conduct a "Keying Conference" to define and document keying system instructions and requirements.
      b. Furnish factory cut, nickel-silver large bow permanently inscribed with a visual key control number as directed by Contracting Officer Representative.
      c. Existing System: Field verify and key cylinders to match North Rim Facility’s existing system.
   8. Key Quantity: Provide the following minimum number of keys:
      a. Change Keys per Cylinder: Two (2)

b. Master Keys (per Master Key Level/Group): Five (5).
c. Construction Keys (where required): Ten (10).
d. Construction Control Keys (where required): Two (2).
e. Permanent Control Keys (where required): Two (2).

9. Construction Keying: Provide temporary keyed construction cores.
10. Key Registration List (Bitting List):
    a. Provide keying transcript list to Contracting Officer Representative in the proper format for importing into key control software.
    b. Provide transcript list in writing or electronic file as directed by the Contracting Officer Representative.

F. KEY CONTROL CABINET
1. Key Control Cabinet: Provide a key control system including envelopes, labels, and tags with self-locking key clips, receipt forms, 3-way visible card index, temporary markers, permanent markers, and standard metal cabinet. Key control cabinet shall have expansion capacity of 150 percent of the number of locks required for the project.

G. MORTISE LOCKS AND LATCHING DEVICES
1. Mortise Locksets, Grade 1 (Heavy Duty): Provide ANSI/BHMA A156.13, Series 1000, Operational Grade 1 Certified Products Directory (CPD) listed mortise locksets. Listed manufacturers shall meet all functions and features as specified herein.

H. LOCK AND LATCH STRIKES
1. Strikes: Provide manufacturer's standard strike with strike box for each latch or lock bolt, with curved lip extended to protect frame, finished to match door hardware set, unless otherwise indicated, and as follows:
2. Flat-Lip Strikes: For locks with three-piece antifriction latchbolts, as recommended by manufacturer.
3. Extra-Long-Lip Strikes: For locks used on frames with applied wood casing trim.
4. Aluminum-Frame Strike Box: Provide manufacturer's special strike box fabricated for aluminum framing.
5. Double-lipped strikes: For locks at double acting doors. Furnish with retractable stop for rescue hardware applications.
6. Standards: Comply with the following:
7. Strikes for Mortise Locks and Latches: BHMA A156.13.
8. Strikes for Bored Locks and Latches: BHMA A156.2.
9. Strikes for Auxiliary Deadlocks: BHMA A156.36.
10. Dustproof Strikes: BHMA A156.16.

I. CONVENTIONAL EXIT DEVICES
1. General Requirements: All exit devices specified herein shall meet or exceed the following criteria:
2. Exit devices shall have a five-year warranty.
3. At doors not requiring a fire rating, provide devices complying with NFPA 101 and listed and labeled for "Panic Hardware" according to UL305. Provide proper fasteners as required by manufacturer including sex nuts and bolts at openings specified in the Hardware Sets.
4. Where exit devices are required on fire rated doors, provide devices complying with NFPA 80 and with UL labeling indicating "Fire Exit Hardware". Provide devices with the proper fasteners for installation as tested and listed by UL. Consult manufacturer’s catalog and template book for specific requirements.
5. Except on fire rated doors, provide exit devices with hex key dogging device to hold the pushbar and latch in a retracted position. Provide optional keyed cylinder dogging on devices where specified in Hardware Sets.
6. Devices must fit flat against the door face with no gap that permits unauthorized dogging of the push bar. The addition of filler strips is required in any case where the door light extends behind the device as in a full glass configuration.

7. Lever Operating Trim: Where exit devices require lever trim, furnish manufacturer's heavy duty escutcheon trim with threaded studs for thru-bolts.
8. Lock Trim Design: As indicated in Hardware Sets, provide finishes and designs to match that of the specified locksets.
9. Where function of exit device requires a cylinder, provide a cylinder (Rim or Mortise) as specified in Hardware Sets.
10. Vertical Rod Exit Devices: Where surface or concealed vertical rod exit devices are used at interior openings, provide as less bottom rod (LBR) unless otherwise indicated. Provide dust proof strikes where thermal pins are required to project into the floor.
11. Dummy Push Bar: Nonfunctioning push bar matching functional push bar.
12. Rail Sizing: Provide exit device rails factory sized for proper door width application.
13. Through Bolt Installation: For exit devices and trim as indicated in Door Hardware Sets.
14. Conventional Push Rail Exit Devices (Heavy Duty): ANSI/BHMA A156.3, Grade 1 Certified Products Directory (CPD) listed exit devices. Listed manufacturers shall

J. SURFACE DOOR CLOSERS
1. All door closers specified herein shall meet or exceed the following criteria:
2. General: Door closers to be from one manufacturer, matching in design and style, with the same type door preparations and templates regardless of application or spring size. Closers to be non-handed with full sized covers.
3. Standards: Closers to comply with UL-10C for Positive Pressure Fire Test and be U.L. listed for use of fire rated doors.
4. Size of Units: Comply with manufacturer's written recommendations for sizing of door closers depending on size of door, exposure to weather, and anticipated frequency of use. Where closers are indicated for doors required to be accessible to the Americans with Disabilities Act, provide units complying with ANSI ICC/A117.1.
5. Closer Arms: Provide heavy duty, forged steel closer arms unless otherwise indicated in Hardware Sets.
6. Closers shall not be installed on exterior or corridor side of doors; where possible install closers on door for optimum aesthetics.
7. Closer Accessories: Provide door closer accessories including custom templates, special mounting brackets, spacers and drop plates as required for proper installation. Provide through-bolt and security type fasteners as specified in the hardware sets.
8. Door Closers, Surface Mounted (Cam Action): ANSI/BHMA 156.4, Grade 1 Certified Products Directory (CPD) listed surface mounted, high efficiency door closers with complete spring power adjustment, sizes 1 thru 6; and fully operational adjustable according to door size, frequency of use, and opening force. Closers to be of the cam and roller design, one piece cast aluminum silicon alloy body with adjustable backcheck and independently controlled valves for closing sweep and latch speed.
9. Door Closers, Surface Mounted (Commercial Duty): ANSI/BHMA 156.4, Grade 1 Certified Products Directory (CPD) listed surface mounted, institutional grade door closers with complete spring power adjustment, sizes 1 thru 6; and fully operational adjustable according to door size, frequency of use, and opening force. Closers to be rack and pinion type, one piece cast iron or aluminum alloy body construction, with adjustable backcheck, closing sweep, and latch speed control valves. Provide non-handed units standard.

K. ARCHITECTURAL TRIM
1. Door Protective Trim
2. General: Door protective trim units to be of type and design as specified below or in the Hardware Sets.
3. Size: Fabricate protection plates (kick, armor, or mop) not more than 2-inch less than door width (LDW) on stop side of single doors and 1-inch LDW on stop side of pairs of doors, and not more than 1-less than door width on pull side. Coordinate and provide proper width and height as required where conflicting hardware dictates. Height to be as specified in the Hardware Sets.

4. Where plates are applied to fire rated doors with the top of the plate more than 16 inches above the bottom of the door, provide plates complying with NFPA 80. Consult manufacturer's catalog and template book for specific requirements for size and applications.
5. Protection Plates: ANSI/BHMA A156.6 protection plates (kick, armor, or mop), fabricated from the following:
6. Stainless Steel: 300 grade, 050-inch thick.
7. Options and fasteners: Provide manufacturer's designated fastener type as specified in the Hardware Sets. Provide countersunk screw holes.

L. DOORSTOPS AND HOLDERS
1. General: Door stops and holders to be of type and design as specified below or in the Hardware Sets.
2. Door Stops and Bumpers: ANSI/BHMA A156.16, Grade 1 door stops and wall bumpers. Provide wall bumpers, either convex or concave types with anchorage as indicated, unless floor or other types of doorstops are specified in Hardware Sets. Do not mount floor stops where they will impede traffic. Where floor or wall bumpers are not appropriate, provide overhead type stops and holders.

M. ARCHITECTURAL SEALS
1. General: Thresholds, silencers, weatherstripping, and gasket seals to be of type and design as specified below or in the Hardware Sets. Provide continuous weatherstrip gasketing on exterior doors and provide smoke, light, or sound gasketing on interior doors where indicated. At exterior applications provide non-corrosive fasteners and elsewhere where indicated.
2. Smoke Labeled Gasketing: Assemblies complying with NFPA 105 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for smoke control ratings indicated, based on testing according to UL 1784.
3. Provide smoke labeled perimeter gasketing at all smoke labeled openings.
4. Fire Labeled Gasketing: Assemblies complying with NFPA 80 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire ratings indicated, based on testing according to UL-10C.
5. Provide intumescent seals as indicated to meet UL10C Standard for Positive Pressure Fire Tests of Door Assemblies, and NFPA 252, Standard Methods of Fire Tests of Door Assemblies.
6. Sound-Rated Gasketing: Assemblies that are listed and labeled by a testing and inspecting agency, for sound ratings indicated.
7. Replaceable Seal Strips: Provide only those units where resilient or flexible seal strips are easily replaceable and readily available from stocks maintained by manufacturer.
8. Silencers:
   a. Hollow metal frames. Provide a minimum of three.
   b. FRP frames. Provide a minimum of three.

2.3 FABRICATION

A. Fasteners: Provide door hardware manufactured to comply with published templates generally prepared for machine, wood, and sheet metal screws. Provide screws according to manufacturers' recognized installation standards for application intended.

2.4 FINISHES

A. Standard: Designations used in the Hardware Sets and elsewhere indicate hardware finishes complying with ANSI/BHMA A156.18, including coordination with traditional U.S. finishes indicated by certain manufacturers for their products.

B. Provide quality of finish, including thickness of plating or coating (if any), composition, hardness, and other qualities complying with manufacturer's standards, but in no case less than specified by referenced standards for the applicable units of hardware

C. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

PART 3 - EXECUTION

3.1 INSTALLATION

3.2 EXAMINATION

A. Examine scheduled openings, with Installer present, for compliance with requirements for installation tolerances, labeled fire door assembly construction, wall and floor construction, and other conditions affecting performance.

B. Notify Contracting Officer Representative of any discrepancies or conflicts between the door schedule, door types, drawings and scheduled hardware. Proceed only after such discrepancies or conflicts have been resolved in writing.

3.3 PREPARATION

A. Hollow Metal Doors and Frames: Comply with ANSI/DHI A115 series.

B. Wood Doors: Comply with ANSI/DHI A115-W series.

C. Aluminum Storefront Doors and Frames: Comply with Manufacturer Standards.

D. Aluminum Hybrid and Fiberglass Doors: Comply with Manufacturer Standards.

E. Fiberglass Doors and Frames: Comply with Manufacturer Standards.

3.4 INSTALLATION

A. Install each item of mechanical and electromechanical hardware and access control equipment to comply with manufacturer's written instructions and according to specifications.
   1. Installers are to be trained and certified by the manufacturer on the proper installation and adjustment of fire, life safety, and security products including hanging devices; locking devices; closing devices; and seals.

B. Mounting Heights: Mount door hardware units at heights indicated in following applicable publications, unless specifically indicated or required to comply with governing regulations:
   1. Standard Steel Doors and Frames: DHI's "Recommended Locations for Architectural Hardware for Standard Steel Doors and Frames."
   2. DHI TDH-007-20: Installation Guide for Doors and Hardware.
   3. Where indicated to comply with accessibility requirements, comply with ANSI A117.1 "Accessibility Guidelines for Buildings and Facilities."
   4. Provide blocking in drywall partitions where wall stops or other wall mounted hardware is located.

C. Thresholds: Exterior and any interior doors scheduled to have a threshold, shall be installed in full bed of sealant complying with requirements of Specification Section 07 92 00.

D. Storage: Provide a secure lock on storage containers that have stored hardware, that has been delivered to the project but has not yet been installed. Control the handling and installation of hardware items so that the completion of the work will not be delayed by hardware losses before and after installation.

3.5 FIELD QUALITY CONTROL

A. Contractor shall produce a final project punch report for each installed door opening indicating compliance with approved submittals and verification hardware is properly installed, operating and adjusted. Include list of items to be completed and corrected, indicating the reasons or deficiencies causing the Work to be incomplete or rejected.

1. Organization of List: Include separate Door Opening and Deficiencies and Corrective Action Lists organized by Mark, Opening Remarks and Comments, and related Opening Images and Video Recordings.

3.6 ADJUSTING

A. Initial Adjustment: Adjust and check each operating item of door hardware and each door to ensure proper operation or function of every unit. Replace units that cannot be adjusted to operate as intended. Adjust door control devices to compensate for final operation of heating and ventilating equipment and to comply with referenced accessibility requirements.

3.7 CLEANING AND PROTECTION

A. Protect all hardware stored on construction site in a covered and dry place. Protect exposed hardware installed on doors during the construction phase. Install all hardware at the latest possible time frame.

B. Clean adjacent surfaces soiled by door hardware installation.

C. Clean operating items as necessary to restore proper finish. Provide final protection and maintain conditions that ensure door hardware is without damage or deterioration at time of building occupancy.

3.8 DEMONSTRATION

A. Instruct Contracting Officer Representative and maintenance personnel on how to adjust, operate, and maintain mechanical and electromechanical door hardware.

3.9 DOOR HARDWARE SETS

A. The hardware sets represent the design intent. They are a guideline only and should not be considered a detailed hardware schedule. Discrepancies, conflicting hardware and missing items should be brought to the attention of the Contracting Officer Representative with corrections made prior to the bidding process. Omitted items not included in a hardware set should be scheduled with the appropriate additional hardware required for proper application and functionality.
   1. Quantities listed are for each pair of doors, or for each single door.
   2. The supplier is responsible for handing and sizing all products.
   3. Where multiple options for a piece of hardware are given in a single line item, the supplier shall provide the appropriate application for the opening.
   4. At existing openings with new hardware the supplier shall field inspect existing conditions prior to the submittal stage to verify the specified hardware will work as required. Provide alternate solutions and proposals as needed.

B. Manufacturer’s Abbreviations:
   1. MK - McKinney
   2. PE - Pemko
   3. RF - Rixson
   4. SU - Securitron
   5. RO - Rockwood
   6. SA - SARGENT
   7. BE - BEST Locks & Closers
   8. YA - ASSA ABLOY ACCENTRA
   9. NO - Norton
   10. OT - Other

3.10 SCHEDULES

A. HARDWARE SETS:

Set: 1.0

Doors: 62-105A

2	Continuous Hinge	DFM__HD1 PT (Prep as required)		PE
2	Electric Power Transfer	EL-CEPT	613E	SU
1	Mullion	L980S	PC	SA
1	Mounting Kit (Stl.)	98-2579		SA
1	Rim Exit Device, Classroom	43 72 8813 ETL	US10BE	SA
1	Rim Exit Device, Exit Only	43 8810 EO	US10BE	SA
2	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Cylinder	72 980C1	US10B	SA
2	Surface Closer	PS2800ST	690	NO
2	Drop Plate	As required	690	NO
2	Door Stop	466-RKW (As required)	Black	RO
1	Rain Guard	346D (As required)		PE
1	Gasketing	5110BL		PE
2	Sweep	315DN		PE
1	Threshold	158D (Or as detailed)		PE
2	ElectroLynx Harness (Door)	QC-CXXXP		MK
2	ElectroLynx Harness (Frame)	QC-CXXX		MK

Notes: Future access control components to be furnished by the end-user.
Perimeter gasketing to be furnished by the storefront door supplier.

Set: 3.0

Doors: 62-110B

6	Hinge, Full Mortise, Hvy Wt	T4A3386 NRP	US10BE(SS)	MK
1	Mullion	L980S	PC	SA
1	Mounting Kit (Stl.)	98-2579		SA
1	Rim Exit Device, Storeroom	43 72 8804 ETL	US10BE	SA
1	Rim Exit Device, Exit Only	43 8810 EO	US10BE	SA
2	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Cylinder	72 980C1	US10B	SA
2	Surface Closer	PS2800ST	690	NO
2	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
2	Door Stop	466-RKW (As required)	Black	RO
1	Gasketing	2891DS		PE

1	Rain Guard	346D (As required)		PE
1	Gasketing	5110BL		PE
2	Sweep	315DN		PE
1	Threshold	158D (Or as detailed)		PE

Notes: Mount PA bracket on top of rigid seal.

Set: 6.0

Doors: 52-101A

3	Hinge, Full Mortise, Hvy Wt	T4A3386 NRP	US10BE(SS)	MK
1	Rim Exit Device, Storeroom	43 72 8804 ETL	US10BE	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Surface Closer	PS2800ST	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Door Stop	466-RKW (As required)	Black	RO
1	Gasketing	2891DS		PE
1	Rain Guard	346D (As required)		PE
1	Sweep	315DN		PE
1	Threshold	158D (Or as detailed)		PE

Notes: Mount rim strike and PA bracket on top of rigid seal.
Field verify existing opening, confirm that hardware as specified will mount properly onto door and/or frame.

Set: 7.0

Doors: 62-111B

6	Hinge, Full Mortise, Hvy Wt	T4A3386 NRP	US10BE(SS)	MK
1	Mullion	L980S	PC	SA
1	Mounting Kit (Stl.)	98-2579		SA
1	Rim Exit Device, Classroom	43 72 8813 ETL	US10BE	SA
1	Rim Exit Device, Exit Only	43 8810 EO	US10BE	SA
2	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Cylinder	72 980C1	US10B	SA
2	Surface Closer	PS2800ST	690	NO
2	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
2	Door Stop	466-RKW (As required)	Black	RO
1	Gasketing	2891DS		PE
1	Rain Guard	346D (As required)		PE
1	Gasketing	5110BL		PE
2	Sweep	315DN		PE

1	Threshold	158D (Or as detailed)		PE

Notes: Mount PA bracket on top of rigid seal.

Set: 9.0

Doors: 62-112B, 62-113B

6	Hinge, Full Mortise, Hvy Wt	T4A3386 NRP	US10BE(SS)	MK
1	Mullion	L980S	PC	SA
1	Mounting Kit (Stl.)	98-2579		SA
1	Rim Exit Device, Exit Only	43 8810 EO	US10BL	SA
1	Rim Exit Device, Classroom	43 72 8813 ETL	US10BL	SA
2	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Cylinder	72 980C1	US10B	SA
2	Surface Closer	PS2800ST	690	NO
2	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
2	Door Stop	466-RKW (As required)	Black	RO
1	Gasketing	2891DS		PE
1	Rain Guard	346D (As required)		PE
1	Gasketing	5110BL		PE
2	Sweep	315DN		PE
1	Threshold	158D (Or as detailed)		PE

Notes: Mount PA bracket on top of rigid seal.

Set: 10.0

Doors: 52-101C

3	Hinge, Full Mortise, Hvy Wt	T4A3386 NRP	US10BE(SS)	MK
1	Rim Exit Device, Classroom	43 72 8813 ETL	US10BE	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Surface Closer	PS2800ST	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Door Stop	466-RKW (As required)	Black	RO
1	Gasketing	2891DS		PE
1	Rain Guard	346D (As required)		PE
1	Sweep	315DN		PE
1	Threshold	158D (Or as detailed)		PE

Notes: Mount rim strike and PA bracket on top of rigid seal.

Set: 12.0

Doors: 62-109C

3	Hinge, Full Mortise, Hvy Wt	T4A3386 NRP	US10BE(SS)	MK
1	Electric Power Transfer	EL-CEPT	613E	SU
1	Rim Exit Device, Classroom	43 72 8813 ETL	US10BL	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Surface Closer	PS2800ST	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Door Stop	466-RKW (As required)	Black	RO
1	Gasketing	2891DS		PE
1	Rain Guard	346D (As required)		PE
1	Sweep	315DN		PE
1	Threshold	158D (Or as detailed)		PE
1	ElectroLynx Harness (Door)	QC-CXXXP		MK
1	ElectroLynx Harness (Frame)	QC-CXXX		MK

Notes: Mount rim strike and PA bracket on top of rigid seal.
Future access control components to be furnished by the end-user.

Set: 13.0

Doors: 62-109D

3	Hinge, Full Mortise, Hvy Wt	T4A3386 NRP	US10BE(SS)	MK
1	Rim Exit Device, Classroom	43 72 8813 ETL	US10BL	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Surface Closer	PS2800ST	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Door Stop	466-RKW (As required)	Black	RO
1	Gasketing	2891DS		PE
1	Rain Guard	346D (As required)		PE
1	Sweep	315DN		PE
1	Threshold	158D (Or as detailed)		PE

Notes: Mount rim strike and PA bracket on top of rigid seal.

Set: 14.0

Doors: 62-100B

6	Hinge, Full Mortise	TA2314 NRP	US10BE(SS)	MK
1	Flush Bolt (Comb Hm)	2845	10BE	RO
1	Dust Proof Strike	570	10BE	RO
1	Storeroom/Closet Lock	72 8204 LNL	US10BE	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Coordinator	2600	Black	RO
2	Mounting Bracket	2601D	Black	RO
2	Surface Closer	PS2800ST	690	NO
2	Kick Plate	K1050 10" x 1" LDW CSK BEV	10BE	RO
2	Door Stop	466-RKW (As required)	Black	RO
1	Astragal	357SP		PE
2	Gasketing	303DS		PE
1	Gasketing	S88BL		PE
1	Rain Guard	346D (As required)		PE
2	Sweep	315DN		PE
1	Threshold	158D (Or as detailed)		PE

Notes: Mount smoke seals at head, rigid seals at jambs.

Set: 15.0

Doors: 52-102B

3	Hinge, Full Mortise	TA2314 NRP	US10BE(SS)	MK
1	Storeroom/Closet Lock	72 8204 LNL	US10BE	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Surface Closer	PS2800ST	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Door Stop	466-RKW (As required)	Black	RO
1	Gasketing	303DS		PE
1	Rain Guard	346D (As required)		PE
1	Sweep	315DN		PE
1	Threshold	158D (Or as detailed)		PE

Notes: Field verify existing opening, confirm that hardware as specified will mount properly onto door and/or frame. (Dr #52-102B)

Set: 17.0

Doors: 62-102A

2	Hinge, Full Mortise	TA2314 NRP	US10BE(SS)	MK
1	Hinge, Full Mortise	TA2314 QC	US10BE(SS)	MK
1	Classroom Lock	72 8237 LNL	US10BE	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Surface Closer	PS2800ST	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Door Stop	466-RKW (As required)	Black	RO
1	Gasketing	303DS		PE
1	Rain Guard	346D (As required)		PE
1	Sweep	315DN		PE
1	Threshold	158D (Or as detailed)		PE
1	ElectroLynx Harness (Door)	QC-CXXXP		MK
1	ElectroLynx Harness (Frame)	QC-CXXX		MK

Notes: Future access control components to be furnished by the end-user.

Set: 19.0

Doors: 62-111A

6	Hinge, Full Mortise, Hvy Wt	T4A3786	US10BE	MK
1	Surface Vert Rod Exit, Passage	43 NB8715 ETL	US10BL	SA
1	Surface Vert Rod Exit, Exit Only	43 NB8710 EO	US10BL	SA
2	Surface Closer	PR8501 M	690	NO
2	Kick Plate	K1050 10" x 1" LDW CSK BEV	10BE	RO
2	Wall Stop (Convex)	406	10BE	RO
2	Silencer	608-RKW		RO

Set: 20.0

Doors: 62-110A

3	Hinge (heavy weight)	T4A3786 NRP	US26D	MK
1	Rim Exit Device, Classroom	43 72 8813 ETL	US10BL	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Surface Closer	PR8501 M	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Wall Stop (Convex)	406	10BE	RO
3	Silencer	608-RKW		RO

Set: 21.0

Doors: 62-112A, 62-113A

3	Hinge (heavy weight)	T4A3786 NRP	US26D	MK
1	Rim Exit Device, Classroom	12 43 72 8813 ETL	US10BL	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Surface Closer	PR8501 M	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Wall Stop (Convex)	406	10BE	RO
3	Silencer	608-RKW		RO

Set: 22.0

Doors: 62-105B

2	Continuous Hinge	DFM__HD1		PE
1	Mullion	L980S	PC	SA
1	Mounting Kit (Stl.)	98-2579		SA
1	Rim Exit Device, Exit Only	43 8810 EO	US10BE	SA
1	Rim Exit Device, Passage	43 8815 ETL	US10BE	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Cylinder	72 980C1	US10B	SA
2	Surface Closer	PR8501	690	NO
2	Drop Plate	As required	690	NO
2	Wall Stop (Convex)	406	10BE	RO

Notes: Perimeter gasketing to be furnished by the storefront door supplier.

Set: 23.0

Doors: 62-108A

3	Hinge, Full Mortise	TA2714 NRP	US10BE	MK
1	Storeroom/Closet Lock	72 8204 LNL	US10BE	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Wall Stop (Convex)	406	10BE	RO
3	Silencer	608-RKW		RO

Set: 24.0

Doors: 62-100A

3	Hinge, Full Mortise	TA2714	US10BE	MK
1	Storeroom/Closet Lock	72 8204 LNL	US10BE	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE

1	Surface Closer	8501	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Wall Stop (Convex)	406	10BE	RO
3	Silencer	608-RKW		RO

Set: 25.0

Doors: 52-102A

3	Hinge, Full Mortise	TA2714 NRP	US10BE	MK
1	Storeroom/Closet Lock	72 8204 LNL	US10BE	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Surface Closer	PR8501	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Wall Stop (Convex)	406	10BE	RO
3	Silencer	608-RKW		RO

Set: 26.0

Doors: 62-108B

3	Hinge, Full Mortise	TA2714 NRP	US10BE	MK
1	Storeroom/Closet Lock	72 8204 LNL	US10BE	SA
1	I.C. Permanent Core	Facility SFIC Standard	613	BE
1	Surface Closer	CPS8501	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
3	Silencer	608-RKW		RO

Set: 31.0

Doors: 62-106A, 62-107A

3	Hinge, Full Mortise	TA2714	US10BE	MK
1	Passage Latch	8215 LNL	US10BE	SA
1	Surface Closer	8501	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Wall Stop (Convex)	406	10BE	RO
3	Silencer	608-RKW		RO

Set: 33.0

Doors: 62-109A

3	Hinge, Full Mortise	TA2314 NRP	US10BE(SS)	MK
1	Passage Latch	8215 LNL	US10BL	SA

1	Surface Closer	P8501 M	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Wall Stop (Convex)	406	10BE	RO
3	Silencer	608-RKW		RO

Set: 36.0

Doors: 62-103A

3	Hinge, Full Mortise	TA2714	US10BE	MK
1	Privacy Lock	8265 LNL	US10BE	SA
1	Surface Closer	8501	690	NO
1	Kick Plate	K1050 10" x 2" LDW CSK BEV	10BE	RO
1	Kick Plate	K1050 4" X 1" LDW CSK BEV	10BE	RO
1	Wall Stop (Convex)	406	10BE	RO
1	Gasketing	S88BL		PE
1	Coat Hook	RM802	10BE	RO

Set: 37.0

Doors: 52-101B, 62-109B

1	All Materials	By Door Manufacturer		OT

HARDWARE SCHEDULE FOR POWER DISTRIBUTION NORTH RIM – VOL 9

Set: 1

Door: 101A

3	Hinge, Hvy Wt	T4A3386 NRP	US32D(SS)	MK
1	Rim Exit Device, Storeroom	43 72 8804 - ETL (US26D Lever)	US32D	SA
1	I.C. Permanent Core	Facility SFIC Standard	630	BE
1	Surface Closer/110 Deg Stop	PS2800ST	689	NO
1	Kick Plate	K1050 10" x 2d" LDW CSK BEV	US32D	RO
1	Gasketing	2891AS		PE
1	Rain Guard	346C (As required)		PE
1	Sweep	315CN		PE
1	Threshold	154A (Or as detailed)		PE

Notes: Mount rim strike and PA bracket on top of rigid seal.

END OF SECTION

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SECTION 08 81 00
GLASS AND GLAZING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Glass and glazing.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 07 92 00 - Joint Sealants.
   2. Section 08 11 00 - Hollow Metal Doors and Frames.
   3. Section 08 11 16 - Aluminum Doors and Frames.
   4. Section 08 14 16 - Flush Wood Doors
   5. Section 08 70 00 - Finish Hardware
   6. Section 10 28 13 - Toilet Accessories (Packaged mirror units)

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Standards Institute (ANSI):
      a. Z97.1, Glazing Materials Used in Buildings - Safety Performance Specifications and Methods of Test.
   2. ASTM International (ASTM):
      a. C864, Standard Specification for Dense Elastomeric Compression Seal Gaskets, Setting Blocks, and Spacers.
      b. C1036, Standard Specification for Flat Glass.
      c. C1048, Standard Specification for Heat-Strengthened and Fully Tempered Flat Glass.
      d. C1376, Standard Specification for Pyrolytic and Vacuum Deposition Coatings on Flat Glass.
      e. E2190, Standard Specification for Insulating Glass Unit Performance and Evaluation.
   3. Code of Federal Regulations (CFR):
      a. Title 16 - Commercial Practices, Chapter ii - Consumer Product Safety Commission (CPSC), Subchapter B - Consumer Product Safety Act Regulations:
         1) 16 CFR 1201, Safety Standard for Architectural Glazing Materials.
   4. Glass Association of North America (GANA):
      a. Glazing Manual.
   5. Insulating Glass Certification Council (IGCC).
   6. Insulating Glass Manufacturers Alliance (IGMA):
      a. TM-3000, North American Glazing Guidelines for Sealed Insulating Glass Units for Commercial & Residential Use.
   7. National Park Service (NPS)
      a. NPS Bird-Collisions-Handbook.
      b. NPS Sustainable Design Principles.
      c. NPS Bird Friendly Building Guidelines (regional guidance where applicable)
   8. American Bird Conservancy (ABC)
      a. Threat Factor Testing.

1.3 DEFINITIONS

A. Glass: Includes prime glass, processed glass, and fabricated glass products. Glass products is hereby defined to include glazing plastics. Glazing includes glass and glass product installation and materials used to install glass and glass products.

B. Manufacturer: A firm that produces primary glass, fabricated glass or framing as defined in referenced glazing publications.

C. Installer or Applicator:
   1. Installer or applicator is the person actually installing or applying the product in the field at the Project site.
   2. Installer and applicator are synonymous.

D. Safety Glazing: Glazing meeting the requirements of the building code and CPSC 16 CFR 1201.

E. Other terms as identified in CSPC 16 CFR 1201.

1.4 SUBMITTALS

A. Manufacturer's Product Data and Shop Drawings:
   1. Product technical data including:
      a. Submit glass / glazing manufacturer's latest edition of product data specifications and installation instructions for each type of glass / glazing required, including other data as may be required to show compliance with specified requirements. Transmit copy of each instruction to the installer.
      b. Glazing sealant manufacturer shall submit preparation data that is based on previous testing of current sealant products for adhesion to and compatibility with submitted glazing materials.
      c. Provide certificates attesting that materials meet requirements and tests specified herein.
         1) Acknowledgement that products submitted meet requirements of standards referenced.
      d. Manufacturer's installation instructions.
         1) Copies of all test criteria.
      e. Bird Friendly glazing pattern, spacing, optical performance, and ABC Threat Factor rating.
         1) Documentation demonstrating compliance with NPS bird friendly design requirements
         2) Confirmation that materials meet NPS sustainability guidelines (low toxicity, durable, low maintenance)
   2. Certification that insulating glass units meet requirements of IGCC and are certified by IGCC to ASTM E2190.

B. Shop Drawings: Submit shop drawings showing glazing details.

C. Samples:
   1. Two, 12-inches x 12-inches sample of each type, color, and thickness specified including bird friendly pattern and orientation of the first surface etching.
      a. Samples are not required for clear monolithic glass.

D. Mockup Required:
      a. Minimum 4 feet by 4 feet installed sample
      b. Review by Contracting Officer for visual impact, wildlife protection, and compatibility with regional landscape.

E. Informational Submittals:
   1. Warranty: Provide manufacturer's written 10-year warranty to cover deterioration of glass, glass units and coating. Insulating glass units shall be warranted against failure of hermetic seal resulting in fogging or film formation on the interior glass surfaces.
      a. Submit written agreement for warranty of hermetic seals signed by the Manufacturer, Installer and Contractor.
   2. Warranty: Provide a 20-year warranty to cover delamination and failure of the structural integrity of the warm edge spacers.

1.5 JOB CONDITIONS

A. Examine framing or glazing channel surfaces, backing, stop design, and conditions under which glazing is to be performed. Do not proceed until unsatisfactory conditions have been corrected.

B. Do not proceed with installation of sealants or glazing under adverse weather conditions, or when temperatures are below or above manufacturer's recommended limitations for installation.

1.6 PROJECT CONDITIONS

A. Field Measurements: Verify actual measurements for openings by field measurements before fabrication. Show recorded measurements on shop drawings. Coordinate field measurements and fabrication schedule with construction progress to avoid construction delays.

1.7 DELIVERY, STORAGE AND HANDLING

A. Deliver materials to specified destinations in manufacturers or distributor’s packaging undamaged, complete with installation instructions.

B. Store off ground, under cover, protected from weather and construction activities and at temperature conditions recommended by manufacturer.

C. Manufactured glass units shall be delivered and stored until installation in the manufacturer’s container’s and shall be clearly marked on the exterior as to type, and quantity of units.

D. Protect bird friendly etched glazing on the first surface from abrasion and contamination.

E. Protect glass in accordance with the manufacturer’s recommendations.

1.8 WARRANTY

A. Warranty on Hermetic Seals: Provide insulating glass manufacturer's written warranty, agreeing to, provide and replace insulating glass units which have defective hermetic seals for a period of ten (10) years**.** Defective hermetic seals shall be defined to include intrusion of moisture or dirt, internal condensation at temperatures above minus twenty (-20) degrees F, deterioration of internal glass coatings, and other visual evidence of seal failure except for glass breakage.

B. Insulating Glass Spacers: Provide a 20-year warranty for delaminating and against failure of structural integrity.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Glass and Glazing:
      a. Vitro Architectural Glass.
      b. Or Equal.
   2. Bird Friendly Glass:
      a. Vitro Architectural Glass - BirdSmart Bird Friendly Glass.
      b. Or Equal.
   3. Warm Edge Spacers:
      a. Viracon - “ExtremEdge”.
      b. Or Equal.

2.2 MATERIALS

A. General:
   1. Glass which shall be subject to wind pressure shall conform to the glass manufacturer's recommendations for maximum allowable areas.

B. Heat Strengthened and Fully Tempered Glass: Comply with ASTM C1048.
   1. General use: Kind HS.
   2. Safety glazing: Kind FT.
      a. Meet requirements of ANSI Z97.1 and CSPC 16 CFR 1201.

C. Low - E, Sputter-Coated Float Glass: Comply with ASTM C1376, float glass with metallic-oxide or -nitride coating deposited by vacuum deposition process after manufacture and complying with other specified requirements.

2.3 MANUFACTURED UNITS

A. Insulating Glass Units:
   1. ASTM E2190, Class A.
   2. Two sheets of 1/4 inch thick glass as specified and separated by a 1/2-inch-thick sealed space containing dry air with minus twenty (-20) degrees F dew point, with Class A sealant - type edge construction to maintain a hermetic seal.
   3. Spacers shall be warm edge type spacers.
      a. Thickness: 1/2-inches.
         1) Color: Black.
      b. Perimeter Sealant: Silicone

2.4 ACCESSORIES

A. Glazing Compounds:
   1. Non-sag, non-stain type.
   2. Pigmented to match frame units not requiring painting.
   3. Compatible with adjacent surfaces.
   4. One- or two-part polyurethane or silicone sealant for use in setting glass.
      a. Provide glazing compounds which will not be affected by chemicals stored in rooms where glazing compounds are used.

B. Sealant Tape: Butyl rubber sealant tape or ribbon having a continuous neoprene shim.

C. Gaskets:
   1. Flexible polyvinyl chloride or neoprene.
      a. ASTM C864.
      b. Provide gaskets which will not be affected by chemicals stored in rooms where gaskets are used.
   2. Extruded of profile and hardness required to receive glass and provide a watertight installation.

D. Setting Blocks and Spacers:
   1. Neoprene or EPDM, compatible with sealants used.
      a. ASTM C864.
   2. Thickness shall be approximately the same as the glass - edge clearance dimension; the length shall be 4-inches minimum.

E. Compressible Filler Stock: Closed - cell or waterproof - jacketed rod stock of synthetic rubber or plastic foam, proven to be compatible with sealants used, flexible and resilient, with 5 - 10 pounds per square inch compression strength for 25 percent deflection.

F. Shims, Clips, Screws and Other Miscellaneous Items: As required by condition.

2.5 CLEANING SOLUTIONS

A. All cleaning solutions shall be compatible with the glazing materials and as recommended by the glass manufacturer or fabricator.

2.6 FABRICATION

A. Fabricate glass and other glazing products in sizes required to glaze openings indicated for Project, with edge and face clearances, edge and surface conditions, and bite complying with recommendations of product manufacturers and referenced glazing standard as required to comply with system performance requirements.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install in accordance with recommendations of manufacturer, GANA Glazing Manual and IGMA TM-3000.

B. Install setting blocks in adhesive or sealant.

C. Where wedge - shaped gaskets are driven into one side of channel to pressurize sealant or gasket on opposite side, provide adequate anchorage to ensure that gasket will not "walk" out when installation is subjected to movement. Anchor gasket to stop with matching ribs, or by proven adhesives, including embedment of gasket tail in cured heel bead.

D. Install spacers inside and out, of proper size and spacing, for all glass sizes larger than 50 united inches, except where gaskets are used for glazing.

E. Provide 1/8-inches minimum bite of spacers on glass.

F. Spacer thickness to equal sealant width.

G. Prevent sealant exudation from glazing channels of insulating glass which is more than 1/2-inches thick; colored, heat absorbing, coated or laminated glass sizes larger than 75 united inches; and other glass more than 9/32-inches thick or larger than 125 united inches.
   1. Leave void at heel (or install filler) at jambs and head.
   2. Do not leave void (or install filler) at sill.

H. Miter cut and bond gasket ends together at corners.

I. Protect glass from breakage immediately upon installation, by use of crossed streamers attached to framing and held away from glass. Do not apply markers to surfaces of glass. Remove nonpermanent labels and clean surfaces. Cure sealants for high early strength and durability.

J. Use polysulfide-based glazing sealants in window assembly and as perimeter sealant around frames in areas which may be exposed to chlorine gas or chlorine liquid splash or spillage.
   1. See Specification Section 07 92 00 for sealants.

3.2 FIELD QUALITY CONTROL

A. Do not install glass with edge damage.

B. Do not apply anything to surfaces of glass.

C. Remove and replace damaged glass.

3.3 CLEANING

A. Maintain glass reasonably clean during construction, so that it will not be damaged by corrosive action and will not contribute to deterioration of other materials.

B. Wash and polish glass on both faces not more than seven days prior to acceptance of work in each area.
   1. Comply with glass manufacturer's recommendations.

C. Temporary labels shall be removed immediately after the glass and glazing work has been approved and prior to final cleaning.

3.4 SCHEDULES

A. General:
   1. Provide safety glazing for all applications where required by the building code and CPSC 16 CFR 1201.
   2. Provide heat strengthened glazing for all general use applications where safety glazing is not required.

B. Glass Type 1: Clear Tempered Float Glass – 1/4-inch-thick complying with ASTM 1048, Type 1, Class 1, Quality q3, Kind FT, Condition A.

C. Glass Type 2: Bronze Tinted Insulating Glass:
   1. Exterior Sheet of Glass shall be Tinted, Transparent float glass complying with ASTM C1036, Type 1, quality q3, Class 1, Kind HS, Condition C with an Exterior Reflectance - 7 percent, SHGC - 0.28, VLT - 42, U - Value Winter - 0.29, LSG - 1.50. The tinted glass shall have the first (1st) surface markers be laser etched permanent, non-reflective, and resistant to UV, abrasion, and weathering, meeting or exceed the National Park Service bird friendly guidelines (2 inches by 2 inches maximum spacing) and achieves American Bird Conservancy Threat Factor of less than or equal to 25. The second (2nd) surface of the tinted glass shall have a Low - E sputtered coated surface.
   2. The Interior Sheet of Glass shall be Clear Float Glass - 1/4-inch-thick complying with ASTM C1036, Type 1, Class 1, Quality q3, Kind HS, Condition A.
   3. Basis-of-Design Product Manufacturer: Glass and Glazing.
      a. Insulating Glass Unit Construction: 1/4-inch-thick Vitro Architectural Glass - Solarbronze glass, Solarban 60 solar control (sputtered) on the second (2) surface and BirdSmart Bird Friendly Glass on the first (1) surface + 1/2-inch air space + 1/4-inch Clear float glass.

D. Glass Type 3: Bronze Tinted Tempered Insulating Glass:
   1. Exterior Sheet of Glass shall be Tinted, Transparent, Tempered - 1/4-inch-thick float glass complying with ASTM C1048, Type 1, Class 1, Quality q3, Kind FT, Condition C with an Exterior Reflectance - 7 percent, SHGC - 0.28, VLT - 42, U - 43, Value Winter - 0.29, LSG – 1.50. The tinted tempered glass shall have the first (1st) surface markers be laser etched permanent, non-reflective, and resistant to UV, abrasion, and weathering, meeting or exceed the National Park Service bird friendly guidelines (2 inches by 2 inches maximum spacing) and achieves American Bird Conservancy Threat Factor of less than or equal to 25. The second (2nd) surface of the tinted glass shall have a Low - E sputtered coated surface.
   2. The Interior Sheet of Glass shall be Clear Tempered Float Glass - Complying with ASTM 1048, Type 1, Class 1, Quality q3, Kind FT, Condition A.
   3. Basis-of-Design Product Manufacturer: Glass and Glazing.
      Insulating Glass Unit Construction: 1/4-inch-thick tempered Vitro Architectural Glass - Solarbronze glass, Solarban 60 solar control (sputtered) on the second (2) surface and BirdSmart Bird Friendly Glass on the first (1) surface + 1/2-inch air space + 1/4-inch Clear tempered float glass.

END OF SECTION

SECTION 08 90 00
LOUVERS AND VENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Louvers and vents.
   2. Exterior galvanized snow hood.

B. Related Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 01 78 23 - Operation and Maintenance Data
   3. Section 07 92 00 - Joint Sealants.
   4. Section 09 06 00 – Color Schedule.
   5. Section 09 96 00 – High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Fenestration & Glazing Industry Alliance (FGIA) (AAMA):
      a. AAMA 2605 - Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels (with Coil Coating Appendix).
   2. Air Movement and Control Association (AMCA).
      a. AMCA 511 - Certified Ratings Program — Product Rating Manual for Air Control Devices.
   3. (ASCE)
      a. ASCE-7 - Minimum Design Loads and Associated Criteria for Buildings and Other Structures.
   4. ASTM International (ASTM):
      a. B221 - Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes.
      b. D822 - Standard Practice for Filtered Open-Flame Carbon-Arc Exposures of Paint and Related Coatings.
      c. D2244 - Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates.
      d. D4214 - Standard Test Methods for Evaluating the Degree of Chalking of Exterior Paint Films.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Drawing showing location of each louver and galvanized snow hood, indicating size and arrangement as required.
   2. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Product Test Reports: For each type of louver, for tests performed by a qualified testing agency.
      d. Color chart showing manufacturers’ full line of colors including exotic and special colors for color selection by Contracting Officer Representative.

B. Samples:
   1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.

2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
    a. Provide actual metal samples (aluminum).
3. Refer to “Specification Section 09 96 00 – High Performance Industrial Coatings” for specified material.

C. Contract Closeout Information:
1. Operation and Maintenance Data:
    a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.4 WARRANTY

A. Manufacturer shall provide standard limited warranty for louver systems for a period of five years (60 months) from date of completion. When notified in writing from the Contracting Officer Representative of a manufacturing defect, manufacturer shall promptly correct deficiencies without cost to the Government.

B. Louver Manufacturer shall provide 20-year limited warranty for fluoropolymer-based finish on extruded aluminum substrates.
1. Finish coating shall not peel, blister, chip, crack or check.
2. Chalking, fading or erosion of finish when measured by the following tests:
    a. Finish coating shall not chalk in excess of 8 numerical ratings when measured in accordance with ASTM D4214.
    b. Finish coating shall not change color or fade in excess of 5 NBS units as determined by ASTM D2244 and ASTM D822.

1.5 QUALITY ASSURANCE

A. Manufacturer Qualifications:
1. The manufacturer shall have implemented the management of quality objectives, continual improvement, and monitoring of customer satisfaction to assure that customer needs and expectations are met.
2. Manufacturer shall be International Organization for Standardization (ISO) 9001 accredited.

B. Product Qualifications:
1. Louvers licensed to bear AMCA Certified Ratings Seal. Ratings based on tests and procedures performed in accordance with AMCA 511 and comply with AMCA Certified Ratings Program. AMCA Certified Ratings Seal applies to air performance and water penetration ratings.
2. Louvers shall be factory engineered to withstand the specified seismic loads.
    a. Minimum design loads shall be calculated to comply with ASCE – 7, or local requirements of Authority Having Jurisdiction (AHJ).

1.6 DELIVERY, STORAGE, AND HANDLING

1. Store products in manufacturer's unopened packaging until ready for installation.
2. Store materials in a dry area indoors, protected from damage and in accordance with manufacturer's instructions.
3. Handling:
    a. Protect materials and finishes during handling and installation to prevent damage.
4. Store and dispose of solvent-based materials, and materials used with solvent-based materials, in accordance with requirements of local authorities having jurisdiction.

1.7 PROJECT CONDITIONS

A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Louvers:
      a. Basis of Design: Ruskin Company – "ELF6425DD".
      b. Or Equal.
   2. Galvanized Snow Hood:
      a. Basis of Design: Canarm HVAC – GH Galvanized Hoods.
      b. Or Equal.
   3. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MANUFACTURED UNITS

A. Louvers:
   1. Frame:
      a. 6 inches deep, unless noted otherwise on the plans and schedule sheets.
      b. ASTM B221 extruded aluminum, alloy 6063-T6, minimum 0.081-inches thick.
      c. Prefinished "PVDF" finish.
   2. Blades:
      a. Drainable with blades at 42-1/2 degrees.
      b. ASTM B221 extruded aluminum, alloy 6063-T6, minimum 0.094-inches thick.
      c. Continuous blade appearance. Prefinished to match Louvers
   3. Minimum free area: 51 percent.
   4. Maximum pressure drop: 0.10-inches of water at 800 feet per minute.
   5. Water penetration: 0.01-ounce per square foot at 873 feet per minute.
   6. AMCA 511 certified:
      a. Air Performance and Water Penetration.
   7. Insect screen:
      a. 18-16 mesh aluminum 0.011-inch wire, mill finish.
      b. Install in standard aluminum removable frame.
   8. Bird screen:
      a. 1/2-inches square aluminum mesh.
      b. Install in standard removable aluminum frame.
   9. Extended Sills:
      a. Extruded aluminum, Alloy 6063-T6. Minimum nominal thickness 0.060 inch. Set on top of aluminum sill flashings. Prefinished to match Louvers.
   10. Aluminum Flashing:
      a. Shop formed aluminum flashings 0.040-inchs thick. Prefinished to match Louvers.
   11. Louver and Accessories Finish:
      a. 70 percent PVDF: Finish shall be applied at 1.2 mil total dry film thickness.
      b. Coating shall conform to AAMA 2605. Apply coating following cleaning and pretreatment. Cleaning: AA-C12C42R1X.
         1) Standard 2-coat.
      c. Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.
      d. 20-year finish warranty.

B. Galvanized Snow Hoods:
   1. 18-Gauge galvanized steel hood construction.
   2. 1-3/4-inch wide flanges for anchoring to exterior wall.

C. Anchors, Fasteners, Reinforcing:
   1. Aluminum or stainless steel.

D. Sizes of Louvers:
   1. See Louver Schedules on Mechanical Drawings.

E. Sizes of Galvanized Snow Hoods:
   1. Hoods to be match length and height of louvers as shown in Detail on Architectural Drawing.

F. Galvanized Snow Hoods Finish:
   1. Refer to “Specification Section 09 96 00 – High Performance Industrial Coatings” for specified material.
   2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Inspect areas to receive louvers. Notify the Contracting Officer Representative of conditions that would adversely affect the installation or subsequent utilization of the louvers. Do not proceed with installation until unsatisfactory conditions are corrected.

B. If opening preparation is the responsibility of another installer, notify Contracting Officer Representative of unsatisfactory preparation before proceeding.

3.2 PREPARATION

A. Clean opening thoroughly prior to installation.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

3.3 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Install louvers at locations indicated on the drawings and in accordance with manufacturer's instructions.

C. Install louvers plumb, level, in plane of wall, and in alignment with adjacent work.

D. The supporting structure shall be designed to accommodate the point loads transferred by the louvers when subject to the design wind loads.

E. Install anchoring and bracing accessories as required.

F. Anchor Galvanized Snow Hoods to exterior wall with appropriate anchors.

G. Seal around perimeter on exterior and interior.
   1. See Section 07 92 00.

3.4 CLEANING

A. Clean louver surfaces in accordance with manufacturer's instructions.

B. Touch-up, repair or replace damaged products before final completion.

END OF SECTION

DIVISION 09

FINISHES

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SECTION 09 06 00

COLOR SCHEDULE

PART 1 - GENERAL

1.1 SUMMARY

A. Exterior materials and color selections.

B. Interior materials and color selections.

1.2 SUBMITTALS

A. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.

PART 2 - PRODUCTS - (NOT USED)

PART 3 - EXECUTION

3.1 SCHEDULES

A. Room Finish Schedule shown on drawings.
   1. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.

B. Exterior Materials and Color Selections:
   1. **AL-1:** Aluminum storefront entrances and frames.
      a. Color: RAL 8025 “Pale Brown”.
   2. **AL-2:** Aluminum hybrid entrances and frames.
      a. Color Aluminum: RAL 8025 “Pale Brown”.
      b. Color Fiber Reinforced Plastic: Special-lite “Sage Brown”.
   3. **AL-3:** Aluminum window frames.
      a. Color: RAL 8025 “Pale Brown”.
   4. **AL-4:** Aluminum rolling overhead doors.
      a. Color: RAL 8025 “Pale Brown”.
   5. **AL-6:** Factory painted louvers.
      a. Color: RAL 8025 “Pale Brown”.
   6. **AL-7:** Factory painted deck, fascia, gutter, downspouts, steel hanger pipe, escutcheons and couplers.
      a. Color: Pac-Clad – “Evergreen”
   7. **MTL-1:** Galvanized steel bollards.
      a. Color: High visibility safety yellow complying with OSHA Specification 1910.144.
   8. **MTL-2:** Galvanized snow hoods.
      a. Color: RAL 8025 “Pale Brown”.
   9. **MTL-3:** Factory painted standing seam metal roofing, fascia, soffit, trims, and flashings (except at MTL-4 locations).
      a. Color: Pac-Clad – “Evergreen”.
   10. **MTL-4:** (Roaring Springs Pumphouse and Solar Shade Canopy Locations Only) Factory painted standing seam metal roofing, fascia, soffit, trims, and flashings.
      a. Color: Pac-Clad – “Weathered Steel”.
   11. **CMU-1:** Factory shot blast-face masonry.
      a. Color: Block Lite - Mix of two colors – (2/3RD’s “R2-C2 Harvest Gold”) and (1/3RD’s “R3-C18 Apache Red”).
   12. **MOR-1:** Colored masonry mortar.
      a. Color: TCC Spec Mix - SM600 “Dark Brown”.

13. **CST-1:** Cast Stone Caps – Artisan Stone – “Sample #3851”.
14. **CBS-1:** Cement board lap siding, trim, fascia and accessories.
    a. Color: James Hardie – “Dream Collection Saddle Brown”.
15. **HPIC-1:** Loose lintles.
    a. Color: RAL 8025 “Pale Brown”.
16. **HPIC-2:** Exterior equipment screen.
    a. Color: RAL 8025 “Pale Brown”.
17. **JS-1:** Exterior Joint sealants.
    a. Color: “Selection to Be Determined Depending on Location and Abutting Materials.”

C. Interior Materials and Color Selections:
1. **ACT-1:** Acoustic Tile Ceiling
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
    b. Ceiling grid shall be standard “White”.
2. **CT-1:** Ceramic Floor tile.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
3. **CT-2:** Ceramic Wall tile.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
4. **CT-3:** Ceramic Wall tile accent band.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
5. **G-1:** Grout for ceramic tile.
    a. Color – Floor Tile: Mapei - 5009 Gray.
    b. Color – Wall Tile: Mapei – 5005 Chamois.
6. **TT-1:** Metal Transition Strips.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
7. **EF-1:** Epoxy flooring.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
8. **RFT:** Static dissipative rubber sheet flooring.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
9. **RB-1:** Rubber wall base.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
10. **P-1:** Paint at walls field color.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
11. **AP-1:** Paint at ceilings
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
12. **P-D1:** Paint dryfall color. Acoustic Metal Decking.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
13. **P-M1:** Paint hollow metal doors and frames color.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
14. **PL-1:** Plastic laminate for millwork cabinetry.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
15. **WS-1:** Roller window shades

a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.

16. **SS-1:** Solid surface material for counter tops, backsplashes, side splashes and windowsills.
    a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
17. **CTS-1:** Countertop support brackets.
    a. Color: Black.
18. **ST-1:** Wood doors stain.
    a. Color: PPG, Olympic – Storm Gray.
19. **JS-2:** Interior Joint Sealants.
    a. Color: “Selection TBD Depending on Location and Abutting Materials.”
20. **AL-X:** Aluminum doors, frames and window frames.
    a. Color: RAL 8025 “Pale Brown”.
21. **FRP-1:** Fiber reinforced plastic doors and frames.
    a. Color: Special-lite “Sage Brown”.
22. **FWP-1:** Cementitious wood fiber acoustic panels.
    a. Color: Factory painted “White”.
23. **SG-1:** Signage.
    a. Color Backer Finish: Savannah.
    b. Color Face Finish: Cloud Hatch.
    c. Color Raised Text, Logo Art and Symbols: Black
24. **MPB-1:** Metal peg board and accessories.
    a. Color: Galvanized (Metallic/Silver) peg board and Black accessories.
25. **MWB-1:** Metal workbench.
    a. Color: Dove Gray.
26. **MWS-1:** Metal shelving.
    a. Color: Dove Gray.
27. **BV-1:** Bench Vice.
    a. Color: High visibility safety yellow complying with OSHA Specification 1910.144.
28. **IVC-1:** Industrial vinyl curtain.
    a. Color: Curtain and Valance “White” Floor Sweep “Black”.
29. **HPIC-3:** Structural steel (columns, beams, trusses, bracing and bar joist, etc. coating).
    a. Color: To be selected by Contracting Officer.
30. **HPIC-4:** Safety Stripping.
    a. Color: High visibility safety yellow complying with OSHA Specification 1910.144.

END OF SECTION

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SECTION 09 22 16
NON-STRUCTURAL METAL FRAMING

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Non-Structural Metal Framing in accordance with provisions of Contract Documents.

B. Completely coordinate with work of other trades.

1.2 RELATED SPECIFICATION SECTIONS:

A. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 01 71 21 – Specialty Engineering Requirements.
   3. Section 05 50 00 – Metal Fabrications.
   4. Section 06 10 00 – Rough Carpentry.
   5. Section 07 84 00 – Firestopping.
   6. Section 07 92 00 – Joint Sealants.
   7. Section 09 29 00 – Gypsum Board.

1.3 REFERENCES

A. The American Iron and Steel Institute (AISI):
   1. AISI S220, North American Standard for Cold-Formed Steel Framing – Nonstructural Members.

B. ASTM International (ASTM):
   1. ASTM A641, Standard Specification for Zinc–Coated (Galvanized) Carbon Steel Wire.
   2. ASTM A653, Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
   3. ASTM A1008, Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Required Hardness, Solution Hardened, and Bake Hardenable.
   4. ASTM B633, Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel.
   5. ASTM C645, Standard Specification for Nonstructural Steel Framing Members.
   6. ASTM C754, Standard Specification for Installation of Steel Framing Members to Receive Screw-Attached Gypsum Panel Products.
   7. ASTM C840, Standard Specification for Application and Finishing of Gypsum Board.
   8. ASTM E1190, Standard Test Methods for Strength of Power-Actuated Fasteners Installed in Structural Members.
   9. ASTM E1512, Standard Test Methods for Testing Bond Performance of Bonded Anchors.

C. Gypsum Association (GA):
   1. GA-600, Fire Resistance and Sound Control Design Manual Gypsum Systems.

1.4 SUBMITTALS

A. Product Data:
   1. Manufacturer’s specifications for each type of material and accessory.
      a. Where fire resistance classification is indicated, submit copies of nationally recognized testing laboratory listings of products proposed for use.
   2. Where EQ coatings are used, submit copies of nationally recognized testing laboratory results showing conformance with ASTM A653 and AISI S220.
      a. Include data required to show specification compliance.

B. Shop Drawings:
   1. Sizes and spacing of typical framing members and opening framing.
   2. Show locations and sizes of typical framing members, wall framing sections, and opening elevations.
   3. Methods of fastening framing members to each other and to supporting systems.
   4. Details of vertical deflection connections to structures.
   5. Locations and spacing of lateral bracing and structural bracing systems.
   6. Accessory products required for complete installation.
   7. Bear seal and signature of Registered Professional Engineer licensed in the State of Arizona.

C. Structural Calculations:
   1. Comply with Section 01 71 21 - Specialty Engineering Requirements.
   2. Submit calculations bearing seal and signature of Registered Professional Engineer licensed in the State of Arizona.
      a. Description of design criteria.
      b. Analysis depicting stress and deflection requirements for each framing application.
      c. Selection of framing components, accessories, fasteners, and welded connection requirements.
   3. Submit concurrent with Shop Drawings.

1.5 QUALITY ASSURANCE

A. Manufacturer Qualifications:
   1. Member of Certified Steel Stud Association (CSSA), Steel Framing Industry Association (SFIA), or Steel Stud Manufacturers Association (SSMA).

A. Code-Compliance Certification of Studs and Tracks:
   1. Provide documentation that framing members are certified according to the product-certification program of the Certified Steel Stud Association and the Steel Framing Industry Association or the Steel Stud Manufacturers Association.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Non-Structural Metal Framing:
      a. Basis of Design: ClarkDietrich Building Systems.
      b. Or Equal.
   2. Isolation Strip Material:
      a. Basis-of-Design: Reflectix, Inc.
      b. Or Equal.
   3. Products proposed for use in fire-rated assemblies:
      a. Approved by nationally recognized testing laboratory.
   4. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 DESIGN CRITERIA

A. Select steel studs in accordance with manufacturer's standard load tables and following design pressures and maximum deflections:

Performance Criteria		
Use Condition [2]	Design Pressure	Maximum Deflection
Wall enclosing stairs, and other vertical shafts	10 psf	L/240
Wall enclosing vestibules and similar spaces subject to intermittent exposure to exterior wind conditions	15 psf	L/240
Walls scheduled with Tile Backer Board, Moisture-Resistant, Impact-Resistant, or Abuse-Resistant Gypsum Wallboard	5 psf	L/360
Walls scheduled to receive Tile, lath and plaster, or veneer plaster. [1]		L/360
Typical Interior Walls/Partitions (those not listed above)	5 psf	L/240
Interior Ceilings, Soffits and Bulkheads	5 psf	L/360

Footnotes

1. Limit deflection to L/360 where wall cladding on either face is any of the following: Ceramic Tile, Stone Tile, Porcelain Tile and similar brittle finishes which are prone to movement induced cracking.
2. Where elements meet multiple conditions; Use most stringent Deflection and Design Pressure values.

2.3 MATERIALS

A. Metal Studs and Floor Tracks:
   1. C-shaped, roll formed studs and tracks conforming to ASTM C645.
   2. Steel design standard: 33KSI.
   3. Galvanized: G40 or G40EQ conforming to ASTM A653 and AISI S220.
   4. Stud and track depths: As indicated by wall type.
   5. Minimum flange width: 1-1/4-inch.
   6. Minimum thickness: 30 mil (20 GA), except as follows:
      a. Increase member thickness to comply with performance criteria.
   7. In lieu of increased member thickness, design may employ diagonal braces above ceiling to reduce overall span.
      a. Coordinate locations with building services items.
      b. Do not employ studs with member thickness less than allowed by fire resistance rated assemblies.
   8. High strength 50 ksi or 70 ksi studs shall comply with design criteria of equivalent thickness standard 33 ksi studs listed.
   9. At walls designated STC 40 and above, use only studs with physical characteristics of studs used in documented STC testing.

B. Flexible Track:
   1. Adjustable segmented track fabricated from same material, gauge, and width as specified for straight track.
   2. Use at curved walls and soffits.

C. Head of Wall Accessories:
   1. Configure to accommodate deflection of superstructure without inducing axial loading on partition wall.
   2. Maintain structural integrity, fire and smoke-resistance, and sound control as required by each wall.
   3. Slotted top deflection track:
      a. Deep leg, vertically slotted track.
      b. Cold-formed sheet steel; galvanized; G60.

c. Thickness: 30 mil (20 GA) minimum.
d. Width: As required for stud sizes indicated.
e. Depth: Minimum 2-1/2-inch down-standing legs with 1/4-inch wide by 1-1/2-inch high slots spaced 1-inch on center.

4. Z-bars, cold-formed channels, and clips:
   a. Accommodate thickness of spray-applied fire-resistive materials.
5. UL-listed fire-resistant components tested for compliance with requirements indicated.
6. Firestopping Materials:
   a. Sealants, sprays, intumescent strips and forming materials.
   b. Coordinate with sealants specified in Section 07 84 00 and Section 07 92 00.

D. Furring Channels:
1. Hat shaped sections.
2. Galvanized: G40 or certified equivalent.
3. Sizes: 7/8-inch and 1-1/2-inch, as indicated.
4. Minimum Thickness: 30 mil (20 GA); Use heavier gauge as dictated by conditions.

E. Metal Backing:
1. General:
   a. See Drawings for applications of backing types listed and further details.
   b. Use heavier gauge as necessary for items to be supported.
   c. Comply with manufacturer's backing requirements if capacity exceeds types listed.
2. Flat Plate (Type A):
   a. Flat, sheet metal stock per ASTM A1008.
   b. G40 galvanized or certified equivalent.
   c. Thickness: 43 mil (18 GA) minimum.
   d. Install at walls with heavy loading requirements.
3. Metal Backing (Type B):
   a. C-shaped modified track runners.
   b. G40 galvanized or certified equivalent.
   c. Backing height: 6 inches minimum.
   d. Flange width: 1-1/4-inch minimum.
   e. Thickness: 30 mil (20 GA) minimum.
   f. May be installed continuously across multiple stud spaces.
   g. Use where no other type of backing is designated.
4. Metal Backing (Type C):
   a. C-shaped modified track runners.
   b. G40 galvanized or certified equivalent.
   c. Backing height: 8 inches minimum.
   d. Flange width: 1-1/4-inch minimum.
   e. Thickness: 54 mil (16 GA) minimum.
   f. Install at single stud space with properly oriented studs.
5. Laboratory casework and wall mounted equipment without specific backing requirements:
   a. Type B backing as listed above with revised minimum thickness.
   b. Thickness: 54 mil (16 GA) minimum.
      1) Use heavier gauge as necessary for items to be supported.

F. Z-Furring:
1. Z-shaped sections attached to structural parent wall.
2. Galvanized: G40 or certified equivalent.
3. Sizes: 1, 1-1/2, and 2 inches.
4. Thickness: 18 mil (25 GA) minimum; Use heavier gauge as dictated by conditions.

G. Accessory Items:
1. Wire Ties:
   a. Minimum thickness: 43 mil (18 GA) soft annealed, galvanized.
2. Track Fasteners:

a. Power driven type, to withstand minimum 190 pounds shear when driven.

H. Support Systems for Gypsum Ceilings:
1. Stud-Framed Ceiling/Soffit Systems:
a. C-shaped studs or joists; roll-formed.
b. Galvanized: G40.
c. Frame member depth: 3-5/8-inch minimum, unless otherwise indicated.
1) Use wider stud sections if ceiling span and support requires.
d. Flange width: 1-1/4-inch minimum.
e. Stud thickness: 33 mil minimum.
2. Tie Wire:
a. ASTM A641, Class 1 zinc coating, soft temper.
b. Diameter, single strand: 62 mil (14 GA) minimum.
c. Diameter, double strand: 42 mil (18 GA) minimum.
3. Wire Hangers:
a. ASTM A641, Class 1 zinc coating, soft temper.
b. Diameter: 97 mils (12 GA) minimum.
4. Anchors in Concrete:
a. Anchors of type and material indicated below, with holes or loops for attaching hangers of type indicated and with capability to sustain, without failure, a load equal to 5 times that imposed by ceiling construction, as determined by testing per ASTM E488 or ASTM E1512 as applicable.
b. Acceptable types: Cast-in-place, post-installed expansion anchors and post-installed bonded anchors.
c. Material: Carbon-steel components zinc plated to comply with ASTM B633, Class Fe/Zn 5 for Class SC 1 service condition.
5. Powder-Actuated Fasteners in Concrete:
a. Fastener system of type suitable for application indicated, fabricated from corrosion-resistant materials, with clips or other accessory devices for attaching hangers of type indicated, and with capability to sustain, without failure, a load equal to 10 times that imposed by ceiling construction, as determined by testing per ASTM E1190.
b. Comply with seismic design requirements where applicable.
6. Other items including suspension wire, tie wire, attachment devices: As specified and indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine supporting structure and conditions under which system will be installed.

B. Correct conditions detrimental to proper installation.

C. Installation constitutes acceptance of responsibility for performance.

3.2 INSTALLATION - GENERAL

A. Layout and install metal framing accurate to dimensions indicated in drawings.

B. Installation Standard: ASTM C754, except comply with framing sizes and spacing indicated.
1. Gypsum Board Assemblies: Comply with additional requirements in ASTM C840 relative to framing installation.

C. Install supplementary framing, and blocking to support fixtures, equipment services, heavy trim, grab bars, toilet accessories, wall stops, furnishings, or similar construction.

D. Install bracing at terminations in assemblies.

E. Do not bridge building control and expansion joints with non-load-bearing steel framing members. Frame both sides of joints independently.

F. Where studs are installed directly against exterior masonry walls or dissimilar metals at exterior walls, install isolation strip between studs and exterior wall.

G. Extend framing full height to structural supports.
   1. Exception: Where partitions are indicated to terminate at, or just above, suspended ceilings.
   2. Continue framing around ducts and similar items which penetrate partitions.

H. Position studs vertically engaging floor track and head of wall deflection track.
   1. Align stud knockouts to facilitate running of wires and conduit.

I. Space studs maximum 16-inches on center.

J. Provide additional studs at corners, partition intersections and terminations of partitions, and at each side of control joints.

K. Positively anchor studs to floor tracks with self-tapping pan head screws, or stud clinching tool per ASTM C754.

L. Anchor studs to deflection track with wafer head screws on both flanges of each stud.
   1. Maintain deflection gap between top of stud and top of slotted track.
   2. Install screws at centerline of slot and secure allowing vertical movement. Do not over secure screws.

M. Anchor fire rated partitions as required by fire resistance design, and firestopping design.

N. Where partitions abut vertical structural elements, provide perimeter relief per Gypsum Association GA-600 Strain Relief System details.

O. Head-of-Wall:
   1. Provide slotted top track for walls extended to structure.
   2. Configure to resist lateral loads while accommodating deflection of overhead building superstructure without inducing axial loading on partition framing.
   3. Secure deflection track to structure in accordance with industry standards and regulatory requirements.
   4. Secure at corners and at ends.
   5. Cut vertical studs 3/4-inch short to create a deflection gap when installed into top track.
      a. Secure vertical studs to top track with framing screw at each stud, screwing through track slots for positive stud connection.
   6. Secure Gypsum Wallboard to vertical studs; do not secure Gypsum Wallboard to top track directly.
   7. Prepare wall for installation of seals, firestopping, or both:
      a. Fire-rated Walls: Prepare for fire-resistive joint assemblies specified in Section 07 84 00.
      b. Non-fire rated partitions including Smoke Partitions: Prepare for Acoustical Sealant specified in Section 07 92 00.

P. Furring Channels:
   1. Attach furring channel systems directly to parent walls.
   2. Install channels at maximum 16-inches on center.
   3. Provide additional framing at openings, cutouts, corners, and control joints.
   4. Space fasteners not more than 24-inches on center, staggered on opposite flanges of furring channels.

Q. Sound Isolation Clips:
   1. Install per manufacturer's instructions.
   2. Where electrical device, outlet, and service boxes are fastened to framing with Sound Isolation Clips:
      a. Leave a gap between the gypsum wallboard and the electrical device, outlet, or service box to avoid flanking of the Sound Isolation Clips.
      b. Seal the gap with Acoustical Sealant.

R. Stud Wall Isolation Strip:
   1. Install per manufacturer's instructions.

3.3 FRAMING AT OPENINGS

A. Control Joints (CJ):
   1. Install additional stud, maximum 1/2-inch from jamb studs.
   2. Do not fasten extra stud to track or jamb stud.
   3. Refer to specification Section 09 29 00 for control joint locations.

B. Prefabricated headers, jambs, and sill framing systems option:
   1. Proprietary opening framing systems may be used as an alternative to conventionally fabricated framing.

C. Door Openings:
   1. Screw vertical studs at jambs to jamb anchor clips on door frames; install runner track section for cripple studs at head and secure to jamb studs. Screw into web of jamb stud.
   2. Unless indicated otherwise, extend jamb studs through suspended ceilings and secure laterally to overhead structure.
   3. Jamb Studs:
      a. Minimum thickness of jamb studs: 30 mil (20 GA) at openings.
      b. Install two studs at each jamb, toe-to-toe unless otherwise indicated.
         1) Securely attach first stud to frame.
         2) Fill cavity between studs with acoustic batt insulation where required by acoustical rating of wall.
         3) Join second stud to first stud on each face with 30 mil (20 GA) screw attached steel straps at 42-inches on center maximum.
   4. Headers:
      a. Openings less than 4-feet-6-inches wide:
         1) Cut-to-length section of floor runner above and below wall openings.
         2) Cut flanges and bend webs at ends.
         3) Overlap and screw attach jamb studs to frames.
      b. Openings over 4-feet-6-inches wide:
         1) Cut-to-length, horizontal box beam studs above and below wall openings.
         2) Design for actual span and loading.
      c. Incorporate miscellaneous steel members, specified in Section 05 50 00, and wood blocking, specified in Section 06 10 00, where indicated.
   5. Control Joints at head of jambs:
      a. Install cripple studs at head adjacent to each jamb stud, with a minimum 1/2-inch clearance from jamb stud to allow for installation of control joint in finished assembly.
      b. Gypsum Wallboard control joints as specified in Section 09 29 00.

D. Other Framed Openings:
   1. Frame openings other than door openings the same as required for door openings, unless otherwise indicated.
   2. Cripple Studs:
      a. Install cut-to-length intermediate vertical studs above and below openings.
      b. Spacing: As indicated for typical full-length studs.
      c. Match stud framing below sills of openings to framing installed above opening.
   3. Incorporate miscellaneous steel members, specified in Section 05 50 00, and wood blocking, specified in Section 06 10 00, where indicated.

3.4 WALL BACKING AND BLOCKING

A. Provide in-wall backing reinforcement where following items are mounted to interior walls and interior face of exterior walls:
   1. Wall-mounted door stops.
   2. Contractor or Government furnished equipment indicated to be wall mounted.
   3. Toilet accessories that do not include proprietary backing devices.

4. Toilet partitions and lockers.
5. Markerboards, tackboards, and chalkboards.
6. Other wall-mounted items where backing is indicated by details or specification.

B. Metal Wall Backing requirements are described as Accessory Items in Part 2 of this section.

C. Wood Wall Blocking may be used where allowed by Code and Manufacturer of item to be supported. Specified in Section 06 10 00.

D. Verify metal stud framing has been installed to support wall-mounted items specified in Section 05 50 00.

E. Coordinate mounting height, location, and coverage with item to be supported.

F. Determine material width according to item to be supported.

G. Provide in-wall metal wall backing material to interior metal stud walls specified herein.

H. Attachment: Minimum (2) #10 sheet metal screws at each stud.

3.5 CEILING FRAMING

A. Install in compliance with manufacturer's recommendations.

B. Provide required items to support and trim out neatly, flush or recessed mechanical and electrical items.

C. Frame openings in ceiling support system to accommodate access panels and similar openings and penetrations.
1. Completely frame openings with closed channel side of stud facing opening for support of recessed mechanical and electrical items.

3.6 CEILING SUPPORT SYSTEMS

A. Install suspension system components in sizes and spacing indicated on Drawings, but not less than required by referenced installation standards for assembly types and other assembly components indicated.

B. Isolate suspension systems from building structure where abutting or penetrated by building structure.

C. Suspend hangers from building structure as follows:
1. Install hangers plumb and free from contact with insulation or other objects that are not part of supporting structural or suspension system.
   a. Splay hangers only where required to miss obstructions and offset resulting horizontal forces by bracing, counter-splaying, or other equally effective means.
2. Where width of ducts and other construction within ceiling plenum interfere with locations of hangers required to support standard suspension system members, install supplemental suspension members and hangers in the form of trapezes or equivalent devices.
   a. Size supplemental suspension members and hangers to support ceiling loads within performance limits established by referenced installation standards.
3. Wire Hangers: Secure by looping and wire tying, either directly to structures or to inserts, eye screws, or other devices and secure fasteners appropriate for substrate.
4. Flat Hangers: Secure to structure, including intermediate framing members, by attaching to inserts, eye screws, or other devices and secure fasteners appropriate for structure and hanger.
5. Do not attach hangers to steel roof deck.
6. Do not attach hangers to permanent metal forms. Furnish cast-in-place hanger inserts that extend through forms.
7. Do not attach hangers to rolled-in hanger tabs of composite steel floor deck.
8. Do not connect or suspend steel framing from ducts, pipes, or conduit.

END OF SECTION

SECTION 09 29 00
GYPSUM BOARD

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Gypsum board.
   2. Moisture - Resistant gypsum wallboard.
   3. Coated tile backer gypsum board.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 06 10 00 - Rough Carpentry.
   2. Section 07 92 00 - Joint Sealants.
   3. Section 09 22 16 - Non-Structural Metal Framing.
   4. Section 09 30 13 - Ceramic Tile (CT).
   5. Section 09 91 00 - Architectural Painting.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American National Specifications Institute (ANSI):
      a. A108.11, Installation of Ceramic Tile.
   2. ASTM International (ASTM):
      a. A653/A653M, Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
      b. C11, Standard Terminology Relating to Gypsum and Related Building Materials and Systems.
      c. C36, Standard Specification for Gypsum Wallboard.
      d. C475/C475M, Standard Specification for Joint Compound and Joint Tape for Finishing Gypsum Board.
      e. C557, Standard Specification for Adhesives for Fastening Gypsum Wallboard to Wood Framing.
      f. C630, Standard Specification for Glass Mat Gypsum Substrate for Use as Sheathing.
      g. C840, Standard Specification for Application and Finishing of Gypsum Board.
      h. C1002, Standard Specification for Steel Self-Piercing Tapping Screws for Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs.
      i. C1047, Standard Specification for Accessories for Gypsum Wallboard and Gypsum Veneer Base.
      j. C1177, Standard Specification for Glass Mat Gypsum Substrate for Use as Sheathing.
      k. C1396/C1396M, Standard Specification for Gypsum Board.
      l. D3273, Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber.
      m. E119, Standard Test Methods for Fire Tests of Building Construction and Materials.
   3. Gypsum Association (GA):
      a. GA-214, Recommended Levels of Gypsum Board Finish.
   4. Underwriters Laboratories, Inc. (UL):
      a. Building Materials Directory.
      b. Fire Resistance Directory.
   5. United States Gypsum Company:
      a. Gypsum Construction Handbook.

1.3 DEFINITIONS

A. Gypsum Board Terminology: Refer to ASTM C11 for definitions of terms for gypsum board assemblies.

B. Wet Area:
   1. Toilet rooms, showers, laboratories, janitor closets, or similar areas.
   2. Areas within 5 feet of emergency showers, eye wash stations, service sinks, or mop sinks.

1.4 QUALITY ASSURANCE

A. Fire - Test - Response Characteristics: For gypsum board assemblies with fire - resistance ratings, provide materials and construction identical to those tested in assembly indicated according to ASTM E119 by an independent testing and inspecting agency acceptable to authorities having jurisdiction.

B. Fire - Resistance - Rated Assemblies: Indicated by design designations from UL's "Fire Resistance Directory." Or GA - 600, "Fire Resistance Design Manual."

1.5 SUBMITTALS

A. Shop Drawings:
   1. Drawings of unusual conditions.
      a. Control joint layout.

B. Product technical data including:
   1. Submit manufacturer’s specifications for each product submitted showing compliance with requirements of standards referenced.
   2. Manufacturer's installation instructions.
   3. Manufacturer's trims, adhesive, joint treatment compound and tape recommendations.

1.6 PRODUCT HANDLING

1. Coordinate delivery with installation to minimize storage periods at the project site. Deliver in manufacturer's unopened containers, bundles or packages, fully identified with manufacturer's name, brand, type, and grade.
2. Store materials inside, under cover and keep them dry and protected against damage from weather, direct sunlight, surface contamination, corrosion, construction traffic and other causes. Neatly stack gypsum boards flat to prevent sagging.
3. Handle gypsum boards to prevent damage to edges, ends and surfaces. Do not bend or otherwise damage metal corner bead and trim.

1.7 JOB CONDITIONS

1. Examine all parts of supporting structure and conditions under which the wallboard is to be installed. Do not proceed with installation until unsatisfactory conditions have been corrected. Coordinate installation with the work of other trades to allow time for the correct installation of their work.
2. Establish and maintain environmental conditions for application and finishing gypsum board to comply with ASTM C840 and with gypsum board manufacturer's written recommendations whichever are more stringent
3. Minimum Room Temperatures: For non - adhesive attachment of gypsum board to framing, maintain not less than 60-degree F. For adhesive attachment and finishing of gypsum board, maintain not less than 60-degree F for 48 hours prior to application and continuously thereafter until drying is complete.
4. Ventilate building spaces to remove water not required for drying joint treatment materials. Avoid drafts during dry, hot weather to prevent materials from drying too rapidly.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Gypsum Board:
      a. Georgia-Pacific Gypsum LLC – Tough Rock Fireguard.
      b. Or Equal.
   2. Moisture - Resistant Gypsum Wallboard.
      a. Georgia-Pacific Gypsum LLC – Tough Rock Fireguard Moisture Guard.
      b. Or Equal.
   3. Coated Tile Backer Gypsum Board:
      a. Georgia-Pacific Gypsum LLC – DensShield Tile Backer.
      b. Or Equal.

2.2 MATERIALS

A. General:
   1. Provide UL Listed materials in fire-resistant rated construction.
   2. Furnish in lengths as long as practicable.

B. Gypsum Board (GB): Interior Locations.
   1. ASTM C36.
   2. ASTM C1396/C1396M.
   3. Edges: Tapered.
   4. Thickness: Unless noted otherwise.
      a. 5/8-inches, Type X.

C. Water and Mold Resistant Gypsum Board (WRGB): Interior Locations.
   1. ASTM C630.
   2. Water-resistant core and treated paper facers.
   3. Smooth face for finishing similar to standard gypsum board.
   4. Mold-resistant: ASTM D3273.
   5. Thickness:
      a. 5/8-inches, Type X.

D. Coated Tile Backer Board: Interior Locations.
   1. ASTM C1177.
   2. Coated water-resistant tile backer board.
      a. Mold Resistant, ASTM D3273.
   3. Thickness:
      a. 5/8-inches, Type X.

2.3 ACCESSORIES

A. Trim:
   1. ASTM C1047.
   2. Galvanized: ASTM A653/A653M G-60, unless noted otherwise.
   3. Interior Trim: Comply with ASTM C 1047.
   4. Shapes:
      a. Corner bead: Metal Corner bead formed from zinc alloy, with flanges knurled and perforated or of fine - mesh expanded metal. Use at outside corners, unless otherwise indicated.
      b. LC - Bead: J - shaped, steel edge trim formed from galvanized steel, complying with type “LC” Bead per Figure 1 of ASTM C840 with exposed long flange which receives joint compound. Use at exposed panel edges.

c. Expansion (Control) Joint: One - piece, rolled zinc with V - shaped slot 1/4-inch wide by approximately 7/16-inch deep with a removable strip covering slot opening. Use where required.

B. Fasteners:
1. Gypsum board, water and moisture resistant gypsum board, and tile backer board:
a. ASTM C1002.
b. Self-drilling, corrosion-resistant bugle head screws.
1) For fastening gypsum board to metal framing: Type S.

C. Adhesive:
1. As recommended by board manufacturer and as follows:
a. Laminating Adhesive: Adhesive or joint compound recommended for directly adhering gypsum panels to continuous substrate.
b. Fastening Adhesive for Wood: Comply with ASTM C557.
c. Fastening Adhesive for Metal: Special adhesive recommended for laminating gypsum boards to steel framing.

D. Joint Tape:
1. ASTM C475/C475M.
2. Recommended by manufacturer for specified board type and location.

E. Joint Treatment Compound:
1. ASTM C475/C475M.
2. Recommended by manufacturer for specified board type and location.
3. Pre - filling: At open joints, rounded or beveled panel edges, and damaged surface areas, use setting - type taping compound.
4. Embedding and First Coat: For embedding tape and first coat on joints, fasteners, and trim flanges, use setting - type taping compound or drying - type, all - purpose compound.
a. Use setting - type compound for installing paper - faced metal trim accessories.
5. Fill Coat: For second coat, use setting - type, sandable topping compound or drying - type, all - purpose compound.
6. Finish Coat: For third coat, use setting - type, sandable topping compound or drying - type, all - purpose compound.
7. Moisture - Resistant Gypsum Board: Use setting - type taping and setting - type, sandable topping compounds.

PART 3 - EXECUTION

3.1 EXAMINATION

A. General:
1. Install materials listed in accordance with manufacturer's specifications and recommendations when not in conflict with specific directions included herein.

B. Installation shall be done by experienced craftsmen trained to this type of work. Examine substrates to which drywall construction attaches or abuts, preset hollow metal frames, cast - in - anchors and structural framing, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of drywall construction. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Suspended Ceilings: Coordinate installation of ceiling suspension systems with installation of overhead structure to ensure that inserts and other provisions for anchorages to building structure have been installed to receive ceiling hangers at spacing required to support ceilings and that hangers will develop their full strength.

B. Furnish concrete inserts and other devises indicated to other trades for installation in advance of time needed for coordination and construction.

C. Coordination of concealed construction, supplementary framing and bracing, and blocking:

D. Be aware of all toilet accessories, toilet partitions, special items, cabinet work, case work or any other items which are to attach to and / or hang from partitions.

E. Coordinate the installation of anchors furnished by suppliers of the above mentioned items as well as blocking and backing as specified under Carpentry Section. Provide all necessary double studs, or studs spaced at 1/2 normal spacing to adequately support the weight of all items to be secured to or hung from the partitions.

F. Provide necessary anchors for items mentioned above that are not normally furnished with those items and adequately secure all anchors to two or more studs before applying board.

G. The completed drywall partitions shall be constructed and adequately braced to withstand the weight of all equipment to be hung on the partitions in addition to all loads to be superimposed on that equipment.

H. Install supplementary framing, blocking and bracing at terminations in the work and for support of mixtures, equipment services, heavy trim, grab bars, toilet accessories, furnishings and similar construction to comply with details indicated and with recommendations of gypsum board manufacturer, or if non-available, with "Gypsum Construction Handbook" published by United States Gypsum Company.

I. When outdoor temperature is below 60 degrees F., heat shall be maintained in the building continuously and uniformly at not less than 60 degrees F. from one week prior to beginning of installation until the wallboard application and joint treatment is completed.

J. Ventilation, either natural or supplied by fans, circulators, or air conditioning system, shall be provided to remove excess moisture during joint treatment.

3.3 INSTALLATION

A. General:
   1. Verify that metal stud framing has been installed plumb, true, and in accordance with the Contract Documents.
   2. Prior to application of gypsum board, ensure that all blocking, backing and bracing has been installed as necessary for the support of appurtenant items.
   3. Install gypsum board in accordance with ASTM C840.
   4. Install board in fire-rated construction in accordance with UL requirements.
      a. Self-adhesive applied fire rated tape is not acceptable for use on board joints in fire rated walls and suspended ceilings.
      b. Tape all joints using conventional fire rated joint tape and joint treatment compound.
   5. Erect all board vertically with edges over supporting members.
      a. See Specification Section 09 22 16 non-structural metal framing.
   6. Secure to each support or framing member with screws.
      a. Provide fasteners of sufficient length to penetrate framing members or studs not less than 3/8-inches.
   7. In areas having gypsum board ceilings and walls, install ceiling first.
   8. Bring boards into contact, but do not force into place.
   9. Fit neatly and carefully.
   10. Stagger edge joints on opposite side of a partition so they occur on different framing members.
   11. Hold board in firm contact with support while fasteners are being driven.
   12. Proceed with attachment from center of board toward ends and edges.
   13. Scribe board prior to cutting.
   14. In fire-rated ceiling assemblies, fasten minimum 4-inches wide drywall strip continuous over all joints on backside of board.
   15. Where gypsum board abuts concrete, masonry, metal deck, exterior doors and windows, or other dissimilar material; provide 3/8-inches joint between edge of gypsum board and abutting material.

a. Provide continuous casing bead trim on edge of board.
b. Seal joint with sealant and backer rod.
c. See Specification Section 07 92 00 for sealant.

16. Use water-resistant gypsum board (WRGB) in wet interior locations not scheduled to receive tile finish.
17. Use Coated Tile Backer Gypsum Board in wet interior locations scheduled to receive tile finish.
    a. Install in accordance with ANSI A108.11 and manufacturer's recommendations.
    b. Use in areas where ceramic tile is scheduled.

B. Installation:
1. Set fasteners between 3/8 and 1/2-inches from edges and -inches in from board corner.
    a. Space maximum of 12-inches on center at edges and in field of board.
    b. Where board butts at wall/ceiling juncture, hold fasteners back 6inches from edges.
    c. Space fasteners closer if required by UL.
2. Install fasteners, in gypsum board, so that head rests in a slight dimple without cutting face paper or fracturing core or as recommended by board/panel manufacturer.

C. Control Joints:
1. Install prefabricated control joints to provide following maximum unjointed lengths or areas:
    a. Partitions: 30 feet, maximum straight run, and at both sides of jamb from head of each door, borrow lite, or window opening to top of partition.
        1) Control joints are not required at openings where the partition forms a cross-corridor condition.
        2) At borrow lites, windows or similar conditions, extend control joints at jamb from sill to floor.
    b. Ceilings:
        1) 50 feet maximum in one direction,
        2) At change of direction or irregular shapes.
        3) Ceiling area: 2,500 square feet, maximum.
2. Seal control joints.
3. Use color to match wall or ceiling color as closely as possible where control or expansion joints occur in fire or sound rated assemblies, install suitable backing material to maintain required rating.
4. Where a partition or ceiling abuts a structural element or dissimilar wall or ceiling, install corner bead, casing bead or other trim as required.

D. Gypsum Board Finishing:
1. Securely attach continuous corner beads to all external corners in accordance with manufacturer's recommendations.
2. Provide the following minimum levels of gypsum board finish in accordance with GA-214.
    a. Areas exposed to view:
        1) Surfaces to receive: Level #4.
    b. Areas not exposed to view:
        1) Fire rated partitions: Level #2 unless a higher grade of finish is required by UL.
        2) Non-fire rated partitions: Level #2.
    c. Provide additional coats of joint compound as required to completely conceal joints, fasteners and accessories.
        1) Joint telegraphing will not be acceptable.
3. Sand each coat to remove excess joint compound.
    a. Avoid roughing paper facing on board.
4. Finish surface shall be smooth and free of tool marks and ridges.
5. Prime gypsum board surfaces in accordance with Specification Section 09 91 00.
    a. After primer has been applied, inspect surfaces and repair and refinish all areas which show defects.
6. Refer to ASTM C840 for additional finishing requirements.

E. Coated Tile Backer Gypsum Board Finishing:
   1. Follow the manufacturer's published installation instructions.
   2. In showers or similar wet areas scheduled to receive a tile finish:
      a. Seal perimeter, joints and penetrations in accordance with manufacturer's instructions to provide a water-tight installation.
      b. Apply fiberglass mesh tape and finish with latex modified mortar per manufacturer's instructions.
      c. Cover fasteners with latex modified mortar.
   3. In wet areas scheduled to receive a painted finish:
      a. Apply paper joint tape with setting-type joint compound.

3.4 FIELD QUALITY CONTROL

A. Above - Ceiling Observation: Before Contractor installs gypsum board ceilings, Contracting Officer Representative shall be informed so that they can conduct an above - ceiling observation and report deficiencies in the Work observed. Do not proceed with installation of gypsum board to ceiling support framing until deficiencies have been corrected.
   a. Notify Contracting Officer Representative seven days in advance of date and time when Project, or part of Project, will be ready for above - ceiling observation.
   b. Before notifying Contracting Officer Representative, complete the following in areas to receive gypsum board ceilings:
      1) Installation of 80 percent of lighting fixtures, powered for operation.
      2) Installation, insulation, and leak and pressure testing of water piping systems.
      3) Installation of air-duct systems.
      4) Installation of air devices.
      5) Installation of mechanical system control - air tubing.
      6) Installation of ceiling support framing.

3.5 CLEANING AND PATCHING

A. Clean exposed drywall surfaces free of soil and stain that would affect finish.

B. Repair or remove and replace defective work.

C. Remove all excess materials and debris from the site.

D. Provide final protection and maintain conditions, in a manner suitable to Installer, which ensures gypsum drywall construction being without damage or deterioration at time of Final completion.

END OF SECTION

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SECTION 09 30 13
CERAMIC TILE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Work under this section consists of furnishing all labor, materials and equipment necessary for and incidental to the complete and proper installation of all ceramic tile and related work as shown on the drawings or specified herein, and in accordance with all applicable requirements of the Contract Documents. Work within this Section shall be completely coordinated with work of all other trades.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 - Submittal Procedures.
   2. Section 01 78 23 - Operation and Maintenance Data.
   3. Section 07 92 00 - Joint Sealants.
   4. Section 09 29 00 - Gypsum Board.
   5. Section 09 06 00 - Color Schedule.

1.2 QUALITY ASSURANCE

A. Reference Standards:
   1. American National Standards Institute (ANSI):
      a. A108.4, Installation of Ceramic Tile with Organic Adhesives or Water Cleanable Tile Setting Epoxy Adhesives.
      b. A108.5, Installation of Ceramic Tile with Dry-Set Portland Cement Mortar or Latex-Portland Cement Mortar.
      c. A108.6, Installation of Ceramic Tile with Chemical Resistant, Water Cleanable Tile - Setting and Grouting Epoxy.
      d. A108.8, Installation of Ceramic Tile with Chemical Resistant Furan Resin Mortar and Grout.
      e. A108.10, Installation of Grout in Tilework.
      f. A118.3, Chemical Resistant, Water Cleanable Tile - Setting Epoxy and Water Cleanable Tile - Setting and Grouting Epoxy Adhesive.
      g. A118.4, Latex - Portland Cement Mortar.
      h. A118.7, Polymer Modified Tile Gouts for Tile Installation.
      i. A118.11, American National Standard Specifications for EGP (Exterior Glue Plywood) Modified Dry-Set Mortar.
      j. A137.1, Ceramic Tile.
   2. ASTM International (ASTM):
      a. C109, Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or 50-mm Cube Specimens).
      b. C348, Standard Test Method for Flexural Strength of Hydraulic - Cement Mortar.
      c. C794, Standard Test Method for Adhesion-in-Peel of Elastomeric Joint Sealants.
      d. C979, Standard Specification for Pigments for Integrally Colored Concrete.
      e. D751, Standard Test Methods for Coated Fabrics.
   3. Tile Council of North America (TCNA):
      a. TCA Handbook for Ceramic, Glass, and Stone Tile Installation.
      b. TCA F113, Floors, Interior, Concrete Subfloor, Latex-Portland Cement Mortar.
      c. TCA W245, Interior Walls, Wood or Metal Studs, Coated Glass Mat Water-Resistant Gypsum Backer Board.

B. Qualifications:

1. The installer must have minimum five years' experience installing similar products with similar substrates.

C. Mock-Ups:
   1. Provide a sample wall and/or floor area minimum 4-feet x 4-feet each using specified tile, grout, special shapes and trim pieces.
      a. Sample area substrate shall be of same construction as actual project condition.
   2. Construct additional sample areas as required until accepted by Contracting Officer Representative.
   3. Sample area will constitute minimum acceptable standard of quality for actual construction.
   4. Sample area shall not be built into permanent construction.
   5. Remove sample when directed by Contracting Officer Representative.

1.3 DEFINITIONS

A. Installer or Applicator:
   1. Installer or applicator is the person actually installing or applying the product in the field at the Project site.
   2. Installer and applicator are synonymous.

B. Ceramic Tile: For purposes of this Specification Section, the term "ceramic tile" is intended to include ceramic, porcelain, granite, and terrazzo tiles.

1.4 SYSTEM DESCRIPTION

A. Manufacturer: Provide products specified in the Interior Finish Legend shown on the Drawings.

B. Tile Products: Provide tiles complying with the Standard Grade requirements of the Tile Council of America - TCA ANSI A137.1 unless indicated otherwise.

C. Handle, store, mix and apply proprietary setting and grouting materials in compliance with manufacturer's instructions.

1.5 SUBMITTALS

A. Product Data:
   1. Submit manufacturer's product technical information and data for each materials specified including:
      a. Ceramic tile, marked to show each type, size, and shape required.
      b. Chemical resistant mortar and grout (Epoxy).
      c. Dry set Portland cement mortar and grout.
      d. Divider and transition strips and cap trims.
      e. Leveling compound.
      f. Latex Portland cement mortar and grout.
      g. Commercial Portland cement grout.
   2. Manufacturer's installation instructions.

B. Samples:
   1. Refer to "Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals" for submission procedures and requirements of Color Samples.
   2. Provide selected colors, types, patterns and sizes of each tile, transition strips and trims specified in the "Interior Finish Legend" shown on the Architectural Finish Schedules Drawings.
      a. Include selected grout color as part of sample.
   3. Refer to "Specification Section 09 06 00 – Color Schedule".

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

D. Informational Submittals:

1. Certification:
    a. Provide Master Grade Certificate for each type of tile and trim specified, signed by Tile Manufacturer and Installer.
2. Letter from grout and adhesive manufacturer stating that grout and tile adhesive being used is compatible with the tile backer board specified in Specification Section 09 29 00.

1.6 PRODUCT HANDLING

A. Deliver packaged materials and store in original containers with seals unbroken and labels intact until time of use, in accordance with manufacturer's instructions.

1.7 JOB CONDITIONS

A. Maintain environmental conditions and protect work during and after installation in accordance with referenced standards and manufacturer's printed recommendations.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
    1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
    1. Ceramic Tile:
        a. Florida Tile, Inc. (FT).
        b. Or Equal.
    2. Adhesives, mortars, grouts and leveling compounds:
        a. LATICRETE International, Inc.
        b. Or Equal.
    3. Epoxy Mortar:
        a. Mapei; Kerapoxy.
        b. Or Equal.
    4. Latex - Portland Cement Mortar:
        a. Mapei - Keralastic System; Kerabond premium dry - set mortar and Keralastic acrylic latex admixture.
        b. Or Equal.
    5. Latex Portland (Unsanded) Cement Grout:
        a. Mapei; KeraColor U.
        b. Or Equal.
    6. Latex Portland (Sanded) Cement Grout:
        a. Mapei - KeraColor S.
        b. Or Equal.
    7. Metal Edge Strips:
        a. Schluter Systems; Reno - U.
        b. Or Equal.
    8. Expansion and Control Joint Sealants:
        a. Laticrete; Latasil and Latasil 9118 Primer.
        b. Or Equal.

2.2 MATERIALS

A. MORTAR MATERIALS

1. Epoxy Mortar: Comply with ANSI A118.3 and ANSI A108.6 specifications.
2. Latex modified portland cement dry - set mortar shall meet or exceed ANSI A118.4, ANSI A118.11 and ANSI A108.8 specifications. Mortar shall be manufactured of white non - re - emulsifying polymers with ethylene vinyl acetate additive. Acrylic Latex Admixture shall be of high solids latex emulsion formulated to improve the performance of flexural and tensile strength of the mortar.

B. GROUT MATERIALS
1. Coloring Pigments shall be pure mineral pigments, lime proof and non - fading, complying with ASTM C979, Coloring pigments shall be added to grout by the manufacturer. Job colored grout is not acceptable. Coloring pigments are required in Latex Portland Cement Grout.
2. Latex Portland (Sanded) Cement Grout: Proprietary compound of portland cement, sanded, non - re - emulsifying polymers with ethylene vinyl acetate additive, factory - blended to decrease shrinkage and increase moisture resistance. Grout shall be used for joints over 1/8-inch wide. Grout shall meet or exceed ANSI A118.7 specifications.
3. Latex Portland (Unsanded) Cement Grout: Proprietary compound of portland cement, unsanded, non - re - emulsifying polymers with ethylene vinyl acetate additive, factory - blended to decrease shrinkage and increase moisture resistance. Grout shall be used for joints up to 1/8-inch wide. Grout shall meet or exceed ANSI A 118.7 specifications.

C. EXPANSION AND CONTROL JOINT SEALANTS
1. Expansion and Control Joint Sealant to be a one component, neutral cure, exterior grade silicone sealant and meet the following requirements:
   a. Tensile Strength (ASTM C794): 280 pounds per square inch.
   b. Hardness (ASTM D751; Shore A): 25 (colored sealant) /15 (clear sealant)
2. Weather Resistance (QUV Weather - ometer): 10,000 hours (no change)

D. METAL EDGE STRIPS
1. Solid brass protective edge trim with integral perforated deform leg for anchorage to mortar bed and substrate by keying to mortar, unless otherwise shown. Transitions between ceramic tile and thinner or thicker flooring materials shall comply with the Americans with Disabilities Act Accessibility Guidelines (ADAAG) requirements.

E. PATCHING AND LEVELING COMPOUND
1. Portland cement base, polymer - modified, self - leveling compound, manufactured specifically for resurfacing and leveling concrete floors. Products containing gypsum are not acceptable. Patching and leveling compounds shall have the minimum following physical properties:
   a. Compressive strength 3500 psig per ASTM C109 / C109M.
   b. Flexural strength 1000 psig per ASTM C348 (28-day value).
2. Patching and leveling compounds shall be capable of being applied in layers up to 12-inches thick without fillers and up to four inches thick with fillers, being brought to a feather edge, and being trowelled to a smooth finish.
3. Primers, fillers, and reinforcement shall be installed when required by manufacturer for application and substrate condition.
4. Compound shall be ready for use in 48 hours after application.

F. CLEANING COMPOUNDS
1. Specifically designed for cleaning masonry and concrete and which will not prevent bond of subsequent tile setting materials including patching and leveling compounds and elastomeric waterproofing membrane and coat.
2. Materials containing acid or caustic material not acceptable.

2.3 MANUFACTURED CERAMIC TILE UNITS

A. Floor Tile: Provide products specified in the Interior Finish Legend shown on the Drawings or Approved Equal.

B. Wall Tile: Provide products specified in the Interior Finish Legend shown on the Drawings or Approved Equal.

Tile Base and Associated Trim: Provide factory made trim shapes for all curbs, caps, corners, bullnose, bases, coves, beads, moldings, etc.

2.4 FINISHES

A. Provide selected colors, types, patterns and sizes of each tile, transition strips, trims and grout specified in the "Interior Finish Legend found on Architectural Drawings".
   1. Also refer to "Specification Section 09 06 00 – Color Schedule".

2.5 MAINTENANCE MATERIALS

A. Extra Material:
   1. Furnish Government with the following extra materials:
      a. 2 percent of each different size, pattern, style and/or color of tile used including trim shapes with minimum of one full size tile of each for maintenance purposes.
      b. Enough extra grout mix of each different color and/or type to allow for full 2 percent of tile to be replaced.

PART 3 - EXECUTION

3.1 ENVIRONMENTAL REQUIREMENTS

A. Maintain ambient temperature of work areas at not less than 60 degrees F, without interruption, for not less than 24 hours before installation and not less than three days after installation.

B. Maintain higher temperatures for a longer period of time where required by manufacturer's recommendation and ANSI Specifications for installation.

C. Do not install tile when the temperature is above 100 degrees F.

D. Do not install materials when the temperature of the substrate is below 60 degrees F.

E. Do not allow temperature to fall below 50 degrees F after the fourth day of completion of tile work.

3.2 ALLOWABLE TOLERANCE

A. Variation in plane of subfloor, including concrete leveling compounds and mortar beds:
   1. Not more than 1/4-inch in 10-feet from required elevation where Portland cement mortar setting bed is used.
   2. Not more than 1/8-inch in 10-feet where dry-set Portland cement, and latex-Portland cement mortar setting beds and chemical-resistant bond coats are used.

B. Variation in Plane of Wall Surfaces:
   1. Not more than 1/4-inch in eight feet from the required plane where Portland cement mortar setting bed is used.
   2. Not more than 1/8-inch in eight feet where dry set or latex Portland cement mortar or organic adhesive setting materials are used.

3.3 SURFACE PREPARATION

A. Cleaning New Concrete or Masonry:
   1. Chip out loose material, clean off all oil, grease dirt, adhesives, curing compounds, and other deterrents to bonding by mechanical method, or by using products specifically designed for cleaning concrete and masonry.
   2. Use self-contained power blast cleaning systems to remove curing compounds and steel trowel finish from concrete slabs where ceramic tile will be installed directly on concrete surface with thin set materials.
   3. Steam cleaning or the use of acids and solvents for cleaning will not be permitted.

B. Patching and Leveling:
   1. Mix and apply patching and leveling compound in accordance with manufacturer's instructions.
   2. Fill holes and cracks and align concrete floors that are out of required plane with patching and leveling compound.
   3. The thickness of compound as required to bring the finish tile system to elevation shown.
   4. At substrate expansion, isolation, and other moving joints, allow joint of same width to continue through underlayment.
   5. Apply patching and leveling compound to concrete and masonry wall surfaces that are out of required plane.
   6. Apply leveling coats of material compatible with wall surface and tile setting material to wall surfaces, other than concrete and masonry that are out of required plane.

3.4 METAL EDGE STRIPS INSTALLATION

A. Provide where shown, and where exposed edge of ceramic tile flooring is to meet vinyl resilient tile, epoxy, sealed concrete, or other floor coverings, and at all other locations as indicated on Drawings. Set metal edge strips in mortar bed to line and level centered under doors, in openings or as shown on Drawings.

3.5 TILE INSTALLATION, GENERAL

A. Comply with ANSI A 108 series of tile installation standards in "Specifications for Installation of Ceramic Tile" applicable to methods of installation.

B. Before commencing work, field pattern and border line locations shall be established and the work shall be centered symmetrically so that no tile need be cut to less than half size. Joints in wall and floor tile shall be aligned vertically and horizontally; staggered joints will not be accepted. Make all cuts on the outer edge of the field. Provide uniform joint widths, unless otherwise shown.

C. New tile work scheduled for existing spaces shall align with the existing tile work unless specified otherwise.

D. Set tile firmly in place with finish surfaces in true planes. Align tile flush with adjacent tile unless shown otherwise.

E. Form intersections and returns accurately.

F. Cut and drill tile neatly without marring surface.

G. Cut or grind edges of tile abutting penetrations, finish, or built in items:

H. Fit tile closely around electrical outlets, piping, fixtures and fittings, so that plates, escutcheons, collars and flanges will overlap cut edge of tile.

I. Seal tile joints watertight with sealant, around electrical outlets, piping fixtures and fittings before cover plates and escutcheons are set in place.

J. Completed work shall be free from hollow sounding areas and loose, cracked or defective tile.

K. Remove and reset tiles that are out of plane or misaligned.

L. Tile Floors:
   1. Extend floor tile beneath casework and equipment, except those units mounted in wall recesses.
   2. Align finish surface of new tile work flush with other and existing adjoining floor finish where shown.
   3. In areas where floor drains occur, slope to drains where shown.
   4. Shove and vibrate tiles over 8-inches square to achieve full support of bond coat.

M. Tile Walls:

1. Cover walls and partitions, including pilasters, furred areas, and freestanding columns from floor to ceiling, or from floor to nominal wainscot heights shown with tile.
2. Finish reveals of openings with tile, except where other finish materials are shown or specified.
3. Finish wall surfaces behind and at sides of casework and equipment, except those units mounted in wall recesses, with same tile as scheduled for the room.

N. Joints:
1. Keep all joints in line, straight, level, perpendicular and of even width unless shown otherwise.
2. Make joints 1/16-inch wide for glazed wall tile and mosaic tile work.
3. Make joints in Porcelain type; maximum 1/8-inch wide.
4. Back Buttering: For installations indicated below, obtain 100 percent mortar coverage by complying with applicable special requirements for back buttering of tile in referenced ANSI A108 series of tile installation standards:
5. Finished tile installation shall display no uneven surfaces or high or low spots in excess of 1/8-inch in 8-feet when measured with an 8-foot straightedge in any direction. Tile floors pitched to a floor drain shall be measured at any point along an 8-foot radius from the floor drain.

3.6 EXPANSION AND CONTROL JOINT SEALANT INSTALLATION

A. Prior to installing Expansion and Control Joint Sealants remove all contaminants and foreign material from joint spaces/surfaces, such as dirt, dust, oil, water, frost, setting/grouting materials, sealers and old sealant/backer. Use manufacturers recommended primer for wet area shower stall applications. Install appropriate backing material (e.g. closed cell backer rod) based on expansion joint design. Apply masking tape to the face of tile. Use a caulking gun, or other applicator, to completely fill joints with expansion and control joint sealant. Within 5-10 minutes of filling joint, tool sealant surface to a smooth finish. Remove masking tape immediately after tooling joint. Wipe excess sealant off all surfaces immediately.

3.7 SCHEDULE OF TILE FLOOR INSTALLATION METHODS

A. For Interior Tile Floors Thin set over concrete slabs on grade comply with TCA Installation Method F113.
1. Bond Coat shall be Epoxy Mortar complying with ANSI A118.3 and 108.6.
2. Grout shall be Latex Portland (Sanded) or (Unsanded) Cement complying with ANSI A118.7.
3. Comply with Installation Specifications ANSI A108.5 for tile and ANSI A108.10 for grout.

3.8 SCHEDULE OF WALL TILE INSTALLATION METHODS

A. For Interior Tile Walls over Coated Tile Backer Gypsum Board where studs do not exceed 16-inches on center comply with TCA Installation Method W245.
1. Bond coat shall be Latex - Portland Cement Mortar.
2. Grout shall be Latex Portland (Sanded) or (Unsanded) Cement complying with ANSI A118.7.
3. Comply with Installation Specifications ANSI A108.4 or ANSI A108.5 for tile and ANSI A108.10 for grout.

3.9 CLEANING

A. Upon completion of placement and grouting, clean all ceramic tile surfaces so they are free of foreign matter.

B. Thoroughly sponge and wash tile. Polish glazed surfaces with clean dry cloths.

C. Methods and materials used shall not damage or impair appearance of tile surfaces.

D. The use of acid or acid cleaners on glazed tile surfaces is prohibited.

3.10 PROTECTION

A. When recommended by tile manufacturer, apply a protective coat of neutral protective cleaner to completed tile walls and floors. Before final inspection, remove protective coverings and rinse neutral cleaner from tile surfaces.

B. Prohibit foot and wheel traffic from using tiled floors for at least 3 days after grouting is completed.

C. Where traffic occurs over tile floor prior to final inspection, cover tile floor. Floor covers shall be maintained in good condition. Do not remove protective cover until time for final inspection. Clean tile of any tape, adhesive and stains.

3.11 REPAIRING DAMAGED WORK

A. Leave finished installation clean and free of cracked, chipped, broken, unbounded, or otherwise defective tile work.

B. Contractor shall repair by removing and replacing damaged and unacceptable portions of completed work with new work to match adjacent surfaces at no additional cost to the Government.

END OF SECTION

SECTION 09 51 00
ACOUSTICAL CEILINGS

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Acoustical Ceiling Systems in accordance with provisions of Contract Documents.

B. Completely coordinate with work of other trades.

C. See INTERIOR FINISH LEGEND on the Drawings for Basis of Design for all Acoustical Ceiling Materials, Sizes and Colors.

D. Related Sections:
1. Section 01 33 23 – Submittal Procedures.
2. Section 01 78 23 – Operation and Maintenance Data.
3. Section 09 06 00 – Color Schedule.
4. Section 23 31 00 – HVAC – Ductwork.
5. Section 26 50 00 – Interior and Exterior Lighting.

1.2 REFERENCES

A. American Society of Civil Engineers (ASCE):
1. ASCE7 Minimum Design Loads and Associated Criteria for Buildings and Other Structures

B. ASTM International (ASTM):
1. ASTM A641/A641M Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire.
2. ASTM B633 Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
3. ASTM C423 Standard Test Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method.
4. ASTM C635/C635M Standard Specification for the Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-In Panels Ceilings.
5. ASTM C636/C636M Standard Specification for Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-In Panels.
6. ASTM E90 Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements.
7. ASTM E413 Classification for Rating Sound Insulation.
8. ASTM E488/E488M Standard Test Methods for Strength of Anchors in Concrete Elements.
9. ASTM E1264 Standard Classification for Acoustical Ceiling Products.
10. ASTM E1414/E1414M Airborne Sound Attenuation Between Rooms Sharing a Common Ceiling Plenum.
11. ASTM E1477 Standard Test Method for Luminous Reflectance Factor of Acoustical Materials by Use of Integrating-Sphere Reflectometers.
12. ASTM E1512 Standard Test Methods for Testing Bond Performance of Bonded Anchors.
13. ASTM F593 Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.
14. ASTM F594 Standard Specification for Stainless Steel Nuts.

C. DIN Standards (DIN)
1. DIN 5036 Radiometric and photometric properties of materials; methods of measurement for photometric and spectral radiometric characteristics.

D. ICC Evaluation Service Report (ICC ESR):
1. ICC ESR1222 ACT Grid and Accessories.
2. ICC ESR 3336 ACT Grid and Accessories.

E. International Organization for Standardization (ISO)
   1. ISO 4622 Paints and Varnishes – Pressure Test for Stackability.

1.3 QUALITY ASSURANCE

A. Site Classification and Seismic Design Categories (as defined in the International Building Code).
   1. Code Requirements for Seismic Category C shall be required.

B. Acoustical Performance:
   1. Materials and construction identical to units whose NRC, CAC, and STC (as applicable) are determined according to ASTM C423, ASTM E1414/E1414M, ASTM E90, and ASTM E413.

1.4 SUBMITTALS

A. Product Data:
   1. Manufacturer's literature indicating products comply with specifications.

B. Shop Drawings:
   1. Submit reflected ceiling plans drawn to scale showing layout of ceiling tiles, baffles and clouds include details of suspension system and seismic details. Include coordinated penetrations and ceiling-mounted items.
   2. Identify the location and direction of the main tees on the submitted reflected ceiling plan. Include any necessary details or drawings from the manufacturer regarding recommended installation.

C. Samples:
   1. Refer to "Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals" for submission procedures and requirements of Color Samples.
   2. Refer to "Room Finish Schedule and Interior Finish Legend found on Architectural Finish Schedules" drawing for color selections.
   3. Refer to "Specification Section 09 06 00 – Color Schedule".

D. Contract Closeout Information:
   1. Maintenance Data: See Section 01 78 23.

1.5 DELIVERY, STORAGE AND HANDLING

A. Delivery of acoustic free-hanging element products will be in the original unopened packages with the manufacturer's label intact.

B. Handling and storage should be in accordance with the manufacturer's Safety Data Sheets (SDS).

C. Individual panels should be handled carefully to avoid damage.

1.6 PROJECT CONDITIONS

A. Environmental Limitations
   1. Install acoustic panels only in conditions that are within the manufacturer's published limits for temperature and humidity.
   2. Areas receiving ceiling panels should be free of construction debris and dust.
   3. Mechanical, sprinkler and electrical trades shall have completed their work above the ceiling structure prior to commencement of the ceiling panel installation.

1.7 COORDINATION

A. Coordinate the installation of the acoustic free-hanging elements with all trades whose work is impacted by that installation

1.8 EXTRA MATERIAL

A. Provide 96-square feet of each type, color, and size of acoustic ceiling tile specified.

B. Provide acoustic free-hanging elements (Clouds and Baffles) – Minimum 5 percent of each type installed.

C. Provide Suspension System Components – Minimum 5 percent of each type installed for acoustic ceiling tiles, clouds and baffles.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Steel Suspension Systems:
      a. Basis of Design: CertainTeed.
      b. Or Equal.
   2. Acoustical Ceiling Tile:
      a. Basis of Design: CertainTeed
      b. Or Equal.

C. See "Interior Finish Legend on Architectural Drawings for Basis of Design - Acoustical Ceiling Materials, Sizes and Colors.

D. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 ACOUSTIC SUSPENSION SYSTEMS

A. General Requirements:
   1. Heavy duty systems, ASTM C635.
   2. Main runner jointing by spliced, interlocking ends, tab locks, pin locks, or other suitable connections.
   3. Cross runners interlocking with main runners.
   4. Include components and accessories necessary to resist seismic loads and dead loads of items such as light fixtures and air diffusers.

B. Steel Hanger Wire:
   1. Pre-stretched, with a yield stress load of at least 5 times design load, but not less than 0.106-inches (12 Gauge).
   2. Utilize continuous lengths, without kinks and splices.
   3. Galvanized Steel:
   4. Galvanized, soft annealed steel wire conforming to ASTM A641.

C. Stainless Steel Wire:
   1. Type 304, soft annealed steel wire conforming to ASTM A641.
   2. Use in the following wet areas: Showers, kitchens, and other wet areas.

D. Attachment Devices:
   1. As required in accordance with the manufacturer's written instructions.
   2. Anchors in Concrete:
      a. Anchors of type and material indicated below, with holes or loops for attaching hangers of type indicated and with capability to sustain, without failure, a load equal to 5 times

that imposed by ceiling construction, as determined by testing per ASTM E488/E488M or ASTM E1512.
        b. Acceptable types: Cast-in-place, post-installed expansion anchors and post-installed bonded anchors.
            (1) Material: Carbon-steel components zinc plated to comply with ASTM B633, Class Fe/Zn 5 for Class SC 1 service condition.
        c. Use stainless steel or for wet/corrosive conditions.
            (1) Material: Stainless-steel components complying with ASTM F593 and ASTM F594, Group 1 Alloy 304 or 316.
    3. Wall Moldings: In accordance with the International Building Code.
        a. Description: 7/8-inch minimum horizontal leg Color to match grid face.
    4. Accessories
        a. Accessories:
            (1) 15/16 Outside Corner Cap – 2 1/4-inch x 2 1/4-inch x 1/64-inch. Color to match grid face.
            (2) 15/16 Inside Corner Cap – 2 1/4-inch x 2 1/4-inch x 1/64-inch. Color to match grid face.

E. Suspension System Types:
    1. CG-1 used with ACT-1: Exposed steel grid, non-rated:
        a. Description: Galvanized, double web steel, main and cross runners.
        b. Rating: Heavy Duty.
        c. Face width: 15/16-inches.
        d. Finish and Color on exposed grid face surfaces: Smooth, flat white.

F. Minimum Seismic Code Requirements
    1. Comply with Seismic Category “C” classification for all suspended ceilings per ASTM C636 and ASCE7.
    2. Comply with the Following Minimum Requirements:
        a. All 4 sides must have a 3/8-inch gap from tee end to the wall.
        b. ICC ESR 1222.
        c. ICC ESR 3336.

2.3 ACOUSTICAL CEILING TILE

A. General Requirements for all Ceiling Tile, Clouds and Baffles.
    1. Scheduled finishes to be factory applied.
    2. Class A incombustible units.
        a. Flame spread: 25 or less.
        b. Smoke developed: 50 or less.
        c. Fuel contributed: 0.
    3. Edges uniformly fabricated, true, square.
    4. Sizes as required to fit scheduled suspension system.

B. Acoustic Ceiling Tile Types:
    1. Basic Mineral Fiber Ceiling Tile (ACT-1):
        a. Description: Wet-formed mineral fiber with acoustically transparent membrane and factory-applied latex paint.
        b. Classification: ASTM E1264, Type IV, Form 2, Pattern E.
        c. Surface Texture: Smooth.
        d. Light Reflectance: Not less than 0.90.
        e. CAC: 40.
        f. NRC: Not less than -0.80.
        g. Edge: Square.
        h. Thickness: Minimum 7/8 inches.
    2. See INTERIOR FINISH LEGEND on the Drawings for Basis of Design for all Acoustical Ceiling Materials, Sizes and Colors.

C. Diffusers and Grilles: See Section 23 31 00.

D. Light Fixtures: See Section 26 50 00.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify suitability of substrate to accept installation.

B. Examine installation site for irregularities having effect on quality and execution of work.

C. Consult other trades involved before start of ceiling work, to determine areas of potential interference.

D. Do not start installation until the interferences have been resolved.

E. Installation constitutes acceptance of responsibility for performance.

3.2 PREPARATION

A. Coordinate ceiling layout with sprinkler head spacing and work penetrating acoustical ceiling systems.

B. Tolerances:
1. Comply with ASTM C635 Standard Specification for the Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-in Panel Ceilings.
2. Deviation from level plane: 1/8-inches in 10-feet with no load applied maximum.
3. Bow: 1/32-inches in 2-feet maximum.
4. Camber: 1/32-inches in 2-feet maximum.
5. Twist: 1 degree in 2 feet maximum.

3.3 INSTALLATION

A. Suspension System for Ceiling Grid “CG-1”.
1. Install suspension system in accordance with manufacturers' instructions.
2. Grid layout:
   a. See Reflected Ceiling Plans and submittal shop drawings.
   b. Install grid based on electrical lighting fixture layout indicated in Electrical Drawings, unless otherwise indicated,
   c. Acoustical panel dimension at perimeter walls: Not less than 6-inches.
   d. Acoustical panel dimension at perimeter walls: Not less than 12-inches.
   e. In case of conflict notify the Contracting Officer Representative.
3. Install grid square with room and with grid or acoustical panel center lines coinciding with center lines of room, each direction.
4. Intersections between main tees and cross tees:
   a. Cut and notch as required.
5. Wall angles:
   a. Install wall angles or moldings where ceilings meet walls, partitions, vertical elements, and other types of ceilings or ceiling fixtures.
   b. Secure angles to wall construction at stud locations.
   c. Maximum spacing from terminal ends: 3-inches.
   d. Draw fasteners tight against vertical surfaces.
   e. Level tolerance: not more than 1 in 1000.
   f. Miter cut inside and outside corners and install pre-manufactured inside or outside corner caps.
   g. Install with leg supporting bottom flange of runners.
   h. Install with min 7/8-inch horizontal leg supporting bottom flange of main tees and cross tees and install “Certainteed - CTSPC2 Clip” or “Armstrong - BERC2 Clip” or “USG - ACM7” seismic clips on all tees.

i. Install on walls where runners are not fixed to runner allow terminal runner end to move 3/4-inches in both directions.

6. Main runners:
   a. Utilize wall angles to align and receive terminal ends of main tees without transferring load to wall angle.
   b. Space main tees as indicated to receive lay-in panels and fixtures.
   c. Support terminal ends of main tees by wires located within 6 inches from boundary walls.
7. Cross runners:
   a. Space cross tees as indicated to receive lay-in panels and fixtures.
   b. Install cross runners with positive interlock.
   c. Tee connections to maintain 60-pound tension and compression.
   d. Keep tee ends from spreading with spreader/stabilizer bars.
   e. Utilize wall angles to align and receive terminal ends of cross tees without transferring load to wall angle.
   f. Support terminal ends of cross tees by wires located within 6-inches from boundary walls.
8. Do not bear runners on walls or partitions.
9. Leave the suspension system ready to accept installation of acoustic materials.
10. Hanger wires:
    a. Provide hangers and inserts necessary to support ceiling suspension systems and ceiling dead loads.
    b. Coordinate location and alignment with work of other trades.
    c. Install hanger wires plumb to main tees and cross tees.
    d. Do not suspend any part of suspension system from ducts, pipes, conduit, cable tray or equipment.
    e. Provide supplementary rough suspension system where necessary to support ceilings beneath pipes, ducts, equipment, and cable trays.
    f. Splay hangers no greater than 30 degrees from vertical to avoid obstructions or other conditions that prevent plumb, vertical installation.
    g. Offset horizontal forces by bracing or counter-splaying.
    h. Space hangers to prevent eccentric deflection and rotation due to loads from items in or on ceiling.
    i. Provide supplemental hangers to support lighting fixtures and within 6 inches from end of main runners and fixtures which exceed manufacturer’s published load data.
    j. Hanger wire shall be attached to the main tees 4-feet on center with 3 turns in 3-inches.
    k. Hanger wires shall be installed no more than 1:6 out of plumb.
    l. Do not bear runners on walls or partitions.
11. Main runners:
    a. Utilize wall angles to align and receive terminal ends of main tees without transferring load to wall angle.
    b. Space main tees as indicated to receive lay-in panels and fixtures.
    c. Support terminal ends of main tees by wires located within 6-inches from boundary walls.
12. Cross runners:
    a. Space cross tees as indicated to receive lay-in panels and fixtures.
    b. Install cross runners with positive interlock.
    c. Tee connections to maintain 60-pound tension and compression.
    d. Keep tee ends from spreading with spreader/stabilizer bars.
    e. Utilize wall angles to align and receive terminal ends of cross tees without transferring load to wall angle.
    f. Support terminal ends of cross tees by wires located within 6-inches from boundary walls.
13. Leave the suspension system ready to accept installation of acoustic materials.
14. Hanger Wires for all Suspended Acoustic Ceiling Grid, Clouds and Baffles.

a. Provide hangers and inserts necessary to support ceiling suspension systems and ceiling dead loads.
b. Coordinate location and alignment with work of other trades.
c. Install hanger wires plumb to main tees and cross tees.
d. Do not suspend any part of the suspension system from ducts, pipes, conduit, cable tray or equipment.
e. Provide a supplementary rough suspension system where necessary to support ceilings beneath pipes, ducts, equipment, and cable trays.
f. Splay hangers no greater than 30 degrees from vertical to avoid obstructions or other conditions that prevent plumb, vertical installation.
g. Offset horizontal forces by bracing or counter-splaying.
h. Space hangers to prevent eccentric deflection and rotation due to loads from items in or on ceiling.
i. Provide supplemental hangers to support lighting fixtures and within 6-inches from end of main runners and fixtures which exceed manufacturer's published load data.
j. Hanger wire shall be attached to the main tees 4-feet on center with 3 turns in 3-inches.
k. Hanger wires shall be installed no more than 1:6 out of plumb.

15. Suspended Lay-In for Acoustic Ceiling Tile (ACT-1):
    a. Install acoustic materials in accordance with manufacturer's written instructions.
    b. Place lay-in panels, fixtures, diffusers, grilles, and similar items in manner not compromising suspension system performance.
    c. Field cut materials to fit grid.
    d. Cut edges to match profile of factory edges and paint to match.
16. Acoustic Ceiling Tile Paint:
    a. Touch-up minor surface scratches and blemishes with manufacturer's ceiling panel touch-up paint.
    b. Paint field cut edges exposed to view with manufacturer's ceiling panel touch-up paint
    c. Provide and install hold down clips to hold down panels where required to comply with seismic requirements.

3.4 CLEANING:

A. Clean all surfaces following installation per manufacturer's cleaning instructions.

B. Maintenance per Manufacturer's finish maintenance instructions.

3.5 PROTECTION:

A. Protect ceilings from damage during the remainder of construction.

B. Finished ceilings shall be without damage. Replace units having scratches, abrasions, or other defects with new units.

END OF SECTION

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SECTION 09 65 36
STATIC DISSIPATIVE RESILIENT TILE FLOORING AND RESILIENT BASE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Static dissipative resilient tile (RFT).
   2. Resilient base (RB).

B. Related Sections:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 09 06 00 – Color Schedule.
   3. Section 26 05 26 – Grounding and Bonding.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Americans with Disabilities Act (ADA):
      a. Accessibility Guidelines for Buildings and Facilities (ADAAG).
   2. ASTM International (ASTM):
      a. ASTM D2047, Standard Test Method for Static Coefficient of Friction of Polish-Coated Floor Surfaces as Measured by the James Machine.
      b. ASTM D2240, Standard Test Method for Rubber Property - Durometer Hardness.
      c. ASTM D3389, Standard Test Method for Coated Fabrics Abrasion Resistance (Rotary Platform, Double-Head Abrader).
      d. ASTM E64, Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source.
      e. ASTM E662, Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials.
      f. ASTM F150, Standard Test Method for Electrical Resistance of Conductive and Static Dissipative Resilient Flooring.
      g. ASTM F710, Standard Practice for Preparing Concrete Floors to Receive Resilient Flooring.
      h. ASTM F970, Standard Test Method for Static Load Limit.
      i. ASTM F925, Standard Test Method for Resistance to Chemicals of Resilient Flooring.
      j. ASTM F1344, Standard Specification for Rubber Floor Tile.
      k. ASTM F1482, Standard Practice for Installation and Preparation of Panel Type Underlayments to Receive Resilient Flooring.
      l. ASTM F1514, Standard Test Method for Measuring Heat Stability of Resilient Flooring by Color Change.
      m. ASTM F1816, Standard Safety Specification for Drawstrings on Children's Upper Outerwear.
      n. ASTM F1869, Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride.
      o. ASTM F2170, Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes.
      p. ASTM F2199, Standard Test Method for Determining Dimensional Stability and Curling Properties of Resilient Flooring after Exposure to Heat.
      q. ASTM F2753, Standard Practice to Evaluate the Effect of Dynamic Rolling Load over Resilient Floor Covering System.
      r. ASTM F3010, Standard Practice for Two-Component Resin Based Membrane-Forming Moisture Mitigation Systems for Use Under Resilient Floor Coverings.
      s. ASTM F3191, Standard Practice for Field Determination of Substrate Water Absorption (Porosity) for Substrates to Receive Resilient Flooring.

t. ASTM F3311, Standard Practice for Mat Bond Evaluation of Performance and Compatibility for Resilient Flooring System Components Prior to Installation.
u. ASTM G21, Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi.

3. National Fire Protection Association (NFPA):
   a. NFPA 253, Test Method for Critical Radiant Flux of Floor Covering Systems Using a Radiant Energy Source.
   b. NFPA 258, Test Method for Specific Density of Smoke Generated by Solid Materials
4. American National Standards Institute (ANSI):
   a. ANSI/ESD STM7.1, Resistive Characterization of Materials - Floor Materials.
   b. ANSI/ESD STM 97.1
   c. ANSI/ESD STM 97.2, Floor Materials and Footwear - Voltage Measurement in Combination with a Person.
5. American Association of Textile Chemists and Colorists (AATCC):
   a. AATCC Test Method 134, Electrostatic Propensity of Carpets.
6. Federal Aviation Administration (FAA):
   a. FAA-STD-019, Lightning and Surge Protection, Grounding, Bonding, and Shielding Requirements for Facilities and Electronic Equipment.
7. Federal Testing Method Standard (FTMS):
   a. FTMS 101c – Method 2065, Puncture Test.
8. Motorola:
   a. R56, Standards and Guidelines for Communication Sites.

1.3 SUBMITTALS

A. Product Data:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Recommendations on adhesives, primers and leveling and patching compounds.

B. Samples:
   1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
   2. Refer to Interior Finish Legend shown on the Drawings for Colors.
   3. Provide samples for each Static Dissipative Rubber Tile Flooring and Rubber Base specified.
   4. Refer to “Specification Section 09 06 00 – Color Schedule”.

C. Operation and Maintenance Manual:
   1. Provide a maintenance and floor care manual with full instructions on how to clean and maintain the static dissipative resilient tile floor.

1.4 MAINTENANCE MATERIALS

A. Extra Materials:
   1. Furnish the following extra material:
      a. Remaining portion of one partially used roll of resilient base material with a minimum of 10-linear foot of each height, color and type.
   2. Package and label extra materials to protect material during storage.

1.5 WARRANTY

A. Static dissipative resilient tile warranty:
   1. Manufacturers 10 years limited product warranty.
   2. Manufacturers 15 year limited electrical resistance warranty.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Static Dissipative Rubber Tile:
      a. Roppe Corporation – ESD Rubber Tile Dissipative.
      b. Or Equal.
   2. Resilient base:
      a. Roppe Corporation.
      b. Or Equal.

2.2 MANUFACTURED UNITS

A. ESD Rubber Dissipative Tile:
   1. Size: 24-inches x 24-inches x 0.080-inches thick.
   2. Tile Performance:
      a. ASTM F1344 – Rubber Tile – Class I, Type B.
      b. ASTM E64 (NFPA 253) – Critical radiant Flux – Class I, greater than or equal to 0.45 W/cm squared.
      c. ASTM E662 (NFPA 258) – Smoke Density – Passes, less than or equal to 450.
      d. ASTM D2047 – Static Coefficient of Friction – greater than or equal to 0.50.
      e. ASTM F970 – Static Load Resistance – Passes less than 0.005 inches indentation @ 250 pounds per square inch and passes less than 0.005 inches indentation @ 1,100 pounds per square inch Modified.
      f. ASTM F2753 - Rolling Load Limit – less than or equal to 500 pounds per inch squared.
      g. ASTM F925 – Chemical Resistance – Has chemical resistance to a listing of chemicals.
      h. ASTM G21 – Fungi Resistance – Passes.
      i. ASTM F2199 – Dimensional Stability – Excellent, less than or equal to 0.15 percent.
      j. ASTM D3389 – Abrasion Resistance – Excellent, less than or equal to 1 g loss (after 1000 cycles).
      k. ASTM F1514 – less than or equal to 8.
      l. ASTM D2240 – Durometer Hardness Indentation – greater than or equal to 85 Shore A.
   3. Electrical Performance:
      a. ASTM F150 – Electrical Resistance
      b. ANSI/ESD STM 7.1 – Resistance – Meets Requirements.
      c. ANSI/ESD STM 97.1 – Resistance – Meets Requirements.
      d. ANSI/ESD STM 97.2 – Body Voltage – Meets Requirements.
      e. Motorola R56, Appendix C.3.3 Flooring – Meets Requirements.
      f. FAA-STD-019f, ESD Control Flooring – Meets Requirements.
      g. FTMS 101c – Surface Resistivity – Passes; 5,000 v to 0 v.
      h. FTMS 101c – Method 4046 – Static Decay < 0.01 Second.
      i. AATCC-134 – Electrostatic Propensity - < 20 v with ESD Shoes.
   4. Grounding Requirements:
      a. One (18-inch) ESD copper grounding strip required for every 2,000 square-feet of continuous tile installed.
   5. Adhesive:
      a. Excelsior PSD-805 Modified Pressure Sensitive ESD Adhesive.
      b. Excelsior USD-810 Urethane Two-Part ESD Adhesive.

B. Resilient Base (RB):
   1. Rubber, ASTM F1861.
      a. Group 1, solid through-color.
      b. Style B, coved.
      c. 1/8-inch thick by 4-inches high.
   2. Factory-formed inside and outside corners.
   3. Continuous rolls, minimum 95-feet long.
   4. Adhesives and primers as recommended by manufacturer.

C. Accessories:
   1. Transition strips and reducing strips tapered to meet abutting materials.
   2. Metal edge strip:
      a. Concealed anchorage: Butt-type metal edge strips.
         1) Material: Extruded aluminum, mill finish.

D. Finishes:
   1. Provide selected colors for ESD Rubber Dissipative Tile, Resilient Base and transition strips specified in the "Architectural Finish Schedule and Interior Finish Legend found on Architectural Drawings".
   2. Refer to "Specification Section 09 06 00 – Color Schedule".

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine surfaces for defects, irregularities and conditions under which flooring is to be installed.

B. Verify substrates are dry, clean, smooth and free of paint, varnish, solvents, wax, oils, or foreign matter.

C. Repair cracks and other surface defects.

D. Verify floors are level or meet indicated slope.

E. Do not proceed with installation until unsatisfactory conditions have been corrected.

F. Installation constitutes acceptance of substrate and responsibility for performance.

3.2 PREPARATION

A. Prepare substrate in accordance with manufacturer's instructions.

B. Concrete substrate flatness tolerance must be within 3/16-inch per 10-feet-0-inches.

C. Fill construction joints and other non-moving joints with products approved by manufacturer of flooring system.

D. Coordinate leveling with vapor emission control system provider.

E. Prepare surfaces in accordance with manufacturer's recommendations and ASTM F710 and ASTM F1482.

F. Concrete substrates must be evaluated in accordance with:
   1. ASTM F2170 RH testing.
   2. ASTM F1869.
   3. ASTM F3010.
   4. ASTM F3191.
   5. ASTM F3441.
   6. ASTM F3311.

G. Acclimate resilient base to the area in which it is to be installed for minimum 72 hours at 68 degrees F prior to installation.

1. Provide manufacturer's recommended relative humidity levels.

3.3 INSTALLATION

A. Static Dissipative Rubber Tile Flooring:
   1. Provide installation under direct supervision of manufacturer's representative. Install in a traditional point-to-point method. Install in a consistency in direction of the sanding lines to prevent any shading or light reflectance issues.
   2. Follow manufacturers written instructions on installation.
   3. Coordinate installation with other floor, wall, and ceiling work.
   4. Do not start work until related work of other trades in same area has been completed.
   5. Lay out tile in a manner to avoid tiles of less than 1/2 tile size.
   6. Mix and apply adhesive as recommended by manufacturer.
   7. Install each tile fully bonded to floor and flush, tight and in true alignment with adjacent tiles, for finished surface free from imperfections.
   8. Fit flooring neatly into breaks and recesses, under equipment, against walls, around pipes, and other installations and obstructions.
   9. Install tile flooring and base as scheduled for rooms under and behind equipment.
   10. Roll flooring using 100-pound smooth roller. Roll floor within 30 minutes of installation, crossing in a perpendicular direction after initial roll. Use hand roller in areas that cannot be reached with a larger roller.
   11. Clean excessive adhesive or adhesive residue from the surface of the material per adhesive instructions.
   12. Install continuous copper grounding strips set in adhesive where recommended by manufacturer for maximum static dissipative performance.
      a. Prior to installing ESD flooring, place a minimum of 9-inches of the copper grounding strip into the freshly applied adhesive and trowel adhesive over the strip to ensure it is fully embedded within the adhesive. The remaining portion of the strip can be attached to the grounding point or connected to additional materials that will be attached to the grounding point.
      b. Coordinate location of leads with equipment grounding conductor interface provided in Section 26 05 26 of the specifications.
      c. All cracks, joints and voids must be bridged with a copper grounding strip to maintain ESD properties. Center copper grounding strip over crack, joint or void and embed copper grounding strip inside the adhesive.

B. Resilient Base:
   1. Install products in accordance with manufacturer's instructions.
   2. Floors to be free of all dust, paint, grease, oils, solvents, curing and hardening compounds, sealers and any other deleterious material which may affect the bonding of the adhesive used to install the floor coverings.
   3. Ensure recommended minimum installation temperatures are maintained before, during and after installation as required by the manufacturer.
   4. General:
      a. Apply primer and adhesive as recommended by manufacturer.
      b. Maintain a minimum temperature of 68 degrees F for a minimum of 72 hours prior to, during and after installation.
      c. Install base after wall material has thoroughly dried out.
      d. Provide base at intersections of floor and all vertical surfaces in areas scheduled to receive base, where intersection is exposed to view.
      e. Set base straight and true.
      f. Fit into breaks and recesses.
      g. Provide factory-formed outside and inside corners.
         1) Make joints tight.
         2) Where door frames are inset in opening, provide factory formed outside corner returned to frame, trim flush with face of frame.
      h. Install with top level and bottom edge in firm contact with floor.

3.4 FIELD TESTING

A. Perform resistance testing of completed static dissipative rubber tile floor per the following:
   1. Testing must be performed by a licensed and qualified electrician.
   2. Conduct field tests in accordance with ASTM F150.
   3. Perform one test for each 2000-square-feet of floor, but not less than one test for each room.
   4. Correct areas failing to comply with electrical resistance requirements, including replacement of flooring if necessary.
   5. Retest flooring until electrical resistance requirements comply.
   6. Submit final test report.

3.5 CLEANING

A. Clean floors as recommended by manufacturer just prior to occupancy.
   1. Remove any protective coverings prior to maintenance.
   2. Use dust mop to remove any dirt, dust, or debris.
   3. Add 4-6 ounces of Excelsior NC-900 Neutral Cleaner to one U.S. gallon of clean, potable water.
   4. Use a clean mop or pump-up sprayer to apply cleaning solution to the area and let stand for 5-10 minutes.
   5. Using either a low-speed rotary floor machine (175 – 350 RPM) or auto-scrubber, wet scrub the floor using a Mal-Grit
   6. Scrub Brush from Malish or equivalent 22 gauge soft-bristled scrubbing brush. A 3M 5100 Red Buffer Pad or equivalent may be used but the brushes will work better for textured surfaces.
   7. Use an auto-scrubber, wet vacuum, or clean mop to remove all excess cleaning solution.
   8. Rinse area with clean, cool water and allow floor to dry completely. Ensure the flooring area is clean and that all cleaning residue has been removed (this may require additional rinsing).
   9. After the area has cooled for a minimum of three (3) hours, dry buff area with a low-speed rotary floor machine (175 – 350 RPM) using a White Tampico Brush from Malish or equivalent polishing brush attachment. A 3M 4100 White Super Polish Pad or equivalent may be used. Ensure to utilize multiple passes in opposite directions.

B. Clean resilient base in accordance with manufacturer's recommendations.

3.6 PROTECTION

A. Provide non-staining protection for finished flooring until building is ready for occupancy.

B. Protect with non-staining, non-sticking building paper as may be necessary to prevent dirt and damage.

C. Replace damaged, scratched or cut resilient base until date of Substantial Completion.

END OF SECTION

SECTION 09 67 26
SEAMLESS EPOXY FLOORING

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Seamless Epoxy Flooring (EF), as indicated, in accordance with provisions of Contract Documents.

B. Completely coordinate with work of other trades.

1.2 RELATED SPECIFICATION SECTIONS

A. Section 01 33 23 – Submittal Procedures.

B. Section 01 78 23 – Operation and Maintenance Data.

C. Section 07 92 00 – Joint Sealants.

D. Section 09 06 00 – Color Schedule.

1.3 REFERENCES

A. ASTM International (ASTM):
1. ASTM C307
2. ASTM C413, Standard Test Method for Absorption of Chemical-Resistant Mortars, Grouts, Monolithic Surfacing, and Polymer Concretes.
3. ASTM C531, Standard Test Method for Tensile Strength of Chemical-Resistant Mortar, Grouts, and Monolithic Surfacings.
4. ASTM C579, Standard Test Methods for Compressive Strength of Chemical-Resistant Mortars, Grouts, Monolithic Surfacings, and Polymer Concretes.
5. ASTM C580, Standard Test Method for Flexural Strength and Modulus of Elasticity of Chemical-Resistant Mortars, Grouts, Monolithic Surfacings, and Polymer Concretes.
6. ASTM D2240, Standard Test Method for Rubber Property—Durometer Hardness.
7. ASTM D2794, Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact).
8. ASTM E648, Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source.
9. ASTM F1869, Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride.
10. ASTM F2170, Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes.

B. International Concrete Repair Institute (ICRI):
1. Technical Guideline 03732, Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, and Polymer Overlays.

C. U.S. Department of Justice:
1. ADA Standards for Accessible Design (ADAAG).

1.4 SUBMITTALS

A. Product Data:
1. Provide copies of manufacturer’s product data on specified products describing physical performance characteristics, chemical resistance, available colors and packaging.
   a. Also provide copies of MSDS “Material Safety Data Sheets” on all materials to be used.
2. Provide complete Installation Instructions published by manufacturer including special procedures, perimeter termination procedures, substrate joint and crack treatment for

penetrations, adjacent vertical wall surface treatment and recommendations and procedures for establishing anti-slip profile on complete surfaces.
3. Submit copies of manufacturers' written instructions for maintenance practices for flooring to Contracting Officer Representative.
4. Submit statement attesting of the approval of the installer by the manufacturer for installation of the specified products.

B. Samples:
1. 1. Refer to "Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals" for submission procedures and requirements of Color Samples.
2. Refer to "Room Finish Schedule and Interior Finish Legend found on Architectural Drawings" for Seamless Epoxy Flooring Materials, Coves and Colors.
3. Refer to "Specification Section 09 06 00 – Color Schedule".
   a. Submit 6 by 6-inch samples of each color and texture specified, applied to a rigid backing.

C. Project Information:
1. Installer Certificates: Signed by manufacturer stating that installers complies with specified requirements.
2. Material Certificates: For each resinous flooring component, from manufacturer.

D. Contract Closeout Information:
1. Maintenance data:
   a. Submit copies of manufacturers' Maintenance Data complying with information provided as outlined in this specification.
   b. See Section 01 78 23.

1.5 QUALITY ASSURANCE

A. Installer Qualifications:
1. Installer shall be manufacturer installer or installer trained and certified by manufacturer for installation of system specified.
2. Installer shall have a minimum five years of experience in application of specified materials for projects of similar size and complexity.

B. Single Limitation: Provide materials manufactured by single supplier to insure compatibility and proper chemical and mechanical bond.

C. Supervisor: Must have been trained by the manufacturer and shall be on site during entire installation.

D. Provide materials which comply with or exceed slip resistance requirements set forth by local, state and national codes and in accordance with the Americans with Disabilities Act Accessibility Guidelines (ADAAG).

E. Floor system thickness verification shall be provided by the Contractor at the Contracting Officer Representatives discretion. The contractor shall take a minimum of 4 (four) 1-inch diameter random cores through the system into the substrate to verify proper system thickness.
1. Cored areas less than specified thickness shall be removed and replaced or increased in thickness by the installing Contractor, in a manner that does not affect the performance or integrity of the system.
2. Cored areas which comply with the recommended system thickness shall be built up to match the surrounding surface elevation prior to applying the seal coat. Cores taken and patched will be noticeable; therefore, cores should be taken from areas where aesthetics are less critical.
3. Cost associate with repair of cored areas shall be a part of this contract and shall be borne by the Contractor.

1.6 DELIVERY, STORAGE AND HANDLING

A. Deliver, unload, store, protect and handle all specified products required on this project. All materials must be delivered to the job site in sealed, undamaged containers.

B. All containers shall have proper DOT labeling, material name, and / or numbers, date of manufacture, manufacturers' name, health and safety information, Chemtrec Emergency Response Information, color, batch number where appropriate, component designation (A, B, etc.), and product mix ratio.

C. Materials should be allowed to stabilize at the storage temperatures for a minimum of three (3) days prior to application.

1.7 JOB CONDITIONS AND ENVIRONMENTAL REQUIREMENTS

A. Temperature in area of application shall be maintained at the ambient temperature recommended by the manufacture or at a minimum of 70 degrees F. The substrate temperature shall be at least 5 degrees F above the dew point during installation.

B. During the application of any materials and for the duration of the prescribed curing time the floor surface shall be protected from the direct air flow of any heat or cooling duct, open window or door.

C. Installer must examine substrates for moisture content, quality, severity of cracking, and other conditions, including the extent of floor repairs needed under which flooring is to be installed, and to notify Contractor in writing of conditions detrimental to proper completion of the work. Do not proceed until unsatisfactory conditions have been corrected.
   1. When existing concrete floor is found to be high in moisture, clean the floors with a five (5) percent solution of muriatic acid or other etching cleaner. Flush the floor with clean water to remove acid, neutralize with ammonia, rinse, allow to dry and vacuum before installing flooring. Test all concrete floors with a moist pH paper and repeat process until pH test results are greater than 11.

D. Clear away debris and scrape up cementitious deposits from surfaces to receive flooring, vacuum clean immediately before installation. Check concrete surface to ensure no "dusting" through installed flooring, apply sealer where required to prevent dusting.

E. The contractor shall exercise care during surface preparation and system installation to protect surrounding substrates and surfaces, as well as in place equipment. The contractor shall prepare the substrate to remove laitance and open the surface. This shall be achieved as specified elsewhere in this specification. Surface profile achieved shall be as specified elsewhere in this specification and shall be free from bond inhibiting contaminants.

F. Maintain lighting at a minimum uniform level of 50 or more-foot candles measured at the floor in areas where the seamless epoxy flooring is being installed. It is the recommendation of the manufacturer that the permanent lighting be in place and working during the installation.

1.8 WARRANTY

A. Applicator will provide a written one-year warranty from date of substantial completion covering both labor and materials. Warranty shall be written on company letter head, signed by an officer of the corporation and notarized.
   1. Warranty shall cover any defects in material or application that produces delamination from the substrate, degradation of the surface and excessive signs of wear.
   2. Warranty from the applicator shall detail the product or system used, the manufacturer's name and address and copy of manufacturer's warranty.

B. Warranty will detail procedures to be used in formal notification of applicator when a suspected defect or problem is to be addressed by the applicator. The contract information for the manufacturer will be furnished to allow the Government to send duplicate of formal notification to the manufacturer.

1.9 MAINTENANCE DATA

A. Some cleaners will affect the color, gloss or texture of seamless epoxy flooring surfaces. Test each cleaner, in a small area, utilizing the type of cleaning technique recommended by the flooring material manufacturer. This precaution will demonstrate the effect of your cleaner and technique. If no deleterious effects are observed, continue with the procedure. If deleterious effects do occur, modify the cleaning material and/or procedure. Use only flooring manufacturers recommended types of cleaners. Provide complete maintenance data from the manufacturer covering recommended periodic maintenance steps, maintenance materials, procedures for regular cleaning, stain removal and spot repairs.

B. Provide recommendations from manufacturer for schedule of periodic inspections by representative of the manufacturer.

C. Submit a list of wet samples of each product that will be supplied at no charge to the Government to be retained for touch - up of damaged areas

1.10 INSTALLATION DESCRIPTION

1. The installer shall be responsible for verifying all measurements of the work and shall use those measurements for all shop drawing submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Seamless Epoxy Flooring:
      a. Basis of Design: Stonhard – Stonclad GS.
      b. Or Equal.
   2. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Seamless Epoxy Flooring System:
   1. Nominal 1/4-inch abrasion-, impact-, and chemical-resistant, high-performance, resin-based, monolithic flooring system designed to produce a seamless floor.

B. Epoxy Flooring System Components: Manufacturer's standard components that are compatible with each other and as follows:
   1. Primer:
      a. Manufacturer's Standard Primer.
      b. Resin: Epoxy.
      c. Formulation Description: (2) two components, 100 percent solids.
      d. Application Method: Squeegee and roller.
      e. Number of Coats: (1) One.
   2. Mortar Base:
      a. Manufacturer's Standard Mortar Base.
      b. Resin: Epoxy.

c. Formulation Description: (3) three components, 100 percent solids.
d. Application Method: Metal Trowel.
1) Thickness of Coats: nominal-1/4-inch.
2) Number of Coats: (1) One.
e. Aggregates: Pigmented Blended aggregate.

3. Topcoat:
a. Manufacturer's Standard Topcoat.
b. Resin: Epoxy.
c. Formulation Description: (2) two components, 100 percent solids.
d. Type: Pigmented.
e. Finish: Standard.
f. Number of Coats: (1) One.

C. System Physical Properties: Provide resinous flooring system with the following minimum physical property requirements when tested according to test methods indicated:
1. Compressive Strength: 10,000 psi after 7 days per ASTM C579.
2. Tensile Strength: 1,750 psi per ASTM C307.
3. Flexural Strength: 4,000 psi per ASTM C580.
4. Water Absorption: < 1 percent per ASTM C413.
5. Impact Resistance: > 160 in. lbs. per ASTM D2794.
6. Flammability: Class 1 per ASTM E648.
7. Hardness: 0.85 to 0.90, Shore D per ASTM D2240.
8. Flexural Modulus of Elasticity: 2.0x106 psi per ASTM C580
9. Thermal Coefficient of Linear Expansion: 1.4x10-5 inch/inch degree F per ASTM C531.

2.3 MISCELLANEOUS MATERIALS

A. Provide materials as specified herein. All products specified for the total coating system installation shall be manufactured by a single manufacturer. All products shall meet or exceed the performance specification of the products specified. The products named are those established as most suitable for the intended use and establish a desired level of quality and performance.

B. Floor patch material shall be manufacturers recommended low odor, three - part, solvent - free 100 percent solids epoxy system specifically designed for economically filling depressions and deep holes or sloping floor surfaces to insure proper drainage before the application of a finish coating or overlayment.

C. Sealants for cracks and joints that are located under the flooring system shall be manufacturer's recommended low odor, 100 percent solid, two - part semi - rigid epoxy. Sealants for control and expansion joints in the concrete floor shall be as specified in Section 07 92 00 Joint Sealants.

D. Finished surface of all materials shall be a slip - resistant profile created in the final coat. Slip resistant surfacing profile shall be type as recommended by the manufacturer of the coating system and as approved by the Contracting Officer Representative.

2.4 FINISHES

A. Refer to "Room Finish Schedule and Interior Finish Legend found on Architectural Drawings" for Seamless Epoxy Flooring Materials, Coves and Colors.

B. Refer to "Specification Section 09 06 00 – Color Schedule".

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine surfaces for defects, irregularities, and conditions under which flooring is to be installed.

B. Verify floors are level or meet indicated slope.

C. Do not proceed with installation until unsatisfactory conditions have been corrected.

D. The following tests and observations shall be made by the applicator to insure the substrate is acceptable for the application of the specified system. All tests and necessary corrections of defects shall be completed by Contractor prior to application of any of the materials specified herein.
   1. Moisture content shall be tested according to ASTM F2170 for moisture in concrete surfaces and per ASTM F1869 for moisture vapor emission rate. Seamless epoxy flooring installation shall not proceed until floor is passes both tests.
   2. The surface temperature shall be checked on any surfaces to be covered to insure the application will be accomplished at the temperature range recommended by the material manufacturer for the material being applied.
   3. A visual examination of the surface to be covered shall be made by the installer to detect areas of skimmed concrete, loose or weak areas. Inspection by sounding with a hammer or dragging a heavy chain followed by marking any defective or hollow surface to identify complete removal of sub - standard substrate.

E. Curing agent or form release materials shall be completely removed by the installer prior to any coating or overlayment application.

3.2 SURFACE PREPARATION

A. The contractor responsible for the installation of the polymer system shall be responsible for providing a “proper substrate” and shall verify that the substrate is clean, durable, saturated surface dry and free of surface contaminants.

B. Contractor shall remove any oil or grease spots with appropriate solvent or chemical degreasing materials followed by pressure washing and insure complete removal of any residue.

C. Contractor shall remove any existing failing / deteriorated coatings, patches, or weak areas completely from the floor as well as equipment bases and curbs, exposing the structural substrate. Contractor shall repair any and all areas found to be deficient in accordance with these specifications and per the flooring manufacturers’ written requirements.

D. Acid etching is acceptable as a means of creating open texture to new concrete and cementitious self - leveling floor underlayment, removal of laitance and chemical removal of some contaminants. After etching, the surface must be thoroughly rinsed until rinse water has a pH of 11 or more. When the floor is dry it must be totally vacuumed to remove all residual dust and concrete particles.

E. All open cracks, joints and construction joints shall be cleaned of all loose debris, dirt, oils, or other contaminants. Prepare joints the same as flooring.

F. Installer shall remove all failing coatings and rust from all metal equipment feet, bases, metal stair feet, supports and other metal surfaces projecting from the concrete to a minimum of 4-inches above the substrate surface.

G. Installer shall cut back and/or remove any joint backing or filler strips in the expansion joints to a minimum of 1-1/2-inch deep.

H. Insert sealant in the joints to prevent filling with the overlayment materials and to allow for accurate location of final sawcuts in the overlayment.

I. All straight-line cracks shall be saw cut to produce a minimum joint 3/4-inch wide and 1-1/4-inch deep centered over the original crack. Remove all dust and debris and fill as specified for expansion joint application.

J. All cracks that are not straight lines or expansion joints, shall be routed out to produce a minimum joint of 1/4-inch x 1/4-inch. These joints shall be filled with slurry of primer and graded silica aggregate.

K. Cut a key at least 1/2-inch x 1/2-inch around the perimeter of all drains, steel posts and any other penetrations and remove all dust, dirt or loose particles. Remove any rust sealants or coating material from the drain ring.

L. Cut a straight and true key at least 1/2-inch x 1/2-inch under doors where EFS transitions to concrete.

M. Saw cut the concrete where there will be a stopping point for the overlayment application that does not meet a rising or vertical surface. The inside edge of the saw cut shall be chipped away at a 45-degree angle to form an undercut key for the edge of the application.

N. Any horizontal surface requiring more than 1/4-inch of fill to restore them to a level condition shall be filled with manufacturers recommended floor patch to bring them level with the surrounding smooth surface of the existing floor. Floor patch shall be applied at any thickness over 1/4-inch in one application, without developing shrinkage cracks. Use only material recommended by flooring manufacturer that will provide adhesion to wood, metal, tile, brick, and damp as well as dry concrete. Comply with manufacturer's instructions on mixing and application.

O. Fill cracks, construction joints and other non-moving joints with elastomeric sealant.

P. Mask and cover adjacent surfaces, fixtures, cabinetwork, equipment, and similar items not receiving coating.

3.3 INSTALLATION

A. After initial preparation has occurred, inspect the concrete for bug holes, voids, fins and other imperfections. Protrusions shall be ground smooth while voids shall be filled with filler as recommended by the flooring manufacturer.

B. Throughout the application process, substrate temperature shall be maintained at the ambient temperature recommended by the manufacture or at a minimum of 70 degrees F. The substrate temperature shall be at least 5 degrees F above the dew point during installation. Applications on concrete substrates should occur while temperature is falling to lessen off - gassing. The material shall not be applied in direct sunlight, if possible.

C. General:
   1. Apply components of resinous flooring system according to manufacturer's written instructions to produce a uniform, monolithic wearing surface of thickness indicated.
   2. Coordinate application of components to provide optimum adhesion of resinous flooring system to substrate, and optimum intercoat adhesion.
   3. Cure resinous flooring components according to manufacturer's written instructions. Prevent contamination during application and curing processes.
   4. At substrate expansion and isolation joints, provide joint in resinous flooring to comply with resinous flooring manufacturer's written recommendations.
      a. Apply joint sealant to comply with manufacturer's written recommendations.

D. Apply primer where required by resinous system, over prepared substrate at manufacturer's recommended spreading rate.

E. Integral Base:
   1. Install mortar base mix to wall surfaces before applying flooring. Apply according to manufacturer's written instructions and details including those for taping, mixing, priming, troweling, sanding, of cove base. Round internal and external corners.
      a. Install Integral Cove Base: 6-inches high.

F. Apply metal trowel single mortar coat in thickness indicated for flooring system. Hand or power trowel and grout to fill voids. When cured, sand to remove trowel marks and roughness.

G. Apply topcoat in number of coats indicated for flooring system and at spreading rates recommended in writing by manufacturer.

3.4 TERMINATIONS

A. Sawcut edges to “lock” the flooring system into the concrete substrate along lines of termination.

B. Penetration Treatment: Lap and seal resinous system onto the perimeter of the penetrating item by bridging over compatible elastomer at the interface to compensate for possible movement.

C. Trenches: Continue flooring system into trenches to maintain monolithic protection. Treat cold joints to assure bridging of potential cracks.

D. Treat floor drains by sawcutting the flooring system to lock in place at point of termination.

E. Finish Work shall match approved samples, be uniform in thickness, sheen, color, pattern, and texture, and be free from defects detrimental to performance.

3.5 MATERIALS APPLICATION - JOINT SEALING

A. Seal the perimeter of all posts, drains and other penetrations with a bead of sealant.

B. Insert compressible closed cell joint backing in all expansion joints using a large enough size to provide 30 percent compression. The joint backing should be forced down in the joint to a depth equal to one-half the joint width, not to exceed 3/8-inch deep.

C. Seal all expansion and control joints with the sealant and tool the bead to insure full and complete bead contact. Finish bead should be neatly aligned with the saw cut joint edge with no gaps, strings or air bubbles.

3.6 PROTECTION

A. Protect Work from damage and wear during the remainder of the construction period. Use protective methods and materials, including temporary covering, recommended in writing by flooring manufacturer.

B. Prohibit traffic on floor finish for 24 hours after installation at ambient temperature of 70 degrees F. (Contact manufacturer for protection times required at temperatures below 50 degrees F.)

C. Barricade area to protect application until fully cured.

3.7 CLEANING

A. Upon completion of the Work, clean up and remove from the premises surplus materials, tools, appliances, empty cans, cartons, and rubbish resulting from the Work. Clean off all spattering and drippings, and all resulting stains.

B. Clean floors as recommended by manufacturer prior to final inspection.

END OF SECTION

SECTION 09 84 00
ACOUSTICAL PANEL SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Acoustical Panel System - Cementitious Wood Fiber, as indicated, in accordance with provisions of Contract Documents.

B. Completely coordinate with the work of other trades.

C. Acoustical Panel System – Cementitious Wood Fiber Panels are to be installed as replacement panels in the Roaring Springs Pump House.

D. Related Sections:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 09 06 00 – Color Schedule.

1.2 REFERENCES

A. Regulatory Requirements and Approvals:
   1. ASTM International (ASTM):
      a. ASTM C423 Standard Test Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method.
      b. ASTM D3273 Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber.
      c. ASTM E84 Standard Test Method for Surface Burning Characteristics of Building Materials.
      d. ASTM E1264 Standard Classification for Acoustical Ceiling Products.
      e. ASTM E1477 Standard Test Method for Luminous Reflectance Factor of Acoustical Materials by Use of Integrating-Sphere Reflectometers.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Layout Plan and Elevations as required to show panel joints and locations of furring.
   2. Attachment details.
   3. Panel cut out details around structural and surface mounted items that surround or penetrate the panels.

B. Product Data:
   1. Manufacturer’s technical data for each type of panel required.

C. Samples:
   1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
   2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
      a. Provide 4-inches x 4-inches pieces of each panel type specified showing color and edges specified.

D. Certifications:
   1. UL certifications that products comply with specified requirements, including laboratory reports showing compliance with specified tests and standards. Acoustic performance, products must be tested to the A and D-20, methods.

1.4 QUALITY ASSURANCE

A. Installer Qualifications:
   1. Experience on projects of similar size and complexity.

B. Performance Requirements:
   1. Surface Burning Characteristics: Acoustical panels to perform as specified when tested in accordance with ASTM E84. Acoustical panel surface burning performance should comply with the International Building Code and other local building code requirements.
   2. Acoustical Characteristics: Acoustical panels to perform as specified when tested in accordance with ASTM C423.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Storage and Handling Requirements:
   1. Handle products carefully to avoid damage or chipping edges.
   2. Store products in a clean, cool, dry place, and out of direct sunlight.
   3. Store products in a space where the ambient temperature and humidity conditions are being maintained at the levels indicated for the project when occupied for its intended use.

1.6 SITE CONDITIONS

A. Ambient Conditions: Maintain ambient temperature and humidity conditions at levels indicated for the project when occupied for its intended use.

B. Existing Conditions: Do not install cementitious wood fiber panels until space is weatherproofed and is completely dry.

1.7 WARRANTY

A. Cementitious Wood Fiber Direct-Attached Acoustic Wall Panels: Submit a written warranty executed by the manufacturer, agreeing to repair or replace panels that fail within the warranty period. Failures include, but are not limited to the following:
   1. Defects in materials or factory workmanship warranty - Thirty (30) years from date of substantial completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Cementitious Wood Fiber:
      a. Basis of Design: Armstrong World Industries, Inc. - Tectum Direct – Attached Wall Panels.
      b. Or Equal.
   2. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 DESIGN CRITERIA

A. Surface Burning Characteristics:
   1. Class I / Class A.
   2. Flame Spread: less than 25.
   3. Smoke Developed: less than 450.

2.3 MATERIALS

A. Cementitious Wood Fiber Acoustical Panel System:
   1. Composition: Aspen wood fibers bonded with inorganic hydraulic cement.
   2. Thickness: (Match Existing Panel Conditions).
      a. 1-1/2-inches on exterior walls.
      b. 1-inch on interior partition walls.
   3. Width:
      a. Match existing panels which appear to be approximately 47-3/4-inches.
   4. Length:
      a. Match existing panel sizes or maximum practical lengths for conditions indicated.
   5. Edges:
      a. Long edge: Beveled.
      b. Short edge: Square.
   6. UL Classified Noise Reduction Coefficient (NRC): ASTM C423 Classified with UL label based on mounting type:
      a. Mounting type A - (0.40) on interior wall partitions.
      b. Mounting type D-20 - (0.45) on exterior walls.
   7. UL Classified Flame Spread: ASTM E1264; Class A. Product must be able to meet this criteria after being painted six times.
   8. Light Reflectance (LR) White Panel: ASTM E1477; (70)
   9. Dimensional Stability/Mold Resistance: HumiGuard Plus and no significant mold growth when tested by ASTM D3273.
   10. USDA Certified Biobased Product, 98 percent.
   11. Finishes: Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

B. Furring Strips:
   1. Wood.
   2. Size to match existing on exterior walls.

C. Construction Adhesive:
   1. As recommended by panel system manufacturer.

D. Painted-Head Drywall Screws:
   1. Steel.
   2. Length as recommended for conditions by Cementitious Wood Fiber Acoustical Panel System manufacturer.
   3. Color to match Cementitious Wood Fiber Acoustical Panel System .

E. Touch-Up Paint:
   1. Color as specified for Cementitious Wood Fiber Acoustical Panel System.

F. Miscellaneous Accessories:
   1. Provide fasteners, furring strips, moldings, trim, and other items as necessary for a complete single source installation.

PART 3 - EXECUTION

3.1 PREPARATION

A. Examine surfaces scheduled to receive suspended or directly attached cementitious wood fiber acoustical panels for unevenness, irregularities and dampness that would affect quality and execution of work.

B. Measure each wall area and establish layout of wall units. Coordinate panel layout with existing conditions that will require cut outs in the panels.

C. Do not proceed with installation of wall panel system until unacceptable conditions are corrected.

3.2 INSTALLATION

A. Comply with the instructions and recommendations of the acoustical wall panel system manufacturer.

B. Furring Strips:
   1. Secure to substrate wall using construction adhesive and fastener type and sizes as appropriate for conditions.
   2. Install on wall surfaces in the direction to match existing panels.

C. Secure Panels using construction adhesives and screws.
   1. Screw heads to be flush with panel surface.

D. Securely affix wall panels by means of splines attached vertically to smooth wall or furring strips.
   1. Apply adhesive where necessary.

E. Cover field cut edges by means of trim or other moldings.

3.3 CLEANING, PROTECTION AND REPAIRS

A. Protect installed work from damage due to subsequent construction activity, including temperature and humidity limitations and dust control.

B. Clean exposed surfaces of acoustical panel, trim, moldings, and suspension members.

C. Touch up any minor finish damage.

D. Remove and replace work which cannot be successfully cleaned or repaired.

END OF SECTION

SECTION 09 91 00
ARCHITECTURAL PAINTING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Surface Preparation.
   2. Field application of:
      a. Architectural Coatings.
      b. Stains and varnishes.
      c. Any other coating, thinner, accelerator, inhibitor, etc., specified or required as part of a complete System specified in this Specification Section.
   3. Environmental controls for field application of coatings.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 - Submittal Procedures.
   2. Section 08 11 00 - Hollow Metal Doors and Frames.
   3. Section 08 14 16 - Flush Wood Doors.
   4. Section 09 06 00 – Color Schedule.
   5. Section 09 29 00 - Gypsum Board.
   6. Section 09 96 00 - High Performance Industrial Coatings.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. D523, Standard Test Method for Specular Gloss.
      b. D4258, Standard Practice for Surface Cleaning Concrete for Coating.
      c. D4259, Standard Practice for Abrading Concrete.
      d. D4261, Standard Practice for Surface Cleaning Concrete Unit Masonry for Coating.
      e. D4262, Standard Test Method for pH of Chemically Cleaned or Etched Concrete Surfaces.
      f. D4263, Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method.
      g. F1869, Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride.
      h. E84, Standard Test Method for Surface Burning Characteristics of Building Materials.
   2. National Fire Protection Association (NFPA):
      a. 101, Life Safety Code.
   3. Steel Door Institute/American National Standards Institute (SDI/ANSI):
      a. A250.10, Test Procedure and Acceptance Criteria For Prime Painted Steel Surfaces for Steel Doors and Frames.
   4. The Society for Protective Coatings (SSPC):
      a. SP 1, Solvent Cleaning.
      b. SP 2, Hand Tool Cleaning.
      c. SP 3, Power Tool Cleaning.
      d. SP 16, Brush-off Blast Cleaning of Non-Ferrous Metals.
   5. The Society for Protective Coatings/NACE International (SSPC/NACE):
      a. SP 6/NACE No. 3, Commercial Blast Cleaning.
      b. SP 7/NACE No. 4, Brush-off Blast Cleaning.
      c. SP 13/NACE No. 6, Surface Preparation of Concrete.
   6. United States Environmental Protection Agency (EPA).

B. Miscellaneous:

1. Coating used in all corridors and stairways shall meet requirements of NFPA 101 and ASTM E84.

1.3 DEFINITIONS

A. Installer or Applicator:
   1. Installer or applicator is the person actually installing or applying the product in the field at the Project site.
   2. Installer and applicator are synonymous.

B. Approved Factory Finish: Finish on a product in compliance with the finish specified in the Specification Section where the product is specified.

C. Exposed Exterior Surface:
   1. Exterior surface which is exposed to view.
   2. Exterior surface which is exposed to weather but not necessarily exposed to view.

D. Finished Area:
   1. An area that is listed in or has finish called for on “Room Finish Schedule found on Architectural Finish Schedules” drawing and “Interior Finish Legend found on Architectural Finish Legend and Notes” drawing.
   2. An area that is indicated on Drawings to be painted.

E. Coating:
   1. A film or thin layer applied to a base material called a substrate. A coating may be a metal, alloy, paint, or solid / liquid suspensions on various substrates (metals, plastics, wood, paper, leather, cloth, etc.). They may be applied by electrolysis, vapor deposition, vacuum, or mechanical means such as brushing, spraying, calendaring, and roller coating.
   2. A coating may be applied for aesthetic or protective purposes or both. The term "coating" as used herein includes fillers, primers, sealers, emulsions, oils, alkyds, latex, enamels, stains, varnishes, sealers, epoxies, thinners and other applied coatings, whether used as primer, intermediate, or finish coat. The terms paint and coating are used interchangeably.

F. DFT:
   1. Dry film thickness, the film thickness of the fully cured, dry paint or coating.

G. Mil / mils:
   1. The English measurement for 0.001 in or one / one - thousandth of an inch, equal to 25.4 microns or 0.0254 millimeter.

H. MPI:
   1. Master Painting Institute.

I. Gloss Range:
   1. Specular gloss measured in accordance with ASTM D523:
      a. Gloss Level – G1 Matte or Flat:
         1) 0 to 5 units at 60 degrees, 10 maximum units at 85 degrees.
      b. Gloss Level – G2 Velvet:
         1) 0 to 10 units at 60 degrees, 10 to 35 units at 85 degrees.
      c. Gloss Level - G3 Eggshell:
      d. 10 to 25 units at 60 degrees, 10 to 35 units at 85 degrees.
      e. Gloss Level - G4 Satin:
         1) 20 to 35 units at 60 degrees, 35 units minimum at 85 degrees.
      f. Gloss Level - G5 Semi – Gloss:
         1) 35 to 70 units at 60 degrees.
      g. Gloss Level - G6 Gloss:
         1) 70 to 85 units at 60 degrees.
      h. Gloss Level – G7 High Gloss:
         1) More that 85 at 60 degrees.

J. Paint includes the following:

1. Architectural paints (AP) include: Acrylic latex or alkyd enamel coatings.
2. Stains and varnish include alkyd stain and polyurethane varnish.

1.4 REGULATORY REQUIREMENTS

A. Environmental Protection:
   1. In addition to requirements specified elsewhere for environmental protection, provide coating materials that conform to the restrictions of the local Air Pollution Control District and regional jurisdiction. Notify Contracting Officer Representative of any paint specified herein which fails to conform.

B. Lead Content:
   1. Do not use coatings having a lead content over 0.06 percent by weight of non - volatile content.

C. Chromate Content:
   1. Do not use coatings containing zinc - chromate or strontium - chromate.

D. Asbestos Content:
   1. Materials shall not contain asbestos.

E. Mercury Content:
   1. Materials shall not contain mercury or mercury compounds.

F. Silica:
   1. Abrasive blast media shall not contain free crystalline silica.

G. Human Carcinogens:
   1. Materials shall not contain ACGIH 0100Doc confirmed human carcinogens (A1) or suspected human carcinogens (A2).

1.5 SUBMITTALS

A. Product Data:
   1. Provide primers and undercoat paint produced by the same (Single - Source) manufacturer as the finish coats. This submittal shall include the name of the corresponding specified item together with the following detail information of the manufacturer:
      a. Spreading rate.
      b. Dry mil thickness.
      c. Non - volatile percent by weight.
      d. Dust free time.
      e. Recoat time.
      f. Chemistry of product.
      g. Indicate VOC content.
   2. Manufacturer's Material Safety Data Sheets: Submit manufacturer's Material Safety Data Sheets for coatings, solvents, and other potentially hazardous materials, as defined in Federal Standard 313.
   3. Provide test reports for materials used indicating fire hazard classification as determined by ASTM E 84. Materials shall meet requirements established in NFPA Number 101, Life Safety Code & International Building Code. Generally, all paint used must maintain a Class "A" fire rating on non-combustible substrates.

B. Certifications:
   1. Submit Applicator's Qualifications.

C. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's surface preparation instructions.
      c. Manufacturer's application instructions.

D. Schedule:

1. Provide a complete schedule indicating where each product specified will be used.
    a. Identify each building separately in the schedule and what surface each product will be applied on along with the corresponding color.

E. Samples:
1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
2. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings” for color selection.
3. Refer to “Specification Section 09 06 00 – Color Schedule”.

F. Informational Submittals:
1. Test results.
2. Applicator's daily records:
    a. Submit daily records at end of each week in which painting work is performed unless requested otherwise by Contracting Officer Representative.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Deliver in original containers, labeled as follows:
1. Name or type number of material.
2. Manufacturer's name and item stock number.
3. Contents, by volume, of major constituents.
4. Warning labels.
5. VOC content.
6. Application instructions.
7. Color name and number. Corresponding to Schedule of Finishes.

B. Protect from freezing or damage.

C. Store all materials in a single place designated by the General Contractor. Keep storage neat and clean. Make good all damage thereto or to surroundings. Remove rags and waste from building daily. Take precautions to avoid danger of fire. Post "No Smoking" signs in storage areas.

D. Maintain storage for paint materials in conformity with fire codes and authorities having jurisdiction. Protect floor from any spilled materials.

E. Paints and thinners shall be stored in accordance with the manufacturer's written directions, and as a minimum, stored off the ground, under cover, with sufficient ventilation to prevent the buildup of flammable vapors, and at temperatures between 40 to 95 degrees F. Do not store paint, polyurethane, varnish, or wood stain products with materials that have a high capacity to adsorb VOC emissions. Do not store paint, polyurethane, varnish, or wood stain products in occupied spaces.

1.7 COORDINATION OF PAINTING REQUIREMENTS

A. The painting subcontractor shall examine other Sections of the Specifications and shall thoroughly familiarize themselves with all their provisions regarding their painting requirements. Except as otherwise specified, surfaces left unfinished by the requirements of other Sections shall be painted or finished as a requirement of this contract.

B. Prime Coat Coordination: Provide finish coats which are compatible with prime paints used. Review other sections of these specifications in which prime paints are to be provided to ensure compatibility of total coating system for various substrates. Upon request from other trades, furnish information on characteristics of finish materials proposed for use, to ensure compatible prime coats are used. Provide barrier coats over incompatible primers or remove and reprime as required at no additional cost to Government. Notify Contracting Officer Representative in writing of any anticipated problems using specified coating systems with substrates primed by others.

1.8 PROJECT CONDITIONS

A. Comply, at minimum, with manufacturer recommendations for space ventilation during and after installation. Isolate area of application from rest of building when applying high - emission paints or coatings.
   1. Supply 100 percent outside air 24 hours a day.
   2. Supply airflow at a rate of 6 air changes per hour, when outside temperatures are between 55 degrees F and 85 degrees F and humidity is between 30 percent and 60 percent.
   3. Supply airflow at a rate of 1.5 air changes per hour, when outside air conditions are not within the range stipulated above.
   4. Avoid painting near temporary heaters which might discolor paint.
   5. Vent exhaust gases to exterior environment.
   6. No exhaust gases shall be allowed to vent into the space being painted or any adjacent space.

B. Coatings - Do not apply coating when air or substrate conditions are:
   1. Less than 5 degrees F above dew point;
   2. Below 50 degrees F or above 95 degrees F unless specifically pre - approved in writing by the Owner’s Representative and the product manufacturer. Under no circumstances shall application conditions exceed manufacturer recommendations.

C. Protection During Cleaning:
   1. Personnel engaged in solvent - cleaning of galvanized metal and aluminum, or cleaning concrete, masonry, or portland cement plaster with 5 to 10 percent solution of hydrochloric acid, shall wear the appropriate personal protective equipment to prevent skin and eye contact and fumes inhalation. Ventilate all work areas properly.
   2. Personnel Protection During Coating Applications:
      a. Personnel painting with high-build glaze coating systems shall wear the appropriate personal protective equipment to prevent skin or eye contact or inhalation. Ensure employees use and maintain solvent - resistant, personal protective equipment for the whole body. Emergency eye wash and water supplies shall be available near the work area for emergency flushing of the eyes and body. Coating applications shall be performed only in areas with good ventilation. Smoking will not be permitted in the area where coating is being applied. Wall and room temperature at the time of coating application and curing shall be in accordance with the manufacturer's instructions.

D. Protection During Application of Polyurethane Paints:
   1. Mix and apply polyurethane paints only in specifically designated areas with local exhaust ventilation and other environmental control measures as recommended on the basis of an on-site industrial hygiene survey. Supply and use air respirators in closed areas where adequate ventilation cannot be obtained.

E. Post - Application:
   1. Vacate space for as long as possible after application. Wait a minimum of 48 hours before occupying freshly painted rooms. Maintain one of the following ventilation conditions during the curing period, or for 72 hours after application:

F. Painting Contractor shall apply all materials under adequate illumination. Contractor shall assure adequate illumination exists in all areas where painting operations are in progress.

G. The painting Contractor is responsible for inspecting the work of others prior to the application of any paint or finishing material. If any surface to be finished cannot be put in proper condition for finishing by customary cleaning, sanding and puttying operations, the painting Contractor will immediately notify the General Contractor and the Contracting Officer, in writing, and shall not proceed with this work until conditions have been corrected and are acceptable.

H. Commencing of work in a specific area constitutes acceptance of surfaces, and responsibility for satisfactory work.

I. Painting Contractor shall provide wet paint signs as required.

J. Paint exposed surfaces whether or not colors are designated in "schedules," except where a surface or material is specifically indicated not to be painted or is to remain natural. Where an item or surface is not specifically mentioned, paint the same as similar adjacent materials or surfaces. If color or finish is not designated, the Contractor shall follow the color selection procedures found in Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals.

K. Contracting Officer Representative's approval is required PRIOR to application of each coat.

L. The minimum numbers of paint coats are specified in the Paint Schedule found elsewhere in this section. The application of additional coats to provide a total color and sheen uniformity on all surfaces shall be provided at no additional cost to the Government. The painting Contractor shall not be entitled to payment in excess of the amount agreed upon in their contract for any extra work over and above that specified unless authorized in writing by a Change Order.

M. Verify that atmosphere in area where painting is to take place is within paint manufacturer's acceptable temperature, humidity and sun exposure limits.

N. Do not apply coatings in snow, rain, fog or mist.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Dry-Fall Ceiling Systems:
      a. Basis of Design: Sherwin Williams – Waterborne Dry Fall Acrylic (Eggshell).
      b. Or Equal.
   2. Hollow Metal Doors and Frames:
      a. Basis of Design: Sherwin Williams – Macropoxy 646 Fast Cure Epoxy (Primer); and Acrolon 218 HS (Finish).
      b. Or Equal.
   3. Gypsum Drywall Systems (Ceilings):
      a. Basis of Design: Sherwin Williams – Harmony Interior Acrylic Latex (Primer); and Sherwin Williams – Harmony Interior Acrylic – Flat (Finish).
      b. Or Equal.
   4. Gypsum Drywall Systems (Walls):
      a. Basis of Design: Sherwin Williams – Harmony Interior Acrylic Latex (Primer); and Sherwin Williams – Harmony Interior Acrylic – Semi-Gloss (Finish).
      b. Or Equal.
   5. Wood Doors:
      a. Basis of Design: Sherwin Williams –Sher-Wood Fast Dry Sanding Sealer (Sealer); Sherwin Williams – Sher-Wood Oil Stain - Wiping Stain (Stain); and Sherwin Williams Wood Classics Waterborne Polyurethane Varnish – Clear Satin (Polyurethane Varnish).
      b. Or Equal.
   6. Submit request for substitution in accordance with Specification Section 01 33 23.
   7. Provide products from a single manufacturer to the greatest extent practicable.

2.2 MATERIALS

A. General:
   1. For unspecified materials such as thinner, provide manufacturer's recommended products.
   2. Unless noted otherwise, products listed are manufactured by the manufacturer listed below.
      a. Products of other manufacturers will be considered for use provided that the product:
         1) Is of the same generic formulation.
         2) Has comparable application requirements.
         3) Meets the same VOC levels or better.
         4) Provides the same finish and color options.
   3. Coatings shall comply with the VOC limits of EPA.
   4. Colors:
      a. Refer to "Room Finish Schedule and Interior Finish Legend found on Architectural Drawings" for color selection.
      b. Refer to "Specification Section 09 06 00 – Color Schedule".

2.3 PAINT SYSTEMS

A. General:
   1. Refer to Specification Section 09 96 00 for:
      a. Items in corrosive or highly corrosive environments.
      b. Items subject to immersion service.
      c. Items subject to exterior exposure.
      d. Any other locations where High Performance Industrial Coatings (HPIC) are required.

B. Schedule:

Substrate	Prime Coat[1]	Intermediate Coat(s)[1]	Finish Coat[1]
Hollow Core Concrete:	2.0 to 4.0 mil Dry-Fall Acrylic	N/A	2.0 to 4.0 mil Dry-Fall Acrylic
Concrete Masonry: See Specification Section 09 96 00 – High Performance Industrial Coatings	N/A	N/A	N/A
Structural Steel and Miscellaneous Metals: See Specification Section 09 96 00 – High Performance Industrial Coatings	N/A	N/A	N/A
Galvanized Structural Steel and Miscellaneous Metals: See Specification Section 09 96 00 – High Performance Industrial Coatings	N/A	N/A	N/A
Galvanized Metal Deck	2.0 to 4.0 mil Dry-Fall Acrylic	XX	2.0 to 4.0 mil Dry-Fall Acrylic
Factory Primed Metal Deck	Factory Primed	2.0 to 4.0 mil Dry-Fall Acrylic	2.0 to 4.0 mil Dry-Fall Acrylic
Wood Doors - Interior indicated to be field stained	400 to 500 square feet/GAL Sanding Sealer	Two coats at 400 to 500 square feet/GAL Alkyd Wood Stain	Two coats at 400 to 500 square feet/GAL Polyurethane Varnish
Hollow Metal Doors and Frames – Interior	4.0 to 5.0 mil DFT Epoxy Barrier Coat	2.0 to 3.0 mil HDP Acrylic[2]	2.0 to 3.0 mil HDP Acrylic[2]
Aluminum and Hybrid Aluminum Doors and Frames – Exterior and Interior: See Specification Section 09	N/A	N/A	N/A

96 00 – High Performance Industrial Coatings			
Aluminum Roll Down Doors: See Specification Section 09 96 00 – High Performance Industrial Coatings	N/A	N/A	N/A
Gypsum Board scheduled to receive "P and AP" Paint	300 to 400 square feet/GAL Acrylic Primer	300 to 400 square feet/GAL Acrylic Latex [2]	300 to 400 square feet/GAL Acrylic Latex [2]

1. Application rates (SF/GAL) shown are for unthinned materials.
2. Sheen as scheduled or selected.

C. Dry-Fall Ceiling Systems:
   1. Primer/Finish Coat - Interior Latex
      a. Waterbased prime/sealer with a minimum solids by volume of 41 percent with a total dry film thickness to develop a dry mil thickness of not less than 2.9 mils excluding primer and to provide total color and sheen uniformity.

D. Hollow Metal Doors and Frames:
   1. Primer:
      a. Modified Polyamidoamine Epoxy with a minimum solids by volume of 72 percent with a total dry film thickness to develop a dry mil thickness of not less than 5 mils.
   2. Intermediate and Finish Coats - Interior Latex Semi Gloss:
      a. Polyester modified, aliphatic, acrylic polyurethane with a minimum solids by volume of 65 percent. Minimum 2 finish coats over a primer with total dry film thickness not less than 3 mils excluding primer and to provide total color and sheen uniformity.

E. Gypsum Drywall Ceiling Systems: (Latex)
   1. Primer:
      a. Acrylic Interior Latex low odor primer / sealer with a minimum solids by weight of 54 percent and solids by volume of 42 percent with a total dry film thickness to develop a dry mil thickness of not less than 1.6 mils.
   2. Intermediate and Finish Coats - Interior Latex Flat:
      a. Interior Latex with a durable flat finish that has washability, with a minimum solids by weight of 54 percent and solids by volume of 42 percent. Minimum 2 finish coats over a primer with total dry film thickness not less than 3 mils excluding primer and to provide total color and sheen uniformity.

F. Gypsum Drywall Wall Systems: (Latex)
   1. Primer:
      a. Acrylic Interior Latex low odor primer / sealer with a minimum solids by weight of 50 percent and solids by volume of 35 percent with a total dry film thickness to develop a dry mil thickness of not less than 1.6 mils.
   2. Interior Latex Semi-Gloss:
      a. Interior Latex with a durable semi-gloss finish that has washability, with a minimum solids by weight of 54 percent and solids by volume of 42 percent. Minimum 2 finish coats over a primer with total dry film thickness not less than 3 mils excluding primer and to provide total color and sheen uniformity.

G. Wood Doors:
   1. First Sanding Sealer Coat: Vinyl modified nitrocellulose sanding sealer with a minimum solids by volume of 14.7 percent.
   2. Second Stain Coat: Penetrating wiping stain with a minimum solids by weight of 21 percent and solids by volume of 15.8 percent.
   3. Third Coat: Water Based Polyurethane Varnish – Satin Finish with a minimum solids by weight of 49 percent and solids by volume of 43 percent.

4. Fourth Finish Coat: Water Based Polyurethane Varnish – Satin Finish with a minimum solids by weight of 49 percent and solids by volume of 43 percent.

PART 3 - EXECUTION

3.1 ITEMS TO BE PAINTED

A. Exterior surfaces, including but not limited to:
   1. Concrete:
      a. Where indicated on Drawings.
   2. Concrete masonry:
      a. Where indicated on Drawings.
   3. Structural steel:
      a. Columns, beams and bracing.
      b. Field welded connections of factory painted structural steel.
   4. Miscellaneous ferrous metal surfaces:
      a. Items specifically noted on Drawings to be painted.
   5. Miscellaneous galvanized steel surfaces:
      a. Items specifically noted on Drawings to be painted.

B. Interior Areas:
   1. Refer to “Room Finish Schedule found on Architectural Finish Schedules” drawing and “Interior Finish Legend found on Architectural Finish Legend and Notes” drawing.
      a. If space is scheduled to be painted, paint all appurtenant surfaces within the space unless specifically noted otherwise.
      b. Provide coating manufacturer’s recommended bonding primer.
      c. Appurtenant surfaces include but are not limited to:
         1) Columns, beams, bracing and similar components.
         2) Underside of roof or floor decks above.
         3) Conduit, boxes, covers and supports.
         4) Ductwork, duct insulation and duct supports.
         5) Piping, pipe insulation and jacketing.
         6) Miscellaneous ferrous metal surfaces.
         7) Hollow core concrete exposed to view surfaces.
   2. Concrete walls and columns.
   3. Concrete masonry.
   4. Doors:
      a. Wood Doors.

3.2 ITEMS NOT TO BE PAINTED

A. General: Do not paint items listed in this Article, unless noted otherwise.

B. Items with Approved Factory Finish: These items may require repair of damaged painted areas or painting of welded connections.

C. Electrical equipment.

D. Moving parts of mechanical and electrical units where painting would interfere with the operation of the unit.

E. Code labels, equipment identification or rating plates and similar labels, tagging and identification.

F. Contact surfaces of friction-type structural connections.

G. Stainless steel surfaces.

H. Aluminum Surfaces Except:
   1. Where specifically shown in the Contract Documents.

2. Where in contact with concrete.
3. Where in contact with dissimilar metals.

I. Fiberglass Surfaces Except:
1. Fiberglass piping where specifically noted to be painted.
2. Piping supports where specifically noted to be painted.

J. Architectural finishes:
1. Exterior concrete indicated to receive another finish.
2. Plastic laminate.
3. Solid surface material.
4. Fiberglass fabrications.
5. Anodized aluminum.
6. PVDF coated metals.
7. Finish hardware.
8. Glass and glazing.
9. Ceramic tile.
10. Acoustical materials.
11. Building specialties.
12. Louvers.
13. Casework and countertops.

3.3 EXAMINATION

A. Concrete:
1. Test pH of surface to be painted in accordance with ASTM D4262.
   a. If surface pH is not within paint manufacturer's required acceptable range, use methods acceptable to paint manufacturer as required to bring pH within acceptable range.
   b. Retest pH until acceptable results are obtained.
2. Verify that moisture content of surface to be painted is within paint manufacturer's recommended acceptable limits.
   a. Test surface to be coated in accordance with ASTM D4263 to determine the presence of moisture.
      1) If moisture is detected, test moisture content of surface to be coated in accordance with ASTM F1869.
      2) Provide remedial measures as necessary to bring moisture content within paint manufacturer’s recommended acceptable limits.
      3) Retest surface until acceptable results are obtained.

B. Concrete Unit Masonry:
1. Test pH of surface to be painted in accordance with ASTM D4262.
   a. If surface pH is not within paint manufacturer's required acceptable range, use methods acceptable to paint manufacturer as required to bring pH within acceptable limits.
   b. Retest pH until acceptable results are obtained.
2. Verify that moisture content of surface to be painted is within paint manufacturer's recommended acceptable limits.
   a. Test surface to be coated in accordance with ASTM D4263 to determine the presence of moisture.
      1) If moisture is detected, test moisture content of surface to be coated in accordance with ASTM F1869.
      2) Provide remedial measures as necessary to bring moisture content within paint manufacturer’s recommended acceptable limits.
      3) Retest surface until acceptable results are obtained.

3.4 PREPARATION

A. General:
1. Prepare surfaces to be painted in accordance with paint manufacturer's instructions and this Specification Section unless noted otherwise in this Specification Section.

a. Where discrepancy between paint manufacturer's instructions and this Specification Section exists, the more stringent preparation shall be provided unless approved otherwise, in writing, by the Contracting Officer Representative.

2. Remove all dust, grease, oil, compounds, dirt and other foreign matter which would prevent bonding of paint to surface.
3. Adhere to manufacturer's recoat time surface preparation requirements.
   a. Surfaces that have exceeded paint manufacturer's published recoat time and/or have exhibited surface chalking shall be prepared prior to additional paint in accordance with manufacturer's published recommendations.

B. Protection:
1. Protect surrounding surfaces not to be coated.
2. Remove and protect hardware, accessories, plates, fixtures, finished work, and similar items; or provide ample in-place protection.
3. Protect code labels, equipment identification or rating plates and similar labels, tagging and identification.

C. Prepare and paint before assembly all surfaces which are inaccessible after assembly.

D. Existing Surfaces:
1. Wherever existing work is cut, patched or modified; repair and repaint to match new work.
2. Where a wall or ceiling is disturbed and patched, paint entire wall or ceiling.

E. Wood:
1. Sandpaper smooth, remove dust.
2. Opaque Finishes:
   a. Seal all knots, pitch and resinous sapwood after prime coat has dried.
   b. Putty holes and imperfections; sand smooth.
3. Transparent Finishes:
   a. Treat wood with compatible wash-coat prior to stain application.
   b. Putty holes and imperfections to match wood color; sand smooth.

F. Ferrous Metal:
1. Complete fabrication, welding or burning before beginning surface preparation.
   a. Chip or grind off flux, spatter, slag or other laminations left from welding.
   b. Remove mill scale.
   c. Grind smooth rough welds and other sharp projections.
2. Solvent clean in accordance with SSPC SP 1 to remove all dust, grease, oil, compounds, dirt and other foreign matter.
3. Exterior exposure:
   a. Commercial blast clean in accordance with SSPC SP 6/NACE No. 3.
4. Interior exposure:
   a. Hand tool cleaning in accordance with SSPC SP 2 and/or power tool cleaning in accordance with SSPC SP 3.

G. Hollow Metal:
1. Solvent clean in accordance with SSPC SP 1 to remove all dust, grease, oil, compounds, dirt and other foreign matter.
2. Lightly sand primed surfaces with fine grit sandpaper as recommended by hollow metal manufacturer.

H. Galvanized Steel and Non-ferrous Metals:
1. Solvent clean to remove all dust, grease, oil, compounds, dirt and other foreign matter.
2. Brush-off blast in accordance with SSPC SP 16 or hand tool cleaning in accordance with SSPC SP 2 to remove surface contaminants.

I. Gypsum Wallboard:
1. Repair minor irregularities left by finishers.
2. Avoid raising nap of paper face on gypsum wallboard.

3. Verify moisture content is less than 8 percent before painting.
4. After application of prime coat and between subsequent coats, inspect surface and repair holes, dents, irregularities or other defects as necessary to provide a smooth, uniform finish.

3.5 APPLICATION

A. General:
   1. Thin, mix and apply paints in accordance with manufacturer's installation instructions.
      a. Where discrepancy exists between manufacturer's instructions and this Specification Section, the more stringent requirement shall apply.
      b. Backroll spray applied paints.
   2. Temperature and weather conditions:
      a. Avoid painting surfaces exposed to hot sun.
      b. Do not paint on damp surfaces.
   3. Evenly spread to provide full, smooth coverage.
      a. All paint systems are "to cover."
         1) When color or undercoats show through, apply additional coats until paint film is of uniform finish and color.
      b. Finished paint system shall be uniform and without voids, bugholes, holidays, laps, brush marks, roller marks, runs, sags or other imperfections.
   4. Work each application of material into corners, crevices, joints, and other difficult to work areas.
   5. When painting rough surfaces, hand brush and backroll paint to work into all recesses.
   6. Smooth out runs or sags immediately, or remove and recoat entire surface.
   7. Allow preceding coats to dry before recoating.
      a. Recoat within time limits specified by paint manufacturer.
      b. If recoat time limits have expired re-prepare surface in accordance with paint manufacturer's printed recommendations.
   8. Allow coated surfaces to cure prior to allowing traffic or other work to proceed.
   9. Finish colors not otherwise indicated shall be selected by Engineer from paint manufacturer's complete offering.

B. Fillers, surfacers or patching compounds:
   1. Provide fillers, surfacers or patching compounds in accordance with manufacturer's recommendations and as specified herein as necessary to provide a smooth, defect free substrate.

C. Prime Coat Application:
   1. Prime all surfaces indicated to be painted.
      a. Apply prime coat in accordance with paint manufacturer's written instructions and as written in this Specification Section.
   2. Ensure field-applied paints are compatible with factory-applied paints or existing coatings.
      a. Employ services of coating manufacturer's qualified technical representative.
         1) Certify through material data sheets.
         2) Perform test patch.
      b. If field-applied coating is found to be not compatible, require the coating manufacturer's technical representative to recommend, in writing, product to be used as barrier coat, thickness to be applied, surface preparation and method of application.
      c. At Contractor's option, coatings may be removed, surface re-prepared, and new coating applied using appropriate paint system complying with this Specification Section.
         1) All damage to surface as result of coating removal shall be repaired to original condition or better by Contractor at no additional cost to Government.

D. Finish Coat Application:
   1. Apply finish coats in accordance with paint manufacturer's written instructions and in accordance with this Specification Section.
   2. Touch up damaged finish coats using same application method and same material specified for finish coat.

3. Hollow metal frames and doors:
    a. Finish coats shall be spray applied only.
    b. Finish edges same as faces of doors.
4. Varnish:
    a. Apply first coat of varnish: Gloss.
        1) Allow to dry a minimum of 48 hours.
    b. Apply second and third coats of varnish: Satin.
        1) Allow a minimum of 48 hours between each coat.
    c. Lightly sand between coats as required and remove dust.

3.6 FIELD QUALITY CONTROL

A. Application Deficiencies:
1. Surfaces showing runs, laps, brush marks, telegraphing of surface imperfections or other defects will not be accepted.
2. Surfaces showing evidence of fading, chalking, blistering, delamination or other defects due to improper surface preparation, environmental controls or application will not be accepted.

B. Provide protection for painted surfaces.
1. Surfaces showing soiling, staining, streaking, chipping, scratches, or other defects will not be accepted.

C. Maintain Daily Records:
1. Record the following information during application of each coat of paint applied:
    a. Date, starting time, end time, and all breaks taken by painters.
    b. For exterior painting:
        1) Sky condition.
        2) Wind speed and direction.
    c. Air temperature.
    d. Relative humidity.
    e. Moisture content and surface temperature of substrate prior to each coat.
    f. Provisions utilized to maintain work area within manufacturer's recommended application parameters including temporary heating, ventilation, cooling, dehumidification and provisions utilized to mitigate wind-blown dust and debris from contaminating the wet paint film.
    g. Record environmental conditions, substrate moisture content and surface temperature information not less than once every four (4) hours during application.
        1) Record hourly when temperatures are below 50 degrees F or above 100 degrees F.
2. Record the following information daily for the paint manufacturer's recommended curing period:
    a. Date and start time of cure period for each item or area.
    b. For exterior painting:
        1) Sky conditions.
        2) Wind speed and direction.
    c. Record environmental conditions not less than once every 12 hours.
        1) Record once every 4 hours when ambient temperature is below 35 degrees F.
    d. Provisions utilized to protect each item or area and to maintain areas within manufacturer's recommended curing parameters.
3. Format for daily record to be computer generated.

D. Measure surface temperature of items to be painted with surface temperature gage specifically designed for such.

E. Measure substrate humidity with humidity gage specifically designed for such.

F. Provide wet paint signs.

3.7 CLEANING

A. Clean paint spattered surfaces.

1. Use care not to damage finished surfaces.

B. Remove masking, adhesive residue or other foreign materials.

C. Upon completion of painting, replace hardware, accessories, plates, fixtures, and similar items.

D. Remove surplus materials, scaffolding, and debris.

3.8 COLOR SCHEDULE

A. Refer to “Room Finish Schedule found on Architectural Finish Schedules” drawing and “Interior Finish Legend found on Architectural Finish Legend and Notes” drawing.

END OF SECTION

SECTION 09 96 00
HIGH PERFORMANCE INDUSTRIAL COATINGS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. High performance industrial coatings (HPIC).
   2. Any other coating, thinner, accelerator, inhibitor, etc., specified or required as part of a complete System specified in this Specification Section.
   3. Minimum surface preparation requirements.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 01 61 03 - Equipment - Basic Requirements.
   3. Section 09 06 00 – Color Schedule.
   4. Section 09 91 00 - Architectural Painting.
   5. Section 40 05 00 - Pipe and Pipe Fittings - Basic Requirements.
   6. Section 43 41 15 – Field Erected Welded Steel Tanks

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. B499, Standard Test Method for Measurement of Coating Thicknesses by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metals.
      b. D3359, Standard Test Methods for Rating Adhesion by Tape Test.
      c. D4258, Standard Practice for Surface Cleaning Concrete for Coating.
      d. D4259, Standard Practice for Abrading Concrete.
      e. D4261, Standard Practice for Surface Cleaning Concrete Masonry Units for Coating.
      f. D4262, Standard Test Method for pH of Chemically Cleaned or Etched Concrete Surfaces.
      g. D4263, Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method.
      h. D4414, Standard Practice for Measurement of Wet Film Thickness by Notch Gages.
      i. D4541, Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers.
      j. D6132, Standard Test Method for Nondestructive Measurement of Dry Film Thickness of Applied Organic Coatings Using an Ultrasonic Gage.
      k. D6677, Standard Test Method for Evaluating Adhesion by Knife.
      l. D7091, Standard Practice for Nondestructive Measurement of Dry Film Thickness of Nonmagnetic Coatings Applied to Ferrous Metals and Nonmagnetic, Nonconductive Coatings Applied to Non-Ferrous Metals.
      m. D7234, Standard Test Method for Pull-Off Adhesion Strength of Coatings on Concrete Using Portable Pull-Off Adhesion Testers.
      n. E337, Standard Test Method for Measuring Humidity with a Psychrometer (the Measurement of Wet- and Dry-Bulb Temperatures).
      o. F1869, Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride.
      p. F2170, Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes.
   2. Environmental Protection Agency (EPA).
   3. International Concrete Repair Institute (ICRI):
      a. 310.2, Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, Polymer Overlays, and Concrete Repair.

4. NACE International (NACE).
    a. SP0188, Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates.
5. National Association of Pipe Fabricators (NAPF):
    a. 500-03, Surface Preparation Standard for Ductile Iron Pipe and Fittings in Exposed Locations Receiving Special External Coatings and/or Special Internal Linings:
        1) 500-03-04, Abrasive Blast Cleaning for Ductile Iron Pipe.
        2) 500-03-05, Abrasive Blast Cleaning for Cast Ductile Iron Fittings.
6. NSF International (NSF).
    a. 61, Drinking Water System Components - Health Effects.
7. The Society for Protective Coatings (SSPC):
    a. PA 2, Procedure for Determining Conformance to Dry Coating Thickness Requirements.
    b. SP 1, Solvent Cleaning.
    c. SP 2, Hand Tool Cleaning.
    d. SP 3, Power Tool Cleaning.
    e. SP 16, Brush-off Blast Cleaning of Coated and Uncoated Galvanized Steel, Stainless Steels, and Non-Ferrous Metals.
8. The Society for Protective Coatings/ NACE International (SSPC/ NACE):
    a. SP 5/ NACE No. 1, White Metal Blast Cleaning
    b. SP 6/ NACE No. 3, Commercial Blast Cleaning.
    c. SP 7/ NACE No. 4, Brush-off Blast Cleaning.
    d. SP 10/ NACE No. 2, Near-White Blast Cleaning.
    e. SP 13/ NACE No. 6, Surface Preparation of Concrete.

B. Qualifications:
1. Coating manufacturer's technical representative shall be a NACE Certified Coatings Inspector, Level 3 minimum.
2. Applicators shall have minimum of 10 years of experience in application of similar products on similar project.
    a. Provide references for minimum of three different projects completed in last five years with similar scope of work.
    b. Include name and address of project, size of project in value (coating) and contact person.
3. NACE inspector shall be NACE Certified Coatings Inspector Level 3 minimum and shall have minimum of five years of experience of conducting inspections and tests as indicated in this Specification Section.

C. Miscellaneous:
1. Furnish coating through one manufacturer unless noted otherwise.

D. Deviation from specified MIL thickness or product type is not allowed without written authorization of Contracting Officer’s Representative.

E. Material shall not be thinned unless approved, in writing, by coating manufacturer's technical representative.

1.3 DEFINITIONS

A. Applicator:
1. Applicator is the person actually installing or applying the product in the field, at the Project site, or at an approved shop facility.

B. Approved Factory Finish: Finish on a product in compliance with the finish specified in the Specification Section where the product is specified or in Specification Section 01 61 03.

C. Appurtenant Surface: Accessory or auxiliary surface attached to or adjacent to a surface indicated to be coated.

D. Corrosive Environment:

1. Immersion in or subject to:
    a. Condensation, spillage or splash of a corrosive material such as water, wastewater or chemical solution.
    b. Exposure to corrosive caustic or acidic agent, chemicals, chemical fumes, chemical mixture, or solutions.
    c. For purposes of this Specification Section, corrosive environments include:
        1) Outdoor areas not otherwise identified as highly corrosive.
        2) Surfaces within 2-feet of high-water level.

E. Outdoor Atmosphere or Surface: Outdoor atmosphere or surface exposed to weather and/or direct sunlight.

F. Finished Area: A room or area that is listed in or has finish called for on Room Finish Schedule or is indicated on Drawings to be coated.

G. Highly Corrosive Environment:
1. Immersion in or subject to:
    a. Condensation, spillage or splash of a highly corrosive material such as wastewater, or chemical solution.
    b. Exposure to highly corrosive caustic or acidic agent, chemicals, chemical fumes, chemical mixture, or solutions.
2. For purposes of this Specification Section, highly corrosive environments include:
    a. Surfaces within 10-feet of chemical fill stations.
    b. Areas within 6-feet of chemical tank vents.
    c. Exposure to hydrogen sulfide (H2S) gas.
    d. Digester cover attic space.
    e. Chemicals storage and feed areas
3. For purposes of this Specification Section, the entire building/spaces of the following buildings will be considered as highly corrosive environments:
    a. Building 052 – Roaring Springs Pumphouse.
    b. Building 062 – North Rim Water Treatment Plant.

H. Holiday:
1. A void, crack, thin spot, foreign inclusion, or contamination in the coating that significantly lowers the dielectric strength of the coating.
2. May also be identified as a discontinuity or pinhole.

I. HPIC: High performance industrial coatings.
1. Epoxies, urethanes, vinyl ester, waterborne vinyl acrylic emulsions, acrylates, silicones, alkyds, acrylic emulsions and any other coating listed as a HPIC.

J. Indoor Atmosphere or Surface: Indoor atmosphere or surface not exposed to weather and/or direct sunlight.

K. Immersion Service:
1. Any surface immersed in water or some other liquid.
2. Surface of any pipe, valve, or any other component of the piping system subject to frequent wetting.
3. Surfaces within two feet above high water level in water bearing structures.

L. Piping System: Pipe, valves, fittings and accessories.

M. Surface Hidden from View:
1. Within pipe chases.
2. Between top side of ceilings and underside of floor or roof structures above.

N. Vapor Space: Interior space within tankage, closed structures, or similar elements that is above the low liquid line and subject to the accumulation of fumes, vapor and/or condensation.

1.4 SUBMITTALS

A. Certifications:
   1. Applicator experience qualifications.
      a. No submittal information will be reviewed until Contracting Officer's Representative has received and approved applicator qualifications.
   2. NACE inspector certification.
   3. NACE inspector experience qualifications.
   4. Certification that High Performance Coating Systems proposed for use have been reviewed and approved by a NACE Certified Coatings Inspector employed by the coating manufacturer.
      a. Submittals not including this certification will be returned without review.

B. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's surface preparation instructions.
      c. Manufacturer's application instructions.
         1) Manufacturer's standard details, including but not limited to penetrations, transitions, and terminations for:
            a) High-build coatings on concrete.
            b) Secondary containment coatings.
            c) Membrane Bioreactor tanks.
            d) UV channels.
            e) Other special conditions as applicable.
      d. If products being used are manufactured by Company other than listed in the MATERIALS Article of this Specification Section, provide complete individual data sheet comparison of proposed products with specified products including:
         1) Application procedure.
         2) Coverage rates.
         3) Certification that product is designed for intended use and is equal or superior to specified product.
      e. Contractor's written plan of action for containing airborne particles created by blasting operation and location of disposal of spent contaminated blasting media.
      f. Coating manufacturer's recommendation on abrasive blasting.
      g. Coating manufacturer's technical representative's written statement attesting that applicator has been instructed on proper preparation, mixing and application procedures for coatings specified.
      h. Manufacturer's recommendation for universal barrier coat.
      i. Manufacturer's recommendation for providing temporary or supplemental heat or dehumidification or other environmental control measures.
   2. Manufacturer's statement regarding applicator instruction on product use.

C. Coating Work Plan:
   1. Submit for any tank larger than 20 feet in diameter.
   2. Shall contain the minimum:
      a. Product Data Sheets (PDS) and Safety Data Sheets (SDS) for each coating, blasting, or other materials required to complete this section.
      b. Safety, fire, health, and environmental plan. Disposal of coatings, blast materials.
      c. Equipment for surface preparation, coating application, and quality assurance testing.
      d. Contractor and worker qualifications and certifications.
      e. Material storage and handling.
      f. Timeline/Schedule.
      g. Daily contractor log, and coating inspector log.
      h. Warranty signed by contractor and coatings manufacturer's representative.

i. Any additional details required to provide complete coating system.

D. Samples:
1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
3. Refer to “Specification Section 10 14 00 – Identification Devices” for the piping system and banding material and refer to the “PIPE SCHEDULES in the Drawings” for the paint and banding colors.

E. Informational Submittals:
1. Approval of application equipment.
2. Applicator's daily records:
a. Submit daily records at end of each week in which coating work is performed unless requested otherwise by Contracting Officer’s Representative on-site representative.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver in original unopened containers, labeled as follows:
1. Name or type number of material.
2. Manufacturer's name and item stock number.
3. Contents, by volume, of major constituents.
4. Warning labels.
5. VOC content.
6. Lot number.
7. Date of manufacture.

B. Store materials in tightly covered unopened containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 degrees F, per manufacturers recommendations.

1.6 PROJECT CONDITIONS

A. Pre-application Conference:
1. Prior to commencement of surface preparation or coating application, the Contractor shall convene a pre-application conference with all affected parties, including but not limited to: the applicator, coating manufacturer's technical representative, and Contracting Officer’s Representative.
2. The meeting shall discuss all aspects of the Project including but not limited to:
a. Coating Work Plan.
b. Schedule.
c. Material storage and handling.
d. Examination of surfaces to be coated.
e. Protection of surfaces not to be coated.
f. Surface preparation.
g. Coating application:
1) Environmental conditions for application of coatings.
2) Temporary environmental controls.
h. Field quality control requirements:
1) Manufacturer's technical representative responsibilities.
2) Contractor performed testing.
a) Instrumentation requirements.
b) Frequency of testing.
c) Record keeping.
3) NACE inspector performed testing.

B. Verify that atmosphere in area where coating is to take place is within coating manufacturer's acceptable temperature, humidity and sun exposure limits.

1. Provide temporary heating, shade and/or dehumidification as required to bring area within acceptable limits.
    a. Provide temporary dehumidification equipment properly sized to maintain humidity levels required by coating manufacturer.
    b. Provide clean heat with heat exchanger type equipment sufficient in size to maintain temperature on a 24-hour basis.
        1) Vent exhaust gases to outdoor environment.
        2) No exhaust gases shall be allowed to vent into the space being coated or any adjacent space.
2. Do not apply coatings in snow, rain, fog or mist.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
    1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
    1. High Performance Industrial Coatings:
        a. The Sherwin-Williams Company.
        b. Or Equal.

C. "Or Equal" Submittals:
    1. Materials by other manufacturers are acceptable provided that they are established as being compatible with and of equal quality to the coatings of the manufacturers listed.
    2. Provide satisfactory documentation from the proposed "or-equal" manufacturer that proposed materials meets or exceeds the following:
        a. Is of the same generic resin.
        b. Requires comparable surface preparation.
        c. Has comparable application requirements.
        d. Meets the same VOC levels or better.
        e. Provides the same finish and color options.
        f. Is suitable for the intended service.
        g. Resistance to abrasion and physical damage.
        h. Resistance to chemical attack.
        i. Resistance to UV exposure.
        j. Ability to recoat in future.
        k. Dry film thickness per coat.
            1) Where manufacturer's product data sheet indicates a minimum MIL thickness per coat that is greater than specified herein, MIL thickness for entire coating system shall be increased proportionately.
        l. Minimum and Maximum time between coats.
        m. Compatibility with other coatings.
        n. Temperature limitations in service and during application.
        o. Type and quality of recommended undercoats and topcoats.
        p. Ease of application.
        q. Ease of repairing damaged areas.
        r. Stability of colors.
    3. The cost of all testing and analyzing of the proposed substitute materials shall be borne by the CONTRACTOR.

2.2 MATERIALS

A. Coatings used for indoor finishes shall meet the requirements of the building code and NFPA 101.
   1. Door and frames finishes: ASTM E84.
      a. Class A: Flame spread index 25 or less; smoke-developed index 450 or less.

B. For unspecified materials such as thinner, provide manufacturer's recommended products.

C. High Performance Industrial Coatings:

COATING CODE	GENERIC DESCRIPTION	MANUFACTURER
		SHERWIN WILLIAMS
AAE	Acrylic/Acrylate Emulsion	Cement Plex 875
AREC	Abrasion-Resistant Epoxy Coating	Duraplate 5900
AAP	Aliphatic Acrylic Polyurethane	Acrolon Ultra
CRU	Corrosion Resistant Urethane	Polylon HP
ESF	Epoxy Surfacer/Filler	Steel Seam FT 910
EMM	Epoxy Modified Cementitious Mortar	Duraplate 2300
EF	Epoxy Flooring	GP3746
GFRE	Glass Flake Reinforced Epoxy	Sher-Glass FF
HRE	H2S-Resistant Epoxy	Duraplate 235
HREM	H2S-Resistant Epoxy Mortar	Duraplate 5900 Mortar
MPE	Multi-Purpose Epoxy	Macropoxy 646
MTEP	Moisture-Tolerant Epoxy Primer	Corobond 100
SCE	Secondary Containment Epoxy	Cor Cote HP
SCEP	Secondary Containment Epoxy Primer	GP3552
STEP	Surface-Tolerant Epoxy Primer	Macropoxy 646
UHSE	Ultra-High Solids Epoxy (NSF 61)	Duraplate UHS
ZRU	Zinc-Rich Urethane	Corothane 1 Galvapak

D.

2.3 COATING SYSTEMS:

A. The following tables indicate coating systems by material and environment unless a specific application is indicated.

B. Ferrous Metals (Structural and Miscellaneous Metals):

Environment/ Application	Surface Preparation	Prime Coat	Intermediate Coats	Finish Coat
Immersion - Wastewater	SSPC-SP 10/ NACE No. 2	3.0 to 4.0 mil MPE		12 to 16 mil GFRE
Immersion - non NSF	SSPC-SP 10/ NACE No. 2	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE
Immersion - NSF	SSPC-SP / NACE No.1	4.0 to 5.0 mil NSFE	4.0 to 5.0 mil NSFE	4.0 to 5.0 mil NSFE
Outdoor atmospheric	SSPC-SP 6/ NACE No. 3	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE	2.5 to 3.5 mil AAP

C. Galvanized Steel:

Environment/ Application	Surface Preparation	Prime Coat	Intermediate Coats	Finish Coat
Indoor atmospheric	SSPC-SP 16	4.0 to 6.0 mil STEP		2.0 to 3.0 mil MPE
Immersion - non NSF	SSPC-SP 16	4.0 to 6.0 mil STEP	2.0 to 3.0 mil MPE	2.0 to 3.0 mil MPE
Outdoor atmospheric	SSPC-SP 16	4.0 to 6.0 mil STEP		2.5 to 3.5 mil AAP
Field cut pipe threads	SSPC-SP 3	4.0 to 6.0 mil STEP	Coat per exposure above	Coat per exposure above

D. Non-Ferrous Metals:

Environment/ Application	Surface Preparation	Prime Coat	Intermediate Coats	Finish Coat
Dissimilar Materials Protection	SSPC-SP 2	4.5 to 5.5 mil MPE		
Indoor atmospheric	SSPC-SP 2	3.0 to 4.0 mil MPE		3.0 to 4.0 mil MPE
Outdoor atmospheric	SSPC-SP 2	4.0 to 6.0 mil MPE		2.5 to 3.5 mil AAP

E. Ferrous Piping:

Environment/ Application	Surface Preparation	Prime Coat	Intermediate Coats	Finish Coat
Indoor atmospheric	SSPC-SP 6/ NACE No. 3	2.5 to 3.5 mil ZRU	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE
Vapor space at covered clarifiers, digesters and similar structures	SSPC-SP 10/ NACE No. 2 min. 3 mil anchor profile	5 to 7 mil HRE		30 to 40 mil AREC
Immersion - Wastewater	SSPC-SP 10/ NACE No. 2	3.0 to 4.0 mil HRE		12 to 16 mil GFRE
Immersion - Wastewater (abrasion resistant)	SSPC-SP 10/ NACE No. 2, min 3 mil anchor profile	15 to 20 mil AREC		15 to 20 mil AREC
Immersion - non NSF	SSPC-SP 5/ NACE No.1	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE
Immersion - NSF	SSPC-SP 5/ NACE No.1	4.0 to 5.0 mil NSFE	4.0 to 5.0 mil NSFE	4.0 to 5.0 mil NSFE
Outdoor atmospheric	SSPC-SP 6/ NACE No. 3	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE	2.5 to 3.5 mil AAP

F. Ductile Iron Piping:

Environment/ Application	Surface Preparation	Prime Coat	Intermediate Coats	Finish Coat
Indoor atmospheric	Pipe: NAPF 500-03-04 Fittings: NAPF 500-03-05	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE
Vapor space at covered clarifiers, digesters and similar structures	Pipe: NAPF 500-03-04 Fittings: NAPF 500-03-05	5 to 7 mil HRE		30 to 40 mil AREC
Immersion - Wastewater	Pipe: NAPF 500-03-04 Fittings: NAPF 500-03-05	3.0 to 4.0 mil HRE		12 to 16 mil GFRE
Immersion - Wastewater (abrasion resistant)	Pipe: NAPF 500-03-04 Fittings: NAPF 500-03-05	15 to 20 mil AREC		15 to 20 mil AREC
Immersion - non NSF	Pipe: NAPF 500-03-04 Fittings: NAPF 500-03-05	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE
Immersion - NSF	Pipe: NAPF 500-03-04 Fittings: NAPF 500-03-05	3.0 to 4.0 mil NSFE		20.0 to 25.0 mil UHSE
Outdoor atmospheric	Pipe: NAPF 500-03-04 Fittings: NAPF 500-03-05	3.0 to 4.0 mil MPE	3.0 to 4.0 mil MPE	2.5 to 3.5 mil AAP

G. Cast Iron Piping:

Environment/ Application	Surface Preparation	Prime Coat	Intermediate Coats	Finish Coat
Immersion – NSF	Pipe: NAPF 500-03-04 Fittings: NAPF 500-03-05	3.0 to 4.0 mil NSFE		20.0 to 25.0 mil UHSE

H. Concrete:
1. For repair of deteriorated existing concrete, provide additional surface preparation as specified in PREPARATION article in this Specification Section.

Environment/ Application	Surface Preparation	Filler/Surfacer	Prime Coat	Intermediate Coat(s)	Finish Coat
Indoor Safety Striping	SSPC-SP 13/ NACE No. 6 ICRI CSP 3	ESF and/or EMM as necessary to fill holes and depressions			6.0 to 8.0 mil EF Pigmented
Immersion - non NSF	SSPC-SP 13/ NACE No. 6 ICRI CSP 5	1/16 to 1/4 inches EMM			16 to 20 mil UHSE
Immersion - Wastewater (Abrasion Resistant)	SSPC-SP 13/ NACE No. 6 ICRI CSP 5	1/16 to 1/4 inches EMM	1/8 inches HREM		15 to 25 mil AREC
Outdoor atmospheric Corrosive Environment	SSPC-SP 13/ NACE No. 6 ICRI CSP 5	ESF and/or EMM as necessary to fill holes and depressions	150 to 175 square feet/GAL AAE		150 to 175 square feet/GAL AAE
Outdoor - Secondary Containment Corrosive	SSPC-SP 13/ NACE No. 6 ICRI CSP 5	ESF and/or EMM as necessary to fill holes and depressions	6.0 to 8.0 mil MTEP	60 to 80 mil SCEP Fiberglass mat Saturated with 8.0 to 12 mils SCE	10 to 12 mil SCE 20. to 3.0 mil CRU

PART 3 - EXECUTION

3.1 ITEMS TO BE COATED

A. Outdoor Surfaces, including but not limited to:
   1. Concrete:
      a. Components of concrete tankage:
         1) Coat interior surfaces of tankage.
      b. Secondary containment enclosures and pits.
   2. Piping, valves, fittings, hydrants and supports:
      a. As scheduled in Specification Section 40 05 00.
      b. Process piping scheduled to be insulated.
      c. Field welded connections of factory coated piping.
   3. Pumps and motors.
   4. New and existing ferrous metal tankage.
      a. Both the internal and external surfaces of the existing Finished Water Tanks and the new Raw Water Tanks.
      b. Internal piping of Raw Water Tanks.
      c. Internal piping of Finished Water Tanks.
         1) Remove existing coatings and recoat.
   5. Ferrous metal process equipment.
      a. Clarifier mechanisms.
      b. Equipment bridges.
      c. Gates and operators.
   6. Structural steel:

a. Columns, beams and bracing.
b. Steel joists or trusses, including bridging.
c. Field welded connections of factory coated structural steel.

7. Miscellaneous ferrous metal surfaces:
   a. Items specifically noted on Drawings to be coated.
8. Miscellaneous galvanized steel surfaces:
   a. Pipe bollards.
   b. Embed plates.
   c. Loose lintels.
   d. Galvanized snow hoods.
   e. Items specifically noted on Drawings to be coated.
9. Miscellaneous nonferrous metal surfaces:
   a. Dissimilar metals.
   b. Aluminum doors and frames.
   c. Aluminum roll down doors.
   d. Aluminum window frames.
   e. Exterior equipment screen.
10. Appurtenant surfaces attached to or adjacent to a surface indicated to be coated:
    a. Conduit, boxes, covers and supports.

B. Indoor Areas:
1. Refer to Room Finish Schedule on Drawings.
   a. If space is scheduled to be coated, coat all appurtenant surfaces within the space unless specifically noted otherwise. Appurtenant surfaces include but are not limited to:
      1) Columns.
      2) Equipment pads.
      3) Equipment supports.
      4) Miscellaneous ferrous metal surfaces.
2. Concrete:
   a. Components of concrete tankage:
      1) Walls, columns, beams:
         a) Coat from 1-foot below low water level to top of component.
      2) Troughs, launders, weirs.
      3) Underside of concrete walkways within 2-feet of high water level.
   b. Chemical storage areas:
   c. Chemical storage areas:
      1) Flooring where scheduled or indicated on Drawings.
      2) Secondary containment enclosures.
3. Piping, valves, fittings, hydrants and supports:
   a. Do not coat piping scheduled to be insulated.
4. Pumps and motors.
5. Ferrous metal tankage.
6. Ferrous metal process equipment.
   a. Reaction Tanks.
   b. Clarifier mechanisms.
   c. Equipment bridges.
   d. Gates and operators.
7. Structural steel:
   a. Columns, beams and bracing.
   b. Steel joists or trusses, including bridging.
   c. Field welded connections of factory coated structural steel.
   d. Items specifically noted on Drawings to be coated.
8. Miscellaneous galvanized steel surfaces:
   a. Pipe bollards.
   b. Embed plates.
   c. Loose lintels.

d. Items specifically noted on Drawings to be coated.
e. Hollow metal window frames.

9. Miscellaneous nonferrous interior finishes:
   a. Aluminum doors and frames.
   b. Aluminum window frames.
10. Safety Striping:
    a. Equipment Pads: Coat vertical face and return 3-inches onto horizontal surface of pad.
    b. Pipe supports, columns, piers and similar vertical elements.
    c. As shown on Drawings.

3.2 ITEMS NOT TO BE COATED

A. General: Do not coat items listed in this Article, unless noted otherwise.

B. Items with Approved Factory Finish: These items may require repair of damaged coated areas or coating of welded connections.

C. Electrical Equipment.

D. Moving parts of mechanical and electrical units where coating would interfere with the operation of the unit.

E. Code labels, equipment identification or rating plates and similar labels, tagging and identification.

F. Contact surfaces of friction-type structural connections.

G. Stainless Steel Surfaces, except:
   1. Dissimilar metals in immersion service.
   2. Piping where specifically noted to be coated.
   3. Banding as required to identify piping.

H. Aluminum Surfaces, except:
   1. Where specifically shown in the Contract Documents.
   2. Where in contact with concrete.
   3. Where in contact with dissimilar metals.
   4. Appurtenant surfaces as described in the ITEMS TO BE COATED article.

I. Fiberglass Surfaces, except:
   1. Fiberglass piping where specifically noted to be coated.
   2. Piping supports where specifically noted to be coated.
   3. Appurtenant surfaces as described in the ITEMS TO BE COATED article.

J. Mechanical piping scheduled to be insulated.

K. PVC Pipe.

L. Interior of Pipe, Ductwork, and Conduits.
   1. See Division 23 for ductwork.
   2. See Division 40 for pipe linings.

M. Galvanized Steel Items, unless specifically noted to be coated.

N. Architectural Finishes:
   1. Outdoor concrete indicated to receive another finish.
   2. Anodized aluminum.
   3. PVDF coated metals.
   4. Finish hardware.
   5. Interior louvers and grilles.
   6. Pipe insulation and jacketing.

N. Cathodic Protection anodes, wiring, and rigging.

3.3 EXAMINATION

A. Concrete:
   1. Test pH of surface to be coated in accordance with ASTM D4262.
      a. If surface pH is not within coating manufacturer's required acceptable range, use methods acceptable to coating manufacturer as required to bring pH within acceptable range.
      b. Retest pH until acceptable results are obtained.
   2. Verify that moisture content of surface to be coated is within coating manufacturer's recommended acceptable limits.
      a. Test surface to be coated in accordance with ASTM D4263 to determine the presence of moisture.
         1) If moisture is detected, test moisture content of surface to be coated in accordance with ASTM F1869 or ASTM F2170.
         2) Provide remedial measures as necessary to bring moisture content within coating manufacturer’s recommended acceptable limits.
         3) Retest surface until acceptable results are obtained.

B. Concrete Unit Masonry:
   1. Test pH of surface to be coated in accordance with ASTM D4262.
      a. If surface pH is not within coating manufacturer's required acceptable range, use methods acceptable to coating manufacturer as required to bring pH within acceptable limits.
      b. Retest pH until acceptable results are obtained.
   2. Verify that moisture content of surface to be coated is within coating manufacturer's recommended acceptable limits.
      a. Test surface to be coated in accordance with ASTM D4263 to determine the presence of moisture.
         1) If moisture is detected, test moisture content of surface to be coated in accordance with ASTM F1869.
         2) Provide remedial measures as necessary to bring moisture content within coating manufacturer’s recommended acceptable limits.
         3) Retest surface until acceptable results are obtained.

3.4 PREPARATION

A. General:
   1. Prepare surfaces to be coated in accordance with coating manufacturer's instructions and this Specification Section unless noted otherwise in this Specification Section.
      a. Where discrepancy between coating manufacturer's instructions and this Specification Section exists, the more stringent surface preparation shall be provided unless approved otherwise, in writing, by the Contracting Officer Representative.
   2. Remove all dust, grease, oil, compounds, dirt and other foreign matter which would prevent bonding of coating to surface.
   3. Adhere to manufacturer's recoat time surface preparation requirements.
      a. Surfaces that have exceeded coating manufacturer's published recoat time and/or have exhibited surface chalking shall be prepared prior to additional coating in accordance with manufacturer's published recommendations.
         1) Minimum SSPC-SP 7/ NACE No. 4 unless otherwise approved by Contracting Officer Representative.

B. Protection:
   1. Protect surrounding surfaces not to be coated.
   2. Remove and protect hardware, accessories, plates, fixtures, finished work, and similar items; or provide ample in-place protection.
   3. Protect code labels, equipment identification or rating plates and similar labels, tagging and identification.

C. Prepare and coat before assembly all surfaces which are inaccessible after assembly.

D. Ferrous Metal:
   1. Prepare ductile iron pipe in accordance with pipe manufacturer's recommendations and NAPF.
      a. All piping, pumps, valves, fittings and any other component used in the water piping system that requires preparation for coating shall be prepared in accordance with requirements for immersion service.
      b. Prepare all areas requiring patch coating in accordance with recommendations of manufacturer and NAPF.
      c. Remove bituminous coating per piping manufacturer, coating manufacturer and NAPF recommendations.
         1) The most stringent recommendations shall apply.
   2. Complete fabrication, welding or burning before beginning surface preparation.
      a. Chip or grind off flux, spatter, slag or other laminations left from welding.
      b. Remove mill scale.
      c. Grind smooth rough welds and other sharp projections.
   3. Solvent clean in accordance with SSPC-SP 1.
   4. Restore surface of field welds and adjacent areas to original surface preparation.

E. Galvanized Steel and Non-ferrous Metals:
   1. Solvent clean in accordance with SSPC-SP 1 followed by brush-off blast clean in accordance with SSPC-SP 16 to remove zinc oxide and other foreign contaminants.
      a. Provide uniform 1 mil profile surface.

F. Concrete:
   1. Cure for minimum of 28 days.
   2. Concrete surfaces shall be cleaned in accordance with ASTM D4258.
   3. Abrasive blast concrete surfaces in accordance with ASTM D4259 and SSPC-SP 13/ NACE No. 6.
      a. Provide profile per ICRI 301.2 as listed in MATERIALS article of this Specification Section.
   4. Test pH and moisture content in accordance with EXAMINATION article in this Specification Section.

G. Concrete Masonry:
   1. Cure for minimum of 28 days.
   2. Remove all mortar spatters and protrusions.
   3. Clean concrete masonry in accordance with Specification ASTM D4261.
   4. Test pH and moisture content in accordance with EXAMINATION article in this Specification Section.

H. Preparation by Abrasive Blasting:
   1. Schedule the abrasive blasting operation so blasted surfaces will not be wet after blasting and before coating.
   2. Provide compressed air for blasting that is free of water and oil.
      a. Provide accessible separators and traps.
   3. Protect nameplates, valve stems, rotating equipment, motors and other items that may be damaged from blasting.
   4. All abrasive-blasted ferrous metal surfaces shall be inspected immediately prior to application of coatings.
      a. Inspection shall be performed to determine cleanliness and profile depth of blasted surfaces and to certify that surface has been prepared in accordance with these Specifications.
   5. Perform additional blasting and cleaning as required to achieve surface preparation required.
      a. Re-blast surfaces not meeting requirements of these Specifications.
      b. Prior to coating, re-blast surfaces allowed to set overnight and surfaces that show rust bloom.

c. Surfaces allowed to set overnight or surfaces which show rust bloom prior to coating shall be re-inspected prior to coating application.

6. Profile depth of blasted surface: Not less than 1 mil or greater than 2 mils unless required otherwise by coating manufacturer.
7. Ensure abrasive blasting operation does not result in embedment of abrasive particles in coating.
8. Confine blast abrasives to area being blasted.
   a. Provide shields of polyethylene sheeting or other such barriers to confine blast material.
   b. Plug pipes, holes, or openings before blasting and keep plugged until blast operation is complete and residue is removed.
9. Abrasive blasting media may be recovered, cleaned and reused providing Contractor submits, for Contracting Officer Representative's review, a comprehensive recovery plan outlining all procedures and equipment proposed in reclamation process.
10. Properly dispose of blasting material contaminated with debris from blasting operation.

I. All Plastic Surfaces:
   1. Sand using 80-100 grit sandpaper to scarify surfaces.

3.5 APPLICATION

A. General:
   1. Thin, mix and apply coatings by brush, roller, or spray in accordance with manufacturer's installation instructions.
      a. Application equipment must be inspected and approved in writing by coating manufacturer.
   2. Temperature and weather conditions:
      a. Do not coat surfaces when surface temperature is below 50 degrees F unless product has been formulated specifically for low temperature application and application is approved in writing by Contracting Officer Representative and coating manufacturer's technical representative.
      b. Avoid coating surfaces exposed to hot sun.
      c. Do not coat damp surfaces.
      d. Apply coating to concrete or masonry surfaces in descending temperatures, in accordance with coating manufacturer's application instructions.
   3. Apply materials under adequate illumination.
   4. Provide complete coverage to MIL thickness specified.
      a. Thickness specified is dry MIL thickness.
   5. Evenly spread to provide full, smooth coverage.
      a. All coating systems are "to cover."
         1) In situations of discrepancy between manufacturer's square footage coverage rates and MIL thickness, MIL thickness requirements govern.
      b. When color or undercoats show through, apply additional coats until coating is of uniform finish and color.
      c. Finished coating system shall be uniform and without voids, bugholes, holidays, laps, brush marks, roller marks, runs, sags or other imperfections.
   6. If directed by Contracting Officer’s Representative, do not apply consecutive coats until Contracting Officer’s Representative has had an opportunity to observe and approve previous coats.
   7. Work each application of material into corners, crevices, joints, and other difficult to work areas.
   8. Provide coating manufacturer’s recommended details at all terminations, penetrations, embedments, cracks, joints and changes in substrate direction.
   9. Avoid degradation and contamination of blasted surfaces and avoid inter-coat contamination.
      a. Clean contaminated surfaces before applying next coat.
      b. Intercoat surface cleanliness shall be inspected and approved by the Contracting Officer Representative prior to application of each coat.

10. Smooth out runs or sags immediately, or remove and recoat entire surface.
11. Allow preceding coats to dry before recoating.
    a. Recoat within time limits specified by coating manufacturer.
    b. If recoat time limits have expired re-prepare surface in accordance with coating manufacturer's printed recommendations.
12. Allow coated surfaces to cure prior to allowing traffic or other work to proceed.
13. Coat all aluminum in contact with dissimilar materials.
14. When coating rough surfaces which cannot be backrolled sufficiently, hand brush coating to work into all recesses provided that the maximum DFT is not exceeded.
15. Backroll surfaces if coatings are spray applied.

B. Employ services of coating manufacturer's technical representative to ensure that field-applied coatings are compatible with factory-applied or existing coatings.
    1. Certify through material data sheets.
    2. Perform test patch.
        a. Prepare existing coating surface to receive specified coating system.
        b. Apply coating to a minimum 1 square feet area and allow to cure in accordance with manufacturer's recommendations.
        c. Evaluate adhesion to existing coating:
            1) Concrete or Masonry substrates: ASTM D4541.
            2) All other substrates: ASTM D6677 and ASTM D3359 (X-cut method).
    3. If field-applied coating is found to be not compatible, require the coating manufacturer's technical representative to recommend, in writing, product to be used as barrier coat, thickness to be applied, surface preparation and method of application.
        a. Perform test patch as described above.
    4. At Contractor's option, coatings may be removed, surface re-prepared, and new coating applied using appropriate coating system listed in the MATERIALS Article, Coating Systems paragraph of this Specification Section.
        a. All damage to surface as result of coating removal shall be repaired to original condition or better by Contractor at no additional cost to Government.

C. Prime Coat Application:
    1. Apply structural steel and miscellaneous steel prime coat in the factory.
        a. Finish coats shall be applied in the field.
        b. Prime coat referred to here is prime coat as indicated in this Specification.
            1) Prime coating applied in factory (shop) as part of Fabricator's standard rust inhibiting and protection coating is not acceptable as replacement for specified prime coating.
    2. Prime all surfaces indicated to be coated.
        a. Apply prime coat in accordance with coating manufacturer's written instructions and as written in this Specification Section.
    3. Prime ferrous metals embedded in concrete to minimum of 1 inch below exposed surfaces.
    4. Apply zinc-rich primers while under continuous agitation.
    5. Brush or spray bolts, welds, edges and difficult access areas with primer prior to primer application over entire surface.
    6. Touch up damaged primer coats prior to applying finish coats.
        a. Restore primed surface equal to surface before damage.
    7. All surfaces of steel lintels and steel components of concrete lintels used in wall construction shall be completely coated with both prime and finish coats prior to placing in wall.

D. Finish Coat Application:
    1. Apply finish coats in accordance with coating manufacturer's written instructions and in accordance with this Specification Section; manufacturer instructions take precedent over these Specifications.
    2. Touch up damaged finish coats using same application method and same material specified for finish coat.

a. Prepare damaged area in accordance with the PREPARATION Article of this Specification Section.

3.6 FIELD QUALITY CONTROL

A. Application Deficiencies:
   1. Surfaces showing runs, laps, brush marks, telegraphing of surface imperfections or other defects will not be accepted.
   2. Surfaces showing evidence of fading, chalking, blistering, delamination or other defects due to improper surface preparation, environmental controls or application will not be accepted.
      a. Epoxy surfaces showing evidence of chalking or amine blush shall be prepared and recoated as follows:
         1) Solvent clean surfaces in accordance with SSPC-SP1 and abrasive blast in accordance with SSPC-SP7/ NACE No. 4.
         2) Recoat with intermediate and finish coats in accordance with coating system specified herein.

B. Provide protection for coated surfaces.
   1. Surfaces showing soiling, staining, streaking, chipping, scratches, or other defects will not be accepted.

C. Contractor Performed Testing:
   1. Provide ongoing testing and inspection, including but not limited to the following:
      a. Measurement and recording of environmental conditions as specified herein.
      b. Measurement and recording of substrate conditions as specified herein.
      c. Thickness Testing:
         1) Wet film thickness during application in accordance with ASTM D4414.
         2) Dry Film Thickness (DFT) in accordance with SSPC-PA 2.
         3) Contracting Officer Representative may measure coating thickness at any time during project to assure conformance with these Specifications.
      d. Bond Strength:
         1) Bond strength testing will be required by the Contracting Officer Representative where there is reason to suspect the integrity of the coating system.
         2) Measure bond strength of the coating in accordance with:
            a) Steel substrate: ASTM D4541.
            b) Concrete substrate ASTM D7234.
         3) The number of test sites and locations to be tested shall be determined by the Contracting Officer Representative after application of coating. The Contractor will apply the dollies, perform the tests and repair the coating in the presence of the Contracting Officer Representative.
            a) For each test that fails, two additional tests shall be performed in the adjacent area.
            b) Further bond tests may be performed to determine the extent of potentially deficient bonded areas at no additional cost to the Government.
         4) Repairs shall be made by applicator in strict accordance with manufacturer's recommendations. Any coated areas that do not pass the bond strength tests shall be removed and replaced at the expense of the Contractor.
   2. Holiday Testing (Spark Testing):
      a. Provide holiday testing for the following:
         1) Interior and exterior surfaces of all ferrous metal tankage.
         2) Interior surfaces of Membrane tanks.
         3) Interior surfaces of Membrane Bioreactor tanks.
         4) Interior surfaces of UV channels.
      b. Test completed coating system in accordance with NACE SP0188.
      c. All detected holidays shall be marked and repaired.
         1) Abrade the coating surface as recommended by the coating manufacturer.

2) After proper abrading and cleaning, apply additional protective coating material to the repair area.
3) All touch-up/repair procedures shall follow the protective coating manufacturer's recommendations.

d. Following manufacturer's recommended cure time for repaired areas, measure dry film thickness:
1) Ensure the specified thickness has been achieved.

e. Retest and recoat as required until area passes test criterion.
f. Provide certification that completed lining system is free of pinholes or other material defects.

D. NACE inspection:
1. The Government reserves the right to retain a NACE Level 3 coating inspector to perform observation, inspection and testing as deemed necessary to document the quality of the Work.
a. All work shall be done to the satisfaction of the Government's inspector.
b. Any portion of the coating that does not satisfactorily pass the inspection and testing requirements shall be repaired or replaced by the Contractor at no additional cost to the Government.
c. Additional testing and/or inspection may be done at the discretion of the Government.
1) The Contractor will provide all equipment, materials, and labor to perform the testing.

2. Inspection, testing or observation by the Government's inspector shall not relieve the Contractor of responsibility for surface preparation, inspection or quality control specified herein.

E. Instrumentation:
1. Provide instrumentation as necessary to measure and record atmospheric and substrate conditions, including but not limited to:
a. Dry Film Thickness Gauge:
1) Ultrasonic: ASTM D6132.
2) Magnetic: ASTM B499.

b. Wet Film Thickness Gauge: ASTM D4414.
c. Sling Psychrometer: ASTM E337.
d. Surface Temperature Gauge.
e. Anemometer.
f. Moisture Meter.
g. Adhesion test apparatus:
1) Steel: ASTM D4541.
2) Concrete: ASTM D7234.

F. Maintain Daily Records:
1. Record the following information during application:
a. Date, starting time, end time, and all breaks taken by applicators.
b. Air temperature.
c. Relative humidity.
d. Dew point.
e. Moisture content and pH level of concrete or masonry substrates prior to coating.
f. Surface temperature of substrate.
g. Provisions utilized to maintain work area within manufacturer's recommended application parameters including temporary heating, ventilation, cooling, dehumidification and provisions utilized to mitigate wind-blown dust and debris from contaminating the wet coating.
h. For outdoor coating, also record:
1) Sky condition.
2) Wind speed and direction.

i. Record environmental conditions, substrate moisture content and surface temperature information not less than once every 4 hours during application.
1) Record hourly when temperatures are below 50 degrees F or above 100 degrees F.

2. Record the following information daily for the coating manufacturer's recommended curing period:
a. Date and start time of cure period for each item or area.
b. For outdoor coating, also record:
1) Sky conditions.
2) Wind speed and direction.
3) Air temperature.
a) Dry Bulb.
b) Wet Bulb.
4) Relative humidity.
5) Dew point.
6) Surface temperatures.
c. Record environmental conditions not less than once every 4 hours.
1) Record hourly when temperatures are below 50 degrees F or above 100 degrees F.
d. Provisions utilized to protect each item or area and to maintain areas within manufacturer's recommended curing parameters.
3. Format for daily record to be computer generated.

G. Provide wet paint signs.

3.7 CLEANING

A. Clean coating spattered surfaces.
1. Use care not to damage finished surfaces.

B. Upon completion of coating, replace hardware, accessories, plates, fixtures, and similar items.

C. Remove surplus materials, scaffolding, and debris.

3.8 COLOR SCHEDULE

A. Pipe Bollards: Refer to Specification Section 09 06 00.

B. Aluminum and Aluminum Hybrid, Doors and Frames: Refer to Specification Section 09 06 00.

C. Rolling Aluminum Doors: Refer to Specification Section 09 06 00.

D. Exterior Equipment Screen: Refer to Specification Section 09 06 00.

E. Galvanized Snow Hoods: Refer to Specification Section 09 06 00.

F. Piping:
1. Refer to the PIPE SCHEDULES in the Drawings (Volumes 5, 6, and 7) for the paint and banding colors.
2. For exposed piping in the Inner-canyon: Tan paint.
a. Provide color palette for final selection by Government.

END OF SECTION

DIVISION 10

SPECIALTIES

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SECTION 10 11 00
LIQUID CHALKBOARD AND TACKBOARDS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Liquid Markerboards (MB).

B. Related Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 06 10 00 – Rough Carpentry.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. A424/A424M, Standard Specification for Steel, Sheet, for Porcelain Enameling.
      b. B209, Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate.

B. Furnish all liquid markerboards from one manufacturer.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Include dimensional plans, elevations and details, large-scale sections of typical members, and other components.
   2. Show anchors, grounds, reinforcement and layout, and indicate finishes.

B. Product Data:
   1. Provide copies of manufacturer's specifications and installation instructions for each type of material and accessory required.
   2. Where fire resistance classification is indicated, submit copies of nationally recognized laboratory testing listings of products proposed for use.
   3. Include data required to show specification compliance.

C. Project Information:
   1. Installer qualifications.

D. Contract Closeout Information:
   1. Maintenance data.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:

1. Liquid Markerboards:
    a. Basis of Design: Claridge Products & Equipment.
    b. Or Equal.
2. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MANUFACTURED UNITS

A. Provide factory-built units.

B. Liquid Markerboards (MB):
1. Balanced, high-pressure laminated, three-ply construction, with facing sheet, core, and backing sheet.
2. Porcelain enamel writing surface over ground primer coat with seal coat on reverse side.
    a. Primer coat thickness: Minimum 0.0025-inches.
    b. Writing surface coating thickness: Minimum 0.0025-inches.
    c. Seal coat thickness: Minimum 0.0025-inches.
3. Face sheet:
    a. Porcelain enamel steel ASTM A424/A424M, minimum 24-gauge.
4. Core:
    a. Minimum 1/4-inches thick plywood or hardboard.
5. Backer:
    a. 26 Gauge zinc plated steel, 28 Gauge random porcelain coil, or 0.015-inch-thick aluminum sheet.
6. Backing panel:
    a. Moisture resistant plywood or hardboard 1/4-inch thick, with 1/4-inch backing; or one-piece 1/2-inch thick.
7. Color: White colored surface suitable for liquid markers.
    a. Light reflectance factor: Minimum 15 percent; maximum 20 percent.
    b. Surface shall be capable of supporting papers by means of magnets.

C. Frames and Trim:
1. Minimum 0.062-inches thick aluminum, ASTM B209.
2. Size and shape as indicated on Drawings.
3. Single-length units to minimize joints.
4. Miter all corners to a neat, hairline closure.
5. Finish: "Anodized clear".
6. Manufacturer's standard "narrow" trim, approximately 1/2-inches wide.
7. Provide structural support accessories as required for liquid markerboards in addition to normal trim, provide such additional support to provide necessary support.

D. Troughs:
1. Provide continuous, full length for each board.
2. Box type, slanted front with cast aluminum end closures.

PART 3 - EXECUTION

3.1 INSPECTION

A. A. Verify suitability of substrate to accept installation.

B. B. Verify that adequate Wall Backing has been installed.
1. Coordinate and direct installation of backing where required.

C. Correct unsatisfactory conditions.

D. Installation constitutes acceptance of responsibility for performance.

3.2 INSTALLATION

A. Comply with manufacturer's instructions and recommendations.

B. Locate with top of markerboard 6-feet-0-inches above finished floor line.

C. Provide additional backing as indicated or necessary to properly stiffen and support boards.

D. Install with concealed hangers, plumb and level.

3.3 MARKERBOARD SCHEDULE

A. MB-1
   1. Size: 24-inches high x 36-inches wide.
   2. Magnetic.
   3. Aluminum Edges.
   4. White surface.
   5. Locations/Quantity:
      a. Building 62: Room 107 (1)

B. MB-2
   1. Size: 36-inches high x 48-inches wide.
   2. Magnetic.
   3. Aluminum Edges.
   4. White surface.
   5. Locations/Quantity:
      a. Building 62: Room 104 (1) and Room 106 (2)

END OF SECTION

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SECTION 10 14 00
IDENTIFICATION DEVICES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Tag, tape and stenciling systems for equipment, piping, valves, pumps, ductwork and similar items.
   2. Hazard and safety signs.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. A13.1, Scheme for the Identification of Piping Systems.
   2. The International Society of Automation (ISA).
   3. National Electrical Manufacturers Association/American National Standards Institute (NEMA/ANSI):
      a. Z535.1, Safety Color Code.
      b. Z535.2, Environmental and Facility Safety Signs.
      c. Z535.3, Criteria for Safety Symbols.
      d. Z535.4, Product Safety Signs and Labels.
   4. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
      b. 704, Standard System for the Identification of Hazards of Materials for Emergency Response.
   5. Occupational Safety and Health Administration (OSHA):
      a. 29 CFR 1910.145, Specification for Accident Prevention Signs and Tags.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Catalog information for all identification systems.
      b. Acknowledgement that products submitted meet requirements of standards referenced.
   2. Identification register, listing all items in PART 3 of this Specification Section to be identified, type of identification system to be used, lettering, location and color.
   3. Schedule of Hazard and Safety Signage indicating text and graphics.

PART 2 - PRODUCTS

Product requirements are to conform with the following salient characteristics.

2.1 MANUFACTURED UNITS

A. Type A1 - Round Metal Tags:
   1. Materials:
      a. Aluminum or stainless steel.
      b. Stainless steel shall be used in corrosive environments.
   2. Size:
      a. Diameter: 1-1/2 inches minimum.
      b. Thickness: 0.035 inches (20 GA) minimum.
   3. Fabrication:
      a. 3/16 inches minimum mounting hole.
      b. Legend: Stamped and filled with black coloring.

4. Color: Natural.

B. Type A2 - Rectangle Metal Tags:
   1. Materials: Stainless steel.
   2. Size:
      a. 3-1/2 inches x 1-1/2 inches minimum.
      b. Thickness: 0.036 inches (20 GA) minimum.
   3. Fabrication:
      a. 3/16 inches minimum mounting hole.
      b. Legend: Stamped and filled with black coloring.
   4. Color: Natural.

C. Type A3 - Metal Tape Tags:
   1. Materials: Aluminum or stainless steel.
   2. Size:
      a. Width 1/2 inches minimum.
      b. Length as required by text.
   3. Fabrication:
      a. 3/16 inches minimum mounting hole.
      b. Legend: Embossed.
   4. Color: Natural.

D. Type B1- Square Nonmetallic Tags:
   1. Materials: Fiberglass reinforced plastic.
   2. Size:
      a. Surface: 2 x 2 inches minimum.
      b. Thickness: 100 mils.
   3. Fabrication:
      a. 3/16 inches mounting hole with metal eyelet.
      b. Legend: Preprinted and permanently embedded and fade resistant.
   4. Color:
      a. Background: Manufacturer standard or as specified.
      b. Lettering: Black.

E. Type B2 - Nonmetallic Signs:
   1. Materials: Fiberglass reinforced or durable plastic.
   2. Size:
      a. Surface: As required by text.
      b. Thickness: 60 mils minimum.
   3. Fabrication:
      a. Rounded corners.
      b. Drilled holes in corners with grommets.
      c. Legend: Preprinted, permanently embedded and fade resistant for a 10 year minimum outdoor durability.
   4. Color:
      a. Background: Manufacturer standard or as specified.
      b. Lettering: Black.
   5. Standards for OSHA signs: NEMA/ANSI Z535.1, NEMA/ANSI Z535.2, NEMA/ANSI Z535.3, NEMA/ANSI Z535.4, OSHA 29 CFR 1910.145.

F. Type C - Laminated Name Plates:
   1. Materials: Phenolic or DR (high impact) acrylic.
   2. Size:
      a. Surface: As required by text.
      b. Thickness: 1/16 inches.
   3. Fabrication:
      a. Outdoor rated and UV resistant when installed outdoors.
      b. Two layers laminated.

c. Legend: Engraved through top lamination into bottom lamination.
d. Two drilled side holes, for screw mounting.

4. Color: Black top surface, white core, unless otherwise indicated.

G. Type D - Self-Adhesive Tape Tags and Signs:
1. Materials: Vinyl tape or vinyl cloth.
2. Size:
a. Surface: As required by text.
b. Thickness: 5 mils minimum.
3. Fabrication:
a. Indoor/Outdoor grade.
b. Weather and UV resistant inks.
c. Permanent adhesive.
d. Legend: Preprinted.
e. Wire markers to be self-laminating.
4. Color: White with black lettering or as specified.
5. Standards for OSHA signs: NEMA/ANSI Z535.1, NEMA/ANSI Z535.2, NEMA/ANSI Z535.3, NEMA/ANSI Z535.4, OSHA 29 CFR 1910.145.

H. Type E - Heat Shrinkable Tape Tags:
1. Materials: Polyolefin.
2. Size: As required by text.
3. Fabrication:
a. Legend: Preprinted.
4. Color: White background, black printing.

I. Type F - Underground Warning Tape:
1. Materials: Polyethylene.
2. Size:
a. 6 inches wide (minimum).
b. Thickness: 3.5 mils.
3. Fabrication:
a. Legend: Preprinted and permanently imbedded.
b. Message continuous printed.
c. Tensile strength: 1750 psi.
4. Color: As specified.

J. Type G - Stenciling System:
1. Materials:
a. Exterior type stenciling enamel.
b. Either brushing grade or pressurized spray can form and grade.
2. Size: As required.
3. Fabrication:
a. Legend: As required.
4. Color: Black or white for best contrast.

K. Underground Tracer Wire:
1. Materials:
a. Wire:
1) 12 GA AWG.
2) Solid.
b. Wire nuts: Waterproof type.
c. Split bolts: Brass.

2.2 ACCESSORIES

A. Fasteners:
1. Bead chain: #6 brass, aluminum or stainless steel.
2. Plastic strap: Nylon, urethane or polypropylene.

3. Screws: Self-tapping, stainless steel.
4. Adhesive, solvent activated.

2.3 MAINTENANCE MATERIALS

A. Where stenciled markers are provided, clean and retain stencils after completion and include in extra stock, along with required stock of paints and applicators.

PART 3 - EXECUTION

3.1 GENERAL INSTALLATION

A. Install identification devices at specified locations.

B. All identification devices to be printed by mechanical process, hand printing is not acceptable.

C. Attach tags to equipment with sufficient surface or body area with solvent activated adhesive applied to back of each tag.

D. Attach tags with 1/8 inches round or flat head screws to equipment without sufficient surface or body area, or porous surfaces.
   1. Where attachment with screws should not or cannot penetrate substrate, attach with plastic strap.

E. Single items of equipment enclosed in a housing or compartment to be tagged on outside of housing.
   1. Several items of equipment mounted in housing to be individually tagged inside the compartment.

F. Tracer Wire:
   1. Attach to pipe at a maximum of 10 feet intervals with tape or tie-wraps.
   2. Continuous pass from each valve box and above grade at each structure.
   3. Coil enough wire at each valve box to extend wire a foot above the ground surface.
   4. 1,000 feet maximum spacing between valve boxes.
   5. If split bolts are used for splicing, wrap with electrical tape.
   6. If wire nuts are used for splicing, knot wire at each splice point leaving 6 inches of wire for splicing.
   7. Use continuous strand of wire between valve box where possible.
      a. Continuous length shall be no shorter than 100 feet.

3.2 SCHEDULES

A. Hazard and Safety Signage:
   1. Permit Required Confined Space signage:
      a. Tag Type: Type B2 - Nonmetallic Signs.
      b. Fastener: Screw or adhesive.
      c. Size: 10 inches x 14 inches.
      d. Location: As indicated on Drawings.
      e. Location: Field located as directed by the COR.
         1) Allowance: Provide 10 signs.
      f. Legend:
         1) OSHA Danger sign.
         2) Description of hazard: “PERMIT REQUIRED CONFINED SPACE DO NOT ENTER”.
   2. Miscellaneous OSHA hazard signage:
      a. Tag Type: Type B2 - Nonmetallic Signs.
      b. Fastener: Screw or adhesive.
      c. Size: 10 inches x 14 inches.
      d. Location: As indicated on Drawings.
      e. Location: Field located as directed by the COR.

1) Allowance: Provide 13 OSHA Danger, Caution, Safety Instruction or Biohazard signs as directed by the COR.

f. Legend:
1) Description of hazard shall be determined by the COR.
2) Provide international graphic symbology where indicated.

3. No Smoking Signage:
a. Tag Type: Type B2 - Nonmetallic Signs.
b. Fastener: Screw or adhesive.
c. Size: 10 inches x 14 inches.
d. Location: On the doors entering into the rooms with flammable gas or other material requiring No Smoking signage, as indicated on the Drawings.
e. Location: Field located as directed by the COR.
1) Allowance: Provide 5 “NO SMOKING” signs.
f. Legend: “NO SMOKING” with international graphic symbology, adhering to the International Fire Code.

4. Hazardous Material Identification Signage:
a. Tag Type: Type B2 - Nonmetallic Signs.
b. Fastener: Screw or adhesive.
c. Size (NFPA Diamond): Per NFPA 704, 10” minimum.
d. Size (Hazardous Material name, with concentration % where applicable): 2 inches minimum letters, directly below corresponding NFPA Diamond.
e. Location: On the doors entering into the rooms with Hazardous Material.
f. Location: Field located as directed by the COR.
1) Allowance: Provide 8 NFPA 704 Diamond signs.
g. Legend:
1) NFPA 704 Diamond hazard numbers: As directed by the COR and appropriate for the Hazardous Material.

B. Process Systems:
1. General:
a. Provide arrows and markers on piping.
1) At 20 feet maximum centers along continuous lines.
2) At changes in direction (route) or obstructions.
3) At valves, risers, "T" joints, machinery or equipment.
4) Where pipes pass through floors, walls, ceilings, cladding assemblies and like obstructions provide markers on both sides.
b. Position markers on both sides of pipe with arrow markers pointing in flow direction.
1) If flow is in both directions use double headed arrow markers.
c. Apply tapes and stenciling in uniform manner parallel to piping.

2. Trenches with piping:
a. Tag type: Type F - Underground Warning Tape
b. Location: Halfway between top of piping and finished grade.
c. Letter height: 1-1/4 inches minimum.
d. Natural gas or digester gas:
1) Color: Yellow with black letters.
2) Legend:
a) First line: “CAUTION CAUTION CAUTION”
b) Second line: “BURIED GAS LINE BELOW”
e. Potable water:
1) Color: Blue with black letters.
2) Legend:
a) First line: “CAUTION CAUTION CAUTION”
b) Second line: “BURIED WATER LINE BELOW”
f. Storm and sanitary sewer lines:
1) Color: Green with black letters.
2) Legend:

a) First line: "CAUTION CAUTION CAUTION"
b) Second line: "BURIED SEWER LINE BELOW"

g. Chemical feed piping (e.g., chlorine solution, caustic solution, etc.):
1) Color: Per Pipe Schedules in Drawings (Volumes 5, 6, and 7).
2) Legend:
a) First line: "CAUTION CAUTION CAUTION"
b) Second line: "BURIED CHEMICAL LINE BELOW"

h. Other piping (e.g., compressed air, irrigation, refrigerant, heating water, etc.):
1) Color: Yellow with black letters.
2) Legend:
a) First line: "CAUTION CAUTION CAUTION"
b) Second line: "BURIED PIPE LINE BELOW"

3. Yard valves, buried, with valve box and concrete pad:
a. Tag type: Type A2 - Rectangle Metal Tags.
b. Fastener: 3/16 inches x 7/8 inches plastic screw anchor with 1 inch #6 stainless steel pan head screw.
c. Legend:
1) Letter height: 1/4 inches minimum.
2) Valve designation as indicated on the Drawings (e.g., "V-xxx").

4. Valves and slide gates:
a. Tag type:
1) Outdoor locations: Type B1 - Square Nonmetallic Tags.
2) Indoor noncorrosive:
a) Type A1 - Round Metal Tags.
b) Type B1 - Square Nonmetallic Tags.
3) Indoor corrosive:
a) Stainless steel Type A1 - Round Metal Tags.
b) Type B1 - Square Nonmetallic Tags.

b. Fastener:
1) Type A1: Chain of the same material.
2) Type B1: Stainless steel chain.

c. Color: Per ASME A13.1 corresponding to the piping system.
d. Legend:
1) Letter height: 1/4 inches minimum.
2) Valve designation as indicated on the Drawings (e.g., "V-xxx").

5. Process equipment (e.g., pumps, pump motors, blowers, air compressors, bar screens, clarifier drive mechanism, etc.):
a. Tag type:
1) Type B2 - Nonmetallic Signs.
2) Type D - Self-Adhesive Tape Tags and Signs.
3) Type G - Stenciling System.

b. Fastener:
1) Self.
2) Screws.
3) Adhesive.

c. Legend:
1) Letter height: 1/2 inches minimum.
2) Equipment designation as indicated on the Drawings (e.g., "Primary Sludge Pump P-xxx").

6. Piping systems:
a. Tag type:
1) Outdoor locations: Type G - Stenciling System.
2) Indoor locations:
a) Type D - Self-Adhesive Tape Tags and Signs.
b) Type G - Stenciling System.

b. Fastener: Self.
c. Color: Per Pipe Schedules in Drawings (Volumes 5, 6, and 7).
d. Legend:
1) Letter height: Manufacturers standard for the pipe diameter.
2) Mark piping in accordance with ASME A13.1.
3) Use piping designation as indicated on the Drawings.
4) Arrow: Single arrow.

7. Process tanks (over 1000 GAL) and basins, (e.g., chemical storage, clarifiers, trickling filters, digesters, etc.):
a. Tag type:
1) Type B2 - Nonmetallic Signs.
2) Type G - Stenciling System.
b. Fastener:
1) Screw.
2) Self.
c. Location as directed by Owner.
d. Legend:
1) Letter height: 4 inches minimum.
2) Equipment designation as indicated on the Drawings (e.g., “Clarifier CL-xxx”).

8. Tanks (less than 1000 GAL) (e.g., break tanks, chemical tanks, hydro-pneumatic tanks, air receivers, etc.):
a. Tag type:
1) Type D - Self-Adhesive Tape Tags and Signs.
2) Type G - Stenciling System.
b. Fastener: Self.
c. Legend:
1) Letter height: 2 inches minimum.
2) Equipment designation as indicated on the Drawings (e.g., “Polymer Storage Tank Txxx”)

9. Equipment that starts automatically:
a. Tag type:
1) Type B2 - Nonmetallic Signs.
2) Type D - Self-Adhesive Tape Tags and Signs.
b. Fastener:
1) Type B2 - Screw or adhesive.
2) Type D - Self.
c. Size: 5 inches x 7 inches
d. Location: As directed by the COR.
e. Legend:
1) OSHA Warning Sign.
2) Description of Warning: “THIS MACHINE STARTS AUTOMATICALLY”.

C. Instrumentation Systems:
1. Instrumentation Equipment (e.g., flow control valves, primary elements, etc.):
a. Tag type:
1) Outdoor locations: Type B1 - Square Nonmetallic Tags.
2) Indoor noncorrosive:
a) Type A1 - Round Metal Tags.
b) Type B1 - Square Nonmetallic Tags.
3) Indoor corrosive:
a) Stainless steel Type A1 - Round Metal Tags.
b) Type B1 - Square Nonmetallic Tags.
b. Fastener:
1) Type A1: Chain of the same material.
2) Type B1: Stainless steel chain.
c. Legend:

1) Letter height: 1/4 inches minimum.
2) Equipment ISA designation as indicated on the Drawings (e.g., “FIT-xxx”).

2. Enclosure for instrumentation and control equipment, (e.g., PLC control panels, etc.):
   a. Tag type: Type C - Phenolic Name Plates.
   b. Fastener: Screws.
   c. Legend:
      1) Letter height: 1/2 inches minimum.
      2) Equipment name (e.g., "PLC CONTROL PANEL PCP-xxx").
3. Components inside equipment enclosure, (e.g., PLC’s, control relays, contactors, and timers):
   a. Tag type: Type D - Self-Adhesive Tape Tags.
   b. Fastener: Self.
   c. Legend:
      1) Letter height: 3/16 inches minimum.
      2) Description or function of component (e.g., "PLC-xxx” or “CR-xxx").
4. Through enclosure door mounted components (e.g., selector switches, controller digital displays, etc.):
   a. Tag type: Type C - Phenolic Name Plates.
   b. Fastener: Screws.
   c. Legend:
      1) Letter height: 1/4 inches minimum.
      2) Component ISA tag number as indicated on the Drawings (e.g., “HS-xxx”).

D. HVAC Systems:
   1. General:
      a. Provide arrows and markers on ducts.
         1) At 20 feet maximum centers along continuous lines.
         2) At changes in direction (route) or obstructions.
         3) At dampers, risers, branches, machinery or equipment.
         4) Where ducts pass through floors, walls, ceilings, cladding assemblies and like obstructions provide markers on both sides.
      b. Position markers on both sides of duct with arrow markers pointing in flow direction.
         1) If flow is in both directions use double headed arrow markers.
      c. Apply tapes and stenciling in uniform manner parallel to ducts.
   2. HVAC Equipment (e.g., unit heaters, exhaust fans, air handlers, etc.):
      a. Tag type:
         1) Type B2 - Nonmetallic Signs.
         2) Type C - Phenolic Name Plates.
      b. Fastener: Screws.
      c. Legend:
         1) Letter height: 1 inch minimum.
         2) Equipment designation as indicated on the Drawings (e.g., "EF-xxx").
   3. Ductwork:
      a. Tag type:
         1) Type D - Self-Adhesive Tape Tags and Signs.
         2) Type G - Stenciling System.
      b. Fastener: Self.
      c. Legend:
         1) Letter height: 1 inch minimum.
         2) Description of ductwork, (e.g., “AIR SUPPLY”).
         3) Arrows: Single arrow.
   4. Enclosure for instrumentation and control equipment, (e.g., fan control panels, etc.):
      a. Tag type: Type C - Phenolic Name Plates.
      b. Fastener: Screws.
      c. Legend:
         1) Letter height: 1/2 inches minimum.

2) Equipment designation as indicated on the Drawings (e.g., "FAN CONTROL PANEL FCP-xxx").

5. Wall mounted thermostats:
   a. Tag type: Type D - Self-Adhesive Tape Tags and Signs.
   b. Fastener: Self.
   c. Legend:
      1) Letter height: 3/16 inches minimum.
      2) Description of equipment controlled (e.g., "UH-xxx" or AHU-xxx").
6. Components inside equipment enclosure, (e.g., controller's, control relays, contactors, and timers):
   a. Tag type: Type D - Self-Adhesive Tape Tags and Signs.
   b. Fastener: Self.
   c. Legend:
      1) Letter height: 3/16 inches minimum.
      2) Description or function of component (e.g., "CR-xxx").
7. Through enclosure door mounted equipment (e.g., selector switches, controller digital displays, etc.):
   a. Tag type: Type C - Phenolic Name Plates.
   b. Fastener: Screws.
   c. Legend:
      1) Letter height: 1/4 inches minimum.
      2) Component tag number as indicated on the Drawings or as defined by contractor (e.g., "HS-xxx").

E. Electrical Systems:
1. Trenches with ductbanks, direct-buried conduit, or direct-buried wire and cable.
   a. Tag type: Type F - Underground Warning Tape.
   b. Letter height: 1-1/4 inches minimum.
   c. Location:
      1) Where trench is 12 inches or more below finished grade: In trench 6 inches below finished grade.
      2) Where trench is less than 12 inches below finished grade: In trench 3 inches below finished grade.
   d. Electrical power (e.g., low and medium voltage):
      1) Color: Red with black letters.
      2) Legend:
         a) First line: "CAUTION CAUTION CAUTION".
         b) Second line: "BURIED ELECTRIC LINE BELOW".
   e. Communications (e.g., telephone, instrumentation, LAN, SCADA):
      1) Color: Orange with black letters.
      2) Legend:
         a) First line: "CAUTION CAUTION CAUTION".
         b) Second line: "BURIED COMMUNICATION LINE BELOW".
2. Exterior pad-mounted equipment (e.g. transformers, switchgear, sectionalizing equipment, shunt reactors, etc.):
   a. Tag type: Type C – Phenolic Name Plates.
   b. Fastener: Screws.
   c. General legend:
      1) Letter height:
         a) First line: 1-1/2IN minimum.
         b) Subsequent lines: 1/2IN minimum.
      2) First line: Equipment name (e.g. "TRANSFORMER Txxx").
      3) Second line: system voltage (e.g. "13,800V:480/277V").
      4) Third line: Date installed (e.g. "INSTALLED JULY 20xx").
   d. Each section/cubicle legend: 1-1/2IN minimum.

e. Description of source or load (e.g. “MAIN DISCONNECT” or “TO SWITCHGEAR SGxxx” or “TO TRANSFORMER Txxx”).

3. Switchgear, switchboards and motor control centers:
   a. Tag type: Type C - Phenolic Name Plates.
   b. Fastener: Screws.
   c. Main equipment legend:
      1) Letter height:
         a) First line: 1 inch minimum.
         b) Subsequent lines: 3/8 inches minimum.
      2) First line: Equipment name (e.g., "MAIN SWITCHBOARD MSBxxx").
      3) Second line:
         a) Source of power (e.g., "FED FROM MCCxxx LOCATED IN ROOM xxx").
         b) Include the building name or number if the source is in another building.
      4) Third line: System voltage and phase (e.g., “480/277 V, 3PH”).
      5) Fourth line: Date installed (e.g., “INSTALLED JULY 20xx”).
   d. Main and feeder device legend:
      1) Letter height: 3/8 inches minimum.
      2) Description of load (e.g., “MAIN DISCONNECT”, "PUMP Pxxx" or "PANELBOARD HPxxx").
4. Panelboards and transformers:
   a. Tag type: Type C - Phenolic Name Plates.
   b. Fastener: Screws.
   c. Legend:
      1) Letter height:
         a) First line: 3/8 inches minimum.
         b) Subsequent lines: 3/16 inches minimum.
      2) First line: Equipment name (e.g., "PANELBOARD LPxxx" or "TRANSFORMER Txxx").
      3) Second line (panelboards only): System voltage and phase (e.g., “208/120V, 3PH”).
      4) Third line:
         a) Source of power (e.g., "FED FROM MCCxxx LOCATED IN ROOM xxx").
         b) Include the building name or number if the source is in another building.
      5) Fourth line: Date installed (e.g., “INSTALLED JULY 20xx”).
5. Transfer switches:
   a. Tag type: Type C - Phenolic Name Plates.
   b. Fastener: Screws.
   c. Legend:
      1) Letter height:
         a) First line: 3/8 inches minimum.
         b) Subsequent lines: 3/16 inches minimum.
      2) First line: Equipment name (e.g., "AUTOMATIC TRANSFER SWITCH ATSxxx").
      3) Second line: Normal source of power (e.g., "NORMAL SOURCE FED FROM MCCxxx").
      4) Third line: Emergency source of power (e.g., "EMERGENCY SOURCE FED FROM SGENxxx").
      5) Fourth line: Date installed (e.g., “INSTALLED JULY 20xx”).
6. Safety switches, separately mounted circuit breakers and motor starters, VFD’s, etc.:
   a. Tag type: Type C - Phenolic Name Plates.
   b. Fastener: Screws.
   c. Legend:
      1) Letter height: 1/4 inches minimum.
      2) First line: Description of load equipment is connected to (e.g., "PUMP Pxxx").
      3) Second line:

a) Source of power (e.g., "FED FROM MCCxxx LOCATED IN ROOM xxx").
b) The source of power room number is only required when there are multiple electrical rooms, if the source is in another building, the building name or number shall be used.

7. Enclosure for instrumentation and control equipment, (e.g., lighting control panels, etc.):
   a. Tag type: Type C - Phenolic Name Plates.
   b. Fastener: Screws.
   c. Legend:
      1) Letter height: 1/2 inches minimum.
      2) Equipment name (e.g., "LIGHTING CONTROL PANEL LCPxxx").
8. Components inside equipment enclosures (e.g., circuit breakers, fuses, control power transformers, control relays, contactors, timers, etc.):
   a. Tag type: Type D - Self-Adhesive Tape Tags and Signs.
   b. Fastener: Self.
   c. Legend:
      1) Letter height: 3/16 inches minimum.
      2) Description or function of component (e.g., "M-xxx", "CR-xxx" or "TR-xxx").
9. Through enclosure door mounted equipment (e.g., selector switches, controller digital displays, etc.):
   a. Tag type: Type C - Phenolic Name Plates.
   b. Fastener: Screws.
   c. Legend:
      1) Letter height: 1/4 inches minimum.
      2) Component tag number as indicated on the Drawings or as defined by contractor (e.g., "HS-xxx").
10. Conductors in control panels and in pull or junction boxes where multiple circuits exist.
    a. Tag type: Type E - Heat Shrinkable Tape Tags.
    b. Fastener: Self.
    c. Tag conductor at both ends.
    d. Legend:
       1) Letter height: 1/8 inches minimum.
       2) Circuit number or wire number as scheduled on the Drawings or as furnished with the equipment.
11. Conductors in cable trays.
    a. Tag type: Type D – Self-Adhesive Tape Tags
    b. Fastener: Self.
    c. Tag all conductors at the same location in the tray at maximum 25FT intervals.
    d. Legend:
       1) Letter height: 3/16IN minimum.
       2) Circuit number or wire number as scheduled on the Drawings.
12. Conductors in handholes and manholes.
    a. Tag type: Type A3 - Metal Tape Tags.
    b. Fastener: Nylon strap.
    c. Tag conductor at both ends.
    d. Legend:
       1) Letter height: 1/8 inches minimum.
       2) Circuit number or wire number as scheduled on the Drawings.
13. Grounding conductors associated with grounding electrode system in accordance with the following:
    a. Tag type: Type D - Self-Adhesive Tape Tags.
    b. Fastener: Self.
    c. Legend:
       1) Letter height: 1/8 inches minimum.
       2) Function of conductor (e.g., "MAIN BONDING JUMPER", "TO GROUND RING", "TO MAIN WATER PIPE").

14. Flash protection for switchboards, panelboards, industrial control panels and motor control centers:
    a. Tag type: Type D - Self-Adhesive Tape Signs.
    b. Fastener: Self.
    c. Legend: Per NFPA 70.
15. Power cable trays:
    a. Tag type: Type D – Self-Adhesive Tape Tags and Signs.
    b. Fastener: Self.
    c. Size: 1-3/4IN x 2-1/2IN minimum.
    d. Location:
        1) Every 20FT maximum interval.
        2) Label each barriered section of tray.
    e. Legend:
        1) OSHA Danger Sign.
            a) Description of Danger (e.g. “UP TO 480 VOLTS”).
            b) “DO NOT STAND” signs.
                (1) On exposed cable trays adjacent to steps, platforms or other locations near personnel access, and at up to 10 additional locations at each site as directed by the Contracting Officer.
16. Control cable trays:
    a. Tag type: Type D – Self-Adhesive Tape Tags and Signs.
    b. Fastener: Self.
    c. Size: 1-3/4IN x 2-1/2IN minimum.
    d. Location:
        1) Every 20FT maximum interval.
        2) Label each barriered section of tray.
    e. Legend:
        1) OSHA Danger Sign.
        2) Description of Danger (e.g. “LESS THAN 50VOLTS, CLASS 2 OR 3”, “ANALOG INSTRUMENTATION”, or “DIGITAL COMMUNICATION”).
17. Entrances to electrical rooms:
    a. Tag type: Type B2 - Nonmetallic Signs.
    b. Fastener: Screw or adhesive.
    c. Size: 5 inches x 7 inches.
    d. Location: Each door to room.
    e. Legend:
        1) OSHA Danger Sign.
        2) Description of Danger: “HIGH VOLTAGE, AUTHORIZED PERSONNEL ONLY”.
18. Entrance to Control System, Network, and Telecommunication Rooms:
    a. Tag type: Type B2 – Nonemetallic Signs.
    b. Fastener: screw or adjesive.
    c. Size: 5IN x 7IN.
    d. Location: Each door to room.
    e. Legend:
        1) OSHE Warning Sign.
        2) Description of Warning: “TELECOM ROOM – AUTHORIZED PERSONNEL ONLY”.
19. Equipment where more than one voltage source is present:
    a. Tag type:
        1) Type B2 - Nonmetallic Signs.
        2) Type D - Self-Adhesive Tape Signs.
    b. Fastener:
        1) Screw or adhesive.
        2) Self.

c. Size: 1-3/4 inches x 2-1/2 inches.
d. Location: Exterior face of enclosure or cubical.
e. Legend:
    1) OSHA Danger Sign.
    2) Description of Danger: “MULTIPLE VOLTAGE SOURCES”.

END OF SECTION

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SECTION 10 14 23
SIGNAGE

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Room identification signs.
   2. Other identification signs:
      a. Fire and/or smoke barrier identification signs.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 09 06 00 – Color Schedule.
   3. Section 10 14 00 - Identification Devices.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Americans with Disabilities Act (ADA):
      a. Accessibility Guidelines for Buildings and Facilities (ADAAG).

1.3 DEFINITIONS

A. Authority Having Jurisdiction (AHJ): Building official, fire chief, fire marshal or other individual having statutory authority.

B. Wet and/or Corrosive Areas: For the purposes of this Specification Section, the following rooms or areas are considered wet and/or corrosive:

1.4 SUBMITTALS

A. Product Data:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.

B. Shop Drawings:
   1. Schedule of all signs indicating text and graphics.
   2. Layout drawings of all signage showing material, construction detail, size, letter style, text, border, finish, and installation/mounting detail.
      a. Provide drawings for:
         1) Each Room, Exit and Stair identification signs.
         2) Fire and/or smoke barrier identification signs.

C. Samples:
   1. Room identification signs.
      a. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
      b. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.
   2. Fire and/or smoke barrier identification signs.
      a. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
   3. Warning signs (Chemical Drum Rooms).
      a. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.

1.5 QUALIFICATIONS

1. Manufacturers specializing in manufacturing the products specified in this section shall have a minimum of five years' experience.
2. All signs shall be obtained from one source and a single manufacturer.

1.6 WARRANTY

1. Provide manufacturers' warranty against defects in materials and workmanship for a minimum of 5 years.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Room signs:
      a. Takeform - Steep FC09 using Fusion construction.
      b. Or Equal.
   2. Fire and/or smoke barrier identification signs:
      a. Seton.
      b. Or Equal.
   3. Warning signs (Chemical Drum Rooms):
      a. Seton.
      b. Or Equal.

2.2 SIGN STANDARDS

A. Engineered and Tested:
   1. The signage system shall have undergone testing to ensure longevity and optimal performance as defined below. Testing shall include environmental testing to ensure that materials can withstand changes in temperature and humidity without distortion as well as testing to ensure resistance to chemicals and UV effects. Further, mechanical testing shall ensure that the tensile and pull-out strength of mounting hardware is adequate to ensure a safe installation. Test data shall be included with submittals.

2.3 TYPOGRAPHY

A. Type style:
   1. Copy shall be a true, clean, accurate reproduction of specified typeface(s). Letter spacing to be normal and interline spacing shall be set by manufacturer.

B. Arrows, symbols and logo art: To be provided in style, sizes, colors and spacing as shown in shop drawings.

C. Raised Braille: Grade II Braille utilizing perfectly round, clear insertion beads.

2.4 SIGNS (ROOM IDENTIFICATION SIGNS)

A. Signage System:
   1. All signs shall have a matching appearance and constructed utilizing the same manufacturing process to ensure a consistent look throughout.

B. Materials (Room Identification Signs):
   1. Sign face shall be 0.035-inches (nominal) standard grade, high pressure surface laminate. A painted sign face shall not be acceptable.
   2. The sign shall incorporate balanced construction, specifically a core material sandwiched between identical laminates to prevent warping. Laminate on the sign face only shall not be acceptable.
   3. Tactile lettering shall be precision machined, raised 1/32-inch, matte PETG (polyethylene terephthalate glycol, thermoplastic material) and subsurface colored for scratch resistance.
   4. Wet and/or corrosive areas: Aluminum or fiberglass suitable for raised lettering and Braille.

C. Construction (Room Identification Signs):
   1. Insert components shall have a 0.080-inch thickness non-glare acrylic window and shall be inlaid flush to sign face for a smooth, seamless appearance.
   2. The laminates (front and back) shall be pressure laminated, and precision machined together to a 90-degree angle. Edges shall be smooth, void of chips, burrs, sharp edges and marks.
   3. The signage shall utilize an acrylic sphere for Grade II Braille inserted directly into a scratch resistant, high pressure laminate sign face. Braille dots are to be pressure fit in high tolerance drilled holes.
   4. Braille dots shall be half hemispherical domed and protruding to a minimum 0.025-inches.
   5. The signage shall utilize a pressure activated adhesive. The adhesive shall be nonhazardous and shall allow for flexing and deflection of the adhered components due to changes in temperature and moisture without bond failure.
   6. The core material shall not contain PVC. It shall also provide dimensional stability in the face of changing humidity and temperature.
   7. All signs shall be provided with appropriate mounting hardware. Hardware shall be finished and architectural in appearance and suitable for the mounting surface.
   8. Provide international graphic symbology and handicap symbol for all toilet and shower rooms or combinations thereof, and for unisex toilet rooms.
   9. Provide international graphic symbology and handicap symbol for all Stairs.

D. Printed Inserts (Room Identification Signs):
   1. The signage shall accommodate polyester inserts (for opaque), polycarbonate inserts (for clear), or printed paper inserts by the Government to allow updating as required.
   2. Manufacturer shall provide a template containing layout, font, color, artwork and trim lines to allow the Government to produce inserts on laser or ink jet printer. The template shall be in a PDF or Word format.

E. Text:
   1. Text Typeface: Sans Serif.
      a. Size: Minimum 3/4-inches high.
      b. Room Numbers and Sign Copy are indicated in the “Interior Signage Schedule” found on the “Architectural Finish Schedules Drawings”.
   2. Finish, and Colors:
      a. Finish: Eggshell.
      b. Color:
         1) Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

2.5 SIGNS (FIRE AND/OR SMOKE BARRIER IDENTIFICATION SIGNS)

A. Materials: (Fire and/or Smoke Barrier Identification Signs):
   1. Self-adhesive tape tags and signs:
      a. Materials: Vinyl tape or vinyl cloth.
         1) Indoor/Outdoor grade.
         2) Weather and UV resistant inks.
         3) Permanent adhesive.
      b. Size:
         1) Surface: As required by text.
         2) Thickness: 5-mils minimum.

c. Color: Standard White with black lettering.
2. Typeface: Helvetica Medium.
3. Text Size:
   a. Height: 3-inches minimum.
   b. Stroke: 3/8-inches minimum.
   c. Text: As indicated in the SCHEDULES in Part 3 of this specification.

2.6 REGULATORY REQUIREMENTS

A. All signs shall comply with the requirements of ADA, state or local authority, if appropriate.

B. Hazard Communication Signage (NFPA and OSHA signage): See Specification Section 10 14 00.

2.7 MAINTENANCE MATERIALS

A. Where stenciled markers are provided, clean and retain stencils after completion and include in extra stock, along with required stock of paints and applicators.

PART 3 - EXECUTION

3.1 DELIVERY, STORAGE, PROTECTION

A. Package to prevent damage or deterioration during shipment, handling, storage and installation. Products should remain in original packaging until removal is necessary. Store products in a dry, indoor location.

3.2 EXAMINATION

A. Installer shall examine signs for defects, damage and compliance with specifications. Installation shall not proceed until unsatisfactory conditions are corrected.

B. General: Installation locations shall be in accordance with ADA specifications. Locate signs where indicated using mounting methods in compliance with manufacturer's written instructions:
   1. The signage contractor shall coordinate installation schedules with the Contracting Officer Representative.

3.3 INSTALLATION

A. Installation shall be performed by manufacturer’s personnel trained and certified in manufacturer’s methods and procedures.

B. Installer to conduct a pre-installation survey prior to manufacturing to verify copy and sign location. Each location shall be noted using a low tack vinyl reproduction of actual sign. Full scale renderings of directories and directionals shall also be provided. Any location discrepancy or message issues shall be submitted to the Contracting Officer Representative for review.

C. Signs shall be level, plumb, and at heights indicated with sign surfaces free from defects.

D. Installation of Room Identification Signs:
   1. Install signs using foam tape for interior signs.
   2. Mounting Locations:
      a. Tactile characters on signs shall be located 48-inches minimum above the finished floor or ground surface, measured from the baseline of the lowest tactile character and 60 inches maximum above the finish floor or ground surface, measured from the baseline of the highest tactile character.
      b. Where a tactile sign is provided at a door, the sign shall be located alongside the door at the latch side. Where a tactile sign is provided at double doors with one active leaf, the sign shall be located on the inactive leaf. Where a tactile sign is provided at double doors with two active leaves, the sign shall be located to the right side of the right-hand door. Where there is no wall space at the latch side of a single door or at the right side of double doors, signs shall be located on the nearest adjacent wall.

c. Signs containing tactile characters shall be located so that a clear floor space of 18-inches minimum by 18-inches minimum, centered on the tactile characters, is provided beyond the arc of any door swing between the closed position and 45-degree open position.

3. Interior signs identifying permanent rooms and spaces shall comply with ADA, state or local authority, if appropriate.

E. Fire and/or Smoke Barrier Identification Signs:
   1. Provide marking and identification in compliance with building code.
   2. Locate in accessible concealed floor, floor-ceiling, or attic spaces.
   3. Locate within 15-feet of the end of each wall or partition and at intervals not exceeding 30-feet horizontally.
   4. Provide a vertical demarcation line on walls where wall types change assemblies. Install appropriate identification signs on both sides of the demarcation lines.

F. Upon completion of the work, signage contractor shall remove unused or discarded materials, containers and debris from site.

3.4 SCHEDULES

A. Room Identification Signs:
   1. Refer to Interior Signage Schedule found on the Architectural Finish Schedules Drawing.

B. Fire and/or Smoke Barrier Identification Signs:

ASSEMBLY	VERBIAGE
Fire Walls	(XXX) HR FIRE WALL - PROTECT ALL OPENINGS
Fire Barriers	(XXX) HR FIRE BARRIER - PROTECT ALL OPENINGS
Fire Partitions	(XXX) HR FIRE PARTITION - PROTECT ALL OPENINGS
Smoke Barriers	(XXX) HR SMOKE BARRIER - PROTECT ALL OPENINGS
Smoke Partitions	(XXX) HR SMOKE PARTITION - PROTECT ALL OPENINGS
Fire and Smoke Barriers	(XXX) HR FIRE AND SMOKE BARRIER - PROTECT ALL OPENINGS
Fire and Smoke Partitions	(XXX) HR FIRE AND SMOKE PARTITION - PROTECT ALL OPENINGS

1. (XXX) Indicates hourly rating of assembly.
2. Verify exact verbiage and mounting locations with AHJ prior to fabricating signage.

END OF SECTION

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SECTION 10 28 13
TOILET ACCESSORIES

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Toilet and Bath Accessories in accordance with provisions of Contract Documents.

B. Completely coordinate with the work of other trades.

C. Related Sections:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 01 78 23 – Operation and Maintenance Data.

1.2 REFERENCES

A. Referenced Standards:
   1. ASTM International (ASTM):
      a. A240/A240M Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip.
      b. C1036, Standard Specification for Flat Glass.
      c. C1503, Standard Specification for Silvered Flat Glass Mirror.
   2. U.S. Department of Justice:
      a. ADA Standards for Accessible Design.

1.3 SUBMITTALS

A. Product Data:
   1. Manufacturer's standard literature indicating systems and products specified.

B. Contract Closeout Information:
   1. Maintenance data.
      a. See Section 01 78 23.
   2. Letter stating that extra material has been delivered.

1.4 QUALITY ASSURANCE

A. Provide accessories from one manufacturer as far as practicable.

1.5 WARRANTY

A. Mirror Glazing: Warranted against silver spoilage for 15 years.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Hooks:
      a. Basis of Design: American Specialties, Inc. (ASI) Model 7340-S.
      b. Or Equal.
   2. Toilet Tissue Dispensers:
      a. Basis of Design: American Specialties, Inc. Model 0264-1A.
      b. Or Equal.
   3. Sanitary Disposal:
      a. Basis of Design: American Specialties, Inc. Model 0852.
      b. Or Equal.
   4. Paper Towel Dispensers:
      a. Basis of Design: American Specialties, Inc. Model 0210.
      b. Or Equal.
   5. Soap Dispensers:
      a. Basis of Design: American Specialties, Inc. Model 0347.
      b. Or Equal.
   6. Mop Holders:
      a. Basis of Design: American Specialties, Inc. Model 1315-4.
      b. Or Equal.
   7. Shower Curtain Rod:
      a. Basis of Design: American Specialties, Inc. Model 1200V.
      b. Or Equal.
   8. Curtain Hooks.
      a. Basis of Design: American Specialties, Inc. Model 1200-SHU.
      b. Or Equal.
   9. Mirrors:
      a. Basis of Design: American Specialties, Inc. Model 0600.
      b. Or Equal.
   10. Shelves:
      a. Basis of Design: American Specialties, Inc. Model 0692.
      b. Or Equal.
   11. Toilet Seat Cover Dispensers:
      a. Basis of Design: American Specialties, Inc. Model 0477-SM.
      b. Or Equal.
   12. Grab Bars:
      a. Basis of Design: American Specialties, Inc. 800 Series.
      b. Or Equal.
   13. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Type 304, 18-8 stainless steel per ASTM A240/A240M, unless noted otherwise.
   1. Finish: Satin 4 on exposed surfaces.

B. Corrosion resistant fasteners and attachment devices, and other fittings necessary to assure function and operation of accessories.

C. See drawings for items, quantities, locations, and required mounting heights.

D. Locks:
   1. Tumbler locks keyed alike.

2.3 TOILET ACCESSORIES

A. Hooks:
   1. TA1A and TA1B:
      a. Single Robe Hook.

B. Toilet Tissue Dispensers:
   1. TA2:
      a. Double Roll Tissue Dispenser.
      b. Cast aluminum with ABS thermoplastic spindle.
      c. Non-controlled delivery.

C. Sanitary Disposal:
   1. TA5:
      a. Surface mounted.

D. Paper Towel Dispensers:
   1. TA7:
      a. Surface mounted.
      b. Capacity: 400 C-fold or 525 multi-fold towels

E. Soap Dispensers:
   1. TA9V:
      a. All-purpose valve.
      b. Surface mounted.
      c. Locate right side of mirror.
      d. Center over sinks on side wall.
      e. Mount with spout at 42-inches above finished floor line.

F. Mop Holders:
   1. TA10A:
      a. The following models are sensor operated.
      b. Four rubber cam mop holders and three stainless steel hooks.
      c. 36-inches wide with 8-inch-deep stainless-steel shelf and drying rod.
      d. Surface mounted.
      e. Mount with shelf at 48-inches above finished floor line.

G. Mirrors:
   1. TA13:
      a. Frame: 18-8, type 304, 18 gauge, 0.048-inches thick) stainless steel. Unitized all welded construction. Corners mitered, welded and polished. Exposed surfaces have Satin Finish. Edges and corners are burr free.
      b. Mirror Glazing: #1 quality, 1/4-inch-thick plate/float, silver coated and hermetically sealed with a uniform coating of electrolytic copper plating. Glass shall comply with ASTM C-1503, and ASTM C-1036.
      c. Filler: Expanded polyethylene microcell foam sheet material, abrasion resistant and shock absorbing, water resistant, 1/8-inch total layer thickness.
      d. Mirror Size: 18-inches wide x 36-inches high. Surface mount bottom of mirrors 40-inches above finished floor line.

H. Toilet Seat Cover Dispensers:
   1. TA20:
      a. Surface mounted.
      b. 250 sheets.

I. Grab Bars:
   1. General:
      a. Bar Diameter: 1-1/2-inch Outside Diameter.
      b. Concealed mounting.
      c. Include anchoring devices to withstand minimum concentrated load of 250 pounds typical and 800 pounds at bariatric locations.
      d. Include 3800M spacer to keep grab bar parallel to wall faces at offset conditions.
   2. TA33V:
      a. 18-inch vertical grab bar.
   3. TA36:

a. 36-inch horizontal grab bar.

4. TA38:
   a. 48-inch horizontal grab bar.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify suitability of substrate to accept installation.

B. Coordinate and direct installation of backing where required for toilet accessories.

C. Verify that adequate wall backing has been installed.
   1. Coordinate and direct installation backing where required for toilet accessories.
   2. Utilize proprietary backing devices where available.

D. Correct deficiencies before proceeding to install accessories.

E. Where an item is mounted on or in a toilet partition, coordinate interior reinforcing location with partition manufacturer.

F. Installation constitutes acceptance of responsibility for performance.

3.2 INSTALLATION

A. Install accessories in accordance with manufacturer's instructions.
   1. Install plumb, level, and rigidly anchored to substrates.

B. Where drawings or schedules require barrier-free accessibility, install accessories in accordance with applicable regulations.

C. Coordinate accessory locations to fit spaces.

D. Coordinate items to avoid mounting conflicts.

E. Mount items with theft-resistant fasteners.

F. Seal grab bar mounting plate to shower walls with silicone sealant or gasket prior to installation of bar.

3.3 ADJUSTING AND CLEANING

A. Protect accessories from damage due to construction.
   1. Remove protective coverings when no longer required.

B. Test accessories and adjust for proper operation.

C. Clean exposed surfaces.

END OF SECTION

SECTION 10 44 33
FIRE PROTECTION SPECIALTIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Portable fire extinguishers.
   2. Fire extinguisher wall brackets.
   3. Fire extinguisher wall signs.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 03 15 19 – Anchorage to Concrete.
   2.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Americans with Disabilities Act (ADA):
      a. 2010 ADA Standards for Accessible Design.
   2. National Fire Protection Association (NFPA):
      a. 10, Standard for Portable Fire Extinguishers.

1.3 DEFINITIONS

A. Authority Having Jurisdiction (AHJ): Building official, fire chief, fire marshal or other individual having statutory authority.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Fire Extinguisher Brackets.
      c. Manufacturer's installation instructions.
      d. Ratings and classification of extinguishers.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver and install filled and charged extinguishers just prior to building occupancy.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Fire extinguishers:
      a. Larsen’s Manufacturing Co. – MP Series – Multi-Purpose Dry Chemical.
      b. Or Equal.

2. Fire extinguisher wall brackets:
    a. Larsen's Manufacturing Co. – Standard Mounting for each type of Fire Extinguisher.
    b. Or Equal.
3. Double Faced Fire extinguisher signs:
    a. Seton - 16551D.
    b. Or Equal.

2.2 MANUFACTURED UNITS

A. Fire Extinguisher (FEXT):
1. Steel bodied, all metal top (head) and valves.
2. Multi-purpose dry chemical extinguisher with hose and nozzle.
3. Provide one listed 10 pound: 4A:80B:C MP extinguisher as indicated in Fire Extinguisher Schedule.
4. Finish: Standard Red with epoxy finish coat.

B. Fire Extinguisher (FEXCA):
1. Steel bodied, all metal top (head) and valves.
2. Multi-purpose clean agent extinguisher with hose and nozzle.
3. Provide one listed: 11 pound: 1A:10B:C HT extinguisher as indicated in Fire Extinguisher Schedule.
4. Finish: Standard Red with epoxy finish coat.

C. Wall Brackets:
1. Provide Standard Bracket type to fit specified extinguisher.
2. Furnish bracket for each extinguisher not in cabinet.
3. Bracket to be finished in standard red enamel.
4. Provide one Wall Bracket as indicated in Fire Extinguisher Schedule.

D. Double Faced Fire Extinguisher Signage:
1. Fire extinguisher signs shall be mounted free from obstructions at all times to enable quick identification of extinguishers
2. Signs shall be heavy gauge 0.063-inches aluminum.
3. Four 7/32-inch corner holes provide secure installation of fasteners to wall surface.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install products in accordance with manufacturer's instructions and NFPA 10.
1. Install wall brackets with extinguisher top not over 48-inches above finished floor.
2. Install wall brackets to concrete or masonry substrate with self-tapping concrete anchors.
    a. See Specification Section 03 15 19.

B. Fire extinguisher locations shown on Drawings are approximate locations.
1. Verify all extinguisher mounting locations with the AHJ.

C. Provide "FIRE EXTINGUISHER" sign at each extinguisher location.
1. Provide a double-faced sign to provide optimum visibility for each extinguisher location.
2. Mount bottom of sign 84-inches above finished floor, centered on fire extinguisher.

3.2 FIRE EXTINGUISHER SCHEDULE

A. Locations are shown on the Life Safety Plan Drawings.

B. 052 – Roaring Springs Pump House:
1. (1 Each) (FEXT) Fire Extinguisher on Wall Bracket – Room 101 Pump Room.
2. (1 Each) (FEXT) Fire Extinguisher on Wall Bracket – Room 103 Electrical Area.

C. 062 – North Rim Water Treatment Plant:
1. (1 Each) (FEXT) Fire Extinguisher on Wall Bracket – Room 100 Mechanical Room.

2. (1 Each) (FEXT) Fire Extinguisher in Fire Extinguisher Cabinet – Room 104 Break/Kitchen.
3. (1 Each) (FEXCA) Clean Agent Fire Extinguisher on Wall Bracket – Room 106 SCADA/IT Closet.
4. (2 Each) (FEXT) Fire Extinguisher on Wall Bracket – Room 109 Process Area.
5. (1 Each) (FEXCA) Clean Agent Fire Extinguisher on Wall Bracket – Room 110 Electrical Room.
6. (1 Each) (FEXT) Fire Extinguisher on Wall Bracket – Room 111 Process Materials Shop.
7. (1 Each) (FEXT) Fire Extinguisher on Wall Bracket – Room 112 Bases Drum Room.

END OF SECTION

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SECTION 10 51 13
METAL LOCKERS AND LOCKER BENCHES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Metal lockers.

B. Related Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 01 78 23 – Operation and Maintenance Data.
   3. Section 09 06 00 – Color Schedule.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Americans with Disabilities Act (ADA):
      a. Accessibility Guidelines for Buildings and Facilities (ADAAG).
   2. ASTM International (ASTM):
      a. A540/A540M, Standard Specification for Alloy-Steel Bolting Materials for Special Applications.
      b. A1008/A1008M, Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Baked Hardenable.
      c. B108/B108M, Standard Specification for Aluminum-Alloy Permanent Mold Castings.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Drawings showing the following:
      a. Location.
      b. Numbering sequence
      c. Anchoring method.
      d. Locking.
      e. Verifying dimensions affecting locker installations.
      f. Locker elevations (Front and End Views).
      g. Show lockers in detail including fillers, trim, base, and accessories.
   2. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer’s technical data and installation instructions.
      c. Manufacturer's installation instructions.

B. Samples:
   1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
   2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Compressible Sealant:
      a. Basis of Design: Lyon LLC – "All Welded" Lockers for Personal Lockers.
      b. Or Equal.
   2. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Steel: ASTM A1008/A1008M.

B. Fasteners: Zinc-Nickel plated steel ASTM A540/A540M.

C. Hooks and Hanger Rods: Zinc-Nickel plated steel ASTM A540/A540M or cast aluminum ASTM B108/B108M.

2.3 ACCESSORIES

A. Hooks:
   1. Provide one double-prong ceiling hook and three single-prong wall hooks for single-tier and double-tier lockers.
   2. One wall hook and one ceiling hook for multi-tier lockers 12-inches and less in width.

B. Provide hat shelf in single-tier lockers.

C. Provide hanger rod minimum 5/8-inches diameter, in lieu of ceiling hook, in single tier lockers 18-inches deep or deeper.

D. Number Plates:
   1. Manufacturer's standard etched, embossed, or stamped, non-ferrous metal number plates.
   2. Three-digit numerals not less than 3/8-inches high.
   3. Sequence numbers as directed by Contracting Officer Representative.
   4. Attach plates centered, near top of each locker door, with two fasteners of same finish as number plate.
   5. Provide on each locker door.

E. Metal Base:
   1. Minimum 14 gauge – 4-inches high closed style (Z-Type).
   2. Cover entire front of base of lockers without additional fastening devices.
   3. Factory-finish base to match lockers.

2.4 FABRICATION

A. General:
   1. Fabricate preassembled lockers square, rigid, and without warp, with metal faces flat and free of dents or distortion for rigidity and durability.
   2. Ease all exposed metal edges.
   3. Weld frames together.

a. No bolts, screws, or rivet shall be used in the assembly of the locker bodies.
4. Grind exposed welds flush.
5. Chemically pretreat metal with degreasing and phosphatizing process.
6. Apply baked-on enamel finish to all surfaces, exposed and concealed, except plated and non-ferrous metal.
   a. Finished film thickness of not less than 0.75 mil for all exterior surfaces and not less than 0.5 mil for all interior surfaces.
   b. Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.

B. Lockers:
1. Frames:
   a. Minimum 16 gauge channels or 12 gauge angles, with corners electrically welded to form a solid one-piece structure.
   b. Provide door stops at door openings.
2. Backs and sides:
   a. Minimum 18 gauge.
   b. Flange backs on vertical edges and sides where they intermember with backs, making double-flanged rear corners.
3. Exposed ends of non-recessed lockers: Minimum 16 gauge.
4. Tops, bottoms, and shelves: Minimum 24 gauge, flanged on all sides.
5. Sloped tops: Continuous, minimum 20 gauge.
6. Base: Continuous, set level with floor.
7. Doors:
   a. One-piece, minimum 14 gauge, flanged at all edges, with corners.
   b. Extra bracing or reinforcing on the inside of doors over 15-inches wide.
   c. Construct to prevent springing when opening or closing.
   d. Door swing of 180 degrees.
   e. Stamped louvered vents in door faces.
      1) Single-tier lockers: Not less than six 6-inches louver openings in top and bottom of each door.
      2) Double-tier lockers: Not less than three 6-inches louver openings in top and bottom of each door.
8. Door Jambs:
   a. Single tier lockers shall have three door jambs.
   b. Double tier lockers shall have two jambs.
   c. Door jambs shall be welded to side of door frames to engage locking device.
   d. Design and gauge of jamb shall prevent freeing of locking device by prying.
   e. Each jamb shall have safety reverse nose to eliminate hazard of sharp - pointed edges protruding into the locker.
   f. Each jamb shall have easily replaceable soft rubber bumper.
9. Door hinges:
   a. Full-loop, five-knuckle, tight pin.
   b. Not less than 0.050-inches thick steel, 2-inches high.
   c. Continuous weld hinges to inside of frame and secure to door with not less than two factory-installed fasteners, completely concealed and tamperproof when door is closed.
   d. Minimum three hinges for each door 42-inches high and over.
   e. Minimum two hinges for each door less than 42-inches high.
10. Latching:
   a. Positive, automatic, pre-locking, pry-resistant latch and pull.
   b. Rubber silencers.
   c. Chromium-plated, vandal proof or kick proof lift-up handle, containing strike and hole for padlock.
   d. Enclose latch on four sides in a boxed receptacle in lock bar channel, and engaging latch hooks on frame opposite hinges.
   e. Three-point latching for single-tier lockers.
   f. Two-point latching for double-tier lockers.

11. Provision for padlock: Latch pull with hole to accept padlock.
12. ADA compliant lockers:
    a. Bottom Locker in a Double tier locker that is manufactured in the same construction as standard lockers specified and provide with the following modifications:
        1) Shelves:
            a) Bottom shelf located not less than 9-inches above finished floor.
                (1) Provide full width closure panel from bottom shelf to locker floor.
            b) Top shelf located not more than 54-inches above finished floor.
        2) Top of hanger rod located not more than 48-inches above finished floor.
        3) Single-prong wall hooks located not more than 46-inches above finished floor.
        4) Door latching device:
            a) ADA Lockers shall be provided with easy to reach, grip and turn handles which comply with the Americans with Disabilities Act for accessing lockers by the disabled.
            b) Operating device to be located 48-inches above finished floor.
            c) Spring loaded.
        5) Keyed lock to be self-locking when door is closed.
    b. Top of locker to align with adjacent lockers.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas to receive lockers. Notify Contracting Officer Representative if areas are not acceptable. Do not begin installation until unacceptable conditions have been corrected.

3.2 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Install plumb, level, square, rigid and flush.

C. Use manufacturer's supplied hardware.

D. Anchor lockers to floors and walls as shown on drawings and shop drawings.

E. Space fasteners not over 48-inches on center:
    1. Install through suitable reinforcing plates where necessary to prevent metal distortion.
    2. Conceal all fasteners wherever possible.

F. Provide and install filler and closure pieces as required using concealed fastners.

G. Install sloping tops and bases using concealed fasteners.

H. Joints: Provide flush hairline joints against adjacent surfaces.

I. Number Plates: Attach number plates to face of doors level with 2 aluminum rivets. Attach in sequence as indicated on the Drawings or as indicated by the Contracting Officer Representative.

3.3 FIELD QUALITY CONTROL

A. Touch-up any damaged finishes or replace as directed by Contracting Officer Representative.
    1. Use only materials and finishes as recommended or furnished by locker manufacturer.

3.4 ADJUSTMENT

A. Adjust doors and latches to operate easily without bind.
    1. Verify satisfactory operation of integral locking devices.

3.5 CLEANING

A. Clean surfaces promptly after installation in accordance with manufacturer's instructions.

B. Do not use harsh cleaning materials or methods that could damage finish.

3.6 PROTECTION

A. Protect installed lockers from damage during construction.

3.7 SCHEDULES

A. Locker Types and Sizes:
   1. Type A: Single tier 15 x 18 x 72 inches. (Provide (2) in Building 062-Water Treatment Building.
   2. Type B: Double tier 15 x 18 x 36 inches. (Provide (1) in Building 062-Water Treatment Building.

END OF SECTION

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DIVISION 11

EQUIPMENT

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SECTION 11 24 26
SHOP AND STORAGE EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Shop and storage equipment:
      a. Shelving.
      b. Metal Peg Board.
      c. Metal Worktables.
      d. Bench Vise.
      e. Industrial Curtain.

B. Related Sections include but are not necessarily limited to:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 01 78 23 – Operation and Maintenance Data.
   3. Section 09 06 00 – Color Schedule.
   4. Section 10 14 00 - Identification Devices.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. National Fire Protection Association (NFPA):
      a. 701 - Standard Methods of Fire Tests for Flame Propagation of Textiles and Films.
   2. Occupational Safety and Health Administration (OSHA):
      a. 1910.144 - Safety color code for marking physical hazards.

1.3 SYSTEM DESCRIPTION

A. Provide single-source responsibility for equipment listed by this Specification Section.

1.4 SUBMITTALS

A. Shop Drawings:
   1. Fabrication and/or layout drawings.
      a. Fabrication details and rough-in diagrams.
      b. Provide drawings indicating the industrial curtain and valance sizes, heights and tracks.
         1) Including mounting and support details for the tracks.
   2. Product technical data including:
      a. Provide manufacturers information and data for shelving, cabinets, and metal pegboard panels.
      b. Provide complete manufacturers information and data for industrial curtains, valances, including tracks and all accessories.
      c. Acknowledgement that products submitted meet the requirements of standards referenced.
      d. Manufacturer's installation instructions.

B. Samples:
   1. Refer to “Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals” for submission procedures and requirements of Color Samples.
   2. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.5 WARRANTY

A. Shelving, Metal Pegboard Panels and Metal Worktable:
   1. Limited Lifetime Warranty which protects against product failure and manufacturing defects, when installed and used correctly.

B. Industrial Curtain:
   1. Provide Manufacturers 5-year Warranty on all curtains.
   2. Provide Manufacturers Lifetime Warranty on all Hardware.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:

B. Acceptable Manufacturers: Contracting Officer will consider Contractor's request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Shelving:
      a. Lyon Metal Products, Inc. - 8000 Series Closed Steel Shelving.
      b. Or Equal.
   2. Metal Pegboard:
      a. Wall Control Pegboard.
      b. Or Equal.
   3. Metal Workbenches:
      a. Lyon Metal Products, Inc. – Adjustable Workbench.
      b. Or Equal.
   4. Bench Vise:
      a. Yost Vises – Model 910-HV 10-Inch.
      b. Or Equal.
   5. Industrial Curtains:
      a. Akon Curtains.
      b. Or Equal.

2.2 SHELVING

A. Industrial Closed Shelving Structure:
   1. Industrial shelving upright assembly structure consists of 16-guage T-posts with a side sheet panel constructed of durable hot-rolled steel.
   2. Side panel sheets are spot welded to the steel shelving post uprights. Shelving upright assemblies are 84-inches high x 24-inches deep x 42-inches wide. The T-post boltless shelving uprights are punched for shelf clips. This allows adjustment of steel shelving on 1-1/2-inch centers.

B. Metal Shelving Sides and Backs:
   1. The closed 20-guage steel sides and backs bolt on to the closed uprights. Backs consist of 2 half panels. They have 3 vertical rows of holes at 1-1/2-inch centers so they can easily bolt to both the uprights and shelves.

C. Box Shelves:

1. The box steel storage shelves have ends that are flanged with a 90 degree return flange. All corners of each shelf are lapped and welded for extra strength. All steel shelves are punched at each of the four corners so you can bolt them to the uprights. Shelves shall be 18-gauge steel.
2. Each shelf shall be supported by 4 shelf clips. The shelf clips shall be designed to quickly lock into place on either side of the uprights.
3. Provide 6 shelves per shelving unit.

D. Finish:
1. Industrial Closed Shelving Structure, Metal Shelving Sides and Backs and Box Shelves shall have a durable powder coat finish.
2. Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.

E. Anchorage:
1. Industrial Closed Shelving Structure shall be anchored to the floor and/or wall. Provide the manufacturers with an upright-to-floor angle and a concrete wedge anchor to anchor each upright to the floor.

F. SCHEDULE:
1. Building 062 – Water Treatment:
   a. Provide (3) Shelves in Process Materials Shop, Room #111 on the north wall where shown on Ground Level Plan.
   b. Provide (3) Shelves in Process Materials Shop, Room #111 on the north wall where shown on Ground Level Plan.

2.3 METAL PEGBOARD

A. Metal wall pegboards panels:
1. 16-inches wide x 32-inches high 20 gauge galvanized metallic steel pegboard panels.
2. Construction of the metal wall panels will not warp, fray, or crack creating an industrial grade tool storage solution.
3. Metal wall pegboard panels accept conventional 1/4-inch pegboard pegs as well as slotted pegboard hooks, brackets, shelves, and supplies for maximum tool storage versatility.
4. Metal wall pegboards accept magnets and magnetic tool organizers such as magnetic socket sets, LED lights, and magnet pads.

B. Metal wall pegboard panels are constructed with a preformed 3/4-inch return flange on four sides that separates the panel storage surface from the wall. Each storage panel is an easy one-piece installation requiring no cleat system or added framework.

C. Provide Mounting Hardware from the Manufacturer.

D. Metal wall pegboard panels mount directly into studs or blocking behind sheet rock with no framework required. Mount top of metal wall pegboard at 6-feet-0-inches above finished floor.

E. Finish:
1. Refer to "Specification Section 09 06 00 – Color Schedule" for color selections.

F. SCHEDULE:
1. Building 062 – Water Treatment
   a. Provide two sets of (3) Galvanized Metallic Steel Pegboard Panels (Covers 48-inchs x 32-inches installed on the North Wall of Process Materials Shop, Room #111 in 062-Water Treatment).
   b. Provide two sets of (3) Galvanized Metallic Steel Pegboard Panels (Covers 48-inchs x 32-inches installed on the South Wall of Process Materials Shop, Room #111 in 062-Water Treatment).

2.4 METAL WORKBENCH

A. Metal workbench shall be heavy duty steel with the following:
1. Steel top.

2. Steel stringer.
3. 2-open panel legs.

B. Legs of the workbench shall be adjustable height:
1. Height range shall be in increments of 12-inches between 32-1/4-inches and 44-1/4-inches.

C. Workbench size shall be 72-inches wide x 30-inches deep.

D. SCHEDULE:
1. Building 062 – Water Treatment:
a. Provide (3) Metal Workbenches in Process Materials Shop, Room #111 on the north wall where shown on Ground Level Plan.
b. Provide (3) Metal Workbenches in Process Materials Shop, Room #111 on the south wall where shown on Ground Level Plan.

2.5 BENCH VICE

A. Bench vice
1. Made of structural cast steel (SC37).
2. Replaceable hardened steel serrated top jaws and built-in pipe jaws.
3. Steel vise shall have a 360-degree swivel base with 2 lockdowns.
4. Permanent pipe jaws and large reinforced anvil.
5. No pinch main screw handle.

B. Finish:
1. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

C. SCHEDULE:
1. Building 062 – Water Treatment:
a. Provide (1) Bench Vice on Metal Workbench in Process Materials Shop, Room #111 on the north wall where shown on Ground Level Plan.
b. Provide (1) Bench Vice on Metal Workbench in Process Materials Shop, Room #111 on the south wall where shown on Ground Level Plan.

2.6 INDUSTRIAL CURTAINS

A. Industrial curtain and valance shall be manufactured from durable 18-ounce vinyl reinforced with a polyester mesh scrim. Vinyl shall meet NFPA 701. The edges of the vinyl shall be hemmed edges. Grommets shall be provided along the top of the curtain and valance at 12-inches on center.
1. Industrial Curtain shall be 14-feet wide and 9-feet-6-inches high which includes 5 percent for proper drape and 3-inches subtracted for overall height for curtain track.
2. Industrial Curtain shall have a replaceable Velcro bottom floor sweep that abuts the floor. The floor sweep shall hang 6-inches below the bottom of the curtain.
3. Industrial Curtain shall be provided with floor strap anchors spaced every 4-feet on center with Velcro strap and anchor hooks.
4. Valance shall be 13-feet long with a sloped top. The short side shall be 8-1/2 inches high, and the long side shall be 5-feet-8-inches high.
5. Provide blockouts in valance where required for structural members penetrating the valance.
6. Provide manufacturers zipper where required and indicated on drawings. Velcro and magnets are not allowed except Velcro is allowed where curtain abuts wall and where called out on plans.
7. Provide vision panels in curtain (2-feet high x maximum width in each curtain).

B. Industrial curtain hardware shall be provided by the manufacturer. The industrial curtain shall be trolley style straight track that allows the curtain to roll from side to side.
1. The track shall be 16-gauge galvanized steel with track stops added to both ends.
2. Track shall be suspended from ceiling/roof with threaded rod mount supports.
3. Industrial curtain shall hang from the trolley style track with manufacturers 1 inch diameter nylon wheels with 16-gauge hot dipped galvanized steel roller hooks.

a. Each hook shall be rated for a minimum of 40-pounds.
b. Provide one hook per each lineal foot of curtain.

C. Industrial valance hardware shall be provided by the manufacturer. The industrial valance shall be trolley style straight track.
1. The track shall be 16-gauge galvanized steel with track stops added to both ends.
2. Track shall be suspended from ceiling/roof.
3. Industrial valance shall hang from the trolley style track with manufacturers with 16-gauge hot dipped galvanized steel fixed hooks.
a. Each hook shall be rated for a minimum of 40-pounds.
b. Provide one hook per lineal foot of curtain.

D. Finish:
1. Refer to “Specification Section 09 06 00 – Color Schedule” for color selections.

2.7 MAINTENANCE MATERIALS

A. Extra Materials:
1. Furnish Contracting Officer two, 12 ounces spray cans of touch-up paint for each equipment color provided.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install in accordance with manufacturer's written instructions and approved shop drawings.

B. Comply with Section 10 14 00 for identification and tagging and for warning or caution signs.

END OF SECTION

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SECTION 11 30 13
APPLIANCES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Residential Grade Appliances.

B. Related Sections include but are not necessarily limited to:
   1. Specification Section 01 26 01 – Contract Modification Procedures.
   2. Specification Section 01 33 23 – Submittal Procedures.
   3. Specification Section 01 78 23 – Operation and Maintenance Data.

C. Completely coordinate with work of other trades.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Americans with Disabilities Act (ADA):
      a. Accessibility Guidelines for Buildings and Facilities (ADAAG).
   2. NSF International (NSF/ANSI):
      a. NSF/ANSI Standard 184 – Residential Dishwashers.

1.3 SUBMITTALS

A. See Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Shop Drawings:
   1. Details of connections, design elements, and relation to adjacent items.
   2. Product Data:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Product data/information for each appliance specified.
      c. Manufacturer's installation instructions.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. Select 01 33 04 for Engineering projects, and 01 78 23 for Architectural projects.
      b. See 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Residential Grade Appliances – Dishwasher:
      a. GE Appliances.

b. Or Equal.
2. Residential Grade Appliances – Microwave and Refrigerator:
   a. LG Electronics USA.
   b. Or Equal.

2.2 EQUIPMENT

A. Residential Grade Appliances:
   1. Microwave Oven:
      a. Countertop style.
         1) Color:
            a) Stainless Steel.
         2) Exterior Design:
            a) 2-Step Open Door.
         3) Capacity:
            a) 2 Cubic Foot.
         4) Controls:
            a) Clock.
            b) LED Display.
            c) Kitchen Timer.
            d) Smooth Touch Glass Controls.
            e) Cooking Complete Reminder.
         5) Exterior Dimensions:
            a) 24-inches wide x 13-3/4-inches high x 19-1/2-inches deep (max).
         6) Features:
            a) 10 Power Levels.
            b) Add 30 Seconds.
            c) Child Lock.
            d) Defrost (Auto and Rapid).
            e) Easy Clean Interior.
            f) Popcorn Key.
            g) 7 Sensor Cook Options.
            h) Beverage, Casserole, Dinner Plate, Pie, Pizza Sensor Reheat Options.
            i) Smart Inverter, 16-inch diameter turntable.
            j) LED Lighting.
         7) Power Rating:
            a) 1200 Watts (Output).
         8) Energy Star® Certified.
         9) Amps / Watts and Power Rating:
            a) 10.0 A and 1200 W.
            b) 120 VAC, Single Phase.
         10) Warranty:
            a) 10 years (Magnetron) and 1 year (Parts and Labor).
   2. Refrigerator.
      a. Freestanding refrigerator with bottom-drawer freezer.
         1) ADA Compliant.
         2) Color:
            a) Stainless steel.
         3) Exterior Dimensions:
            a) 32-3/4- inches wide x 70-inches high x 31-1/4-inches deep.
            b) Depth with drawer open – 46-inches.
            c) Depth with door open – 42-inches.
         4) Capacity:
            a) 20.8 cubic feet total.
            b) Freezer – 6.6 cubic feet.
            c) Refrigerator – 14.2 cubic feet.

5) Energy:
    a) Energy Star.
    b) Energy Consumption – 574 kWh/Year.
6) Icemaker:
    a) Factory installed.
    b) Cubed
    c) 2.5 pounds, Freezer ice maker – daily production.
    d) 5.1 pounds Freezer ice storage capacity.
7) Features:
    a) Display type – White LED.
    b) Temperature Controls – Electronic/Digital.
    c) Door Alarm.
    d) Child Lock.
    e) Smart Diagnosis.
    f) Auto Closing Door Hinge.
    g) Wi-Fi Enabled.
    h) Proactive Smart Alerts.
8) Refrigerator Construction:
    a) 2 full width tempered glass with white décor trim shelves.
    b) 2 crisper bins.
    c) Pantry.
    d) Light.
    e) 3 door bins on each of the two doors.
    f) Bar type handle.
    g) Hidden Hinges.
    h) Insulation – Low GWP and Zero ODP Blowing Agent.
9) Freezer Construction:
    a) Pull Drawer type door.
    b) Handle built into door.
    c) 2 tier drawer / shelves.
    d) Fixed drawer divider.
    e) Upper LED light and lower pocket light.
10) Warranty:
    a) Parts and Labor 1 year.
    b) Sealed System (Parts and Labor) 5 years.
    c) Compressor (Parts and Labor) 5 years.
    d) Linear / Inverter Compressor (Parts Only) 6 – 10 years.

3. Dishwasher
    a. Undercounter dishwasher
        1) ADA Compliant.
        2) Color:
            a) Stainless Steel.
        3) Exterior Style:
            a) Stainless Full-Length Flat Door.
            b) Recessed Pocket Handle – Stainless Steel Color.
        4) Exterior Dimensions:
            a) 23-3/4-inches wide x 32-1/4-inches high x 23-1/2-inches deep.
        5) Flush Cabinet Installation.
        6) Door Insulated.
        7) Outer Wrap with U-Blanket.
        8) Motor:
            a) Single Speed.
        9) Sealed toe Kick.
        10) 51 dBA rated for sound.
        11) Features:

a) Connected Capabilities – (Amazon Alexa, The Google Assistant, IFTTT, and Sonos).
b) Top controls, fully integrated electronic, 4 push buttons.
c) 3 Digit countdown display with 1 “Clean” light / 1 “Sanitized” light Cycle Progress Indicators, Cycle countdown, Audible End-of Cycle Signal, Last Cycle memory and Start/Reset Pad.
d) Auto Sense, Heavy Wash and Light Wash Dishwasher Cycles.
e) Condensate Dry.
f) Options - Start/Reset, Child Lock, Display Start 1-12 Hours, Temp Boost, and Sani (Certified to NSF Standard 184).
g) Leveling Legs.
h) Center Pull Latch.
i) Nylon Lower Rack.
j) 3 Cycles.
k) 12 Place Settings.
l) Clean Sensor.
m) Stainless Steel Tub and Door Liner.
n) Standard 3-Level Wash System.
o) Automatic Temperature Control, Automatic Hot Start, Hard Food Disposer, Flood Control System and Dual Pump.

12) Racking System:
a) 2 Piece Silverware Basket with Cell Covers, 1 Plastic Utility Shelf – Upper, ball Tipped Tines, Stem Safe.
b) Nylon Adjustable Upper Rack.

13) Wash Arms:
a) 2 with upper nozzles.

14) Water Filtration System:
a) 100 percent Filtration with Hard Food Disposer and Removable Filter.

15) Provide with WIFI Connection.

16) Electrical Requirements:
a) 120-volt, 60 Hz and 9 amp (max) and 1070 watts (max).

17) Warranty:
a) 1 – year for Labor and Parts.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install appliances in accordance with manufacturer's installation instructions and applicable Code(s).

B. Connect to mechanical, plumbing and/or electrical in accordance with applicable Code(s).

C. Adjust to be level and plumb.

END OF SECTION

SECTION 11 53 00
LABORATORY INSTRUMENTATION - TABLE-TOP AND HANDHELD

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Laboratory table-top and handheld instrumentation for the North Rim WTP (Volume 6).

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Laboratory instrument for monitoring turbidity in water accordance/compliance with DIN EN ISO 7027 and USEPA method 10258.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See Specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. General:
         1) Materials.
         2) Sizes, dimensions and electrical requirements.
         3) Accessories including sampling reagents and consumables.
         4) Installation/setup.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specifications.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.

B. Submit request for product substitution in accordance with Specification Section 01 33 23.

2.2 TURBIDIMETER

A. Models:
   1. Hach TU5200 (Basis of Design).
   2. Or equal.

B. Quantities and Locations
   1. North Rim WTP: 1 unit.

C. Design

1. In accordance with DIN EN ISO 7027.
    a. Measuring Range:
        1) 0 to 1000 NTU / FNU / TE/F / FTU.
        2) 0 to 250 EBC.
        3) 0 to 100 mg/L.
    b. Accuracy:
        1) ±2% of reading ±0.01 NTU from 0 to 40 NTU based on Formazin primary standard at 25°C.
        2) ±10% of reading from 40 to 1000 NTU based on Formazin primary standard at 25°C.
    c. Repeatability:
        1) 1% or ± 0.002 NTU, whichever is greater, from 0 to 40 NTU based on Formazin primary standard at 25°C.
        2) 3.5% at turbidity levels greater than 40 NTU, based on Formazin primary standard at 25°C.
    d. Resolution:
        1) 0.0001 NTU / FNU / TE/F / FTU / EBC / mg/L.

2. In accordance with USEPA Method 10258.
    a. Measuring Range:
        1) 0 to 700 NTU / FNU / TE/F / FTU.
        2) 0 to 175 EBC.
        3) 0 to 100 mg/L.
    b. Accuracy:
        1) ±2% of reading ±0.01 NTU from 0 to 40 NTU based on Formazin primary standard at 25°C.
        2) ±10% of reading from 40 to 700 NTU based on Formazin primary standard at 25°C
    c. Repeatability:
        1) 1% or ± 0.002 NTU, whichever is greater, from 0 to 40 NTU based on Formazin primary standard at 25°C.
        2) 3.5% at turbidity levels greater than 40 NTU, based on Formazin primary standard at 25°C.
    d. Resolution:
        1) 0.0001 NTU / FNU / TE/F / FTU / EBC / mg/L.

3. The analyzer shall have the option of an integrated RFID module that allows for measurement tracking between online and laboratory instrumentation.

2.3 PORTABLE MULTI-METER

A. Models:
    1. Hach HQ2200 (Basis of Design).
    2. Or equal.

B. Quantities and Locations
    1. North Rim WTP: 1 unit.

C. Design
    1. Parameters include pH, Conductivity, TDS, Salinity, Dissolved Oxygen, ORP.
    2. DO Measurement Range: 0.1 – 20.0 mg/L.
    3. Operating Temperature: 0 - 60 °C.
    4. pH Measurement Range: 0 – 14 pH.
    5. Salinity Measurement Range: 0 – 42 (ppt).
    6. TDS Measurement Range: 0.00 mg/L – 0.1 g/L.

2.4 SPECTROPHOTOMETER

A. Models:
    1. Hach DR 6000 (Basis of Design).

2. Or equal.

B. Quantities and Locations
   1. North Rim WTP: 1 unit.

C. Design
   1. The power requirements for the spectrophotometer 100/240 Vac, 50/60 Hz.
   2. The wavelength capability is 190-1100 nm.
   3. The characteristics of the optical system are:
      a. Accuracy 5 mAbs at 0.0-0.5 Abs, <1% at 0.5-2.0 Abs at 546 nm.
      b. Linearity: 0.01 nm at <2 Abs (with neutral glass at 546 nm).
      c. Bandwidth: 2nm.
      d. Wavelength accuracy: +/- 1 nm.
      e. Wavelength Reproducibility: < 0.1 nm.
   4. Readout modes are:
      a. Transmittance (%).
      b. Absorbance.
      c. Concentration.
      d. Scanning wavelength.
   5. Equipped with a USB interface, type A (2), USB type B.
      a. RFID module optional.
   6. Storage up to 5000 data points (date, time, results, sample ID, user ID) and 200 user-programs.
   7. Capable of measuring the following parameters.
      a. Alachlor: 0.1 to 0.5 ppb
      b. Alkalinity: 250 to 400 mg/l
      c. Aluminum: 0.002 to 0.800 mg/L
      d. Ammonia, nitrogen: 0.015 to 50.0 mg/L
      e. Arsenic: 0.020 to 0.200 mg/L
      f. Atrazine: 0.5 to 3.0 ppb
      g. Barium: 2 to 100 mg/L
      h. Benzotriazole: 0.2 to 16.0 mg/L
      i. Boron: 0.02 to 14.0 mg/L
      j. Bromine: 0.05 to 4.50 mg/L
      k. Cadmium: 1.3 µg/L to 0.30 mg/L
      l. Carbohydrazide: 5 to 600 µg/L
      m. Chloramine, mono-: 0.04 to 10.0 mg/L
      n. Chloride: 0.1 to 25.0 mg/L
      o. Chlorine dioxide: 0.01 to 1000 mg/l
      p. Chlorine, free: 0.02 to 10.0 mg/L
      q. Chlorine, total: 2 µg/L to 10.0 mg/L
      r. Chromium, hexavalent: 0.010 to 1.00 mg/L
      s. Chromium, total: 0.01 to 0.70 mg/L
      t. Cobalt: 0.01 to 2.00 mg/L
      u. Color: 0 to 500 units
      v. COD (Chemical Oxygen Demand): 0.7 to 15,000 mg/L
      w. Copper: 1 µg/L to 8.00 mg/L
      x. Cyanide: 0.002 to 0.240 mg/L
      y. Cyanuric Acid: 5 to 50 mg/l
      z. DEHA (Diethyl hydroxylamine): 3 to 450 µg/L
      aa. Dissolved oxygen: 6 µg/L to 40 mg/L
      bb. Erythorbic acid (isoascorbic acid): 13 to 1500 µg/L
      cc. Fluoride: 0.02 to 2.00 mg/L
      dd. Formaldehyde: 2 to 500 µg/L
      ee. Hardness, total (calcium and magnesium as $CaCO_3$): 4 µg/L to 4.00 mg/L
      ff. Hydrazine: 4 to 600 µg/L
      gg. Hydroquinone: 9 to 1000 µg/L

hh. Iodine: 0.07 to 7.00 mg/L
ii. Iron, ferrous: 0.02 to 3.00 mg/L
jj. Iron, total: 0.009 to 6.00 mg/L
kk. Lead: 3µg/L to 2 mg/l
ll. Manganese: 0.006 to 20.0 mg/L
mm. Mercury: 0.1 to 2.5 µg/L
nn. Methylethylketoxime (MEKO): 15 to 1000 µg/L
oo. Molybdenum, molybdate: 0.02 to 40.0 mg/L
pp. Nickel: 0.006 to 6 mg/L
qq. Nitrate, nitrogen: 0.01 to 35.0 mg/L
rr. Nitrite, nitrogen: 0.002 to 250 mg/L
ss. Nitrogen Simplified Total Kjeldahl: 0 to 16 mg/l
tt. Nitrogen, total: 0.5 to 150 mg/L
uu. Nitrogen, total inorganic: 0.2 to 25.0 mg/L
vv. Nitrogen, total Kjeldahl: 1 to 150 mg/L
ww. Ozone: 0.01 to 1.50 mg/L
xx. PCB (polychlorinated biphenyls): 1 to 50 ppm
yy. Phenols: 0.002 to 0.200 mg/L
zz. Phosphonates: 0.02 to 125.0 mg/l
aaa. Phosphorus, acid hydrolysable: 0.06 to 100 mg/L
bbb. Phosphorus, reactive (orthophosphate): 19 µg/L to 100 mg/L
ccc. Phosphorus, total: 0.06 to 100 mg/L
ddd. Potassium: 0.1 to 7.0 mg/L
eee. Quaternary ammonium compounds: 0.2 to 5.0 mg/L
fff. Selenium: 0.01 to 1.00 mg/L
ggg. Silica: 3 µg/L to 100 mg/L
hhh. Silver: 0.005 to 0.700 mg/L
iii. Sulfate: 2 to 900 mg/L
jjj. Sulfide: 5 to 800 µg/L
kkk. Surfactants, anionic: 0.002 to 0.275 mg/L
lll. Suspended solids: 5 to 750 mg/L
mmm. Tannin and lignin: 0.10 to 0.90 mg/L
nnn. TOC (Total Organic Carbon): 0.3 to 700 mg/L
ooo. Tolyltriazole: 1.0 to 20.0 mg/L
ppp. Toxicity: 0 to 100% Inhibition
qqq. TTHM (trihalomethanes, total): 10 to 600 µg/L
rrr. TPH (Total Petroleum Hydrocarbons): 2 to 200 ppm threshold
sss. Volatile Acids: 27 to 2800 mg/L
ttt. Zinc: 0.01 to 3.00 mg/L

8. Manufacturer shall provide high quality pre-mixed reagents and standards.
9. Instrument includes:
   a. Universal Cell Adapter.
   b. Manual.
   c. Power supply.
   d. Dust Cover.
   e. 1-inch square matched glass sample cells.

2.5 TOTAL ORGANIC CARBON (TOC) ANALYZER

A. Models:
   1. Suez/Sievers M5310C (Basis of Design).
   2. Or equal.

B. Quantities and Locations
   1. North Rim WTP: 1 unit

C. Design

1. Sample Range: 4 ppb – 50 ppm.
2. Sample Temperature Range: 5-95 °C.
3. Ambient Temperature Range: 5-40 °C.

2.6 GANG STIRRER

A. Models:
   1. Phipps and Bird PB-900 paddle (Basis of Design).
   2. Or equal.

B. Quantities and Locations
   1. North Rim WTP: 1 unit.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Setup equipment per manufacturer's recommendations.

3.2 FIELD QUALITY CONTROL

A. Provide manufacturer's field services.

END OF SECTION

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SECTION 11 53 13
LABORATORY FUME HOODS AND EXHAUST DEVICES

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Laboratory Fume Hoods in accordance with provisions of Contract Documents.

B. SECTION INCLUDES
   1. Chemical Fume Hoods, including bench mounted hoods.
   2. Fume Extractor Arms (Snorkels).

C. Completely coordinate with the work of other trades.

1.2 RELATED SECTIONS

A. Section 01 81 33 - Cyber Security Requirements.

B. Canopy hoods and low slotted exhausts are specified in Section 12 35 53 under Stainless Steel Fabrications.

1.3 DESCRIPTION

A. Provide fume hoods and exhaust devices, as indicated, complete with accessories as described herein, and shown on Laboratory Furnishings drawings.

B. Fume hoods with accessories shall be pre-piped and pre-wired.
   1. Pre-pipe service fittings to single point connection for each service at 150mm 6 inches above top of hood or as otherwise shown. Cup sink tailpiece shall be provided with fume hood. Refer to Section 11 53 43 and details on Laboratory Furnishings drawings for service fittings. P-trap, waste piping and tailpiece extensions for cup sinks, if required, shall be furnished and installed by Division 22. Comply with Division 22 requirements for piping and installation requirements.
   2. Pre-wire electrical devices to junction box at top of hood. Provide wire terminal blocks and terminal identification. Comply with Division 26 requirements for electrical work. Lighting fixtures, electrical outlets, switches, wiring, terminal blocks, terminal boxes, safety alarms and other electrical devices mounted on or in fume hoods shall be approved for use in Class 1, Division 2 locations indicated on the drawings.
   3. Work of this Section requires close coordination with Work of Divisions 22, 23, 25 and 26, as well as installation of Government furnished components and Work specified in other Sections. Sequence Work to ensure an orderly progress in the project without removal of previously installed Work and so as to prevent damage to finishes and products.

C. Exhaust Requirements:
   1. Refer to Exhaust Schedule for requirements.
   2. Fume hoods and exhaust devices shall be designed to operate safely within the values provided on the Exhaust Schedule. The airflow values provided on the Exhaust Schedule represent the total airflow through the fume hood or exhaust device, including the airflow through the sash or work opening, airfoil, bypass, and leakage, respectively, as they apply to particular devices. Exhaust devices shall operate at specified face velocity within total airflow scheduled.
   3. Proposed modifications or corrections shall be reviewed and approved by Laboratory Planner and Mechanical Contracting Officer Representative for devices that requires adjustment to operate within specified design requirements.

D. Accessibility:

1. Where indicated on Laboratory Furnishings drawings, fume hoods shall be furnished and installed in a manner to make them accessible in accordance with the Americans with Disabilities Act and state or local building codes or regulation having jurisdiction. The height of the highest point of access to the work surface above finished floor shall not exceed 863mm 34 inches. Fittings for piped services and electrical receptacles shall be of a design and in a location to satisfy accessibility requirements.

1.4 REFERENCES

A. American Society of Heating, Refrigeration, and Air conditioning Engineers:
   1. ASHRAE 110, latest edition, Method of Testing Performance of Fume Hoods.

B. Scientific Equipment and Furniture Association:
   1. SEFA 1 Fume Hoods.
   2. SEFA 2 Installation.
   3. SEFA 3 Laboratory Work Surfaces.
   4. SEFA 7 Fixtures.
   5. SEFA 8 W Laboratory Grade Wood Casework.
   6. SEFA 8 M Laboratory Grade Metal Casework.
   7. SEFA 8 P Laboratory Grade Polypropylene Casework.

C. Occupational Safety and Health Administration:
   1. Federal Register 29 CFR Part 1910 Occupational Exposures to Hazardous Chemicals in Laboratories.

D. State OSHA regulations, if developed for state of the project location.

E. American National Standards Institute/American Industrial Hygiene Association:
   1. ANSI/AIHA Z9.5 Standard for Laboratory Ventilation.

F. National Fire Protection Association:
   1. NFPA 45 Standard on Fire Protection for Laboratories Using Chemicals.

G. American Conference of Government Industrial Hygienists:
   1. ACGIH Industrial Ventilation.

1.5 SUBMITTALS

A. Shop Drawings:
   1. Submit complete shop fabrication and installation drawings, including plans, elevations, sections, dimensions, materials and metal gauge sizes, details, fittings, duct connections, schedules, and steam table piping and vents from cabinets below where applicable.
   2. Show relationship to adjoining materials and construction.
   3. Identify connection points, locations, and sizes to building services and systems. Provide clear identification where equipment requirements deviate from the service/utility provisions in the Construction Documents.
   4. Coordinate shop drawing submittals of both this Section and Section 12 35 53 so that each recognizes and incorporates each other's products.
   5. Provide piping, wiring, and/or control diagrams, including connection points and sizes to building services and systems. Provide flow rates, pressure drops, voltage and amperage, etc.
   6. Identify where equipment requirements deviate from service/utility provisions identified in the Construction Documents.

B. Product Data:
   1. Description of hoods and exhaust devices, including construction details, materials, gauges, sash lock and release procedure, hardware cut sheets, piping of equipment and description of re-lamping procedures.
   2. Statement giving face velocity, operating volume and pressure drop at operating sash position for each size hood or exhaust device.
   3. Description of proposed factory dynamic testing procedures.

4. Submit complete materials list, including catalog data of materials, equipment, fan curves, test designs, performance charts, and products for Work specified in this Section.

C. Samples:
   1. Manufacturer's color charts showing the full range of colors, textures, and patterns available for each type of finish, for selection or verification by Contracting Officer Representative.
   2. Submit two (2) samples of each type of specified finish and color range available, or as identified in the Finish Schedule.
   3. Submit two (2) samples of fume hood sash safety label.

D. Project Information:
   1. As Manufactured (AM) Fume Hood Testing in Manufacturing Facility: Provide certification that each type and size of fume hood has achieved an AM performance rating equal or better than 0.05 ppm with 4.0 Lpm tracer gas release rate when tested in accordance with ASHRAE 110.
   2. Factory Authorized Testing (FAT) Procedures:
      a. Requirements and Documentation.
      b. Include calibration documentation.
   3. Fume Hood Sound Level Certification: Provide certification of fume hood compliance with design criteria for maximum allowable noise within laboratories.
      a. Provide test data of octave band analysis verifying fume hood is capable of a 50 NC value when connected to a 50 NC HVAC source. Measurements shall be taken 36 inches in front of full open sash, 60 inches above the floor at the specified face velocity.

E. Refer to Section 01 81 33 - Cyber Security Requirements for required cyber security related submittals.

F. Contract Closeout Information:
   1. Structural calculations for Laboratory Fume Hoods and Exhaust Devices indicating design conforms to specified design criteria, sealed by the Specialty Structural Engineer.
      a. Submit concurrent with Shop Drawings.
   2. Certification: Submit certification by an independent testing company stating that equipment is installed per applicable and referenced codes and standards, adjusted and balanced for design operations, and is complete and ready for intended function.
      a. Certify that fume hoods will not exceed design maximum at specified operating conditions.
      b. Certify that perchloric acid fume hood alarm is compatible for use in a perchloric acid and wash down atmosphere.
   3. Operation and Maintenance data:
      a. Operating and maintenance manuals that describe proper operating procedures.
      b. Maintenance and replacement schedules.
      c. Component parts list.
      d. Wiring diagrams.
      e. Closest factory representative for components and service.
      f. See Section 01 78 23.

1.6 QUALITY ASSURANCE

A. Coordinate work of this Section with Section 12 35 53 Laboratory Casework and Furnishings.
   1. Unless specified otherwise, scope of work described in this Section shall be provided by supplier of Section 12 35 53.

B. Chemical fume hoods shall be the product of a single manufacturer .

C. Manufacturer Qualifications:
   1. Work in this Section, with respect to the specific equipment specified herein, shall be manufactured by a firm having a minimum eight (8) years documented experience, and an established organization and production facilities including all tools, equipment, and special machinery necessary for specializing in the fabrication and installation of the type of

equipment required with skilled personnel, factory trained workmen and an experienced engineering department. Each shall have the demonstrated knowledge, ability, and the proven capability to produce the specified equipment of the required quality and the proven capacity to complete an installation of this size and type within the required time limits. Upon request, manufacturers shall produce evidence of financial stability and bonding capacity required to perform on this project.

2. Manufacturer shall maintain a factory test facility which provides variable exhaust and make-up air control. Test facility shall contain, as permanent equipment, ANSI/ASHRAE 110 testing equipment as specified for performance testing.

D. Installer Qualifications:
   1. Installed by manufacturer or installers approved in writing by fume hood or exhaust device manufacturer for installation of specified products.

E. Provide interface products of style, material, finish, and color in order to produce a homogenous installation.

F. Fume Hoods shall be UL tested and labeled and conform to Class A requirements of ANSI Z9.5 Laboratory Ventilation.
   1. Fume Hood Testing shall be in accordance with ASHRAE 110.
   2. Fume Hood Testing shall be in accordance with the National Institutes of Health Design Requirements Manual.

1.7 PRODUCT HANDLING

A. Protect work before, during and after installation including installed work and materials of other trades. Maintain protective covering until start-up.

B. Deliver laboratory equipment after wet operations in building are complete.

C. Laboratory equipment shall be stored in a ventilated area, protected from weather, with relative humidity of 50 percent or less at 70 deg F.

D. Replace, repair, and restore damaged work to original condition.

E. At no time shall worker use the installed equipment as a work bench, scaffolding, or for other uses.

PART 2 - PRODUCTS

2.1 CHEMICAL FUME HOODS

A. Manufacturers:
   1. Base:
      a. Kewaunee Scientific Corporation.
   2. Optional:
      a. Air Master Systems Corp.
      b. Bedcolab Ltd.
      c. CiF Lab Solutions LP.
      d. Mott Manufacturing Limited.
      e. Jamestown Metal Products, Inc.
      f. Lab Crafters Inc.
   3. Other manufacturers desiring approval comply with Section 01 67 00.
   4. Substitutions are not permitted.

B. Refer to Section 01 81 33 - Cyber Security Requirements for cyber security related product requirements.

C. Fume hoods shall be UL 1805 classified.
   1. Attach label to face of each fume hood indicating classification to the UL 1805 standard for Laboratory Fume Hoods.

D. Materials:
   1. Steel:
      a. Recycled Steel Content: A minimum of 20 percent of the steel used in the fabrication of laboratory cabinets, fume hoods and modular laboratory systems shall consist of the sum of post-consumer recycled content plus one-half of the pre-consumer content, based on the cost of the total value of materials.
      b. ASTM A1008 high quality, cold rolled steel, and free from rust, scale, scratches, buckles, ragged edges, and other defects.
      c. Minimum Thickness: 18 GA.
   2. Stainless Steel:
      a. Recycled Steel Content: A minimum of 20 percent of the stainless steel used in the fabrication of laboratory cabinets, fume hoods and modular laboratory systems shall consist of the sum of post-consumer recycled content plus one-half of the pre-consumer content, based on the cost of the total value of materials.
      b. Type 304, ASTM A240/A240M or ASTM A666, with exposed surfaces ground and polished to a No. 4 finish.
      c. Type 316, ASTM A240/A250M or ASTM A666, with exposed surfaces ground and polished to a No. 4 finish. Provide type 316L when specified.
      d. Minimum Thickness: 16 GA.
      e. Welding: Stainless steel welding material shall be of similar type to sheet material. Welds shall be made without discoloration, ground, polished, and passivated to blend with a No. 4 finish.
   3. Liner and Baffle:
      a. Flame Spread: 25 or less in accordance with ASTM E84 or ANSI/UL 723.
      b. Glass-reinforced polyester panel, flame-retardant and self-extinguishing with smooth finish and white color.
         1) Flexural strength: 14,000 psi.
         2) Liner thickness: 3/16 inches.
         3) Baffle thickness: 1/4 inches, minimum.
      c. Radioisotope Hoods: 16 GA Type 304 stainless steel. Continuously weld liner top to sides and grind welds smooth. Radius vertical corners 1/2 inches.
      d. Perchloric Acid Hoods: 16 GA Type 316L stainless steel. Continuously weld liner top to sides and grind welds smooth. Radius vertical corners 1/2 inches.
      e. Polypropylene and/or Trace Metals Hoods: White polypropylene skin bonded to a 1/4 inches thick rigid, chemical resistant base.
      f. Baffle shall be same material as liner.
   4. Glass: 7/32 inches laminated safety glass. Glass shall not be etched with manufacturer's name, logo, or other permanent markings, other than to identify the glass as safety glass. Light fixture lens may be tempered safety glass. Safety glass shall be in compliance with ANSI Z97.1.
   5. Sash Guides: Extruded PVC.
   6. Sash pull: Full width corrosion-resistant steel with chemical resistant powder coating, stainless steel, or plastic.
   7. Gaskets: White 70 durometer PVC for interior access panels. Gasket interior access panels to eliminate air leakage and to retain liquids inside hood.
   8. Fasteners:
      a. Exterior structural member attachments: Sheet metal screws, zinc plated.
      b. Exposed exterior fastening devices shall be corrosion-resistant, non-metallic material; exposed screws are not acceptable.
      c. Interior fastening devices: Except where specifically allowed by this Specification, interior fastening devices shall be concealed; exposed screws are not acceptable. (Screw head "caps" not acceptable.)
   9. Instruction Plate: Corrosion resistant or plastic plate attached to the fume hood exterior with condensed information addressing the recommended locations for apparatus and accessories, baffle settings, if adjustable, and use of sash.

E. Construction:
   1. Superstructure: Rigid, self-supporting assembly of double wall construction, maximum 4-7/8 inches thick. Wall shall consist of a sheet steel outer shell and a corrosion resistant inner liner, and shall house and conceal steel framing members, piping, wiring, attaching brackets, and remote operating service fixture mechanisms and services. Panels shall be attached to a full frame construction, 14 GA, minimum, galvanized members. Panels and brackets attached to eliminate screw heads and metallic bracketry from hood interior. Front panels shall be factory-punched for service fittings, electrical, control, and monitoring equipment. Provide stainless steel or epoxy- or urethane-coated plug buttons for holes not used.
   2. Access Panel: Access to fixture valves and piping concealed in wall shall be through flush access panels on the inside liner walls, or through removable front posts. Panels shall be secured with fitted gasket, tamperproof, epoxy- or urethane-coated, countersunk, flat head screws, or similar method, providing a tight fit. Hook and loop type attachments and panels held by gravity are not acceptable.
   3. Baffle Adjustment: In standard chemical fume hoods, if adjustable baffles are provided, the adjustment control shall be operable from outside the hood without exposing the user to the hood interior environment and, for ADA compliant hoods, shall be within the reach of a wheelchair bound operator. Baffles, supports, and brackets to be non-metallic. Baffles may be made fixed. Baffles shall be removable for cleaning.
   4. Baffles: Baffles shall be fixed and non-adjustable. Baffles shall be removable for cleaning.
   5. Baffle Adjustment: Baffles shall be remotely adjustable. The adjustment control shall be operable from outside the hood without exposing the user to the hood interior environment and, for ADA compliant hoods, shall be within the reach of a wheelchair bound operator. Baffles shall be removable for cleaning.
   6. Baffles in radioisotope hoods shall be fixed and non-adjustable. Baffles shall be removable for cleaning.
   7. Baffles in Perchloric acid hoods shall be fixed and non-adjustable. Baffles shall be removable for cleaning.
   8. Ceiling Closure Panels: Provide 18 GA steel paneled enclosure from top of hood to the ceiling. Enclosure shall include hinged door to access hood lighting fixture. Finish shall match superstructure exterior. Panels shall terminate 1 inches below finished ceiling, where provided, or at 9 feet above finished floor where there is no finished ceiling.
   9. Vertical Sash Enclosure:
      a. Fume hoods shall be provided with sash enclosures, gasketed or sealed penetrations and connections, as required, which limit air leakage to 1 percent of the design air flow quantities indicated on the Exhaust Schedule. Fume hood manufacturer shall remain responsible for achieving the required capture velocity at the specified operating sash height.
      b. Provide sheet metal enclosure to completely encase vertically operated or combination sashes when sash is in the fully open position.
      c. Enclosure shall prevent air of fume leakage above the fume hood.
      d. Enclosure shall be easily removed and replaceable to allow access to light fixture housing.
   10. Bypass Grill: Low-resistant type 18 GA steel with upward directional louvers. Alternative bypass solutions are acceptable; manufacturer shall notify Contracting Officer Representative of alternative solutions and verify solution is coordinated.
   11. Trim and Side Panels: Provide matching steel trim and side panels, as required, to finish openings around and between hoods. Finish shall match superstructure exterior.
   12. Finished Back: Provide for fume hoods where back of hood is exposed to view. 18 GA steel sheet. Finish shall match superstructure exterior.
   13. Exhaust Collar:
      a. Provide contoured 20 GA exhaust collar and transition piece, if necessary, to receive circular exhaust duct connection by Division 23. Collar and transition piece shall receive urethane powder coating. Collar shall include duct transition if collar diameter

is less than the duct diameter indicated on the HVAC ductwork drawings. Collar shall be 20 GA stainless steel if hood has stainless steel liner.

14. Cup Sink:
    a. Oval with raised rim, material and color to match work surface, sizes in accordance with drawings. Comply with Section 11 53 43 requirements.
    b. Rectangular with raised rim, color to match work surface, size in accordance with drawings. Comply with Section 11 53 43 requirements.
    c. For floor-mounted hoods, wall-mounted oval molded black epoxy resin complete with strainer, outlet, and wall mounting bracket. Comply with Section 11 53 43 requirements.
    d. For stainless steel fume hood work surfaces, oval stainless steel cup sink integral with work surface with raised rim, size in accordance with drawings. Comply with Section 11 53 43 requirements.
    e. Raised Rim Height: 1/4 inches.
    f. Raised Rim Height: 1/2 inches.
15. Piping shall be as specified in Division 22 for respective system.
16. Service Fittings: As shown on Laboratory Furnishings Drawings and specified in Section 11 53 43, factory-installed and complete with all gaskets, grommets, and sleeves. Fittings shall be mounted on the front post(s).
17. Alarm and Controls: Coordinate cut outs for fume hood alarm and controls to be provided under Division 25. All cut outs for alarm and controls shall be made in the factory; field cutting is not acceptable.
18. Electrical Receptacles:
    a. Flush mounting, 120V/20A duplex type, single gang, NEMA 5-20R, 3-wire, grounding type receptacle, one per side, or as indicated on the Laboratory Furnishings Drawings, with brushed stainless steel cover plate.
        1) GFCI type receptacle at fume hoods with cup sinks.
    b. Flush mounting, 208V/20A single gang, NEMA 6-20R, 3-wire, grounding type receptacle, one per side, or as indicated on the Laboratory Furnishings Drawings, with brushed stainless steel cover plate.
    c. Variacs (Variable Voltage Transformer): Where indicated on the Laboratory Electrical or Laboratory Furnishings drawings, provide fume hood with variacs, as follows:
        1) Manufacturers:
            a) Laboratory Craftsmen Inc.
            b) Other manufacturers desiring approval comply with Section 01 67 00 (no known equal).
            c) Substitutions are not permitted.
        2) Base Product: Laboratory Craftsmen “Powermite” Model EB-1F, or equal.
        3) Description:
            a) Variacs receptacle unit shall be factory-installed in the front fume hood post as indicated on the drawings. The fume hood manufacturer shall mount the unit and factory-install branch circuit conduit and wiring through the fume hood to a single point of connection junction box at the top of the fume hood. Variacs shall be wired separately from the other fume hood receptacles, lighting, and other fume hood power requirements for connection to a separate branch circuit.
        4) Features:
            a) Unit shall be solid state, for flush mounting, with integral fuse holder, housed is a cast aluminum enclosure.
            b) On/off/variable voltage dial and two pilot lights.
            c) Integral NEMA 5-15R receptacle.
                (1) GFCI type receptacles at fume hoods with cup sinks.
            d) Stainless steel cover plate with black painted calibrated dial with markings from 5 percent through 90 percent of full voltage, in increments of not exceeding 20 percent between markings.

e) Unit shall have a continuously adjustable power output range of 5 percent to 90 percent, 1000 W capacity at full 120 V, 60 Hz.

d. Color: Receptacles shall be brown with hoods painted dark colors and white for hoods painted white, off white, grey, yellow, or similar colors.

19. Interior Hood Lighting:
    a. Lighting within hood shall be provided by a protected fluorescent lighting fixture with two lamps (32W T8, electronic ballast, rapid start) operated by an exterior switch with stainless steel cover plate located on the face of the fume hood. Lamp size shall not exceed 48 inches; provide multiple fixtures as required.
    b. Provide safety glass panel cemented and vapor-tight sealed to the hood roof.
    c. Light level: Average light level on the work surface shall be 80 footcandles, minimum.
    d. Relamping shall be achieved from outside the hood enclosure.
    e. Light fixture shall be U.L. listed.
    f. Color: Switch shall be brown with hoods painted dark colors and white for hoods painted white, off white, grey, yellow, or similar colors.
20. Safety label: Provide self-adhesive polyester label, as described on the drawings. Labels shall include graphic indicator for safe operating conditions with respect to fume hood sash position. Labels solely indicating face velocity sash position are not acceptable.
21. Hood Finish: Fume hood finish shall comply with SEFA 8M Cabinet Surface Finish performance requirements.
22. Exterior Color: As selected by Contracting Officer Representative from manufacturer's full color line and complying with finish requirements.

F. Bench Mounted Chemical Fume Hoods:
1. Style: General purpose.
    a. Base Product: Kewaunee Scientific Corporation Supreme Air H05 Fume Hood, or equal, satisfying the identified requirements, characteristics, and features as specified herein.
2. Depth: 25-1/2 inches interior, 36 inches exterior (base), nominal.
3. Special Fume Hoods:
    a. High Clearance:
        1) Provide where indicated on Drawings.
        2) Work Area Height: 60 inches.
        3) Sash Opening Height: 36 inches.
    b. Pass-Through Demonstration Hoods:
        1) Provide where indicated on Drawings.
        2) Fume hoods shall be designed for use from opposite (front and rear) sides.
        3) Restricted bypass.
        4) Two full-view vertically rising sashes.
        5) Sash interlock allowing only one sash opened at a time.
        6) Fixed baffle.
4. Design:
    a. Restricted bypass/constant volume (CV) type, with air foil.
    b. Restricted bypass fume hoods for variable air volume or constant volume exhaust systems with airfoil. Bypass shall be sufficient in size to allow 25 percent flow with sash closed. Bypass must be achieved through low resistance opening at top of front lintel panel. Bypass shall be designed to provide a smooth down flow effect.
    c. Bypass/constant volume (CV) type, with air foil. Provide sufficient bypass capacity so that the face velocity with a 6 inches sash opening does not exceed three times the face velocity with the sash fully opened.
    d. Design fume hoods for consistent and safe air flow through the hood face. Negative variations of face velocity shall not exceed 20 percent of the average face velocity at designated measuring point as defined in this section.
        1) Fume hoods shall be designed to operate safely at face velocities of 100 fpm to 125 fpm.

2) Fume hoods shall be designed to operate safely at face velocities between 60 fpm and 100 fpm.

5. Work Surface: 1-1/4 inches dished epoxy resin, in compliance with Section 12 35 53 requirements. Color: Black.
6. Work Surface: Stainless steel with seamless die-formed 1/2 inches marine edge, in compliance with Section 12 35 53 requirements. Reinforce work surface with structural channels to support lead bricks to a load of 200 pounds per square foot.
7. Work Surface: 1-1/4 inches dished solid polyester/acrylic resin, in compliance with Section 12 35 53 requirements. Provide with 1/4 inches wide red safety warning line inside sash opening.
8. Downdraft Bypass: Low resistant type, 18 GA steel chamber; with directional louvers.
9. Sash Cable: Uncoated, 1/8 inches, stainless steel cable; Military Spec ML-W-83420D-3.
10. Pulley assembly for sash cable: 50mm 2 inches diameter, zinc dichromate finish, ball bearing type, with cable retaining device. Nylon tire not acceptable.
11. Airfoil: The airfoil shall allow ample room for electrical hospital grade cords to fit beneath the airfoil. Sill must pivot forward to provide cord and trough access. Bottom horizontal foil shall provide nominal 1 inch bypass when sash is in the closed position. Bottom foil shall not be removable without use of special tools. Airfoil shall be steel with urethane or epoxy powder coating.
    a. Air foil and sill shall be flush with the work surface at the bottom of the sash opening; airfoil sills that are not flush with the top plane of the work surface dish are not acceptable.
    b. Sill shall be rounded or chamfered on front edge with right angle corners radiused or angled.
12. Fume hood sash (Vertical): Full-view, with clear, unobstructed, side-to-side view of fume hood interior and service fixture connections. Sash to have a 35-inch nominal sightline and a 28-1/2-inch nominal vertical access height.
    a. Counterbalance system: Single weight, cable and pulley, counterbalance system to prevent sash tilting and permit ease of operation along full width pull. Maximum 7 pounds pull required to raise or lower sash throughout its full length of operating sash opening. Design system to hold sash without creep and prevent sash drop in the event of cable failure.
    b. Sash shall have the capability to be raised to full 28-1/2-inch nominal vertical opening for loading or unloading of large apparatus.
    c. Sash Stop: To allow manual override with automatic reset for an 18-inch sash opening.
13. Sash Stop: Sash stops shall not be provided with the fume hoods.

G. Fume Hood Liner Test: Polyresin
1. Test No. 1: Spills and Splashes:
    a. Suspend a 42 inches horizontal by 12 inches vertical panel in a position to expose the surface to be tested in a vertical plane. Divide the panel vertically into 3/4 inches spaces.
    b. Using an eyedropper, apply five drops of each reagent as listed.
    c. Liquid reagents shall be applied at the top of the panel and permitted to flow down full panel height. (CAUTION! Flush away any reagent drops.)
2. Test No. 2: Fumes and Gases:
    a. Prepare a panel 24 inches by 12 inches by dividing panel into 2 inches squares. Using 100ml beakers, place 25ml (approximately 1/2 inches of reagent) into each beaker. Place beakers in position so that test panel may be placed over beaker tops in the proper sequence. Place panel over beakers. Note: Beaker pouring lip permits atmospheric oxygen to enter and participate in the reaction of the reagent fumes.
    b. After a 24 hour time period has elapsed, remove panel, flush off with water, clean with naphtha and detergent, rinse and wipe dry. Evaluate.
3. Evaluating Ratings:

0	No effect	No detectable change in the material surface.

1	Excellent	Slight detectable change in color or gloss but no change in function or life of the surface.
2	Good	A clearly discernable change in color or gloss but no significant impairment of surface life or function.
3	Fair	Objectionable change in appearance due to discoloration or etch, possibly resulting in deterioration of function over an extended period of time.
4	Failure	Pitting, cratering, or erosion of the surface. Obvious and significant deterioration.

4. Performance: Test results shall equal or exceed the following:

Reagent	percent by wt.	Spills	Fumes
Acetic acid, glacial		0	0
Acetone		1	0
Acid dichromate		1	0
Ammonium hydroxide	28	0	0
Amyl acetate		1	0
Benzene		1	0
Butyl alcohol		0	0
Carbon tetrachloride		0	0
Chloroform		1	0
Chromic acid, saturated		2	0
Cresol		1	0
Dichloro acetic acid	93	1	1
Dimethyl formamide		1	0
Dioxane		0	0
Ethyl acetate		0	0
Ethyl alcohol		0	0
Ethyl ether		1	0
Formaldehyde	37	0	0
Formic Acid	88	0	0
Furfural		3	0
Gasoline		1	0
Hydrochloric acid	37	0	1
Hydrofluoric acid	48	0	4
Hydrogen peroxide	30	0	0
Methyl alcohol		0	0
Methyl ethyl ketone		1	0
Methylene chloride		0	0
Monochlorobenzene		1	0
Naphthalene		1	0
Nitric acid	20	1	0
Nitric acid	30	1	0
Nitric acid	70	1	0
Phenol	85	0	1
Phosphoric acid	85	0	0
Silver Nitrate	10	1	0
Sodium Hydroxide	10	1	0
Sodium Hydroxide	20	1	0
Sodium Hydroxide	40	1	0
Sodium Hydroxide Flake		0	0
Sodium Sulfide, saturated		2	0
Sulfuric acid	33	0	0
Sulfuric acid	77	1	0

Reagent	percent by wt.	Spills	Fumes
Sulfuric acid	93	2	0
Sulfuric acid/Nitric acid, equal parts	77/70	0	1
Tincture of Iodine		1	1
Trichloroethylene		1	0
Toluene		1	0
Xylene		1	0
Zinc Chloride		0	0

Note: Maximum concentration is to be understood unless a lower concentration is shown in the table.

2.2 FUME EXTRACTOR ARMS (SNORKELS)

A. Manufacturers:
   1. All snorkel fume extractors shall be the product of a single manufacturer.
   2. Base:
      a. Nederman Inc.
   3. Optional:
      a. Airflow Systems, Inc.
      b. Alsident System, distributed by Laboratory Enterprises, Inc.
      c. Enviroflex International Inc.
      d. Movex Inc.
      e. Plymovent Corporation.
   4. Other manufacturers desiring approval comply with Section 01 67 00
   5. Substitutions are not permitted.

B. Base Product: Nederman Original Fume Extractor Arm. 6-1/4 inches hose diameter, hinged, self-supporting air extractor arm assembly, or equivalent.
   1. Arm shall consist of 360 degree swivel elbow, support flange, internal support, pre-set joints with adjustable wear discs, white fiberglass-reinforced PVC hose, hood, and flange for 6 inches exhaust duct. Provide protective grille to prevent objects from being drawn into arm.
   2. Fume extractor shall be a constant volume device with no manual damper.
   3. Mounting Bracket: wall bracket.
   4. Mounting Bracket: 3 feet ceiling bracket. Manufacturer shall coordinate location of duct connection with the work of Division 23. Provide ceiling escutcheon.
   5. Mounting Bracket: 3 feet wall bracket.
   6. Mounting Bracket: 7 feet wall bracket.
   7. Mounting Bracket: 7 feet ceiling bracket. Manufacturer shall coordinate location of duct connection with the work of Division 23. Provide ceiling escutcheon.
   8. Bracket Support: Provide slotted channel framing as required to suspend and support ceiling bracket from structure above.
   9. Arm Length:
      a. Arm Length: 5 feet; mainly vertical joints.
      b. Arm Length: 7 feet; mainly vertical joints.
      c. Arm Length: 10 feet; horizontal and vertical joints.
      d. Arm Length: 14 feet; mainly horizontal joints.
      e. Arm Length: 14 feet; mainly vertical joints.
   10. Silencer.
   11. Hood Type:
      a. White polycarbonate hood.
      b. Extension hood.
      c. Metal hood.
   12. Hood light, including switch and transformer, if required; UL listed.
   13. Fan and fan switch.
   14. Automatic damper.

C. Base Product: Nederman Miniarm FX, self-supporting tubing and inlet accessory, or approved equal, satisfying the identified requirements, characteristics, and features as specified herein.
   1. Provide protective grille to prevent objects from being drawn into arm.
   2. Fume extractor shall be a constant volume device with no manual damper.
   3. Ceiling Mounting:
      a. White powder coated steel ceiling column with side-mounted 4 inches duct flange. Ceiling column shall be 4 inches diameter by 40 inches long.
      b. Ceiling column extension shall be supported from structural deck with slotted channel framing.
      c. Manufacturer shall coordinate the location of duct connection with the work of Division 23.
      d. Provide ceiling escutcheon.
   4. Wall Bracket: White powder coated steel.
   5. Extraction Arm Diameter:
      a. 3 inches diameter.
   6. Extraction Arm Length:
      a. 57 inches long tubing with three mechanical joints.
   7. Materials:
      a. Anodized aluminum pipes with white polypropylene joints.
      b. White polypropylene pipes and joints.
      c. ESD anodized aluminum pipes and conductive polypropylene joints. Base shall be conductive polypropylene. Provide a connecting ground cable.
   8. Inlet Accessory:
      a. Angle Nozzle: Tube with angled end. Tube material shall match the extractor arm tube.
      b. Metallic: White, powder coated aluminum, 10 inches diameter cone hood.
      c. Flange Nozzle: Transparent plastic, 7-1/2 inches by 11-3/4 inches.
      d. Dome Hood: Transparent plastic, 13-3/4 inches diameter by 4-3/4 inches high.
      e. Square Hood: Transparent plastic, 14 inches with 12 inches face height.

D. Refer to Exhaust Schedule for airflow rate.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Prior to installation of the Work of this Section, carefully inspect the installed Work specified in other sections and verify that Work is complete to the point where this installation may properly commence.

B. Verify that Work has been installed in complete accordance with the original design, received submittals, and the manufacturer's recommendations.

C. In the event of discrepancy, immediately notify the Contracting Officer Representative. Do not proceed with installation in areas of discrepancy until discrepancies have been fully resolved.

3.2 INSTALLATION

A. Work in this Section requires close coordination with Work specified in Divisions 22, 23, 25 and 26, as well as installation by Government of Government furnished components. Coordinate Work to ensure an orderly process in the Project, without removal of previously installed Work, and to prevent damage to finishes and products.

B. Coordinate location and alignment of fume hoods and cabinets for proper connection of piping and duct work.

C. Install equipment in accordance with manufacturer’s written instructions, applicable codes and regulations, accepted Shop Drawings, and as necessary for a complete operating system.

D. Install equipment plumb, square, and straight with no distortion and securely anchored, as required.

E. Where cup sinks are indicated, coordinate alignment of fume hood cup sink(s) so that sink is centered below water fitting outlet.

F. Coordinate with Section 12 35 53 for venting corrosives storage cabinets behind rear baffle of fume hood.

3.3 FACTORY TESTING

A. Manufacturer shall shop calibrate and field verify instruments.

B. Functional tests shall be performed at the factory for supplied equipment through the PLC by Manufacturer.

C. Manufacturer is responsible for correcting deficiencies identified during shop testing procedures.

D. Test results shall be made available to the Government upon request.

E. Factory Acceptance Test (FAT)
   1. Equipment requiring FAT:
      a. Polypropylene fume hoods.
      b. Custom fume hoods as indicated.
   2. Fume hoods shall be provided with service identical to those that will be provided on site, including piped services connections, flows and temperature, as applicable, and electrical service.
   3. The manufacturer shall:
      a. Issue Factory Acceptance Test protocols and notification to the Government at least 30 days in advance of scheduled testing for Government comment and approval.
      b. Execute tests and measurements identified in Part 1 of these specifications.
         1) Fume hoods shall be tested for ‘as manufactured’ and ‘as installed’ conditions.
      c. Analyze test results.
      d. Describe deviations in test report and make necessary corrections. Retest/repeat procedures, if required.
      e. Check that registers deviations are complete, and documentation is correct.
      f. Verify calibration of test equipment.
      g. Check the conclusion of the test conformity.
      h. Sign test report after completion and submit it to Government.
   4. Government's authorized representative(s) shall have the right to inspect work during and after fabrication completion and shall perform and witness testing.
      a. This inspection shall occur at the vendor’s manufacturing facility.
      b. Facility shall have utilities as required to perform full testing of the equipment.
      c. The representative’s decisions to reject material or workmanship for non-compliance with the Specification, Drawings and purchase order shall be final.
   5. Government's representative(s) inspection shall include, but not be limited to the following items:
      a. Manufacturer’s adherence to the terms of Purchase Order, particularly in the areas of schedule and required documents.
      b. Manufacturer’s adherence to the Contract Documents, including revisions.
      c. Manufacturer’s adherence to fabrication procedures as mandated by Code, Specifications and Drawings.
      d. Fabrication within tolerances as established by Code, Specifications, and Drawings, whichever is most stringent.
      e. Full operation of the equipment including proper functioning of the following items:
         1) Controls.
         2) Sensors.
         3) Alarms.
         4) Exhaust flow.
         5) Service fittings.
         6) Electrical components.
         7) Sash operation.

8) Safety features.
9) All other operational components of the equipment.

f. Overall workmanship of equipment and accessories.

6. FATs shall include the following:
   a. Verification of equipment and scope of delivery.
   b. Verification of the pipe work.
   c. Verification of mechanical installation.
   d. Verification of electrical installation.
   e. Verification of utilities.
   f. Verification of documentation.
   g. Test protocol run analysis.
7. Manufacturer shall prepare and maintain a Deviation Tracking List identifying deviations or faults that occur during the FAT.
8. Acceptance by the Government's representative shall not relieve the manufacturer of the responsibility for workmanship conforming to the Contract Documents and the Purchase Order.

3.4 FIELD TESTING: CHEMICAL FUME HOODS

A. Provide two-week advance notice of scheduled testing.

B. Fume hood field tests shall be performed by a qualified independent testing company on each hood to determine face velocity and air flow patterns.

C. Balance, test and certify each fume hood in accordance with ASHRAE 110 (AI) for Flow Visualization, Face Velocity, and Tracer Gas Containment Testing Requirements.
   1. Fume hoods shall achieve an “As Installed” (AI) performance rating equal or better than 0.10 ppm with 4.0 Lpm tracer gas release rate when tested in accordance with ASHRAE 110.

D. Balance, test and certify each fume hood in accordance with the requirements and specifications for the National Institutes of Health on Site Testing of Fume Hoods, as identified in the NIH Design Requirements Manual.

E. Balancing of the system is in the scope of work of Division 23.

F. Verify exhaust air quantity does not exceed design, plus allowable leakage.

G. Verify hood pressure drop does not exceed design.

H. Adjust and retest hoods that do not meet specified performance.

I. Replace hoods which do not meet standards after repetitive testing.

3.5 CLOSEOUT ACTIVITIES

A. Refer to Section 01 81 33 - Cyber-Security Requirements for cyber security related closeout requirements.

3.6 ADJUSTING, CLEANING, AND PROTECTION

A. Repair or remove and replace defective work as approved by the Contracting Officer Representative upon completion of installation.

B. Adjust all moving or operating parts to function within their design parameters.

C. Clean equipment, touch up as required.

D. Protect all units before, during, and after installation. Damaged materials due to improper protection shall be cause for rejection.

3.7 DEMONSTRATION

A. Recorded video showing recommended operation and maintenance of each type of hood specified.

END OF SECTION

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SECTION 11 53 43
LABORATORY SERVICE FITTINGS AND FIXTURES

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Laboratory Service Fittings and Fixtures, as indicated, in accordance with provisions of Contract Documents.

B. Section includes:
   1. Laboratory service fittings, valves, electrical pedestal boxes, and related components.
   2. Laboratory emergency plumbing fixtures.
   3. Emergency shower tempering system.
   4. Laboratory sinks.
   5. Recessed Valve Box.

C. Completely coordinate with work of other trades.

1.2 DESCRIPTION

A. Work includes but is not necessarily limited to furnishing to the project site for installation by Division 22, laboratory fittings, emergency plumbing fixtures, and fixtures described herein and shown on the Laboratory Furnishings Drawings.
   1. Electrical work associated with emergency equipment and benchtop electrical pedestals shall be installed by Division 26.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Submit complete shop fabrication and installation drawings, including plans, elevations, sections, dimensions, details, and schedules.
   2. Show relationship to adjoining materials and construction.

B. Product Data:
   1. Submit complete materials list, including catalogue data of materials, equipment, and products for Work in this Section.

C. Samples:
   1. Submit 2 samples of each type of specified finish and color.

D. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. Operating and maintenance manuals describing operating and maintenance procedures.
      b. Maintenance and replacement schedules.
      c. Component parts list.
      d. Nearest local factory representative for components and repairs.
      e. See Section 01 78 23.

1.4 QUALITY ASSURANCE

A. Unless specified otherwise, because of special coordination requirements, the scope of work described in this Section shall be provided by the supplier of the Section 12 35 53 scope of work.

B. Work in this Section requires close coordination with Work in Section 12 35 53, Division 22 Plumbing, Division 23 HVAC and Division 26 Electrical.
   1. Coordinate Work to assure an orderly progress in the Project, without removal of previously installed Work, and so as to prevent damage to finishes and products.

C. Review conditions of installation, procedures, and coordination with related Work.

D. Carefully inspect the installed Work specified in other Sections and verify that Work is complete and ready for the installation of this Work to properly commence.

E. Install Work in accordance with original design, reviewed submittals, and manufacturer's recommendations.

F. Manufacturer and Installer Qualifications:
   1. 5 years of documented experience specializing in manufacture of type of equipment specified, with demonstrated ability to produce the specified equipment of required quality and quantity for complete installation in a project of this type and size within required time limits.
   2. Upon request, manufacturers shall produce evidence of financial stability and bonding capacity required to perform on this project.

G. Conform to the recommended practices for laboratory service fittings and fixtures published by the Scientific Equipment and Furniture Association (SEFA) 7: Fixtures, current version.

H. Emergency plumbing fixtures shall comply with requirements of ANSI Standard Z358.1: American National Standard for Emergency Eyewash and Shower Equipment, current edition.

I. Where identified, service fittings and sinks shall be accessible in compliance with the requirements of the federal Americans with Disabilities Act (ADA), ADA Accessibility Guidelines (ADAAG), and state accessibility regulations.

J. Emergency plumbing fixtures shall be accessible in compliance with requirements of federal Americans with Disabilities Act (ADA), ADA Accessibility Guidelines (ADAAG), and state accessibility regulations.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver fittings and fixtures to job site in recommended packaging, with each fitting individually packaged, marked, and scheduled for point of use.

B. Inventory fittings, at job site, verify that type and quantity are correct, and re-package until installed.

PART 2 - PRODUCTS

2.1 GENERAL

A. Service fittings and emergency plumbing fixtures shall be designed for laboratory use.

B. Service fittings and emergency plumbing fixtures shall be factory pre-assembled including the assembly of valves to turrets, mounting shanks to turrets, etc., and individually factory tested.

C. Laboratory service fittings shall be the product of one service fitting manufacturer to assure ease of replacement and maintenance.

D. Emergency plumbing fixtures shall be the product of one manufacturer to assure ease of replacement and maintenance.

E. Service valves, fittings, and accessories shall be of cast brass with a minimum copper content of 85 percent, except for items which are to be brass forging or bar stock or are specifically specified to be of another material.

F. Provide fittings at locations indicated on the Laboratory Furnishings Drawings. Refer to Service Fitting Schedule.

G. Assembly components and operating parts such as valve stems, renewable units, packing nuts, outlet nozzles and straight serrated hose ends shall be made from solid brass stock.

H. Replaceable seats, needle cones, valve disc screws and other accessories shall be Monel or stainless steel alloys especially selected for use intended.

I. Fittings shall be factory tested and shall be supplied with nipples, lock nuts, shanks, etc.

J. Serrated tip fittings shall have 3/8 inches IPS thread with the hose end being tapered. Diameter of orifice in serrated tip shall be 1/8 inches, except where otherwise specified.

K. Turrets shall be brass drop forging of design indicated in detail shown elsewhere in the Section and shall be one or two-way, as required, with 3/8 inches IPS female inlet thread for connections.
   1. Units shall be furnished with brass shanks, brass locknuts, and washers.

L. Fittings located on the same plane shall have their handles project the same distance from the plane of reference to present a uniform related appearance, regardless of valve type construction.

M. Flanges shall be brass forging of approved design with 3/8 inches IPS female inlet and outlet.

N. Goosenecks shall provide full thread for attachment of anti-splash outlet fittings, serrated tips, and filter pumps.

O. Hot water/cold water gooseneck mixers and wall-mounted cold water goosenecks shall swivel.
   1. Swivel point shall be at turret or at valve level if wall mounted.
   2. Swing joints shall have heavy Teflon type packings.
   3. O-rings will not be permitted.
   4. Cold water goosenecks at cup sinks shall be rigid.

P. Fittings shall have plastic colored service index buttons.

Q. Provide vacuum breakers for hot and cold water fittings including water fittings at fume hoods.

R. Provide plug and socket (2-piece) quick connect service fittings for chilled water supply and return fittings, compressed air fittings for pressures greater than 30 psi, and other services as indicated in this Section and on the Laboratory Equipment Drawings.
   1. Manufacturers:
      a. Base:
         1) Swagelok Company.
      b. Optional:
         1) Hansen Manufacturing (Air-Oil Systems, Inc.),
         2) Tomco Products, Inc.
      c. Other manufacturers desiring approval comply with Section 01 67 00.

S. Fittings and fixtures designated to be accessible (ADA) with operable parts shall be operable with one hand and shall not require tight grasping, pinching, or twisting of the wrist.
   1. Force required to activate operable parts shall be 5 pounds, maximum.

2.2 LABORATORY SERVICE FITTINGS

A. Laboratory service fittings specified in this section shall be provided by a single manufacturer.

B. Manufacturers:
   1. Base:
      a. Water Saver Faucet Co.
   2. Optional:
      a. T&S Brass and Bronze Works, Inc.
      b. The Chicago Faucet Company.
   3. Other manufacturers desiring approval comply with Section 01 67 00.

C. Body Profile:
   1. Service fittings shall have cylindrical body profiles.
      a. Mushroom profile not allowed.
      b. Turrets shall be provided without deck flanges.

D. Handles:
   1. Locations designated to be accessible (ADA).
      a. Water faucets: wrist-blade handles with screw on index identification discs.

b. Laboratory gases, air, and vacuum valves: ball valves fitted with lever-type handles and screw on index identification discs.

2. Other fittings shall be fitted with four arm handles.

E. Finish: Polished chrome, with clear, acid-resistant (epoxy) coating; refer to Finishes section below for requirements.

F. Water Valves:

1. General:
   a. Water fixtures shall be fully assembled and factory tested at 80 psi water pressure.
   b. Water valves shall include a renewable unit containing working parts which are subject to wear, including stainless steel or monel seat, monel screw and heavy duty seat disk and Teflon packing, and an integral adjustable volume control.
   c. Unit shall be convertible from compression to self-closing, and vice versa, without disturbing faucet body proper and shall also be convertible from water construction to needle valve or steam valve construction having outside packing gland without disturbing faucet body.
   d. Unit shall be sealed in valve body with special composition gasket.
      1) Metal-to-metal or ground joint type of sealing is not acceptable.
   e. Exposed metal shall be finished as specified elsewhere in this Section.
   f. Water service fixtures shall satisfy the requirements of ANSI/ASME A 112.18.1M and be certified by the Canadian Standards Association (CSA) under Standard CAN/CSA B.125.M89.
2. HCW4: Bench-mounted hot and cold water faucet.
   a. Base Product: Water Saver LA422-8VB-BH.
   b. Wrist blade handles .
   c. Renewable water valves and valve body.
   d. Swing gooseneck with vacuum breaker and 8 inches spread.
   e. Removable aerator.
      1) Aerator should terminate approximately 3 inches above deck.
   f. Threaded mounting with locknut, washer, and coupling nuts.
   g. Adjustable volume control.
3. CW10: 1/2 inches ball valve for equipment cooling water.
   a. Base Product: Water Saver L4303.
   b. Lever handle with colored plastic index button.
   c. 1/2 inches IPS female inlet.
   d. Quick connect fitting with 3/8 inches NPT male plug.
   e. Provide mounting flange, all applications:
      1) At wall-mounted applications: Flange shall be threaded in a manner to be installed tight to wall surface and allow fitting to be installed plumb.
   f. Length of valve and quick connect fitting shall be 4-7/8 inches, nominal.
4. FHW1: Fume hood-mounted remote control laboratory water valve for water service.
   a. Base Product: Water Saver L739xW-L051WSA-BO358B or L740xW-L051WSA-BO358B.
   b. Panel-mounted valve.
   c. Four-arm handle with colored plastic index button.
   d. Locking ring.
   e. Forged brass valve body located behind panel at front of hood.
   f. Epoxy coated, panel-mounted turret base with serrated hose end.
   g. Adjustable volume control fitting with hex wrench knob to be attached between turret base and serrated hose end.
   h. Mounting shank.
   i. End of serrated hose end shall be 2-3/4 inches, nominal, from turret centerline.
      1) Centerline of serrated hose end shall be 2-1/16 inches, nominal, from panel.
      2) Handle shall be 2-3/4 inches long from panel to index button.

G. High Purity Water Valves:

1. General:
    a. Suitable for purified water.
    b. Components in contact with purified water shall be polypropylene.
    c. Valve stem and bonnet shall be brass.
    d. Elastomeric diaphragm.
    e. Exposed metal shall be finished as specified elsewhere in this Section.
    f. Water fixtures shall be fully assembled and factory tested.
2. PW2: Panel-mounted purified water fixture.
    a. Base Product: Water Saver L7840SC.
    b. Forged brass valve body and 8 inches spread riser with polypropylene interior and lining.
    c. Self-closing lever.
    d. Serrated, polypropylene hose end.
    e. Provide mounting flange, all applications:
        1) At wall-mounted applications: Flange shall be threaded in a manner to be installed tight to wall surface and allow fitting to be installed plumb.
    f. Mounting shank.
    g. End of serrated hose end shall be 5-1/2 inches, nominal, above centerline of panel mounting.

H. Needle Valves:
1. General:
    a. Fully assembled and factory tested at 225 psi air pressure.
    b. Gas, air, vacuum, and steam needle valve fittings shall have stainless steel replaceable floating cone that is precision ground and self-centering which shall seat against a stainless steel or monel renewable valve seat.
    c. Valve shall be for standard control under pressure up to 150 psi and shall have easily replaceable, subject-to-wear parts.
    d. Provide pressure regulators designed for use with the appropriate service at locations indicated on the Laboratory Furnishing drawings.
    e. Exposed metal shall be finished as specified elsewhere in this Section.
    f. Needle valves for natural laboratory gas service shall be certified for use with natural gas by the Canadian Standards Association under ANSI Z21.15-1997/CGA9.1-M97.
    g. Needle valves in fume hoods shall be mounted on the front panel of the fume hood, with components subject to wear accessible from the exterior face of hood.
2. G1: Bench-mounted, single straight pattern, needle valve fitting.
    a. Base Product: Water Saver L2880-131WSA.
    b. Four-arm handle with colored plastic index button.
    c. Turret base.
    d. Serrated hose end.
        1) Outlets shall be 5-1/8 inches, nominal, from turret centerline.
        2) Centerline of serrated hose ends shall be 2-1/16 inches, nominal, above bench top.
    e. Mounting shank.
3. FHG1: Fume hood-mounted, remote control, laboratory needle valve for gas service.
    a. Base Product: Water Saver L739xN-L022WSA or L740N-022WSA.
    b. Panel-mounted valve.
    c. Four-arm handle with colored plastic index button.
        1) Handle shall be 2-3/4 inches long from panel to index button.
    d. Locking ring.
    e. Valve body to be located behind panel at front of fume hood.
    f. Panel-mounted, color epoxy coated brass flange with angled serrated hose end.
        1) Outlet shall be 3-3/8 inches, nominal, from panel.
        2) Mounting shank for outlet.

I. Service Fitting Color Index: for colored plastic index buttons:

Service Name	Disc Color	Letters	Letter Color
Lab Air	Orange	AIR	Black
Compressed Air	Orange	AIR60,90,100	White
Vacuum	Yellow	VAC	Black
Industrial (Laboratory) Cold Water	Dark Green	ICW	White
Cold Water (Potable)	Dark Green	CW	White
Hot Water (Potable)	Red	HW	White
High Purity Water	White	DI	Black

2.3 LABORATORY SAFETY EQUIPMENT

A. Manufacturers:
   1. Laboratory safety fixtures specified in this section shall be provided by a single manufacturer.
   2. Base:
      a. Water Saver Faucet Co.
   3. Optional:
      a. Haws Corporation.
      b. Guardian Equipment.
   4. Other manufacturers desiring approval comply with Section 01 67 00.

B. Laboratory safety equipment shall be certified in accordance with ANSI Z358.1, current edition.

C. General: Recessed cabinet in concrete masonry unit walls shall be fabricated so that piping does not interfere with the block face shell (outer wall).

D. SSEW4: Barrier-free safety station with swing-down eye/face wash, drain pan and emergency shower actuation valve in stainless steel cabinet for recessed mounting.
   1. Basis for design: Water Saver Model No. SSBF2152.
   2. Ceiling-mounted exposed showerhead.
      a. Nipple length shall be as required for a complete installation; verify finished ceiling height.
      b. Showerhead shall have perforated stainless steel spreader.
      c. Exposed piping, showerhead, nipple, and escutcheon shall be stainless steel.
   3. Eyewash heads shall be ABS plastic with flip-top dust covers.
   4. Eyewash components and safety shower actuating arm shall be mounted in a flanged, recessed-mounted 18 GA stainless steel cabinet with No. 4 finish.
      a. A stainless steel drain pan shall be integral with eyewash components and shall direct eyewash water to daylight drain outlet in face of recessed mounting cabinet.
      b. Safety shower actuating arm shall be stainless steel.
      c. Eyewash flow shall be activated by swing-down actuation valve connected to eyewash piping.
      d. Stay-open brass ball valves concealed behind stainless steel/access panel housing.
   5. Fixture shall be furnished with green plastic sign with graphic symbol for safety shower/eyewash.

E. DH1: Dual-purpose eye wash/drench hose, deck mounted.
   1. Basis for design: Water Saver Model No. EW1022-BP.
   2. Dual-purpose eye wash/drench hose unit with dual gentle spray outlet heads and squeeze handle/valve with locking clip.
   3. Heads shall be equipped with flip top dust cover that automatically releases with water pressure.
   4. Flexible, reinforced PVC hose shall be furnished with swivel fitting at inlet end.
   5. Furnish with in-line backflow preventer at the hose inlet.
   6. Mounting flange and squeeze valve shall be chrome-plated brass with clear epoxy coating.
   7. Mounting shank.

2.4 FINISHES

A. Colored Epoxy Finish:
   1. Applicable to:
      a. Fume hood service fittings.
   2. Preparation:
      a. Surfaces to be coated shall be polished to a uniform fine-grained surface and cleaned with a phosphoric acid solution to thoroughly remove oil, grease, and other foreign substances.
   3. Epoxy Finish:
      a. Following cleaning, coating material shall be electrostatically applied to exposed surfaces.
      b. After application, coating shall be fully baked to permit curing.
      c. Surfaces shall have a minimum finished coating thickness of 3 mils.
         1) Coating material shall be free-flowing epoxy powder with particle size of 35 to 70 microns.
   4. Color:
      a. White.
      b. Fittings inside fume hoods shall have a colored finish color-coded to match the fitting service index color.
      c. Surfaces of emergency fixtures that are not PVC or stainless steel shall be Green.
      d. Surfaces of emergency fixtures that are not PVC or stainless steel shall be Red.

B. Polished Chrome Finish with Clear Epoxy Coating:
   1. Applicable to:
      a. Laboratory service fittings except fittings inside fume hoods.
   2. Polished Chrome Finish:
      a. Exposed brass surfaces shall be polished and buffed, then electroplated with one layer of nickel and one layer of chrome.
      b. Each layer of plating shall completely cover visible areas.
   3. Clear epoxy coating:
      a. Following chrome plating, exposed surfaces shall be cleaned and degreased.
      b. Clear epoxy coating shall be applied to exposed surfaces and then baked to permit curing.
      c. Surfaces shall have a minimum coating thickness of 3 mils.

C. Performance requirements for coated finishes:
   1. Chemical resistance:
      a. Fume Test:
         1) Suspend coated samples in a container of at least 6 cubic feet capacity, approximately 12 inches above open beakers, each containing 100ml of 70 percent nitric acid, 94 percent sulfuric acid and 35 percent hydrochloric acid, respectively.
         2) After exposure to these fumes for 150 hours, the finish on the samples shall show no discoloration, disintegration, or other effects.
      b. Direct Application Test:
         1) Subject coated samples to the direct action of the following reagents and solvents at a temperature of 25 deg C dropping from a burette at the rate of 60 drops per minute for ten minutes.
         2) Finish on samples shall not rupture, though slight discoloration or temporary softening is permissible.

Reagent	Concentration
Acetic Acid	98 percent
Acetone	
Ammonium Hydroxide	28 percent
Amyl Acetate	

Reagent	Concentration
Amyl Alcohol	
Benzene	
Butyl Alcohol	
Calcium Hypochlorite	
Carbon Disulfide	
Carbon Tetrachloride	
Chloroform	
Chromic Trioxide Acid	
Cresol	
Crude Oil	
Dioxane	
Distilled Water	
Ether	
Ethyl Acetate	
Ethyl Alcohol	
Ethyl Ether	
Formaldehyde	37 percent
Formic Acid	90 percent
Gasoline	
Glacial Acetic Acid	99.5 percent
Glycerine	
Hydrochloric Acid	38 percent
Hydrofluoric Acid	48 percent
Hydrogen Peroxide	5 percent
Isopropyl Alcohol	
Lactic Acid	10 percent
Kerosene	
Methanol	
Methyl Alcohol	
Methyl Ethyl Ketone	
Methylene Chloride	
Mineral Oil	
Monochlor Benzene	
N-Hexane	
Naphthalene	
Nitric Acid	70 percent
Perchloric Acid	70 percent
Phenol	
Phosphoric Acid	75 percent
Sea Water	
Silver Nitrate	30 percent
Sodium Bichromate	saturated
Sodium Carbonate	10 percent
Sodium Chloride	20 percent
Sodium Hydroxide	50 percent
Sodium Hypochlorite	
Sodium Sulfide	
Sulfuric Acid	87 percent
Toluene	
Trichlorethylene	
Turpentine	
Urea	saturated
Xylene	
Zinc Chloride	saturated

2. Mar and abrasion resistance:

a. Coating material shall have a pencil hardness of 2H – 4H with adhesion substantial enough to withstand both direct and reverse impacts of 160 inch-LBS.
b. Coating shall have excellent mar resistance and be capable of withstanding scuffing, marring and other ordinary wear.

3. Reparability:
a. Scratches and other localized surface damage shall be field-repairable.

2.5 EMERGENCY SHOWER TEMPERING SYSTEM

A. Manufacturers:
1. Base:
a. Guardian Equipment.
2. Optional:
a. Haws Corporation.
b. Encon Safety Products.
3. Other manufacturers desiring approval comply with Section 01 67 00.

B. Thermostatic mixing valve and recessed stainless steel cabinet with door and lock for emergency shower and safety station.
1. Basis for design: Guardian Equipment G3807.
2. Connections for hot and cold water inlets.
3. Delivered water temperature range 60 deg F to 90 deg F.
a. Initial setting to be 85 deg F unless otherwise instructed by Contracting Officer Representative.

2.6 LABORATORY SINKS

A. General:
1. Inside dimensions of sink bowl are provided on drawings.
2. Provide tailpiece compatible with waste piping system for sinks unless otherwise specified.
a. Refer to Division 22 for piping requirements.

B. Epoxy Resin:
1. Acceptable Manufacturer: Manufacturer shall be the manufacturer of the epoxy resin work surfaces specified in Section 12 35 53.
2. Laboratory Sinks:
a. Drop-in Type:
1) Drop-in installation by Division 11 in epoxy resin work surface.
2) Color to match work surface.
b. Comply with the requirements of Section 12 35 53 for epoxy resin.
c. Exposed edges shall be radiused not less than 1/4 inches.
d. Drain grooves in top, when indicated on drawings.
1) Sink shall be set 1/8 inches below the lowest drain groove level.
e. Tops without drain grooves: Sink shall be set 1/8 inches below the level of the adjacent surface.
f. Provide epoxy resin sink outlet with strainer, stopper, and open-end overflow, and install with continuous bead of silicone sealant.
3. Cup Sinks:
a. Fume Hood Locations:
1) Provide cup sinks at fume hoods as described in Section 11 53 13.
b. Laboratory Work Surface Installations:
1) Raised rim, color to match work surface, sizes as indicated on drawings, with integral threaded tailpiece.
2) Flush with work surface, color to match work surface, sizes as indicated on drawings, with integral outlet and threaded tailpiece.
c. Comply with the requirements of Section 12 35 53 for epoxy resin.
d. Provide strainer for cup sinks.
e. Provide mounting bracket for wall-mounted cup sinks.

2.7 RECESSED VALVE BOX

A. Stainless Steel Valve Box Hot and Cold Water:
   1. Manufacturers:
      a. Base:
         1) Bradley.
      b. Other manufacturers desiring approval comply with Section 01 67 00.
   2. Base Product: Bradley Model 7906VB recessed hose box with hot and cold water valves.
   3. Description:
      a. Recessed wall box.
      b. Fabricate of 18 GA Type 304 stainless steel.
         1) Exposed surfaces shall be polished to a No. 4 finish.
         2) Seams shall be heliarc welded and ground smooth.
      c. Approximate overall dimensions: 10-1/4 inches wide by 12-1/4 inches tall by 4-1/2 inches deep.
      d. Stops and valve bodies shall be cast brass.
      e. Equip with two 1/2 inches threaded inlet connections and 3/4 inches male hose thread outlets.
      f. Provide vacuum breakers.
      g. Provide cutout for 2 inches drain connection.

PART 3 - EXECUTION

3.1 SITE CONDITIONS

A. Inspection:
   1. Prior to installation of fixtures, carefully inspect the installed Work specified in other Sections and verify that Work is complete to the point where this installation may properly commence.
   2. Verify that Work has been installed in complete accordance with the original design, approved submittals, and the manufacturer's recommendations.

B. Discrepancy:
   1. In the event of discrepancy, immediately notify the Contracting Officer Representative.

3.2 INSTALLATION

A. General:
   1. Rough-in fixture piping connections in accordance with minimum sizes indicated in fixture rough-in schedule for particular fixtures.
   2. Install fixtures plumb and level.
   3. Provide piping connections to fixtures with valves and escutcheons as specified in Division 22.
   4. Verify that fixtures and trim are tight, leak-free and function properly.

B. Set internal volume control on cup sink water fittings so water does not splash out of sink.

C. Set sinks in chemical resistant sealing compound, secure and support, as recommended by the manufacturer.

D. Safety Stations, Emergency Showers, and Emergency Eyewash:
   1. Emergency showers must be installed with a minimum of 16 inches of clearance from obstruction to the center of the showerhead.
      a. Eyewash or eye/face wash fixture is not considered an obstruction.
   2. Install emergency showers with showerhead positioned not lower than 82 inches or above 96 inches measured from the operational standing surface.
      a. Refer to drawings for mounting height within that range.
      b. If no range is provided, shower head shall be installed 96 inches above operational standing surface.

3. Recessed wall cabinets in fire rated partitions shall be installed in accordance with UL approved design.

END OF SECTION

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DIVISION 12

FURNISHINGS

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SECTION 12 24 13
ROLLER WINDOW SHADES

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Roller Shades, as indicated, in accordance with provisions of Contract Documents.

B. Completely coordinate with work of other trades.

C. Related Sections:
   1. Section 01 33 23 – Submittal Procedures.
   2. Section 09 06 00 – Color Schedule.

1.2 REFERENCES

A. ASTM International (ASTM):
   1. ASTM G21, Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi
      a. No growth for fungi ATCC9642, ATCC 9644, ATCC9645.

B. Window Covering Manufacturer's Association (WCMA):
   1. ANSI/WCMA A100.1, American National Standard for Safety of Window Covering Products.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Plans, elevations, sections, product details, installation details, operational clearances, wiring diagrams and relationship to adjacent work.

B. Product Data:
   1. Manufacturer's data sheets on each product to be used, including:
      a. Preparation instructions and recommendations.
      b. Styles, material descriptions, dimensions of individual components, profiles, features, finishes and operating instructions.
      c. Storage and handling requirements and recommendations.
      d. Mounting details and installation methods.
      e. Window treatment schedule:
         1) Use same room designations as indicated on Drawings and include opening sizes and key to typical mounting details.

C. Samples:
   1. Refer to "Specification Section 01 33 23 – Submittal Procedures, Section 2.4 Color Submittals" for submission procedures and requirements of Color Samples.
   2. Refer to "Specification Section 09 06 00 – Color Schedule".
   3. Refer to "Room Finish Schedule and Interior Finish Legend found on Architectural Finish Schedules" drawing for color selection.
   4. Provide samples of Shadecloth and aluminum finish.

D. Contract Closeout Information:
   1. Warranty.
   2. Maintenance Data:

1.4 QUALITY ASSURANCE

A. Manufacturer Qualifications:
   1. Obtain roller shades through one source from a single manufacturer.

2. Minimum 20 years of experience in manufacturing products comparable to those specified in this section.

B. Installer Qualifications:
   1. Installer trained and certified by manufacturer with a minimum of 10 years of experience installing products comparable to those specified in this section.

C. Mock-up:
   1. Provide a mock-up, for manual shades only, of one roller shade assembly for evaluation of mounting, appearance and accessories.
   2. Locate mock-up at opening designated by Contracting Officer Representative.
   3. Do not proceed with remaining work until, mock-up is accepted by Contracting Officer Representative.

1.5 WARRANTY

A. Roller Shade Hardware: Manufacturer's standard non-depreciating 25-year limited warranty.

B. Shadecloth: Manufacturer's standard non-depreciating 25-year limited warranty.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Basis-of-Design Product Manufacturer: Roller Shades:

B. Acceptable Manufacturers: Contracting Officer will consider Contractor’s request for comparable products when the following conditions are satisfied. If the following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed and other manufacturers not listed, but capable of meeting this Specification Section, are expected to provide complete information showing full compliance with specified products.
   2. Use of Manufacturers not listed may require additional design changes which shall be provided by the Contractor at no additional cost to the Government.
   3. Compressible Sealant:
      a. Basis of Design: Mecho/5 - MechoShade Systems.
      b. Or Equal.
   4. Submit request for substitution in accordance with Specification Section 01 33 23.

2.2 MATERIALS

A. Transparent Single-Fabric Shadecloth (Shade Type 1):
   1. Vinyl/polyester yarn complying with ASTM G21 and ANSI/WCMA A100.1.
   2. Fabric thickness: 0.025-inches.
   3. Basket Weave: EcoVeil Screen 1350 series, 5 percent open.
   4. Color:
      a. Refer to “Room Finish Schedule and Interior Finish Legend found on Architectural Drawings”.
      b. Refer to “Specification Section 09 06 00 – Color Schedule”.

2.3 COMPONENTS

A. Manual Operated Chain Drive Hardware and Brackets (Shade Type 1):
   1. Adjustable universal, regular and offset drive capacity.
   2. Removable fascia with concealed fasteners.
   3. Provide shade hardware system that allows for removable regular and/or reverse roll fascia’s to be mounted continuously across two or more shade bands without requiring exposed fasteners of any kind.
   4. Provide multiple shade operation by a single chain operator.

5. Minimum 1/8-inches thick plated steel.
6. Drive Bracket and Brake Assembly:
7. Model M5:
8. SnapLoc fascia, room darkening side / sill channels, center supports and connectors for multi-banded shades.
9. M5 drive sprocket and brake assembly shall rotate and be supported on a welded 3/8-inches steel pin.
    a. Brake pull force: 50 pounds in stopped position.
    b. Permanently lubricated assembly.
    c. Mount on the steel support bracket, independent of shade tube assembly.

B. Drive Chain:
1. No.10 stainless steel chain
2. Rating: 90 pounds, minimum.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Do not begin installation until substrates have been properly prepared.

B. If substrate preparation is the responsibility of another installer, notify Contracting Officer Representative of unsatisfactory preparation before proceeding.

3.2 PREPARATION

A. Clean surfaces thoroughly prior to installation.

B. Prepare surfaces using the methods recommended by manufacturer for achieving the best result for the substrate under the project conditions.

3.3 INSTALLATION

A. Install roller shades level, plumb, square, and true according to manufacturer's written instructions, and located so shade band is not closer than 2-inches to interior face of glass.

B. Allow clearances for window operation hardware.

C. Adjust and balance roller shades to operate smoothly, easily, safely, and free from binding or malfunction throughout entire operational range.

D. Clean roller shade surfaces after installation, according to manufacturer's written instructions.

E. Train Government's maintenance personnel to adjust, operate and maintain roller shade systems.

3.4 PROTECTION

A. Protect installed products until completion of project.

B. Touch-up, repair or replace damaged products before Substantial Completion.

3.5 SCHEDULE

A. Shade Type 1:
1. Manual operating, chain drive.
2. Sunscreen roller shades.
3. Provide at exterior windows identified as “W1” in rooms and spaces shown on Drawings.
4. Provide a single sunscreen roller shade at each individual punched window and a single sunscreen roller shade at pairs of windows.

END OF SECTION

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SECTION 12 35 53
LABORATORY CASEWORK AND OTHER FURNISHINGS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Metal Laboratory Casework, Tables, and Casework Systems:
      a. Metal Laboratory Casework.
      b. Adaptable Laboratory Furniture Systems.
   2. Stainless Steel Laboratory Casework and Tables.
   3. Cabinet Hardware.
   4. Special Purpose Cabinets:
      a. Corrosives Storage Cabinet.
      b. Flammable Liquid Storage.
   5. Laboratory Work Surfaces.
   6. Shelving Assemblies.
   7. Cylinder Restraint Assembly.
   8. Metal Pipe Drop Enclosure.
   9. Drying Rack.
   10. Metal Fabrications and Finish Requirements.
   11. Stainless Steel Fabrications:
      a. Work Surfaces.
      b. Laboratory Sinks and Cup Sinks.
      c. Canopy Hoods.
   12. Sealant.

B. Completely coordinate with work of other trades.

1.2 REFERENCES

A. Scientific Equipment and Furniture Association (SEFA):
   1. SEFA 2 Installation.
   2. SEFA 3 Work Surfaces.
   3. SEFA 7 Fixtures.
   4. SEFA 8-M Laboratory Grade Metal Casework.

B. American National Standards Institute (ANSI):
   1. ANSI A208.1 Particleboard, current edition.
   2. ANSI A208.2 Medium Density Fiberboard (MDF) for Interior Applications, current edition.
   3. ANSI/HVPA HP-1 American National Standard for Hardwood and Decorative Plywood, with the Hardwood Veneer Plywood Association, current edition.
   4. ANSI/BHMA A156.9 Cabinet Hardware Standards, Grade 1.
   5. ANSI/NEMA LD 3 High-Pressure Decorative Laminates.

C. State of California Air Resources Board (CARB):
   1. State of California Air Resources Board “Airborne Toxic Control Measure to Reduce Formaldehyde Emissions from Composite Wood Products,” CCR Title 17, Sections 93120 – 93120.12

1.3 SUBMITTALS

A. Product Data:
   1. Submit materials list, including catalogue data, of materials, equipment, and products for work in this section.

B. Shop Drawings:

1. Shop fabrication and installation drawings, including plans, elevations, sections, details, and schedules.
2. Show relationship to adjoining materials and construction.
3. Submit wood, plastic laminate, and solid phenolic casework shop drawings after accepted by AWIQCP reviewer.
4. Indicate surface grain directions on wood casework.

C. Samples:
1. Accompanying Materials List, submit four (4) samples of each of following items for Contracting Officer Representative's approval:
   a. Laboratory Work Surfaces: 4 inches by 4 inches sample of each type specified.
   b. Painted Surfaces, Plastic Laminate, Solid Phenolic, Polypropylene: 3 inches by 5 inches sample of each available standard color, or specified color if identified in Finish Schedule, as is applicable.
   c. Wood Panel: 12 inches by 12 inches by 3/4 inches thick sample with stain and finish. Samples shall represent range of color and grain expected to be provided.
   d. Hardware: Submit samples of each hardware item that will be visible at exposed surfaces when the job is complete.

D. Project Information:
1. Structural:
   a. Submit detailed anchorage and attachment drawings and calculations to show compliance with seismic restraint requirements.
   b. Engineering design shall be performed and sealed by registered Contracting Officer Representative, licensed to practice structural engineering in the state of the project location.
   c. Refer to the Structural Drawings for seismic design criteria.
2. Wood products and painted metal finish:
   a. Provide letter from a third-party testing agency, verifying independent chemical resistance test results.
3. Statement of Installer Qualifications.
4. Load Tests: Provide on request, load test results certified by an independent testing laboratory for cabinet box, drawers, doors, suspensions slides, and unit shelving as identified in SEFA 8.
5. Certificates:
   a. Certify that factory tests specified for mechanical service fixtures have been performed and that products or systems meet or exceed specified requirements.
   b. As a condition of acceptance, submit certification stating that equipment is complete and ready for intended function.
6. Independent inspection shop drawing reviews and reports prepared under the AWI/QCP shall be submitted for information.

E. Contract Closeout Information:
1. Operations/Maintenance Manuals: Accompanying certification, submit for Contracting Officer Representative 's review and Government's use, including:
   a. Complete operating and maintenance manuals that describe proper operating procedures.
   b. Maintenance and replacement schedules.
   c. Component parts list.
   d. Nearest factory representative for components and service.

1.4 QUALITY ASSURANCE

A. Qualifications:
1. Work in this section shall be performed by a company having a minimum of eight (8) years documented experience manufacturing the respective products specified herein, and an established organization and production facilities including tools, equipment, and special machinery necessary for the fabrication and installation of the type of equipment required,

with skilled personnel, factory trained workmen and an experienced engineering department. Each shall have the demonstrated knowledge, ability, and the proven capability to produce the specified equipment of the required quality and the proven capacity to complete an installation of this size and type within the required time limits. Upon request, manufacturers shall produce evidence of financial stability and bonding capacity required to perform on this project.

2. Casework manufacturer must have at least one project in past twelve (12) months where the value of the laboratory casework was within 20 percent of the cost of the laboratory casework for this project.
3. Casework installers shall be approved in writing by the casework manufacturer for the installation of specified products.
4. Work shall be in accordance with the Grade or Grades specified of the Architectural Woodwork Standards for applicable products.
5. Fabricators of wood, plastic laminate, and solid phenolic casework shall be accredited through the AWI Quality Certification Program (QCP).

B. Undivided Responsibility:
1. Unless specified otherwise, because of special coordination requirements, the supplier of scope of work described in this Section shall also provide scope of work described in following Sections:
2. Section 11 53 00 Laboratory Instrumentation – Table-Top and Handheld
3. Section 11 53 13 Fume Hoods and Exhaust Devices.
4. Section 11 53 43 Laboratory Service Fittings and Fixtures.

C. Conform to recommended practices of the Scientific Equipment and Furniture Association (SEFA), current version, except as superseded by this specification:

1.5 DELIVERY, STORAGE, AND HANDLING

A. Contractor shall schedule the delivery of casework and furnishings when spaces are sufficiently complete so materials can be installed immediately following delivery.

B. Delivery of laboratory casework shall be made only when the area of operation is enclosed, plaster and concrete work is dry, and the area broom clean.

C. Product shall be stored in a clean storage area.

1.6 FIELD CONDITIONS

A. Field verify all dimensions prior to fabrication of casework.

B. Casework and furnishings shall not be delivered and installed prior to completion of the following items:
1. Windows and doors shall be installed, and the building shall be weather-tight.
2. Finished ceilings, if specified, overhead ductwork, piping, electrical, and lighting work shall be installed.
3. Painting shall be complete.
4. Flooring shall be installed, except when an integral base is specified to be installed over the casework toe kick.

C. Maintain interior building temperature between 65 deg F and 80 deg F.

D. Maintain ambient relative humidity between 25 and 55 percent prior to delivery, during installation, and for the remainder of the project.

E. Frequent and/or excessive changes in temperature and/or humidity levels during casework installation, or once casework is installed, must be avoided to prevent damage to materials.

PART 2 - PRODUCTS

2.1 CASEWORK DESIGN AND PERFORMANCE CRITERIA

A. Cabinets may be manufacturer's standard height and depth, provided such standard is not more or less than 1/2 inches of the height and depth indicated on the Drawings, unless noted otherwise in this Section.

B. System Structural Performance:
   1. Laboratory casework and support framing system shall withstand the effects of the following loads and stresses without permanent deformation, excessive deflection, or binding of drawers and doors:
      a. Casework, doors, drawers, work surfaces and shelving shall comply with SEFA requirements for the respective casework material.
      b. Work Surfaces: In addition to SEFA test requirements, work surface spans without continuous base cabinet support shall support 50 psf; deflection shall be limited to 1/180 of the length of the span, not to exceed 1/4 inches.
      c. Shelving: In addition to SEFA test requirements:
         1) Shelf spans without continuous support shall support 50 psf.
            a) Polypropylene shelving shall comply with SEFA loading requirements.
         2) Deflection shall be limited to 1/180 of the length of the span, not to exceed 1/4 inches.
         3) Deflection shall be limited to 1/144 of the length of the span, not to exceed 1/4 inches.
      d. Seismic Anchor: Provide seismic anchor for freestanding cabinets and cabinets located below fume hoods designated to be removable for access for the disabled. Seismic anchors may be floor or wall attachments but shall not attach to adjacent casework or work surfaces. Seismic anchors shall be accessible without removal of laboratory casework, furnishings, or equipment. Anchor attachment shall not void UL listing.
   2. Suspended and laser shelving systems:
      a. Satisfy lateral loading requirements identified in Division 1 or on Structural Drawings.
      b. Design to support framing and live load requirement identified for casework above.
   3. Radioisotope Base Cabinets Structural Requirements: Base cabinets for radioisotope use shall be capable of supporting:
      a. 50 psf of work surface.
      b. 200 psf on the work surface.
      c. 75 psf within the cabinets.
      d. Weight of fume hood, where applicable.

2.2 METAL LABORATORY CASEWORK, TABLES AND CASEWORK SYSTEMS

A. Manufacturers:
   1. Laboratory Casework:
      a. Kewaunee Scientific Corporation.
      b. A.T. Villa USA, Inc.
      c. Bedcolab Ltd.
      d. Mott Manufacturing Limited.
      e. Other manufacturers desiring approval comply with Section 01 67 0001 67 00.
      f. Substitutions are not permitted.

B. Components:
   1. Cold rolled sheet steel:
      a. Recycled Steel Content: A minimum of 20 percent of the steel used in the fabrication of laboratory cabinets and modular laboratory systems, if specified, shall consist of the sum of post-consumer recycled content plus one-half of the pre-consumer content, based on the cost of the total value of materials.

b. Fabricators Scrap: Fabricators shall provide documentation that manufacturing fall-off is recycled to respective steel mills and neither enters the solid waste system nor becomes a product of landfill space.
c. Prime grade, roller leveled and treated at the mill to be free of scale, ragged edges, deep scratches, or other injurious effects. Thickness of metal used in construction of cases shall be 18 GA, except as follows:
1) 20 GA: Solid door interior panels, drawer fronts, scribe strips, filler panels, enclosures, drawer bodies, shelves, security panels, and sloping tops.
2) 16 GA: Top front rails, top rear gussets, intermediate horizontal rails, table legs and frames, leg rails and stretchers.
3) 14 GA: Drawer suspensions, door and case hinge reinforcements, and front corner reinforcements.
4) 12 GA: Table leg corner brackets and gussets for leveling screws.
d. Metal Casework Finish Requirements: Refer to Painted Metal Finish Performance Requirements elsewhere in this Section
2. Glass:
a. Type: Laminated with clear PVB inner layer, ANSI Z97.1, ASTM C1036 or C1048.
b. Thickness: Two layers of 1/8 inches glass with 30 mil interlayer.
c. Without imperfections or marred surfaces.
d. Cut or drill to receive hardware.
e. Polished edges where unframed doors are indicated.
3. Hardware: As specified elsewhere in this Section.

C. Design Requirements:
1. Door and Drawer Design:
a. Inset/Flush:
1) Square edged, inset door and drawer flush front construction with front surfaces above the toe space in the same plane.
2) Front width of end panels shall be 3/4 inches and front height of top and bottom members shall be 1 inch.
3) Material: Painted Metal.
b. Full flush overlay:
1) Provide applied panels in areas such as sink cabinets and knee spaces with apron panels to provide a full flush overlay appearance.
a) Apron panels shall align with plane of adjacent doors and drawers for full flush overlay appearance.
2) Material:
a) Painted Metal.
b) Wood.
c) Plastic Laminate.
2. Pulls on door and drawers shall be mounted horizontally.
3. Pulls on doors shall be mounted vertically and on drawers horizontally.
4. Tall Storage Cabinets:
a. Toe space height and depth to match base units.
5. Fume Hood Cabinets:
a. Provide false front panel above door at cup sink locations when flush overlay design is specified.
b. Cabinets shall be in alignment with adjacent casework and toe kicks.

D. Fabrication:
1. General:
a. Cabinets shall be constructed and finished to be suitable for use as stand-alone units and to permit future rearrangement without the need for additional parts or finish.
b. Units shall have a cleanable smooth interior. Front and rear posts, reinforcing members or channel uprights shall be enclosed full heights on cabinet openings.

c. Exterior corners: shall be spot and arc welded with gussets at exterior corners. Face joints shall be arc welded and ground smooth to provide a continuous flat plane.
d. Units less than 49 inches tall: Provide internal reinforcing and rear posts for end panels and cabinet backs.
e. Units 49 inches tall and greater: Provide formed end panels with front and rear reinforcing posts. Back shall be formed steel panel, recessed 3/4 inches for mounting purposes.
f. Posts: Front post fully closed with full height reinforcing upright to facilitate cleaning.
g. The front edge shall be formed to provide a strike for doors and drawers and shall be pre-drilled for intermediate rails and hinge screws.
h. Intermediate Vertical Uprights: shall be furnished to enclose cabinets when used in a unit in combination with a half width bank of drawers. However, to allow storage of large or bulky objects, no upright of any type shall be used at the center of double door cupboard units.
i. End Uprights shall be formed into not less than a channel formation at top, bottom, back and front.
j. An upright filler shall be screwed in place in wall cabinet units to close the back of the channel at front of the upright and to provide a smooth interior for the wall cabinet.
k. The upright filler shall be perforated with shelf adjustment holes at no more than 2 inches centers.
l. The inside front of the upright shall be further reinforced with a full height 14 GA hinge reinforcement angle.
m. Exposed fasteners are not allowed without prior approval of the Contracting Officer Representative. If allowed, exposed fasteners in BSL-3, ABSL-3, BSL-3Ag, BSL-4 or ABSL-4 shall be Type 304 or 316 stainless steel or protected with epoxy paint or sealant.

2. Drawer Construction:
   a. Drawer bodies shall be made in one-piece construction including the bottom, two sides, back and inner front. Front and back faces should be spot welded to sides or center section.
   b. Fully coved at interior bottom on all four sides for easy cleaning.
   c. Full height sides.
   d. 1/2 inch side clearance to frame opening.
   e. Drawers shall be a minimum of 18 inches front to back.
   f. Drawers shall be easily removable in the field without the use of special tools.
   g. Drawers shall be sized on a modular basis for interchange to satisfy varying storage requirements.
   h. Drawer Suspension: Refer to Drawer Slides under Hardware section.
   i. Drawer Heads:
      1) Thickness to match doors.
      2) Widths of drawer front in aprons shall be lesser of the width of the apron or 24 inches.
      3) Two-piece sheet steel assembly of 3/4 inches overall thickness to consist of an inner pan formed as an extension of the drawer body, an outer pan having a channel formation on all four sides welded and ground to eliminate exposure of sharp raw edges, and the interior space filled with a non-organic sound deadening material at the time of assembly.
         a) Welds shall be ground smooth.
         b) Painted inside and out prior to assembly.

3. Doors: Doors shall be readily removable and hinges easily replaceable. Hinges shall be applied to the case and door with screws. Welding of hinges to either case or door will not be acceptable.
   a. Two-piece sheet steel assembly of 3/4 inches overall thickness to consist of an inner pan formed as an extension of the drawer body, an outer pan having a channel formation on all four sides welded and ground to eliminate exposure of sharp raw

edges, and the interior space filled with a non-organic sound deadening material at the time of assembly.
1) Welds shall be ground smooth.
2) Painted inside and out prior to assembly.

b. Framed Glazed Doors: Framed glazed door construction shall match construction and quality of solid panel doors. Inner head shall include top, bottom, and side framing members which are removable for installation and replacement of glass. Continuous vinyl retainer shall be provided to receive glass.
c. Framed Sliding Doors: Design for tilt-out removal after removal of bottom guide. Doors shall be hung with nylon-tired sleeve bearing rollers in formed steel top track and shall close against rubber bumpers.
d. Unframed Sliding Glass Doors: Glass with ground edges shall be set in extruded Aluminum shoe with integral pulls, wheel assemblies and top and bottom extruded Aluminum track. Provide rubber bumpers at fully opened and closed-door position.

4. Cabinet Bottom:
   a. Case bottom and bottom rail shall be formed of one piece of metal except in corner units and shall have both sides and back formed up or down and shall be rabbeted in front for drawers and swinging doors.
5. Cabinet Top:
   a. Wall and Tall Case Top: One-piece, with front edge formed into front rail.
   b. Provide sloping tops when indicated on the Drawings.
6. Security Panels: Provide security panels where keyed different locks are specified. Security panels shall be removable in cabinets with removable back panels from the underside of the panel.
7. Cabinet Back, Unexposed: Cabinet back shall consist of a top and bottom rail, channel formed for maximum strength and welded to back and top flange of end uprights, with space between left open for access to plumbing lines.
   a. Provide removable back panels for all fixed base cabinets. Removable panels shall not be provided in BSL-3, ABSL-3, BSL-3Ag, BSL-4, and ABSL-4 installations.
   b. Drawer cabinets shall have split back panel allowing for removal from cabinet.
   c. Sink base back shall be full half-height construction to allow for plumbing and sink waste connection.
   d. Back panels shall be removable from inside cabinet without the use of special tools.
8. Toe Bases, Kicks, and Sleepers:
   a. Toe Space Rail:
      1) Provide 3 inches deep and 4 inches high formed steel base with corner gussets. Whenever the base is omitted for units to be set on building bases or separate metal bases, the toe space rail shall extend back 4-1/2 inches.
      2) Provide 3/8 inches diameter leveling screw with integral bottom flange of minimum 0.56 inches2 area at each corner, accessible through openings in toe space.
   b. Peninsula and Island Bench Arrangements; Provide toe kick at front and sides of island and peninsula end cabinets when turned perpendicular to length of bench. Side toe kick shall be continuous with adjacent cabinets.
9. Shelves:
   a. Full depth turned down 1 inch on four sides, and returned under front and back 1 inch, nominal.
   b. Shelves over 36 inches in length shall be additionally reinforced by a flanged channel shaped member electro-welded to underside of shelf.
   c. Shelves shall be adjustable.
   d. Shelf adjustment posts shall be perfectly aligned for level setting, with holes for shelf adjustment at 1/2 inches on center.
10. Mobile Units:
    a. Where indicated on the Drawings, provide cabinets mounted on four swivel casters instead of levelers; the front two casters shall be locking.

b. Finish back and side panels as exposed surface.
c. Omit sub-base and provide casework with finished bottom panels.
d. Design mobile units to prevent tipping and allow only one drawer to open at a time.

11. Closures:
    a. Knee Space/Opening Panels: Where specified, shall be of the same finish as cabinets, and shall be furnished at open spaces under countertop where no cabinets occur. They shall be easily removable and shall cover piping from underside of top of service ledge to floor.
    b. Closure/Filler Panels:
        1) Provide filler panels where required between cabinets, at corner intersections of cabinets, between cabinets and walls and wherever else required for a complete finished installation.
        2) For tall cabinets: provided for vertical face and top.
        3) For wall cabinets: provided for vertical face, top and bottom. When wall-hung cabinets are installed to ceiling, provide continuous metal filler between top of cabinet and ceiling, to permit door to swing clear of ceiling.
12. Leg Assemblies:
    a. Legs: Not less than 2 inches by 2 inches by 16 GA square tubular steel sections.
    b. Leg rails, where required: Not less than 1-1/4 inches by 2-1/2 inches by 16 GA steel sections, reinforced as necessary for leg attachment.
    c. Each leg shall have a recessed leveling screw and a leg shoe.
13. Apron:
    a. Typical: 1-1/2 inches by 4-1/2 inches by 16 GA channel steel sections, reinforced as necessary for leg attachment.
    b. For work surfaces set at 32 inches above finished floor, or less: 1-1/2 inches by 2 inches by 16 GA channel steel sections, reinforced as necessary for leg attachment.
    c. Apron drawers:
        1) Where indicated on the Laboratory Furnishing drawings, provide support rails; drawer unit, hardware and suspension as specified for base unit drawers.
14. Metal-Framed Laboratory Tables:
    a. Legs: Not less than 2 inches by 2 inches by 16 GA square tubular steel sections with welded leg brackets.
        1) Table legs shall be telescoping to allow vertical height adjustment of work surface from 30 inches to 36 inches above finished floor.
    b. Leg rails and spreader rail: Not less than 1-1/4 inches by 2-1/2 inches by 16 GA steel sections, reinforced as necessary for leg attachment.
    c. Aprons and Rails: 1-1/2 inches, minimum, by 4-1/2 inches, maximum, 16 GA channel steel sections, reinforced as necessary for leg attachment. Provide 1-1/2 inches by 2 inches by 16 GA channel steel sections, reinforced as necessary for leg attachment, where no drawers are required.
    d. Work Surface: Refer to Laboratory Furnishing drawings for laboratory work surface materials, described in the Laboratory Work Surfaces section of this specification.
    e. Electrical receptacles: Where indicated on the Laboratory Electrical drawings provide cutouts for electrical receptacles as work of this Section and coordinate with Division 26.
    f. Cord and plug shall be provided under the work of Division 26.
    g. Leveling Glide and Leg Shoe: Each leg, other than those fitted with casters, shall have a recessed leveling screw and a leg shoe.
    h. Casters: Where indicated on Laboratory Furnishing drawings or specifications, as specified under Cabinet Hardware.
    i. Table Drawers: Where indicated on the drawings, provide front and back rails; drawer unit, hardware, and suspension same as specified for casework base unit drawers.
    j. Low level shelf:
        1) Epoxy, as described in the Laboratory Work Surfaces section of this specification. Thickness: 1 inch.

2.3 CABINET HARDWARE

A. General:
   1. Special metal cabinets, such as corrosives storage, flammable liquid and solvent storage, rock storage, map storage, museum storage, radioisotope storage, and narcotics locker, may be provided with the manufacturer's standard hardware.
   2. Exception: Door and drawer pulls shall match, regardless of type of casework, except for polypropylene casework, if specified.
      a. Polypropylene casework hardware is specified under Polypropylene Casework.

B. Hinges:
   1. General:
      a. Satisfy ANSI/BHMA Grade 1 requirements.
   2. Five Knuckle Hinges:
      a. Characteristics:
         1) Institutional grade butt hinge.
         2) Hospital tip.
         3) Edge mounted.
         4) Height: 70mm 2-3/4 inches, nominal.
         5) Thickness: 2.4mm 0.095 inches.
         6) Material:
            a) Type 302 or 304 stainless steel with stainless steel screws.
            b) Steel with US26 bright chrome brushed finish with zinc-plated screws.
            c) Steel with US26D satin chrome finish with zinc-plated screws.
            d) Steel with US28 clear anodized finish with zinc-plated screws.
            e) Steel with epoxy powder coating with zinc-plated screws.
               (1) Color:
                  (a) Gold.
                  (b) Black.
                  (c) White.
                  (d) Off-white.
                  (e) Gray.
   3. Semi-Concealed Hinges for Flush Overlay Doors:
      a. Three-way adjustable clip-on institutional quality hinge with built in catch.
         1) With door in closed position, hinge site line shall be limited to the 5mm diameter knuckle; remainder of hinge should be concealed.
         2) Back mounted.
         3) Roller catch is not required when hinge has built-in catch.

C. Shelving Pilaster Standards and Clips:
   1. Manufacturers:
      a. Knape & Vogt Manufacturing Co.
      b. Bainbridge Manufacturing, Inc.
      c. The Engineered Products Company (Epco).
      d. Fixture Hardware Manufacturing.
      e. Rockford Process Control, Inc.
      f. Sugatsune America, Inc
      g. Other manufacturers desiring approval comply with Section 01 67 00.
   2. The following systems are acceptable:
      a. Surface Mounted Pilaster Standard for painted metal, stainless steel casework, only:
         1) 23 GA high strength steel.
         2) 21mm 13/16 inches wide by 3/16 inches deep.
         3) Pre-drilled or pre-punched mounting holes.
         4) Zinc finish.
         5) Shelf clip with vertical upper tab and horizontal lower tab to engage two slots in pilaster standard.
            a) Clip shall be profiled to engage sides of pilaster standard.

b) Zinc finish.

b. Mortised Mounted Pilaster Standard for wood, plastic laminate, solid phenolic casework, only:
   1) 23 GA high strength steel.
   2) 5/8 inches wide by 3/16 inches deep.
   3) Pre-drilled or pre-punched mounting holes.
   4) Zinc finish.
   5) Shelf clip with vertical upper tab and horizontal lower tab to engage two slots in pilaster standard.
      a) Clip shall install flush against face of pilaster standard.
      b) Zinc finish
c. Die-formed, zinc-plated steel, designed to engage in shelf adjustment holes in side of cabinet.
d. Where casework requires seismic restraint:
   1) Metal shelves: Adjustable shelf clip support with lockdown clips designed to engage in shelf adjustment holes in side of cabinet or pilaster standards, or shelves may be screwed to shelf clip. Shelves shall be vertically adjustable on 1/2 inches centers.
e. Shelf hardware shall support loading in accordance with Part 1.

D. Ball Bearing Drawer Slides:
   1. Manufacturers:
      a. Base:
         1) Accuride.
      b. Optional:
         1) Waterloo Furniture Components Inc.
         2) Fulterer USA.
         3) Other manufacturers desiring approval comply with Section 01 67 00.
   2. Typical Drawers:
      a. Satisfy ANSI/BHMA Grade 1HD requirements.
      b. Base Product:
         1) For drawer widths up to 18 inches: Accuride 4033.
         2) For drawer widths greater than 18 inches and up to 27 inches: Accuride 4032.
         3) For drawer widths greater than 27 inches and up to 42 inches: Accuride 930.
      c. Full extension.
      d. Dynamic Load Rating: 100 pounds/pair capacity, minimum:
   3. File Drawers:
      a. Satisfy ANSI/BHMA Grade 1HD requirements.
      b. Base Product: Accuride 3641.
      c. Provide with interlock, rail, and pocket mount, and overtravel.
      d. Dynamic Load Rating: 180 pounds/pair capacity.

   4. Base Metal:
      a. Clear, zinc-coated steel.
      b. Black, zinc-coated steel.
      c. Electrocoat white steel.
      d. Sheet metal to match cabinet construction.

E. Hanging File Suspension System:
   1. Manufacturers:
      a. Base:
         1) Julius Blum, Inc
      b. Optional:
         1) Hettich America, L.P.
   2. Other manufacturers desiring approval comply with Section 01 67 00.
   3. Base Product: Blum Metafile, or Hettich MultiTech Hanging File Frame Kit.

4. Description:
    a. File hangers shall be fastened and secured to drawer construction and shall not be freestanding units set inside the drawer.
    b. Provide in file drawers.

F. Locks:
1. Swinging Doors and Drawers:
    a. Manufacturers:
        1) National Cabinet Lock.
        2) Illinois Lock Company.
        3) Other manufacturers desiring approval comply with Section 01 67 00.
2. General:
    a. Provide locks on file cabinet drawers.
    b. Provide locks at other locations as indicated on the drawings.
3. Provide chain bolts 3 inches long, with an 18 inches pull and an angle strike to secure inactive door on cabinets over 72 inches in height.
4. 5- or 8-tumbler locks are acceptable.
5. Locks shall be provided with removable cores and stamped with identifying numbers.
6. Finish: Satin nickel or satin chrome finish.
7. Keys: Stamped brass keys available from manufacturer or locksmith and supplied in the following quantities, unless otherwise specified:
    a. Two keys for each different lock.
    b. Three keys for each group keyed alike locks.
    c. Two keys for each master key system.
8. Keying:
    a. Keyed differently with up to 225 primary key changes. Master key one level with the potential of 40 different, non-interchangeable master key groups.
    b. Keyed differently with up to 2000 primary key changes. Master key one level with built in flexibility to accommodate, if required, three levels, one grandmaster, 59 master groups, and 70 sub-master groups with 13 primary changes under each.
9. Framed sliding door locks shall be plunger type.

G. Roller Catch:
1. Manufacturers:
    a. The Engineered Products Company (Epco).
    b. Amerock.
    c. Ives.
    d. Other manufacturers desiring approval comply with Section 01 67 00.
2. Description:
    a. Adjustable, spring-loaded dual polyethylene roller with a steel strike plate.
    b. Double doors without locks: Provide a catch on each door.
    c. Tall cabinets: Locate catch on the upper and lower part of each door.
    d. Cabinets equipped with locks: Right hand door shall have a roller catch; left hand door shall have a positive catch.

H. Levelers:
1. Manufacturers:
    a. Blum, Inc
    b. Camar, distributed by Peter Meier Inc.
    c. Häfele America Co.
    d. Hettich America LLP.
    e. Other manufacturers desiring approval comply with Section 01 67 00.
2. Provide levelers consisting of sockets and levelers.
    a. Load Rating: 330 pounds minimum capacity.
    b. Quantity: four per cabinet, minimum.

c. Service Access: Levelers shall be accessible through a hole in bottom of cabinet. Provide plastic cap for each leveler hole.
d. Provide clip or other hardware for attachment of toe kick.

I. Casters:
1. Location: Where indicated on Laboratory Furnishing drawings or specifications.
2. Description:
a. Self-lubricating precision swivel casters with locking brakes.
b. Wheel Material: Urethane or polyurethane.
c. Wheel Material: High temperature phenolic, nylon, or synthetic rubber designed to be autoclaved.
d. Load Rating: 200 pounds minimum, each.
e. Wheel Diameter and Load Height:
1) Tables: 4 inches diameter. Load Height: 5 inches. Stem mount.
2) Cabinets: 3 inches. Load Height: 4 3/8 inches. Plate mount.
f. Caster Housing: Heavy gauge cold rolled steel with bright zinc plating.
g. Caster Housing: Heavy gauge cold rolled steel with epoxy powder coating.
h. Caster Housing: Type 304 stainless steel.

J. Pulls:
1. Drawer and hinged doors: Full width hardwood pulls. Hardwood species to match casework exposed hardwood selection.
2. Drawer and hinged doors: Attach with machine screws. Two pulls shall be furnished on drawers wider than 28 inches.
a. Type:
1) Pulls shall be round wire.
b. Material and Finish:
1) Extruded Aluminum with US26 bright chrome brushed finish.
2) Plastic pulls or other types subject to breakage are not acceptable.
3) Back mounted.
c. Length: 4 inches center to center of screw holes.
d. Diameter: 1/4 inches.

K. Drawer Stops: Provide integral stops or drawer bumpers on each side of drawer body to prevent drawer head impact with cabinet body.

L. Cabinet Door Bumper Pads: Non-staining, non-marring, clear polyurethane pads with pressure-sensitive, adhesive backing for sound and vibration dampening, preventing direct contact between door and cabinet. Pad should have raised tip in middle of pad.
1. Location:
a. Doors 36 inches and less in height: At top and bottom corners along pull edge.
b. Doors greater than 36 inches in height: At top and bottom corners and at an intermediate location along pull edge.
c. Drawers: At center of each side.

M. Door Stops: Provide chain door stops for any tall cabinet door, which will strike an obstruction when opened between 90 degrees and 135 degrees.
1. Provide #30 zinc-plated steel sash chains; cut to length to allow door to open 1-1/2 inches from obstruction.
2. Provide zinc chromate wire screw eyes. Open eye as required to attach chain. Through-bolting not allowed.

N. Glides: Provide non-marring material on movable tables, unless otherwise indicated.
1. Diameter: 1 inch diameter, minimum.
2. Vertical Adjustment: 5/8 inches, minimum.

O. Leveling devices: Provide leveling bolt with floor clip at each table leg.
1. Bolt Diameter: 3/8 inches, minimum.
2. Nylon Pad: 2 inches diameter.

P. Corner Base Guards: 4 inches high Type 304 stainless steel corner guards with No. 4 finish.

Q. Floor clips: Provide leg assemblies and fixed table legs with floor clips securely fastened to the floor after shimming.

R. Support Struts and Service Ledging: Refer to specifications for slotted channel framing in this Section.

S. Light-tight Covers: 20 GA steel paintlock sheet, factory painted.

T. Light-tight Covers: 20 GA stainless steel sheet.

2.4 SPECIAL PURPOSE CABINETS

A. Corrosives Storage Cabinets:
   1. Manufacturers:
      a. Manufacturer of laboratory casework.
      b. Justrite Manufacturing Company.
      c. Eagle Manufacturing Company.
      d. Other manufacturers desiring approval comply with Section 01 67 00.
      e. Substitutions are not permitted.
   2. Description: Purpose-designed cabinet completely lined with a polypropylene liner with sealed or seamless intersections between panels.
      a. Provide removable, polypropylene shelf.
      b. Construction shall be equivalent to laboratory casework of that construction.
   3. Cabinet Construction:
      a. Below Worksurface Locations:
         1) Painted metal.
         2) Stainless steel.
         3) Wood.
         4) Plastic laminate.
         5) Match respective laboratory casework.
      b. Cabinet Construction, Freestanding Cabinets:
         1) Painted metal.
   4. Label:
      a. "CORROSIVES" in conspicuous silk-screened lettering. Stick-on decals are not acceptable.
      b. Size and style of lettering shall match the Flammable Liquid/Solvent Storage Cabinet label.
   5. Door: Inset design to create a tight (near airtight) seal around face of cabinet when closed.
   6. Hardware:
      a. Locks: Provide key locks for cabinet doors.
      b. Hinge: Provide 5-knuckle or piano stainless steel hinge.
      c. Casters: Provide locking casters on cabinets identified to be mobile.
   7. Venting:
      a. Cabinets below or adjacent to fume hoods:
         1) 2 inches polypropylene vent pipe extending 2 inches above dished worktop behind baffle in hood.
         2) Provide hole through fume hood work surface behind cabinet to accommodate vent pipe.
         3) Seal gap around penetration with clear sealant.
      b. Cabinets not below and not adjacent to fume hoods:
         1) 2 inches polypropylene vent pipe to run horizontally in the chase space behind the casework to nearest pipe drop enclosure and rise vertically to 6 inches above ceiling level, or 9 feet in rooms with no finished ceiling.
         2) Connection to exhaust duct system shall be by Division 23.
   8. Seismic Attachment:
      a. Provide restraints at the following locations:

1) Movable cabinets located below fume hoods designated to be accessible to the disabled.
2) Freestanding corrosives storage cabinets.

b. Attachment may be to floor or wall but shall not attach to adjacent casework or work surfaces.

c. Attachment shall be accessible without removal of laboratory casework, furnishings, or equipment.

B. Flammable Liquid/Solvent Storage Cabinets:

1. Manufacturers:
   a. Manufacturer of laboratory casework.
   b. Justrite Manufacturing Company.
   c. Eagle Manufacturing Company.
   d. Other manufacturers desiring approval comply with Section 01 67 00.
   e. Substitutions are not permitted.
2. Description: Purpose-designed cabinet for the storage of flammable, combustible, and solvent liquids.
   a. Provide removable, adjustable shelf.
   b. Construction shall be equivalent to laboratory casework of that construction.
3. Cabinet Construction:
   a. Below Worksurface Locations:
      1) Painted metal.
      2) Match respective laboratory casework.
   b. Freestanding Cabinets:
      1) Painted metal.
   c. Cabinet construction shall comply with the requirements of:
      1) OSHA 29 CFR 1910.106
      2) NFPA 1 (latest edition): Fire Code.
      3) NFPA 30 (latest edition): Flammable and Combustible Liquids Code.
      4) International Fire Code (latest edition).
4. Regulations, Approvals, and Listings:
   a. Factory Mutual (FM Global) Approval Standard for Storage Cabinets (Flammable and Combustible Liquids) Class Number 6050.
   b. Underwriters Laboratories (UL) Listed: UL 1275.
5. Label:
   a. FLAMMABLE - KEEP FIRE AWAY in conspicuous silk-screened lettering. Stick-on decals are not acceptable.
   b. Size and style of lettering shall match that of the Corrosives Storage Cabinet label.
6. Floor pan: 2 inches deep, removable, powder coated or stainless steel, liquid tight pan with continuously welded corners to cover the entire bottom of the cabinet to contain liquid leaks and spills.
7. Doors:
   a. Inset design to create a tight (near airtight) seal around face of cabinet when closed.
   b. Well fitting, self-closing and self-latching.
8. Hardware:
   a. Hinges: full length piano hinge or other hinge that shall not lose holding capacity when subjected to fire exposure.
   b. Locks: Doors shall have key locks.
   c. Casters: Provide locking casters on cabinets identified to be mobile.
9. Cabinets shall not be vented; seal vent openings with bungs as provided by manufacturer.
10. Venting
    a. Remove two metal bungs from cabinet outlets and replace with flash arrestors provided by manufacturer.

b. Connection with 2 inches steel vent piping to the HVAC system as shown on drawings shall be by Division 23. Vents from multiple cabinets shall not be manifolded prior to connection to the building system.

11. Electrical grounding:
    a. Provide an externally mounted grounding conductor screw terminal for up to #8 AWG conductor, mounted at the top of each cabinet.
    b. Connection to ground shall be by Division 26.
12. Seismic Attachment:
    a. Provide restraints at the following locations:
        1) Movable cabinets located below fume hoods designated to be accessible to the disabled.
        2) Freestanding flammable liquid storage cabinets.
    b. Attachment may be to floor or wall but shall not attach to adjacent casework or work surfaces.
    c. Attachment shall be accessible without removal of laboratory casework, furnishings, or equipment.
    d. Anchor attachment shall not void UL or FM listing when required.

2.5 LABORATORY WORK SURFACES

A. Epoxy Resin:
1. Countertop Grade: AWS Premium Grade.
    a. Modifications: In addition to satisfying the specified grade, products shall incorporate the identified modified requirements, characteristics, and features as specified herein.
2. Manufacturers:
    a. Durcon Inc.
    b. Epoxyn Products.
    c. Kewaunee Scientific Corporation.
    d. Other manufacturers desiring approval comply with Section 01 67 00.
    e. Substitutions are not permitted.
3. Thickness:
    a. Typical work surface: 1 inch.
    b. Fume hood work surfaces:
        1) 1-1/4 inches thick at outer edge, indented 1/4 inches, nominal, to provide a raised rim around exposed edges 1 inch wide, minimum, worksurface or as to allow for the fume hood sash.
        2) Front top edge of the raised rim and exposed vertical corners of the top shall be rounded or chamfered to a 1/8 inches radius.
        3) Juncture between the raised rim and the top surface shall be coved or chamfered to a 1/4 inches radius.
4. Color:
    a. Black.
    b. Dark gray (Platinum gray as supplied by Durcon Inc. or similar).
    c. Color sample shall be submitted for approved by Contracting Officer Representative .
5. Provide the following:
    a. Drip Grooves:
        1) Provide under work surface exposed edges, unless noted otherwise on the Laboratory Furnishing Drawings.
        2) Where the top overhangs 1 inch: 1/2 inches from the edge.
        3) Where the top overhangs 1/2 inches: 1/4 inches from the edge.
    b. Edge profile:
        1) All exposed upper edges and corners: 1/4 inches radius, or 1/8 inches bevel.
    c. Marine edges:
        1) Where indicated on the Laboratory Furnishing Drawings.
        2) 1 inch wide and 1/4 inches high with chamfered or radiused transition to and an integral part of the work surface.

d. Indented areas:
   1) Where indicated on the Laboratory Furnishing Drawings.
   2) Profile: 1/4 inches deep with chamfered or radiused sides.
   3) Internal and external corners: 1/4 inches to 1/2 inches radius.
   4) Marine edges formed around indented areas shall not be less than 1 inch wide.
e. Sink Mounting:
   1) Drop-in Sink Cutouts:
      a) Profiled to provide support for the sink, and to ensure that the rim of the installed sink is 1/8 inches below the surrounding work surface level or bottom of drain grooves if present.
      b) Top edge of the cutout shall have 1/8 inches bevel.
      c) No gaps between the installed sink rim and work surface.
f. Curbs and Splashes:
   1) Curbs and Splashes: 1 inch thick.
   2) Curbs and Splashes: 3/4 inches thick.
   3) Height: 4 inches, unless noted otherwise on Laboratory Furnishing Drawings.
   4) Bonded to the surface of the top to form a square joint.
g. Provide holes and cutouts as required for built-in equipment and mechanical and electrical service fixtures. Verify size of opening with actual size of equipment to be used prior to making openings. Form inside corners to a radius of not less than 1/8 inches. After sawing, rout and file cutouts to ensure smooth, crack-free edges. Seal exposed edges after cutting with a waterproofing material recommended by the manufacturer.

6. Physical Properties:
   a. Chemical resistance:
      1) Organic solvents test:
         a) Saturate a cotton ball with the test chemical; place; in a one ounce bottle with a reservoir of liquid above the ball.
         b) Invert the container on the test material surface for a period of 24 hours.
         c) Test temperature: 23 deg C ±2 deg C.
      2) Other test chemicals test:
         a) Place five drops (1/4 cc) of the test chemical on the test material surface.
         b) Cover the chemical with a 1 inch diameter watch glass for a period of 24 hours.
         c) Test temperature: 23 deg C ±2 deg C.
      3) Evaluation:
         a) After 24 hours exposure: Wash exposed areas with water, then with a detergent solution, finally with naphtha, then rinse with distilled water, dry with a cloth, and rate as follows:

0	No effect	No detectable change in the material surface.
1	Excellent	Slight detectable change in color or gloss but no change in function or life of the surface.
2	Good	A clearly discernible change in color or gloss but no significant impairment of surface life or function.
3	Fair	Objectionable change in appearance due to discoloration or etch, possibly resulting in deterioration of function over an extended period of time.
4	Failure	Pitting, cratering, or erosion of the surface. Obvious and significant deterioration.

      4) Test results:

Test chemical	Concentration Percent	Black	Dark gray	Light gray	Beige
Chromic acid	40	3	2	2	2
Hydrochloric acid	10	0	0	0	0
Hydrochloric acid (conc.)	37	0	0	0	0
Nitric acid	40	0	0	0	0
Nitric acid (conc.)	70	0	0	0	0
Sulfuric acid	60	0	0	0	0
Sulfuric acid (conc.)	96	4	4	4	4
Acetic acid	5	0	0	0	0
Acetic acid (glacial)		0	0	0	0
Citric acid	1	0	0	0	0
Oleic acid		0	0	0	0
Phenol solution	5	0	0	0	0
Ammonium hydroxide	10	0	0	0	0
Sodium carbonate sol.	20	0	0	0	0
Sodium hydroxide sol.	60	0	0	0	0
Sodium hypochlorite sol.	4	0	0	0	0
Acetone		1	1	1	1
Benzene		1	1	1	1
Carbon tetrachloride		1	1	0	0
Diethyl ether		0	0	1	1
Dimethyl formamide		0	0	0	0
Ethyl acetate		0	1	1	0
Ethyl alcohol	95	0	0	0	0
Ethylene dichloride		0	0	0	0
Heptane		0	0	1	0
Isooctane		0	0	0	0
Kerosene		0	0	0	0
Methyl alcohol		0	0	0	0
Toluene		0	0	0	0
Aniline		0	0	0	0
Mineral oil		0	0	0	0
Olive oil		0	0	0	0
Soap solution	1	0	0	0	0
Transformer oil		0	0	0	0
Turpentine		0	0	0	0

b. Heat resistance tests:
   1) High temperature test:
      a) Heat a porcelain crucible to a dull red color, place on the test material, and allow to cool to ambient temperature.
      b) Result: No observable surface deformation.
   2) Flame test:
      a) Overturn a 3/8 inches Bunsen burner, adjusted to a quiet flame, with a 1-1/2 inches inner cone, on the test material, and allow to stay for 5 minutes.
      b) Result: no observable surface deformation.

c. Physical properties:

Physical Properties		
Compressive strength	ASTM D695	31,400 psi

Tensile strength	ASTM D638	8,000 psi
Flexural strength	ASTM D790	11,700 psi
Rockwell hardness M	ASTM D785	122
Specific density	ASTM D792	122.4 psf
Water absorption	ASTM D570	0.01 percent
Fire Resistance	ASTM D635	ATB (sec)=0
Heat deflection at 264 psi (1.82 MPa)	ASTM D648	342 deg F

B. Stainless Steel: Refer to Stainless Steel Fabrications section of this specification.

C. Solid Lumber Core:
1. Edge grain, hard maple strips, 45mm 1-3/4 inches wide, with edge grain exposed, glued with water-resistant resin under heavy pressure side to side and end to end.
2. Round top edges and corners and sand smooth.
3. Thickness: 1 inch.
4. Thickness: 45mm 1-3/4 inches.
5. Thickness: 57mm 2-1/4 inches.
6. Finish:
   a. Fine sand.
   b. Two coats of boiled linseed oil well rubbed into surfaces.

D. Galvanized Steel:
1. 24 GA hot galvanized steel bonded to a solid core of suitable material.
2. Completely cover top, sides and edges. Edges shall be flanged down the same dimension as adjacent non-metal tops with 1 inch being the minimum dimension and return a minimum of 1 inch on underside of top.
3. Solder corners, clean and remove burrs and sharp edges.
4. Provide integral 4 inches high coved side and back splashes at walls and vertical adjacent surfaces.

2.6 SHELVING ASSEMBLIES

A. Shelf Material:
1. Metal Shelving:
   a. Provide 18 GA steel shelves with integral seismic lip and hat-section stiffener with 14 GA, minimum, bookend brackets.
   b. Rear of bracket shall be profiled to fit into slots of shelf standards as described below.
   c. Refer to details on Laboratory Furnishings Drawings.
   d. Provide safety edging at front of shelf.
   e. Provide safety edging at rear of shelf.

B. Adjustable Wall Shelves:
1. Shelving:
   a. Metal shelving as specified above.
   b. High-Pressure Decorative Laminate shelving as specified above.
   c. Solid phenolic shelving as specified above.
2. Double Slot Shelf Standards:
   a. Manufacturers:
      1) Base:
         a) Knape & Vogt Manufacturing Co.
      2) Optional:
         a) Fixture Hardware Manufacturing.
         b) Other manufacturers desiring approval comply with Section 01 67 00.
         c) Substitutions are not permitted.
   b. Base Product: Knape & Vogt 85 ANO series uprights. Length as indicated on the drawings.
3. Shelf standards shall be slotted channel framing as specified elsewhere in this Section.
4. Shelf Brackets: 16 GA cold formed steel, bookend type, as detailed on drawings.

a. Unless indicated otherwise on Drawings:
1) Brackets shall be full depth of shelf,
2) Height: 5 inches above top of shelf.
3) Radius: 3 inches.
4) Rear edge of bracket shall be profiled for attachment to shelf standards.

5. Finish: Factory finish standards and brackets, and metal shelving when specified, with epoxy powder coating.
a. Color to be selected by the Contracting Officer Representative.

2.7 GAS AND LIQUID CYLINDER RESTRAINT ASSEMBLY

A. Cylinder Rack Assembly (Steel Tube Assembly):
1. Manufacturers:
a. Safe-T-Rack Systems, Inc.
b. Spectra Gases, Inc.
c. Matheson Tri·Gas.
d. Air Liquide.
e. Other manufacturers desiring approval comply with Section 01 67 00.
f. Substitutions are not permitted.
2. Frame members: 2 inches by 2 inches by 1/8 inches square steel tube.
3. Construction:
a. All welded construction.
b. Weld cover plates to close exposed tube ends.
c. Grind and polish welds to produce smooth surface with no visible evidence of welding when painted.
d. Frame shall have welded chain link or similar hardware for attachment for each end of chain.
e. Finish:
1) Powder coated factory finish.
2) Color selected by Contracting Officer Representative.
4. Retainer Rods:
a. Provide 1/4 inches diameter stainless steel retainer rods with turned down ends.
b. Drill 9/32 inches diameter holes in top rails to receive retaining rods.
5. Chain and Associated Hardware:
a. All chain and hardware shall be Type 304 or Type 316 stainless steel.
b. Chain: 1/4 inches diameter, welded chain. Two chains per cylinder at high and low points.
c. Provide threaded connector chain link at each end and one harness clip at the one end: McMaster-Carr Supply Company, Suncor Stainless, Inc. Other manufacturers desiring approval comply with Section 01 67 00.

B. Gas and Liquid Cylinder Restraint Assembly:
1. Slotted Framing Channel, Fittings, Swivel Hangers, and End Caps:
a. Slotted channel framing as specified elsewhere in this Section.
1) Color: To be selected by Contracting Officer Representative.
b. Provide two swivel hangers per gas cylinder or liquid cylinder per wall bracket.
2. Chain and Associated Hardware:
a. All chain and hardware shall be Type 304 or Type 316 stainless steel.
b. Chain:
1) 1/4 inches diameter welded chain.
2) Chain shall be long enough to secure 9 inches gas cylinder or 20 inches liquid cylinder, as indicated on drawings.
c. Provide one threaded connector chain link at each end, and one harness clip at one end: McMaster-Carr Supply Company, Suncor Stainless, Inc. Other manufacturers desiring approval comply with Section 01 67 00.
d. Provide two assemblies per gas or liquid cylinder to anchor at a high and low point.

2.8 METAL PIPE DROP ENCLOSURE

A. Fabricate pipe drop enclosures from minimum 18 GA galvanized steel, per details shown on the Laboratory Furnishing drawings, except as noted.
   1. Pipe drop enclosures in Controlled Temperature Rooms shall be fabricated of 18 GA, Type 304 stainless steel with a No. 4 finish.

B. Seal joints between dissimilar metals and at panel seams with clear sealant for typical applications. Do not seal removable panel to fixed elements.

C. Seal joints between dissimilar metals, panel seams, and removable panel to fixed elements with clear sealant for enclosures in BSL-3 applications.

D. Materials and finish shall be as specified under Metals Fabrications in this Section.

2.9 DRYING RACK

A. Stainless Steel Drying Rack with White Polypropylene Pegs:
   1. Manufacturers:
      a. Base:
         1) Inter Dyne Systems, Inc.
      b. Optional:
         1) Other manufacturers desiring approval comply with Section 01 67 00 (no known equal).
   2. Base Product: Inter Dyne Systems V- Victoria Series, modified as indicated on the drawings.
      a. Body: One-piece design.
         1) Material 20 GA Type 304 stainless steel with a No. 4 finish.
         2) Construction:
            a) The top shall have two 90-degree bends, and sides shall have one 90 degree bend.
            b) Punch with T-shaped holes to accommodate the peg arrangement shown on the drawings. Holes shall be designed to support pegs for holding single or multiple utensil drip trays, drain shelves, funnel racks or pipette holders.
      b. Drip Trough: Integral, full-width
         1) Material: 20 GA, Type 304 stainless steel.
         2) Construction:
            a) Stainless steel drain tube.
            b) Continuously welded at corners.
            c) Screen Insert: Full-length, Type 304 stainless steel wire mesh. Turn down on sides.
      c. Pegs:
         1) Material: Injection molded white polypropylene.
         2) Peg base shall have a T-shaped protrusion to allow easy removal and replacement without the need for tools. Pegs shall fit into the punched holes in the rack body.
      d. Wall Hangers: Provide for each rack, designed to enable the removal and replacement of the entire rack for cleaning without the need for tools.
      e. Hardware: Provide stainless steel fixing screws of appropriate type for attachment to support structure.
      f. Drain Tube: Clear, tight-fitting hose to drain from drip trough drain tube into sink.
      g. Back Panel: Provide finished stainless steel back panel when any portion of the back of drying rack is exposed.

2.10 METAL FABRICATIONS AND FINISH REQUIREMENTS

A. Applicability:
   1. Applies to metal fabrications specified in this Section, including, but not limited to, pipe drop enclosures, radioisotope storage cabinets, shelving support systems, metal-framed laboratory tables, metal-framed balance tables, cylinder racks, and other miscellaneous

brake-formed and shop fabricated components and trim, such as required for overhead service carriers.

B. Materials:
   1. Cold rolled sheet steel: Prime grade, roller leveled, and treated at the mill to be free of scale, ragged edges, deep scratches, or other injurious effects.
      a. Recycled Steel Content: Post-consumer recycled content plus one-half of the pre-consumer content shall be 20 percent, minimum, of the cost of the total value of materials.

C. Finish Requirements:
   1. Description:
      a. Finish Type: Dry powder coating.
      b. Performance Requirements: Meet or exceed SEFA 8 M Cabinet Surface Finish requirements.
      c. Operator Protection:
         1) Convenient and easily mastered through robotic application plus manual detailing.
         2) Process shall be contained and have no solvent odor and be performed in an air conditioned room.
      d. Overspray:
         1) Overspray shall be captured and re-sprayed.
         2) 99 percent efficiency in coating usage, reducing waste generated.
         3) A closed collection system shall be used for overspray that is not reused.
         4) Powder overspray shall not escape the facility and shall be collected in bulk, eliminating the need for daily replacement/disposal of filter media.
      e. VOC Emissions:
         1) Sprayed and baked with a near zero (0.29 pounds/gallon maximum) VOC (Volatile Organic Compounds) emissions.
         2) Comply with the GS (Green Seal Standard) 11 allowable emissions.
      f. Off gassing:
         1) Finish shall be firm and stable after final cure.
         2) No further emissions or off gassing/decomposition vapors shall occur at room temperature.
   2. Metal Casework Color: Selected by the Contracting Officer Representative from manufacturer's full color line.

2.11 STAINLESS STEEL FABRICATIONS

A. Applicability:
   1. Applies to stainless steel fabrications specified in this Section, including, but not limited to, work surfaces, drying racks, canopy hoods, sinks, low slot exhausts, and other miscellaneous brake-formed and shop fabricated components and trim.

B. Materials:
   1. Stainless Steel Sheet: Type 304, unless otherwise noted, and of gauge indicated on Laboratory Furnishing drawings or this specification.
      a. Recycled Steel Content:
         1) Post-consumer recycled content plus one-half of the pre-consumer content shall be 20 percent, minimum, of the cost of the total value of materials.
   2. Stainless steel nuts, screws, bolts, and rivets, etc.: Same type stainless steel as in the sheet material and shall have a tumbled finish closely resembling that of a No. 4 finish.
   3. Welds:
      a. Welding material shall be of type similar to the sheet material or a richer quality.
      b. All welds shall be made without discoloration and shall be ground, polished, and passivated to blend harmoniously with the specified finish.
      c. Joints in stainless steel tops and work surfaces shall be welded, unless noted otherwise.

C. Finish: Exposed surfaces shall be ground and polished to a No. 4 finish, unless noted otherwise for one-piece, seamless appearance.

D. Fabrication:
   1. Form to shapes indicated with straight lines, sharp angles, and smooth curves.
   2. Drill or punch holes with smooth edges for temporary field connections and attachment of work by other trades.
   3. Make permanent shop and field connections with continuous fillet type welds.
   4. Grind exposed welds smooth.
   5. Shop-fabricate in as large assemblies as practicable.
   6. Qualify welding processes and welding operators in accordance with American Welding Society.
   7. Supply items required to complete construction and installation.

E. Work Surfaces:
   1. Thickness:
      a. Typical: 16 GA.
      b. Tops with Field-Welded Joints: 14 GA.
   2. Fabrication:
      a. Edges:
         1) Turn down flange in the same dimension as the adjacent non-stainless steel top, if any, with 1 inch being a minimum.
         2) Return 1/2 inches, minimum, over reinforcing channels.
      b. Reinforcement:
         1) General: 16 GA stainless steel channels or equivalent at formed edges, rear and middle top, below surface, to ensure rigidity and prevent buckling, warping, or oil canning.
         2) At bench mounted fittings: Provide reinforcement to allow for rigid, secure mounting of fittings.
      c. Undercoating: Waterborne and non-flammable in liquid state, with no volatile organic compounds (VOCs), providing sound deadening and to prevent condensation.
         1) Undercoating shall not be provided at any BSL-3, ABSL-3, BSL-3Ag, BSL-4, and ABSL-4 locations.
      d. Stainless steel side and back-splashes:
         1) Integrally welded to top, coved, and finished as indicated above.
         2) Back side of exposed backsplashes shall be finished to match front and sides.
      e. Joints: Electrically weld shop joints; grind smooth and polish. Field joints may be bolted with full length supported, resulting in a hairline seam with flat, level surfaces on each side of joint.
      f. Provide holes and cutouts as required for built-in equipment and mechanical and electrical service fixtures.
         1) Verify size of opening with actual size of equipment to be used prior to making openings.
         2) Form inside corners to a radius of not less than 1/8 inches.
         3) After sawing, rout and file cutouts to ensure smooth, crack-free edges with no burrs.
      g. Marine edges: Where indicated on the Laboratory Furnishing Drawings:
         1) 1 inch wide and 1/4 inches high with chamfered or radiused transition to and be an integral part of the work surface.
         2) Seamless die-formed edges.
   3. Tops with Sinks:
      a. Integral, fabricated with a marine edge, pitched to sink bowl for proper drainage.
      b. Provide sanitary radii where sink bowls meet drainboards.
   4. Wall-Supported Benchtop:
      a. Provide stainless steel wall support and bracket angles as indicated on Laboratory Furnishings Drawings.

b. Refer to Wall Hung Benchtop requirements elsewhere in this Section.

F. Cup Sink:
1. Integral, one piece construction with top.
2. Cup sinks at fume hoods shall have 1/4 inches high lip.
3. Undercoating: Waterborne and non-flammable in liquid state, with no volatile organic compounds (VOCs), providing sound deadening and to prevent condensation.

G. Laboratory Sink: Integral one piece construction with stainless steel work surface.
1. Thickness: 16 GA, unless otherwise noted.
2. Construction:
a. Provide sufficient reinforcement to prevent oil canning.
b. Joints:
1) Butt-welded, ground smooth by the heliarc welding process.
2) Inside radii shall be 1 inch, minimum, 1 3/4 inches maximum.
3) Pitch bottom to the drain indent.
4) No soldering will be permitted in connection with sink construction.
c. Sink bowl dimensions provided are inside dimensions.
d. Undercoating Waterborne and non-flammable in liquid state, with no volatile organic compounds (VOCs), providing sound deadening and prevent condensation.
1) Undercoating shall not be provided at any BSL-3, ABSL-3, BSL-3Ag, BSL-4, and ABSL-4 locations.

H. Canopy Hood: Provide stainless steel canopy with hangers and miscellaneous hardware at locations and sizes as indicated on the Laboratory Furnishing drawings.
1. Canopy hood shall be constructed to avoid interference with autoclave pressure gauges if located in fascia panel above autoclave door. Coordinate with autoclave manufacturer.
2. Materials:
a. Type 304 stainless steel.
1) Finish, exposed to view components: No. 4.
b. Canopy Hood, Baffle, and Fascia Panel Thickness: 18 GA, minimum.
c. Hood Support Brackets; 16 GA, 1-1/4 inches wide.
d. Baffle Support:
1) Stainless steel slotted channel framing.
2) Brackets: 16 GA, 1-1/4 inches wide.
e. Threaded Rod: 3/8 inches minimum.
f. Hardware and Fasteners: Stainless steel nuts and washers
3. Construction:
a. Provide reinforcing necessary to prevent oil canning or deflection of panel between supports.
b. Continuously weld corners and joints, grind smooth, free of defects. Welded joints with visible burn marks will not be accepted.
c. Form with 1/2 inches deep condensation gutter with continuous welds to be watertight.
d. Provide welded 18 GA exhaust collar with no open seams.
4. Accessories: Provide stainless steel hangers and miscellaneous support hardware as required for a complete installation.
5. Light fixtures and wiring shall be provided under Division 26.
a. Holes for electrical work shall be made by canopy hood fabricator.
6. Provide exhaust duct transition pieces for mechanical connection above the ceiling.
7. Refer to Exhaust Schedule for exhaust rate and Mechanical Drawings for duct connection sizes.

2.12 SEALANT

A. Refer to Section 07 92 00.

B. Sealant shall be installed by installer of the work of this Section.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Prior to installation of the work of this Section, verify installed work is complete to the point where this installation may properly commence.

B. Verify that work may be installed in complete accordance with the original design, reviewed submittals, and the manufacturer's recommendations.

C. Verify adequacy and proper location of any required backing or support framing.

D. Verify that mechanical, electrical, plumbing, and other building components affecting work in this Section are in place and ready.

E. Installation of the Work constitutes Contractor's acceptance of underlying work, substrates, and field conditions prevailing at the time of the Work.

3.2 INSTALLATION

A. Installation of items specified in this Section shall be performed by installers experienced in the installation of the respective item as determined by the respective manufacturer.

B. Coordinate work with any Government furnished and/or installed components indicated on drawings.

C. Casework:
   1. Set casework items square, plumb, and level. Shim as required, using concealed shims and securely anchor.
   2. Where required, assemble units into one integral unit with joints flush, tight, and uniform.
   3. Provide matching filler pieces where casework abuts walls or columns or should be closed off.
      a. All wood, plastic laminate, solid phenolic and polypropylene work abutting other building components shall be properly scribed.
   4. Mechanical fasteners used at exposed and semi-exposed surfaces, excluding installation attachment screws and those securing cabinets end to end, shall be countersunk.
   5. Cutouts for equipment, mechanical, electrical, or plumbing services shall be made by the casework manufacturer or casework installer.
   6. Install hardware uniformly and precisely. Set hinges snug and flat in mortises. Adjust doors, drawers, hardware, fixtures and other moving or operating parts to function smoothly and correctly, and so that doors and drawers operate without warp or bind, and contact points meet accurately.
   7. Secure base cabinets to service struts and ledging, partition framing, wood blocking, or reinforcements in partitions with fasteners spaced not more than 24 inches on center. Bolt adjacent floor mounted cabinets together with joints flush, tight, and uniform.
   8. Floor mounted casework shall receive top set or integral base as specified under Division 09 and on the Finishes drawings.

D. Laboratory Tops:
   1. Scribe tops as necessary for close and accurate fit.
   2. Field Joints: Factory-prepared and identical to factory joints, locate only where indicated on approved Shop Drawings. Field processing of top and edge surfaces is not acceptable, except as described by manufacturer in approved Submittal Data. Provide full length, one-piece tops and backsplashes wherever possible, and keep field joints to an absolute minimum.
   3. Abut top and edge surface in one true plane, with internal supports placed to prevent any deflection. Joints in top units shall be flush and the narrowest for the respective materials of construction.
   4. Epoxy Resin: Cement joint in accordance with the manufacturers' specifications.
   5. Stainless Steel: Field-weld joints in stainless steel tight, without open seams. Finish material to match adjacent to weld.

E. Tolerances: Casework shall not exceed the following tolerances:
   1. Variation of Bottoms of Wall Cabinets from Level: 1/8 inches in 10 feet.
   2. Variation of Cabinet Faces from a True Plane: 1/8 inches in 10 feet.
   3. Variation of Adjacent Surfaces from a True Plane: 1/32 inches.
   4. Variation in Alignment of Adjacent Door and Drawer Edges: 1/16 inches.
   5. Variation of Work Surfaces from Level: 1/16 inches in 10 feet.

F. Laboratory Sinks:
   1. Epoxy Resin, Undermount: Sinks shall be set in work surface with chemical-resisting sealing compound, secured and supported in accordance with manufacturer's instructions. Adjust sink and securely support to prevent movement. Remove excess sealant or adhesive while still wet and finish joint for neat appearance.
   2. Epoxy Resin, Drop In: Set sink in adhesive and fill remainder of rabbet with sealant or adhesive. Use procedures recommended by sink and work surface manufacturers. Remove excess adhesive and sealant while still wet and finish joint for neat appearance.
   3. Epoxy Resin and Polypropylene Cup Sinks, Drop In: Set sink in adhesive and fill remainder of rabbet with sealant or adhesive. Use procedures recommended by sink and work surface manufacturers. Remove excess adhesive and sealant while still wet and finish joint for neat appearance.
   4. Epoxy Resin and Polypropylene Cup Sinks, Top Mount: Set sink in adhesive. Use procedures recommended by sink and work surface manufacturers. Remove excess adhesive and sealant while still wet and finish joint for neat appearance.
   5. Stainless Steel, Top Mount: Before setting, apply sink and work surface manufacturers' recommended sealant under rim lip and along work surface. Remove excess sealant while still wet and finish joint for neat appearance.

G. Miscellaneous Furnishings and Accessories:
   1. Install in accordance with manufacturer's instructions.
   2. Securely fasten wall mounted adjustable shelving supports, stainless steel shelves, drying racks, etc. to partition framing, wood blocking, or reinforcements in partitions.
   3. Install shelf standards plumb and at heights to align shelf brackets for level shelves. Install shelving level and straight, closely fitted to other work where indicated.
   4. Tighten screws to seal flat; do not drive.

H. Sealant:
   1. Caulk edges of tops, backsplashes, and side splashes to adjacent wall surface, and around work surface penetrations, with sealant.
   2. Sealant application shall be in accordance with manufacturer's published recommendations.

I. Repair or remove and replace defective work as approved by the Contracting Officer Representative at no additional cost to the Government.
   1. Where repair is approved by Contracting Officer Representative, touch-up finishes applied to damaged surfaces shall have a VOC content of no more than 250 g/L in accord with SCAQMD Rule #1168.

3.3 CLEANING

A. Clean counter tops with diluted dishwashing liquid and water leaving tops free of grease and streaks. Use no wax or oils.

B. Before completion, adjust moving and operating parts to function smoothly and correctly.

C. Damaged work shall be replaced, repaired, or restored to original condition to the approval of the Contracting Officer Representative at no additional cost to the Government.
   1. Nicks, chips, and scratches in the finish shall be filled and retouched. Damaged items that cannot be repaired shall be replaced.

3.4 PROTECTION

A. Protect work of this section before, during and after installation including installed work and materials of other trades.

B. Cover work surfaces with 1/4 inches corrugated cardboard, secured in place, after installation for protection against scratching, soiling, and deterioration during remainder of construction period. Remove protection prior to final cleaning.

C. Standing or staging work on protected or unprotected work surfaces is not allowed.

D. Replacement: Damaged work shall be replaced, repaired, and restored to original condition to the approval of the Contracting Officer Representative at no additional cost or inconvenience to the Government.

END OF SECTION

DIVISION 13

SPECIAL CONSTRUCTION

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SECTION 13 47 13.17
CATHODIC PROTECTION OF STEEL WATER TANKS - INTERNAL GALVANIC

PART 1 - GENERAL

1.1 DESCRIPTION

A. This section includes material, design, installation, and testing for a complete galvanic cathodic protection (CP) system for internal submerged surfaces of steel water tanks including:
   1. New Raw Water Tanks at the North Rim WTP – Volume 6.
   2. Existing Finished Water Tanks at the North Rim WTP – Volume 6.

B. The system shall have ice-free anodes/rope support system. The system will use diodes to control anode output limiting the ON potentials. The system is designed to operate 20 years with no maintenance.

C. CP installer will protect coatings to prevent damage during installation and coordinate with coatings contractor on timing when the coatings are ready for CP installation. Clips for anode support shall be installed before coatings are applied.

D. Reference Specifications, Codes, and Standards.
   1. AWWA, American Water Works Association:
      a. D100, Welded Carbon Steel Tanks for Water Storage.
      b. D102, Coating Steel Water-Storage Tanks.
   2. Association for Materials Protection and Performance (AMPP) formerly NACE International (NACE):
      a. SP0196, Galvanic Anode Cathodic Protection of Internal Submerged Surfaces of Steel Water Storage Tanks.
      b. SP0169, Standard Recommended Practice Control of External Corrosion on Underground or Submerged Metallic Piping Systems.
   3. National Fire Protection Association (NFPA):
      a. 70, National Electric Code (NEC).
   4. Cordage Institute
      a. CI 1500-02 International Standard Test Method for Ropes
   5. MIL-R-24677 Rope, Fibrous, Double-Braded (Polyester)
   6. UL
      a. UL 486C Splicing Wire Connectors
      b. UL 467 Grounding and Bonding Equipment

1.2 QUALIFICATIONS

A. Corrosion Expert: Accredited or certified by AMPP formally NACE International Accredited Corrosion Specialist or Certified Cathodic Protection Specialist or registered professional engineer. Corrosion expert shall have five years of experience in corrosion control of metallic surfaces of the type under this contract.

B. Corrosion Technician: Accredited or certified by AMPP as a Cathodic Protection Technician with experience in the installation and maintenance, of the type under this contract.

C. The Contractor shall have a minimum of five years experience installing and servicing internal water tank CP.

1.3 SYSTEM DESCRIPTION

A. Obtain the services of a Corrosion Expert to supervise, inspect, and test the installation and performance of the cathodic protection system in complete compliance with:
   1. AMPP/NACE 0196 AND SP0169.

2. NFPA 70.
3. All applicable Federal, State and Local regulations.
4. Applicable AMPP/NACE standards.
5. Minimum requirements of this Specification Section and additional Specifications under this contract.

1.4 SUBMITTALS

A. Certifications:
   1. Corrosion Expert qualifications.
   2. Corrosion technician qualifications.

B. Shop Drawings:
   1. Submitted by the Corrosion Expert.
   2. Include the following:
      a. Anodes.
      b. Reference electrodes.
      c. Anode support rope.
      d. Anode support clips.
      e. Wire, and splicing materials.
      f. Anode Junction box, shunts, and electronic components.
      g. Pressure entrance fittings.
      h. Drawings depicting system design and configuration for alternate or modifications of the design submitted with calculations.
      i. Sketch and documentation for any modification of installation details.

C. Record Drawings
   1. Show the exact locations of all buried wires, rectifiers, anode wells, and CP test boxes using dimensional ties to existing structures or survey monuments.
   2. Depict tank layout including at a minimum; access hatches, ladders, support columns, drain and fill lines, and CP system for any variance from the drawings.
   3. Anode and anode support system layout.
   4. Location of reference electrodes.
   5. Electrical wire, and splices.
   6. Pressure entrance fittings.
   7. Junction boxes.

D. FINAL REPORT
   1. Include the following:
      a. Statement that system was installed per approved shop drawings and submittals and tank meets AMPP/NACE criteria for CP without overprotection.
      b. Measured tank potentials and anode output measured during installation, native, On, and polarized protected potential with the anodes interrupted. Record these potentials with reference electrodes and portable reference electrode at 5-foot intervals from the tank floor to the water surface.
      c. Photos of installation showing key components.

1.5 DESIGN

A. The criteria for protection shall be based on a tank-to-water polarized potential, within a range of -0.850 volts to -1.050 volts relative to a copper-copper sulfate reference electrode.

B. The anode system shall be designed for uniform current distribution.
   1. Small tanks shall use a single fixed anode hoop system.
   2. Larger tanks shall have additional anode hoops as required. System shall have a minimum anode design life of 20 years.

C. A minimum of two reference electrodes shall be used at locations where minimum and maximum current densities are expected.

D. The anodes shall be designed for "ice-free" operation with all rigging located and attached to the interior wall as indicated in the Drawings.
   1. Rigging and wiring supported from, or which penetrates the roof, are not permitted.

1.6 DELIVERY, STORAGE AND HANDLING

A. All materials shall be protected from damage from weather, vandalism, and work.

1.7 WARRANTY

A. One year from CP system startup and acceptance by Government. If the system is tested and not left operating, the warranty shall start when the system is left in operation after any time for the coating to "cure".

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturers listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable.
   1. Cathodic protection system:
      a. Farwest Corrosion.
      b. Or equal.

2.2 MANUFACTURED UNITS

A. COMPONENTS
   1. Pressure entrance fitting shall provide leak-free interface between the internal and external wiring.
      a. Pressure entrance fitting shall be installed in accordance with AWWA D100-96, Section 3.13.
      b. The pressure entrance fitting shall accommodate anode and reference electrode lead wires at the base of the tank per Drawings.
      c. The fitting shall be manufactured to prevent leakage through the fitting and to prevent water migration through the wire insulation.
   2. Anodes extruded rod magnesium 1.6 to 1.7V (with respect to copper-copper sulfate reference electrode). Solid Steel core extending a minimum of 3 5/8-inch from anode extrusion. Length per plans 0.9-inch diameter nominal, 0.516 lb./ft, and 0.135 in diameter steel core. Anodes suitable for use in fresh water with no mercury or other hazardous metals.
   3. Chemical composition:

Material	Mg (%)	Al (%)	Mn (%)	Zn (%)	Cu (%) max	Ni (%) max	Fe (%) max	Other (%) max	
								Each	Total
Ultra Mag	Balance	0.010 max	0.50 to 1.3	--	0.02	0.001	0.03	0.05	0.3

4. Split bolts, high strength bronze alloy. UL listed to UL486C, UL 467.
5. Reference Electrode(s):
    a. The reference electrode shall consist of a copper-copper sulfate electrode which is manufactured to remain stable (plus or minus 10mV) for a minimum of 10 years. The reference electrode to lead wire connection shall be encapsulated to prevent water migration.
6. Anode rope support shall be double braid polyester 5/8 in. Minimum tensile strength 16,000 lb., with a 2% elongation at 20% load with CI 1500-02 and MIL-R-24677 certifications.
7. Quick links shall be 5/16 zinc plated steel with a working load limit (ULL) of 1,540 lb. with a threaded closure that allows for secure attachment and detachment of chains, ropes, or cables. Rounded body to evenly distribute loads, minimizing stress on connected components.
8. Heat shrink tube shall be semi-rigid, adhesive-lined, flexible polyolefin. UL Recognized 600 Volts and 125 ºC maximum. The operating temperature range is from -55ºC (-67ºF) to 110 ºC (230ºF), with a shrink temperature minimum of 121ºC (250 ºF).
9. Optional splice kits two (2)-part resin power cable designed for direct bury, 3M Scothchcast or equal. Dielectric strength (ASTM D149) 500 V/mill, tensile strength (ASTM D412) 4900 psi (33.8MPa), and glass transition temperature (ASTM E1356-03) 129F (54C).
10. Anode hoop support clips and reference electrode supports shall be fabricated from 3/8 in. mild steel rod. Ensure loop diameter will fit quick clip.
11. Junction box shall be 16-guage galvanized steel rated NEMA 3R with a hasp for padlock. Finish shall be ANSI 61 grey polyester powder finish inside and out over galvanized steel.
    a. Shunts, HOLLOEWAY type SS 0.01ohm rated 25 Amp.
    b. Buss bar ¼ in. x ½-inch copper.
    c. Diode for current control shall be Vishay P600M or equal, 1000 V maximum repetitive peak reverse Voltage, maximum average forward current 6.0 Amp at 25C.
12. Wiring, all wiring shall be stranded copper.
    a. Internal tank wiring:
        1) Wiring shall be compatible for wet and submerged applications and have no nicks or defects in the insulation. Wiring shall be RHH/RHW-2 or XLPE insulation color per the drawings. All internal splicing shall be completed with splice kits.
    b. Reference electrode wiring shall be #14AWG stranded copper with yellow RHH/RHW-2 insulation.
    c. External tank wiring:
        1) Structure, #10AWG stranded copper with RHH/RHW-2 black insulation.
        2) Anode, #10AWG stranded copper with RHH/RHW-2.

2.3 MISCELLANEOUS COMPONENTS

A. Wire markers:
    1. All wires shall have insulation per color code, see section 2.02.
    2. Panduit wire markers shall be used on each wire terminated in the junction box identifying reference electrode location and anode location.

2.4 ANSI/NSF 61

A. All materials in contact with the water or exposed to the interior of the tank shall be classified in accordance with ANSI/NSF 61 "Drinking Water System Components."

PART 3 - EXECUTION

3.1 PERFORMANCE

A. All Work shall be in accordance with the following requirements.

B. Components of the cathodic protection system shall be installed in the manner and at the locations as shown on the Drawings or approved modifications submitted by the Corrosion Expert.

C. Pressure entrance fitting shall be installed in accordance with AWWA D100-96, Section 3.13.

D. Welding, cutting, and coating shall be in accordance with AWWA Standards D100, D102, and D105. Steel work shall be completed before any tank coatings are applied.

E. Materials and equipment shall be inspected prior to installation. Any defective component shall be repaired or replaced.

F. Lead wires shall be installed to prevent damage from abrasion. Route internal wires along support ropes to pressure entrance fitting from the anodes and reference electrodes.

G. Electrical connections within the tank shall be sealed to prevent water migration.

H. The junction box shall be mounted at a convenient height (3-5 ft) above grade for monitoring and service purposes.

I. Disinfection shall be the responsibility of the Contractor.

3.2 INITIAL INSTALLATION

A. Before any coating work is complete for tanks or during welding repairs of existing tanks where new supports or pressure fitting is required, install:
   1. The anode support hoops, reference electrode supports shall be welded to the tank shell interior.
   2. A 1-inch internal threaded fitting shall be welded into the shell near the junction box for each required pressure entrance fitting. Weld all around interior and exterior.
   3. Finished water tank (larger diameters with multiple anode hoops) will require two (2) pressure entrance fittings. Place plug into threads to prevent coating buildup.

3.3 ANODE INSTALLATION

A. Install pressure entrance fitting into pipe threaded fitting welded into tank shell with sealant for a watertight connection after all tank coating is complete and accepted.

B. Install anode rope supports per drawings, use cinch knots to apply tension to all ropes. Leave adequate free rope at knots and secure with half-hitches and tape to prevent pull through. Field verify exact dimensions and locations of tank support columns.

C. Make loop in anode steel core after applying heat shrink to the inner anode location leaving space to connect jumper wires between anodes.

D. Clip anodes to rope supports and make wiring connections as shown on the Drawings.

E. Route anode wiring to pressure entrance fitting using rope support to secure wiring.

F. Connect anodes and reference electrodes to pressure entrance fitting wiring with splits bolts and heat shrink or wire splice kits to encapsulate all copper.

G. Install junction box at wiring feed thru and complete wiring from anodes terminating anodes at shunts through diode control. Provide structure connection exothermically welded to tank and terminate on the copper buss bar.

3.4 FIELD QUALITY CONTROL

A. The Contracting Officer's Representative shall have the option to witness the installation and energization of the system.
   1. Contractor shall notify the Contracting Officer's Representative seven (7) days in advance of the installation and energization of the system.

3.5 STARTUP SERVICE / ENERGIZING SYSTEM

A. After the system is installed and the tank is filled, the Contractor shall provide startup service which includes energizing, testing, and adjusting the system for optimum performance of the cathodic protection system. This startup service shall be coordinated with the Contracting Officer's Representative.

B. All tank-to-water potential measurements shall be conducted with a calibrated portable copper-copper sulfate reference electrode and a portable high impedance voltmeter.

C. Native potentials shall be recorded for each measurement in Parts D and E before the final connection of the structure to the anodes in the anode junction box. The Contractor shall give the system 24hours to polarize before recording final potentials in Parts D, E, and F.

D. Polarized potentials shall be measured with a portable calibrated reference electrode at 5 ft intervals from the floor to the water surface with the anode current interrupted. Coordinate with the Contracting Officer's Representative to conduct final tests with the tanks full if possible. All test data shall be reviewed and evaluated by the Corrosion Expert.

E. Criteria for protection shall be -850 mV structure to water measured with a copper-copper sulphate reference electrode. Or 100 mV polarization per NACE SP0169. Polarized potentials from any measurement, portable reference electrode or fixed reference electrodes shall not exceed -980 mV for well coated tanks.

F. Measure and record rectifier junction box shunts, DC Voltage anode to tank.

3.6 TRAINING:

A. See Section 01 79 00 for training requirements.

3.7 PERFORMANCE VERIFICATION

A. The final test and adjustment of the system shall be conducted 12 months after the startup service.

END OF SECTION

DIVISION 22

PLUMBING

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SECTION 22 05 03
PIPE AND PIPE FITTINGS - PLUMBING SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Plumbing piping material and accessories as shown on the plumbing drawings.
   2. Plumbing piping accessories:
      a. Flex connectors.
      b. Pressure Gauges.
      c. Wye strainers.
      d. Dielectric fittings.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 73 20 - Openings and Penetrations in Construction.
   2. Section 09 96 00 - High Performance Industrial Coatings.
   3. Section 22 05 23 - General Duty Valves for Plumbing Piping.
   4. Section 22 05 29 - Hangers and Supports for Plumbing Piping and Equipment.
   5. Section 40 42 00 - Pipe, Duct and Equipment Insulation.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B1.2, Gages and Gaging for Unified Screw Threads.
      b. B31.9, Building Services Piping.
      c. B40.100, Pressure Gauges and Gauge Attachments.
   2. ASTM International (ASTM):
      a. A53, Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
      b. A74, Standard Specification for Cast Iron Soil Pipe and Fittings.
      c. A126, Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings.
      d. A234, Standard Specification for Pipe Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service.
      e. B32, Standard Specification for Solder Metal.
      f. B88, Standard Specification for Seamless Copper Water Tube.
      g. C564, Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings.
      h. D1784, Standard Classification System and Basis for Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds.
      i. D1785, Standard Specification for Poly (Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120.
      j. D2466, Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40.
      k. D2665, Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings.
      l. D2855, Standard Practice for the Two-Step (Primer and Solvent Cement) Method of Joining Poly (Vinyl Chloride) (PVC) or Chlorinated Poly (Vinyl Chloride) (CPVC) Pipe and Piping Components with Tapered Sockets.
      m. F493, Standard Specification for Solvent Cements for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe and Fittings.
   3. American National Standards Institute (ANSI/ASTM):
      a. B16.3, Malleable Iron Threaded Fittings.

b. B16.18, Cast Copper Alloy Solder Joint Pressure Fittings.
c. B16.22, Wrought Copper and Bronze Solder - Joint Pressure Fittings.
d. B16.26, Cast Copper Alloy Fittings for Flared Copper Tubes.

4. American Water Works Association (AWWA):
   a. B300-10, Standard for Hypochlorites.
   b. C651, Disinfecting Water Mains.
5. American Welding Society (AWS):
   a. A5.8M/A5.8, Specification for Filler Metals for Brazing and Braze Welding.
6. Cast Iron Soil Pipe Institute (CISPI):
   a. 301, Standard Specification for Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications.
7. National Sanitation Foundation (NSF):
   a. 61/372, Drinking Water System Components.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Product technical data including:
      a. Manufacturer catalog data sheets for pipe fittings and accessories furnished under this section indicating materials of construction, dimensions, pressure and temperature ratings.
      b. Acknowledgement that products submitted meet requirements of standards referenced.
      c. Copies of manufacturer's written directions regarding material handling, delivery, storage and installation.
      d. Separate schedule sheet for each piping system scheduled in this Specification Section showing compliance of all system components.
         1) Attach technical product data on gaskets, pipe, fittings, and other components.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

C. Informational Submittals:
   1. Qualifications of lab performing disinfection analysis on water systems.
   2. Test reports:
      a. Copies of pressure test results on all piping systems.
      b. Disinfection test report.
      c. Notification of time and date of piping pressure tests.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Piping Systems:
   1. Piping systems are scheduled by service in PART 3 of this specification.
   2. Piping systems scheduled herein are applicable to piping systems shown on the plumbing drawings.

B. Coatings:
   1. Where piping systems are specified to be coated, provide pipe coatings in accordance with Section 09 96 00.

C. Pipe and Tube:
   1. Copper:
      a. Type K or L tube per ASTM B88.

b. Utilize only annealed (soft) type tubing where flared joints are used and drawn temper (hard) type tubing where soldered or brazed joints are used.
c. Fittings, per System Type:
    1) Cast copper or bronze (pressure, solder): Per ASTM B16.18.
    2) Wrought copper or bronze (pressure, solder): ASTM B16.22.
d. Joints:
    1) Soldered or Brazed:
        a) Above ground below 180 degrees F: ASTM B32 solder with a tin/antimony ratio of 95/5 and non-corrosive flux.
        b) Above ground 180 degrees F and above: use brazing alloy with melting temperature above 1000 degrees F and suitable flux.
        c) Buried: Silver solder per AWS A5.8M/A5.8.
        d) Provide unions and valves at equipment.
e. Unions: Class 150, bronze.
f. Provide a polyethylene pipe wrap for copper piping located below ground:
    1) Minimum 8 mil thickness, compliant with AWWA C-105.

2. Steel:
    a. Schedule 40 or 80 per ASTM A53.
    b. Finish: Hot-dip galvanized per ASTM A53.
    c. Fittings: Per System Type:
        1) Pressure rating: Class 150.
        2) Malleable iron: Per ASTM B16.3.
        3) Forged Steel: Per ASTM A234.
    d. Joints: Per System Type:
        1) Threaded.
            a) With unions and valves and equipment.
        2) Flanged with rubber gaskets.
        3) Socket or butt welded.
    e. Unions: Class 150, malleable iron, threaded.
3. Cast Iron Soil Pipe
    a. Service weight, ASTM A74.
    b. Fittings, per System Type:
        1) Hubless per CISPI 310.
        2) Hub and spigot per ASTM A74.
    c. Joints, per System Type:
        1) Neoprene gaskets and stainless steel clamp and shield assemblies per CISPI 310.
        2) Rubber gasket joint devices per ASTM C564.
    d. Coatings:
        1) Bituminous.
4. Plastic:
    a. PVC:
        1) Per ASTM D1785 for schedule 40 or 80.
        2) Per ASTM D2665 for Drain, Waste, Vent (DWV), schedule 40.
        3) Fittings, per System Type:
            a) PVC per ASTM D2466 or ASTM D2467 for pressure pipe.
            b) PVC per ASTM D2665 for DWV fittings.
        4) Joints:
            a) Solvent weld per ASTM D2855.
            b) Hub and spigot with solvent weld per ASTM D2855.
    b. CPVC:
        1) Per ASTM D2846, schedule 40.
        2) Fittings: CPVC per ASTM D2846.
        3) Joints: Solvent weld per ASTM F493.

2.2 MANUFACTURED UNITS

A. Unions:
   1. Copper pipe:
      a. Copper ground joint unions for pipe sizes 2 inches and smaller.
      b. Brass flanged unions for pipe sizes larger than 2 inches.

2.3 ACCESSORIES

A. Flexible connectors:
   1. Provide flexible braided bronze connectors at piping connections to water heaters, pumps and vibrating equipment.
   2. Bronze exterior braid and bronze internal corrugated hose.
   3. Solder or threaded end connections, size to match equipment connections.

B. Wye strainers:
   1. 2 inch and smaller:
      a. Bronze body and plug, 150 psi working pressure.
      b. Threaded or soldered end connections
   2. 2-1/2” and larger:
      a. Cast or ductile iron body.
      b. Class 125 flanged connections.
   3. Removable type 304 stainless steel mesh strainer, 0.062 inch perforations.

C. Pressure gauges:
   1. Stainless steel housing.
   2. Dial size: 2-1/2” diameter.
   3. Piping connection size: 1/2 inch, unless noted otherwise.
   4. Liquid filled.
   5. NSF-61/372 compliant.
   6. Units of measure: PSI
   7. Pressure range suitable for the application.

D. Dielectric Fittings:
   1. Lead free components.
   2. Dielectric unions:
      a. Threaded or soldered end connections.
      b. Rated at 175 psi, 180 degrees F.
      c. Provide dielectric gaskets suitable for continuous operation at union rated temperature and pressure.
   3. Dielectric flange kits:
      a. Flat faced.
      b. 1/8 inches thick dielectric gasket, phenolic, non-asbestos.
      c. Suitable for 175 psi, 180 degrees F.
      d. 1/32 inches wall thickness bolt sleeves.
      e. 1/8 inches thick phenolic insulating washers.

E. Pipe Sleeves:
   1. Refer to spec 01 73 20 for pipe sleeves and penetrations through construction.

PART 3 - EXECUTION

3.1 PREPARATION

A. Ream pipe and tube ends.

B. Remove burrs.

C. Remove scale and dirt on inside and outside before assembly.

D. Prepare piping connections to equipment with flanges and unions.

E. Deep open ends of pipe free from scale and dirt.

F. Protect open ends with temporary plugs or caps.

3.2 EXTERIOR BURIED PIPING INSTALLATION

A. Unless otherwise shown on the Drawings, provide a minimum of 3-1/2 feet and maximum of 8 feet earth cover over exterior buried piping systems and appurtenances conveying water, fluids, or solutions subject to freezing.

B. Enter and exit through structure walls, floors, and ceilings by using penetrations and seals specified in Specification Section 01 73 20 and as shown on Drawings.

C. Install expansion devices as necessary to allow expansion and contraction movement.

D. Laying Pipe in Trench:
   1. Excavate and backfill trench in accordance with Specification Section 31 23 33 - Trenching and Backfilling.
   2. Clean each pipe length thoroughly and inspect for compliance to specifications.
   3. Grade trench bottom and excavate for pipe bell and lay pipe on trench bottom.
   4. Install gasket or joint material according to manufacturer's directions after joints have been thoroughly cleaned and examined.
   5. Except for first two joints, before making final connections of joints, install two full sections of pipe with earth tamped along side of pipe or final with bedding material placed.
   6. Lay pipe in only suitable weather with good trench conditions.
      a. Never lay pipe in water except where approved by the Contract Officer Representative.
   7. Seal open end of line with watertight plug if pipe laying is stopped.
   8. Remove water in trench before removal of plug.

E. Lining Up Push-On Joint Piping:
   1. Lay piping on route lines shown on Drawings.
   2. Deflect from straight alignments or grades by vertical or horizontal curves or offsets.
   3. Observe maximum deflection values stated in manufacturer's written literature.
   4. Provide special bends when specified or where required alignment exceeds allowable deflections stipulated.
   5. Install shorter lengths of pipe in such length and number that angular deflection of any joint, as represented by specified maximum deflection, is not exceeded.

F. Anchorage and Blocking:
   1. Provide reaction blocking, anchors, joint harnesses, or other acceptable means for preventing movement of piping caused by forces in or on buried piping tees, wye branches, plugs, or bends.
   2. Place concrete blocking so that it extends from fitting into solid undisturbed earth wall.
      a. Concrete blocks shall not cover pipe joints.
   3. Provide bearing area of concrete in accordance with drawing detail.

G. Install underground hazard warning tape per Specification Section 10 14 00.

H. Install dielectric unions or flanges where dissimilar metals are joined together.

3.3 INTERIOR PIPING INSTALLATION

A. Install piping in vertical and horizontal alignment as shown on Drawings.

B. Install piping according to Drawing intent and with clearance and allowance for:
   1. Expansion and contraction.
   2. Operation and access to equipment, doors, windows, hoists, moving equipment.
   3. Headroom and walking space for working areas and aisles.
   4. System drainage and air removal.

C. Enter and exit through structure walls, floor and ceilings using penetrations and seals specified in Specification Section 01 73 20 and as shown on the Drawings.

D. Install vertical piping runs plumb and horizontal piping runs parallel with structure walls.

E. Pipe Supports:
   1. Use methods of piping support as shown on Drawings and as required in Specification Section 22 05 29.
      a. In general, pipe supports for piping smaller than 12 inches diameter are not shown on the Drawings, except for supports shown in the standard details.
      b. Where not shown on the drawings or details, the Contractor is responsible for the design of pipe supports, in accordance with Specification Section 22 05 29.
   2. Where pipes run parallel and at same elevation or grade, they may be grouped and supported from common trapeze-type hanger, provided hanger rods are increased in size as specified for total supported weight.
      a. The pipe in the group requiring the least maximum distance between supports shall set the distance between trapeze hangers.
   3. Size pipe supports with consideration to specific gravity of liquid being piped.

F. Locate and size sleeves and castings required for piping system.
   1. Arrange for chases, recesses, inserts or anchors at proper elevation and location.

G. Use reducing fittings throughout piping systems.
   1. Bushings will not be allowed unless specifically approved.

H. Equipment Drainage:
   1. Provide drip pans and piping at equipment where condensation may occur.
   2. Avoid routing piping over electrical components such as motor control centers, panelboards, etc.
      a. If piping must be so routed, utilize 16 GA, 316 stainless steel drip pan under piping and over full length of electrical equipment.
      b. Hard pipe drainage to nearest floor drain.

I. Miscellaneous Piping:
   1. If system is not otherwise specified, provide stainless steel tubing, size to handle application.

J. Unions:
   1. Install in position which will permit valve or equipment to be removed without dismantling adjacent piping.
   2. Mechanical type couplings may serve as unions.
   3. Additional unions are not required at flanged connections.

K. Install expansion devices as necessary to allow expansion/contraction movement.

L. Provide full face gaskets on all systems.

M. Anchorage and Blocking:
   1. Block, anchor, or harness exposed piping subjected to forces in which joints are installed to prevent separation of joints and transmission of stress into equipment or structural components not designed to resist those stresses.

N. Equipment Pipe Connections:
   1. Make piping connections to plumbing and HVAC equipment, including but not limited to installation of fittings, strainers, pressure reducing valves, flow control valves and relief valves provided with or as integral part of equipment.
   2. Furnish and install sinks, fittings, strainers, pressure reducing valves, flow control valves, pressure relief valves, and shock absorbers which are not specified to be provided with or as integral part of equipment.
   3. For each water supply piping connection to equipment, furnish and install union and isolation valves.

a. Provide stop valves at each plumbing fixture cold and hot water supply connections.
b. Minimum size: 1/2 inches.

4. Furnish and install "P" trap for each waste piping connection to equipment if waste is connected directly to building sewer system.
   a. Size trap as required by the International Plumbing Code.
5. Stub piping for equipment, sinks, lavatories, supply and drain fittings, key stops, "P" traps, miscellaneous traps and miscellaneous brass through wall or floor and cap and protect until such time when later installation is performed.

O. Provide dielectric fittings with insulating components where dissimilar metals are joined together.

P. Install piping concealed in finished structures such as administration and office facilities and at locations shown on Drawings.

3.4 CONNECTIONS WITH EXISTING PIPING

A. Where connection between new work and existing work is made, use suitable and proper fittings to suit conditions encountered.

B. Perform connections with existing piping at time and under conditions which will least interfere with service to customers affected by such operation.

C. Undertake connections in fashion which will disturb system as little as possible.

D. Provide suitable equipment and facilities to dewater, drain, and dispose of liquid removed without damage to property.

E. Where connection involves potable water systems, provide disinfection methods as prescribed in this Specification Section.

3.5 ACCESS PROVISIONS

A. Comply with the access doors or panels requirements as specified in section 08 31 00 - Access Doors.

3.6 CATHODIC PROTECTION

A. Isolate, dielectrically, all piping from all other metals including reinforcing bars in concrete slabs, other pipe lines, and miscellaneous metal.

3.7 INSTALLATION – PIPE SYSTEM SPECIFIC

A. Potable Water Piping Installation:
1. Install drain tees with capped nipples at low points.
   a. If low point occurs in concealed piping, provide approved flush access panel.
   b. These drains are not shown on Drawings.
2. Install all threaded piping with clean-cut tapered threads and with ends thoroughly reamed after cutting to remove burrs.
   a. Pipe joint cement permitted only on external threads.
3. For screwed nipples for connections to flush valves, lavatory supplies, and other equipment with threaded connections use iron, copper, or brass pipe.
4. Install unions between valves and connections to each piece of equipment, and install sufficient number of unions throughout piping system to facilitate installation and servicing.
   a. On copper pipe lines, install wrought, solder-joint, copper to copper unions for lines 2 IN and smaller and, for lines 2-1/2 inches and over install brass flange unions.
5. Construct and equip plumbing fixtures and equipment with anti-siphon devices as to entirely eliminate any danger of siphoning waste material into potable water supply system.
6. Where exposed pipes 6 inches in size and smaller pass through floors, finished walls, or finished ceilings, fit with nickel or chrome-plated plates large enough to completely close hole around pipes.
   a. Secure plates to pipe by set screw in approved manner.

7. Size supply branches to individual fixtures as scheduled or indicated on Drawings.
8. Install piping so as to be free to expand with proper loops, anchors and joints without injury to system or structure.
9. Provide branches to wall hydrants or hose bibbs in exterior locations with interior shutoff.
10. Provide approved type vacuum breaker and backflow preventer installations indicated or as required by Code.

B. Soil and Waste Piping Installation:
1. Install horizontal soil or waste lines less than 4 inches diameter with a slope of not less than 1/4 IN/FT or 2% toward the point of disposal.
2. Piping 4 inches and larger may be sloped at 1/8 INFT.
3. Install as close to construction as possible to maintain maximum head room.
4. Make changes of direction with 1/8 bends and junctions with wye fittings.
5. Use short wye fittings in vertical pipe only.
6. Install cleanouts at dead ends, at changes of direction and at 50 feet intervals on horizontal runs.
   a. Where cleanouts occur in concealed spaces, provide with extensions to floors above or to walls as required.
7. For hub and spigot joints, install hub facing flow.

C. Vent Piping Installation:
1. Run vent stack parallel to each soil or waste stack to receive branch vents from fixtures.
2. Where possible, combine soil, waste or vent stacks before passing through roof so as to minimize roof openings.
3. Offset pipes running close to exterior walls away from such walls before passing through roof to permit proper flashing.
4. Terminate each vent with approved frost proof jacket.
5. Carry vent stacks 4 inches and larger full size through roof.
6. Extend vent stacks at least 12 inches above roofing.

3.8 JOINING

A. Install products in accordance with manufacturer's instructions.

B. Joining Methods - Flanges:
1. Joining method:
   a. Leave 1/8 to 3/8 inches of flange bolts projecting beyond face of nut after tightening.
   b. Coordinate dimensions and drillings of flanges with flanges for valves, pumps, equipment, tank, and other interconnecting piping systems.
   c. When bolting flange joints, exercise extreme care to assure that there is no restraint on opposite end of pipe or fitting which would prevent uniform gasket compression or cause unnecessary stress, bending or torsional strains being applied to cast flanges or flanged fittings.
      1) Allow one flange free movement in any direction while bolts are being tightened.
   d. Do not assemble adjoining flexible coupled, mechanical coupled or welded joints until flanged joints in piping system have been tightened.
   e. Gradually tighten flange bolts uniformly to permit even gasket compression.
   f. Do not overstress bolts to compensate for poor installation.

C. Joining Method - Couplings:
1. Mechanical coupling:
   a. Arrange piping so that pipe ends are in full contact.
   b. Groove and shoulder ends of piping in accordance with manufacturer's recommendations.

D. Joining Method – Soldering
1. Tubing shall be cut with square ends and reamed to prevent burrs, out-of-round or improperly sized ends.

2. After cutting, all surfaces to be soldered shall be thoroughly cleaned to a metal-bright finish, free from dirt, grease or other material before fluxing and soldering. This cleaning shall be performed by using emery cloth, sandpaper or steel wool. Clean the outside end of the tubing for a length of 1/2-in greater than the depth of the fitting. The inside of the fittings shall be cleaned in a similar manner.
3. Apply non-corrosive flux and assemble the joint. Acid solder or acid flux will not be allowed. Flux compound shall be certified by NSF in accordance with NSF Standard Nos. 14 and 61.
4. The surfaces to be joined shall be heated up slowly and uniformly to the melting point of the solder. The surface being soldered shall be maintained above the melting point of the solder for sufficient time to draw the solder completely into the joint. When the solder congeals to a plastic state the excess metal shall be removed with a cloth brush, leaving a fillet around the end of the fitting.
5. Full penetration of the solder uniformly throughout the entire socket is required. The soldered joints shall be allowed to cool in still air. Quenching will not be permitted.
6. Any type of crack, pinhole, area of incomplete penetration, or similar defect will not be accepted. Peening for closing up defects will not be permitted.
7. Remove all external and internal loose solder and flux after joint cools.

E. Joining Method - Threaded and Coupled (T/C):
1. Provide T/C end conditions that meet ASME B1.2 requirements.
2. Furnish pipe with factory-made T/C ends.
3. Field cut additional threads full and clean with sharp dies.
4. Leave not more than three pipe threads exposed at each branch connection.
5. Ream ends of pipe after threading and before assembly to remove burrs.
6. Use Teflon thread tape on male thread in mating joints.

3.9 FIELD QUALITY CONTROL

A. Pipe Testing - General:
1. Test piping systems as follows:
   a. Test exposed, non-insulated piping systems upon completion of system.
   b. Test exposed, insulated piping systems upon completion of system but prior to application of insulation.
   c. Test concealed interior piping systems prior to concealment and, if system is insulated, prior to application of insulation.
   d. Test buried piping (insulated and non-insulated) prior to backfilling and, if insulated, prior to application of insulation.
2. Isolate equipment which may be damaged by the specified pressure test conditions.
3. Perform pressure test using calibrated pressure gages and calibrated volumetric measuring equipment to determine leakage rates.
   a. Select each gage so that the specified test pressure falls within the upper half of the gage's range.
   b. Notify the Contract Officer 24 hours prior to each test.
4. Completely assemble and test new piping systems prior to connection to existing pipe systems.
5. Acknowledge satisfactory performance of tests and inspections in writing to Contracting Officer prior to final acceptance.
6. Bear the cost of all testing and inspecting, locating and remedying of leaks and any necessary retesting and re-examination.

B. Pressure Testing:
1. Testing medium:
   a. Water systems: Water.
   b. Waste and drain systems: Water.
   c. Compressed air and vacuum systems: Air.
2. Testing pressure:

a. See below for gravity systems.
b. For pressurized water and air systems, test at 80 PSI, or 1.25 times the operating pressure, whichever is greater, for a duration of 1 hr.

3. Allowable leakage rates:
   a. All exposed piping systems, all pressure piping systems and all buried piping systems which are hydrostatically pressure tested shall have zero leakage at the specified test pressure throughout the duration of the test.
   b. Non-hazardous gas and air systems which are tested with air shall have a maximum pressure drop of 5% of the specified test pressure throughout the duration of the test.
4. Hydrostatic pressure testing methodology:
   a. General:
      1) All joints, including welds, are to be left exposed for examination during the test.
      2) Provide additional temporary supports for piping systems designed for vapor or gas to support the weight of the test water.
      3) Provide temporary restraints for expansion joints for additional pressure load under test.
   b. Soil, waste, drain and vent systems:
      1) Test at completion of installation of each stack or section of piping by filling system with water and checking joints and fittings for leaks.
      2) Eliminate leaks before proceeding with work or concealing piping.
      3) Minimum test heights shall be 10 feet above highest stack inlet.

C. Dielectric Testing Methods and Criteria:
1. Provide electrical check between metallic non-ferrous pipe or appurtenances and ferrous elements of construction to assure discontinuity has been maintained.
2. Wherever electrical contact is demonstrated by such test, locate the point or points of continuity and correct the condition.

3.10 CLEANING, DISINFECTION AND PURGING

A. Cleaning:
1. Clean interior of piping systems thoroughly before installing.
2. Maintain pipe in clean condition during installation.
3. Before joining piping, thoroughly clean and wipe joint contact surfaces and then properly dress and make joint.
4. Immediately prior to pressure testing, clean and remove grease, metal cuttings, dirt, or other foreign materials which may have entered the system.
5. At completion of work and prior to Final Acceptance, thoroughly clean work installed under these Specifications.
   a. Clean equipment, fixtures, pipe, valves, and fittings of grease, metal cuttings, and sludge which may have accumulated by operation of system, from testing, or from other causes.
   b. Repair any stoppage or discoloration or other damage to parts of building, its finish, or furnishings, due to failure to properly clean piping system, without cost to Government.

B. Disinfection of Potable Water Systems:
1. After favorable performance of pressure test and prior to Final Acceptance, thoroughly flush entire potable water piping system including supply, source and any appurtenant devices and perform disinfection as prescribed.
2. Perform work, including preventative measures during construction, in full compliance with AWWA C651.
3. Perform disinfection using sodium hypochlorite complying with AWWA B300-10.
4. Flush each segment of system to provide flushing velocity of not less than 2.5 feet per second.
5. Drain flushing water to sanitary sewer.
   a. Do not drain flushing water to receiving stream.
6. Use continuous feed method of application.

a. Tag system during disinfection procedure to prevent use.

7. After required contact period, flush system to remove traces of heavily chlorinated water.
8. After final flushing and before placing water in service, obtain an independent laboratory approved by the Government to collect samples and test for bacteriological quality.
   a. Repeat entire disinfection procedures until satisfactory results are obtained.
9. Secure and deliver to Contracting Officer's Representative, satisfactory bacteriological reports on samples taken from system.
   a. Ensure sampling and testing procedures are in full compliance to AWWA C651, local water purveyor and applicable requirements of the State of Arizona.

3.11 SCHEDULES

A. The piping systems scheduled herein are applicable to piping systems shown on the plumbing drawings.

B. System 1 – Potable and Non-Potable Water:
   1. Piping symbol and service:
      a. CW – Potable Water Cold.
      b. HW – Potable Water Hot.
      c. HWC – Potable Hot Water, Return.
      d. NPW – Non-Potable Water.
      e. TW – Tempered Water.
   2. Above ground, 4 inch diameter and less:
      a. Pipe: Type L Copper.
      b. Fittings: Wrought Copper.
      c. Joints: Soldered.
   3. Buried, 3 inch diameter and less:
      a. Pipe: Type K Copper.
      b. Fittings: Cast or Wrought Copper.
      c. Joints: Brazed or flared.
      d. Provide polyethylene pipe wrap for corrosion protection of underground copper pipe.
   4. Coating:
      a. Provide coating on exposed pipe in accordance with 09 96 00.

C. System 2 – Laboratory Compressed Air and Vacuum:
   1. Piping symbol and service:
      a. LA – Laboratory Air.
      b. LV – Laboratory Vacuum.
   2. Above ground, all sizes:
      a. Pipe: Type L Copper.
      b. Fittings: Wrought Copper.
      c. Joints: Soldered.
   3. Coating:
      a. Provide coating on exposed pipe in accordance with 09 96 00.

D. System 3 – Sanitary Waste and Vent:
   1. Piping symbol and service:
      a. SAN – Sanitary Sewer.
      b. V – Vent.
   2. Above Ground and Below Ground (gravity drainage, all sizes):
      a. Pipe: PVC, schedule 40.
      b. Fittings: PVC, DWV fittings.
      c. Joints: Solvent weld.
   3. Coating:
      a. Provide coating on exposed pipe in accordance with 09 96 00.

E. System 4 – Process Area Drainage and Vent:
   1. Piping symbol and service:

a. D – Drain.
b. V – Vent.

2. Above Ground and Below Ground (gravity drainage, all sizes):
   a. Pipe: PVC, schedule 40.
   b. Fittings: PVC, DWV fittings.
   c. Joints: Solvent Weld.
3. Coating:
   a. Provide coating on exposed pipe in accordance with 09 96 00.

F. System 5 - Condensate and Equipment Drains:
1. Piping symbol and service:
   a. CD – Condensate Drain.
2. Above Ground (all sizes):
   a. Pipe: CPVC, schedule 40.
   b. Fittings: CPVC, schedule 40 fittings.
   c. Joints: Solvent Weld.

G. System 6 – Sump Pump Discharge
1. Piping symbol and service:
   a. SPD – Sump Pump Discharge
2. Above ground, 3 inch and less in diameter:
   a. Pipe: Schedule 40, Galvanized Steel.
   b. Fittings: Malleable iron.
   c. Joints: Threaded.
3. Coating:
   a. Provide coating on exposed pipe in accordance with 09 96 00.

END OF SECTION

SECTION 22 05 23
GENERAL-DUTY VALVES FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Gate valves.
   2. Ball valves.
   3. Butterfly valves.
   4. Globe Valves
   5. Check valves.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 61 03 - Common Work Results for Equipment.
   2. Section 09 96 00 - High Performance Industrial Coatings.
   3. Section 22 05 03 - Pipe and Pipe Fittings - Plumbing Systems.

1.2 QUALITY ASSURANCE

A. Reference Standards:
   1. Manufacturers Standardization Society of the Valve and Fittings Industry (MSS):
      a. SP 67, Butterfly Valves.
      b. SP 70, Cast Iron Gate Valves, Flanged and Threaded Ends.
      c. SP 71, Cast Iron Swing Check Valves, Flanged and Threaded Ends.
      d. SP 80, Bronze Gate, Globe, Angle and Check Valves.
      e. SP 110, Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends.
      f. SP 139, Copper Alloy Gate, Globe, Angle, and Check Valves for Low Pressure/Low Temperature Plumbing Applications.

B. National Sanitation Foundation (NSF):
   1. NSF-61/372, Drinking Water System Components – Health Effects.

C. For drinking water service, provide valves complying with NSF 61 and NSF 372.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Product Data: Submit manufacturers catalog information for each type of valve indicating materials of construction and pressure ratings for each service.
   3. Manufacturer's Installation Instructions: Submit hanging and support methods, joining procedures.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 GATE VALVES

A. 3 inches and smaller: MSS SP 80 or MSS SP 139, 300 PSI cold working pressure, bronze body, bronze trim, threaded bonnet, non-rising stem, hand-wheel, inside screw, solid wedge disc, alloy seat rings, solder or threaded ends.

B. Larger than 3 inch: MSS SP 70, Class 125, cast iron body, bronze trim, bolted bonnet, non-rising stem, hand-wheel, outside screw and yoke, solid wedge disc with bronze seat rings, flanged ends. Furnish chain-wheel operators for valves 6 inches and larger mounted over 8 feet above floor.

2.2 BALL VALVES

A. 2 inches and Smaller: MSS SP 110, 400 psi WOG, two piece bronze body, chrome plated brass ball, full port, teflon seats, blow-out proof stem, solder or threaded ends, quarter-turn lever handle, provide extended lever handle for insulated pipes.

2.3 BUTTERFLY VALVES

A. 2-1/2 inches and Larger: MSS SP 67, Class 150.
   1. Body: Cast or ductile iron, lug ends, stainless steel stem, extended neck.
   2. Disc: Nickel-plated ductile iron, Aluminum bronze, or stainless steel.
   3. Seat: Resilient replaceable EPDM or Buna N.
   4. Handle and Operator: 10 position lever handle.

2.4 GLOBE VALVES

A. 3 inch and smaller:
   1. Comply with MSS SP 80.
   2. 300 PSI cold working pressure.
   3. Body, bonnet, disc holder and nut: Bronze.
   4. Stem: Copper silicon alloy.
   5. Packing: Teflon impregnated fiber.
   6. Disc: Teflon.
   7. Union bonnet, rising stem, renewable disc, handwheel operator.

2.5 CHECK VALVES

A. Horizontal Swing Check Valves:
   1. 2 inches and Smaller: MSS SP 80, 200 psi cold working pressure, bronze body and cap, bronze seat, Buna-N or teflon disc, solder or threaded ends.
   2. 2-1/2 inches and Larger: MSS SP 71, Class 125, cast iron body, bolted cap, bronze or cast iron disc, renewable disc seal and seat, flanged ends.

B. Spring Loaded Check Valves:
   1. 2 inches and Smaller: MSS SP 80, 200 psi cold working pressure, bronze body, in-line spring lift check, silent closing, Buna-N or teflon disc, integral seat, or threaded ends.

C. Rubber Flap Check Valves (for sump pump discharge applications)
   1. ASME B16.1, flanged connections, Class 125.
   2. Materials:
      a. Body and cover: ASTM A536 ductile iron. Internally and externally lined for corrosion protection.
      b. Flapper: Buna-N, steel reinforced.
      c. Bolted cover for access to internal flap.
      d. Stainless steel fasteners.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install valves with stems upright or horizontal, not inverted.

B. Install brass male adapters each side of valves in copper piped system. Solder adapters to pipe.

C. Install 3/4 inches ball valves with cap for drains at main shut-off valves, low points of piping, bases of vertical risers, and at equipment.

D. Install valves with clearance for installation of insulation and allowing access.

E. Provide access where valves and fittings are not accessible. Coordinate size and location of access doors with Section 08 31 00.

F. Refer to Section 22 05 29 for pipe hangers.

G. Refer to Section 40 42 00 for insulation requirements for valves.

H. Refer to Section 22 05 03 for plumbing piping materials applying to various system types.

3.2 VALVE APPLICATIONS

A. Install ball or gate valves for drain service at locations indicated on Drawings in accordance with this Section.

B. Install ball or gate valves for shut-off and to isolate equipment, part of systems, or vertical risers.

C. Install ball or globe valves for throttling, bypass, or manual flow control services.

D. Install spring loaded check valves on discharge of water pumps.

E. Install check valves on discharge of pumps in pumped drainage piping.

F. Install lug end butterfly valves adjacent to equipment when functioning to isolate equipment.

G. Install ball or gate valves in domestic water systems for shut-off service.

H. Install ball and globe valves in domestic water systems for throttling service.

3.3 SCHEDULES

SYSTEM DESCRIPTION	VALVE SERVICE		
	SHUTOFF	THROTTLING	CHECK
Domestic Cold and Hot Water	Ball or Gate	Globe	Swing check

END OF SECTION

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SECTION 22 05 29
HANGERS AND SUPPORTS FOR PLUMBING PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
1. Pipe hangers and supports.
2. Seismic restraints.
3. Hanger rods.
4. Inserts.
5. Sleeves.
6. Mechanical sleeve seals.
7. Formed steel channel.

B. Related Sections include but are not necessarily limited to:
1. Section 01 61 03 - Equipment – Basic Requirements.
2. Section 01 81 10 - Wind and Seismic Design Criteria.
3. Section 03 31 30 - Concrete Materials and Proportioning.
4. Section 07 84 00 - Firestopping.
5. Section 07 92 00 - Joint Sealants.
6. Section 09 96 00 - High Performance Industrial Coatings.
7. Section 22 05 03 - Pipe and Pipe Fittings - Plumbing Systems.

1.2 REFERENCES

A. Reference Standards:
1. American Society of Mechanical Engineers (ASME):
   a. B31.9, Building Services Piping.
2. ASTM International (ASTM):
   a. E84, Standard Test Method for Surface Burning Characteristics of Building Materials.
   b. E119, Standard Test Methods for Fire Tests of Building Construction and Materials.
   c. E814, Standard Test Method for Fire Tests of Through Penetration Fire Stops.
   d. F708, Standard Practice for Design and Installation of Rigid Pipe Hangers.
   e. E1966, Standard Test Method for Fire-Resistive Joint Systems.
3. American Welding Society (AWS):
   a. D1.1, Structural Welding Code - Steel.
4. FM Global (FM):
   a. Approval Guide, A Guide to Equipment, Materials & Services Approved By Factory Mutual Research For Property Conservation.
5. Manufacturers Standardization Society of the Valve and Fittings Industry (MSS):
   a. SP 58, Pipe Hangers and Supports - Materials, Design and Manufacturer.
   b. SP 69, Pipe Hangers and Supports - Selection and Application.
   c. SP 89, Pipe Hangers and Supports - Fabrication and Installation Practices.
6. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA):
   a. Seismic Restraint Manual, Guidelines for Mechanical Systems.
7. Underwriters Laboratories Inc. (UL):
   a. 263, Fire Tests of Building Construction and Materials.
   b. 723, Tests for Surface Burning Characteristics of Building Materials.
   c. 1479, Fire Tests of Through-Penetration Firestops.
   d. 2079, Tests for Fire Resistance of Building Joint Systems.
   e. Fire Resistance Directory.
8. Intertek Testing Services (Warnock Hersey Listed):
   a. WH - Certification Listings.

B. Perform Work in accordance with AWS D1.1 for welding hanger and support attachments to building structure.

1.3 SUBMITTALS

A. Action Submittals: Submit the following:
   1. Shop Drawings:
      a. Indicate system layout with location including critical dimensions, sizes, and pipe hanger and support locations and detail of trapeze hangers.
   2. Product Data:
      a. Hangers and Supports: Submit manufacturers catalog data including load capacity.
      b. Firestopping: Submit data on product characteristics, performance and limitation criteria.

B. Informational Submittals: Submit the following:
   1. Manufacturer's Certificate: Certify products meet or exceed specified requirements.
   2. Design Data: Indicate load carrying capacity of trapeze, multiple pipe, and riser support hangers. Indicate calculations used to determine load carrying capacity of trapeze, multiple pipe, and riser support hangers.
   3. Manufacturer's Installation Instructions:
      a. Hangers and Supports: Submit special procedures and assembly of components.

1.4 QUALITY ASSURANCE

A. Manufacturer: Company specializing in manufacturing Products specified in this section with minimum three years of experience.

B. Installer: Company specializing in performing Work of this section with minimum three years' experience.

1.5 ENVIRONMENTAL REQUIREMENTS

A. Materials of construction for pipe supports differs depending on the environment in which they are installed. Corrosive areas require corrosion resistant materials and hardware as specified herein.

B. Corrosive areas include the following:
   1. Chemical rooms at the Water Treatment Building.

C. Non-corrosive areas include the following:

   1. Roaring Springs Pump House.
   2. All areas of the Water Treatment Building, except for the Chemical Rooms.

1.6 RESPONSIBILTY

   1. Where pipe supports and seismic bracing are not shown or detailed on the drawings, Contractor shall design pipe supports and bracing.
   2. Provide all labor, materials, equipment, and incidentals as shown, specified, and required to design, furnish and install the system of hangers, supports, bracing, guidance, anchorage and appurtenances for piping systems shown on the drawings.
   3. General piping support details may be indicated on the Drawings for certain conditions. Incorporate those details with requirements of this specification section to provide the piping support system.

1.7 FIELD MEASUREMENTS

A. Verify field measurements prior to fabrication.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Pipe hangers and supports:
      a. Anvil International.
      b. Or equal.
   2. Mechanical sleeve seals:
      a. Link-seal
      b. Or equal.
   3. Formed steel channel:
      a. Unistrut Service Company.
      b. Or equal.

2.2 PIPE HANGERS AND SUPPORTS

A. General Requirements:
   1. Galvanized components:
      a. Hot-dip galvanized components:
         1) General: ASTM 123/123M.
         2) Fasteners and hardware: ASTM 153/153M.
   2. Dissimilar metals protection:
      a. Similar metals in contact, including galvanized-to-galvanized and galvanized-to-aluminum: No protection required.
      b. All other dissimilar metals in contact, including galvanized-to-dissimilar metal, or stainless-to-dissimilar metal: Provide rubber or plastic isolation between metals, such as neoprene or nylon pads, shims, grommets, or furnish supports with rubber or PVC lining to prevent contact.
   3. Allow free expansion and contraction of the piping to prevent excessive stress resulting from operating or testing conditions or from weight transferred from the piping or attached equipment.
   4. Design supports and hangers to allow for proper slope of pipes.
   5. Support vertical pipe runs in pipe chases at base of riser.
      a. Support pipes for lateral movement with clamps or brackets.

B. Provide 20 GA pipe saddle for plastic pipe support points to ensure minimum contact width of 4 inches.
   1. Material: match support.

C. Piping Supports:
   1. Conform to ASME B31.9, MSS SP58.
   2. Hangers for Pipe Sizes 1/2 to 1-1/2 inches:
      a. Adjustable swivel, split ring.
      b. Materials:
         1) Non-corrosive areas: Galvanized steel.
         2) Corrosive areas: Type 316 stainless steel.
   3. Hangers for Pipe Sizes 2 inches and Larger:
      a. Adjustable clevis.
      b. Materials:

1) Non-corrosive areas: Galvanized steel.
2) Corrosive areas: Type 316 stainless steel.

4. Multiple Pipe or Trapeze Type Hangers:
   a. Non-corrosive areas: Galvanized steel channel or Unistrut and galvanized steel hanger rod.
   b. Corrosive areas: Type 316 stainless steel channel or Unistrut and 316 stainless steel hanger rod.
5. Wall Support: wall-mounted Unistrut with pipe clamp.
   a. Materials:
      1) Non-corrosive areas: Galvanized steel.
      2) Corrosive areas: Type 316 stainless steel.
6. Floor Support:
   a. Vertically-adjustable pipe saddle, locking nut, nipple, and vertical support pipe welded to floor flange and anchored to concrete slab.
   b. Materials:
      1) Non-corrosive areas: Galvanized steel.
      2) Corrosive areas: Type 316 stainless steel.

D. Seismic Restraints:
1. Provide seismic bracing of piping in accordance with SMACNA Seismic Restraint Manual, the International Building Code and Section 01 81 10.
2. Aircraft cables, angles, channels and hardware:
   a. Non-corrosive areas: Galvanized steel.
   b. Corrosive areas: Type 316 stainless steel.
3. Pre-stretch steel aircraft cables to establish a certified minimum modulus of elasticity.
   a. Design cable restraints to resist seismic tension loads.
   b. Design steel restraints to resist both tension and compression loads with a minimum safety factor of 2.
4. Design restraint and connection steel assemblies that swivel to the final installation angle.
5. Do not mix cable and steel angle restraints to brace the same system.
6. Clamp steel angles to the threaded hanger rods at the seismic sway restraint locations utilizing a minimum of two ductile iron clamps.
7. The restraint system: Per Anchorage Preapproval "R" Number from OSHPD in the State of California verifying its capability to resist seismic forces.
8. Provide transverse bracing at 40 feet-0 inches maximum spacing unless noted otherwise.
9. Provide longitudinal bracing at 80 feet-0 inches maximum spacing unless noted otherwise.

E. Insulated Piping Supports:
1. Hangers are to be installed on outside of pipe insulation such that insulation is continuous.
2. Use a pipe covering protection saddle for insulated pipe at support point.
3. Insulated piping 1-1/2 inches and less:
   a. Provide a 9 inches length of high density perlite or high density calcium silicate at saddle.
4. Insulated piping over 1-1/2 inches: Provide a 12 inches length of high density perlite or high density calcium silicate at saddle.
5. See Specification Section 40 42 00 for pipe insulation.

2.3 ACCESSORIES

A. Hanger Rods:
1. Non-corrosive areas:
   a. Mild steel threaded both ends, threaded on one end, or continuous threaded.
   b. Electro-galvanized or zinc-nickel plated after threads are cut.
2. Corrosive areas:
   a. Type 316 stainless steel threaded rods, per ASTM A276.

B. Nuts, Bolts and Washers:
1. Non-corrosive areas: Galvanized steel.

2. Corrosive areas: Type 316 stainless steel.

C. Inserts: Malleable iron, zinc-plated, with threaded connection; size inserts for required hanger rod size.

2.4 FLASHING

A. Metal Flashing: 26 gage minimum thickness

B. Counterflashing: 22 gauge minimum thickness.

C. Materials:
1. Non-corrosive areas: galvanized steel.
2. Corrosive areas: Type 316 stainless steel.

D. Colors to match adjacent architectural finish.

E. Lead Flashing:
1. Waterproofing: 5 pound/SQFT sheet lead.
2. Soundproofing: 1 pound/SQFT sheet lead.

F. Caps: Steel, 22 gage minimum; 16 gage at fire resistant elements.

2.5 SLEEVES

A. Sleeves for Pipes Through Non-fire Rated Floors: 18 gage thick galvanized steel.

B. Sleeves for Pipes Through Non-fire Rated Beams, Walls, Footings, and Potentially Wet Floors: Galvanized steel pipe or 18 gage thick galvanized steel.

C. Sealant: Refer to Section 07 92 00.

2.6 MECHANICAL SLEEVE SEALS

A. Description: Modular mechanical type seal, consisting of interlocking synthetic rubber links shaped to continuously fill annular space between pipe and sleeve, connected with bolts and pressure plates causing rubber sealing elements to expand when tightened, providing watertight seal and electrical insulation.

B. Materials: Furnish with type 316 stainless steel bolts and hardware and EPDM sealing elements.

2.7 FORMED STEEL CHANNEL

A. Product Description: 12 gage thick steel channel. With holes 1-1/2 inches on center.

B. Materials:
1. Non-corrosive areas: Galvanized steel.
2. Corrosive areas: Type 316 stainless steel

2.8 FIRESTOPPING

A. Firestopping Materials: Comply with requirements of Section 07 84 00.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify openings are ready to receive sleeves.

3.2 INSTALLATION - INSERTS

A. Install inserts for placement in concrete forms.

B. Install inserts for suspending hangers from reinforced concrete slabs and sides of reinforced concrete beams.

C. Provide hooked rod to concrete reinforcement section for inserts carrying pipe 4 inches and larger.

D. Where concrete slabs form finished ceiling, locate inserts flush with slab surface.

3.3 INSTALLATION - PIPE HANGERS AND SUPPORTS

A. Install in accordance with ASME 31.9, MSS SP 58.

B. Support horizontal piping as scheduled.

C. Install hangers with minimum 1/2 inches space between finished covering and adjacent work.

D. Place hangers within 12 inches of each horizontal elbow.

E. Use hangers with 1-1/2 inches minimum vertical adjustment.

F. Support horizontal cast iron pipe adjacent to each hub, with 5 feet maximum spacing between hangers.

G. Support vertical piping at every floor. Support vertical cast iron pipe at each floor at hub.

H. Where piping is installed in parallel and at same elevation, provide trapeze type hangers where possible.

I. Support riser piping independently of connected horizontal piping.

J. Provide copper plated or PVC lined hangers and supports for copper piping.

K. Design hangers for pipe movement without disengagement of supported pipe.

L. Provide clearance in hangers and from structure and other equipment for installation of pipe insulation.

3.4 SEISMIC RESTRAINTS

A. Provide seismic restraints on all piping with the exception of the following:
   1. Piping less than 1 inch diameter when used for gas and compressed air.
   2. Piping less than 2-1/2 inches diameter for all other cases.
   3. Piping suspended by individual hangers where the distance from the top of the pipe to the bottom of the support for the hanger is 12 inches or less.
   4. Seismic restraints shall be capable of resisting seismic loads as defined in Section 01 81 10 and the International Building Code.

3.5 INSTALLATION - FLASHING

A. Provide metal flashing and counterflashing where piping penetrates weather or waterproofed walls, floors, and roofs.

B. Install flashing in accordance with roofing or siding manufacturer requirements and in accordance with section 07 62 00.

C. Flash floor drains in floors with topping over finished areas with lead, 10 inches clear on sides. Fasten flashing to drain clamp device.

D. Seal floor, shower, mop sink drains watertight to adjacent materials.

E. Adjust storm collars tight to pipe with bolts; seal around top edge. Use storm collars above roof jacks.

3.6 INSTALLATION - SLEEVES

A. Exterior watertight entries: Seal with mechanical sleeve seals.

B. Set sleeves in position in forms. Provide reinforcing around sleeves.

C. Size sleeves large enough to allow for movement due to expansion and contraction. Provide for continuous insulation wrapping.

D. Extend sleeves through floors 2 inches above finished floor level, unless noted otherwise on the drawings.

E. Where piping penetrates floor, ceiling, or wall, close off space between pipe and adjacent work with insulation and seal airtight.

F. Install chrome plated steel escutcheons at finished surfaces.

3.7 SCHEDULES

PIPE HANGER SPACING		
PIPE MATERIAL	MAXIMUM HANGER SPACING (FT)	HANGER ROD DIAMETER (IN)
Cast Iron (All Sizes)	5	5/8
Cast Iron (All Sizes) with 10 feet length of pipe	10	5/8
CPVC, 1 inch and smaller	3	1/2
CPVC, 1-1/4 inches and larger	4	1/2
Copper Tube, 1-1/4 inches and smaller	6	1/2
Copper Tube, 1-1/2 inches and larger	10	1/2
PVC (All Sizes)	4	3/8
Steel, 3 inches and smaller	10	1/2

END OF SECTION

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SECTION 22 11 26
PACKAGED DOMESTIC WATER PRESSURE BOOSTER SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:
   1. Packaged variable speed water pressure booster system for potable water service.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 61 03 - Common Work Results for Equipment.
   2. Section 22 05 03 - Pipe and Pipe Fittings - Plumbing Systems.
   3. Section 22 05 23 - General Duty Valves for Plumbing Piping.
   4. Division 26 - Electrical.

1.2 QUALITY ASSURANCE

A. Reference Standards:
   1. American Society of Mechanical Engineers (ASME):
      a. B31.9 Building Services Piping.
      b. Boiler and Pressure Vessel Code (BPVC) Section IX.
   2. American Society for Testing Materials (ASTM):
      a. A36 Standard Specification for Carbon Structural Steel.
      b. A53 Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
      c. A234 Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service.
      d. A312 Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes.
      e. A351 Standard Specification for Castings, Austenitic, for Pressure-Containing Parts.
      f. A403 Standard Specification for Wrought Austenitic Stainless Steel Piping Fittings.
   3. Hydraulic Institute.
   4. NEMA - National Electrical Manufacturers Associations.
   5. UL - Underwriters Laboratories.
   6. NEC - National Electric Code.
   7. National Sanitation foundation:
      a. NSF-61 Drinking Water System Components – Health Effects.
      b. NSF-372 Drinking Water System Components – Lead Content.

B. Manufacturer Qualifications:
   1. UL listed as control panel manufacturer.
   2. Written and operational Quality Assurance program.
   3. Minimum ten (10) years experience in manufacturing and application of pumping systems.

C. Factory Tests:
   1. Hydrostatically test assembled pumping systems to 125 psi for a minimum of 1 hour.
   2. Test power and control systems, and simulate control sequences and alarms.

1.3 SUBMITTALS

A. Shop Drawings:
   1. Packaged domestic-water pressure-booster system.
      a. System arrangement and dimensioned drawings to scale.

B. Product Data:
   1. Packaged domestic-water pressure-booster system.
      a. System design information including manufacturer equipment data sheets.

b. Description of system operation, including sequences of operation, shutdown modes, safety shutoffs, alarms and monitoring points.
c. Packaged pump system construction drawings indicating equipment dimensions and materials of construction.
d. Pump curves and performance information indicating design points, speed and operating horsepower.
e. Control panel wiring diagrams. Indicate electrical power supply requirements, voltage, phase, power draw, etc.
f. Provide control panel interior and exterior elevations.
g. Manufacturer catalog information on piping, valves and strainers including materials of construction and pressure ratings.
h. Manufacturer data sheets for Variable Frequency Drives (VFDs).

C. Contract Closeout Information:
1. Operation and Maintenance Data.
a. System design information sheet.
b. Packaged system dimension and general arrangement drawing.
c. Electrical power and control wiring diagram.
d. Bill of materials.
e. System operating and maintenance instructions.
2. Factory-test reports.

1.4 DELIVERY OF EQUIPMENT

A. Configure system to be deliverable and installable through passages and doorways as indicated on the drawings.
1. Factory disassemble systems for shipping and field assembly as required.
2. Provide field assembly instructions.

B. Remove drain plugs from equipment where possibility of freeze damage may exist.

C. Comply with manufacturer's instructions for rigging, unloading, and transporting equipment.

1.5 WARRANTY

A. Equipment furnished under this section shall be warranted by the manufacturer to be free from defects in materials and workmanship for a period of two years from date of equipment startup.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Packaged Domestic Water Pressure-Booster System:
a. Flowtherm.
b. Or equal.
2. 2. Pumps:
a. Goulds eSV series.
b. Or equal.

2.2 MATERIALS

A. Packaged System:
   1. Provide components factory assembled, piped, and wired. Include the following:
      a. Steel base frame, constructed of ASTM A-36 steel.
      b. Electrical control panel.
      c. Duplex pump assembly consisting of two pumps for lead/lag operation.
      d. Variable Frequency Drives (VFDs).
      e. Interconnecting piping, fittings, and valves arranged as shown on the contract drawings.
      f. Instrumentation and control components as required to allow the system to perform in accordance with the sequences of operation noted herein and on the drawings.
      g. All components and interconnecting control and power wiring assembled and installed on the skid such that the system is delivered as a single packaged unit.
   2. Provide a single electrical power point of connection.
   3. All parts of the packaged pumping system in contact with the conveyed fluid shall be lead free, NSF-61 and NSF-372 compliant.
   4. If the package needs to be broken down to allow transportation to the installed location, construct it to be broken down and assembled modularly.
      a. Modular construction:
         1) Field reconnections must be by unions, flanges, bolts, screws, and electrical plugs.
         2) Reconnection requiring cutting, welding, soldering, multiple connections to screw terminals, is not allowed.
   5. Assemble package so ample room exists within the package for servicing of components.

B. Pumps:
   1. Type:
      a. Vertical inline, centrifugal, multi-stage pumps. Piped in parallel arrangement.
   2. Construction:
      a. Pumps shall consist of a base and pump head, with an impeller stack and outer sleeve secured between the base and pump head with stay bolts. Pumps shall be ground-mounted in a vertical orientation with the motor mounted vertically on top of the pump head directly coupled to the pump shaft.
      b. Pumps shall be equipped with cartridge-type mechanical shaft seals with silicone-carbide bearing rings.
      c. Pump base and head shall be cast iron. Impellers, pump shaft, chamber and outer sleeve shall be stainless-steel.
      d. All parts in contact with the water conveyed shall be type 304 stainless steel.
      e. Pumps shall be furnished with permanently lubricated ball bearings. Cast iron surfaces shall be coated with the manufacturer's standard factory applied coating.
      f. Designed so that pump can be serviced without disconnecting piping.
      g. Pumps shall not overload at any point on the pump curves.
   3. Capacity:
      a. Water Pressure Booster Pump Skid:
         1) Refer to the schedule on the drawings for required capacity.

C. Pipe, Fittings, Manual Valves, and Piping Specialties:
   1. Piping: Type 304 stainless steel, Schedule 40, per ASTM A312.
   2. Joints: Threaded or welded.
   3. Fittings: Threaded, type 304 stainless steel per ASTM A351, Class 150 pressure rating; or type 304 ASTM A-403 butt-welded fittings for welded joints.
   4. Provide isolation valves on suction and discharge from each pump.
   5. Provide spring-loaded check valve on discharge from each pump.
   6. Provide wye type strainer on pump suction side. Or provide strainer separately upstream of pump skid.
   7. Supports:
      a. Provide supports for the following:
         1) Suction header.

2) Suction and discharge piping on each pump.
3) Discharge header.

b. Support piping independently of pump flange connections.

8. Pressure gauges on suction and discharge header.
9. Shut-off valves on control sensing lines and gauges.
10. Welded piping shall conform to ASME B31.9 Building Services Piping and ASME BPVC Section IX.
11. Packaged pump systems shall be hydrostatically leak tested at 125 psi for a duration of one hour.

D. Electrical Control Panel:
1. UL listed.
2. NEMA 1 enclosure.
3. Disconnect switch accessible from exterior of panel enclosure.
4. Individual circuit breakers for each pump VFD.
5. Hand-off-auto switches for each pump accessible from exterior of panel.
6. Lead pump selector with automatic and manual alternation.
7. Microprocessor based PLC controls to monitor all sensor inputs.
   a. Control pump speed based on pressure feedback.
   b. Field programmable.
   c. Data entry in engineering units by keypad or digital touchscreen interface.
   d. PID control.
   e. Digital graphic display for monitoring pump status, alarms and adjusting system setpoints.
   f. Digital display shall be visible and accessible from exterior of panel.
   g. Control circuit transformer with protected primary and secondary.
   h. Monitor suction and discharge pressure at pressure transducers by means of a hardwired 4-20 mA DC or 0-10 VDC signals.
8. Power on light.
9. Run light for each pump, and contact for remote output.
10. Elapsed run-time meter for each pump.
11. Low-suction and high discharge pressure cutout with alarm and contact for remote output.
12. Pump failure alarm light, reset pushbutton, and contact for remote output.
13. Pressure/Temperature transmitters:
   a. Self-contained, variable capacitance type.
   b. Install pressure transducers in suction and discharge side of each pump factory wired to the control panel.
   c. Temperature transducers on the discharge header of each pump factory wired to control panel.
14. Integrate variable frequency drives with control panel and pumps.

2.3 SEQUENCES OF OPERATION

A. Provide HAND-OFF-AUTO switches for each pump for manual or automatic operation.

B. Provide lead pump selector switch with automatic and manual alternation.

C. When in HAND mode, the pump shall run continuously.

D. When in AUTO mode, the pump speed shall automatically adjust as needed to match system pressure to the discharge pressure setpoint (adjustable) in the discharge header.

E. Lead pump shall start on a drop in system pressure. When system pressure drops below setpoint (adjustable), the lead pump sleep timer shall start. After the sleep time period (adjustable) has expired, the lead pump shall start. The lead pump shall stop when the system pressure reaches setpoint, or after an adjustable minimum run time.

F. When the lead pump is at maximum speed and the system pressure continues to drop, the lag pump sleep timer shall start. After the sleep time period (adjustable) has expired, the lag pump

shall start and operate with the lead pump. The lag pump shall stop when the system pressure reaches setpoint, or after an adjustable minimum run time.

G. Alarms:
   1. Low suction pressure alarm with automatic pump shutoff and contacts for remote output.
   2. High discharge pressure alarm with automatic pump shut-off and contacts for remote output.
   3. High discharge temperature with automatic pump shut-off with contacts for remote output.

H. Refer to the drawings for additional control requirements.

2.4 DEMONSTRATION

A. System manufacturer or manufacturer's representative: Provide start-up and adjustment service for the packaged pumping system.

B. System manufacturer or manufacturer's representative: Provide a minimum 1 hour of training for the Contracting Officers designated personnel on the operation and maintenance of the packaged pumping system.

PART 3 - EXECUTION

3.1 INSTALLATION

A. The Contractor shall start up each piece of equipment and system and shall make all adjustments so that the system is placed in proper operating conditions to the satisfaction of the Contracting Officer.

B. Complete installation and startup checks according to manufacturer's instructions. Ensure that the pump system is operating in accordance with control sequences noted on the drawings and in accordance with manufacturer recommendations.

C. The Contractor shall not install any equipment or materials until the Contracting Officer has approved all submittals. If any equipment or materials are installed prior to approval of the submittals, it shall be at the Contractor's risk.

D. Properly maintain and service all equipment and systems until the particular equipment or the system has been accepted by the Contracting Officer.

E. Manufacturer's representatives who have complete knowledge of the proper operation and maintenance of the equipment shall be provided to instruct Operations staff on proper operation and maintenance.

F. Test and adjust controls, instrumentation and safeties. Replace damaged and malfunctioning controls and equipment.

3.2 DEMONSTRATION

A. System manufacturer or manufacturer's representative: Provide start-up and adjustment service for the packaged pumping system.

B. System manufacturer or manufacturer's representative: Provide 2 hour of training to the designated personnel on the operation and maintenance of the packaged pumping system. Coordinate training sessions with designated personnel at least one week in advance.

END OF SECTION

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SECTION 22 20 00
PLUMBING FIXTURES AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Plumbing fixtures, trim, and equipment.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 61 03 - Equipment - Basic Requirements.
   2. Section 22 05 03 Pipe and Pipe Fittings – Plumbing Systems.
   3. Section 22 05 23 General Duty Valves for Plumbing Piping.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Americans with Disabilities Act (ADA):
      a. Accessibility Guidelines for Buildings and Facilities (ADAAG).
   2. American National Standards Institute (ANSI):
      a. Z358.1, Emergency Eyewash and Shower Equipment.
   3. American Society of Heating, Refrigerating and Air Conditioning Engineers/Illuminating Engineering Society of North America (ASHRAE/IESNA):
      a. 90.1, Energy Standard for Buildings Except Low-Rise Residential Buildings.
   4. American Society of Mechanical Engineers (ASME):
      a. A112.19.3, Stainless Steel Plumbing Fixtures (Designed for Residential Use).
   5. American Society of Sanitation Engineers (ASSE):
      a. 1011, Performance Requirements for Hose Connection Vacuum Breaker.
   6. Canadian Standards Association (CSA).
   7. National Sanitation Foundation International (NSF).
      a. 61/372, Drinking Water System Components.
   8. Underwriters Laboratories, Inc. (UL).

1.3 SUBMITTALS

A. Shop Drawings:
   1. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Color selection charts for color selection.
   3. Fabrication and/or layout drawings:
      a. Layout plan(s) showing dimensions, elevations, etc.
      b. Details showing connections, installation, rough-in locations, etc.
   4. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Chemical-resistance data.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Plumbing fixtures (vitreous china water closets, lavatories):
      a. Sloan.
      b. Or equal.
   2. Stainless steel sinks:
      a. Just Manufacturing.
      b. Or equal.
   3. Premolded mop sinks:
      a. Fiat.
      b. Or equal.
   4. Utility sink:
      a. Fiat.
      b. Or equal.
   5. Premolded shower:
      a. Fiat.
      b. Or equal.
   6. Water closet seats:
      a. Sloan.
      b. Or equal.
   7. Lavatory fittings:
      a. Chicago Faucets.
      b. Or equal.
   8. Utility sink fittings:
      a. Chicago Faucets.
      b. Or equal.
   9. Mop sink fittings:
      a. Chicago Faucets.
      b. Or equal.
   10. Sink fittings:
      a. Chicago Faucets.
      b. Or equal.
   11. Laboratory Sink Fittings:
      a. Chicago faucets.
      b. Guardian Equipment.
      c. Or equal.
   12. Shower fittings:
      a. Symmons.
      b. Or equal.
   13. Flush valves:
      a. Sloan.
      b. Or equal.
   14. Emergency shower and eyewash:
      a. Haws.
      b. Or equal.

15. Drains, roof drains, carriers, and shock absorbers:
    a. Jay R Smith.
    b. Or equal.
16. Trap primer:
    a. Precision Plumbing Products.
    b. Or equal.
17. Domestic water heaters:
    a. A. O. Smith.
    b. Or equal.
18. Reduced pressure backflow preventer:
    a. Watts.
    b. Or equal.
19. Hot water circulating pump:
    a. Bell and Gossett.
    b. Or equal.
20. Mixing valves for Emergency Fixtures:
    a. Haws.
    b. Or equal.
21. Mixing valves for sinks and lavatories:
    a. Watts.
    b. Or equal.
22. Expansion tanks:
    a. Amtrol.
    b. Or equal.

2.2 MANUFACTURED UNITS

A. Plumbing Fixtures (Vitreous China Water Closets, Lavatories):
    1. Water closet (WC):
        a. Wall mounted or floor mounted as required by the application.
        b. 1.6 GAL per flush.
        c. Siphon jet.
        d. Elongated bowl.
        e. Standard white finish.
        f. 1-1/2 inches top spud.
        g. Bolt caps.
        h. ADA accessibility compliant.
        i. Type:
            1) Refer to the plumbing fixture schedules for additional specifications.
    2. Lavatory (L):
        a. Wall hung.
        b. Standard white finish.
        c. Front overflow.
        d. ADA accessibility compliant.
        e. Type:
            1) Refer to the plumbing fixture schedules for additional specifications.

B. Sinks (S):
    1. Stainless steel sink:
        a. Type 304, 18-8 stainless steel.
        b. Self-rimming, drop-in mount.
        c. Drain punch centered in each bowl.
        d. Strainer and tailpiece.
        e. 3-faucet holes on 4 inches centers.
        f. ASME A112.19.3.
        g. Type:
            1) Refer to the plumbing fixture schedules for additional specifications.

2. Mop sink (MS):
    a. Precast terrazzo (marble chips cast in Portland cement, ground and polished with all air voids grouted).
    b. One-piece.
    c. Drop front.
    d. 2 inches wide shoulder.
    e. Stainless steel threshold.
    f. Integral drain body, removable strainer and 3 inches drain pipe.
    g. Type:
        1) Refer to the plumbing fixture schedules for additional specifications.

C. Showers (SH):
1. SH-1 shower:
    a. Precast shower floor:
        1) One-piece molded stone with slip-resistant surface.
        2) Shoulder not less than 3-1/16 inches high.
        3) Tiling flange not less than 1-1/2 inches above shoulder.
        4) Barrier free threshold for wheelchair accessibility with stainless steel threshold plate.
        5) Factory installed drain stainless steel with removable stainless steel strainer plate.
        6) Provide drain with caulked joint for 2 inches drain pipe.
        7) Basin of dimensions and shape as shown on Drawings.
        8) Color as selected by Contracting Officer.
        9) Refer to the plumbing fixture schedules for additional specifications.

D. Toilet Seat:
1. Molded plastic.
2. Open front.
3. No cover.
4. Stainless steel hinge with check.
5. Elongated bowl.

E. Lavatory Fittings:
1. Lavatory faucets:
    a. Individual lever handles for cold and hot water.
    b. Renewable washer.
    c. Renewable seats.
    d. Non-aerating laminar spray.
    e. Stopper and 1-1/4 inches tail piece.
    f. Coupling nuts.
    g. Chrome plated.
    h. ADA compliant.
    i. Refer to the plumbing fixture schedules for additional specifications.

F. Mop Sink Fittings:
    a. Refer to the plumbing fixture schedules for additional specifications.

G. Sink Fittings:
1. Lever handle.
2. Renewable washer.
3. Renewable seats.
4. Aerator.
5. Stopper and 1-1/2 inches tail piece.
6. Coupling nuts.
7. Chrome plated.
8. Refer to the plumbing fixture schedules for additional specifications.

H. Shower Fittings:

1. Shower valves and accessories:
    a. Concealed.
    b. Integral volume control and stops.
    c. 2 gpm flow restrictor, unless noted otherwise on the drawings.
    d. Shower head, arm and flange.
    e. 60 IN flexible metal hose.
    f. 30 IN vertical metallic slide bar with adjustable shower head mount.
    g. Install in accordance with ADA accessibility guidelines.
    h. Set limit stop to limit hot water temperature to a maximum of 110 DEGF.
    i. Type:
        1) Refer to the plumbing fixture schedules for additional specifications.

I. Manual Flush Valve:
1. Rubber diaphragm type with filtered bypass.
2. Non-hold-open handle.
3. Screw driver Bak-Chek angle stop.
4. Adjustable tailpiece.
5. Vacuum breaker flush connection and spud coupling.
6. Wall and spud flanges.
7. Spud size as required for the fixture.
8. Gallon per flush as required per fixture.
9. Refer to the plumbing fixture schedules for additional specifications.

J. Emergency Fixtures:
1. Emergency shower and eye/face wash (ESEW):
    a. Comply with ANSI Z358.1.
    b. Coated galvanized steel or PVC stanchion pipe with anchor base plate.
    c. 1-1/4” tepid water supply connection.
    d. 1-1/2” waste connection.
    e. Provide flow switch for remote SCADA monitoring where indicated on the drawings:
        1) Rating: 125/250 V, 5 A.
        2) Single pole, double throw.
        3) UL listed.
    f. Deluge shower head:
        1) Ball valve: Stay-open type; type 316 stainless steel construction for corrosion resistant models.
        2) Pull-chain.
        3) PVC or ABS plastic showerhead construction.
    g. Eye/face wash:
        1) Stay-open, full port ball valve, type 316 stainless steel construction for corrosion resistant models.
        2) Push handle control for eye/face wash.
        3) Supply line strainer for eye/face wash.
        4) PVC or ABS plastic eye wash bowl.
        5) Provide dust covers that fall off when water is activated.
    h. Mixing Valve:
        1) Provide a mixing valve complying with ANSI Z358.1 for tepid water delivery and scald protection.
    i. Type:
        1) Refer to the plumbing fixture schedules for basis of design make/models.

K. Carriers:
1. Water closets (wall hung):
    a. Vertical stacks: Wade W-330 or 340 or equal.
    b. Horizontal discharge: Wade W-310 or equal.
2. Lavatories:
    a. Wade W-520 series or equal.

L. Drains:
    1. Floor drain (FD):
        a. Bottom outlet.
        b. Clamping seepage flange.
        c. Seepage openings.
        d. Type: Cast iron body.
        e. Refer to the plumbing fixture schedules for basis of design make/models and drain sizes.

M. Traps:
    1. Floor and equipment drains:
        a. Same material and coating as the piping system.
        b. 3 inches minimum seal.
    2. Fixture drains:
        a. 2 inches minimum seal.
        b. Cast brass.
        c. Chrome plated.
    3. Condensate drains for HVAC equipment: Provide adequate seal depth to maintain seal against the static pressure of the equipment.

N. Trap Primer:
    1. Mechanical type:
        a. All brass body construction.
        b. 1/2 inches male NPT inlet.
        c. 1/2 inches female NPT outlet.
        d. Stainless steel debris screen.
        e. Vacuum breaker with air gap
        f. Brass piston.
        g. Trap primer distribution:
            1) Up to 4 traps.
            2) 2 inches copper body.
            3) Brass outlet.
    2. Electronic trap primers:
        a. Furnished assembled in a wall mounted recessed or surface mounted streel enclosure.
        b. Electrical timer-operated solenoid valve distributes water at pre-set intervals every 24-hours.
        c. Atmospheric vacuum breaker with air gap.
        d. 24-hour timer, with manual test/override.
        e. 120V single point power connection with integral circuit breaker.
        f. Calibrated manifold for equal water distribution to all drains.
        g. Copper pipe and fittings.
        h. Provide manual isolation valve upstream of each unit.
        i. Connection size as noted on the plumbing fixture schedule.
        j. Refer to plumbing fixture schedule for basis of design make and model.

O. Cleanouts (CO):
    1. Cleanouts for cast iron pipe:
        a. Cast iron body with adjustable anchor flange and secured scoriated cast iron or nickel bronze cover.
        b. Extra heavy brass neoprene seal screw plug with solid hexagonal nut.
    2. Cleanouts for steel pipe: Extra heavy brass screw plug in drainage fittings.
    3. Cleanouts turning out through walls and up through floor shall be made by long sweep ells or "y" and 1/8 bends with plugs and face or deck plates to conform to architectural finish in room.
        a. Where definite finish is not indicated, wall plates shall be chrome-plated cast-brass and floor plates polished brass.
    4. Installation requirements:

a. Provide cleanouts of same size as pipe up to 4 inches and not less than 4 inches for larger pipes.
b. Provide wall plates with chrome plated cast-brass round cleanout cover with flanged ring.
c. Provide screws which match cover plate material.
d. Cleanouts installed in completely accessible pipe chases or where piping is exposed do not require special covers.

5. Refer to the plumbing fixture schedules for basis of design make/model.

P. Hose Bibb (HB-1):
1. Provide vacuum breaker meeting the requirements of the ASSE 1011.
2. Refer to the plumbing fixture schedules and plumbing details for sizes and additional specifications.

Q. Hose Rack (HR):
1. HR-1:
a. Wall-mounted.
b. Capacity for 50 feet of 3/4 inches industrial grade rubber hose.
c. Isolation valve at inlet.
d. Type 304 stainless steel construction. Fabricated to meet the dimensions as shown on the drawings.
e. For hose bibs supplied with non-potable water, provide signage indicating “WARNING NON-POTABLE WATER DO NOT DRINK”.

R. Domestic Water Heater (DWH):
1. Electric tank type:
a. Factory-assembled, packaged commercial water heater with storage tank, heating elements, controls, and other components as indicated.
b. Completely factory assemble water heater so that installation involves only setting, leveling, anchoring, and connection of piping and electrical services.
c. Designed for potable water service.
d. UL listed and NSF approved.
e. National Electrical Code (NEC) compliant.
f. Internal surfaces:
1) Glass-lined with alkaline borosilicate composition fused-to-steel.
2) Provide magnesium anode rods rigidly supported for cathodic protection.
g. Low watt density immersed heating elements with zinc-plated copper sheath.
h. Provide water heaters with multiple heating elements, and a thermostat with each heating element:
1) Adjustable temperature range from 120-160°F.
i. High temperature cutoff with manual reset.
j. Low water cutoff.
k. Integral magnetic contactors.
l. All internal circuits shall be fused.
m. Electrical cabinet with heavy duty terminal blocks and hinged access door.
n. Provide ASME rated temperature and pressure relief valve.
o. Provide brass or copper alloy drain valve.
p. Insulate tank with vermin-proof glass fiber or foam insulation or equal.
q. Heavy gage steel jacket with baked enamel finish.
r. Warranty against corrosion for three year period.
s. Water heaters shall meet the standby loss requirements of ASHRAE 90.1.
t. Refer to plumbing schedule sheets for water heater sizes, capacity and models.

S. Reduced Pressure Backflow Preventer:
1. Backflow preventers consist of two check valves, test cocks and relief valve, all assembled as an integral unit.
2. NSF-61/372 certified.

3. Rated for 150 psi working pressure.
4. Pressure loss through backflow preventer not exceeding 12 PSI at design flow.
5. Provide wye strainer upstream to protect valve seats.
6. Provide dielectric fittings or flanges when connecting to dissimilar pipe materials.
7. Provide air gap fitting at relief valve and pipe discharge to nearest floor drain as indicated on the drawings, terminate with 1" air gap.
8. Type:
    a. For line sizes 2-inch and smaller:
        1) Bronze body construction, stainless steel seats, quarter turn ball valves, threaded end connections,
        2) Watts LF909 Small, or equal.
    b. For line sizes 2-1/2 inch and larger:
        1) Cast iron valve body construction, stainless steel seats, resilient seated gate valves, non-rising stem, flanged end connections.
        2) Watts LF909 Large, or equal.

T. Domestic Hot Water Circulating Pump:
1. In-line centrifugal, designed for potable hot water circulation, lead-free compliant with NSF-61/372.
2. Pump casing: Bronze, cast iron or stainless steel.
3. Impeller: Bronze, stainless steel, cast iron or plastic.
4. Rated for 150 psi operating pressure and 180°F operating temperature.
5. Electrically commutated motor (ECM).
6. Motors: Open drip proof (ODP) enclosure with thermal overload protection.
7. Solid one-piece stainless steel shaft.
8. Permanently lubricated bearings.
9. Mechanical seals, carbide or silicon carbide.
10. Provide pump with aquastat and automatic timer kit:
    a. Equal to Bell and Gossett TC-1.
11. Refer to the plumbing schedule sheets for additional information, including pump capacities and electrical information.

U. Mixing Valves:
1. Integral screens and check-stops for cross flow protection.
2. Adjustable temperature setpoint with lock down.
3. Lead free, bronze or brass body, NSF-61/372 certified.
4. Mixing valves for lavatories and sinks:
    a. ASSE 1070 rated.
5. Mixing valves for emergency fixtures:
    a. ASSE 1071 rated.
    b. ANSI Z358.1 certified.
    c. Antiscald protection.
    d. Low cold water pressure valve closure.
    e. Cold water bypass on low hot water pressure.
6. Refer to mixing valve schedules on the drawings for additional specifications including design temperatures, pressure drop, flow rates and basis of design make/model.

V. Expansion Tanks:
1. Designed and constructed to ASME Boiler and Pressure Vessel Code, Section VIII Division 1.
2. Welded steel construction, coated with primer for corrosion resistance.
3. Rated for 150 psi working pressure.
4. Heavy-duty butyl bladder or diaphragm meeting NSF 61/372 requirements for potable water application. Tank shall be pre-charged with compressed air to minimum fill pressure.
5. Provide Schrader air valve for charging.
6. Provide isolation valve at piping connection to the tank.
7. Support tank independently from the piping. Provide seismic bracing for tanks.

8. Refer to the drawings for tank sizes, connection sizes, and basis of design make/model.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Cross Connection: Do not install any plumbing components that will provide a cross connection between potable and non-potable or drainage systems.

B. Fixtures:
   1. Install fixtures at locations indicated on Drawings and in compliance with local Codes.
   2. Connect plumbing supply, drain and vent line sizes as shown on Drawings.
   3. Set proper grounds to form secure base for each fixture and rigid setting.
   4. Install fixtures except water closets with water supply above rim and with Code approved backflow preventers.
   5. Seal fixture joints abutting walls and floors with silicone sealant.
   6. Connect exposed traps and supply pipes for fixtures and equipment to rough piping systems at wall, unless otherwise specified.
   7. Install emergency fixtures in accordance with ANSI Z358.1.

C. Drains:
   1. Install drains at locations indicated on Drawings and in compliance with local Codes.
   2. In uncovered concrete slabs:
      a. Install at the low points of surface areas to be drained or as indicated.
      b. Set tops of drains flush with the finished floor.
      c. Install drain flashing collar or a flange so that no leakage occurs between the drain and the adjoining surfaces.
      d. Maintain the integrity of waterproof membranes, where penetrated.

D. Hose Racks:
   1. Install adjacent to hose bibbs, top of rack 36 inches above finished floor or grade.
   2. Concrete or masonry walls: Mount with 5/8 inches x 2-1/2 inches stainless steel expansion anchors.
   3. Handrail:
      a. Bolt hose rack to 24 inches x 24 inches x 1/4 inches aluminum plate with 5/8 inches stainless steel bolts.
      b. Attach to handrail with 3/8 inches stainless steel through bolt at each corner of the plate.
   4. Pedestal:
      a. Bolt hose rack to 24 inches to 24 inches x 1/4 inches aluminum plate with 5/8 inches stainless steel bolts.
      b. Attach to pedestal with two, 1/2 inches stainless steel bolts through handrail.

E. Hose Bibbs:
   1. Install 36 inches above finished floor.
   2. In exterior locations, provide interior isolation valve.

F. Shock Absorbers:
   1. Install on hot and cold water lines adjacent to each battery of fixtures or other equipment where indicated on Drawings.
   2. Size as recommended by manufacturer for length of pipe served.
   3. Locations having two fixtures or less, install capped air chamber 12 inches long on hot and cold water runouts to each fixture, same size as runout.
   4. Runouts to hose bibbs and wall hydrants do not require air chambers.
   5. Install units vertically on top of pipe or as detailed on the Drawings.

G. Cleanouts:
   1. Install cleanouts:
      a. Above floor in each vertical riser that connects to horizontal branch below floor.

b. At test tee to receive proper test plugs in each vertical riser at least every other floor.
c. As required by local Code.

H. Wall Plates and Escutcheons: Install as specified in Specification Section 22 05 03 or this Specification Section.

I. Water Heater:
1. Install all water heaters in accordance with details, manufacturer's recommendations, and applicable Codes.
2. For units located on concrete pads, plumb level and orient to allow access to the controls, elements and other items requiring service.
3. Connect hot and cold water piping to the unit with line-size, isolation valves and dielectric unions.
4. Connect recirculating hot water to cold inlet piping with unions and valves at check valves as detailed.
5. Start up the unit and adjust all controls for proper temperature control and maximum efficiency.
6. Silicone seal all piping and wiring penetrations.

J. Reduce Pressure Backflow Preventer: Install on water lines as required by Code and in the locations noted on the drawings.

3.2 FIELD QUALITY CONTROL

A. Test piping and fixtures for leaks per Specification Section 22 05 03.

3.3 TRAINING

A. Provide the service of a qualified manufacturer’s representative to thoroughly train designated personnel in the operation and maintenance of systems and equipment furnished and installed under this section.

B. Refer to Section 01 79 00 for Operator training requirements.

C. Provide on-site training sessions, within the contract period, to train personnel designated by the Contracting Officer.

D. Coordinate training sessions with the Contracting Officer.

END OF SECTION

DIVISION 23

HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

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SECTION 23 05 48
VIBRATION AND SEISMIC CONTROLS FOR HVAC PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Vibration isolators.

B. Related Sections include but are not necessarily limited to:
   1. Section 01 81 10 - Wind and Seismic Design Criteria: All vibration and seismic component design and installation must comply with Section 01 81 10 unless otherwise dictated by a more stringent requirement in this section.

1.2 QUALITY ASSURANCE

A. Reference Standards:
   1. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE):
      a. Handbook - HVAC Applications.
   2. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA):
      a. HVAC Duct Construction Standards - Metal and Flexible.

1.3 SUBMITTALS

A. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.

B. Product Data:
   1. Submit a schedule of vibration isolator type with location and load on each.
   2. Submit manufacturer catalog information indicating materials of construction, dimensional data, and load rating.

1.4 QUALIFICATIONS

A. Manufacturer: Company specializing in manufacturing products specified in this Section with minimum 5 years' documented experience.

B. Installer: Company specializing in performing Work of this Section with minimum 5 years' documented experience.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Inspection: Accept materials on Site in manufacturer's original packaging and inspect for damage.

B. Store materials according to manufacturer instructions.

C. Protection:
   1. Protect materials from moisture and dust by storing in clean, dry location remote from construction operations areas.
   2. Provide additional protection according to manufacturer instructions.

1.6 EXISTING CONDITIONS

A. Field Measurements:
   1. Verify field measurements prior to fabrication.
   2. Indicate field measurements on Shop Drawings.

1.7 WARRANTY

A. See Section –01 67 00 – Product Requirements.

PART 2 - PRODUCTS

2.1 PERFORMANCE AND DESIGN CRITERIA

A. Provide vibration isolation devices on motor-driven equipment over 0.5 hp, plus connected piping and ductwork.

B. Minimum Static Deflection of Isolators:
   1. Under 20 hp:
      a. Under 600 rpm: 1.0 inches.
      b. 600 to 800 rpm: 0.5 inches.
      c. Over 800 rpm: 0.2 inches.
   2. Over 20 hp:
      a. Under 600 rpm: 2.0 inches.
      b. 600 to 800 rpm: 1.0 inches.
      c. Over 800 rpm: 0.5 inches.

2.2 VIBRATION ISOLATORS

A. Open Spring Isolators:
   1. Spring Isolators:
      a. Exterior and Humid Areas: Furnish hot-dip galvanized housings and neoprene-coated springs.
      b. Code: Color-code springs based on load carrying capacity.
   2. Springs:
      a. Minimum Horizontal Stiffness: 75% of vertical stiffness.
      b. Working Deflection: Between 30 and 60% of maximum deflection.
   3. Spring Mounts: Furnish leveling devices, minimum 0.25 inches thick neoprene sound pads, and zinc chromate-plated hardware.
   4. Sound Pads:
      a. Size: Based on minimum deflection of 0.05 inches.
      b. As specified for neoprene pad isolators.

B. Closed Spring Isolators:
   1. Spring Isolators:
      a. Exterior and Humid Areas: Furnish hot-dip galvanized housings and neoprene-coated springs.
      b. Code: Color-code springs based on load carrying capacity.
   2. Type: Closed spring mount with top and bottom housing separated by neoprene rubber stabilizers.
   3. Springs:
      a. Minimum Horizontal Stiffness: 75% of vertical stiffness.
      b. Working Deflection: Between 30 and 60% of maximum deflection.
   4. Housings: Incorporate neoprene isolation pad meeting requirements for neoprene pad isolators and neoprene side stabilizers with minimum clearance of 0.25 inches.

C. Spring Hangers:
   1. Spring Isolators:
      a. Exterior and Humid Areas: Furnish hot-dip galvanized housings and neoprene-coated springs.
      b. Code: Color-code springs based on load carrying capacity.
   2. Springs:
      a. Minimum Horizontal Stiffness: 75% of vertical stiffness.
      b. Working Deflection: Between 30 and 60% of maximum deflection.
   3. Housings: Incorporate rubber hanger with threaded insert.
   4. Misalignment: Capable of 20-degree hanger rod misalignment.

D. Neoprene Pad Isolators:
   1. Rubber or neoprene-waffle pads.

a. Hardness: 30 durometer.
b. Minimum Thickness: 1/2 inches.
c. Maximum Loading: 40 psi.
d. Rib Height: Not greater than 0.7 times width.

2. Configuration: Single layer or 1/2 inches-thick waffle pads bonded to each side of 1/4 inches-thick steel plate.

E. Rubber Mount or Hanger:
1. Material: Molded rubber.
2. Deflection: 0.5 inches.
3. Insert: Threaded.

F. Seismic Snubbers:
1. Description: Non-directional and double-acting unit consisting of interlocking steel members restrained by neoprene elements.
2. Neoprene Elements:
a. Replaceable.
b. Minimum Thickness: 0.75 inches.
3. Capacity: Four times assigned load to mount groupings at 0.4 inches deflection.
4. Attachment Points and Fasteners: Capable of withstanding three times rated load capacity of seismic snubber.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install isolation for motor-driven equipment. Install equipment level.

B. Install spring hangers without binding.

C. Isolators:
1. Install isolators so they are free of binding through full range of deflection.
2. Closed Spring Isolators: Adjust such that side stabilizers are clear under normal operating conditions.
3. Prior to making piping connections to equipment with operating weights substantially different from installed weights, block up equipment with temporary shims to final height; when full load is applied, adjust isolators to load to allow shim removal.

D. Snubbers:
1. Provide resiliently mounted equipment, piping, and ductwork with seismic snubbers.
2. Provide each inertia base with minimum of four seismic snubbers located close to isolators.
3. Snubbers: Provide clearance between 0.15 and 0.25 inches.

END OF SECTION

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SECTION 23 05 93
HVAC SYSTEMS - BALANCING AND TESTING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Adjusting, balancing, and testing of all heating, ventilating and air conditioning (HVAC) systems, including the following systems:
      a. Building supply and exhaust fans.
      b. Air handling units.
      c. Fan Coil Units.
      d. Laboratory exhaust systems, including fume hoods and fume exhaust fans.

B. Related Sections include but are not necessarily limited to:
   1. Section 23 09 00 - Instrumentation and Control for HVAC Systems.
   2. Section 23 31 00 - HVAC - Ductwork.
   3. Section 23 34 00 - HVAC - Fans
   4. Section 23 80 00 - HVAC - Equipment.
   5. Section 23 81 29 - Variable Refrigerant Flow Heat Pump Systems.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Associated Air Balance Council (AABC):
      a. National Standards for Total System Balance.
   2. American Industrial Hygiene Association (AIHA):
      a. Z9.5, Laboratory Ventilation.
   3. American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE):
      a. HVAC Applications Handbook, Chapter entitled "Laboratories".
      b. HVAC Systems and Equipment Handbook, Chapter entitled "Testing, Adjusting, and Balancing".
   4. National Environmental Balancing Bureau (NEBB):
      a. Procedural Standards for Testing Adjusting Balancing of Environmental Systems.

B. Qualifications:
   1. Work of this Section to be accomplished by an independent testing and balancing firm certified by one (1) of the following:
      a. Associated Air Balance Council (AABC).
      b. National Environmental Balancing Bureau (NEBB).
      c. Other certification entity approved by the Contracting Officer Representative.
   2. The independent firm shall not be the same firm as the firm installing the HVAC equipment, nor under contract to the firm installing the equipment.
   3. The independent testing and balancing firm shall have successfully completed at least five (5) projects of similar size and scope in accordance with AABC or NEBB requirements and recommendations.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Certifications:
      a. Letter stating the name and qualifications of the firm proposed.
   3. Report forms:

a. Procedures and forms to be used in calibrating of test instruments, balancing systems, and recording and reporting test data.

B. Informational Submittals:
1. Completed test reports and data forms upon completion of installation, balance and testing of HVAC systems.
a. Insert recorded information on report forms required by specifications and approved for use on project.
b. Additional written verification and other related information clearly identifying project, date and specifics of verification.
c. Utilize report forms similar to those shown in Section V of AABC Standard.
d. Provide forms typed and signed by the testing and balancing firm.
2. Submit report on fume hood testing.

PART 2 - PRODUCTS - (NOT USED)

PART 3 - EXECUTION

3.1 PREPARATION

A. Secure approved Shop Drawings of all HVAC equipment.

B. Procedures and Forms:
1. Submit procedures and forms to be used in calibration of test instruments, balancing systems, and recording and reporting test data.
2. Obtain approval before beginning balancing and testing.

C. Do not begin balancing and testing until HVAC systems are complete and in full working order.
1. Place HVAC systems into full operation and continue their operation during each working day of balancing and testing.

D. Provide qualified heating and ventilating Engineer(s) to supervise and perform balancing and testing.

E. Review design Drawings, specifications, approved Shop Drawings and other related items to become thoroughly acquainted with the design of HVAC systems.

F. Check all installed systems against Contract Drawings, Specifications and Shop Drawings to see that system is installed as required.
1. Report deficiencies to the Contracting Officer Representative.
2. Report deficiencies to Contractor for remedial action including providing corrective measures required in the function of any part of system to complete balancing.

G. Make necessary adjustments as required to balance the systems.

3.2 FIELD QUALITY CONTROL

A. Balance and Test Air Systems:
1. Adjust equipment RPM to design requirements.
2. Report motor full load amperes.
3. Obtain design CFM at fans.
a. Make pitot tube traverse of main supply and exhaust ducts within 5%.
4. Test and record system static pressures, suction and discharge.
5. Obtain design CFM for recirculated air.
6. Obtain design CFM outside air.
7. Test and record entering air temperatures, (DB, heating and cooling).
8. Test and record leaving air temperatures, (DB, heating and cooling).
9. Test and record leaving air temperatures, (WB, cooling).
10. Adjust dampers in supply, exhaust and return air ducts to design CFM.

11. Test diffusers, grilles, and registers as follows:
    a. Adjust to comply with design requirements within 10%.
    b. Identify location and area of each.
    c. Adjust face velocity to establish required CFM.
        1) Retest after initial adjustments.
    d. Adjust to minimize drafts and to ensure uniform air distribution in all areas.
12. Identify and list size, type and manufacturer of diffusers, grilles, registers, and HVAC equipment.
    a. Record nameplate manufacturer and model number of installed equipment.
    b. Use manufacturer's nameplate ratings on equipment to make required calculations.
13. Adjust and assure that the operation of automatically operated dampers are as specified.
    a. Check and calibrate controls.
14. Prepare and submit reports.

B. Balance and Test Fume Hood(s):
    1. Provide hood performance tests outlined in ASHRAE, AABC, and AIHA Z9.5 standards.
    2. Upon completion of these tests, provide a report including the following:
        a. Exhaust fan operating characteristics including speed, static pressures, motor amperages, and total exhaust CFM.
        b. CFM in exhaust duct connected to hood.
        c. Position of hood door or face plate.
        d. Make-up air total CFM and position of make-up air fan.
        e. Actual face velocity readings taken at hood face opening, in 4 inches square increments over entire face area.
        f. Results of smoke spillage tests.
        g. Operating condition of surrounding area air conditioning or supply air system.

END OF SECTION

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SECTION 23 09 00
INSTRUMENTATION AND CONTROL FOR HVAC SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
1. Instrumentation and control for HVAC systems.
2. Control panels and accessories.
3. Application specific Direct Digital Controllers (DDC).
4. Damper actuators.
5. Temperature and pressure devices.
6. Miscellaneous components.

B. Related Specification Sections include but are not necessarily limited to:
1. Section 01 61 03 - Equipment Basic Requirements.
2. Section 10 14 00 - Identification Devices.
3. Section 23 31 00 - HVAC - Ductwork.
4. Section 23 80 00 - HVAC - Equipment.
5. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables.
6. Section 26 05 33 - Raceways and Boxes.
7. Section 40 67 00 - Control System Equipment Panels and Racks.

1.2 QUALITY ASSURANCE

A. See Specification Section 01 61 03.

B. Referenced Standards:
1. ASTM International (ASTM):
   a. D1693, Standard Test Method for Environmental Stress-Cracking of Ethylene Plastics.
2. The International Society of Automation (ISA):
   a. S5.1, Instrumentation Symbols and Identification.
   b. S5.4, Standard Instrument Loop Diagrams.
3. National Electrical Manufacturers Association (NEMA):
   a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
4. National Fire Protection Association (NFPA):
   a. 70, National Electrical Code (NEC).
5. Underwriters Laboratories, Inc. (UL).
   a. 508A, Standard for Industrial Control Panels.

C. Miscellaneous:
1. Controls to be in compliance with Section 26 05 00 for NEMA and NFPA 70 enclosure class requirements unless noted or specified otherwise.
2. Unless specifically noted otherwise, components of systems shall be industrial duty suitable for moist, corrosive environments.

1.3 SYSTEM DESCRIPTION

A. Provide a complete system of automatic temperature controls, including control panels, thermostats, sensors, relays, damper actuators, and other associated controls and appurtenances required to operate in accordance with the sequence of operation described herein and on Drawings.
1. Assemble control system with complete system of wiring and conduit to fulfill requirements of the Contract Documents.

B. Install systems using competent technicians under direct supervision of the control system supplier.

C. The control sequences noted herein and on the drawings are designed to illustrate operating functions only.
   1. These minimum specified items, and any additional controls, not indicated but required to meet performance requirements as outlined in the Contract Documents, shall be furnished and installed at no additional cost to the Contracting Officer to make a complete system.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Wiring and control diagrams for each control panel showing point to point terminations and auxiliary interlocks for each item in each control loop. Provide 11x17 PDF prints of each drawing. Include control panel interior and exterior layout drawings.
   3. Control panel bill of materials and electrical load calculations for each panel.
   4. Manufacturer catalog data sheets and specifications for all instrumentation and control equipment furnished under this section.
   5. Instrumentation list for each controlled system. Each element of the controlled system shall be listed in table format. The table shall show element name, type of device, manufacturer, model number, and product data sheet number.
   6. A points list for each system control panel including I/O, point number, the controlled device associated with the I/O point, and the location of the I/O device. Setpoint ranges, alarm points, etc.
   7. Type-written descriptions of equipment sequences of operation.

B. Quality Control Submittals:
   1. Secure from equipment manufacturers, detailed and complete control and power wiring diagrams, word descriptions of controls provided as part of the HVAC equipment or equipment interfaced or interlocked thereto, and submit with equipment manufacturer's submittals.
      a. Provide the above information to control system manufacturer.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.5 SITE CONDITIONS

A. Unless stated otherwise, the environment and air streams may include varying concentrations of the following chemical components:
   1. H_2SO_4 - Sulfuric acid.
   2. H_2S - Hydrogen sulfide.
   3. Condensation.

1.6 COORDINATION WITH OTHER DIVISIONS

A. HVAC instrumentation and control equipment specified under this section and shown on the HVAC controls drawings is intended to be furnished and installed by the Division 23 Controls Contractor.
   1. This includes but is not limited to thermostats, temperature sensors, electric valve and damper actuators, flow switches, control panels and associated control wiring and conduit.

B. Wiring, conduit and raceways shall comply with Division 26 requirements for materials, sizes and installation.

C. Control panel enclosures shall comply with section 40 67 00 for materials, fabrication and installation.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Instrumentation and control systems and equipment:
      a. Johnson Controls Inc.
      b. Or equal.
   2. Motorized damper actuators:
      a. Belimo.
      b. Or equal.

2.2 EQUIPMENT

A. Dampers:
   1. Refer to Specification Section 23 31 00.

B. Electric Damper Actuators:
   1. Provide actuators of proper size and number to secure true throttling or two-position action as required by the application.
   2. Provide actuators with sufficient power and torque to assure tight closing of dampers on fan shutdown or in the fail-safe position. Coordinate with torque requirements of the actual dampers provided.
   3. Actuator functionality is to be consistent with control sequences described herein and on the drawings.
   4. Actuators shall be capable of operating in temperatures between 0°F-100°F.
   5. Actuator housings shall be cast aluminum with a NEMA rating appropriate for the environment in which it is installed. Actuator gear train shall be fully immersed in oil.
   6. Actuators shall be installed outside of ductwork and connected to an extended damper shaft.
   7. Actuators shall have electronic overload protection or digital rotation sensing circuitry to prevent actuator damage throughout the entire rotation. End switches to deactivate the actuator at the end of rotation or magnetic clutches are not acceptable.
   8. Actuators for two-position service shall have a 30-second maximum motor run time to fully open the damper and a 30-second maximum run time to fully close the damper.
   9. Provide spring return actuators to return the damper to its normal position, unless noted otherwise on the drawings or in the control sequences. The spring return mechanism shall be internal and built into the actuator housing.
   10. Actuators shall have a manual crank to allow manual positioning of the damper when the actuator is not powered.
   11. If a single damper operator cannot meet torque requirement, provide sectional dampers and actuators to meet the torque requirement.
   12. Provide separate electrical circuits for damper operators with no more than four operators on a single circuit.
   13. Actuators shall be furnished with end-switches where required by the control descriptions for interlocking damper operation with fans or equipment operation.

C. Industrial Thermostats:
   1. For the following applications:
      a. Located within process spaces, hazardous spaces, humid areas, outdoors and within mechanical rooms.

b. Used for low-limit freeze-protection.

2. Materials:
   a. Stainless steel bulb and coiled capillary, non-toxic oil filled.
   b. Enclosure rating:
      1) Non-Hazardous locations: NEMA 4X
      2) Hazardous locations: NEMA 7, Class I, Division 1, Group D rated.
   c. Temperature adjustment knob or dial.
   d. Temperature sensing range: 40-90°F.
   e. Deadband: 1-2°F
   f. Voltage and amp rating: as required for the application.

D. Programmable Thermostats:
   1. 7-day programmable with independent heating and cooling setpoints for occupied and non-occupied modes.
   2. Automatic changeover between heating and cooling modes.
   3. Setpoint adjustment: 50 to 90°F
   4. Deadband: 1-2°F.
   5. Display Features:
      a. Illuminated LCD display.
      b. System status display in large icons or font.
      c. Room temperature and setpoint display.
      d. Heat/Cool, On/Off modes as required by the application.
      e. Fan On-Auto control modes.

E. Temperature sensors:
   1. RTD or thermistor type.
   2. Operating range: 50 to 90 DEGF.
   3. Duct sensors shall be single point or averaging as shown on the drawings. Averaging sensors shall be a minimum of 5 FT in length per 10 square feet of duct cross section.
   4. Immersion sensors shall be provided with a separable stainless steel well. The well must be rated for the velocities, pressures and temperatures of the pipe in which it is installed.
   5. Provide matched temperature sensors for differential temperature measurement.
   6. Label temperature sensors with identification tag of HVAC equipment controlled using nameplates in accordance with Specification Section 10 14 00.

F. Differential Pressure Switches:
   1. UL listed, SPDT snap-acting, pilot duty rated (125 VA minimum), with scale range and differential suitable for intended application.
   2. User adjustable setpoint.
   3. Repeatability of plus/minus 3.0 percent.

G. Current Switches:
   1. Current-operated switches shall be self-powered, solid-state with adjustable trip current.
   2. The switches shall be selected to match the current of the application and input requirements of the control panel to which it is connected.

H. Transformers:
   1. Provide transformers for supplying current to control equipment operating at less than supply voltage, and where required by manufacturer's automatic control system, capable of supplying 125% of energy requirements of equipment connected for not less than 1 HR.
      a. Enclose transformers in UL listed cabinets with conduit connections.
      b. Provide fused disconnect switches on both primary and secondary sides.

I. Application Specific Controllers (ASC):
   1. Controllers shall be capable of monitoring and controlling both digital and analog electric systems, as required by the system being controlled.
   2. Microprocessor-based controllers, dedicated for controlling a specific piece of equipment, customized for operation within the confines of the equipment they are designed to serve.

3. Each ASC shall be capable of stand-alone operation and shall continue to provide control functions without being connected to a network.
4. Each ASC shall contain sufficient I/O capacity for the system being controlled.
5. Application controllers shall have sufficient memory to support their operating system, database, and programming requirements.
6. Electronic controllers shall perform within the following minimum performance and application criteria:
   a. Setpoint adjustment: 0 to 110% of span.
   b. Repeatability: Setpoint repeats within +0.1% of span.
   c. Dead band: 1-5% of span, adjustable.
   d. Rate: 5 to 30 seconds, adjustable.
   e. Response level: 50 milliseconds for a step change of 1% of span beyond setpoints.
7. Controllers shall be permanently mounted in a control panel unless otherwise indicated.
8. Provide controllers with a local operator interface where local setpoint adjustment and monitoring of parameters is required.
9. Programming shall not be lost in the event of a power failure or controller failure.

J. Local Temperature Control Panels:
1. Panels shall be fabricated and installed in accordance with UL 508A and NFPA 70.
2. Panel enclosures:
   a. All indoor control cabinets shall be fully enclosed with hinged door, three-point latch, key-lock quarter turn handle and removable subpanels.
   b. Panel materials and installation shall comply with Section 40 67 00.
3. Panels shall be floor or wall-mounted and be sized to accommodate the required electrical and control components and required protective devices.
4. Interconnections between internal and face-mounted devices shall be pre-wired with color-coded stranded conductors neatly installed in plastic troughs and/or tie-wrapped. Terminals for field connections shall be UL listed for 600 V service, individually identified per control drawings, with adequate clearance for field wiring. Control terminations for field connection shall be individually identified per the control drawings.
5. All switches, indicators and instruments shall be suitably tagged with their function to match the control drawings and equipment schedules.
6. Points for monitoring and control are indicated on the mechanical control sheets.
7. Unless noted otherwise on the Drawings, provide disconnect and overcurrent protection for power sources at each local panel.

2.3 FABRICATION

A. Corrosion Protection:
1. Protect metal parts of controls, instrumentation and related items from corrosive atmosphere by either applying protective coatings or selecting corrosion resistant materials.
   a. Aluminum and stainless steel require no further protection.
2. Provide NEMA 4X enclosures with tempered glass windows and vapor tight gaskets, illustrated in Hoffman Bulletin A-50, for protection of controls from corrosive environment.
   a. Install control instruments inside enclosure and extend remote stainless steel sensing elements through enclosure wall.
   b. Provide vapor tight seals for penetrations of enclosure.
3. Provide in each enclosure industrial corrosion inhibitors, Hoffman Corrosion Inhibitors, as illustrated in Hoffman's technical Bulletin HCI.
4. Protect metal accessory items such as mounting brackets and fasteners that are not stainless steel, fiberglass or aluminum by epoxy or phenolic coatings.
5. Protect electric motor operator with corrosion inhibitors inside enclosure.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Comply with requirements of Section 26 05 19 and Section 26 05 33.

B. Identification: See Section 10 14 00.

C. Connect control devices to perform functions indicated herein and to perform in accordance with sequence of operations noted on the drawings.

D. Use remote element temperature transmitters for points of temperature measurement occurring in air ducts or shafts, or in mechanical piping system.

E. In general, locate thermostats for room control immediately inside door, above light switch, unless shown otherwise on the drawings.
   1. Where light switch is in an entryway to room, locate thermostat on wall within room so it is capable of sensing true space conditions.
   2. Prior to installation, coordinate thermostat location with other trades and Contracting Officer Representative.

F. Unless shown otherwise on the drawings, mount local control panels adjacent to equipment served. Locate panels so visual observation and adjustment can be accomplished from floor level.

3.2 COORDINATION

A. Coordinate and schedule work with all other work in the same area, or with work that is dependent upon other work, to facilitate mutual progress and adequate space conditions for all trades.

B. Coordination with controls specified in other Sections or Divisions: Other sections and/or divisions of this specification include controls and control devices that are to be part of or interfaced to the control system specified in this section. These controls shall be integrated into the system and coordinated by the contractor as follows:
   1. Each supplier of a controls product is responsible for configuration, programming, start-up, and testing of that product to meet the sequences of operation described on the Drawings and in the Specifications.
   2. Coordinate and resolve any incompatibility issues that arise between the control products provided under this section and those provided under other Sections or Divisions of the Specifications.
   3. The Contractor shall be responsible for the integration of indicated controls and control devices into the system specified in this section, and the installation of any control devices or appurtenances required by, but not provided with, equipment standalone controls.

3.3 GENERAL WORK QUALITY

A. Install equipment, piping, and wiring/raceway parallel to building lines (horizontal, vertical, and parallel to walls) where possible.

B. Provide sufficient slack with flexible connections to allow for vibration of piping and equipment.

C. Install all equipment in readily accessible locations as defined by Chapter 1, Article 100, Part A of the National Electric Code (NEC).

D. Verify integrity of all wiring to ensure continuity and freedom from shorts and grounds.

E. All equipment, installation, and wiring shall comply with acceptable industry specifications and standards for performance, reliability, and compatibility and be executed in strict adherence to local codes and standard practices.

F. All control and interlock wiring shall comply with national and local electrical codes and Specification Section 26 05 19.

G. Communication conduit and wiring shall comply with national and local electrical codes and Specification Section 26 05 19.

3.4 INSTALLATION OF SENSORS

A. Install sensors in accordance with the manufacturer's recommendations.

B. Mount transducers in a location accessible for service without use of ladders or special equipment.

C. Mount sensors rigidly and adequately for the environment within which the sensor operates.

D. Temperature and Pressure Sensors:
   1. Room sensors shall be installed in junction boxes properly supported by the wall framing.
   2. All wires attached to sensors shall be air sealed in their conduit or raceways to stop air transmitted from other areas affecting sensor readings.

3.5 ACTUATORS

A. Mount and link actuators according to manufacturer's instructions.

B. Check operation of damper and valve actuator to confirm that actuator modulates damper or valve smoothly throughout stroke to both open and closed positions.

C. Provide all mounting hardware and linkages for actuator installation.

D. Dampers:
   1. Actuators shall be direct mounted on damper shaft or jackshaft unless shown as a linkage installation.
   2. Damper blades, axles, and linkage must operate without binding. Before system operation and after installation, cycle damper to ensure proper operation. On multiple section assemblies, all sections must open and close simultaneously.
   3. Provide a visible and accessible indication of damper position on the drive shaft end.

E. Valves:
   1. Actuators shall be connected to valves with adapters approved by the actuator manufacturer.
   2. Control valves shall be installed so that they are accessible and serviceable and so that actuators may be serviced and removed without removing the valve from the piping and without interference from the structure or other pipes and/or equipment.

3.6 CONTROL SYSTEM CHECKOUT AND TESTING

A. Start-up Testing: All testing listed in this article shall be performed by the contractor and shall make up part of the necessary verification of an operating control system. This testing shall be completed before the Contracting Officer is notified for system training.
   1. Furnish all labor and testing equipment required to calibrate and prepare for service all instruments, controls, and accessory equipment furnished under this specification.
   2. Verify that all control wiring is properly connected and free of all shorts and ground faults. Verify that terminations are tight.
   3. Enable the control systems and verify calibration of all input devices individually. Perform calibration procedures according to manufacturers' recommendations.
   4. Verify that all binary output devices (relays, solenoid valves, two-position actuators and control valves, magnetic starters, etc.) operate properly and that the normal positions are correct.
   5. Verify that all analog output devices (I/Ps, actuators, etc.) are functional, that start and span are correct, and that direction and normal positions are correct. Check all control valves and automatic dampers to ensure proper action and closure. Make any necessary adjustments to valve stem and damper blade travel.
   6. Verify that the system operation adheres to the sequences of operation. Simulate and observe all modes of operation by overriding and varying inputs and schedules. Tune all control loops and optimum start/stop routines.
   7. Verify all alarms and interlocks:

a. Check each alarm separately by including an appropriate signal at a value that will trip the alarm.
b. Interlocks shall be tripped using field contacts to check the logic, as well as to ensure that the fail-safe condition for all actuators is in the proper direction.
c. Interlock actions shall be tested by simulating alarm conditions to check the initiating value of the variable and interlock action.

3.7 CONTROL SYSTEM TRAINING

A. Provide the service of a qualified manufacturer's representative to thoroughly train designated personnel in the operation and maintenance of HVAC control systems furnished and installed under this section.

B. Refer to Section 01 79 00 for Operator training requirements.

C. Provide on-site training sessions, within the contract period, to train personnel designated by the Contracting Officer.

D. Each session must be coordinated with the Contracting Officer's representatives.

E. Train the designated staff, to enable them to do the following:
1. Understand system operation, control and optimizing routines.
2. Proficiently operate the systems.
3. Understand control system components and configuration.
4. Log on and off systems with password protected access.
5. Access graphics, point reports, and logs.
6. Adjust and change system set points, time schedules, and holiday schedules.
7. Recognize malfunctions of the system by observation of graphical visual signals.
8. Understand system drawings and the Operation and Maintenance manual.
9. Understand the system layout and location of control components.
10. Access data from DDC controllers or ASCs.

END OF SECTION

SECTION 23 31 00
HVAC - DUCTWORK

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. HVAC ductwork and accessories.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 08 90 00 - Louvers and Vents.
   2. Section 01 61 03 - Equipment - Basic Requirements.
   3. Section 23 09 00 - Instrumentation and Control for HVAC Systems.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE):
      a. 52, Method of Testing Air Conditioning Devices Used in General Ventilation for Removing Particulate Matter.
   2. National Fire Protection Association (NFPA).
   3. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA):
      a. Ducted Electric Heat Guide for Air Handling Systems.
      b. HVAC Duct Construction Standards - Metal and Flexible.
      c. Seismic Restraint Manual, Guidelines for Mechanical Systems.
   4. Underwriters Laboratory, Inc. (UL):
      a. 555, Standard for Safety Fire Damper and Ceiling Fire Damper.
      b. 555S, Standard for Safety Leakage Rated Dampers for Use in Smoke Control Systems.
      c. Building Materials Directory.

B. Qualifications:
   1. Fabricator: Firms regularly engaged in the manufacture of the specific product, of type, size required, whose products have been in use in similar service for not less than three years.
   2. Installers: Firm with at least five years installation experience on products similar to that required for this Project.

1.3 DEFINITIONS

A. Installer or Applicator:
   1. Installer or applicator is the person actually installing or applying the product in the field at the Project site.
   2. Installer and applicator are synonymous.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. See Specification Section 01 61 03.
   3. Scaled ductwork drawings showing equipment location, duct and accessory layout and support locations.
   4. Submit manufacturer technical data for ductwork and accessories furnished under this section, indicating materials of construction, dimensions and performance information.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:

a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.5 CONTRACTOR RESPONSIBILITY

A. Duct Support System Design:
   1. Contractor shall design, provide and install duct supports, seismic restraints and anchors for all ductwork shown on the mechanical drawings in accordance with SMACNA HVAC Duct Construction Standards and SMACNA Seismic Restraint Manual.
   2. Seismic bracing shall be designed to comply with the International Building Code and project specific seismic design criteria provided in Section 01 81 10.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

B. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

C. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Transverse joints (factory fabricated aluminum):
      a. Ductmate Industries, Inc.
      b. Or equal.
   2. Flexible ducts:
      a. Thermaflex.
      b. Or equal.
   3. Turning vanes:
      a. Ductmate Industries, Inc.
      b. Or equal.
   4. Flexible duct connections:
      a. Vent Fabrics.
      b. Or equal.
   5. Access doors in ductwork:
      a. Vent Fabrics.
      b. Or equal.
   6. Backdraft dampers:
      a. Greenheck.
      b. Or equal.
   7. Barometric Relief Dampers:
      a. Greenheck.
      b. Or equal.
   8. Fire dampers:
      a. Air Balance.
      b. Or equal.
   9. Ceiling diffusers:
      a. Titus.
      b. Or equal.
   10. Grilles and registers:
      a. Titus.
      b. Or equal.
   11. Automatic and manual (volume) dampers:

a. Air Balance.
b. Or equal.

12. Duct sealers:
    a. United McGill.
    b. Or equal.

2.2 COMPONENTS

A. Duct and Fittings:
1. Materials:
    a. 3003 H-14 aluminum alloy:
        1) Comply with ASTM B209.
2. Fabrication (aluminum):
    a. Minimum sheet material thickness:
        1) Ducts with largest side or diameter to 30 inches: 0.05 inches thick.
        2) Ducts with largest side or diameter greater than 30 inches: 0.08 inches thick.
    b. Utilize SMACNA HVAC Duct Construction Standards for minimum of 2 inches water gage static pressure for the minimum sheet material thickness specified herein.
        1) Heavier gage sheet material may be used with associated reinforcement as an alternate to minimum thickness specified.
        2) Lighter gage sheet material with associated reinforcement shall not be used as an alternate to minimum thickness specified.
    c. Longitudinal seams:
        1) 0.050 material:
            a) Pittsburgh seam.
            b) Continuously welded.
        2) 0.080 material: Continuously welded.
    d. Continuously weld seams on factory assembled units.
    e. Transverse joints (Alternate A):
        1) SMACNA T-22 companion flange.
        2) Gasketed.
        3) Rigidity class:
            a) Ducts with largest side or diameter to 30 inches: SMACNA Class D (1-1/2 x 1-1/2 x 1/8 inches angles).
            b) Ducts with largest side or diameter greater than 30 to 54 inches: SMACNA Class H (2-1/2 x 2-1/2 x 3/16 inches angles).
    f. Transverse joints (Alternate B):
        1) Materials and fabrication:
            a) Angles: Aluminum.
                (1) Ductmate 35.
            b) Corners: Aluminum.
                (1) Ductmate DC 35.
            c) Snap cleats: Aluminized or stainless steel.
            d) Gaskets: Closed cell neoprene.
            e) Bolts: Stainless steel.
            f) Sheet metal screws: Self-drilling stainless steel with unthreaded section under head.
        2) Fabrication:
            a) Rigidity class: SMACNA Class H.
            b) 3/8 inches diameter x 1 inch bolts.

B. Ductwork Pressure Classification:
1. Supply, return and exhaust ductwork: +/- 2 IN W.G.
2. Hazardous exhaust ductwork: +/- 2 IN W.G.

C. Supports and Hangers:
1. Materials for aluminum duct:

a. Support angles, Unistrut and miscellaneous metal: Aluminum or stainless steel.
b. Hanger rods and hardware: Stainless steel.
c. Anchors: Stainless steel.

2. Strap hangers and cable hangers are not allowed. Cable may be used for transverse or longitudinal seismic bracing.

D. Turning Vanes:
1. Materials: Same as duct.
2. Fabrication:
a. Fabricate double vane units.
b. Pressure drop through elbows: Maximum 20% of velocity pressure.

E. Flexible Connections:
1. Materials: Hypalon, double coated closely woven glass fabric.
2. Fabrication: Withstand 4.5 inches water column, positive and negative pressure.

F. Access Doors:
1. Materials:
a. Inner panel, out panel and frame: Same as duct.
b. Gaskets: Closed cell neoprene.
c. Insulation: 1 pound density fiberglass.
d. Hinges: Stainless steel.
e. Latches:
1) Aluminum-zinc alloy.
2) Outside lever handle.
3) Adjustable spacer.
4) Beveled inside flange.
5) Studs:
a) Minimum 3/8 inches diameter stud for doors up to 24 inches wide x 48 inches high.
b) Minimum 1/2 inches diameter stud for doors larger than 24 x 48 inches.

2. Fabrication:
a. Provide four-side continuous gaskets.
b. Utilize continuous piano hinges.
c. Latches required:
1) 12 inches in any direction: One.
2) Up to 18 x 18 inches: Two.
3) Up to 24 x 48 inches: Two with inside handles.
4) Up to 24 x 72 inches: Three with inside handles.
5) Minimum door size: 12 x 12 inches.

G. Flexible Duct:
1. Material: Continuous steel supporting spiral covered with 100% continuous filament fiberglass with nonporous fiberglass/vinyl liner and reinforced Mylar/neoprene outer cover.
2. UL listed, Class 1 with flame spread of 25 or less and smoke development rating not to exceed 50.
3. Provide insulated flexible ductwork for conditioned supply or return air, equal to R-8 fiberglass insulation.
4. A maximum of 5 FT in length of insulated flexible duct may be used to connect sheet metal ductwork to diffusers.

H. Drain Pan:
1. Materials: Aluminum or type 304 stainless steel.
2. Fabrication: 0.080 inches minimum thickness.
3. Provide 3/4” male threaded drain connection, unless noted otherwise.

I. Acoustical Liners:
1. Materials:

a. Thickness: 1 inch.
b. 3 pounds/CF density fiberglass insulation.
c. UL listed, Class 1, non-combustible.

J. Backdraft Dampers:
1. Material:
a. 6063 T5 aluminum frame and blades.
b. Blade edge seals: Extruded vinyl, silicone or PVC.
2. Fabrication:
a. Leakage: Not to exceed 20 CFM/SF at 1.0-IN W.G.
b. Minimum 0.080-IN thick frame and blade thickness.
c. Shaft: 1/2-IN diameter, aluminum.
d. Linkage: 1/2 inch tie bars.
e. Bearings: Synthetic, non-metallic.
f. Flanged duct connections.

K. Barometric Relief Dampers:
1. Material:
a. 6063 T5 aluminum frame and blades.
b. Blade edge seals: Extruded vinyl, silicone or PVC.
2. Fabrication:
a. Provide counterweighted blades with adjustable counterweights to allow pressure relief at pressures as low as 0.01-IN W.G..
b. Leakage: Not to exceed 20 CFM/SF at 1.0-IN W.G.
c. Minimum 0.080-IN thick frame and blade thickness.
d. Shaft: 1/2-IN diameter, aluminum.
e. Linkage: 1/2 inch tie bars.
f. Bearings: Synthetic, non-metallic.
g. Flanged duct connections.

L. Fire Dampers:
1. Materials:
a. Frame, blades, enclosure: Galvanized steel.
2. Fabrication:
a. Frame: 20 GA, G60.
b. Blades:
1) Curtain type.
2) 24 GA, G60.
c. Enclosure: 20 GA, G60.
d. Fusible link: 212 degrees F, UL listed.
e. Fire rating: 1-1/2 hour per UL 555.
f. Mounting: Vertical or horizontal.
g. Design with blade package out of air stream.

M. Diffusers:
1. Furnish in the sizes, capacities and throw patterns noted on the drawings and in the diffuser schedules.
2. Materials:
a. Body: Extruded aluminum, unless noted otherwise.
b. Ceiling or duct mounted diffusers: Sponge rubber gaskets.
3. Fabrication:
a. Furnish square diffusers with removable modular core, unless noted otherwise in the diffuser schedules.
b. Provide key operated opposed blade damper mounted in neck where noted in the diffuser schedules.
c. Finish:
1) Rectangular and round diffusers: White, baked enamel, unless noted otherwise.

2) Perforated supply and return diffusers: Paint interior with flat black paint.

N. Manual (volume and isolation) Dampers for Round Ductwork:
1. Type: Aluminum frame and blades, quarter-turn single-blade.
2. Fabrication:
a. 0.080-IN thick frame and blades.
b. 0.5-IN diameter aluminum shaft.
c. Bronze, synthetic, or nylon bearings.
d. Provide 1/4-IN thick closed cell neoprene blade seal. (Blade seal not required for balancing dampers).
e. Actuators:
1) Manual (volume) dampers: Quarter-turn actuator with locking wing nut. Lock damper into position during balancing.
2) Automatic dampers: See section 23 09 00 for actuator specifications.
f. Furnish with fasteners, duct collars and transitions as required.

O. Automatic and Manual (Volume) Dampers for Rectangular Ductwork:
1. Type: Aluminum frame and blades.
a. Opposed or parallel blades for 2-position (open/closed) service.
b. Opposed blade for balancing or throttling service.
2. Fabrication:
a. 0.080-IN thick frame and blades.
b. 0.5-IN diameter aluminum shaft.
c. Flanged duct connections.
d. Linkage concealed in frame.
e. Bronze, synthetic or nylon bearings.
f. Silicone, PVC, or vinyl blade seals.
g. Isolation dampers (open/closed service):
1) Class 1 damper, maximum leakage rate of 4 CFM/SF at 1.0-IN W.G.
h. Motorized dampers: Install actuator outside of airstream.
i. Actuators:
1) Manual (volume) dampers: Quarter-turn actuator with locking wing nut. Lock damper into position during balancing.
2) Automatic dampers: See section 23 09 00 for actuator specifications.
j. Furnish with fasteners, duct collars and transitions as required.

P. Grille and Register Assembly:
1. Materials:
a. Assembly: Extruded aluminum.
b. Gaskets: Sponge rubber.
2. Fabrication:
a. Supply registers: Two sets individually adjustable louvers.
b. Exhaust and return registers: 45 degrees deflection front blades.
c. Dampers: Key-operated opposed blade, where noted on the grille schedules.
d. Screws, duct collars, and transitions as required.
e. Finish:
1) Manufacturer's standard factory applied finish.
2) Color: White.

Q. Air Filter Enclosure:
1. Housing:
a. Factory fabricated.
b. Minimum 16 GA aluminum.
c. Bracing to eliminate racking.
d. Access doors.
2. Access doors:
a. Minimum 16 GA aluminum.

b. Replaceable positive sealing latches.
c. Replaceable hinges.
d. Neoprene door gasket.
e. Holding frame to door gasket: Polyurethane foam.

3. Tracks:
   a. Field adjustable.
   b. Anodized extruded aluminum.
   c. Polypropylene seal between holding frame and track.
4. Holding frame:
   a. Aluminum.
   b. Multiple fastener lances.
   c. Polyurethane foam gasket.
      1) Internally.
      2) Frame sides.
   d. Accommodate the filter sizes noted on the drawings without modifications filters, frame or housing.

R. Air Filters:
1. Materials:
   a. Filters: Pleated, disposable type.
   b. Holding frame and housing: Aluminum.
2. Fabrication:
   a. Factory built and assembled unit.
   b. Minimum Efficiency Reporting Value (MERV) per ASHRAE 52.
   c. MERV-8, unless noted otherwise on the drawings.
   d. 2 inches thickness minimum.
   e. Efficiency: 20%.
   f. Air velocity: 500 fpm maximum.
   g. Clean pressure drop: 0.25 inches WG maximum.
   h. Size, capacity, and type: As indicated on Drawings.

S. Duct sealer:
1. NFPA rating of "Non-Combustible".
2. Flame spread rating: 25 or lower, in dry condition.
3. Smoke developed rating: 50 or lower, in dry condition.
4. Resistant to water and water vapors.
5. Comply with UL 181.
6. Pressure rupture rating: 16 inches WG, minimum.

T. Louvers: See specification section 08 90 00.

2.3 MAINTENANCE MATERIALS

A. Extra Materials:
1. Furnish the following extra materials:
   a. 3 complete filter media change(s) for each type of filter.
   b. Filter media used during construction is in addition to this requirement.

PART 3 - EXECUTION

3.1 INSTALLATION

A. See Specification Section 01 61 03.

B. Metal Ductwork:
1. Install with longitudinal seams sealed for zero leakage.
2. Install gaskets at each transverse joint and fasten sections together with bolts.
   a. Tighten for zero leakage.

3. Install duct supports, hangers and seismic bracing in accordance with SMACNA HVAC Duct Construction Standards and SMACNA Seismic Restraint Manual.
4. Install turning vanes in square elbows:
    a. Unsupported vane length not to exceed 48 inches.
    b. Position vanes at proper angle to meet specified pressure drop.
5. Install flexible connections at fans:
    a. Locate as close as possible to fan.
    b. Allow 1 inch of slack to prevent vibration transmission.
6. Install access doors where indicated on Drawings and at smoke and fire damper in accordance with NFPA requirements.

C. Flexible Ductwork:
1. Install in concealed areas between: low velocity duct work and diffusers, return air grilles or exhaust outlets and ducts.
2. Use low loss fittings for connection to duct.
3. Connect to metal duct collars by means of non-combustible synthetic rubber sealing compound and stainless steel drawband.
4. Install with maximum length of 5 feet.

D. Drain Pans:
1. Install at cooling coils, and at other sources of moisture.
2. Install drain pans beneath ductwork or piping that passes over the top of electrical or control panels.
3. Connect drain piping to pans and discharge at nearest floor drain or equipment drain with 1-IN air gap.

E. Dampers:
1. Install where indicated on the Drawings and of the sizes shown.
2. Install fire dampers in ductwork passing through 1 hour or higher fire-rated construction.
    a. Install in wall and floor openings utilizing steel sleeves, angles and other materials following practices required to provide installation in accordance with local building codes.

F. Diffusers, Registers and Grilles:
1. Install where shown on Drawings of size and capacities scheduled on Drawings.

G. Air Filters:
1. Install where shown on Drawings of size and capacity scheduled on Drawings.
2. Do not operate equipment during construction without filters.

END OF SECTION

SECTION 23 34 00
HVAC - FANS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Building Ventilation Fans and Accessories.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 61 03 - Equipment - Basic Requirements.
   2. Section 23 05 93 - HVAC Systems - Balancing and Testing.
   3. Section 23 09 00 - Instrumentation and Control for HVAC Systems.
   4. Section 23 31 00 - HVAC - Ductwork.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Air Movement and Control Association (AMCA).
      a. AMCA Publication 203 "Field Performance Measurement of Fan Systems".
      b. ANSI/AMCA 210 "Laboratory Methods of Testing Fans for Aerodynamic Performance Rating".
   2. Air Conditioning and Refrigeration Institute (ARI).
   3. American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE):
      a. HVAC Applications Handbook, Chapter entitled "Sound and Vibration Control."
      b. 52.2, Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size.
   4. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   5. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   6. Underwriters Laboratories, Inc. (UL):
      a. 507, Standard for Electric Fans.

B. Miscellaneous:
   1. Gage thickness specified herein shall be manufacturer's standard gage for steel and Brown and Sharpe gage for non-ferrous metals.
   2. Corrosion protection of equipment to be as specified herein.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Wiring diagrams.
      d. Control diagrams.
      e. Manufacturer's catalog cut sheets and technical data.
      f. Corrosion-protection information.
      g. Fan curves indicating airflow and static pressure, fan speed, power consumption and motor rated horsepower.
      h. Sound data.
      i. Vibration isolation.
      j. Performance data on all equipment.

k. Electrical power requirements.

3. Certifications:
   a. Fan systems have been tested in accordance with AMCA Standard 210 or 260, and are licensed to bear the AMCA Certified Ratings Seal.

B. Factory Performance test for any fan having a flow rate greater than 1,000 cfm and/or a total static pressure rating equal to or greater than 1.5 inches WC.
   1. Pursuant to AMCA Publication 203 or 210 with no plus tolerances on Power and no minus tolerances on flow or pressure.

C. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. Vibration isolation assemblies:
      a. Mason Industries.
      b. Or equal.
   2. Corrosion-protective coatings:
      a. Heresite and Chemical Co.; "Heresite."
      b. Or equal.
   3. In-line Centrifugal fans:
      a. Greenheck.
      b. Or equal.
   4. Wall-mounted centrifugal exhaust fans:
      a. Greenheck.
      b. Or equal.
   5. Wall-mounted propeller-type fans:
      a. Greenheck.
      b. Or equal.
   6. Toilet room exhaust fans:
      a. Greenheck.
      b. Or equal.

2.2 GENERAL

A. All Manufactured Units:
   1. Comply with Specification Section 01 61 03.
   2. Factory wired and assembled.
   3. Use fasteners made of same material as unit.
   4. Fabricate motor assemblies and unit housings with vibration isolation assemblies:
      a. Type: As per Table 47, Chapter 48, ASHRAE HVAC Applications Handbook.

B. Where indicated in the fan schedules on the drawings, fans shall be constructed with corrosion-resistant materials or have corrosion-resistant coating.

1. Type:
    a. Corrosion-resistant materials:
        1) Aluminum.
        2) Stainless steel.
        3) FRP.
    b. Corrosion-resistant coating:
        1) Phenolic-based coating.
        2) 3 mil minimum dry thickness, air-dried coating, for surfaces exposed to temperatures less than 150 degrees F.
        3) 5 mil baked-on coating for heat transfer surfaces and surfaces exposed to temperatures greater than 150 degrees F.
        4) Factory applied.
        5) Provide factory certification of application.

2.3 MANUFACTURED UNITS

A. In-line Centrifugal Fans:
1. AMCA certified.
2. Provide inlet venturi matched to fan wheel.
3. Provide removable side panels for maintenance.
4. Construct fan to allow complete removal of motor, drives, and fan wheel when side panel is removed.
5. Provide universal mounting brackets to allow for horizontal or vertical fan orientation.
6. Provide 1-inch thick fiberglass foil faced insulation liner in fan housing. Do not expose fiberglass to moving airstream.
7. Materials:
    a. Wheel and blades: Aluminum non-overloading wheel with backward-inclined blades, statically and dynamically balanced. Wheels attached to shaft with keys and set screws.
    b. Housing and inlet bell: Aluminum or stainless steel with continuously gasketed contact surfaces. Provide insulated fan housing.
    c. Drive shaft: Solid stainless steel.
8. Bearings:
    a. Manufacturer's standard, heavy duty, field-lubricated pillow block ball type, based on fan size and mounting orientation.
    b. Minimum L-50 Bearing Life: 500,000 hours at maximum operating speed.
9. Motor:
    a. See Specification Section 01 61 03 for additional motor requirements.
    b. Motor enclosures shall be Totally Enclosed Fan Cooled (TEFC) or Totally Enclosed Air Over (TEAO), unless noted otherwise.
    c. Provide premium efficiency motor.
    d. Provide ECM motors with fan speed control where indicated on the fan schedules on the drawings.
    e. Vibration isolated motor assembly.
10. Drive type as noted on the schedules.
11. Belt Drive Units:
    a. Drive Components: V-Belt drive, rated for minimum 150 percent of motor nameplate horsepower, with machined, cast-iron pulleys, and heat resistant, oil resistant, static-free V-belts.
    b. Driver and driven sheaves:
        1) Keyed hub type.
        2) Drive sheaves:
            a) Shipped with variable pitch diameter sheave.
            b) Install fixed pitch sheave, size based on approved test and balance reports.
        3) Automatic drive belt tensioner.
12. Accessories:
    a. Belt guard.

b. Refer to the fan schedule on the drawings for additional accessories to be provided.

13. Size and capacity as scheduled on Drawings.

B. Wall-Mounted Centrifugal Exhaust Fans:
1. AMCA certified.
2. Non-overloading horsepower capabilities.
3. Materials:
a. Housing: Spun aluminum.
b. Wheel: Aluminum.
c. Drive shaft: Stainless steel.
d. Minimum 10 GA motor mounting plate.
4. Backward inclined blades.
5. Statically and dynamically balanced wheel.
6. Bearings:
a. Cast iron pillow blocks.
b. Concentric bearing locking collar for drive shafts 1 inch and larger.
1) SKF "ConCentra."
2) Dodge "D Lock."
c. Regreaseable.
d. 200,000 hour average life.
7. Weathertight compartment for motor and drives.
a. Separated from airstream.
8. Motor:
a. See Specification Section 01 61 03 for additional motor requirements.
b. Motor enclosures shall be Totally Enclosed Fan Cooled (TEFC) or Totally Enclosed Air Over (TEAO), unless noted otherwise.
c. Provide premium efficiency motor.
d. Provide ECM motors with fan speed control, where indicated on the fan schedules on the drawings.
e. Vibration isolated motor assembly.
9. Drive type as noted on the schedules.
10. Belt Drive Units:
a. Drive Components: V-Belt drive, rated for minimum 150 percent of motor nameplate horsepower, with machined, cast-iron pulleys, and heat resistant, oil resistant, static-free V-belts.
b. Driver and driven sheaves:
1) Keyed hub type.
2) Drive sheaves:
a) Shipped with variable pitch diameter sheave.
b) Install fixed pitch sheave, size based on approved test and balance reports.
3) Automatic drive belt tensioner.
11. Accessories:
a. Aluminum birdscreen.
b. Backdraft damper.
c. Extended grease lines and fittings.
d. Refer to the fan schedules on the drawings for accessories to be provided with each fan.
12. Size and capacity as scheduled on Drawings.

2.4 SPARE PARTS

A. Provide two spare belts for each type of belt driven fan installed on the project.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install in accordance with Specification Section 01 61 03.

B. Install fixed pitched drive sheave after sheave has been sized based on accepted test and balance report.

C. Do not operate fans for any purpose until ductwork is clean, filters are in place, bearings are lubricated, and fan has been test run under observation.

D. Install fans in accordance with manufacturer's installation instructions.

E. The Contractor shall not install any equipment or materials until submittals have been favorably reviewed. If any equipment or materials are installed prior to approval of the submittals, it shall be at the Contractor's risk.

F. The Contractor shall start up all equipment and systems in accordance with manufacturer recommendations, and shall make all adjustments so that the system is placed in proper operating condition.

3.2 FIELD QUALITY CONTROL

A. Test and balance air systems in accordance with Specification Section 23 05 93.

B. Properly maintain and service all equipment and systems until the equipment or the system has been accepted by the Contracting Officer and their representative.

C. This maintenance shall include compliance with the manufacturers operating and maintenance instructions as well as periodic checking and cleaning of the filters and the lubrication of moving parts and all required adjustments.

D. Tests and inspections:
1. After installing the fans and after electrical circuitry has been energized, test the units for compliance with performance requirements and proper motor rotation.
2. Adjust fan belts to proper alignment and tension.
3. Adjust fan speed or sheaves to deliver design airflows in accordance with test and balance specifications.
4. Complete installation and startup checks according to manufacturer's written instructions.
5. Inspect fans for visible damage.
6. Verify that labels are clearly visible.
7. Verify that clearances have been provided for servicing.
8. Verify that controls are connected and operable.
9. Verify lubrication on fan and motor bearings has been applied.
10. Inspect fan-wheel rotation for movement in correct direction without vibration and binding.
11. Install new filters on units which have been running prior to acceptance of Project.

3.3 TRAINING:

A. Provide the service of a qualified manufacturer's representative to thoroughly train designated personnel in the operation and maintenance of equipment and systems furnished and installed under this section.

B. Refer to Section 01 79 00 for Operator training requirements.

C. Provide on-site training sessions, within the contract period, to train personnel designated by the Contracting Officer.

D. Each session must be coordinated with the Contracting Officer's representatives.

E. Train the designated staff to enable them to do the following:
1. Understand system operation and control.
2. Understand control system components and configuration.
3. Adjust and change system set points, time schedules, occupied and non-occupied schedules.
4. Recognize malfunctions of the system by observation of graphical or visual signals.
5. Understand system drawings and the Operation and Maintenance manual.
6. Understand the system layout and location of key components.

END OF SECTION

SECTION 23 74 36
REFRIGERANT PIPING SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Refrigeration piping for split system heat pumps and air conditioning units.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 40 05 07 - Pipe Support Systems.
   2. Section 23 80 00 - HVAC - Equipment.
   3. Section 23 81 29 - Variable Refrigerant Flow Heat Pump Systems.
   4. Section 40 42 00 - Pipe, Duct and Equipment Insulation.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE):
      a. 15, Safety Code for Mechanical Refrigeration.
   2. ASTM International (ASTM):
      a. B280, Standard Specification for Seamless Copper Tube for Air Conditioning and Refrigeration Field Service.
   3. Federal Specification (FS):
      a. WW-T-799, Tube, Copper, Seamless, Water (For Use With Solder-Flared or Compression-Type Fittings).

1.3 SUBMITTALS

A. Shop Drawings:
   1. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
   3. Test reports:
      a. A dated declaration of the test of the refrigerant piping for each system shall be provided.
         1) The dated declaration shall include the information outlined in Article 12.3 of ASHRAE 15.
      b. Test reports of the refrigerant piping leak tests for all refrigerant piping systems installed.
      c. The test reports shall contain the following information:
         1) System refrigerant and high and low side pressure used.
         2) Listing of the necessary repairs made before the refrigerant piping system passed the leak test.
         3) Identification of specific system by referencing specific equipment identification numbers.
         4) Leak testing media used.
         5) Suction and discharge refrigerant gas pressures and temperatures taken after the refrigerant system has been charged.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.4 WARRANTY

A. The completed refrigerant piping system shall be guaranteed to be sufficiently free from leaks so that the loss of refrigerant for 18 months from the date of final payment shall not exceed 5 percent.

PART 2 - PRODUCTS

2.1 REFRIGERANT PIPING AND FITTINGS

A. Refrigerant Piping:
   1. Copper tubing conforming to ASTM B280 and/or FS WW-T-799, dehydrated for refrigerant use, with high-temperature soldered joints and wrought copper (400 psig) fittings.
      a. For underground use: Type K.
      b. For aboveground use: Type L.

B. Piping Joints:
   1. Joints between copper tubing and fittings to be high temperature soldered (melting point not less than 1000 degrees F, but less than that of the metal being joined) with phos-copper alloys.
   2. Joints between copper and brass, steel, etc., shall be silver soldered only.

C. Precharged Line Sets: Size per manufacturer's recommendations.

D. Field Assembled Units:
   1. Size refrigeration lines according to manufacturer's published tables using pressure or temperature drops as follows:
      a. Suction lines: 2 degrees F.
      b. Liquid lines: 1 degree F or 2 psi.
      c. Hot gas lines: 1 degree F or 3.6 psi.
      d. Size discharge and hot gas risers for positive oil return to compressors.

E. Hangers: As specified in Specification Section 40 05 07.

F. Insulation: As specified in Section 40 42 00.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Precharged Line Sets: Install per manufacturer's recommendations.

B. Field Assembled Lines:
   1. Refrigerant piping:
      a. Purge refrigerant piping of all air while connections of refrigerant piping are being made.
   2. Testing:
      a. Refrigerant piping systems: Follow general testing guidelines of ASHRAE 15, except as modified herein.
      b. Pressurize the high and low pressure sides of the piping system after completion of the refrigerant piping.
         1) Pressurize at the test pressures specified in ASHRAE 15 for the refrigerant type to be used in the system.
      c. Repair any leaks and repeat tests until no further leaks are found and the system passes a static leak test at test pressure for a duration of 24 hours.
   3. Cleaning:
      a. After complete system is tested, disconnect suction and discharge lines from compressor for cleanup.

b. Clean uncontaminated system(s) by evacuation and purging procedures currently recommended by refrigerant and refrigerant equipment manufacturers, and as specified herein, to remove small amounts of air and moisture. Systems containing moderate amounts of air, moisture, contaminated refrigerant or any foreign matter shall be considered contaminated systems.
c. Restore contaminated system(s) to clean condition including disassembly, component replacement, evacuation, flushing, purging, and re-charging using current refrigerant and refrigeration manufacturer's procedures. Restoring contaminated systems shall be at no additional cost to the Government.

4. Evacuation and Drying:
   a. After tests and cleaning have been completed and system proved tight, charge each circuit with dry clean refrigerant to gas pressure as recommended by the equipment manufacturer.
   b. Evacuate to 100 micron Hg and hold for 72 hours.
      1) Use laboratory type vacuum pump capable of holding absolute pressure of 50 micron Hg.
      2) Check the vacuum with a suitable mercury column gage.
   c. Admit another drying charge of refrigerant and allow 4 to 6 hours to absorb moisture and install dryer cores.
   d. Use second evacuation to remove all refrigerant and moisture.
   e. After second evacuation, charge system with refrigerant.
   f. Charge the system with refrigerant as required after final evacuation.

END OF SECTION

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SECTION 23 80 00
HVAC - EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes the following Heating, Ventilating and Air Conditioning equipment:
   1. Electric Unit Heaters.
   2. Electric Duct Heaters.
   3. Split System Air Conditioning Units.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 61 03 - Equipment - Basic Requirements.
   2. Section 23 05 93 - HVAC Systems - Balancing and Testing.
   3. Section 23 09 00 - Instrumentation and Control for HVAC Systems.
   4. Section 23 31 00 - HVAC - Ductwork.
   5. Section 23 74 36 - Refrigerant Piping Systems.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Air Movement and Control Association (AMCA).
   2. Air Conditioning and Refrigeration Institute (ARI).
   3. American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE):
      a. HVAC Applications Handbook, Chapter entitled "Sound and Vibration Control."
      b. 20, Methods of Testing for Rating Remote Mechanical-Draft Air-Cooled Refrigerant Condensers.
      c. 52.2, Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size.
   4. Canadian Standards Association (CSA).
   5. National Electrical Manufacturers Association (NEMA):
      a. 250, Enclosures for Electrical Equipment (1000 Volts Maximum).
   6. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code (NEC).
   7. National Roofing Contractors Association (NRCA).
   8. Underwriters Laboratories, Inc. (UL):
      a. 507, Standard for Electric Fans.

B. Miscellaneous:
   1. Gage thickness specified herein shall be manufacturer's standard gage for steel and Brown and Sharpe gage for non-ferrous metals.
   2. Corrosion protection of equipment to be as specified herein.

1.3 SUBMITTALS

A. Shop Drawings:
   1. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Fabrication and/or layout drawings.
   3. Product technical data including:
      a. Acknowledgement that products submitted meet requirements of standards referenced.
      b. Manufacturer's installation instructions.
      c. Wiring diagrams.
      d. Control diagrams.

e. Manufacturer's catalog cut sheets and technical data indicating performance, heating and cooling capacities, airflow capacity, motor sizes, power requirements, dimensions, materials of construction, weight, controls and accessories to be provided.
f. Corrosion-protection information.
g. Fan curves indicating airflow, static pressure, fan speed, efficiency and power consumption.
h. Sound data.
i. Vibration isolation.
j. Control description.

B. Contract Closeout Information:
1. Operation and Maintenance Data:
a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.4 CONTRACTOR RESPONSIBILITY

A. Equipment Supports:
1. Equipment supports and anchorage not detailed on the drawings shall be designed and provided by the Contractor or equipment manufacturer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor’s proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
1. Cooling coils - direct expansion:
a. Carrier.
b. Or equal.
2. Coil Coatings:
a. Luvata Electrofin (E-coat).
b. Or equal.
3. Unit heaters - electric:
a. Indeeco.
b. King Electric (ceiling recessed or surface mounted type).
c. Or equal.
4. Duct heaters - electric:
a. Indeeco.
b. Or equal.
5. Air-cooled condensing units - split systems, cooling only:
a. Carrier.
b. Or equal.
6. Air-handling units - split systems, cooling only:
a. Carrier.
b. Or equal.

2.2 GENERAL

A. All Manufactured Units:
1. Comply with Specification Section 01 61 03.

2. Factory wired and assembled.
3. Use fasteners made of same material as unit.
4. Fabricate motor assemblies and unit housings with vibration isolation assemblies:
    a. Type: As per Table 47, Chapter 48, ASHRAE HVAC Applications Handbook.

B. Where indicated in the equipment schedules on the drawings, equipment shall be constructed with corrosion-resistant materials or have corrosion-resistant coating.
1. Type:
    a. Corrosion-resistant materials:
        1) Aluminum.
        2) Stainless steel.
        3) FRP.
    b. Corrosion-resistant coating:
        1) Phenolic-based coating:
            a) 3 mil minimum dry thickness, air-dried coating, for surfaces exposed to temperatures less than 150 degrees F.
            b) 5 mil baked-on coating for heat transfer surfaces and surfaces exposed to temperatures greater than 150 degrees F.
            c) Factory applied.
            d) Provide factory certification of application.

2.3 MANUFACTURED UNITS

A. Equipment Coils:
1. Cooling coils - direct expansion:
    a. ARI certified.
    b. Material:
        1) Copper tubes with aluminum plate fins mechanically bonded to the tubes.
    c. Fin spacing: Minimum 80 fins per foot.
    d. Minimum standard operating limit: 250 psi.
    e. Size and capacity as scheduled on the drawings.
    f. Provide factory applied corrosion resistant coating on the coils where indicated on equipment schedules.
2. Heating coil - electric:
    a. ARI certified.
    b. 80% nickel, 20% chromium elements.
    c. Maximum heating density: 35 watts/SQIN.
    d. Built-in thermal protection.
    e. Airflow switch.
    f. Built-in circuit fusing.
    g. Control voltage transformer.
    h. Terminal block.
    i. Magnetic contactor.
    j. Fused disconnect switch.
    k. Multi-stage or SCR control as scheduled on the drawings.
    l. Single point electrical connection.
    m. Size and capacity as scheduled on Drawings.

B. Unit Heater - Electric: (Non-hazardous, non-corrosive locations)
1. Type: Forced-air, horizontal or vertical discharge as required by the application.
2. UL listed for non-rated areas.
3. Material:
    a. Cabinet: Galvanized steel with manufacturers standard powder coat finish (unless noted otherwise in the equipment schedules).
    b. Heating elements: Copper-clad steel.
4. Fan motors:
    a. See Specification Section 01 61 03.

b. Built-in automatic reset overload protection.
5. Dynamically balanced fan.
6. Built-in automatic reset cutout protection.
7. Built-in magnetic contactor, control transformer and control circuit.
8. Single or multi-stage as indicated on the equipment schedules on the drawings.
9. Accessories:
   a. Mounting bracket.
   b. Provide unit -mounted or wall-mounted thermostats as indicated in the schedules on the drawings. Thermostats shall be adjustable from 40°F to 90°F, with minimum 2°F deadband.
   c. Provide diffuser for vertical discharge heaters, where scheduled on the drawings.
   d. Refer to heater schedules on the drawings for additional accessories.
10. Electrical, fan motor, and airflow data as scheduled on Drawings.

C. Unit Heater - Electric: (Hazardous Locations)
1. Forced-air, horizontal discharge.
2. UL listed for Class 1, Group D, Division 1 areas (explosion proof construction).
3. Material:
   a. Heater housing: Type 316 stainless steel.
   b. Heat Exchanger: Type 316 stainless steel tubes with stainless steel or aluminum fins.
   c. Fan: Non-sparking aluminum.
   d. Junction box and control enclosure: Cast aluminum.
   e. Corrosion resistant hardware.
4. Fan motor:
   a. See Specification Section 01 61 03.
   b. Built-in overload protection.
   c. Explosion proof.
5. Built-in over temperature protection.
6. Built-in magnetic contactor, control transformer and control circuit.
7. Integral terminal block.
8. Accessories:
   a. Mounting bracket.
   b. Refer to heater schedules on the drawings for additional accessories.
9. Electrical, fan motor and airflow data as scheduled on Drawings.

D. Unit Heater - Electric: (Chemical Rooms)
1. UL listed, corrosion-resistant washable.
2. Material:
   a. Heater housing: Type 304 or 316 stainless steel.
   b. Heat exchanger: Type 316 stainless steel tubes with stainless steel or aluminum fins.
   c. Fan: Non-sparking aluminum.
   d. Junction box: NEMA 4X.
3. Fan motor:
   a. See Specification Section 01 61 03.
   b. Built-in overload protection.
   c. Totally enclosed.
4. Louvered outlet grille.
5. Built-in over temperature protection.
6. Built-in magnetic contactor, control transformer and control circuit.
7. Accessories:
   a. Mounting bracket: Stainless steel.
   b. Refer to heater schedules on the drawings for additional accessories.

E. Duct Heaters - Electric:
1. UL listed for zero clearance.
2. Furnish with control panel, factory wired and tested.
3. Materials:

a. Frame: Stainless steel.
b. Heating elements: 80% nickel and 20% chromium.
c. Element terminals: Stainless steel.
d. Insulators and bracket bushings: Non-porous ceramic.
e. Terminal box: Aluminized steel.

4. Heating elements:
   a. Coils: See paragraph(s) in Article 2.3, Equipment Coils.
5. Automatic reset thermal cutout.
6. Standard, manually replaceable, thermally operated secondary cutout.
7. Built-in magnetic contactor, control transformer and control circuit.
8. Multi-stage or SCR control as scheduled on the drawings.
9. Duct mounting: Flanged or slip-in type.
10. Size and capacity as scheduled on Drawings.
11. Accessories:
    a. Fan airflow switch for proof of airflow.
    b. Furnish with manufacturer's duct mounted thermostat or temperature sensor.
    c. Refer to heater schedules on the drawings for additional accessories.

F. Air-Cooled Condensing Units - Split System:
1. Condensers shall consist of coils, compressors, interconnecting piping, unit casing and controls. Units shall be factory assembled and run tested.
2. ARI rated.
3. UL listed.
4. Materials:
   a. Casing: Galvanized steel, with enamel finish.
   b. Mounting/lifting rails: Steel.
   c. Outdoor coil: Seamless copper tubing and aluminum fins.
   d. Fan blades: Aluminum, statically and dynamically balanced.
5. Weatherproof casing:
   a. Provide louvered coil guard to protect coil from hail and other flying debris.
   b. Bolted access panels.
6. Compressor:
   a. Hermetically sealed scroll type.
   b. Internal high and low pressure protection.
   c. Crankcase heater.
   d. Internal spring or neoprene rubber isolated mounts.
   e. Centrifugal oil pump.
   f. Built-in overload protection.
7. Condenser fans and motors:
   a. Vertical discharge.
   b. Direct drive.
   c. Statically and dynamically balanced.
   d. Motor:
      1) See Specification Section 01 61 03.
      2) Permanently lubricated bearings.
      3) Built-in current and thermal overload protection.
8. Built-in refrigerant filter dryer.
9. Built-in liquid line and gas line service valves with gage ports.
10. Outdoor coil:
    a. Fins mechanically bonded to tubing.
    b. Pressure tested to 450 psi.
    c. Provide factory applied e-coating (epoxy polymer) on condenser coils for corrosion protection, where scheduled on the drawings.
11. 24 V factory-wired and tested controls to include fusing and control power transformer.
12. Size and capacity as scheduled on Drawings.

G. Air-Handling Unit - Split Systems:
    1. Packaged air handling unit for direct expansion systems, consisting of unit cabinet supply fan, evaporator coil, expansion valve and integral controls, factory assembled, wired and tested. Unit orientation shall be as indicated on the drawings.
    2. ARI certified.
    3. UL listed.
    4. Unit Construction:
        a. Fabricate unit with galvanized steel channel posts and panels secured with mechanical fasteners, finished with enamel coating.
            1) Seal all panels, access doors, and ship sections with permanently applied bulb-type gasket.
            2) Loose shipped gasketing is not allowed.
        b. Construct panels and access doors at 2 inches nominal thickness; thermal broke double wall assembly, injected with foam insulation for an R-value of not less than R-13.
            1) Construct the outer panel and the inner liner of G90 galvanized steel.
        c. Panel deflection:
            1) Do not exceed L/240 ratio at 125% of design static pressure, maximum positive or negative 8 inches of static pressure.
            2) Measure deflection at the midpoint of the panel height.
        d. Casing leakage rate: Do not exceed 0.5 cfm per square foot of cabinet area at 6 inches of negative static pressure or 5 inches of positive static pressure.
        e. Module to module assembly:
            1) Provide an overlapping, full perimeter, insulated, internal splice joint sealed with bulb type gasketing on both mating modules to minimize on-site labor and meet indoor air quality standards.
        f. Entire unit: Provide a 4 inches full perimeter base rail or sub-base frame for structural rigidity and condensate trapping.
        g. Access doors:
            1) Flush mounted, bolted or hinged gasketed panels for access to filters, and internal serviceable components.
        h. Construct drain pans from stainless steel with cross break and double sloping pitch to drain connection.
            1) Provide drain pans under cooling coil section.
            2) Drain connection centerline: A minimum of 3 IN above the base rail to aid in proper condensate trapping.
            3) Drain connections that protrude from the base rail are not acceptable.
            4) Provide 2 inch thick insulation under drain pan.
    5. Fan section:
        a. Fans:
            1) Double-width, centrifugal type, forward curved fan blades.
            2) Statically and dynamically balanced.
            3) Solid steel shaft.
            4) 200,000 hour grease lubricated ball-bearings.
    6. Motors:
        a. See Specification Section 01 61 03.
        b. Internal spring or rubber isolated mounts.
    7. V-belts and drives sized for 150% motor capacity.
    8. Cooling coil section:
        a. Direct expansion: See paragraph(s) in Article 2.3, Equipment Coils.
        b. Factory installed and integrated thermostatic expansion valve.
    9. Filter section:
        a. Sized for 2-IN thick disposable pleated filters.
        b. Access doors for filter removal.
    10. Insulated drain pan with drain connections.
    11. Type: Draw through

12. Size, capacity and configuration as scheduled on Drawings.

2.4 SPARE PARTS

A. Provide two spare belts for each type of belt driven fan.

B. Provide three complete filter media changes for each type of filter. Filter media used during construction is in addition to this requirement.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install in accordance with Specification Section 01 61 03.

B. The Contractor shall not install any equipment or materials until submittals have been favorably reviewed. If any equipment or materials are installed prior to approval of the submittals, it shall be at the Contractor's risk.

C. Do not operate equipment without filters. Do not run equipment with dirty filter pressure drop more than twice clean filter pressure drop. Provide two complete sets of filters to be used during construction. The first set is to be used for start-up, testing and balancing. The second set shall be installed after final cleanup and acceptance by the Contracting Officer.

D. The Contractor shall start up all equipment and systems in accordance with manufacturer recommendations and shall make all adjustments so that the system is placed in proper operating condition.

E. Install fixed pitched drive sheave after sheave has been sized based on accepted test and balance report.

3.2 FIELD QUALITY CONTROL

A. Test and balance systems in accordance with Specification Section 23 05 93.

B. Properly maintain and service all equipment and systems until the particular equipment or the system has been accepted by the Contracting Officer and their representative.

C. This maintenance shall include compliance with the manufacturers operating and maintenance instructions as well as periodic checking and cleaning of the filters and the lubrication of moving parts and all required adjustments.

D. Test and Inspections:
1. After installing HVAC equipment and after electrical circuitry has been energized, test units for compliance with performance requirements.
2. After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.
3. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
4. Adjust and set supply airflow, supply air temperature setpoints, room temperature setpoints and other user defined parameters to match design values.
5. Complete installation and startup checks according to manufacturer's written instructions.
6. Inspect for visible damage to unit casing.
7. Inspect for visible damage to compressor, coils, and fans.
8. Inspect for visible damage to internal insulation.
9. Verify that labels are clearly visible.
10. Verify that clearances have been provided for servicing.
11. Verify that controls are connected and operable.
12. Verify that filters are installed.
13. Clean condenser coils and inspect for construction debris.
14. Inspect operation of motorized dampers.

15. Verify lubrication on fan and motor bearings.
16. Inspect fan-wheel rotation for movement in correct direction without vibration and binding.
17. Adjust fan belts to proper alignment and tension.

3.3 TRAINING:

A. Provide the service of a qualified manufacturer's representative to thoroughly train designated personnel in the operation and maintenance of systems and equipment furnished and installed under this section.

B. Refer to Section 01 79 00 for Operator training requirements.

C. Provide on-site training sessions, within the contract period, to train personnel designated by the Contracting Officer.

D. Coordinate training sessions with the Contracting Officer's representative.

E. Train the designated staff to enable them to do the following:
1. Understand system operation, control and optimizing routines.
2. Proficiently operate the systems furnished under this section.
3. Understand control system components and configuration.
4. Log on and off systems with password protected access.
5. Adjust and change system set points, time schedules, occupied and non-occupied schedules.
6. Recognize malfunctions of the system by observation of graphical or visual signals.
7. Understand system drawings and the Operation and Maintenance manual.
8. Understand the system layout and location of key components.

END OF SECTION

SECTION 23 81 29
VARIABLE REFRIGERANT FLOW HEAT PUMP SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Split system heat pumps (multi-zone and single zone systems):
      a. Outdoor heat pump units (heating or cooling model).
      b. Indoor ceiling cassette units.
      c. Indoor wall-mounted fan coil units.
      d. Indoor ducted air handling units and dedicated outside air units.

B. Related Specification Sections include but are not necessarily limited to:
   1. Section 01 61 03 - Equipment - Basic Requirements
   2. Section 23 05 93 - Testing, Adjusting and Balancing.
   3. Section 40 42 00 - Pipe, Duct and Equipment Insulation.
   4. Section 23 09 00 - Instrumentation and Controls for HVAC Systems
   5. Section 23 74 36 - Refrigerant Piping Systems.

1.2 QUALITY ASSURANCE

A. Referenced Standards:
   1. Air Movement and Control Association (AMCA):
      a. 99, Standards Handbook.
      b. 210, Laboratory Methods of Testing Fans for Rating (Unit shall bear AMCA Certified Rating Seal).
      c. 300, Reverberant Room Method for Sound Testing of Fans (Unit shall bear AMCA Certified Rating Seal).
      d. 301, Methods for Calculating Fan Sound Ratings from Laboratory Test Data.
      e. 500, Test Method for Louvers, Dampers, and Shutters.
   2. American National Standards Institute (ANSI):
      a. 9, Load Ratings and Fatigue Life for Ball Bearings.
      b. 11, Load Ratings and Fatigue Life for Roller Bearings.
      c. 900, Test Performance of Air Filter Units.
   3. Air-Conditioning Heating and Refrigeration Institute (AHRI):
      a. 1230, Variable Refrigerant Flow Multi-Split Air-Conditioners and Heat Pumps.
   4. American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE):
      a. 15, Safety Code for Mechanical Refrigeration.
   5. National Electrical Manufacturers Association (NEMA).
   6. National Fire Protection Association (NFPA):
      a. 70, National Electrical Code.
      b. 90A, Standard for the Installation of Air Conditioning and Ventilating Systems.
      c. 90B, Standard for the Installation of Warm Air Heating and Air Conditioning Systems.
   7. Sheet Metal and Air Conditioning Contractors' National Association, Inc. (SMACNA).
      a. Comply with applicable SMACNA standards including "HVAC Duct Construction Standards - Metal and Flexible".
   8. Underwriters Laboratories, Inc. (UL): Electrical Laboratories (ETL).

B. Manufacturer's Qualifications:
   1. Provide equipment that is the standard product of a manufacturer regularly engaged in the production of such units who issues complete catalog information on such products.
   2. Units shall not be fabricated by the Contractor.

1.3 SYSTEM DESCRIPTION

A. Heat Pumps:
   1. Multi-zone and single zone systems using variable speed inverter driven compressors.
   2. Multi-zone outdoor heat pumps shall be a VRV/VRF series heat or cool model.
      a. The system shall consist of multiple evaporators, condensing unit, interconnecting refrigerant piping, and PID DDC (direct digital controls).
      b. All zones are each capable of operating separately with individual temperature control.
   3. Operation of the system shall permit either cooling or heating of all the indoor units simultaneously.
   4. Each indoor unit or group of indoor units shall have independent temperature adjustment via a local wall-mounted system controller.

1.4 SUBMITTALS

A. Shop Drawings:
   1. See specification Section 01 33 23 for requirements for the mechanics and administration of the submittal process.
   2. Product technical data including:
      a. Submit manufacturer's technical product data for fan coil units, air handling units, and outdoor condensing units showing dimensions, weights, heating and cooling capacities, efficiency ratings, fan performance data, electrical requirements, materials of construction, installation instructions and sound data.
      b. Submit manufacturer cut sheets for accessories to be provided, including filter boxes and wall-mounted controllers.

B. Contract Closeout Information:
   1. Operation and Maintenance Data:
      a. See Specification Section 01 78 23 for requirements for the mechanics, administration, and the content of Operation and Maintenance Manual submittals.

1.5 ENVIRONMENTAL REQUIREMENTS

A. Do not operate units for any purpose, temporary or permanent, until ductwork is clean, filters are in place, bearings lubricated, and fan has been test run under observation.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Deliver Products to Site under Provisions of Division 01.
   1. Deliver units to the site in containers with manufacturer's stamp or label affixed.

B. Store/Protect Products under Provisions of Division 01 and According to Manufacturer’s Recommendation:
   1. Protect units against dirt, water, chemical, and mechanical damage.
   2. Do not install damaged units.
      a. Remove damaged equipment from the project site.

1.7 EXTENDED WARRANTY

A. For compressors only, under normal use and maintenance for comfort cooling and conditioning applications, the products will be free from defects in material or workmanship for a six-year period.
   1. This extended warranty for compressors is limited in duration to six years from the earlier to occur of:
      a. The date of original installation, whether or not actual use begins on that date, or;
      b. 24 months from the date of shipment, and applies to the compressor and compressor parts only.
   2. The effective date of this extended warranty shall be established as above.

1.8 CONTRACTOR RESPONSIBILITY:

A. Equipment Supports:
   1. Equipment supports and anchorage not detailed on the drawings shall be designed and provided by the Contractor or equipment manufacturer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers: Contracting Officer will consider Contractor's proposal of equivalent products when the specified salient characteristics are satisfied. If the specified salient characteristics are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
   1. Manufacturer listed, and other manufacturers not listed, but capable of meeting the requirements of this Specification Section, are expected to provide complete information showing full compliance with the specified salient characteristics.

B. Subject to compliance with the Contract Documents, the following manufacturers are acceptable:
   1. LG, Multi V.
   2. Or equal.

2.2 OUTDOOR AIR-COOLED HEAT PUMP CONDENSING UNIT

A. General:
   1. The condensing unit shall be factory assembled and pre-wired with all necessary electronic and refrigerant controls.
      a. The refrigeration circuit of the condensing unit shall consist of scroll compressors, motors, fans, condenser coil, electronic expansion valves, solenoid valves, 4-way valve, distribution headers, capillaries, filters, shut off valves, oil separators, service ports and refrigerant regulator.
   2. The outdoor unit shall have a high efficiency oil separator plus additional logic controls to ensure adequate oil volume in the compressor is maintained.
   3. For multi-zone systems, the connection ratio of indoor fan coil units to condensing unit shall follow manufacturer requirements.
   4. The system will automatically restart operation after a power failure and will not cause any settings to be lost, thus eliminating the need for reprogramming.
   5. The unit shall incorporate an auto-charging feature.
   6. The condensing unit shall be modular in design and should allow for side-by-side installation with minimum spacing.
   7. The following safety devices shall be included on condensing units: high pressure sensor and switch, low pressure sensor, control circuit fuses, crankcase heaters, overload relay, inverter overload protector, thermal overload protectors for compressor and fan motors, over current protection for the inverter and anti-recycling timers.
   8. To ensure the liquid refrigerant does not flash when supplying the various indoor units, the circuit shall be provided with a sub-cooling feature.
   9. Each system shall maintain continuous heating during oil return operation.
   10. The condensing unit shall be capable of heating operation at 0 degrees F dry bulb ambient temperature without additional low ambient controls or an auxiliary heat source.
   11. Except for motors, provide electrical components required as part of units, which have been listed and labeled by UL. This includes condensate pumps and other ancillary devices required for operation. The units shall be listed by ETL and bear the ETL label. See Specification Section 01 61 03 - Equipment - Basic Requirements for additional requirements.

B. Unit Cabinet:
   1. The condensing unit shall be completely weatherproof and corrosion resistant.

2. The unit shall be constructed from galvanized mild steel panels coated with a baked enamel finish.
3. Provide unit with base pan and manufacturer's base pan heater to collect condensation and minimize ice formation in sub-freezing conditions.

C. Fan:
1. The condensing unit shall consist of one or more propeller type, direct-drive fan motors that have variable speed operation via a DC (digitally commutating) inverter.
2. The fan discharge shall be horizontal or vertical discharge as indicated on the drawings and as required by the application.
3. The fan motor shall have inherent thermal overload protection and permanently lubricated bearings.
4. The fan motor shall be provided with a fan guard to prevent contact with moving parts.

D. Condenser Coil:
1. The condenser coil shall be manufactured from copper tubes expanded into aluminum fins to form a mechanical bond.
2. The heat exchanger coil shall be of a waffle louver fin and rifled bore tube design to ensure high efficiency performance.
3. Heating and cooling capacity shall be rated based on AHRI standards.
4. Coils shall be factory pressure and leak tested.
5. Coils shall be covered with a factory-applied anti-corrosion finish.

E. Compressor:
1. The inverter driven scroll hermetic compressors shall be variable speed, capable of changing the speed to follow the variations in total cooling and heating load as determined by the suction gas pressure measured in the condensing unit.
2. The capacity control range shall be as low as 4%, up to 100%.
3. Each non-inverter compressor shall also be of the hermetically sealed scroll type.
4. Each compressor shall be equipped with a crankcase heater, high pressure safety switch, and internal thermal overload protector.
5. Oil separators shall be standard in the equipment together with an intelligent oil management system.
6. The compressor shall be spring or neoprene rubber mounted to avoid transmission of vibration.

F. Electrical:
1. The power supply to the condensing unit shall be as scheduled on the drawings.
2. The control wiring shall be a two-wire transmission system, making it possible to connect multiple indoor units to one condensing unit with one 2-cable wire, thus simplifying the wiring installation.

2.3 INDOOR FAN COIL UNITS

A. Non-Ducted Ceiling Cassette Units:
1. General:
a. The indoor unit shall be completely factory assembled and tested.
b. The unit consists of supply fan, coil, filter, factory wiring, piping, electronic proportional expansion valve, control circuit board, fan motor thermal protector, flare piping connections, condensate drain pan, condensate drain pump, condensate safety shutoff and alarm, self-diagnostics, auto-restart function, 3-minute fused time delay, and test run switch.
c. Refer to the equipment schedules on the drawings for size, capacity and accessories to be provided.
2. Indoor Unit:
a. Ceiling cassette units shall be suitable for recessed ceiling installation, or suspended below the ceiling as indicated on the drawings.

b. Indoor unit and refrigerant pipes will be charged with dehydrated air prior to shipment from the factory.
c. Both refrigerant lines shall be insulated from the outdoor unit.
d. Where gravity drainage of condensate is not possible, the indoor units shall be equipped with a condensate pump.
1) The condensate pump shall provide up to 21 inches of lift and include a built in safety shutoff and alarm.
e. The indoor units shall be equipped with a return air thermistor.

3. Unit Cabinet:
a. The cabinet shall be constructed of galvanized steel with sound absorbing foamed polystyrene and polyethylene insulation.
b. Four-way air distribution type unless noted otherwise on the drawings. Air grilles shall be white, impact resistant plastic with a washable decoration panel for finished spaces.
c. Supply air is distributed via motorized louvers which can be horizontally and vertically adjusted from 0 to 90 degrees.
d. The 4-way supply air flow can be field modified to 3-way and 2-way airflow to accommodate various installation configurations including corner installations.

4. Fan:
a. The fan shall be direct-drive type with statically and dynamically balanced impeller.
b. Fans shall have multiple speeds available, user adjustable.
c. The fan motor shall have thermal overload protection.

5. Filter:
a. The return air shall be filtered by means of a washable long-life filter with mildew proof resin.

6. Coil:
a. Coils shall be of the direct expansion type constructed from copper tubes expanded into aluminum fins to form a mechanical bond.
b. The coil shall be of a waffle louver fin and high heat exchange, rifled bore tube design to ensure highly efficient performance.
c. The coil shall be a 2-row cross fin copper evaporator coil completely factory pressure tested.
d. The refrigerant connections shall be flare connections.
e. A thermistor will be located on the liquid and gas line.

7. Electrical:
a. Refer to schedules on the drawings for electrical power requirements.

8. Controls:
a. Furnish with manufacturer’s remote wall mounted controller.
b. Controls shall integrate with outdoor heat pump unit microprocessor controls.

9. Accessories:
a. Refer to the drawing schedules for additional accessories to be provided.

B. Ducted Air Handling Units:
1. General:
a. Indoor air handling units, configured for horizontal or vertical discharge as shown on the drawings.
b. Refer to the equipment schedules on the drawings for size, capacity and accessories required.
c. All units must have AMCA certified performance data for fans tested in the unit casings.
d. Ductwork shall comply with Section 23 31 00 - HVAC - Ductwork.

2. Indoor Unit:
a. The indoor unit shall be completely factory assembled and tested.
b. Air handling units shall consist of supply fan, coil, drain pan, condensate pump, factory wiring, piping, electronic proportional expansion valve, control circuit board, fan motor thermal protector, flare connections, self-diagnostics, auto-restart function, 3-minute fused time delay, and test run switch.

c. Indoor unit and refrigerant pipes will be charged with dehydrated air prior to shipment from the factory.
d. Both refrigerant lines shall be insulated from the outdoor unit.
e. The indoor units shall be equipped with a return air thermistor.
f. The indoor units shall be equipped with a corrosion resistant condensate drain pan and condensate pump.
g. The condensate pump shall provide up to 18 inches of lift from the center of the drain outlet.

3. Unit Cabinet:
   a. Constructed of a galvanized steel casing with baked enamel coating.
   b. Provide hanger brackets for horizontal suspended units.
   c. Minimum 1/2 inch thick, 1.5 pound internal insulation, foil faced.
      1) Insulation shall be attached with adhesive with all exposed edges coated to prevent erosion or use an insulation type not requiring protection.
   d. Insulated access panels for access to filters and serviceable components.
   e. Panels and joints sealed with permanent gasketing.
   f. Where required on the drawings, unit cabinet shall include space for optional electric heat strips.
4. Filter:
   a. The unit cabinet shall include a filter rack sized with 1-inch thick disposable pleated filters.
   b. Refer to drawing schedules for MERV type.
5. Fan:
   a. The fan shall be variable speed, direct-drive, with statically and dynamically balanced impeller.
   b. Multiple fan speed settings shall be available, user adjustable.
   c. The fan motor shall have thermal overload protection.
   d. The fan motor shall be equipped with adjustable external static pressure (ESP) settings.
6. Coil:
   a. Coils shall be ARI certified of the direct expansion type.
   b. All coil casings shall be galvanized steel or stainless steel.
   c. Coils shall be of the direct expansion type constructed from copper tubes expanded into aluminum fins to form a mechanical bond.
   d. The coils shall be pressure tested at the factory.
   e. The refrigerant connections shall be flare connections.
   f. A thermistor will be located on the liquid and gas line.
   g. A condensate pan shall be located under the coil.
7. Electrical:
   a. Refer to the schedules on the drawings for electrical power requirements.
8. Control:
   a. Furnish with manufacturer’s remote wall mounted controller.
   b. Controls shall integrate with outdoor heat pump unit microprocessor controls.
9. Accessories:
   a. Where indicated on the schedules, provide electric strip heater for supplemental heat with integrated controls.
   b. Where indicated on the schedules, provide a condensate pump with an 18 inch minimum lift, located in the condensate pan with a built-in safety alarm.
   c. Refer to the drawing schedules for additional accessories to provided.

C. Non-Ducted Wall-Mounted Fan Coil Units:
1. General:
   a. The indoor unit shall be completely factory assembled and tested.
   b. Refer to the equipment schedules on the drawings for size, capacity and accessories.
   c. Units shall include supply fan, coil, drain pan, condensate pump, factory wiring, piping, electronic proportional expansion valve, control circuit board, fan motor thermal overload protection, flare connections, auto-restart function.

2. Indoor Unit:
    a. The unit shall have an auto-swing louver which ensures efficient air distribution.
    b. Indoor unit and refrigerant pipes will be charged with dehydrated air prior to shipment from the factory.
    c. Both refrigerant lines shall be insulated from the outdoor unit.
3. Unit Cabinet:
    a. Constructed molded ABS plastic, with standard white finish.
    b. The cabinet shall be affixed to a factory supplied wall/ceiling hanging brackets and located in the conditioned space.
    c. The cabinet shall be constructed with sound absorbing foamed polystyrene and polyethylene insulation.
4. Fan:
    a. The fan shall be a direct-drive cross-flow fan, statically and dynamically balanced impeller, constructed of molded ABS plastic.
    b. Fans shall have multiple speeds available, user adjustable.
    c. Fan motor shall have thermal overload protection.
5. Filter:
    a. The return air shall be filtered by means of a washable long-life filter with mildew resistant resin.
6. Coil:
    a. Coils shall be of the direct expansion type constructed from copper tubes expanded into aluminum fins to form a mechanical bond.
    b. The coil shall be of a waffle louver fin and high heat exchange, rifled bore tube design to ensure highly efficient performance.
    c. The coil shall be a 2-row cross fin copper evaporator coil, factory pressure and leak tested.
    d. The refrigerant connections shall be flare connections.
    e. A thermistor will be located on the liquid and gas line.
7. Condensate drain pan:
    a. A corrosion resistant polystyrene condensate drain pan shall be included as standard.
8. Electrical:
    a. Refer to the drawing schedules for electrical power requirements.
9. Control:
    a. Furnish with manufacturer's remote wall mounted controller.
    b. Controls shall integrate with outdoor heat pump unit microprocessor controls.
10. Accessories:
    a. Where indicated on the schedules, provide a condensate pump with an 18 inch minimum lift, located in the condensate pan with a built-in safety alarm.
    b. Refer to the drawing schedules for additional accessories to provided.

2.4 CONTROLS

A. General:
    1. Provide wall-mounted controllers necessary to support a fully operating system.

B. Remote Controllers (Suitable for a Small System):
    1. Controllers shall maintain the optimal operation of the connected indoor units.
    2. Remote controllers shall be hard-wired to the equipment which they control.
    3. No addressing shall be required with the remote controllers.
    4. Basic Operation:
        a. Local remote controller shall control the following group of operations:
            1) On/off, operation mode (Cool, Heat, Fan, and Auto).
            2) Independent cooling and heating setpoints in the occupied mode.
            3) Independent cooling and heating setback setpoints in the unoccupied mode.
            4) Fan speed.
            5) Airflow direction.
            6) Function key lockout.

5. Programmability:
    a. Controller shall support 7-day programmable scheduling.
    b. The controller shall support auto-changeover mode allowing the optimal room temperature to be maintained by automatically switching the indoors unit's mode between cool and heat according to the room temperature and temperature setpoint.
    c. The room temperature shall be capable of being sensed at either, the remote controller, the indoor unit return air temperature sensor (default), or a remote temperature sensor.
6. Display Features:
    a. LED display.
    b. The controller shall display Operations Mode, Setpoint, and Fan Speed.
    c. System Status icons in large font.
    d. Room temperature display.
    e. On/Off status.
    f. Error codes displayed in the event of a system abnormality/error.
    g. Optional:
        1) The following system temperatures can be displayed to assist service personnel in troubleshooting:
            a) Return air temperature.
            b) Liquid line temperature.
            c) Gas line Temperature.
            d) Discharge Air Temperature (depending on fan coil).
            e) Temperature used for indoor unit control.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify that mounting surfaces are ready to receive work.

B. Verify that proper power supply is available.

3.2 INSTALLATION

A. Install in accordance with manufacturer's instructions.

B. Provide Layout Drawings of units, locations and power requirements to electrical installer.

C. Install air filters in unit during installation phase.
    1. Do not operate the unit without filters in place.

D. In the case of suspended units, mount the fan coil units on springs or from spring hangers as required and as shown on Drawings.

E. Unless noted otherwise on the drawings, provide 4 inches high concrete pad extending 6 inches beyond edge of condensing unit on all sides.
    1. Attach condensing unit to concrete pad per manufacturer's instructions.

F. Install 2 inch flexible duct connection at inlets and outlets of ducted units.

G. Install condensate drain piping and traps in accordance with manufacturer's instructions and as shown on the Drawings.
    1. Where drainage to gravity waste is not possible, provide condensate pumps.
    2. Do not route condensate lines over electrical cabinets.

H. Copper refrigerant piping shall be installed per the system manufacturer's instructions. Copper tubing and brazing requirements shall be per Specification Section 23 74 36 - Refrigerant Piping Systems. Refrigerant piping shall be insulated per 40 42 00 - Pipe, Duct and Equipment Insulation.

I. Control Wiring:

1. Communication wiring shall be terminated in a daisy chain design at the outdoor unit, which is then daisy chained to each indoor unit in the system and terminating at the farthest indoor unit.
2. The termination of the wiring shall be non-polar.
3. The remote control wiring shall run from the indoor unit control terminal block to the remote controller connected with that indoor unit.
4. Wiring shall be non-shielded, 2-conductor sheathed vinyl cord or cable, and 18 AWG stranded copper wire.

J. Electrical installer shall install all line voltage power wiring and conduit.
   1. Coordinate with Division 26 – Electrical work.

K. Install a new set of filters prior to final air balance and substantial completion.

3.3 MANUFACTURER'S START-UP SERVICES

A. The Manufacturer shall provide start-up service in the form of a factory trained Service Technician.
   1. The Service Technician shall verify correct installation, correct unit mounting, verify correct refrigerant pipe routing, fan rotation, verify spring isolator adjustments, verify correct control wiring and power wiring installation, start-up the fans, and check for proper operation.
   2. The Service Technician shall provide final adjustments to meet the specified performance requirements.
   3. System shall be tested and balanced per Specification Section 23 05 93, Testing, Adjusting and Balancing.

B. Fully staffed parts and service personnel shall be within eight hours travel from the job site.

3.4 TRAINING

A. Provide the service of a qualified manufacturer’s representative to thoroughly train designated personnel in the operation and maintenance of the systems and equipment furnished and installed under this section.

B. Refer to Section 01 79 00 for Operator training requirements.

C. Provide on-site training sessions, within the contract period, to train personnel designated by the Contracting Officer.

D. Coordinate training sessions with the Contracting Officer’s representative.

E. Train the designated staff to enable them to do the following:
   1. Understand system operation, control and optimizing routines.
   2. Proficiently operate the system.
   3. Understand control system components and configuration.
   4. Log on and off systems with password protected access.
   5. Access graphics, point reports, and logs.
   6. Adjust and change system set points, time schedules, occupied and non-occupied schedules.
   7. Recognize malfunctions of the system by observation of graphical or visual signals.
   8. Understand system drawings and the Operation and Maintenance manual.
   9. Understand the system layout and location of key components.

END OF SECTION

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